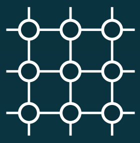
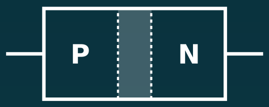
NATIONAL SCHOOL OF ENGINEERS OF SOUSSE (ENISO)
Department of Industrial Electronics

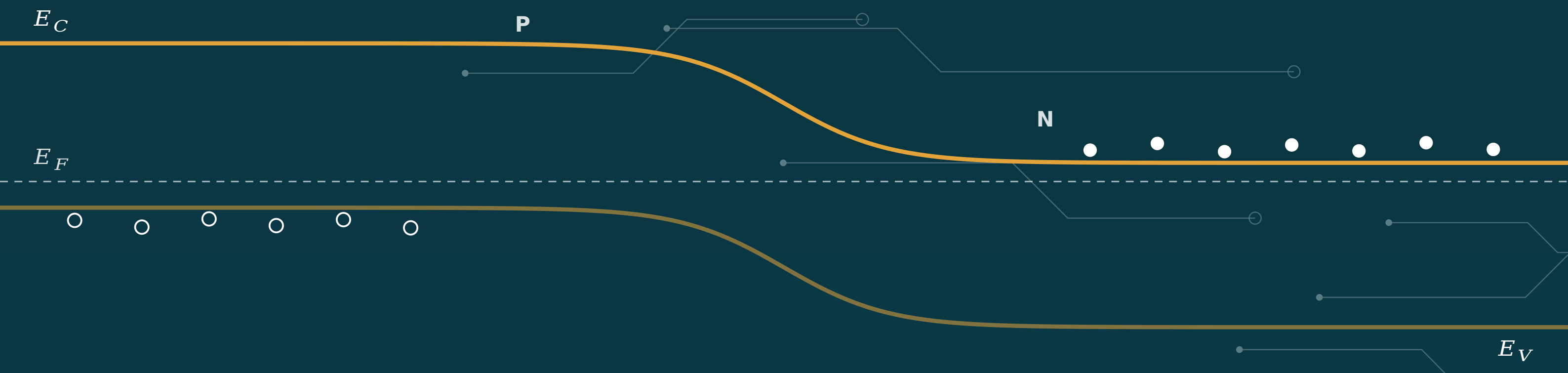


Crystal



Junction

Transistor



COURSE NOTES

Semiconductor Devices Physics

From the silicon crystal to the bipolar transistor

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Chapters 0 to 8: why semiconductor devices? · transport phenomena · semiconductors at equilibrium · energy bands and the Fermi level · carriers out of equilibrium · the PN junction at and out of equilibrium · the capacitances of the junction · the bipolar junction transistor

Annexes A and B: energy, one thread through the course · a uniformly biased intrinsic silicon rod

Contents

0 Why Semiconductor Devices? Goals, History and Technology 1

0.1 Physics as a tool, devices as the goal 3
0.2 Many devices, one material 3
0.3 Beneath the cartoon: the engineer’s level 3
0.4 A framework, not a catalogue 5
0.5 The three cornerstones of the course 5
0.6 From vacuum tubes to transistors: a short history 6
0.7 Modern devices, technologies and markets 10
0.8 How a chip is made 12
Summary 14
Questions for reflection 14
Answers 15
References 15

1 Transport Phenomena: Drift, Diffusion and Continuity 17

1.1 Introduction: who carries the current? 19
1.2 Current density: $J = q\,n\,v$ 20
1.3 Drift: the Drude model 21
1.4 From the local to the macroscopic Ohm’s law 24
1.5 How fast do the carriers really move? 25
1.6 Diffusion 25
1.7 The Einstein relation 27
1.8 Total current and the continuity equation 28
1.9 Case study: switching off a battery 29
Summary 33
Comprehension questions 34
Exercises 35
Answers 35
References 36

2 Pure and Doped Semiconductors at Equilibrium: A Qualitative View 37

2.1 What we observe 40
2.2 Bonding at absolute zero: metal and semiconductor 42
2.3 What metals teach us 43
2.4 Heat breaks bonds: electron–hole pairs 43
2.5 How many pairs? Reference values 45
2.6 The hole: a new carrier, a new kind of electricity 47
2.7 Recombination 49
2.8 Equilibrium: generation balances recombination 50
2.9 n and p doping: the picture at absolute zero 51
2.10 The solid imposes its rules: a hydrogen atom inside silicon 53
2.11 Doping a pure semiconductor: do the concentrations simply add? 56

2.12 Three laws for two unknowns: majority and minority carriers 57
2.13 What is still missing 60
Summary 61
Comprehension questions 63
Exercises 63
Answers 63
References 64

3 Pure and Doped Semiconductors at Equilibrium: Energy Bands and the Fermi Level 66

3.1 Two kinds of bonding, recalled 69
3.2 The metal: free electrons in a wire 70
3.3 A periodic potential: bands and gaps 71
3.4 Metal, insulator, semiconductor 72
3.5 Fermi–Dirac statistics and the Fermi level 76
3.6 Effective mass and density of states 77
3.7 From bands to densities: n and p 80
3.8 The pure semiconductor: the mass-action law, n_i and E_i 82
3.9 Intrinsic density and temperature 83
3.10 Where is the Fermi level of a pure semiconductor, and how does it move with temperature? 84
3.11 Donor and acceptor levels: the ionised donors N_D^+ 86
3.12 Doping leaves the bands unchanged: only E_F moves 88
3.13 Neutrality, N_D^+, n and p: the doped sample at any temperature 88
3.14 The position of the Fermi level in doped material 91
3.15 Clarifications 94
Summary 96
Comprehension questions 97
Exercises 98
Answers 98
References 99

4 Carriers Out of Equilibrium 100

4.1 Three situations: equilibrium, steady state, transient 103
4.2 The return to equilibrium 104
4.3 Light in: absorption and photoconductivity 106
4.4 Mechanisms: beyond the generic model 110
4.5 Carriers in space: electrons, holes and the continuity equations 115
4.6 Case studies in space: the illuminated surface 119
Summary 124
Comprehension questions 125
Exercises 126
Answers 126
References 127

5 The PN Junction at Equilibrium 129
5.1 What we want to explain 131
5.2 Two pieces apart 133
5.3 Contact: the depletion layer forms 134
5.4 The band diagrams of the two pieces 137
5.5 At equilibrium, the Fermi level is constant 138
5.6 The junction in bands 140
5.7 Carrier densities across the junction 143
5.8 The built-in voltage: two routes, one result 144
5.9 The electrostatics of the layer: width and peak field 146
5.10 Clarifications 150
Summary 151
Comprehension questions 152
Exercises 153
Answers 153
References 154

6 The PN Junction Out of Equilibrium 155
6.1 What changes when a voltage is applied 157
6.2 Where the voltage drops 158
6.3 The barrier and the four crossing fluxes 159
6.4 The bands under bias 162
6.5 Low injection and quasi-equilibrium in the SCR 162
6.6 The minority densities at the edges 165
6.7 Densities across the junction, forward and reverse 166
6.8 From the lamp to the junction 167
6.9 The neutral regions: minority carriers diffuse 169
6.10 The Shockley law of the long diode 170
6.11 Worked example: the short diode 171
6.12 The current inside the device 173
6.13 The characteristic and its limits 174
6.14 The bias scenario: everything together 175
6.15 The photovoltaic effect 178
6.16 The light-emitting diode 180
6.17 A short neutral region, and the transistor of 1947 182
Summary 183
Comprehension questions 184
Exercises 184
Answers 184
References 185

7 The Capacitance Effects of the Junction 186

7.1 Why a junction stores charge 188
7.2 The SCR under reverse bias 189
7.3 The transition capacitance C_T 190
7.4 Measuring a junction with $1/C_T^2$ 192
7.5 Engineering $C(V)$: the varicap 193
7.6 The stored minority charge 195
7.7 The diffusion capacitance C_D 197
7.8 C_T and C_D across the bias range 198
7.9 The small-signal diode 199
7.10 Switching a diode off 201
7.11 Transistors and logic circuits 202
Summary 203
Comprehension questions 203
Exercises 204
Answers 205
References 205

8 The Bipolar Junction Transistor 207

8.1 Technology and design constraints 209
8.2 The transistor at equilibrium: two junctions sharing a base 211
8.3 How the two junctions work together: the physical picture 212
8.4 The band diagram at equilibrium 215
8.5 Forward-active bias: barriers, SCRs and fields 216
8.6 Long base and short base, side by side 219
8.7 Several V_{BE}, several V_{CB} 221
8.8 From carrier transport to transistor action 223
8.9 Assumptions and conventions 224
8.10 The currents in forward-active operation 225
8.11 Both junctions at once: the Ebers–Moll equations 227
8.12 I_C against V_{CB}: the plateau 230
8.13 The BC junction forward biased: the knee 231
8.14 I_C against V_{CE}: a change of axis 232
8.15 The operating regions 233
8.16 Base-width modulation 235
8.17 The Early voltage and the output resistance 236
8.18 The two capacitances of the transistor 237
Summary 238
Comprehension questions 239
Exercises 240
Answers 240
References 241

ANNEXES

A **Energy: One Thread Through the Course** **242**
A.1 Energy describes a state; work describes a transfer 243
A.2 The zero is chosen: what a negative energy means 245
A.3 From trajectories to quantum states 247
A.4 From atomic levels to crystal bands 248
A.5 The band edge: what $E - E_C$ means 250
A.6 Allowed states are not occupied states 252
A.7 Transitions: where does the energy go? 253
A.8 What to retain 255
Summary 256
Comprehension questions 257
Exercises 257
Answers 257
References 258

B **A Uniformly Biased Intrinsic Silicon Rod** **259**
B.1 The question and a physical picture 260
B.2 Start from thermal equilibrium 261
B.3 Apply a voltage: the band edges tilt 261
B.4 State the boundary condition 261
B.5 Continuity: a steady current cannot accumulate charge 263
B.6 Combine continuity with drift–diffusion 263
B.7 The current is a drift current 264
B.8 What each assumption does 265
B.9 What happens to the Fermi level? 266
Summary 267
Comprehension questions 267
Exercises 267
Answers 268
References 268

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 0

Why Semiconductor Devices? Goals, History and Technology

Physics as a tool · one material, many devices · the three cornerstones of the course · from vacuum tubes to nanometre transistors · how a chip is made

WHERE WE ARE

This is the opening chapter of the course. It contains no derivation that later chapters need, but it says **what the course is for, how it is built, and where its subject comes from**. It can be read on its own, as a short introduction to semiconductor devices for engineers; it is also the map that the following chapters fill in.

CHAPTER OUTLINE

0.1 Physics as a tool, devices as the goal
0.2 Many devices, one material
0.3 Beneath the cartoon: the engineer’s level
0.4 A framework, not a catalogue
0.5 The three cornerstones of the course
0.6 From vacuum tubes to transistors: a short history
0.7 Modern devices, technologies and markets
0.8 How a chip is made
Summary · Questions for reflection · Answers · References · Credits

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Explain why a course on devices is a course on **semiconductors**, and why physics is used here as a **tool** rather than studied for itself.
- Position the course with respect to simplified explanations: what it adds, and why an engineer needs it.
- Present the **framework** of the course: two prototype devices and three cornerstones (currents, carrier concentrations, continuity equation).
- Tell the story that led from the vacuum tube to the integrated circuit, and give the orders of magnitude of today's industry and technology.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- classify devices by family (analog, digital, power) and by function (rectify, switch and amplify, interact with light, store), and say what they have in common;
- state the three cornerstones of the course and say which chapter delivers each;
- describe what a vacuum diode and a triode do, and why the transistor replaced them;
- place on a timeline the main inventions, from the 1874 rectifying contact to the integrated circuit and the MOS transistor, with their inventors;
- name the four building blocks of semiconductor devices and the main steps of chip fabrication, and explain what EUV lithography does and what a '3 nm' node is (and is not).

HOW THIS CHAPTER WORKS

- **A narrative, not a formula sheet.** The chapter tells why and how; the equations it shows are previews of what later chapters derive.
- **History with its physics.** Each step of the history is attached to the physical idea that made it possible.
- **Facts rebuilt and cited.** Tables and figures are drawn for this course from published facts; the sources are given at the end.

0.1 Physics as a tool, devices as the goal

> **THE QUESTION**
>
> *The course is called 'Semiconductor Devices Physics'. Is it a physics course?*

The word *physics* in the title can be misread. **Physics is not the goal of this course; it is its toolbox.** The goal is to understand **what happens inside the devices**: why a diode lets current through in one direction only, why a transistor amplifies, why a junction emits light or turns light into electricity. To answer these questions we need physical tools, carrier transport, statistics, generation and recombination, and we shall take from physics exactly what the devices require, no more.

This choice shapes everything that follows. A concept enters the course when a device needs it, and its treatment stops when the device is explained. The test of each chapter is therefore not 'is the physics complete?' but 'can we now say what the device does, and compute it?'

0.2 Many devices, one material

> **THE QUESTION**
>
> *What do a microprocessor, a solar panel, a phone charger and a white LED have in common?*

The word *device* covers an enormous family. Sorted by **field of electronics**, there are the devices of **analog** electronics (amplifiers, filters, sensors, radio-frequency circuits), of **digital** electronics (logic, processors, memories) and of **power** electronics (converters, motor drives, chargers). Sorted by **function**, some devices **rectify** (diodes), some **switch and amplify** (transistors), some **interact with light** (light-emitting diodes, lasers, solar cells, photodetectors), some **store information** (DRAM and flash memories).

The answer to the question is that all of them are **made of semiconductors**: mostly silicon, and for specific needs germanium, gallium arsenide, gallium nitride, silicon carbide or indium phosphide (Fig. 0.1). The devices are many and different; the material underneath is shared. **This is why the semiconductor itself is the central subject of the course**: understand the material and its few basic structures, and the devices become variations on a common theme.

0.3 Beneath the cartoon: the engineer's level

> **THE QUESTION**
>
> *There are excellent videos explaining how a transistor works. Why a whole course?*

A first picture of how a device works is easy to obtain today. Animated videos (the *Branch Education* channel is a good example), introductory books, or a question put to an AI assistant give, in a few minutes, a clear and largely correct **working model**: electrons flow here, a barrier forms there, the gate opens a channel. These pictures are useful, and we recommend them as a first contact.

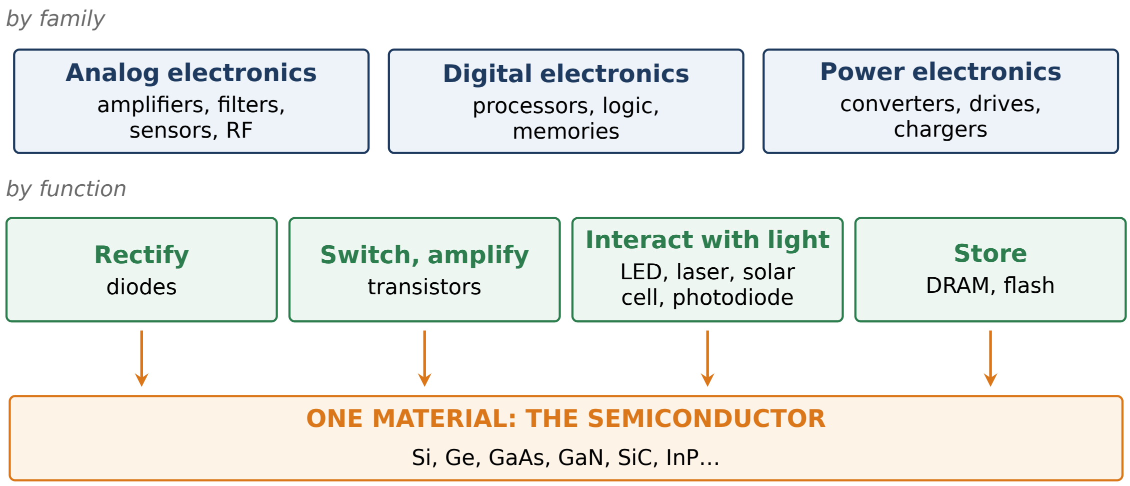
by family
Analog electronics
amplifiers, filters, sensors, RF
Digital electronics
processors, logic, memories
Power electronics
converters, drives, chargers
by function
Rectify
diodes
Switch, amplify
transistors
Interact with light
LED, laser, solar cell, photodiode
Store
DRAM, flash
ONE MATERIAL: THE SEMICONDUCTOR
Si, Ge, GaAs, GaN, SiC, InP...

Figure 0.1 – *The device landscape. Sorted by family or by function, devices rest on one class of materials, the semiconductors.*

They are not what this course does. A cartoon tells **what** happens; it cannot tell **how much**, **how fast**, **under which conditions** it stops being true, or **what to change** to make the device better. Those questions need the **driving mechanisms** and their **equations**. An engineer designs, sizes, diagnoses and improves; for that a working model is not enough, a solid scientific background is required. This is the level at which the course is written.

THE CARTOON AND THE ENGINEER

- **The cartoon** gives a qualitative picture of the mechanism. It is the right first step, and a good way to check one's intuition.
- **The engineer** needs the mechanism *and* its quantitative description: which carriers, how many, which currents, which equation governs them, and within which assumptions.
- The course goes from the first to the second: every mechanism is introduced with a picture, and then turned into an equation that can be solved.

0.4 A framework, not a catalogue

> **THE QUESTION**
>
> *Will the course study every device?*

No. What the course provides is a **framework for studying devices**, not a systematic review of all of them. A framework is a way of asking the questions and a set of tools to answer them; once acquired, it applies to devices the course never mentions.

The framework is built on **prototype examples**. The central one is the **PN junction**, the simplest structure in which the ideas of the course all appear together, and the basis of diodes, solar cells and LEDs. The second is the **transistor effect**, treated in the spirit of the **bipolar** transistor, which is two junctions sharing a thin region.

The **MOS effect**, on which today's digital circuits rely, is deliberately **not** treated. This is a choice, not a gap: a student who has mastered the framework, carriers, currents, junctions and the continuity equation, will find the MOS transistor accessible in a later course or in industry, where it is the daily tool of microelectronics and of the semiconductor industry.

0.5 The three cornerstones of the course

> **THE QUESTION**
>
> *Whatever the device, which questions shall we always ask?*

All the work of the course revolves around **three concepts**, which the next four chapters establish one after the other (Fig. 0.2).

- **Currents.** Wherever there is a device, there is a current; it is what the device delivers to the circuit. Chapter 1 builds the description of currents by drift in a field and by diffusion along a gradient, for instance for electrons:

$$J_n = q\,n\,\mu_n\,\mathcal{E} + q\,D_n\,\frac{\partial n}{\partial x} \tag{0.1}$$

- **Carrier concentrations.** A current is carried by mobile charges, whose concentrations n (electrons) and p (holes) must be known. Chapter 2 explains what the carriers of a semiconductor are and how many there are; Chapter 3 shows what fixes them in equilibrium, and Chapter 4 how they return to it.
- **A governing equation.** Concentrations change because carriers move (the divergence of the current) and because they are created and destroyed (generation G and recombination R). This balance is the **continuity equation**:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\,\frac{\partial J_n}{\partial x} + G - R \tag{0.2}$$

Developed for the excess minority carriers in a nearly neutral region, it becomes the **ambipolar equation** of Chapter 4, which describes how injected carriers spread and disappear:

$$\frac{\partial\,\Delta p}{\partial t} = D_a\,\frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{\tau} + g \tag{0.3}$$

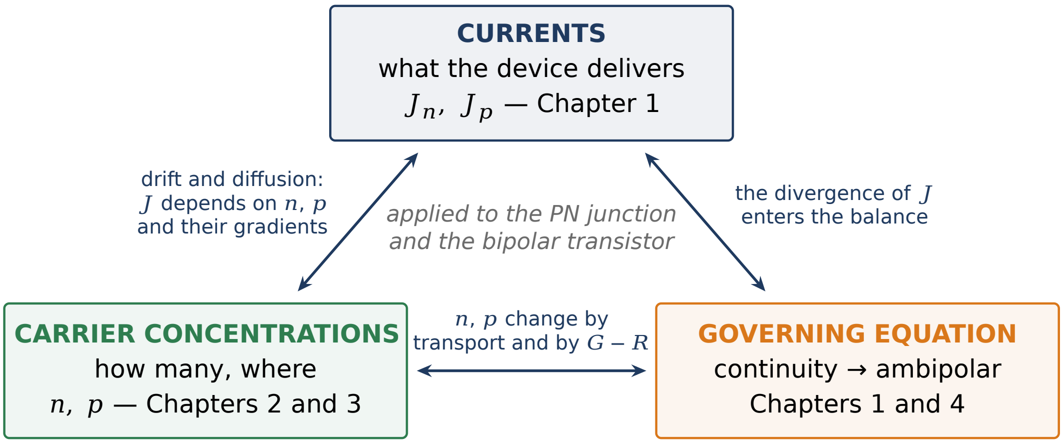
CURRENTS
what the device delivers
$J_n, \ J_p$ — Chapter 1
drift and diffusion:
J depends on n, p
and their gradients
applied to the PN junction
and the bipolar transistor
the divergence of J
enters the balance
CARRIER CONCENTRATIONS
how many, where
$n, \ p$ — Chapters 2 and 3
n, p change by
transport and by $G - R$
GOVERNING EQUATION
continuity → ambipolar
Chapters 1 and 4

Figure 0.2 – *The three cornerstones and their links. Every device of the course is analysed by asking the same three questions: which currents, which carrier concentrations, which equation connects them.*

Chapter	Question	Key result
0	Why devices, why semiconductors?	the framework and its history (this chapter)
1	How do carriers move?	drift and diffusion currents; Einstein relation; continuity equation
2	What are the carriers, and how many?	electrons and holes; doping; mass-action law
3	What fixes them in equilibrium?	bands; n and p of the pure semiconductor; Fermi level of doped material
4	How do they relax?	lifetime; ambipolar equation
next	Applying the framework	the PN junction (Chapters 5 to 7), then the bipolar transistor (Chapter 8)

Table 0.1 – *Road map of the course.*

0.6 From vacuum tubes to transistors: a short history

THE QUESTION

Before 1947 there were radios, long-distance telephones and even computers. What did they use, and why was it replaced?

0.6.1 The vacuum-tube era

Before 1947, the devices that performed electronic functions were **vacuum tubes**. They did two things: they acted as **valves**, letting current flow one way or blocking it, which makes them **switches** for digital computation; and they acted as **amplifiers**, making a weak signal strong.

The story starts with a lamp. In **1879 Thomas Edison** produced a practical incandescent lamp, a hot filament in a vacuum bulb. Around **1883** he noticed that a metal plate placed in the bulb collected a current when it was positive with respect to the filament, and none when it was negative: an **asymmetry**. The hot filament emits electrons (**thermionic emission**), and electrons can only travel from the filament to the plate.

In **1904 John Ambrose Fleming** turned this asymmetry into a device, the **thermionic valve** or vacuum **diode** (Fig. 0.3a). It was a breakthrough because it detected radio signals reliably and made **wireless telecommunication** practical. In **1906 Lee de Forest** placed a **grid** between filament and plate: a small voltage on the grid controls a large current to the plate. This is the **triode** (Fig. 0.3b), the first electronic amplifier and the first electronic switch. Amplification allowed AT&T to open the first transcontinental telephone line in 1915, with triode repeaters overcoming the attenuation of long lines; switching allowed the first electronic computers (ENIAC, 1945, with some 17 000 tubes) and IBM's vacuum-tube machines of the early 1950s.

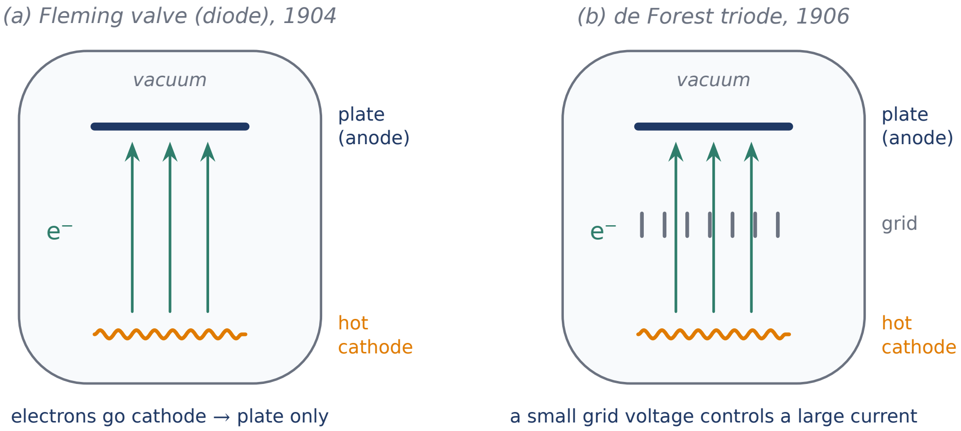
(a) Fleming valve (diode), 1904
vacuum
plate
(anode)
e⁻
hot
cathode
electrons go cathode → plate only
(b) de Forest triode, 1906
vacuum
plate
(anode)
grid
e⁻
hot
cathode
a small grid voltage controls a large current

Figure 0.3 – *(a) The vacuum diode: electrons emitted by the hot cathode reach the plate, never the reverse. (b) The triode: the grid controls the flow; this is the principle of amplification and switching.*

The current of a vacuum diode follows a law found by Child (1911) and Langmuir (1913), the first quantitative law of an electronic device: for a plate at a distance d from the cathode, with q and m the charge and mass of the electron,

$$J = \frac{4\varepsilon_0}{9}\sqrt{\frac{2q}{m}}\,\frac{V^{3/2}}{d^2} \tag{0.4}$$

Tubes worked, but they were **large, fragile, power-hungry** (the filament must be heated), slow to warm up, and limited in frequency. A computer of thousands of tubes failed every few hours.

0.6.2 A crystal that worked without being understood

Meanwhile a much smaller device existed. In **1874 Ferdinand Braun** discovered that a contact between a metal and certain crystals conducts better in one direction than in the other: the **metal–semiconductor rectifying contact**. From the early 1900s, notably with G. W. Pickard's silicon detector patented in **1906**, crystal detectors with a fine metal wire pressed on the crystal, the **cat's whisker**, were used in crystal radio sets. They rectified, sometimes well, but **nobody understood why**, and their behaviour depended on finding the right spot on the crystal. A device that is not understood cannot be designed. This is exactly the situation the physics of semiconductors would change.

0.6.3 The transistor, 1947

After the Second World War, Bell Laboratories, the research centre of AT&T, set up a group led by **William Shockley** to build a **solid-state replacement for the vacuum tube**. The idea was to imitate the grid of the triode: to control the current in a semiconductor with a field applied from outside, a **field effect**. It did not work, because charges trapped at the surface screened the field. Studying this failure, **John Bardeen and Walter Brattain** found a different effect: in **December 1947**, two gold contacts pressed very close together on a germanium crystal gave power gain. This **point-contact transistor** worked by **injecting minority carriers** across a junction, not by a field effect: it was the first **bipolar** transistor. In 1948–1949 Shockley worked out the theory of the p–n junction and of the **junction transistor**, which was realised in 1951. The three men received the Nobel Prize in Physics in 1956.

The transistor was immediately used to **amplify** and to **switch**, and it transformed the industry: devices became **smaller, cheaper, more reliable and more power-efficient**, needed no warm-up, and worked at **higher frequencies**.

0.6.4 Integration, the MOS transistor, light and memory

The next step was to put many transistors on one piece of semiconductor. In **1958 Jack Kilby** (Texas Instruments) built the first **integrated circuit**, on germanium, with its components connected by fine wires. In **1959 Robert Noyce** (Fairchild), using the **planar process** of Jean Hoerni, made a **monolithic** silicon integrated circuit whose interconnections are deposited on the chip itself: this is the way all chips are made today. Also in **1959**, at Bell Labs, **Mohamed Atalla and Dawon Kahng** made the first working **MOS transistor** (metal–oxide–semiconductor), reported in 1960. It realised the field effect that had failed in 1947, thanks to the clean silicon–oxide interface: a gate voltage controls the current through an insulator, a solid-state version of the vacuum-tube grid. Its complementary version (CMOS, 1963) became the basis of digital electronics.

Other effects followed in parallel. On the side of **light**: the first **silicon solar cell** (Bell Labs, 1954), the first visible **LED** and the semiconductor **laser** (1962), the bright **blue LED** (early 1990s) that completed white lighting. On the side of **memory**: the one-transistor **DRAM** cell (R. Dennard, 1966), the first commercial DRAM chip (1970), the floating-gate memory (1967) that led to **flash** memory in the 1980s. Figure 0.4 gathers the main steps.

0.6.5 The equations behind the story

Each era of this story has its law (Fig. 0.5). **Ohm's law** (1827) describes conduction globally; Chapter 1 writes it locally, with the current density **J** and the electric field $\mathcal{E}$:

$$U = R\,I \qquad \Longleftrightarrow \qquad \vec{J} = \sigma\,\vec{\mathcal{E}} \tag{0.5}$$

Drude (1900) explained the conductivity by free electrons of concentration n that collide every τ_c on average, the starting point of Chapter 1:

$$\sigma = \frac{n\,q^2\,\tau_c}{m} \tag{0.6}$$

The **Child–Langmuir** law (1911–1913) gave the current of the vacuum tube, equation (0.4). And **Shockley** (1949) derived the current of the p–n junction, the law of the

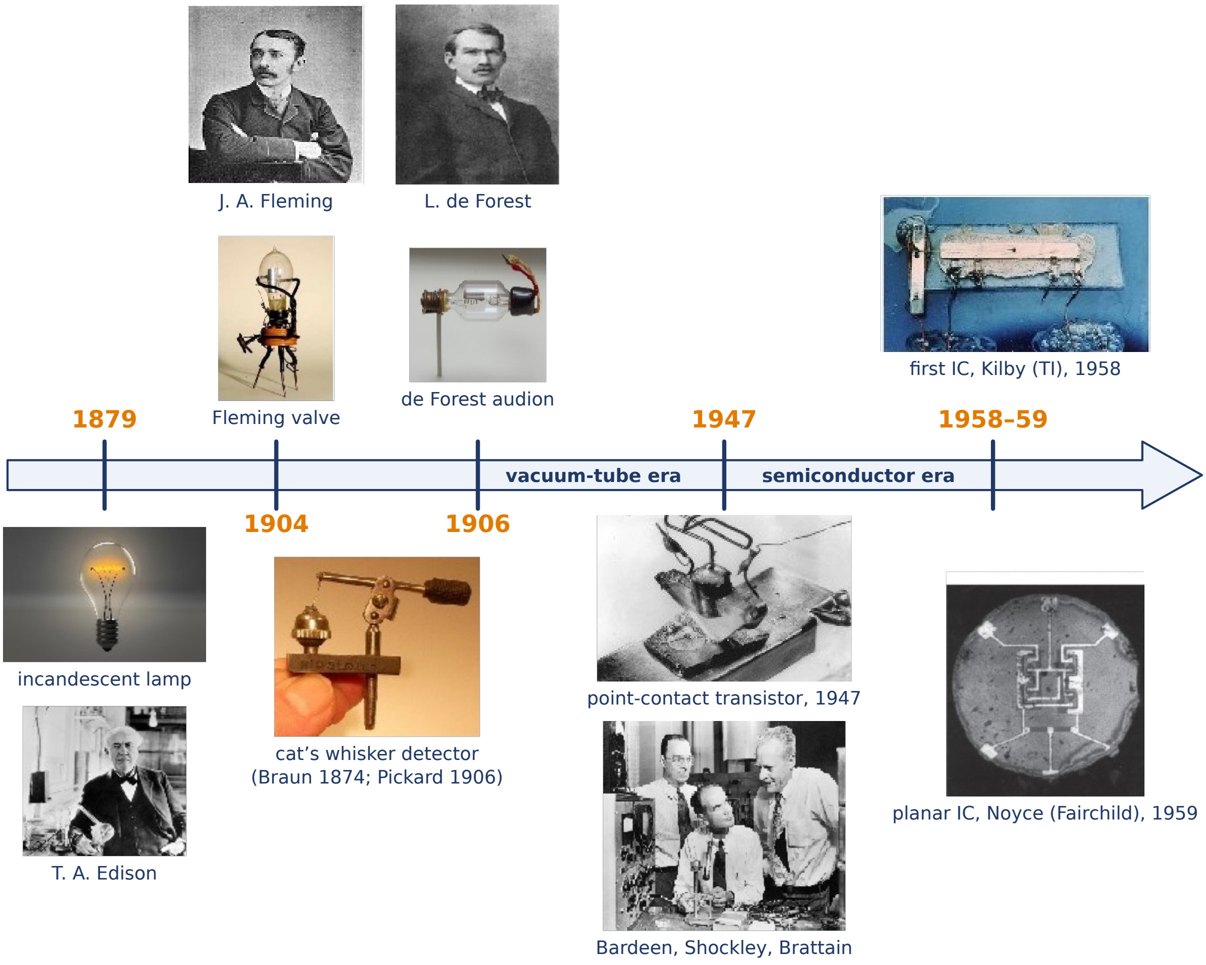
J. A. Fleming
L. de Forest
first IC, Kilby (TI), 1958
Fleming valve
de Forest audion
1879
1947
1958–59
vacuum-tube era
semiconductor era
1904
1906
incandescent lamp
cat’s whisker detector
(Braun 1874; Pickard 1906)
point-contact transistor, 1947
planar IC, Noyce (Fairchild), 1959
T. A. Edison
Bardeen, Shockley, Brattain

Figure 0.4 – *From the vacuum tube to the integrated circuit. Above the axis, the vacuum-tube devices and the first integrated circuit; below, the precursors and the transistor.*

semiconductor diode, which this course aims to understand and derive; η is an ideality factor close to 1 and V_T the thermal voltage:

$$I = I_s\left[\exp\left(\frac{V}{\eta V_T}\right) - 1\right], \qquad V_T = \frac{kT}{q} \approx 26\ \mathrm{mV} \tag{0.7}$$

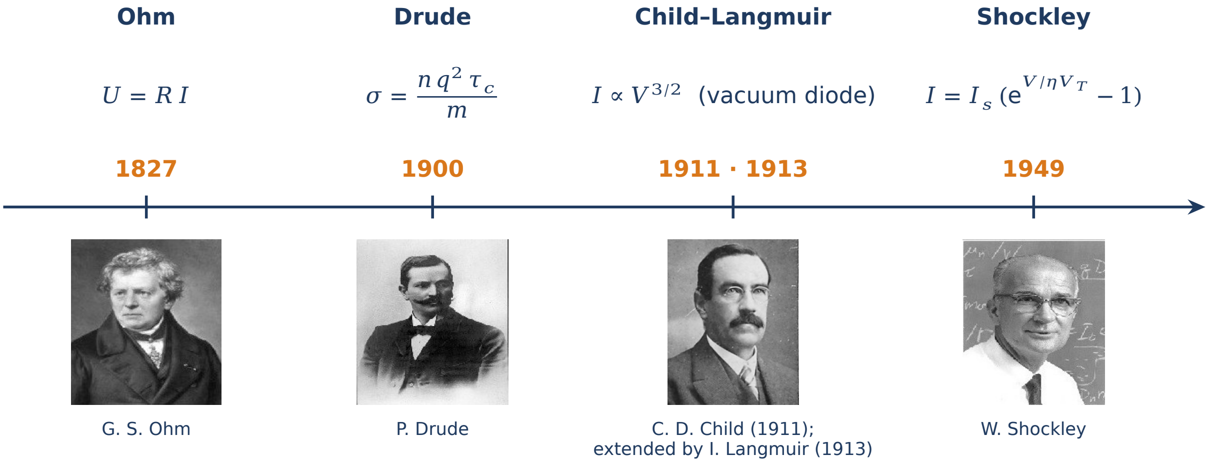
Ohm
Drude
Child–Langmuir
Shockley
$U = R\,I$
$\sigma = \frac{n\,q^2\,\tau_c}{m}$
$I \propto V^{3/2}$ (vacuum diode)
$I = I_s\,(e^{V/\eta V_T} - 1)$
1827
1900
1911 · 1913
1949
G. S. Ohm
P. Drude
C. D. Child (1911);
extended by I. Langmuir (1913)
W. Shockley

Figure 0.5 – *Four laws, four steps in the understanding of electronic conduction.*

0.7 Modern devices, technologies and markets

> ***THE QUESTION***
>
> *Since 1947, which devices and which manufacturing techniques have built the industry, and how large has it become?*

Tables 0.2 and 0.3 list the main **devices** and the main **technologies**, with their inventors. They are drawn up for this course from the historical record, as compiled in the references (Neamen; Sze and Lee).

Year	Device	Inventors
1874	Metal–semiconductor rectifying contact	F. Braun
1907	Electroluminescence of a crystal (first LED effect, SiC)	H. J. Round
1947	Point-contact transistor	J. Bardeen, W. Brattain
1949	p–n junction theory; junction (bipolar) transistor	W. Shockley
1954	Silicon solar cell	D. Chapin, C. Fuller, G. Pearson
1957	Heterojunction bipolar transistor (concept)	H. Kroemer
1958	Tunnel diode	L. Esaki
1958–59	Integrated circuit; monolithic planar IC	J. Kilby; R. Noyce
1959	MOS field-effect transistor	M. Atalla, D. Kahng
1962	Semiconductor laser; visible LED	R. Hall et al.; N. Holonyak
1963	CMOS logic	F. Wanlass, C. T. Sah
1966	One-transistor DRAM cell	R. Dennard
1967	Floating-gate non-volatile memory	D. Kahng, S. M. Sze
1969	Charge-coupled device (image sensor)	W. Boyle, G. Smith
1971	Microprocessor (Intel 4004)	M. Hoff, F. Faggin, S. Mazor, M. Shima
1993	Bright blue LED (GaN)	S. Nakamura; I. Akasaki, H. Amano

Table 0.2 *– Key semiconductor devices.*

The four building blocks. Behind this long list, very few structures are at work (Fig. 0.6). The **metal–semiconductor** contact (rectifying or ohmic), the **p–n junction**, the **heterojunction** between two different semiconductors, and the **metal–oxide–semiconductor** structure. Almost every device is one of them, or a combination: a bipolar transistor is two p–n junctions, a MOS transistor is a MOS structure with two junctions, a laser diode is a set of heterojunctions. This is why a course that understands the p–n junction already holds the key to most of the field.

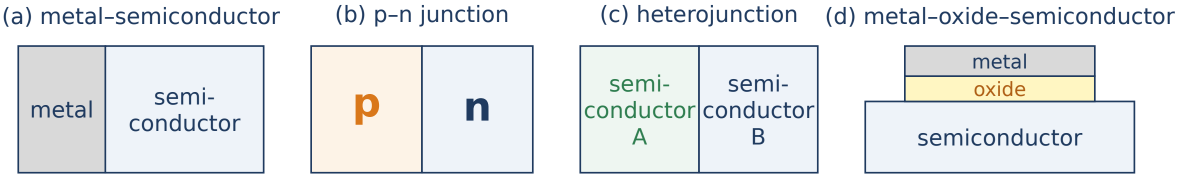

Figure 0.6 *– The four building blocks of semiconductor devices.*

A market that entered every industry. Semiconductors are no longer confined to electronics: they are in cars, energy, medicine, industry and telecommunications. World

Year	Technology	Inventors / first users
1918	Czochralski crystal growth	J. Czochralski
1925	Bridgman crystal growth	P. W. Bridgman
1952	Zone refining (purification)	W. G. Pfann
1957	Photoresist lithography	J. Andrus
1957	Oxide masking for diffusion	C. Frosch, L. Derick
1957	Epitaxial growth by chemical vapour deposition	J. J. Sheftal et al.
1959	Planar process	J. Hoerni
1969	Self-aligned polysilicon gate	R. Kerwin et al.
1969	Metal-organic chemical vapour deposition (MOCVD)	H. Manasevit
1971	Molecular-beam epitaxy	A. Y. Cho
1971	Dry (plasma) etching	S. Irving et al.
1989	Chemical–mechanical polishing (CMP)	B. Davari et al.
1997	Copper interconnects in production	IBM
2011	FinFET (tri-gate) transistors in production	Intel
2019	EUV lithography in volume production	ASML; TSMC, Samsung
2022	Gate-all-around (nanosheet) transistors in production	Samsung

Table 0.3 – *Key semiconductor technologies.*

sales have grown by a factor of about twelve in 34 years (Fig. 0.7).

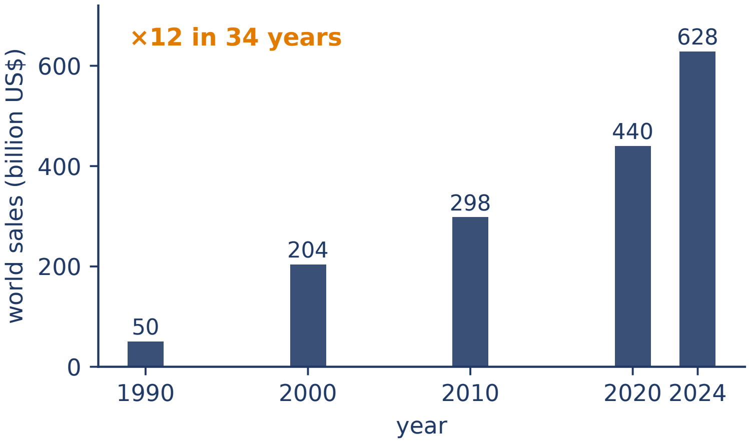

Figure 0.7 – *World semiconductor sales (approximate values, WSTS and SIA data).*

Shrinking dimensions. In 1965 Gordon Moore observed that the number of components on a chip doubled every year; in 1975 he revised the pace to every two years. This **Moore’s law** is not a law of physics but an industrial roadmap, kept for half a century by reducing the dimensions: the minimum feature size went from about 10 µm in 1971 to a few tens of nanometres, and the capacity of a DRAM chip rose from 1 kbit to tens of gigabits (Fig. 0.8).

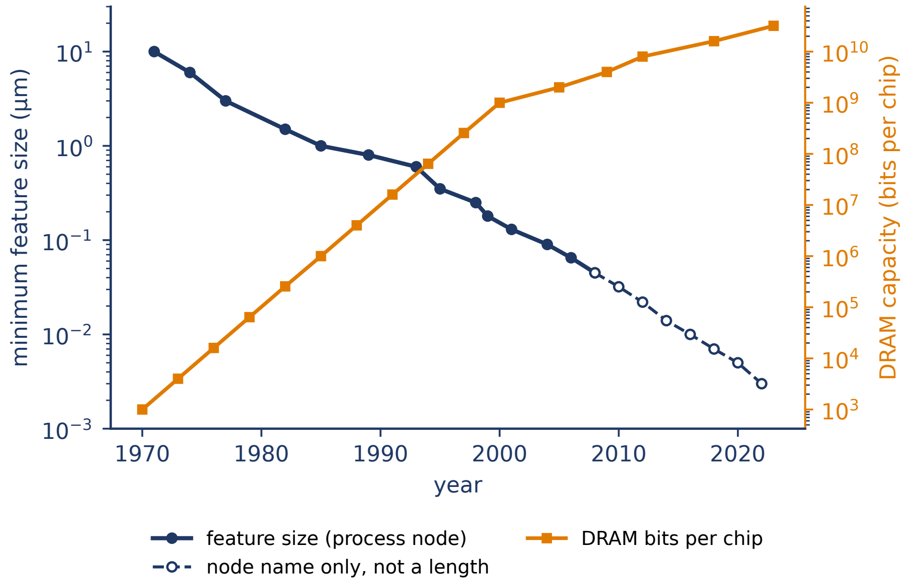

Figure 0.8 – *Minimum feature size (left) and DRAM capacity per chip (right) against year. After about 2010 the node names (dashed) no longer correspond to a physical length.*

0.8 How a chip is made

> ***THE QUESTION***
>
> *How do we go from sand to a chip carrying billions of transistors?*

The flow is summarised in Fig. 0.9. **Silicon** is extracted from quartz sand (SiO_2), then purified into **polysilicon** of extreme purity, about one foreign atom per billion. It is melted and pulled into a **single crystal** by the **Czochralski** method: a seed crystal is dipped into the melt and slowly raised while rotating, producing an **ingot** up to 300 mm in diameter. The ingot is sliced into **wafers**, which are polished to a mirror finish.

Devices are then built **at the surface** of the wafer, layer by layer. The **front end** creates the transistors by repeating a few operations many times: **oxidation** (growing SiO_2), **lithography** (printing a pattern), **etching** (removing material where the pattern says), **doping** (by diffusion or ion implantation), **deposition** of thin films, and **chemical-mechanical polishing** to flatten the surface. The **back end** builds the copper **interconnects**, a dozen or more metal levels that wire the transistors together. Finally the chips are tested, cut and packaged.

Lithography is the key step, because it fixes the dimensions (Fig. 0.10). A light-sensitive **resist** is spread on the wafer and exposed through a **mask** carrying the pattern; the exposed (or unexposed) resist is dissolved; the pattern is then transferred into the layer underneath by **etching**, and the resist is removed. A chip needs several dozen such cycles.

The smallest pattern that light can print is of the order of its wavelength divided by the numerical aperture of the optics. The industry has therefore moved to shorter and shorter wavelengths (Fig. 0.11): mercury-lamp lines at 436 and 365 nm, then excimer lasers at 248 and 193 nm, then 193 nm light through water (immersion), and finally **extreme ultraviolet (EUV) light at 13.5 nm**, in volume production since 2019. EUV light is absorbed by air and by glass: it is produced by a tin plasma and focused by mirrors in

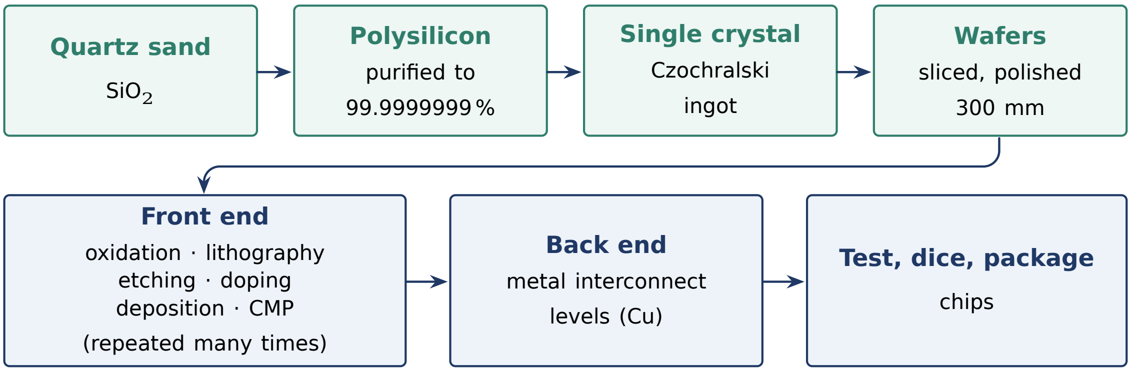
Quartz sand
SiO_2
Polysilicon
purified to
99.9999999 %
Single crystal
Czochralski
ingot
Wafers
sliced, polished
300 mm
Front end
oxidation · lithography
etching · doping
deposition · CMP
(repeated many times)
Back end
metal interconnect
levels (Cu)
Test, dice, package
chips

Figure 0.9 *– From sand to chip: the main steps of fabrication.*

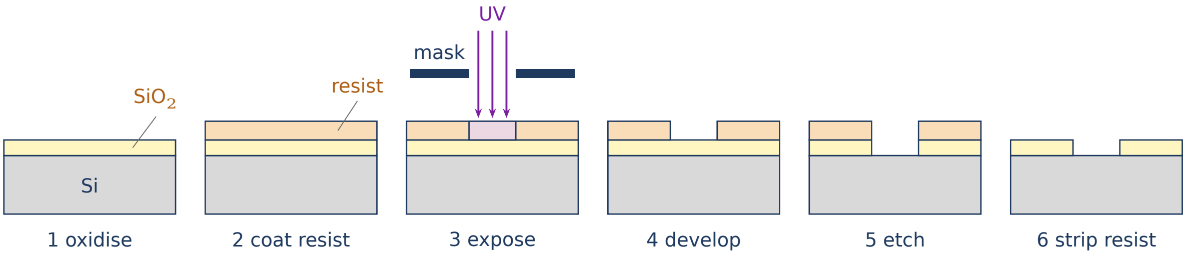
SiO_2
Si
resist
mask
UV
1 oxidise
2 coat resist
3 expose
4 develop
5 etch
6 strip resist

Figure 0.10 *– A lithography cycle: oxidise, coat with resist, expose through a mask, develop, etch the oxide, remove the resist.*

vacuum. The only manufacturer of these machines is the Dutch company **ASML**.

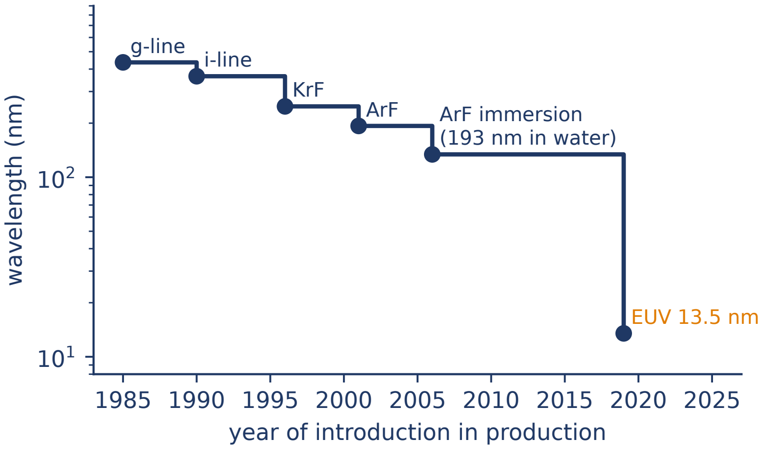

Figure 0.11 *– Wavelength of the light used in production lithography.*

BE AWARE OF…

Two points of accuracy. First, the EUV machine performs the **lithography**, i.e. the exposure of the pattern; **etching** is a separate step done afterwards in other equipment. Second, a name such as ‘3 nm’ designates a **technology generation** (a node), not a dimension of the transistor. In current ‘3 nm’ processes the contacted gate pitch is of the order of 45–50 nm and the tightest metal pitch of the order of 20–25 nm; no feature is 3 nm long. Since about 2010, node names have followed the historical sequence rather than a physical length.

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Purpose.** The course aims to understand devices; physics is the toolbox, used as far as the devices require.
- **One material.** Analog, digital and power devices, and devices that rectify, amplify, interact with light or store, are all made of semiconductors: the semiconductor is the central subject.
- **Level.** The course goes beneath the qualitative ‘cartoon’ to the mechanisms and their equations, the background an engineer needs.
- **Framework.** Prototype devices: the PN junction and the bipolar transistor. The MOS effect is not treated but becomes accessible through the framework.
- **Cornerstones.** Currents (Chapter 1), carrier concentrations (Chapters 2 and 3), continuity equation $\partial n/\partial t = (1/q)\,\partial J_n/\partial x + G - R$, developed into the ambipolar equation (Chapter 4).
- **History.** Edison lamp (1879) and Edison effect; Fleming diode (1904); de Forest triode (1906); Braun’s rectifying contact (1874) and the crystal detector; point-contact transistor (Bardeen, Brattain, 1947) and junction transistor (Shockley, 1949–1951); integrated circuit (Kilby 1958, Noyce 1959); MOS transistor (Atalla, Kahng, 1959).
- **Industry.** Four building blocks (metal–semiconductor, p–n, heterojunction, MOS); sales multiplied by about 12 since 1990; feature sizes divided by more than 100 since 1971.
- **Fabrication.** Sand → polysilicon → Czochralski ingot → wafers → front end (oxidation, lithography, etching, doping, deposition, CMP) → back end (interconnects) → chips. EUV lithography uses 13.5 nm light; node names are not lengths.

Questions for reflection

These questions check the ideas of the chapter; answers are given below.

Q1. Why is a course on semiconductor devices mainly a course on semiconductors?

Q2. What can an animated explanation of a transistor not tell an engineer?

Q3. Why does the course study the PN junction and the bipolar transistor rather than a long list of devices? Why is the MOS transistor not a gap?

Q4. Name the three cornerstones of the course and the chapter that delivers each.

Q5. What asymmetry did Edison observe, and which device did Fleming build on it? What did de Forest add?

Q6. The 1947 team was trying to build a field-effect device. What did they obtain instead, and when was the field effect finally realised?

Q7. List four advantages of the transistor over the vacuum tube.

Q8. What does an EUV machine do, and what does the name '3 nm node' mean?

Answers

Q1. Because devices of every family and function are made of the same class of materials; understanding the material and its few basic structures explains the devices.

Q2. How much current, how fast, under which conditions the picture fails, and what to change to improve the device: these need the mechanisms and their equations.

Q3. The course gives a framework; the PN junction contains all its ideas and the bipolar transistor is two junctions. The MOS transistor relies on the same carriers, currents and junctions, so it becomes accessible once the framework is mastered.

Q4. Currents (Chapter 1); carrier concentrations (Chapters 2 and 3); the continuity equation, developed into the ambipolar equation (Chapters 1 and 4).

Q5. A plate in the lamp collects current only when positive with respect to the hot filament (thermionic emission). Fleming built the vacuum diode (1904); de Forest added a grid (1906), giving the triode, an amplifier and a switch.

Q6. They obtained bipolar action, minority-carrier injection, in the point-contact transistor (December 1947). The field effect was realised in 1959 by Atalla and Kahng with the MOS transistor.

Q7. Smaller, cheaper, more reliable, more power-efficient (no heated filament), no warm-up, higher frequencies.

Q8. It performs lithography with 13.5 nm light, i.e. prints the pattern; etching is a separate step. '3 nm' is the name of a technology generation, not a physical dimension.

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- M. Riordan and L. Hoddeson, *Crystal Fire: The Invention of the Transistor and the Birth of the Information Age*, Norton, 1997.
- G. E. Moore, 'Cramming more components onto integrated circuits', *Electronics*, 38(8), 1965.
- World Semiconductor Trade Statistics (WSTS), market reports (www.wsts.org).
- Branch Education (video channel), animated explanations of electronic devices, recommended as a first contact.

CREDITS AND ACKNOWLEDGEMENTS

Images. The portraits and photographs of Figures 0.4 and 0.5 are historical photographs; their sources must be confirmed before any public distribution of this text. All other figures and all tables were drawn for this course from published facts, cited above.

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WHERE THIS LEADS

Chapter 1 opens the toolbox with the first cornerstone: how carriers move, how currents are written, and how the continuity equation expresses the balance of carriers. The devices of this chapter will come back, explained, once the framework is in place.

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 1

Transport Phenomena: Drift, Diffusion and Continuity

Current density · the Drude model · Ohm's law · thermal velocity · diffusion · Einstein relation · continuity equation · switching off a battery

WHERE WE ARE

Chapter 0 told the story of semiconductor devices and drew the technological landscape. Whatever a device does, rectify, amplify, switch, detect or emit light, it does it through **currents**: currents are what we measure at its terminals and what we design. This chapter comes immediately after the introduction for that reason. Before asking what the carriers of a semiconductor are (Chapter 2) and how many of them there are (Chapters 2 and 3), we fix the road map of the quantities involved and the laws that turn them into currents.

We therefore treat a **generic carrier**, of charge q, density n and mass m, without saying what it is. Most of our intuition comes from metals, where it is a free electron; the same tools will apply to the electrons and holes of semiconductors and to the ions of a liquid. Two causes make carriers flow, a **field** (drift) and a **concentration gradient** (diffusion), and one bookkeeping law, **continuity**, keeps track of where they go.

CHAPTER OUTLINE

1.1 Introduction: who carries the current?
1.2 Current density: $J = q\,n\,v$
1.3 Drift: the Drude model
  1.3.1 Free acceleration and collisions
  1.3.2 Local Ohm's law, conductivity and mobility
  1.3.3 How fast is the steady state reached?
1.4 From the local to the macroscopic Ohm's law
1.5 How fast do the carriers really move?
1.6 Diffusion
1.7 The Einstein relation
1.8 Total current and the continuity equation
1.9 Case study: switching off a battery
  Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Relate the current to what the carriers are and do: derive $J = q\,n\,v$ from the definition $i = \mathrm{d}q/\mathrm{d}t$.
- Explain with the Drude model why a constant field gives a constant velocity, and derive the local Ohm's law, the conductivity and the mobility, stressing the role of

the **collision time**.

- Connect the local law $J = \sigma\mathcal{E}$ to the familiar $V = RI$.
- Compare the drift velocity with the thermal velocity, and show that the random thermal motion is the origin of **diffusion**.
- Derive the diffusion current, the **Einstein relation** and the **continuity equation** for a fixed number of carriers, and use them, with Gauss's law and the inertia of the carriers, to follow what happens when a battery is switched off.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- name the carriers in a metal, a semiconductor and an electrolyte, and say why the course is interested in currents;
- derive $J = q\,n\,v$ with a wire, a cross-section and the distance $v\,\mathrm{d}t$;
- write the Drude equation of motion, derive $v_d = q\tau_c\mathcal{E}/m$, $\mu = |q|\tau_c/m$, $\sigma = n|q|\mu$ and $J = \sigma\mathcal{E}$, and compute τ_c from a measured conductivity;
- go from $J = \sigma\mathcal{E}$ to $R = L/(\sigma S)$;
- estimate the thermal velocity from $\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}kT$ and the mean free path, and compare them with the drift velocity;
- derive the diffusion flux $\Phi = -D\,\mathrm{d}n/\mathrm{d}x$ and $D \approx v_{th}\,l$, and the Einstein relation $D/\mu = kT/|q|$ in two ways;
- write the continuity equation for a fixed number of carriers, $\partial c/\partial t = -\partial\Phi/\partial x$, and deduce charge conservation $\partial\rho/\partial t + \nabla\cdot\vec{J} = 0$;
- explain why the field in a wire survives the opening of a switch, combine Drude (with inertia), continuity and Gauss into $\ddot{\rho} + \dot{\rho}/\tau_c + \omega_p^2\rho = 0$, and tell RC relaxation ($\tau_d = \varepsilon/\sigma$) from plasma oscillation with the number $\omega_p\tau_c$; explain why the bulk stays neutral while it carries a current, and write the equation of the charge on the ends, $L_k\ddot{Q} + R\dot{Q} + Q/C = E$.

HOW THIS CHAPTER WORKS

- **Question before formula.** Every section opens with the question it answers, and the derivations proceed by small questions: what crosses the section? what stops the acceleration? how can random motion produce a net flow?
- **Generic carrier.** The charge q is algebraic ($q = -e$ for an electron, $+e$ for a hole or a positive ion). The laws are written once and specified later (Chapter 4 for electrons and holes). From Chapter 2 on, q denotes the elementary charge itself, a positive number, and the sign of each carrier is written explicitly: $-q$ for an electron, $+q$ for a hole.
- **Metals as the reference.** Numbers are given for copper, the conductor everyone knows; semiconductors and electrolytes appear as comparisons.

Symbol	Meaning	Remark
q, e	algebraic charge of a carrier; elementary charge	$q = -e$ (electron), $+e$ (hole); $e = 1.602 \times 10^{-19}$ C
n (or c)	carrier density (concentration)	m^{-3} or cm^{-3}
m	carrier mass	effective mass m^* in a crystal (Ch. 3)
i, J	current; current density	A; A/m² (or A/cm²)
$\mathcal{E}$, V	electric field; potential, voltage	V/m; V
v_d, v_{th}	drift velocity; thermal velocity	$v_d \ll v_{th}$
τ_c, l	mean time between collisions; mean free path	$l = v_{th}\tau_c$; not the lifetime τ of Ch. 2, 4
μ, σ, ρ	mobility; conductivity; resistivity	$\mu > 0$; $\sigma = n\lvert q\rvert\mu$; $\rho = 1/\sigma$
Φ, D	particle flux; diffusion coefficient	$\mathrm{m}^{-2}\,\mathrm{s}^{-1}$; m²/s (or cm²/s)
ε, τ_d, ω_p	permittivity; dielectric relaxation time; plasma frequency	$\tau_d = \varepsilon/\sigma$; $\omega_p^2 = nq^2/(\varepsilon m)$
k, T	Boltzmann constant; temperature	$kT/e = 25.9$ mV at 300 K

Notation. *Copper at room temperature:* $n = 8.5 \times 10^{28}\ \mathrm{m}^{-3}$ *(one free electron per atom),* $\sigma = 5.96\times10^{7}$ *S/m. Electron mass* $m_0 = 9.11\times10^{-31}$ *kg,* $k = 1.381\times10^{-23}$ *J/K,* $\varepsilon_0 = 8.854\times10^{-12}$ *F/m.*

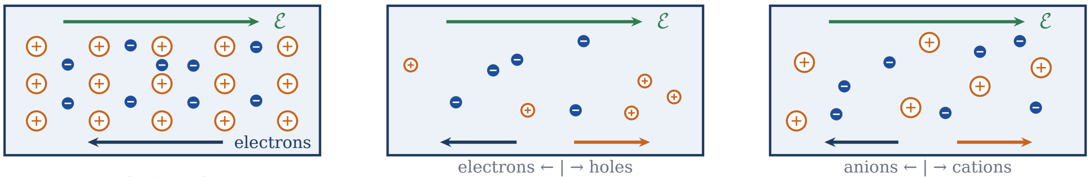
1.1 Introduction: who carries the current?

THE QUESTION

A current is charge in motion. In a given material, which charges move, and what do we need to know about them to compute a current?

The answer depends on the material (Fig. 1.1).

- In a **metal**, each atom releases one or more valence electrons to the whole crystal. These **free electrons** carry the current; the positive ions stay in place.
- In a **semiconductor**, there are two kinds of mobile carriers, **electrons** and **holes**, of opposite charges. Where they come from and how many there are is the subject of Chapters 2 and 3.
- In a **liquid electrolyte** (salt water, the acid of a battery), the carriers are **ions**, positive and negative, which drift in opposite directions.


ℰ
electrons
ℰ
electrons ← | → holes
ℰ
anions ← | → cations

(a) metal: free electrons (fixed cations)

(b) semiconductor: electrons and holes (Ch. 2, 3)

(c) electrolyte: positive and negative ions

Figure 1.1 *– Three conductors, three kinds of carriers. (a) Metal: free electrons among fixed cations. (b) Semiconductor: electrons and holes. (c) Electrolyte: positive and negative ions. In all three, a field makes positive carriers drift along it and negative carriers against it.*

In all three cases the same few quantities decide the current: **how many** carriers there are per unit volume, **what charge** they carry, and **how fast** they move on average. The velocity in turn has two possible causes, an electric field and a non-uniform concentration. Figure 1.2 is the road map of the chapter; we shall fill each box in turn. Our primary interest is the current in devices, so every section ends on a current density.

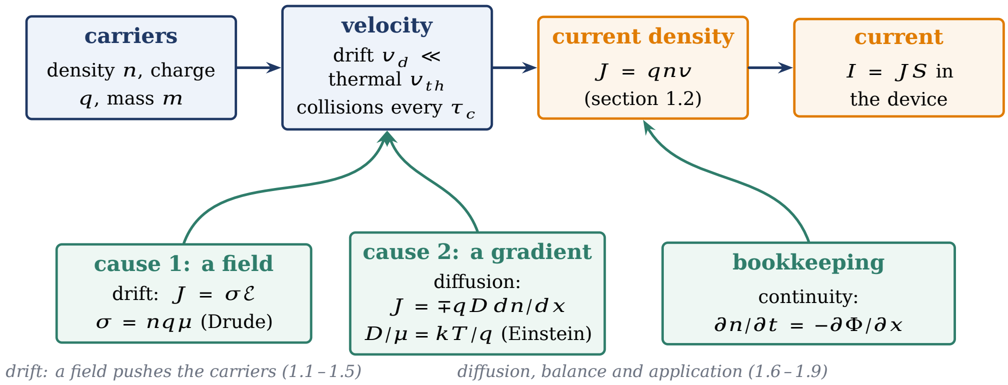

carriers
density n, charge q, mass m
velocity
drift $v_d \ll$ thermal v_{th}
collisions every τ_c
current density
$J = qnv$
(section 1.2)
current
$I = JS$ in the device
cause 1: a field
drift: $J = \sigma \mathcal{E}$
$\sigma = nq\mu$ (Drude)
cause 2: a gradient
diffusion:
$J = \mp q D\, dn/dx$
$D/\mu = kT/q$ (Einstein)
bookkeeping
continuity:
$\partial n/\partial t = -\partial \Phi/\partial x$
drift: a field pushes the carriers (1.1 – 1.5)
diffusion, balance and application (1.6 – 1.9)

Figure 1.2 *– The road map of the chapter. Carriers (top left) acquire a velocity; a velocity gives a current density, and the current density gives the current measured at the terminals. Two causes produce a velocity, a field (drift) and a gradient (diffusion); the continuity equation keeps the count of a fixed number of carriers.*

1.2 Current density: $J = q\,n\,v$

THE QUESTION

Carriers of charge q, density n, all moving at the velocity v along a wire. What current flows through a cross-section?

Start from the definition of current: the charge that crosses a surface per unit time,

$$i = \frac{\mathrm{d}q_{\text{tot}}}{\mathrm{d}t}. \tag{1.1}$$

Take a cross-section S of the wire (Fig. 1.3) and a short interval $\mathrm{d}t$. *Which carriers cross S during* $\mathrm{d}t$? Those that are close enough: during $\mathrm{d}t$ each carrier travels the distance $v\,\mathrm{d}t$, so exactly the carriers contained in the cylinder of base S and length $v\,\mathrm{d}t$ just upstream of S cross it. *How many are they?* The cylinder has volume $S\,v\,\mathrm{d}t$, so it contains $\mathrm{d}N = n\,S\,v\,\mathrm{d}t$ carriers, which carry the charge

$$\mathrm{d}q_{\text{tot}} = q\,\mathrm{d}N = q\,n\,S\,v\,\mathrm{d}t. \tag{1.2}$$

Dividing by dt gives the current, and dividing by S the **current density** $J = i/S$:

$$i = q\,n\,v\,S \quad\Longrightarrow\quad J = q\,n\,v \qquad (\vec{J} = q\,n\,\vec{v}). \tag{1.3}$$

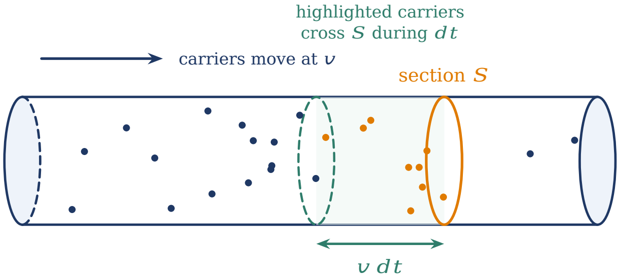


Figure 1.3 *– Counting the charge that crosses a section. During* dt*, the carriers inside the cylinder of length* v dt *(highlighted) cross the section* S*:* $\mathrm{d}N = n\,S\,v\,\mathrm{d}t$*.*

Three remarks make this formula a tool.

- **Signs.** q is algebraic. For electrons, $q = -e$: the current density points **opposite** to their velocity. Positive and negative carriers moving in opposite directions give currents in the **same** direction.
- **Several kinds of carriers** add their contributions: $J = \sum_k q_k n_k v_k$. In a semiconductor, $J = J_n + J_p$; in an electrolyte, cations plus anions.
- v **is an average.** Individual carriers move in all directions (Section 1.5); in (1.3) v is the mean velocity of the population, the only one that carries a net charge across S.

EXAMPLE 1.1 — HOW FAST DO THE ELECTRONS DRIFT IN A COPPER WIRE?

A current of 1 A in a copper wire of cross-section 1 mm²: $J = 10^6$ A/m². With $n = 8.5\times10^{28}$ m^{-3}, $v = J/(en) = 10^6/(1.602\times10^{-19}\times8.5\times10^{28}) = 7.3\times10^{-5}$ m/s, **less than a tenth of a millimetre per second**. A huge number of carriers moving very slowly makes an ordinary current. Keep this number in mind for Section 1.5.

1.3 Drift: the Drude model

THE QUESTION

A field $\mathcal{E}$ acts on the carriers. What velocity do they acquire, and why does a constant voltage give a constant current?

1.3.1 Free acceleration and collisions

What would a free carrier do? The force $q\mathcal{E}$ gives it the acceleration $q\mathcal{E}/m$, and its velocity would grow without limit: $v = q\mathcal{E}t/m$. Then by (1.3) the current would grow with time under a constant voltage. This is not what we observe: a resistor connected to a battery carries a **constant** current. Something must stop the growth.

Drude's answer (1900) is **collisions**. The carriers do not fly freely: they collide with the lattice (its vibrations and its defects). The model rests on two assumptions.

- Between two collisions, the carrier moves freely and is accelerated by the field.
- A collision makes the carrier **forget** its velocity: just after it, the velocity points in a random direction and its average over many carriers is zero. A carrier has the probability $\mathrm{d}t/\tau_c$ of colliding during $\mathrm{d}t$; τ_c is the **mean time between two successive collisions**, the key parameter of the model.

Each carrier therefore gains velocity along the field, loses it at a collision, starts again (Fig. 1.4). *What is the average velocity of the population?* Follow the mean momentum $p = m\langle v\rangle$ during $\mathrm{d}t$. A fraction $\mathrm{d}t/\tau_c$ of the carriers collides and loses, on average, all its momentum; the others keep their momentum and gain $q\mathcal{E}\,\mathrm{d}t$:

$$p(t+\mathrm{d}t) = \left(1 - \frac{\mathrm{d}t}{\tau_c}\right)\left(p(t) + q\mathcal{E}\,\mathrm{d}t\right) \quad \Longrightarrow \quad \frac{\mathrm{d}p}{\mathrm{d}t} = q\mathcal{E} - \frac{p}{\tau_c}$$

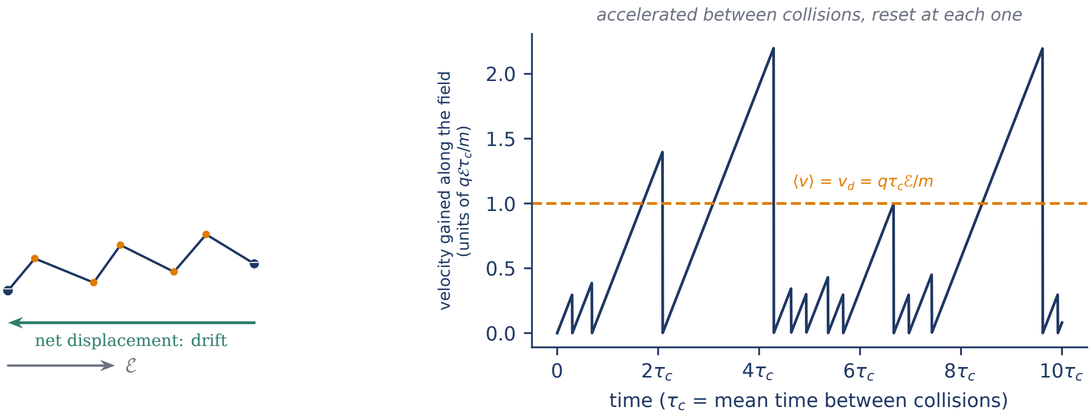
to first order in $\mathrm{d}t$. This is the **Drude equation**: Newton's law with a friction term,

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = q\mathcal{E} - \frac{mv}{\tau_c}. \tag{1.4}$$

The collisions act as a friction force proportional to the velocity. In a constant field the velocity stops growing when friction balances the force, $\mathrm{d}v/\mathrm{d}t = 0$:

$$v_d = \frac{q\,\tau_c}{m}\,\mathcal{E}. \tag{1.5}$$

This is the **drift velocity**: proportional to the field, and to the time τ_c during which the field can act before a collision wipes out the gain. The longer the carrier flies freely, the faster it drifts.



(a) The path of one electron: straight flights interrupted by collisions (dots), with a slow net displacement against the field.

(b) The velocity gained along the field grows linearly during each free flight and is reset at each collision; on average it is $q\mathcal{E}\tau_c/m$.

Figure 1.4 *– The Drude picture. (a) The path of one electron: straight flights interrupted by collisions (dots), with a slow net displacement against the field. (b) The velocity gained along the field grows linearly during each free flight and is reset at each collision; on average it is $q\mathcal{E}\tau_c/m$.*

1.3.2 Local Ohm’s law, conductivity and mobility

Insert the drift velocity (1.5) into $J = q\,n\,v$:

$$J = q\,n\,\frac{q\tau_c}{m}\,\mathcal{E} = \frac{n\,q^2\tau_c}{m}\,\mathcal{E}. \tag{1.6}$$

The current density is proportional to the field: this is **Ohm’s law in its local form**. Two quantities are worth naming.

- The **mobility** μ, the drift velocity per unit field. We define it positive, $\mu = |q|\tau_c/m$, so that $v_d = +\mu\mathcal{E}$ for a positive carrier and $v_d = -\mu\mathcal{E}$ for a negative one.
- The **conductivity** σ, the current density per unit field, and its inverse the resistivity $\rho = 1/\sigma$.

$$J = \sigma\,\mathcal{E}, \qquad \sigma = n\,|q|\,\mu = \frac{n\,q^2\tau_c}{m}, \qquad \mu = \frac{|q|\,\tau_c}{m}. \tag{1.7}$$

Notice q^2: whatever the sign of the carriers, the current flows **along** the field. The conductivity separates what the material offers, **how many** carriers (n), from how **easily** they move (μ, hence τ_c). Semiconductors will be interesting precisely because n can be changed by many orders of magnitude.

EXAMPLE 1.2 — THE COLLISION TIME AND THE MOBILITY OF COPPER

From the measured conductivity, $\tau_c = \sigma m/(ne^2) = 5.96\times10^7\times9.11\times10^{-31}/(8.5\times10^{28}\times(1.602\times10^{-19})^2) = 2.5\times10^{-14}$ s. Between two collisions an electron of copper flies for about **25 femtoseconds**.

Its mobility is $\mu = e\tau_c/m = 4.4\times10^{-3}$ m²/(V·s) = 44 cm²/(V·s). Compare with electrons in silicon, 1350 cm²/(V·s) (Chapter 2): copper conducts far better because it has far more carriers, about 10^7 times more than silicon doped with 10^{16} donors per cm³ and 10^{13} times more than pure silicon (Chapter 2), not because they move more easily.

1.3.3 How fast is the steady state reached?

Equation (1.4) also says how the drift is established. Switch the field on at $t = 0$ with $v(0) = 0$: the solution is

$$v(t) = v_d\left(1 - \mathrm{e}^{-t/\tau_c}\right), \tag{1.8}$$

and if the field is switched off when the carriers drift at v_d, $v(t) = v_d\,\mathrm{e}^{-t/\tau_c}$ (Fig. 1.5). The same τ_c that sets the value of the mobility sets the **response time** of the carriers. With $\tau_c \approx 10^{-14}$ s, the steady state is reached in a few tens of femtoseconds: for any signal slower than about 10^{12} Hz, Ohm’s law holds at every instant. (Switching the field off is an idealisation here: opening a switch does *not* remove the field in a wire, as the case study of Section 1.9 shows.)

BE AWARE OF…

τ_c is a collision time, not a lifetime. Here a carrier does not disappear at a collision; it only loses the memory of its velocity. In Chapter 2 the same kind of reasoning (probability $\mathrm{d}t/\tau$ of an event during $\mathrm{d}t$) will be used with a different event, recombination, where the carrier really disappears. The two times are unrelated and differ by many orders of magnitude.

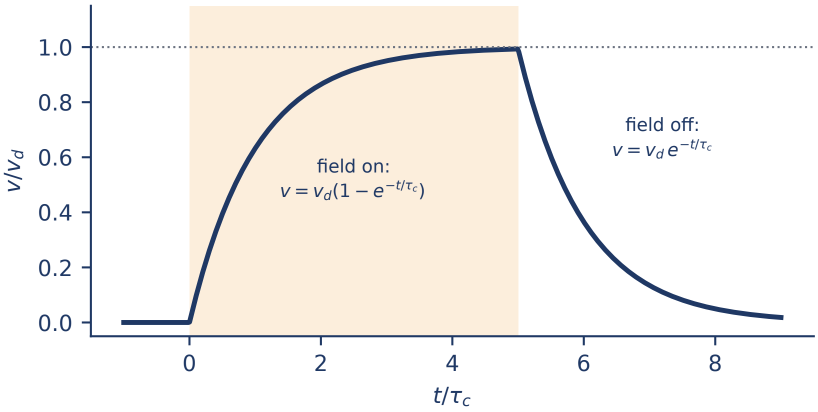


Figure 1.5 – *Response of the mean velocity to a field switched on at* $t = 0$ *and off at* $t = 5\tau_c$. *The steady drift velocity is reached, and lost, in a few collision times.*

1.4 From the local to the macroscopic Ohm’s law

THE QUESTION

The local law relates J *and* $\mathcal{E}$ *at a point. How does it become the* $V = RI$ *of circuits?*

Take a uniform conductor of length L and cross-section S with a voltage V between its ends (Fig. 1.6). *What is the field?* In a uniform conductor the field is uniform and, since V is the integral of the field along the conductor, $\mathcal{E} = V/L$. *What is the current?* The current density is uniform, so $I = JS = \sigma\mathcal{E}S$. Combining,

$$I = \frac{\sigma S}{L} V \quad \Longleftrightarrow \quad V = RI, \qquad R = \frac{L}{\sigma S} = \rho\frac{L}{S}. \tag{1.9}$$

The resistance combines a **material** property, ρ, and a **geometry**, L/S: double the length, double the resistance; double the section, halve it.

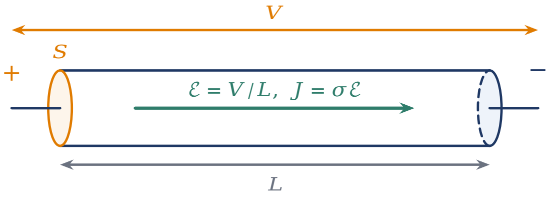


Figure 1.6 – *From local to global: a uniform conductor of length* L *and section* S *under a voltage* V. *The field is* V/L *everywhere, the current density* $\sigma V/L$, *the current* $\sigma SV/L$.

EXAMPLE 1.3 — ONE METRE OF COPPER WIRE

$L = 1$ m, $S = 1$ mm²: $R = 1/(5.96 \times 10^7 \times 10^{-6}) = 0.017\ \Omega$. At 1 A the voltage drop is 17 mV and the field inside the copper only 17 mV/m: a tiny field is enough because n is huge.

1.5 How fast do the carriers really move?

THE QUESTION

The drift velocity in a copper wire is less than a tenth of a millimetre per second. Is that the speed of the electrons?

No. The drift is only the **average**. Each carrier also has a fast, random motion due to the temperature, which is what makes it collide so often. Its order of magnitude follows from the **equipartition of energy**: at temperature T, the mean kinetic energy of a free particle is $\frac{3}{2}kT$ ($\frac{1}{2}kT$ per direction of space),

$$\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}kT \quad\Longrightarrow\quad v_{th} = \sqrt{\frac{3kT}{m}}. \tag{1.10}$$

For an electron at 300 K, $v_{th} = \sqrt{3 \times 1.381 \times 10^{-23} \times 300/9.11 \times 10^{-31}} = 1.2 \times 10^5$ m/s, about 100 km/s. Three consequences.

- **The drift is a tiny bias on a fast random motion**: in Example 1.1, $v_d/v_{th} \approx 7 \times 10^{-5}/1.2 \times 10^5 \approx 10^{-9}$. The random motion carries no net current because its average is zero.
- **The distance between collisions**, the **mean free path**, is $l = v_{th}\tau_c \approx 1.2 \times 10^5 \times 2.5 \times 10^{-14} \approx 3$ nm, about ten atomic spacings.
- **Ohm's law is a weak-field law.** The drift is a small perturbation of the thermal motion as long as $v_d \ll v_{th}$. In very strong fields (in semiconductor devices, above about 10^4 V/cm) the drift velocity no longer grows with the field and saturates near v_{th}.

BE AWARE OF…

The classical estimate and metals. Formula (1.10) treats the carriers as a classical gas. It is the right description for the carriers of a semiconductor (with their effective mass, Chapter 3), which are few and nondegenerate. In a metal the electrons are so dense that the Pauli principle (Chapter 3) forces most of them to move much faster, at the Fermi velocity, about 1.6×10^6 m/s in copper, and the mean free path is then about 40 nm. The conclusion of this section does not change: the drift is a minute bias on a fast random motion.

This random motion, which carries no current in a uniform material, is not useless: if the concentration is **not** uniform, it produces a net flow. That is the next section.

1.6 Diffusion

THE QUESTION

With no field at all, carriers move at random with zero average velocity. Can they nevertheless produce a net flow, and hence a current?

Yes, if their concentration varies in space. We follow the argument of Neamen (*Semiconductor Physics and Devices*, ch. 5) on a one-dimensional model (Fig. 1.7). The concentration $n(x)$ decreases with x; the carriers move at the thermal velocity v_{th}, half of them to the right and half to the left, and travel on average a mean free path l between collisions.

Which carriers cross the plane $x = 0$ from left to right during one collision time? Those that are within a distance l on the left and moving right: half of the carriers of the slab $[-l, 0]$, whose concentration is about $n(-l)$. The flux (carriers per unit area and per unit time) from left to right is therefore $F_1 = \frac{1}{2}n(-l)\,v_{th}$; in the same way, from right to left, $F_2 = \frac{1}{2}n(+l)\,v_{th}$. *Why is there a net flow?* Because more carriers start from the crowded side: each carrier moves at random, but the numbers are unequal. The net flux is

$$\Phi = F_1 - F_2 = \frac{1}{2}v_{th}\,[n(-l) - n(+l)].$$

The mean free path is short compared with the scale over which n varies, so a first-order expansion is accurate: $n(\pm l) \approx n(0) \pm l\,\mathrm{d}n/\mathrm{d}x$. Hence

$$\Phi = -D\,\frac{\mathrm{d}n}{\mathrm{d}x}, \qquad D = v_{th}\,l \quad \text{(1D model).} \tag{1.11}$$

This is **Fick's law**: the particles flow **down** the concentration gradient, with a flux proportional to it. The **diffusion coefficient** D (m²/s, usually cm²/s) is the product of how fast the carriers move and how far they go between collisions. (A three-dimensional average changes the numerical factor, not the physics: D is of the order of $v_{th}l$.)

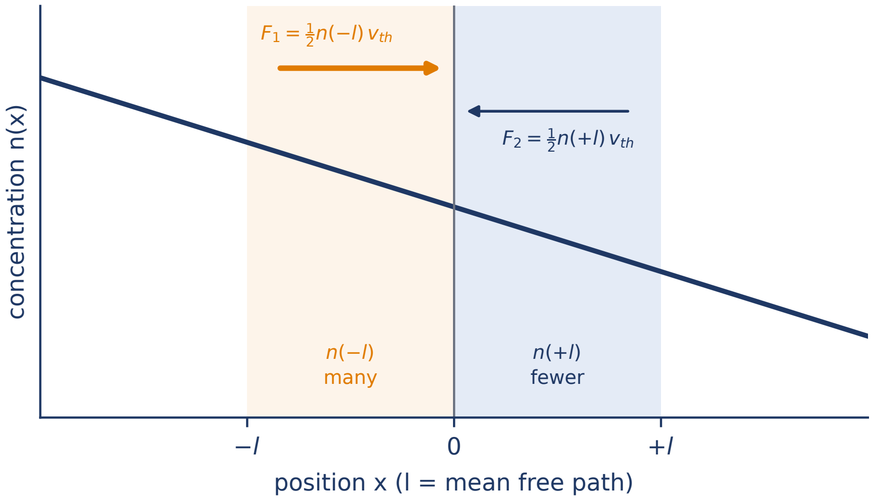

$F_1 = \frac{1}{2}n(-l)\,v_{th}$
$F_2 = \frac{1}{2}n(+l)\,v_{th}$
concentration n(x)
$n(-l)$
many
$n(+l)$
fewer
$-l$
0
$+l$
position x (l = mean free path)

Figure 1.7 *– The diffusion argument (after Neamen). Within one mean free path of the plane $x = 0$, half of the carriers move toward it. More of them start from the crowded side: $F_1 > F_2$, a net flux down the gradient.*

Each carrier carries the charge q, so the **diffusion current density** is

$$J_{\text{diff}} = q\,\Phi = -q\,D\,\frac{\mathrm{d}n}{\mathrm{d}x}. \tag{1.12}$$

Be careful with the sign: the particles always flow down the gradient, but the current follows them for positive carriers and opposes them for electrons ($q = -e$ gives $J = +eD\,\mathrm{d}n/\mathrm{d}x$). In a metal the electron density is uniform and fixed, so diffusion plays no role; in a semiconductor, where concentrations can vary by orders of magnitude over a micrometre, it is often the dominant current.

1.7 The Einstein relation

THE QUESTION

Drift and diffusion both come from the same random motion interrupted by collisions. Are μ and D independent, or must they be related?

A first answer from the models. Along one direction, equipartition gives $\frac{1}{2}m\langle v_x^2\rangle = \frac{1}{2}kT$, so the one-dimensional thermal velocity is $v_{th} = \sqrt{kT/m}$. With $l = v_{th}\tau_c$, the 1D diffusion model gives $D = v_{th}^2\tau_c = kT\,\tau_c/m$, while Drude gives $\mu = |q|\tau_c/m$. The collision time, the hardest parameter to know, cancels in the ratio:

$$\frac{D}{\mu} = \frac{kT\,\tau_c/m}{|q|\,\tau_c/m} = \frac{kT}{|q|}. \tag{1.13}$$

A general answer from equilibrium. The models are rough; is the result? Consider carriers in a potential $V(x)$ that pushes them one way (Fig. 1.8), at thermal equilibrium. Two facts are certain. First, at equilibrium **no current flows**: the drift flux due to the field exactly cancels the diffusion flux due to the gradient that the field has built. Second, at equilibrium the carriers are distributed according to Boltzmann statistics, $n(x) \propto \exp[-qV(x)/kT]$, so $\mathrm{d}n/\mathrm{d}x = -(q/kT)\,n\,\mathrm{d}V/\mathrm{d}x = (q/kT)\,n\,\mathcal{E}$, with $\mathcal{E} = -\mathrm{d}V/\mathrm{d}x$. Writing that the total flux vanishes, with $v_d = (q/|q|)\mu\mathcal{E}$,

$$n\,\frac{q}{|q|}\,\mu\,\mathcal{E} - D\,\frac{q}{kT}\,n\,\mathcal{E} = 0 \qquad \Longrightarrow$$

$$\frac{D}{\mu} = \frac{kT}{|q|} \qquad \text{(Einstein relation).} \tag{1.14}$$

This argument uses no detail of the collisions: the relation is general for nondegenerate carriers. Its meaning: **a carrier that drifts easily also diffuses easily**, because both are the same random walk, biased either by a force or by unequal numbers. At 300 K, $kT/e = 25.9$ mV: electrons in silicon with $\mu_n = 1350$ cm²/(V·s) have $D_n = 35$ cm²/s.

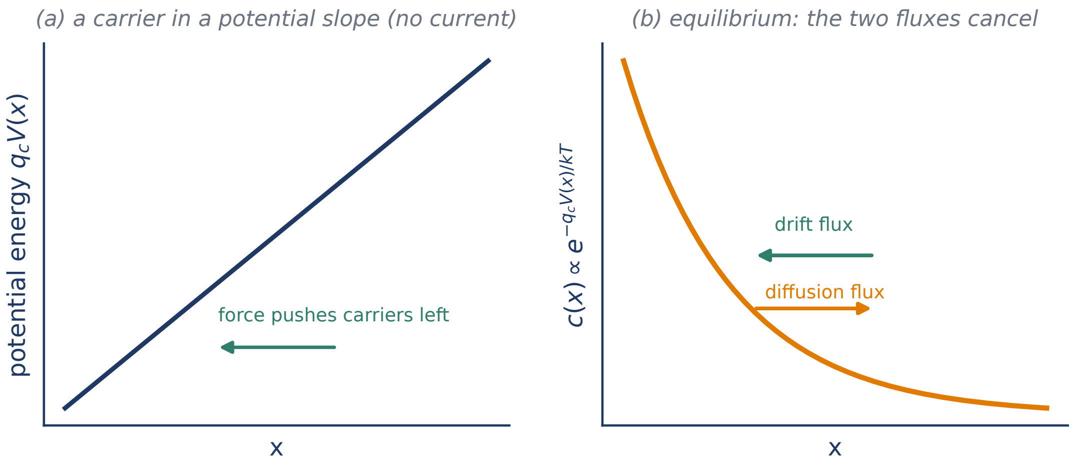

(a) a carrier in a potential slope (no current)
(b) equilibrium: the two fluxes cancel
potential energy $q_cV(x)$
force pushes carriers left
x
$c(x) \propto e^{-q_cV(x)/kT}$
drift flux
diffusion flux
x

Figure 1.8 – *The equilibrium argument for the Einstein relation. (a) A potential slope pushes the carriers one way. (b) They pile up following Boltzmann's law, and the diffusion flux that this gradient creates exactly cancels the drift flux.*

1.8 Total current and the continuity equation

THE QUESTION

A carrier may drift and diffuse at the same time. How do we write the total current, and how do we follow the concentration in space and time?

The general flux. For a generic species of concentration c and charge q_c, the two fluxes add:

$$\Phi = c\,v_d - D\,\frac{\partial c}{\partial x}, \qquad v_d = \pm\mu\mathcal{E}\ (\text{sign of } q_c), \qquad J = q_c\,\Phi. \tag{1.15}$$

The total current density is the sum over all species. This single expression, specified for electrons and holes, will give all the currents of the course (Chapter 4).

The continuity equation. *How can the number of carriers in a small volume change?* In a metal the number of free electrons is **fixed**: each atom has given its valence electrons to the crystal once and for all, and no carrier is created or destroyed. The number of carriers in a volume can then change in one way only: carriers enter or leave it. Take a thin slice between x and $x + \mathrm{d}x$, of unit area (Fig. 1.9). During $\mathrm{d}t$, $\Phi(x)\,\mathrm{d}t$ carriers enter at x and $\Phi(x + \mathrm{d}x)\,\mathrm{d}t$ leave at $x + \mathrm{d}x$. The change of the number $c\,\mathrm{d}x$ is the difference:

$$\frac{\partial c}{\partial t}\,\mathrm{d}x = \Phi(x) - \Phi(x + \mathrm{d}x) = -\frac{\partial \Phi}{\partial x}\,\mathrm{d}x,$$

where the last step is the first-order expansion $\Phi(x + \mathrm{d}x) \simeq \Phi(x) + (\partial\Phi/\partial x)\,\mathrm{d}x$, exact as $\mathrm{d}x \to 0$. Dividing by $\mathrm{d}x$,

$$\frac{\partial c}{\partial t} = -\frac{\partial \Phi}{\partial x} \qquad \left(\text{3D: } \frac{\partial c}{\partial t} = -\nabla \cdot \vec{\Phi}\right). \tag{1.16}$$

If more carriers enter than leave ($\partial\Phi/\partial x < 0$), the concentration rises; if as many leave as enter, it stays constant even though carriers are moving. This is the continuity equation of a **conserved species**, whose carriers move but are never created or destroyed. The general balance adds a source term, the creation minus the disappearance of carriers per unit volume and time; it is not needed for a metal.

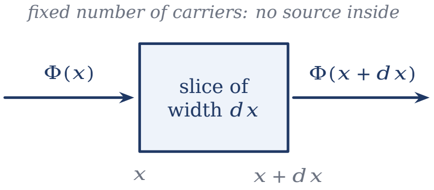


Figure 1.9 – *Balance of carriers in a thin slice: the number inside changes only by what enters at x minus what leaves at $x + \mathrm{d}x$.*

With (1.15) and uniform μ and D, the continuity equation becomes an equation for c alone, the **drift–diffusion equation**:

$$\frac{\partial c}{\partial t} = -\frac{\partial (c\,v_d)}{\partial x} + D\,\frac{\partial^2 c}{\partial x^2}. \tag{1.17}$$

Charge conservation. Multiply (1.16) by q_c. With the charge density of the carriers $\rho_c = q_c c$ and $J = q_c\Phi$, $\partial\rho_c/\partial t = -\partial J/\partial x$. The fixed positive ions add a charge density that does not change in time and carry no current, so the total charge density ρ obeys the same equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0. \tag{1.18}$$

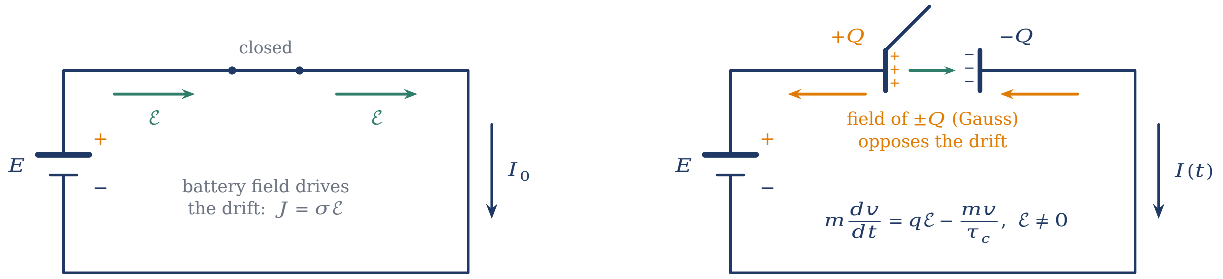
Since the number of carriers is fixed, so is the charge they carry: **charge is neither created nor destroyed, it only moves**. This is the form used in the case study of the next section.

Generation and recombination of carriers, and the recombination rate, are introduced in Chapter 2; the source term they add to this equation is written in Chapter 4.

1.9 Case study: switching off a battery

> ***THE QUESTION***
>
> *A copper wire connects the terminals of a battery and carries the current I_0. At $t = 0$ a switch in the loop opens. Does the field in the wire vanish? Does the current stop at once, and where does the charge go?*

The field does not vanish. It is tempting to argue: once the switch is open the battery no longer acts, the field is zero, and Drude's equation reduces to $m\,\mathrm{d}v/\mathrm{d}t = -mv/\tau_c$, a decay in a few τ_c (Section 1.3.3). This is wrong, because it forgets where a field comes from: **charges**. Follow the carriers during the first instants (Fig. 1.10).


closed
$\mathcal{E}$
$\mathcal{E}$
E
+
−
I_0
battery field drives the drift: $J = \sigma\mathcal{E}$
$+Q$
$-Q$
field of $\pm Q$ (Gauss) opposes the drift
E
+
−
$I(t)$
$m\frac{dv}{dt} = q\mathcal{E} - \frac{mv}{\tau_c}$, $\mathcal{E} \neq 0$

(a) $t < 0$: switch closed

(b) $t > 0$: switch open, the field survives the cut

Figure 1.10 – *Switching off a battery. (a) Switch closed: the field set up by the battery drives the current I_0. (b) Just after the opening the carriers are still drifting; they deposit $+Q$ and $-Q$ on the two ends of the cut. By Gauss's law this charge makes a field which, inside the wire on both sides of the cut, opposes the drift (and in the gap points from $+Q$ to $-Q$). The field does not vanish, so Drude's equation keeps its field term.*

1. **The carriers are still moving.** At the instant of the opening the carriers drift at v_d; nothing has stopped them yet. The current keeps bringing charge to the cut, which it cannot cross.
2. **Charge piles up on the two ends.** The cut leaves two wire segments, each with an end. The conventional current deposits $+Q$ on the upstream end; on the downstream end the electrons, still arriving and unable to cross, accumulate as $-Q$. It sits at the very end of the metal: on the scale of the wire it is a surface charge.
3. **This charge makes a field** (Gauss). Inside the wire, on both sides of the cut, the field of $\pm Q$ points **against** the current: it opposes the drift. The field that the battery

had set up is still present at $t = 0^+$, since the charges that produce it cannot vanish instantly. The field in the wire survives the cut.

4. **Drude's equation keeps its field term**, $m\,\mathrm{d}v/\mathrm{d}t = q\mathcal{E} - mv/\tau_c$. The carriers are slowed by the collisions *and* by the field of the charge that their own current has deposited on the ends. The new velocity sets the new current, hence the rate at which charge piles up, and the loop starts again.

The loop has two memories: the **charge**, which remembers how much current has flowed (as in a capacitor), and the **velocity**, which remembers how the carriers were moving (an inertia). With two memories we expect a second-order equation, like that of an RLC circuit.

The equation. We make the loop exact at every point of a homogeneous conductor (uniform n, m, τ_c and permittivity ε), with three laws already met.

- **Drude with its inertia** (Section 1.3). Multiply (1.4) by nq/m and use $J = nqv$ and $\sigma = nq^2\tau_c/m$:

$$\frac{\partial \vec{J}}{\partial t} = \frac{nq^2}{m}\,\vec{\mathcal{E}} - \frac{\vec{J}}{\tau_c}. \tag{1.19}$$

  In a steady state the left side vanishes and we recover $\vec{J} = \sigma\vec{\mathcal{E}}$; when the current changes, part of the field is spent on changing the velocity of the carriers.

- **Continuity** (Section 1.8): $\partial\rho/\partial t = -\nabla\cdot\vec{J}$.
- **Gauss**: $\nabla\cdot\vec{\mathcal{E}} = \rho/\varepsilon$.

Differentiate the continuity equation once more in time and insert the divergence of (1.19); then use Gauss for $\nabla\cdot\vec{\mathcal{E}}$ and continuity again for $\nabla\cdot\vec{J}$:

$$\frac{\partial^2\rho}{\partial t^2} = -\nabla\cdot\frac{\partial\vec{J}}{\partial t} = -\frac{nq^2}{m}\,\nabla\cdot\vec{\mathcal{E}} + \frac{1}{\tau_c}\,\nabla\cdot\vec{J} = -\frac{nq^2}{m\varepsilon}\,\rho - \frac{1}{\tau_c}\,\frac{\partial\rho}{\partial t}.$$

$$\frac{\partial^2\rho(\vec{r},t)}{\partial t^2} + \frac{1}{\tau_c}\,\frac{\partial\rho(\vec{r},t)}{\partial t} + \omega_p^2\,\rho(\vec{r},t) = 0, \qquad \omega_p^2 = \frac{n\,q^2}{\varepsilon\,m} \quad \text{(plasma frequency)}. \tag{1.20}$$

This is the equation of a **damped oscillator**, the same as that of a series RLC circuit, $L\ddot{Q} + R\dot{Q} + Q/C = 0$. The permittivity plays the capacitance (it stores the field energy), the collisions the resistance (they dissipate), and the **mass of the carriers** the inductance (it stores the kinetic energy of the drift).

Where has the space dependence gone? Gauss's law and the continuity equation both contain a divergence, and in a homogeneous conductor (n, m, τ_c, ε uniform) both turn back into ρ itself: no space derivative of ρ survives. (In a non-uniform material, terms such as $\nabla\sigma\cdot\vec{\mathcal{E}}$ would remain and couple neighbouring points.) Two consequences follow.

- **Every point evolves on its own**, as a damped oscillator with the same ω_p and τ_c everywhere. If the carriers start at rest, the solution separates, $\rho(\vec{r},t) = \rho_0(\vec{r})\,f(t)$: all the space dependence is in the initial profile $\rho_0(\vec{r})$. The **shape** of the charge never changes; only its amplitude decays (or oscillates), by the same factor everywhere. The charge does not travel outward as a front: it disappears *in place*, everywhere at once, while the field $\vec{\mathcal{E}}(\vec{r},t)$ and the current $\vec{J}(\vec{r},t)$, which do depend on position, carry it away to the surface.

- **A neutral point stays neutral.** If at some point $\rho = 0$ and $\partial\rho/\partial t = -\nabla\cdot\vec{J} = 0$, the oscillator starts at rest at zero and stays there: $\rho = 0$ for all t. Charge can appear only where $\nabla\cdot\vec{J} \neq 0$ or where the material is not homogeneous, that is, on its surfaces.

One identity links the three times. With the dielectric relaxation time $\tau_d = \varepsilon/\sigma$ and $\sigma = nq^2\tau_c/m$, $\tau_d = \varepsilon m/(nq^2\tau_c) = 1/(\omega_p^2\tau_c)$:

$$\omega_p^2\,\tau_c\,\tau_d = 1, \qquad \text{so} \qquad \omega_p\tau_c = \sqrt{\tau_c/\tau_d}. \tag{1.21}$$

A single dimensionless number, $\omega_p\tau_c$, decides the behaviour.

Relaxation or oscillation. Try $\rho \propto \mathrm{e}^{rt}$: $r^2 + r/\tau_c + \omega_p^2 = 0$, so

$$r = -\frac{1}{2\tau_c} \pm \sqrt{\frac{1}{4\tau_c^2} - \omega_p^2}. \tag{1.22}$$

The sign under the root separates two regimes, with the critical value $\omega_p\tau_c = 1/2$ (Fig. 1.11, drawn for a charge ρ_0 with carriers initially at rest).

- **Overdamped**, $\omega_p\tau_c \ll 1/2$ (collisions faster than the oscillation). Both roots are real and negative. The fast one, $r_1 \simeq -1/\tau_c$, is the transient of Section 1.3.3: the carriers reach their drift velocity. The slow one, expanding the root, is $r_2 \simeq -\omega_p^2\tau_c = -1/\tau_d$ by (1.21). After a few τ_c only the slow one remains:

  $$\rho(\vec{r},t) \simeq \rho_0(\vec{r})\,\mathrm{e}^{-t/\tau_d}, \qquad \tau_d = \frac{\varepsilon}{\sigma} \qquad \text{(dielectric relaxation time).} \tag{1.23}$$

  This is the limit where inertia is negligible and the local Ohm's law $J = \sigma\mathcal{E}$ holds at every instant. It has a circuit meaning: a block of length L and section S has $R = L/(\sigma S)$ and, between its faces, $C = \varepsilon S/L$, so that

  $$RC = \frac{L}{\sigma S}\cdot\frac{\varepsilon S}{L} = \frac{\varepsilon}{\sigma} = \tau_d. \tag{1.24}$$

  The dielectric relaxation time is the **RC time constant of the material itself**: the conductor discharges its own internal charge through its own resistance.

- **Underdamped**, $\omega_p\tau_c \gg 1/2$ (oscillation faster than collisions). The roots are complex: $\rho = \rho_0\,\mathrm{e}^{-t/2\tau_c}\,[\cos\omega t + \sin(\omega t)/(2\omega\tau_c)]$, with $\omega = \sqrt{\omega_p^2 - 1/4\tau_c^2} \simeq \omega_p$. The carriers rush to neutralise the charge, **overshoot**, the charge changes sign, they come back: a **plasma oscillation**, damped within a few τ_c. The time τ_d then has no physical meaning: the charge disappears on the scale of τ_c.

Medium	τ_c (s)	$\tau_d = \varepsilon/\sigma$ (s)	$\omega_p\tau_c$	Behaviour
Sea water (ions)	1×10^{-14}	1.4×10^{-10}	0.009	RC relaxation
Si, 10^{14} cm^{-3}	2×10^{-13}	4.8×10^{-11}	0.065	RC relaxation
Si, 10^{16} cm^{-3}	2×10^{-13}	4.8×10^{-13}	0.65	near critical
Copper	2.5×10^{-14}	1.5×10^{-19}	≈ 400	plasma oscillation

Table 1.1 *– Times and regimes. Electrons in silicon:* $m^* = 0.26\,m_0$, $\mu = 1350$ *cm²/(V·s);* $\omega_p\tau_c = \sqrt{\tau_c/\tau_d}$.

Back to the switch: the bulk. The field of $\pm Q$ drives a current in the wire: it first slows the drift and, in an oscillating case, reverses it. So the bulk does carry a current;

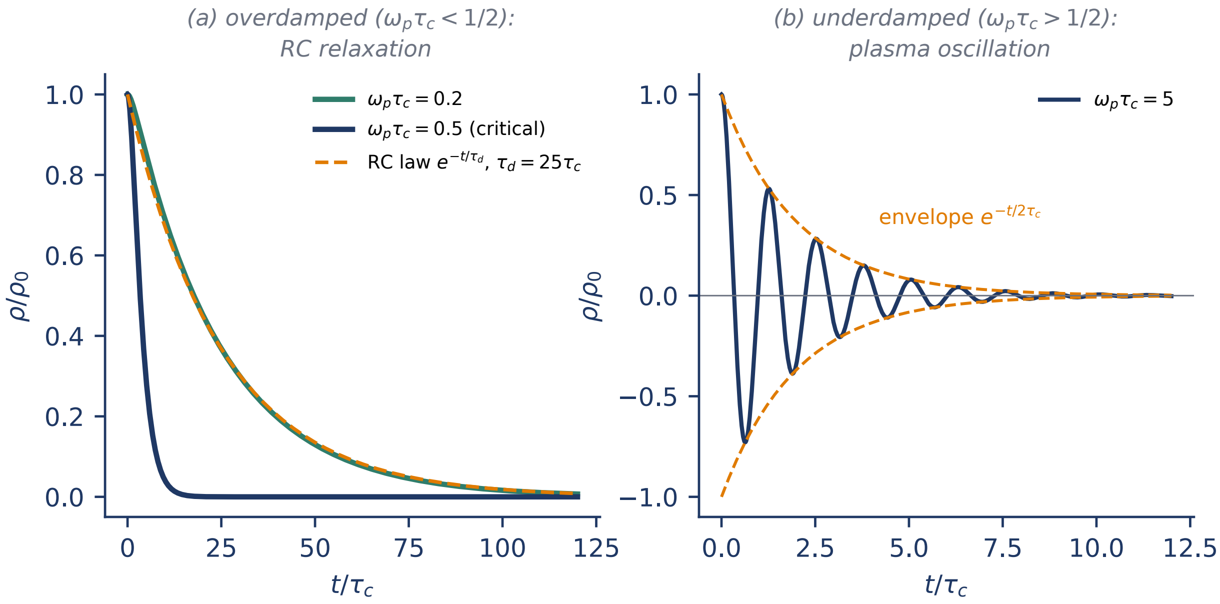


Figure 1.11 *– Response of an excess charge ρ_0 inside a conductor, carriers initially at rest. (a) Overdamped: after a short transient of a few τ_c the curve follows the RC law e^{-t/τ_d}; at the critical value the decay is fastest without oscillation. (b) Underdamped: the charge oscillates at ω_p inside an envelope $\mathrm{e}^{-t/2\tau_c}$.*

but **a current does not create charge, only a current that varies along the wire does**. By continuity, $\partial\rho/\partial t = -\partial J/\partial x$. In the bulk the current $J(t)$ is the same at every x: it changes in time, but a slice gains through one side exactly what it loses through the other. Before the opening the bulk is neutral, $\rho(x,0) = 0$, and the current is uniform, $\partial\rho/\partial t(x,0) = 0$; by the second consequence above, $\rho(x,t) = 0$ at every interior point for all t.

This neutrality is not imposed: it is **restored dynamically** by the loop of (1.20). If a small charge appeared somewhere inside, because the current became slightly non-uniform, its field (Gauss) would drive current outward (Drude), and the current would remove it (continuity): a spring of stiffness $\omega_p^2 = nq^2/(\varepsilon m)$, enormous in a metal. Bulk neutrality and a uniform current are the same condition. In this one-dimensional picture Gauss's law reads $\mathrm{d}\mathcal{E}/\mathrm{d}x = \rho/\varepsilon$. Where $\rho = 0$, in the bulk, $\mathrm{d}\mathcal{E}/\mathrm{d}x = 0$: the field is uniform along the wire. It changes only across an end, where it jumps by the charge per unit area that the end carries, $\Delta\mathcal{E} = Q/(\varepsilon S)$. Charge appears only where the field is discontinuous, at the two ends, and the field of these two charged ends is the uniform field that keeps the current uniform.

The ends: what stops the current. Equation (1.20) tells us where charge can be: only at the ends, where J falls abruptly from I/S to zero. It does not tell us how fast the current stops. That is decided at the ends, by the same loop at the scale of the whole wire (length L, section S). Continuity at an end gives $\mathrm{d}Q/\mathrm{d}t = I$. Integrating the inertial Drude equation (1.19) along the wire, with $I = JS$, and neglecting magnetic induction for now, the voltage spent on accelerating the carriers and on the collisions equals the battery voltage minus the voltage across the gap, Q/C, where C is the small capacitance between the two ends:

$$L_k \frac{\mathrm{d}I}{\mathrm{d}t} + R\,I = E - \frac{Q}{C}, \qquad L_k = \frac{m\,L}{n\,q^2 S}, \qquad R = \frac{L}{\sigma S}, \tag{1.25}$$

$$L_k\,\ddot{Q} + R\,\dot{Q} + \frac{Q}{C} = E. \tag{1.26}$$

This is the same structure as (1.20): a capacitance, a resistance and an inertia (the **kinetic inductance** L_k, the mass of the carriers seen at the scale of the wire). The capacitance is now the geometric capacitance of the two ends rather than the permittivity of the material, and the same reading applies, with $L_k/R = \tau_c$ in place of τ_c and $1/\sqrt{L_k C}$ in place of ω_p. The final state is $\dot{Q} = I = 0$ and $Q = CE$: the bulk is neutral, the field inside the metal is zero, and the voltage E of the battery appears across the gap, held by $\pm Q$ on the ends. In between, the field never vanished: it changed origin, from the charges set up by the battery to the charge piled up on the ends, and it is this field, together with the collisions, that stopped the carriers.

BE AWARE OF…

Copper: no real ringing. For copper, $\omega_p = \sqrt{nq^2/(\varepsilon_0 m)} = 1.6 \times 10^{16}$ rad/s (a period of 0.4 fs) and $\omega_p \tau_c \approx 400$: formally a plasma oscillation, but it is damped within a few τ_c, about 100 fs, and $\tau_d = \varepsilon_0/\sigma = 1.5 \times 10^{-19}$ s is shorter still. Both τ_c and ε/σ are vanishingly small compared with any time we can observe in a circuit: the bulk of the copper becomes neutral, and its own transient is over, practically at once. There is no real ringing of the metal. (Outside the model of this chapter, which neglects magnetic induction: in a real loop the magnetic inductance L_m of the circuit adds to L_k in (1.26) and dominates it: for 1 m of copper wire of 1 mm², $L_k \approx 4 \times 10^{-16}$ H against $L_m \sim 1$ µH. Any ringing seen on an oscilloscope when a switch opens, typically around 100 MHz, comes from this circuit, L_m with the gap capacitance, not from the inertia of the carriers.) The lasting conclusion holds for every conductor: **the bulk of a conductor is neutral almost instantly**. This is the **quasi-neutrality** used throughout Chapter 4.

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Carriers**: free electrons in metals, electrons and holes in semiconductors, ions in electrolytes. The course is about currents in devices.
- **Current density**: $J = q\,n\,v$, from $i = \mathrm{d}q/\mathrm{d}t$ and the charge in the cylinder $S\,v\,\mathrm{d}t$; q algebraic; several species add.
- **Drude**: $m\,\mathrm{d}v/\mathrm{d}t = q\mathcal{E} - mv/\tau_c$; collisions every τ_c act as friction; $v_d = q\tau_c\mathcal{E}/m$, reached in a few τ_c ($\sim 10^{-14}$ s in metals).
- **Ohm**: $J = \sigma\mathcal{E}$, $\sigma = n|q|\mu$, $\mu = |q|\tau_c/m$; globally $V = RI$ with $R = L/(\sigma S)$.
- **Speeds**: $\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}kT$ gives $v_{th} \approx 10^5$ m/s for electrons at 300 K, against $v_d \sim 10^{-4}$ m/s in a wire; mean free path $l = v_{th}\tau_c$; Ohm's law is a weak-field law.
- **Diffusion**: $\Phi = -D\,\mathrm{d}n/\mathrm{d}x$ with $D \approx v_{th} l$; $J_{\text{diff}} = -qD\,\mathrm{d}n/\mathrm{d}x$.
- **Einstein**: $D/\mu = kT/|q|$ (25.9 mV at 300 K), from equilibrium and Boltzmann statistics.

- **Transport and continuity**: $\Phi = c\,v_d - D\,\partial c/\partial x$, $J = q_c\Phi$; with a fixed number of carriers, $\partial c/\partial t = -\partial\Phi/\partial x$; for charge, $\partial\rho/\partial t + \nabla\cdot\vec{J} = 0$. Generation and recombination: Chapters 2 and 4.
- **Switching off a battery**: the field survives the cut, sustained by the charge piling up on the ends (Gauss); the bulk carries a uniform current and stays neutral; Drude with inertia + continuity + Gauss give $\ddot\rho + \dot\rho/\tau_c + \omega_p^2\rho = 0$, $\omega_p^2\tau_c\tau_d = 1$: RC relaxation with $\tau_d = \varepsilon/\sigma$ if $\omega_p\tau_c < 1/2$, plasma oscillation if $\omega_p\tau_c > 1/2$. The current is stopped by the charge on the ends, $L_k\ddot Q + R\dot Q + Q/C = E$. In copper τ_c and τ_d are vanishingly small: the bulk is neutral at once (quasi-neutrality).

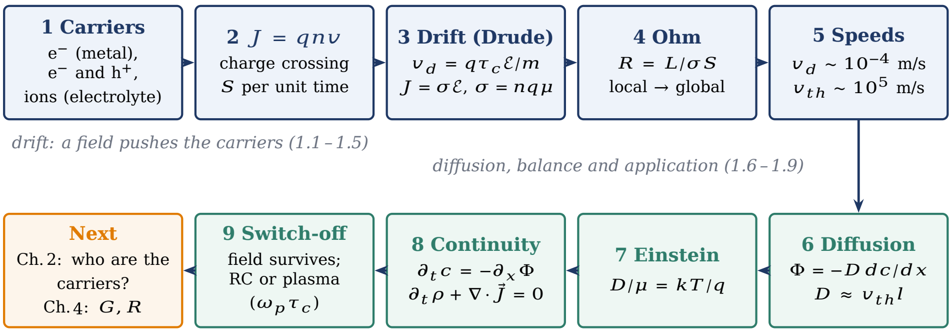

1 Carriers
e⁻ (metal), e⁻ and h⁺, ions (electrolyte)
2 $J = qnv$
charge crossing S per unit time
3 Drift (Drude)
$v_d = q\tau_c\mathcal{E}/m$
$J = \sigma\mathcal{E}$, $\sigma = nq\mu$
4 Ohm
$R = L/\sigma S$
local → global
5 Speeds
$v_d \sim 10^{-4}$ m/s
$v_{th} \sim 10^5$ m/s
drift: a field pushes the carriers (1.1 – 1.5)
diffusion, balance and application (1.6 – 1.9)
6 Diffusion
$\Phi = -D\,dc/dx$
$D \approx v_{th} l$
7 Einstein
$D/\mu = kT/q$
8 Continuity
$\partial_t c = -\partial_x\Phi$
$\partial_t\rho + \nabla\cdot\vec{J} = 0$
9 Switch-off
field survives; RC or plasma ($\omega_p\tau_c$)
Next
Ch. 2: who are the carriers?
Ch. 4: G, R

Figure 1.12 *– The logic of the chapter: drift (boxes 1–5) and diffusion, balance and application (6–9), leading to the specification of the carriers in Chapters 2 and 3.*

Comprehension questions

No new physics here: answer by reading the figures and the equations of the chapter. The answers are at the end.

Q1. Negative and positive ions drift in opposite directions in an electrolyte. Do their currents add or subtract? Use $J = q\,n\,v$.

Q2. Without collisions, what would a constant voltage do to the current in a wire? Which observation does this contradict?

Q3. Why does the conductivity contain q^2, and what does this imply for the direction of the drift current?

Q4. Copper has a lower mobility than silicon, yet it conducts much better. Why?

Q5. The electrons of a wire move at about 10^5 m/s, yet they drift at 10^{-4} m/s. Why does the fast motion carry no current?

Q6. In the diffusion argument, each carrier is equally likely to go left or right. Where does the net flux come from?

Q7. Why must D and μ be related? What cancels in the ratio D/μ of the models?

Q8. In the continuity equation of this chapter, the number of carriers is fixed. What is then the only way the number of carriers in a slice can change, and why does charge conservation follow directly?

Q9. Why is the dielectric relaxation time the RC time constant of the material, and why does it not depend on the size of the block?

Q10. When the switch opens, why does the field in the wire not vanish, and why must Drude's equation keep its field term?

Q11. After the opening the wire still carries a current. Why does the bulk nevertheless stay neutral, and where does the charge accumulate?

Q12. Which single number decides between relaxation and oscillation of an excess charge? Why is no ringing of the metal observed in a copper wire, although $\omega_p \tau_c \approx 400$?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 1, with their answers; the worked solutions are in the **Solutions** document. The 10 exercises that used to close this chapter are Problems 1.10 to 1.19 there; the other problems of the set (1.1 to 1.9) go further.

Answers

Q1. They add: $J = q_+ n_+ v_+ + q_- n_- v_-$, and both products qv point along the field because q and v change sign together.

Q2. The velocity, hence the current, would grow linearly with time. This contradicts the constant current of a resistor under a constant voltage (Ohm's law).

Q3. One q comes from $J = qnv$, the other from $v_d = q\tau_c \mathcal{E}/m$. Their product is positive: the drift current always flows along the field, for positive and negative carriers alike.

Q4. $\sigma = n|q|\mu$: copper has about 10^{29} free electrons per m^3, about 10^7 times more than silicon doped with 10^{16} donors per cm^3 and 10^{13} times more than pure silicon. The number of carriers wins over their mobility.

Q5. The thermal velocities point in all directions; their average is zero, so the charge they carry across a section in one direction is compensated by the charge carried in the other. Only the average, the drift, carries a net current.

Q6. From unequal numbers: more carriers start from the crowded side, so more cross the plane in that direction than in the other.

Q7. Both result from the same random walk with collisions, biased by a force (drift) or by unequal numbers (diffusion). In the models, the collision time τ_c cancels, leaving $kT/|q|$.

Q8. Only by transport through its faces: what enters minus what leaves, $-\partial\Phi/\partial x$. Each carrier carries the fixed charge q_c and the ions do not move, so multiplying by q_c gives $\partial\rho/\partial t + \nabla\cdot\vec{J} = 0$ with nothing to cancel.

Q9. A block of length L and section S has $R = L/(\sigma S)$ and $C = \varepsilon S/L$; in the product RC the geometry cancels, leaving ε/σ. The charge relaxes by the same loop as an RC discharge: charge, field, current, less charge.

Q10. A field is produced by charges, and the charges do not vanish at the opening: the carriers keep drifting and pile up $\pm Q$ on the ends of the cut, whose field (Gauss) opposes the drift. Setting $\mathcal{E} = 0$ would ignore this charge; Drude's equation must keep $q\mathcal{E}$, which couples it to Gauss and continuity.

Q11. Charge appears only where the current varies along the wire, $\partial\rho/\partial t = -\partial J/\partial x$. In the bulk the current is uniform: each slice gains as much as it loses, and any small charge that would appear is removed in femtoseconds by its own field (Gauss, Drude, continuity). The current stops abruptly only at the ends, where $\pm Q$ accumulates at the rate $\mathrm{d}Q/\mathrm{d}t = I$.

Q12. $\omega_p\tau_c = \sqrt{\tau_c/\tau_d}$, compared with 1/2. In copper the oscillation has a period of 0.4 fs and dies in a few τ_c (~100 fs), and $\tau_d \sim 10^{-19}$ s: both are far below any observable time, so the metal is neutral practically at once. Ringing seen in a real circuit comes from the wiring inductance and the gap capacitance.

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- N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Saunders, 1976 — ch. 1 (Drude theory, its successes and failures, the Fermi velocity of metals, the plasma frequency).
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- A. Einstein, "Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen", *Annalen der Physik* 17, 549–560 (1905) — the relation between diffusion and mobility.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, 3rd ed., Cambridge University Press, 2013 — ch. 4 (current, conductivity, charge relaxation in a conductor).
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WHERE THIS LEADS

We now have the language of transport: current density, drift and mobility, diffusion and the diffusion coefficient, the Einstein relation, the continuity equation for a fixed number of carriers and charge conservation. Switching off a battery showed how inertia, collisions and the field of the charges make the bulk of a conductor neutral almost instantly. All of it was written for a generic carrier. Chapters 2 and 3 ask what the carriers of a semiconductor are and how many there are; Chapter 4 specifies these laws for electrons and holes and adds to the continuity equation what a metal does not have: the generation and recombination of carriers.

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 2

Pure and Doped Semiconductors at Equilibrium: A Qualitative View

Observations · bonds · electron–hole pairs · the hole · generation and recombination · n and p doping · hydrogen-like model · mass-action law · neutrality

WHERE WE ARE

Chapter 0 gave the technological landscape: the devices we want to understand. Chapter 1 gave the tools of transport for a generic carrier in a metallic conductor: drift, the Drude model, diffusion, the Einstein relation, and the continuity equation of a species whose number does not change. Those tools said nothing about what the carriers of a semiconductor are or how many there are. That is the job of this chapter and the next, for a sample left to itself at a fixed temperature: a sample at **thermal equilibrium**.

This chapter gives a **generic view**, with no band theory and no quantum statistics. From what is observed, it identifies the carriers, shows that they are created and destroyed in pairs, explains what a pure, an n-doped and a p-doped semiconductor are, and deduces the mass-action law from the balance between creation and disappearance. With charge neutrality, it gives the carrier concentrations of a doped sample. Everything is told with bonds, pairs and balances. Chapter 3 will make this picture quantitative with the band structure and Fermi–Dirac statistics; Chapter 4 will take the sample out of equilibrium.

CHAPTER OUTLINE

2.1 What we observe
PART A — The pure semiconductor
2.2 Bonding at absolute zero: metal and semiconductor
2.3 What metals teach us
2.4 Heat breaks bonds: electron–hole pairs
2.5 How many pairs? Reference values
2.6 The hole: a new carrier, a new kind of electricity
2.7 Recombination
2.8 Equilibrium: generation balances recombination
PART B — The doped semiconductor
2.9 n and p doping: the picture at absolute zero
2.10 The solid imposes its rules: a hydrogen atom inside silicon
2.11 Doping a pure semiconductor: do the concentrations simply add?
2.12 Three laws for two unknowns: majority and minority carriers
2.13 What is still missing
Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Start from the observed properties of semiconductors and turn them into questions: where does the conduction come from, which entities carry it, how many are they, what is a doped semiconductor?
- Compare, at absolute zero, the bonding of a metal and of a semiconductor, and deduce why the carriers of a semiconductor must be created.
- Explain, with the bond picture, why electrons and holes are created in **pairs**, what a hole is and why it is less mobile than an electron, why pairs recombine, and why equilibrium is a **balance** between the two.
- Describe n and p doping at 0 K and at room temperature, and estimate the ionisation energy of a donor with the **model of a hydrogen atom** in a dielectric.
- Deduce from the balance, with stated assumptions, the **mass-action law**, and combine it with total ionisation and **charge neutrality** to obtain the majority and minority carrier concentrations.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- contrast the temperature dependence of the resistivity of a metal and of a semiconductor, and say what each implies about the number of carriers;
- draw the bond picture of silicon at 0 K, with a broken bond, with a donor and with an acceptor, at 0 K and at 300 K, and say what is mobile and what is fixed;
- explain why bonds of 1.12 eV break at 300 K, where $kT \approx 26\,\text{meV}$, and why the number of pairs is nevertheless set by a balance;
- quote the intrinsic concentrations of Si, Ge and GaAs at 300 K and compute the resistivity of the pure material;
- explain how a hole moves and why it is less mobile than an electron in silicon;
- state the generic recombination law $R = \alpha np$ and derive $n_0 p_0 = n_i^2$ from $G = R$, naming the assumptions;
- estimate the binding energy and the orbit radius of a donor electron;
- compute n_0 and p_0 of n-type, p-type and compensated samples, and say when the majority approximation holds.

HOW THIS CHAPTER WORKS

- **Observation first.** Every section starts from something we see or measure and asks what must be true for it to happen.
- **Generic before quantitative.** No band theory, no Fermi level: bonds, pairs and balances. The energy E_g is here only the measured energy needed to free an electron from a bond; Chapter 3 will show that it is the width of a forbidden band.
- **What belongs to the model is said.** Every statement is either derived, or a

model **input** stated as such (a measured value, an imported result, an assumption), or a fact **outside the model**, given afterwards and marked.

- **What is deliberately left out.** The value of n_i and its temperature dependence, the ionisation law of dopants and the position of the Fermi level: Chapter 3. What happens when the equilibrium is disturbed (light, injection) and the motion of carriers in space: Chapter 4.

Symbol	Meaning	Remark
n, p; n_0, p_0	concentrations of free electrons and of holes; at thermal equilibrium	cm^{-3}; subscript 0 = equilibrium
n_i	intrinsic concentration (pairs per cm³ in the pure material)	$n_0 = p_0 = n_i$ in the pure material
E_g	energy needed to free an electron from a bond	measured input; 1.12 eV for Si
kT	typical thermal energy	25.9 meV at 300 K
G, R	generation and recombination rates	pairs per cm³ per second
α	generic recombination coefficient	cm³/s; an effective parameter
τ_n, τ_p	lifetimes of an electron, of a hole before recombination	not to be confused with τ_c
μ_n, μ_p; τ_c	mobilities; collision time (Drude)	cm²/(V·s); s
σ, ρ	conductivity, resistivity	$\rho = 1/\sigma$
N_D, N_A	donor and acceptor concentrations	cm^{-3}
N_D^+, N_A^-	ionised donors, ionised acceptors	fixed charges $+q$, $-q$
E_d, E_a	ionisation energy of a donor, of an acceptor	45 meV for P and B in Si
ε_r; m^*	relative permittivity; effective mass	11.7 and $0.26\,m_0$ for Si

Table 2.1 *– Notation of the chapter.*

Data. Silicon at 300 K throughout: $kT = 25.9\,\text{meV}$ (rounded to 26 meV in the text), $E_g = 1.12\,\text{eV}$, $\mu_n = 1350$ and $\mu_p = 460$ cm²/(V·s), $\varepsilon_r = 11.7$, 5×10^{22} atoms per cm³. **Reference value** $n_i = 10^{10}\,\text{cm}^{-3}$, used in all examples of the course (published values range from 7×10^9 to $1.5 \times 10^{10}\,\text{cm}^{-3}$). Germanium: $E_g = 0.66\,\text{eV}$, $n_i = 2.4 \times 10^{13}\,\text{cm}^{-3}$; gallium arsenide: $E_g = 1.42\,\text{eV}$, $n_i = 1.8 \times 10^6\,\text{cm}^{-3}$ (Table 2.3). Elementary charge $q = 1.6 \times 10^{-19}$ C. **Notation:** from this chapter on, q is the elementary charge, a positive number; Chapter 1 used q for the algebraic charge of a generic carrier, which is now written $-q$ for an electron and $+q$ for a hole.

2.1 What we observe

> ***THE QUESTION***
>
> *What does a semiconductor do that neither a metal nor an insulator does?*

We begin as the history of the subject began, with what is seen on the bench. The film in which Walter Brattain of AT&T Bell Laboratories explains the essentials of semiconductors is a good starting point for the class; the observations it makes can be summarised in five items (figure 2.1).

- **Resistivity against temperature.** In a metal the resistivity ρ rises slowly with temperature (about 0.4% per kelvin for copper). In a pure semiconductor it falls, and it falls fast: by orders of magnitude over a range of a few hundred kelvin.
- **Resistivity against light (photoconductivity).** Shine light on a semiconductor and its resistivity drops; switch the light off and it comes back. A metal shows nothing comparable.
- **Rectification.** A semiconductor junction lets current through in one direction and blocks it in the other.
- **Light emission.** A suitable semiconductor diode emits light when a current is sent through it (the LED).
- **Photovoltaic effect.** The same kind of junction, illuminated, produces a voltage and delivers electrical power.

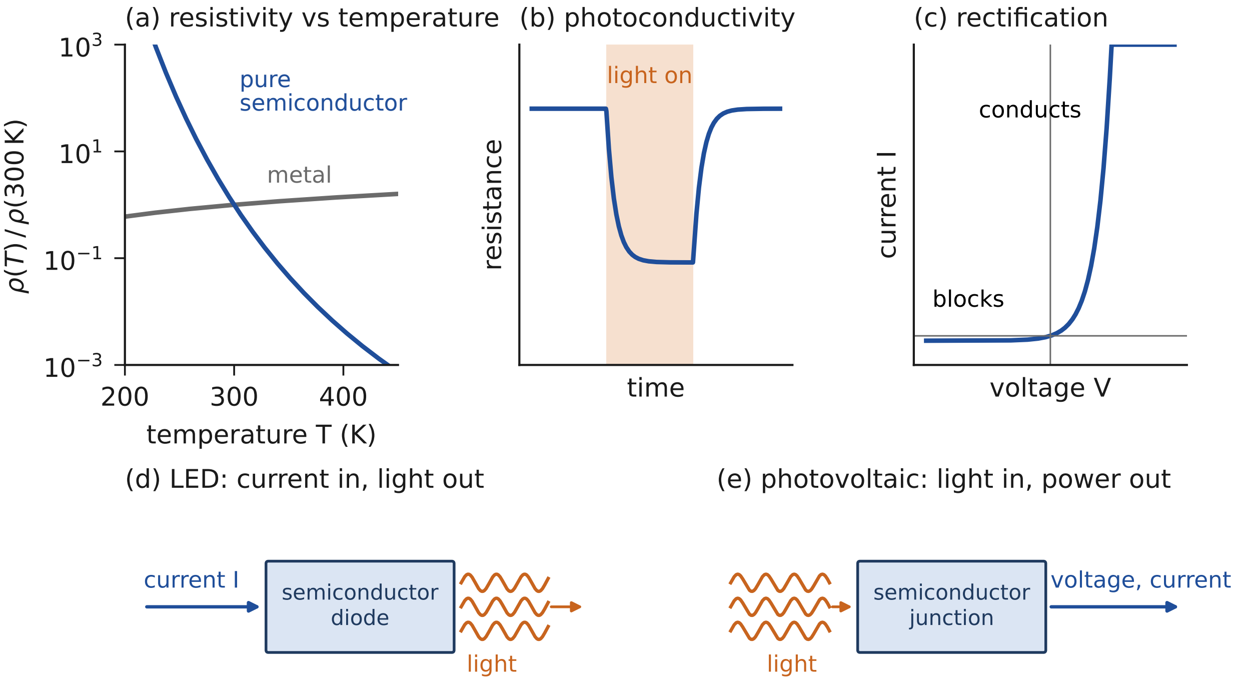


Figure 2.1 *– The five observations. (a) Resistivity against temperature, normalised to its 300 K value, on a logarithmic axis: schematic for a metal and for a pure semiconductor. (b) Photoconductivity: the resistance drops while the sample is illuminated and recovers when the light is switched off. (c) Rectification: the current depends on the sign of the voltage. (d) A light-emitting diode. (e) A photovoltaic junction.*

Two further facts frame the picture. First, the resistivity of a semiconductor is **extremely sensitive to traces of impurities**: one foreign atom in 10^7 changes it by more than five orders of magnitude (Example 2.4). Second, semiconductors sit **between metals and insulators** in resistivity, with no sharp boundary on either side (Table 2.2).

Material	Resistivity at 300 K (Ω·cm)	Class
Copper	1.7×10^{-6}	metal
Pure germanium	≈ 45	semiconductor
Pure silicon	$\approx 3 \times 10^5$	semiconductor
Diamond, glass	10^{10} to $> 10^{13}$	insulator

Table 2.2 *– Orders of magnitude. The values for Ge and Si are those computed in Example 2.1.*

These observations raise the questions that organise the chapter:

THE QUESTIONS OF THE CHAPTER

- Where does the conduction of a semiconductor come from, and which entities carry it? (Part A)
- What fixes their number at equilibrium? A balance between creation and disappearance. (Part A)
- What is an n-doped semiconductor, and a p-doped one? How many carriers of each kind does each have? (Part B)
- The effects of light, of junctions and of injected carriers (observations b to e) need the sample out of equilibrium: Chapter 4 and the junction chapters.

In the rest of this chapter we understand semiconductors through a **generic view**: the pure semiconductor, the n-doped semiconductor, the p-doped semiconductor, and the mass-action law that links their carriers, **without band theory**. The quantitative treatment, which computes n_i and follows the concentrations with temperature, is the subject of Chapter 3. What happens out of equilibrium, under light or by injection of carriers, is the subject of Chapter 4.

PART A

The pure semiconductor

What are the carriers, how are they created and destroyed, how many are there? Bonds and balances, no band theory.

2.2 Bonding at absolute zero: metal and semiconductor

THE QUESTION

At absolute zero, where are the valence electrons of a metal, and those of silicon? Which of them can carry a current?

The metal. A metal is a regular lattice of **cations** immersed in a **cloud of free electrons** (figure 2.2a): each atom has released its valence electron or electrons to the whole crystal. The bond itself is the electron cloud: it is what holds the positive cations together. The free electrons are therefore built in: roughly one per atom, about 10^{22} to 10^{23} cm^{-3}, **at every temperature, 0 K included**. An electric field sets them moving: a metal conducts at absolute zero, and even better and better as it is cooled.

Silicon and germanium. They belong to the fourth column of the periodic table: four valence electrons. Each atom shares one electron with each of four neighbours; each pair of neighbours is held by two electrons, one from each atom. This is the **covalent bond**, and it gives the tetrahedral lattice of the diamond structure, drawn flat in figure 2.2b. Every valence electron sits in a bond, and every bond is full.

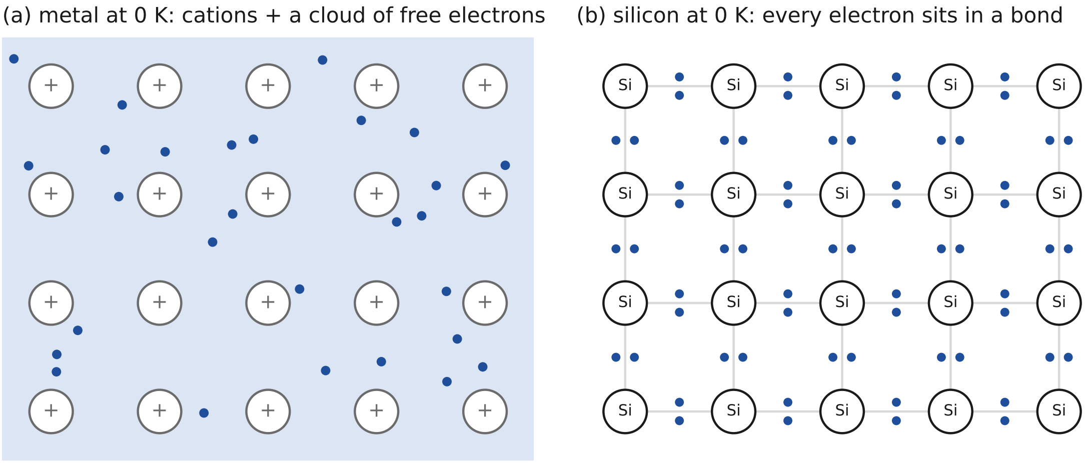

(a) metal at 0 K: cations + a cloud of free electrons
(b) silicon at 0 K: every electron sits in a bond

Figure 2.2 – *The two kinds of bonding at absolute zero. (a) A metal: cations and a cloud of free electrons. The bonding itself supplies the carriers. (b) Silicon, drawn flat: four neighbours per atom, two electrons per bond. No electron is free.*

At absolute zero, then, a crystal of silicon has **no free electron**: an electric field cannot make the bonding electrons move through the crystal, because they are held in place. Pure silicon and germanium are **insulators at 0 K**. So is diamond, which has the same structure with carbon atoms, and for the same reason; but its bond is much stronger.

The energy needed to free one electron from a bond is a **measured input** at this stage: about 1.12 eV in silicon, 0.66 eV in germanium, and 5.5 eV in diamond. We write it E_g;

Chapter 3 will show that it is the width of a forbidden energy gap, hence the letter. At this level of description it is the only difference between the three crystals, and it decides everything.

> **IN SHORT, AT 0 K**
>
> Metal: carriers supplied by the bonding, it conducts. Silicon, germanium, diamond: every electron in a bond, they insulate. The difference in behaviour at room temperature must therefore come from what heat does to the bonds.

2.3 What metals teach us

> **THE QUESTION**
>
> *Chapter 1 explained the conduction of metals. What can we keep of that explanation, and what does it tell us about a semiconductor?*

In Chapter 1 the Drude model gave the conductivity of a material in which free carriers of concentration n and charge q are scattered with a mean collision time τ_c:

$$\sigma = n\,q\,\mu, \qquad \mu = \frac{q\,\tau_c}{m}, \qquad \rho = \frac{1}{\sigma}. \tag{2.1}$$

In a metal, n is fixed by the bonding (Section 2.2): it does not depend on temperature.

The Drude model takes τ_c as a parameter: it does not say how τ_c depends on temperature. We take this from measurement, as an **input**: when a crystal is heated its lattice vibrates more, the collisions become more frequent, and τ_c and μ fall. In a metal, since n does not change, ρ **rises**. This is the metal curve of figure 2.1a.

Now apply the same reasoning to a semiconductor. Its mobility also falls when it is heated (same input). Yet its resistivity falls, and by orders of magnitude. Equation (2.1) leaves only one way out: n **must increase with temperature**, faster than μ decreases. In a semiconductor the free carriers are not built in: at 0 K there are none. They must be **created**, and heat creates them. The new question is how.

2.4 Heat breaks bonds: electron–hole pairs

> **THE QUESTION**
>
> *What can break a bond, what is left behind when it breaks, and how can a few tens of meV of thermal agitation break bonds of 1.12 eV?*

2.4.1 Breaking a bond

Thermal agitation can break a bond. If a bond receives enough energy, at least E_g, one of its two electrons leaves it and wanders through the crystal: it is now a **free electron**, which will carry current like the electrons of a metal (figure 2.3a). But the bond it came from is now incomplete: there is a missing electron, a vacancy. This vacancy is the **hole**. It is something new that has no counterpart in a metal; Section 2.6 will show that it behaves as a carrier in its own right.

For now, keep the counting consequence: a broken bond gives **one free electron and one hole**, always together. Electrons and holes are created **in pairs**. The process is called the thermal **generation** of electron–hole pairs.

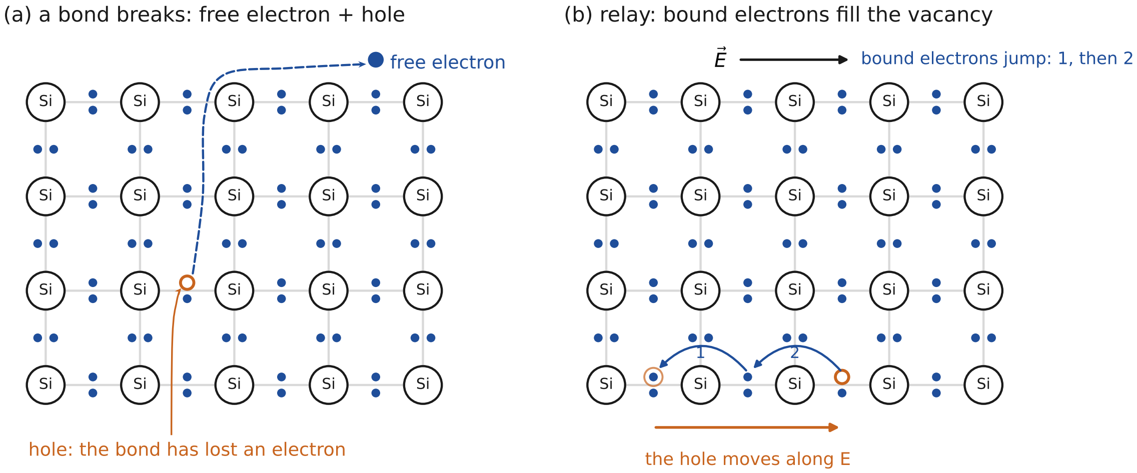

(a) a bond breaks: free electron + hole
free electron
Si
hole: the bond has lost an electron
(b) relay: bound electrons fill the vacancy
$\vec{E}$
bound electrons jump: 1, then 2
1
2
the hole moves along E

Figure 2.3 – *(a) A bond is broken: the electron is free (heat is the agent) and the bond is left incomplete, a hole. (b) The hole moves by a relay: an electron of a neighbouring bond fills the vacancy (jump 1), then another one fills the new vacancy (jump 2). The bound electrons jump to the left, against the field $\vec{\mathcal{E}}$; the vacancy moves to the right, along the field, like a positive charge (Section 2.6).*

2.4.2 The apparent paradox: the statistical argument

The energy needed to break a bond in silicon is 1.12 eV. The thermal energy available at room temperature is of the order of $kT = 26\,\text{meV}$, forty times smaller (the ratio E_g/kT is 43). How can heat break bonds at all?

The answer is in the word **statistical**. The energy of the atoms is not shared equally. At temperature T it fluctuates from atom to atom and from instant to instant, and the probability that a given fluctuation reaches an energy E is not zero. We take from statistical physics, as an **input**, the Boltzmann law: this probability decreases exponentially with E/kT,

$$P(\varepsilon \geq E) \propto \exp\left(-\frac{E}{kT}\right). \tag{2.2}$$

The probability is small, but a crystal is huge. Let us count the bonds. Silicon has 5×10^{22} atoms per cm³; each atom has four bonds, and each bond is shared between two atoms, which makes two bonds per atom:

$$5 \times 10^{22} \text{ atoms/cm}^3 \times \frac{4}{2} \text{ bonds/atom} = 10^{23} \text{ bonds/cm}^3.$$

Even a minute probability, of the order of $\mathrm{e}^{-43} \approx 2 \times 10^{-19}$ for one bond and one fluctuation, means that bonds keep breaking somewhere in the crystal, all the time. **A few meV of typical agitation still breaks 1.12 eV bonds somewhere, all the time**: it is the rare fluctuations, those of the tail of the distribution, that do it (figure 2.4).

In diamond the energy to reach is five times larger ($E_g/kT = 212$). Since it sits in an exponent, the probability is not five times smaller but smaller by a factor $\mathrm{e}^{-(212-43)} \approx 10^{-73}$, and practically no bond breaks: diamond is still an insulator at 300 K. The other way round, germanium ($E_g/kT = 25$) breaks its bonds far more easily than silicon. And since the tail depends exponentially on T, a little heating is enough: from 300 to 350 K, the probability of reaching 1.12 eV is multiplied by about 450 (figure 2.4b).

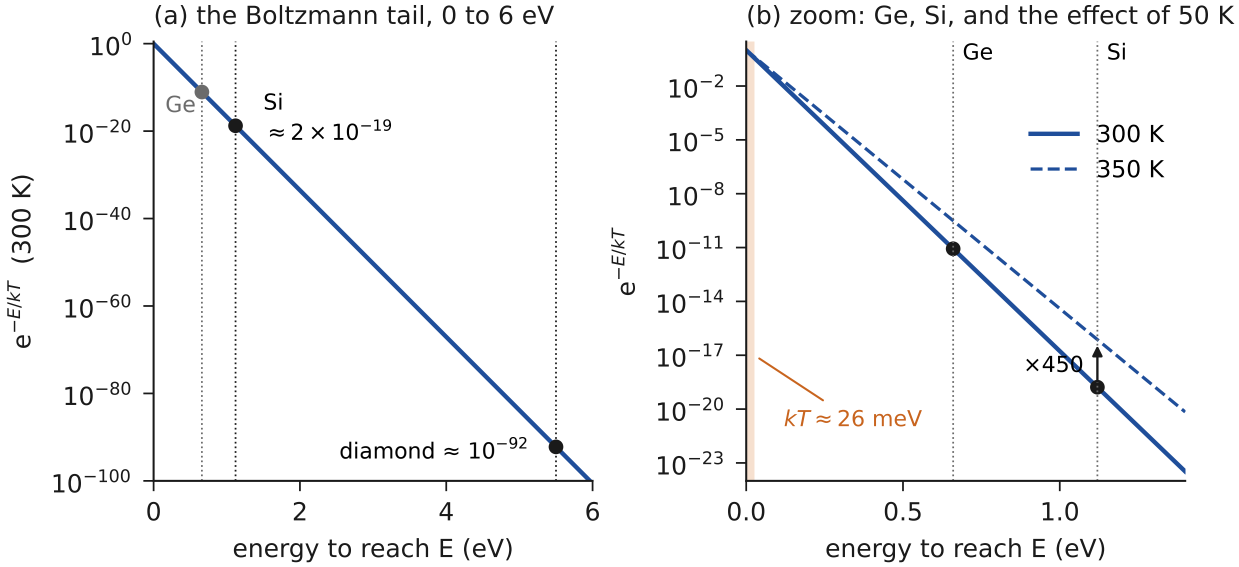


Figure 2.4 *– The Boltzmann tail* $e^{-E/kT}$ *on a logarithmic scale. (a) From 0 to 6 eV at 300 K: the dots mark the energies needed to break a bond in Ge (0.66 eV), Si (1.12 eV) and diamond (5.5 eV). (b) Zoom: the coloured band on the left is* kT *wide; at 350 K (dashed), the probability of reaching 1.12 eV is about 450 times larger than at 300 K.*

A probability is not a number of pairs. Equation (2.2) says how likely a fluctuation is, hence at what rate bonds break; it does not say how many free electrons are present at a given instant. A broken bond does not stay broken: a free electron that meets a hole falls back into the bond (Section 2.7). The number of pairs present is therefore set by a **balance between their creation and their disappearance** (Section 2.8).

BE AWARE OF… A COMMON SLIP

It is tempting to say that the thermal energy ‘is a few meV’ and cannot break a bond of 1 eV. The quantity $kT = 26\,\text{meV}$ is the *typical* energy of an agitation mode, not the maximum. It is the exponential tail of the distribution that matters: what is rare for one bond is frequent in a crystal of 10^{23} of them per cm³, and since the tail depends exponentially on T, so does the rate at which pairs are created. This is why the resistivity falls so fast (figure 2.1a).

2.5 How many pairs? Reference values

THE QUESTION

How many electron–hole pairs does a pure semiconductor contain at room temperature, and how does this number explain the fall of the resistivity?

In a pure crystal each broken bond gives one electron and one hole: at equilibrium, the concentration of free electrons equals that of holes. This common value is called the **intrinsic concentration** n_i (*intrinsic*: belonging to the material itself, without impurity).

Computing it requires the energy structure of the crystal and the statistics of electrons, that is, Chapter 3. Here we take it from **measurement**, as an input (Table 2.3).

Material	E_g (eV)	n_i at 300 K (cm^{-3})	Broken bonds
Germanium (Ge)	0.66	2.4×10^{13}	1 in 4×10^9
Silicon (Si)	1.12	1.5×10^{10} (reference: 10^{10})	1 in 10^{13}
Gallium arsenide (GaAs)	1.42	1.8×10^6	—

Table 2.3 *– Intrinsic concentrations at 300 K, inputs (Neamen, Table 4.2). For Si, published values range from 7×10^9 to 1.5×10^{10} cm^{-3}. The right-hand column uses the reference value 10^{10} cm^{-3} and 10^{23} bonds per cm³ for Si, and 8.8×10^{22} bonds per cm³ for Ge (Problem 2.9 of the exercise collection).*

THE REFERENCE VALUE OF THE COURSE

Published measurements of n_i in silicon at 300 K range from 7×10^9 to 1.5×10^{10} cm^{-3} (the exact value depends on details that Chapter 3 will discuss). We adopt the round value $n_i = 10^{10}$ cm^{-3} as the **standard value of the course**: it is used in all examples and exercises of this chapter and of the following ones.

Three things are worth keeping from this table. First, the order of magnitude: 10^{10} pairs per cm³ among 10^{23} bonds per cm³, that is **one broken bond in** 10^{13} in silicon at 300 K. The crystal is almost perfect; it is its size that makes the number. Second, the order of the materials: the smaller E_g, the more pairs, and the dependence is very strong (0.46 eV less from Si to Ge gives 2400 times more pairs). Third, these numbers are **balances**: generation, which follows the Boltzmann tail, is compensated by recombination (Section 2.8).

From pairs to resistivity. Each pair brings two carriers, an electron and a hole, which both conduct (Section 2.6). Applied to both carriers, the Drude expression (2.1) gives the conductivity of the pure material:

$$\sigma_i = q\, n_i\, (\mu_n + \mu_p). \tag{2.3}$$

EXAMPLE 2.1 — HOW MANY PAIRS, AND WHAT RESISTIVITY? (300 K)

Silicon: $n_i = 10^{10}$ cm^{-3}, i.e. one broken bond in 10^{13}. $\sigma_i = 1.6 \times 10^{-19} \times 10^{10} \times (1350 + 460) = 2.9 \times 10^{-6}$ S/cm, so $\rho \approx 3.4 \times 10^5$ Ω·cm.
Germanium: $n_i = 2.4 \times 10^{13}$ cm^{-3}, $\mu_n + \mu_p = 3900 + 1900 = 5800$ cm²/(V·s), $\rho \approx 45$ Ω·cm. The gap of germanium is smaller by 0.46 eV: its n_i is 2400 times larger and its resistivity 7700 times smaller.
Gallium arsenide: $n_i = 1.8 \times 10^6$ cm^{-3}, $\mu_n + \mu_p = 8500 + 400 = 8900$ cm²/(V·s), $\rho \approx 3.9\times10^8$ Ω·cm: pure, it is almost an insulator, and it is used as a ‘semi-insulating’ substrate.

More heat, many more pairs, a lower resistivity. Heating speeds up generation (the Boltzmann tail thickens), and the equilibrium number of pairs rises. An **input**, measured and derived in Chapter 3: between 300 and 350 K, n_i of silicon is multiplied by about 28. A second input: in pure silicon near room temperature, the mobility varies roughly as $T^{-3/2}$ (lattice scattering), so it falls by about 20%. The conductivity (2.3) is multiplied by $28/1.26 \approx 22$: the resistivity of the pure material falls by a factor of 22 for 50 K more.

This is the semiconductor curve of figure 2.1a, understood: in a pure semiconductor the number of carriers wins by far over the mobility.

Note that n_i is multiplied ‘only’ by 28 while the probability of reaching 1.12 eV is multiplied by 450: a number of pairs is not a probability. When there are more pairs, they also recombine more often, and it is the balance that sets their number; Chapter 3 will give the exact law.

And light? A photon with an energy above E_g can also break a bond: light creates pairs in addition to those created by heat, and the resistivity drops. This is **photoconductivity** (observation b). When the light is switched off, recombination brings the number of pairs back to its equilibrium value. This out-of-equilibrium regime is the subject of Chapter 4.

2.6 The hole: a new carrier, a new kind of electricity

THE QUESTION

We now know that there are free electrons. But the broken bond leaves a hole behind: can it carry a current, with what charge, and why is it less mobile than an electron?

2.6.1 A particle of charge $+q$

A hole behaves as a particle. A neighbouring bond can lend one of its electrons to fill the vacancy; the vacancy then sits in the neighbouring bond (figure 2.3b). Repeating this, the vacancy travels through the crystal in the direction opposite to the electrons that fill it. Since a missing electron is a missing negative charge, the vacancy has an **effective charge** $+q$, and it conducts like a positive carrier: under a field $\vec{\mathcal{E}}$, the bonding electrons jump against the field and the vacancy moves along the field, exactly as a positive charge would.

This is a new kind of electricity. In a metal one species carries the current; in a semiconductor the current is carried by **two kinds of entities**, negative free electrons and positive holes, of opposite charge. Under the same field they move in opposite directions, but their currents add up, since charge and velocity change sign together. Hall-effect measurements, which give the sign of the carriers, confirm it: a sample in which holes dominate gives a Hall voltage of the opposite sign to that of an ordinary metal (outside the model of this chapter).

BE AWARE OF… WHAT A HOLE IS

The hole is not a positron, nor a particle from elsewhere: it is an **incomplete bond**, and its motion is the collective motion of the bonding electrons that fill it in turn. Describing this current by the motion of the vacancy alone, with a charge $+q$, is an exact and much more economical way of counting: one vacancy instead of 10^{23} bonding electrons.

2.6.2 Why the hole is less mobile: the picture

Compare the two motions. The free electron, once out of its bond, no longer belongs to any bond: it moves through the whole crystal and is interrupted only by the collisions of Chapter 1. The hole moves only when a **bound electron** of a neighbouring bond jumps into the vacancy. Its motion is a **relay of bound electrons**, each confined to its bond and taking only one step from one bond to the next. A relay of electrons that stay attached to the bonds responds less readily to a force than an electron that moves freely: under the same field, the vacancy moves more slowly. The hole is **less mobile** than the electron.

> **A PICTURE, NOT A PROOF**
>
> This picture tells in which direction the effect goes; it does not tell by how much, and it cannot explain every material. The real reason is that each carrier responds to forces with an **effective mass** that belongs to the crystal, larger for holes than for electrons in Si, Ge and GaAs. Chapter 3 will define it; here the mobilities are measured inputs.

2.6.3 By how much: the measured mobilities

Table 2.4 gives the mobilities measured at 300 K in pure crystals.

Material	μ_n (cm²/(V·s))	μ_p (cm²/(V·s))	μ_n/μ_p
Silicon	1350	460	2.9
Germanium	3900	1900	2.1
Gallium arsenide	8500	400	21

Table 2.4 *– Electron and hole mobilities at 300 K in pure materials (inputs, Neamen, Table 5.1; for holes in silicon Neamen gives 480, and this course uses 460 cm²/(V·s) throughout). In all three materials the hole is the slower carrier.*

In silicon the hole is about three times less mobile than the electron; in germanium, twice; in gallium arsenide, twenty times, because the electron there is exceptionally light (Chapter 3). This is one of the reasons why fast transistors favour conduction by electrons, and why GaAs is the material of microwave circuits (outside the model).

2.6.4 The pure crystal: $n = p$

Since carriers are born in pairs, a pure crystal contains as many free electrons as holes:

$$n_0 = p_0 = n_i \qquad \text{(pure material, at equilibrium).} \tag{2.4}$$

Its conductivity is the sum of the two contributions: this is expression (2.3). In pure silicon the electrons carry 1350/1810 = 75% of the current and the holes 25%: at equal concentrations, the mobility makes the difference.

2.7 Recombination

THE QUESTION

Pairs are created all the time: why does their number not grow without limit? How can their disappearance be described without going into the details of the mechanisms?

Pairs are created, and pairs disappear: a free electron that meets a hole can fill it, which restores the bond and removes both carriers. This is **recombination**. Both processes are counted in pairs per unit volume and per unit time: the **generation rate** G and the **recombination rate** R, in $\mathrm{cm}^{-3}\,\mathrm{s}^{-1}$. Before any microscopic mechanism, can a simple model be built from the meaning of a carrier lifetime alone?

What do we mean by the lifetime of a carrier? Consider an electron with a mean lifetime τ_n before it recombines. During a short interval $\mathrm{d}t$ the probability that it recombines is $\mathrm{d}t/\tau_n$. Among the n electrons per cm³, $n\,\mathrm{d}t/\tau_n$ recombine during $\mathrm{d}t$, so the recombination rate is

$$\mathrm{d}P_n = \frac{\mathrm{d}t}{\tau_n} \implies \mathrm{d}N_R = n\frac{\mathrm{d}t}{\tau_n} \implies R = \frac{n}{\tau_n}. \tag{2.5}$$

What fixes τ_n? An electron can recombine only with a hole. If no hole is available nothing happens, and the more holes there are the more opportunities. The simplest model therefore makes the probability per unit time proportional to the hole concentration, $1/\tau_n = \alpha p$, where α (in cm³/s) measures how effective an encounter is. The recombination rate becomes the **generic recombination law**

$$\frac{1}{\tau_n} = \alpha\, p \qquad \implies \qquad R = \alpha\, n\, p. \tag{2.6}$$

Read it as a product: n electrons, each with a recombination probability per unit time αp. More electrons provide more possible events, more holes provide more opportunities for them, and the simplest model makes the rate proportional to both. The same reasoning seen from the side of the holes gives $1/\tau_p = \alpha n$ and the same rate $R = p/\tau_p = \alpha n p$: each recombination removes one electron *and* one hole. This is the structure of the **law of mass action** of chemistry, where the rate of a reaction between two species is proportional to the product of their concentrations; it is where the name of the law of Section 2.11 comes from.

BE AWARE OF… A STARTING MODEL, NOT A UNIVERSAL LAW

$R = \alpha n p$ is an **assumption of the model**, chosen for its simplicity. Real recombination goes through several physical mechanisms (Chapter 4), and the effective coefficient depends on the mechanism, the defects, the temperature and the concentrations. We use one coefficient α throughout this chapter, so that nothing in it depends on which mechanism is at work.

What the lifetime is, and is not. Individual recombination events are random: the lifetime is not the instant at which each carrier disappears. It describes a population: if a population decays as $\mathrm{e}^{-t/\tau}$, τ is the time for it to fall to $1/\mathrm{e}$ of its initial value, about 36.8%. It is also unrelated to the collision time τ_c of Chapter 1: a collision only resets the velocity of a carrier, a recombination removes it.

And generation? Generation is the breaking of a bond by the thermal energy of the lattice. It needs **no partner**: no encounter is required. We therefore make the following assumption of the model: **thermal generation is set by the material and the temperature; it does not depend on how many free carriers are already present.**

2.8 Equilibrium: generation balances recombination

> ***THE QUESTION***
>
> *What does 'equilibrium' mean for a semiconductor, and what thermal generation rate maintains it?*

A sample left alone at a fixed temperature, with no light, no applied voltage and no current, reaches **thermal equilibrium**: its concentrations stop changing. In a uniform sample with nothing flowing in or out, a concentration can change only through creation and disappearance, and pairs are created and destroyed together:

$$\frac{\mathrm{d}n}{\mathrm{d}t} = \frac{\mathrm{d}p}{\mathrm{d}t} = G - R. \tag{2.7}$$

This is the bookkeeping of the continuity equation of Chapter 1, for a uniform sample, with one difference: the number of carriers is no longer fixed, so a source term $G - R$ appears. Chapter 4 will write it in full, with transport. At equilibrium the concentrations are constant (subscript 0), so

$$G_0 = R_0. \tag{2.8}$$

This does not mean that generation and recombination stop: bonds keep breaking and pairs keep recombining, and the two average rates cancel. Equilibrium is a **dynamic balance**.

The pure material: the thermal generation rate. In the pure material $n_0 = p_0 = n_i$ (2.4). Recombination is then $R_i = \alpha n_i^2$, and equilibrium (2.8) fixes the thermal generation rate:

$$G_i = R_i = \alpha\, n_i^2. \tag{2.9}$$

This result reads both ways. The generation rate G_i is a property of the material and the temperature (assumption of Section 2.7); the number of pairs n_i is the value at which recombination exactly compensates it. If n_i (Table 2.3) and α are known, G_i is known: with an effective value $\alpha = 10^{-11}$ cm³/s, pure silicon creates and destroys $G_i = 10^{-11} \times 10^{20} = 10^9$ pairs per cm³ per second (Problem 2.11 of the exercise collection).

Is this balance stable? Suppose np were above n_i^2: recombination, which grows with np, would exceed generation, which does not, and the number of pairs would fall. Below n_i^2, generation would win and it would rise. Either way the system returns to the crossing point (figure 2.5): the balance is stable, and it is what fixes the number of carriers.

> **WHAT PART A HAS ESTABLISHED**
>
> - At 0 K a metal has free electrons supplied by its bonding; a semiconductor has none, all its electrons sit in bonds. Its carriers must be **created**.
> - Heat creates them **in pairs**, a free electron and a hole, by breaking bonds, thanks to the exponential tail of the thermal distribution and to the huge number of bonds (10^{23} per cm³).

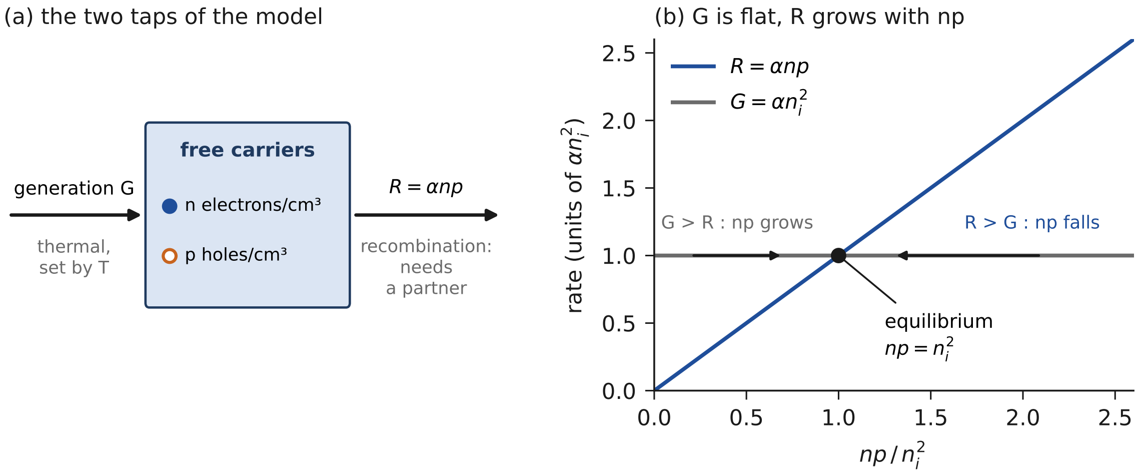


Figure 2.5 – *(a) The 'two-tap' model: generation fills the reservoir of free carriers at a rate set by temperature, recombination empties it at a rate that needs both carriers. (b) Generation does not depend on np; recombination grows with it. They cross where* $np = n_i^2$ *and the crossing is stable: above it recombination wins and np falls, below it generation wins.*

- The pure material contains $n_0 = p_0 = n_i$ carriers of each kind; $n_i = 10^{10}\,\mathrm{cm}^{-3}$ for Si (course reference), 2.4×10^{13} for Ge, 1.8×10^{6} for GaAs. Its conductivity is $qn_i(\mu_n + \mu_p)$, and it rises very fast with T.
- The hole is an incomplete bond that moves by a relay of bound electrons: a carrier of charge $+q$, less mobile than the electron.
- Pairs recombine, at the rate $R = \alpha np$ in the generic model. Equilibrium is a dynamic balance, $G_0 = R_0$, and in the pure material it fixes the thermal generation rate $G_i = \alpha n_i^2$.

PART B

The doped semiconductor

Replacing a few silicon atoms: what is bound at 0 K, what is freed at 300 K, and the laws that fix the concentrations.

2.9 n and p doping: the picture at absolute zero

THE QUESTION

What changes if a few silicon atoms are replaced by atoms of another element? What is bound and what is free, at 0 K and then at room temperature?

A donor: phosphorus. Replace a silicon atom by a phosphorus atom (column V, five valence electrons). Four of its electrons make the four bonds with the silicon neighbours, exactly like a silicon atom. The fifth electron **has no bond to go to**. It is not free for all that: the phosphorus nucleus has one more positive charge than the silicon nucleus, and the phosphorus core, seen from a distance, carries a charge $+q$ that the four bonds do not compensate. The fifth electron is attracted by this charge. **At 0 K it stays bound to its donor**, on an orbit around the phosphorus atom (figure 2.6a). Nothing is free, and the crystal is neutral atom by atom: phosphorus-doped silicon is also an insulator at 0 K.

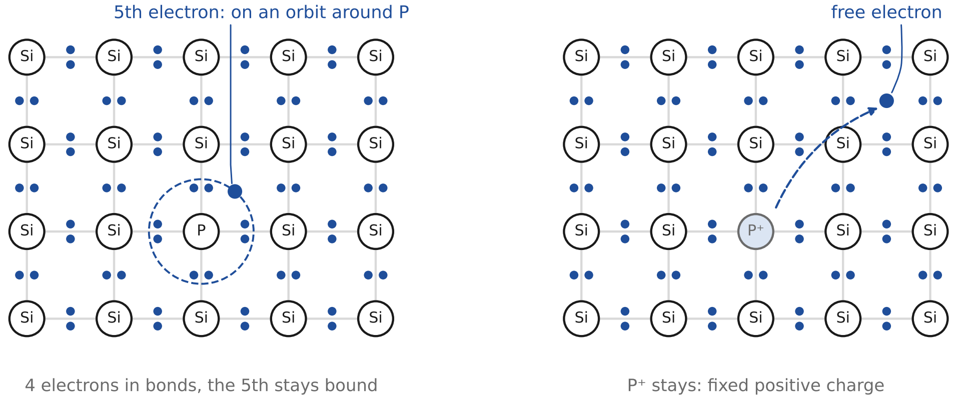
The same donor at room temperature. The fifth electron is only loosely held. How loosely? Measurement gives the energy needed to free it, a **measured input** at this stage: $E_d = 45\,\text{meV}$ for phosphorus in silicon (arsenic 54 meV, antimony 39 meV); Section 2.10 will estimate it. Compared with the 1.12 eV of a bond, the fifth electron is about 25 times less bound than a bonding electron, and its binding energy is comparable to kT: at room temperature thermal agitation frees it easily. The phosphorus atom, called a **donor**, has given a free electron and has become a **fixed positive ion** P^+ (figure 2.6b).

An acceptor: boron. Replace a silicon atom by a boron atom (column III, three valence electrons). One bond out of four lacks an electron: it is a hole. **At 0 K this hole stays bound to the boron**: if it moved away, it would leave behind a negatively charged boron that would attract it back (figure 2.6c). Nothing is free, the crystal is neutral.

The same acceptor at room temperature. An electron of a neighbouring bond completes the boron bond, at a small cost, $E_a \approx 45\,\text{meV}$ (measured); the vacancy moves away: a **free hole**. The boron atom, called an **acceptor**, has accepted an electron and has become a **fixed negative ion** B^- (figure 2.6d).

(a) donor P at 0 K: nothing is free

(b) donor P at 300 K: ionised


5th electron: on an orbit around P
Si
P
free electron
P⁺
4 electrons in bonds, the 5th stays bound
P⁺ stays: fixed positive charge

(c) acceptor B at 0 K: nothing is free

(d) acceptor B at 300 K: ionised

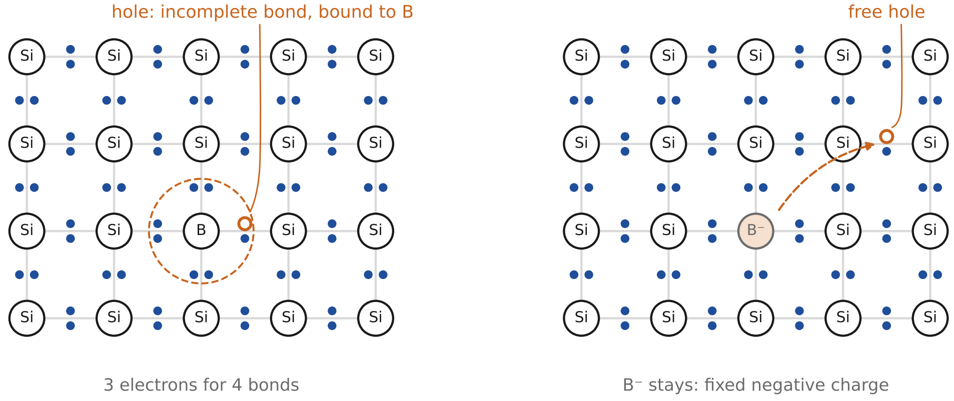

hole: incomplete bond, bound to B
Si
B
free hole
B⁻
3 electrons for 4 bonds
B⁻ stays: fixed negative charge

Figure 2.6 – *Donor and acceptor in silicon. (a) Phosphorus at 0 K: four electrons go into bonds; the fifth stays bound to the donor, on an orbit around it (in reality of radius about ten interatomic distances, Section 2.10). (b) At 300 K this fifth electron is freed: a free electron, a fixed* P^+ *ion. (c) Boron at 0 K: an incomplete bond, the hole, stays bound to the boron. (d) At 300 K a neighbouring electron completes the boron bond: a fixed* B^- *ion and a free hole.*

What is free and what is not. After ionisation the electron (or the hole) moves, the ion does not: it is fixed in the lattice. The crystal as a whole remains **electrically neutral**, since each mobile charge has left a fixed opposite charge behind. If the crystal contains N_D donors per cm³, the concentration of free electrons is of the order of N_D, and the crystal is **n-doped**; with N_A acceptors per cm³ it has about N_A holes and it is **p-doped**. Sections 2.11 and 2.12 make these 'abouts' precise.

BE AWARE OF... IONISATION IS NOT BOND BREAKING

Ionising a donor breaks no bond: the fifth electron had none. It therefore creates **a single carrier**, an electron, not a pair; the compensating charge is a fixed ion, not a hole. Likewise, ionising an acceptor creates a hole alone. This is what allows doping to produce an excess of carriers of one kind only.

2.10 The solid imposes its rules: a hydrogen atom inside silicon

THE QUESTION

Why is the fifth electron of phosphorus held by only a few tens of meV, when the electron of a hydrogen atom is held by 13.6 eV? And what fraction of the donors is ionised at a given temperature?

2.10.1 The hydrogen-like model

An electron bound to a charge $+q$: this is the hydrogen atom, with a binding energy of 13.6 eV and a radius of 0.053 nm (the Bohr radius). The donor electron and the P^+ ion form such an atom, with one essential difference: they are not in vacuum but **inside silicon**, and the solid imposes its rules. We model the situation, as an input, by a hydrogen atom with two changes.

First change: the dielectric. Between the ion and the electron there is silicon, whose bonding electrons polarise and screen the charge of the ion. The Coulomb force is divided by the relative permittivity $\varepsilon_r = 11.7$. The binding energy of hydrogen is proportional to the square of the Coulomb coupling (it varies as $1/\varepsilon_r^2$ when ε_0 is replaced by $\varepsilon_r\varepsilon_0$): it is divided by $\varepsilon_r^2 = 137$,

$$\frac{13.6\,\text{eV}}{\varepsilon_r^2} = \frac{13.6}{137} \approx 0.10\,\text{eV}.$$

Second change: the mass. In the crystal the electron does not respond to the force of the ion with its vacuum mass m_0, but with a smaller **effective mass**, $m^* = 0.26\,m_0$ in silicon (an **input**; Chapter 3 will explain where it comes from). The binding energy of hydrogen is proportional to the mass of the electron. The hydrogen-like estimate is therefore

$$E_d \approx 13.6\,\text{eV} \times \frac{m^*/m_0}{\varepsilon_r^2} = 13.6 \times \frac{0.26}{137} \approx 26\,\text{meV}, \tag{2.10}$$

and the orbit has the Bohr radius multiplied by ε_r and divided by m^*/m_0:

$$a_d \approx 0.053\text{ nm} \times \varepsilon_r \frac{m_0}{m^*} = 0.053 \times \frac{11.7}{0.26} \approx 2.4\text{ nm}. \tag{2.11}$$

The model justifies its own assumption. Using ε_r, a macroscopic property of silicon, makes sense only if the orbit contains many atoms. It does: 2.4 nm is about ten interatomic distances (0.235 nm), and the sphere of the orbit contains nearly 3000 silicon atoms (figure 2.7). The electron does 'see' a continuous medium.

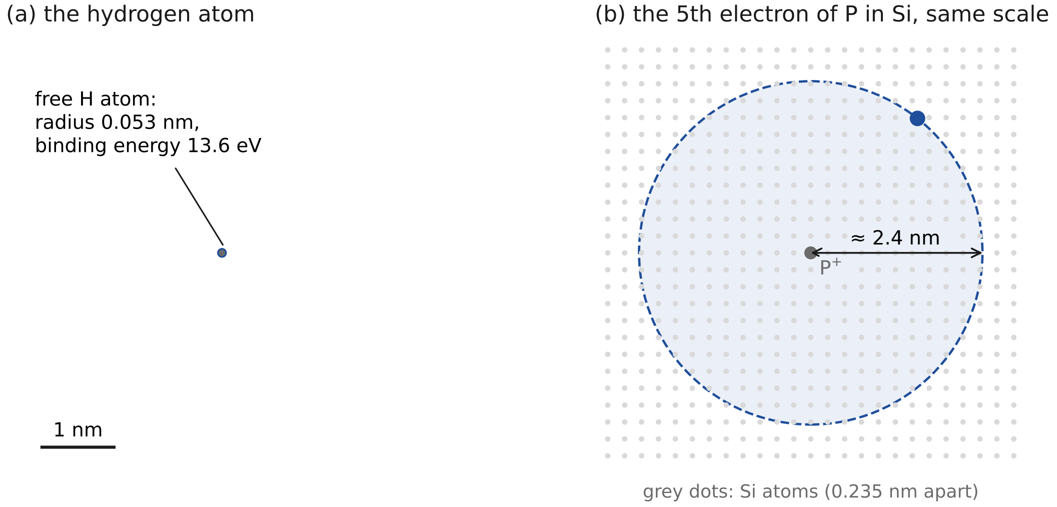


Figure 2.7 *– The hydrogen-like model to scale. (a) The free hydrogen atom: an orbit of 0.053 nm, barely visible. (b) The fifth electron of a donor in silicon, at the same scale: an orbit of radius ≈ 2.4 nm, which encloses thousands of atoms. The grey dots represent schematically the silicon atoms, spaced by the nearest-neighbour distance.*

Comparison with measurement. The model gives the order of magnitude, a few tens of meV, of the measured values (Table 2.5). The differences between impurities come from the core of each impurity atom, which the model ignores. The acceptor is the mirror image: the hole bound to the B^- ion is a 'hydrogen' of opposite charge, and its ionisation energy is of the same order.

Dopant	Type	Measured ionisation energy in Si (meV)
Phosphorus (P)	donor	45
Arsenic (As)	donor	54
Antimony (Sb)	donor	39
Boron (B)	acceptor	45

Table 2.5 *– Ionisation energies of the usual dopants of silicon (inputs, Sze and Lee, ch. 1, Fig. 20). The hydrogen-like estimate (2.10) gives 26 meV for a donor.*

2.10.2 What fraction of the donors is ionised?

These energies are **comparable to** kT (26 meV at 300 K): donors must ionise at room temperature. But 45 meV is not small compared with 26 meV, and one might expect partial ionisation. Figure 2.8 shows what the full calculation gives, for 45 meV donors in silicon.

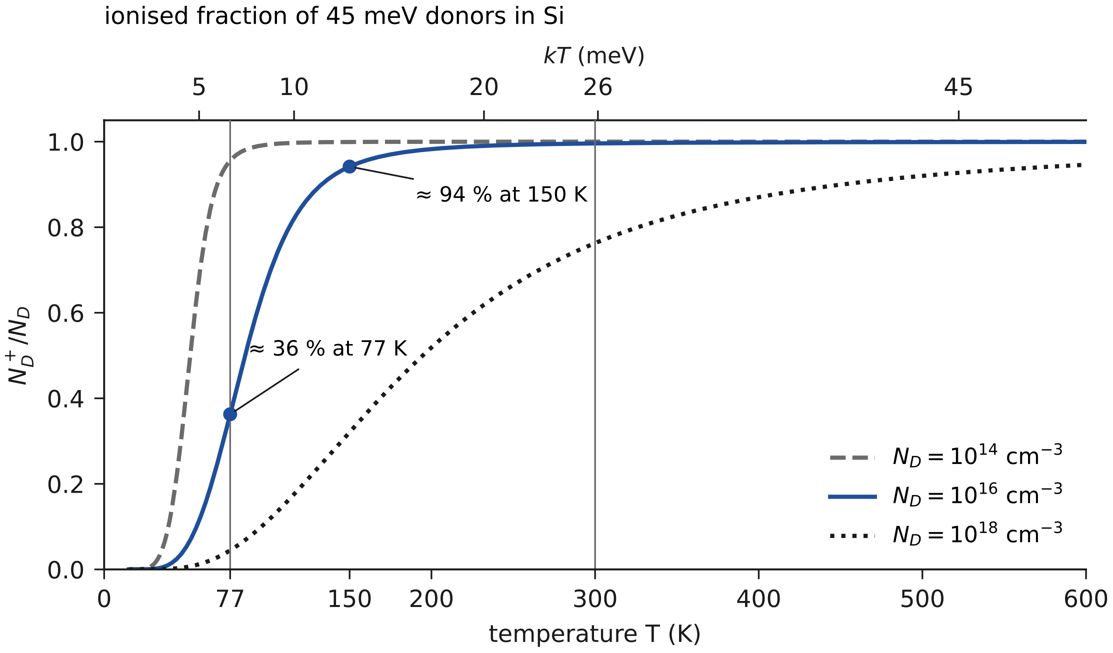


Figure 2.8 *– Ionised fraction of donors (45 meV) in silicon against temperature, for three doping levels. The top axis gives kT in meV. The curves are computed with the ionisation law and charge neutrality; this law will be derived in Chapter 3.*

INPUT: THE LAW BEHIND THIS CURVE

The law that gives the ionised fraction against temperature requires the statistics of electrons and the energy structure of the crystal; it will be **derived in Chapter 3**. Here we read the curve as data.

Three observations follow from it.

- **At low temperature the donors are not ionised** (freeze-out): almost all the electrons stay on their donors and the sample conducts poorly. At 77 K (liquid nitrogen) only about 36%, a third, of $10^{16}\,\mathrm{cm}^{-3}$ donors are ionised: $kT = 6.6\,\mathrm{meV}$ is seven times smaller than 45 meV.
- **Above about 200 K, ionisation is practically complete** for $N_D \lesssim 10^{16}\,\mathrm{cm}^{-3}$. For $10^{16}\,\mathrm{cm}^{-3}$ the ionised fraction is already about 94% at 150 K, 98% at 200 K and 99.6% at 300 K.
- **Heavier doping ionises later.** For $N_D = 10^{17}\,\mathrm{cm}^{-3}$ about 96% of the donors are ionised at 300 K; for $10^{18}\,\mathrm{cm}^{-3}$ only about 76%. But at this last doping the donors are only about 10 nm apart, a few orbit radii: neighbouring orbits nearly touch and the model of isolated donors itself stops being valid (outside the model). The curve should then be read as a trend.

BE AWARE OF... WHY 45 MEV DONORS ALMOST ALL IONISE WITH $kT = 26$ MEV

It is again a **balance**, as for the pairs. Thermal ionisation of a donor needs no partner; capture, which brings a free electron back onto an ionised donor, requires an encounter. Donors are rare, one for millions of atoms: a free electron has the whole crystal at its disposal and seldom finds an empty donor. The balance therefore leans very strongly towards ionisation, far more than the factor $e^{-45/26}$ alone would suggest. Chapter 3 will put numbers on this argument.

For the rest of the chapter we keep: **near room temperature and for moderate doping, all donors and all acceptors are ionised**, $N_D^+ = N_D$ and $N_A^- = N_A$. This is **total ionisation**, an assumption whose domain is given by figure 2.8.

2.11 Doping a pure semiconductor: do the concentrations simply add?

THE QUESTION

Add $N_D = 10^{16}\,\mathrm{cm}^{-3}$ *donors to pure silicon, where* $n_i = 10^{10}\,\mathrm{cm}^{-3}$. *Do the concentrations simply add, with* $n = n_i + N_D$ *and* $p = n_i$*?*

The naive answer. The thermal pairs were there before doping, and the donors add their electrons. One is tempted to add up: $n = n_i + N_D \approx 10^{16}\,\mathrm{cm}^{-3}$ and $p = n_i = 10^{10}\,\mathrm{cm}^{-3}$. For the electrons this is not far from the truth. **For the holes it is wrong**, by a factor of a million.

What goes wrong. Neutrality does not object: $n = p + N_D^+$ is satisfied. It is the **balance** that forbids it. With these concentrations recombination would be $R = \alpha np = \alpha \times 10^{16} \times 10^{10} = \alpha \times 10^{26}$, while thermal generation stays at $G_i = \alpha n_i^2 = \alpha \times 10^{20}$ (2.9). Recombination would win by a factor of a million: this is not an equilibrium. Each hole is now surrounded by abundant electrons and finds a partner a million times faster; pairs disappear until recombination falls back to the level of generation. The electrons lose only a minute fraction; the holes collapse.

The mass-action law. Where does the system stop? Compare the same material at the same temperature, pure and doped. For the pure material, (2.9) gives $G_i = \alpha n_i^2$. For the doped one, equilibrium gives $G_0 = R_0 = \alpha n_0 p_0$. Now make two assumptions: doping leaves the thermal generation rate unchanged, $G_0 = G_i$ (the assumption of Section 2.7, extended to the dopant atoms), and it leaves the coefficient α unchanged. Then

$$\alpha\, n_0\, p_0 = R_0 = G_0 = G_i = \alpha\, n_i^2 \qquad \Longrightarrow \qquad n_0\, p_0 = n_i^2. \tag{2.12}$$

The product of the two concentrations is the same in every sample of a given material at a given temperature: raising n_0 lowers p_0 by the same factor. This is the **mass-action law**. The logic is worth keeping in mind: generation and recombination in balance, recombination needs both carriers, hence a product law.

In the example of the question, $n_0 \approx 10^{16}\,\mathrm{cm}^{-3}$ and therefore $p_0 = n_i^2/n_0 = 10^{20}/10^{16} = 10^4\,\mathrm{cm}^{-3}$, not 10^{10}. **The concentrations do not simply add**: n doping adds electrons *and removes* holes.

THE LIMIT OF THIS ARGUMENT

The equality $G_0 = G_i$ and the constancy of α do not follow from equilibrium: they are **added assumptions** of an ideal model. Real dopants can bring defects, change lifetimes and, when very dense, the structure of the crystal itself. The argument therefore explains why a product law should exist; it does not prove that its value

> is exactly n_i^2 in every sample, nor say where it stops holding. Chapter 3 derives the mass-action law independently, from the statistics of electrons, and gives its value and its domain of validity (doping not too heavy).

Majority and minority carriers. In n-type material $n_0 \gg p_0$: electrons are the **majority carriers** and holes the **minority carriers** (the reverse in p-type material). **Why are minority carriers fewer?** They are still created by heat, at the same rate G_i as in the pure material; there are fewer of them because each one finds a partner much sooner among the abundant majority carriers: its lifetime $\tau_p = 1/(\alpha n_0)$ is very short. The balance $\alpha n_0 p_0 = G_i$ then holds them down to $p_0 = n_i^2/n_0$.

The vocabulary is now fixed. A **pure** (or **intrinsic**) semiconductor has no deliberate impurity: its carriers come in pairs from broken bonds. A **doped** (or **extrinsic**) semiconductor has donors or acceptors, which supply a majority carrier; the minority carrier is still produced in pairs by heat but is far outnumbered.

2.12 Three laws for two unknowns: majority and minority carriers

> **THE QUESTION**
>
> *Mass action fixes only the product $n_0 p_0$. What other relations give the two numbers separately, for a sample doped n, p, or both at once?*

2.12.1 The three laws

The second relation is **charge neutrality**. A homogeneous sample at equilibrium carries no net charge: the negative charges (free electrons and ionised acceptors) balance the positive ones (holes and ionised donors). With mass action and total ionisation, we have three laws:

> **THE THREE LAWS OF THE SEMICONDUCTOR AT EQUILIBRIUM (GENERIC VIEW)**
>
> $$\begin{aligned} &\text{(1) mass action:} && n_0\,p_0 = n_i^2 \\ &\text{(2) total ionisation:} && N_D^+ = N_D, \qquad N_A^- = N_A \\ &\text{(3) charge neutrality:} && n_0 + N_A^- = p_0 + N_D^+ \end{aligned} \tag{2.13}$$
>
> Law (1) comes from the balance $G = R$ (Section 2.11); law (2) is the assumption of Section 2.10.2, valid near room temperature; law (3) states that a homogeneous sample has no net charge. Unknowns: n_0 and p_0. Data: n_i, N_D, N_A.

2.12.2 n-type: the exact root and the majority approximation

Take an n-type sample doped with donors only ($N_A = 0$). Neutrality reads $n_0 = p_0 + N_D$. Substituting $p_0 = n_i^2/n_0$ from mass action gives the quadratic

$$n_0^2 - N_D\,n_0 - n_i^2 = 0, \tag{2.14}$$

whose positive root is

$$n_0 = \frac{N_D}{2} + \sqrt{\frac{N_D^2}{4} + n_i^2}, \qquad p_0 = \frac{n_i^2}{n_0}. \tag{2.15}$$

The majority approximation. When $N_D \gg n_i$ the square root is close to $N_D/2$ and the result simplifies to the familiar form

$$\text{n-type } (N_D \gg n_i): \qquad n_0 \simeq N_D, \qquad p_0 \simeq \frac{n_i^2}{N_D}. \tag{2.16}$$

How large is '$\gg$'? Expanding the root, $n_0 \approx N_D + n_i^2/N_D$: the relative error on the majority concentration is about $(n_i/N_D)^2$, so it falls below 1% as soon as $N_D \geq 10\,n_i$ (figure 2.9). In this regime the majority concentration is controlled by the dopants alone and the semiconductor is called **extrinsic**; when thermally generated carriers dominate instead ($N_D \lesssim n_i$), it behaves as the pure material and is called **intrinsic**.

p-type is the mirror image. Doped with acceptors only, neutrality reads $p_0 = n_0 + N_A$; the exact root follows by exchanging $n_0 \leftrightarrow p_0$ and $N_D \leftrightarrow N_A$ in (2.15), and

$$\text{p-type } (N_A \gg n_i): \qquad p_0 \simeq N_A, \qquad n_0 \simeq \frac{n_i^2}{N_A}. \tag{2.17}$$

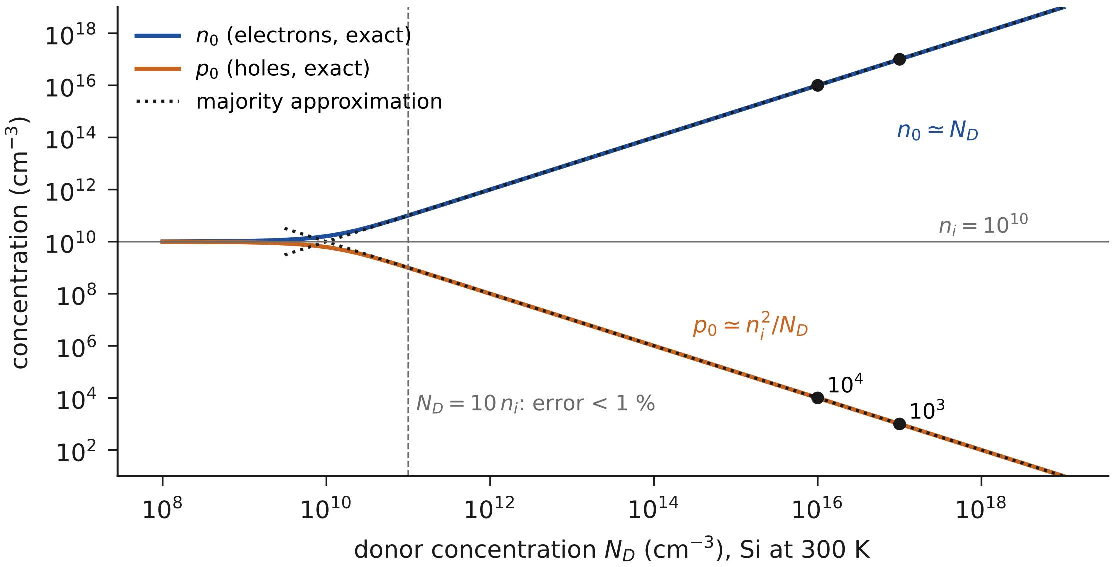


Figure 2.9 *– Exact concentrations (2.15) against donor concentration for silicon at 300 K (the p-type side is the mirror image with n_0 and p_0 exchanged). Dotted: majority approximations (2.16). The curves merge with n_i when $N_D \lesssim n_i$: the sample then behaves as the pure material. The dots correspond to Example 2.2.*

2.12.3 Compensated material

A sample may contain both donors and acceptors: this is common in technology, where a region already doped one way is doped the other way. The material is then called **compensated**. The electrons given up by the donors first go to complete the bonds of the acceptors: the two dopings partly cancel. Neutrality (3) reads $n_0 - p_0 = N_D - N_A$, and everything happens as if the sample were doped by the difference alone. If $N_D > N_A$ and $N_D - N_A \gg n_i$,

$$n_0 \simeq N_D - N_A, \qquad p_0 \simeq \frac{n_i^2}{N_D - N_A}. \tag{2.18}$$

The exact root is (2.15) with N_D replaced by $N_D - N_A$; if $N_A > N_D$ the sample is p-type with $p_0 \simeq N_A - N_D$.

2.12.4 Orders of magnitude and examples

Table 2.6 gives the usual doping levels. Even the heaviest doping stays dilute: dopant atoms are counted in parts per million.

Doping	N (cm^{-3})	One dopant per	Typical uses
Light	10^{14} to 10^{15}	5×10^8 to 5×10^7 Si atoms	substrates, detectors, regions that must withstand a high voltage
Moderate	10^{16} to 10^{17}	5×10^6 to 5×10^5 atoms	wells, transistor channels and bases, solar cells
Heavy	$\geq 10^{18}$	5×10^4 atoms or fewer	ohmic contacts, emitters, sources and drains

Table 2.6 *– Usual doping levels in silicon (5×10^{22} atoms per cm³) and their uses (outside the model: the devices come in later chapters). Above about 10^{18} cm^{-3} the formulas of this chapter become approximate.*

EXAMPLE 2.2 — N-TYPE SILICON, $N_D = 10^{16}$ AND 10^{17} CM^{-3}, 300 K

$N_D = 10^{16} \gg n_i$, so $n_0 \simeq 10^{16}\,\text{cm}^{-3}$ and $p_0 = n_i^2/n_0 = 10^{20}/10^{16} = 10^4\,\text{cm}^{-3}$. Silicon has 5×10^{22} atoms per cm³: one donor per 5×10^6 atoms multiplies the electron concentration by 10^6 with respect to the pure material and divides the hole concentration by 10^6. The product stays at n_i^2.

Ten times more donors, $N_D = 10^{17}\,\text{cm}^{-3}$: $n_0 \simeq 10^{17}\,\text{cm}^{-3}$ and $p_0 = 10^{20}/10^{17} = 10^3\,\text{cm}^{-3}$. When the doping increases, the minority concentration decreases by the same factor.

EXAMPLE 2.3 — P-TYPE GERMANIUM, $N_A = 10^{17}$ CM^{-3}, 300 K

$N_A = 10^{17} \gg n_i = 2.4 \times 10^{13}\,\text{cm}^{-3}$: $p_0 \simeq 10^{17}\,\text{cm}^{-3}$ and $n_0 = n_i^2/N_A = (2.4 \times 10^{13})^2/10^{17} = 5.76 \times 10^{26}/10^{17} \approx 5.8 \times 10^9\,\text{cm}^{-3}$. At the same doping germanium keeps far more minority carriers than silicon (10^3), because its n_i is 2400 times larger, and n_i^2 six million times.

EXAMPLE 2.4 — WHAT A TRACE OF PHOSPHORUS DOES TO SILICON

One phosphorus atom per 10^7 silicon atoms is $5 \times 10^{22}/10^7 = 5 \times 10^{15}$ donors per cm³. With total ionisation and $N_D \gg n_i$, $n_0 \simeq 5 \times 10^{15}\,\text{cm}^{-3}$, against 10^{10} in the pure material, and $p_0 = 2 \times 10^4\,\text{cm}^{-3}$, negligible. With $\mu_n = 1350$ cm²/(V·s), $\rho = 1/(qn_0\mu_n) = 1/(1.6 \times 10^{-19} \times 5 \times 10^{15} \times 1350) = 0.93\ \Omega$·cm, compared with $3.4 \times 10^5\ \Omega$·cm for pure silicon (Example 2.1): a factor of about 4×10^5 for one atom in ten million. This is what makes semiconductors controllable, and what makes purity the central difficulty of their technology.

EXAMPLE 2.5 — A COMPENSATED SILICON

A sample contains $N_D = 10^{16}$ and $N_A = 4\times10^{15}\,\text{cm}^{-3}$. It is n-type, with $n_0 \simeq N_D - N_A = 6\times10^{15}\,\text{cm}^{-3}$ and $p_0 = 10^{20}/(6\times10^{15}) \approx 1.7\times10^{4}\,\text{cm}^{-3}$. Its 1.4×10^{16} ionised dopants per cm³ supply only 6×10^{15} free electrons: the others went to complete the bonds of the acceptors.

BE AWARE OF THE DOMAIN OF VALIDITY

Four conditions stand behind these results: **thermal equilibrium** (no light, no injection), **total ionisation** (not too cold, figure 2.8), $N \gg n_i$ for the majority approximation (not too hot: n_i grows very fast with T), and **doping not too heavy** (below about $10^{18}\,\text{cm}^{-3}$ in silicon, so that the dopants remain isolated atoms and mass action keeps its value n_i^2).

2.13 What is still missing

THE QUESTION

Do we have everything needed to compute the carrier concentrations of any sample, at any temperature?

We now have a clear qualitative picture. We know what the carriers of a semiconductor are, free electrons and holes; where they come from, bonds broken by heat and ionised dopants; why their number is set by a balance between creation and disappearance; and which relations link the concentrations of majority and minority carriers. With three simple laws we can compute these concentrations in a doped sample at room temperature.

But two numbers of this chapter were taken from measurement without being explained: the intrinsic concentration, ten billion pairs per cubic centimetre in silicon, and the ionisation energy of the dopants, forty-five thousandths of an electronvolt, which the hydrogen model approaches only within a factor of two. And two questions remain open: how does the intrinsic concentration vary with temperature, and how do the concentrations of a doped sample evolve when it is cooled down to carrier freeze-out or heated up to the return of the intrinsic regime? The ionisation curve of Section 2.10.2 was shown, not derived.

To answer, two ingredients are needed that the generic view does not contain. The first is the **band structure** of the crystal: which energies an electron may have in the solid, how many states there are at each energy, and why the energy needed to break a bond is the width of a forbidden band. The second is **Fermi–Dirac statistics**: how electrons, which cannot be two in the same state, are distributed among these states at a given temperature. With these two ingredients, Chapter 3 will compute the intrinsic concentration and its temperature dependence, recover the mass-action law by an independent route, derive the ionisation law of dopants, and follow the concentrations of a doped sample from absolute zero up to high temperatures.

WHAT PART B HAS ESTABLISHED

- A donor (P, column V) has a fifth electron with no bond; an acceptor (B, column III) has an incomplete bond. At 0 K the electron or the hole stays **bound to its dopant** and nothing is free; at 300 K it is freed and leaves a **fixed ion** behind (P^+, B^-). Ionising a dopant creates a single carrier.
- The solid imposes its rules: the fifth electron is a hydrogen atom in a dielectric, with an effective mass. $E_d \approx 13.6\,\text{eV} \times (m^*/m_0)/\varepsilon_r^2 \approx 26\,\text{meV}$, orbit of 2.4 nm; measured: 45 meV (P). Comparable to kT: ionisation is practically complete above about 200 K for moderate doping (law derived in Chapter 3).
- The concentrations do not simply add: the balance $G = R$, with $G_0 = G_i$ and α unchanged, gives the **mass-action law** $n_0 p_0 = n_i^2$. n doping adds electrons and removes holes.
- Mass action, total ionisation and neutrality give n_0 and p_0: for $N_D \geq 10\,n_i$, $n_0 \simeq N_D$ and $p_0 \simeq n_i^2/N_D$; for a compensated material, $n_0 \simeq N_D - N_A$.
- What is missing, the value of n_i and the temperature dependence, needs the band structure and Fermi–Dirac statistics: Chapter 3.

WHICH OBSERVATIONS ARE NOW EXPLAINED

- **Resistivity against temperature** (observation a): in the pure material the number of pairs rises very fast with T and wins over the fall of the mobility; ρ falls (Section 2.5). In a doped sample, once the dopants are ionised, $n_0 \simeq N_D$ no longer depends on T as long as $n_i \ll N_D$: near room temperature ρ follows the mobility, as in a metal. The full law is in Chapter 3.
- **Sensitivity to impurities**: one donor in 10^7 atoms changes ρ by more than five orders of magnitude (Example 2.4).
- **Still open**: photoconductivity (b) needs carriers added to equilibrium and their lifetime (Chapter 4); rectification (c), light emission (d) and the photovoltaic effect (e) need, in addition, junctions (later chapters).

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Observations**: ρ of a metal rises with T, that of a semiconductor falls; light lowers ρ; a trace of impurity changes ρ by orders of magnitude; rectification, light emission and the photovoltaic effect exist in junctions.
- **At 0 K**: a metal has free electrons supplied by its bonding and conducts; in Si, Ge and diamond every electron is in a covalent bond: insulators. E_g = measured energy to free an electron from a bond (1.12 eV Si, 0.66 eV Ge, 5.5 eV C).
- **Metals**: n fixed, μ falls with T (input), ρ rises. Semiconductor: ρ falls, so n must be created.

- **Pairs**: a broken bond gives a free electron and a hole. The tail $e^{-E/kT}$ and the 10^{23} bonds per cm³ make 1.12 eV bonds break at 300 K ($E_g/kT = 43$; diamond 212).
- **Values**: $n_i = 10^{10}\,\mathrm{cm}^{-3}$ (Si, reference), 2.4×10^{13} (Ge), 1.8×10^{6} (GaAs); one bond in 10^{13} in Si. $\sigma_i = qn_i(\mu_n + \mu_p)$: $\rho \approx 3.4 \times 10^{5}$ Ω·cm (Si), 45 Ω·cm (Ge).
- **The hole**: an incomplete bond, charge $+q$, moves by a relay of bound electrons, hence less mobile; $\mu_n/\mu_p = 2.9$ (Si), 2.1 (Ge), 21 (GaAs); exact reason: effective mass (Chapter 3). Pure material: $n = p = n_i$.
- **Recombination**: $R = n/\tau_n$, $1/\tau_n = \alpha p$, hence $R = \alpha np$ (generic model). Generation needs no partner.
- **Equilibrium**: $\mathrm{d}n/\mathrm{d}t = G - R$, $G_0 = R_0$, dynamic and stable; pure material: $G_i = \alpha n_i^2$.
- **Doping**: at 0 K the fifth electron of P (the hole of B) stays bound to the dopant; at 300 K it is free, the P^+ (B^-) ion stays fixed. Hydrogen-like model: $E_d \approx 13.6\,\mathrm{eV}\,(m^*/m_0)/\varepsilon_r^2 \approx 26\,\mathrm{meV}$, $a_d \approx 2.4$ nm; measured 45 meV. Ionisation practically complete above about 200 K.
- **Mass action**: $n_0 p_0 = n_i^2$ (assumptions: $G_0 = G_i$, α unchanged). The concentrations do not simply add.
- **Three laws**: mass action, total ionisation, neutrality $n_0 + N_A^- = p_0 + N_D^+$. n-type: $n_0 \simeq N_D$, $p_0 \simeq n_i^2/N_D$ (within 1% if $N_D \geq 10\,n_i$); compensated: $n_0 \simeq N_D - N_A$.
- **What is missing**: band structure and Fermi–Dirac statistics (Chapter 3).

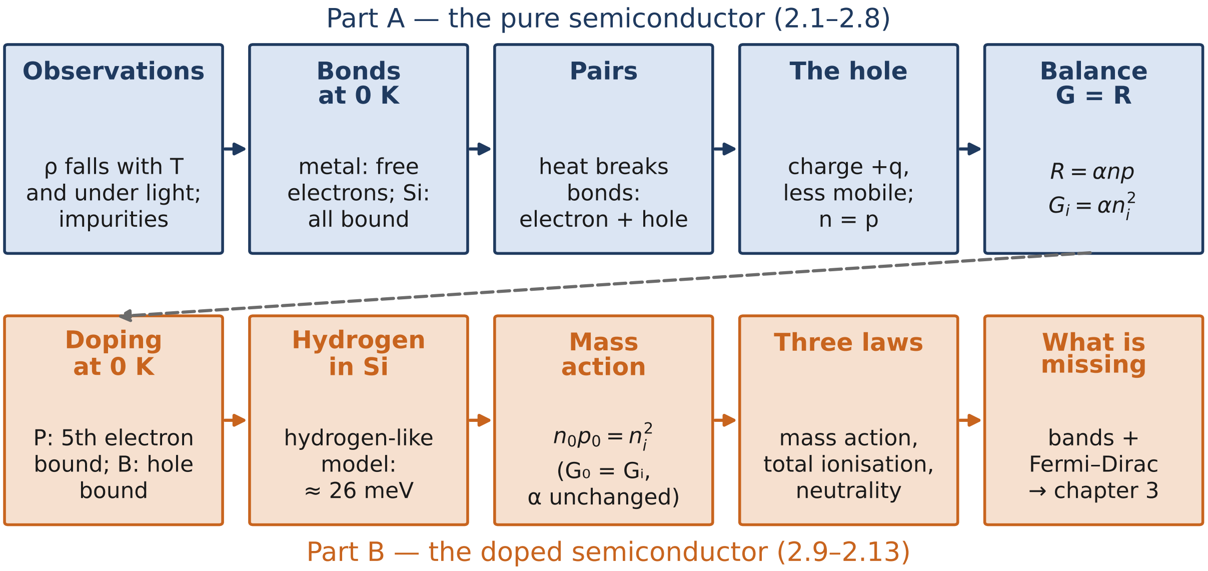

Part A — the pure semiconductor (2.1–2.8)
Observations
ρ falls with T and under light; impurities
Bonds at 0 K
metal: free electrons; Si: all bound
Pairs
heat breaks bonds: electron + hole
The hole
charge +q, less mobile; n = p
Balance G = R
R = αnp
Gᵢ = αnᵢ²
Doping at 0 K
P: 5th electron bound; B: hole bound
Hydrogen in Si
hydrogen-like model: ≈ 26 meV
Mass action
n₀p₀ = nᵢ²
(G₀ = Gᵢ, α unchanged)
Three laws
mass action, total ionisation, neutrality
What is missing
bands + Fermi–Dirac → chapter 3
Part B — the doped semiconductor (2.9–2.13)

Figure 2.10 *– The logic of the chapter. Part A (blue): observations, bonds and a balance of generation and recombination explain where the carriers of the pure material come from. Part B (orange): doping, the hydrogen-like model and three laws give the concentrations of a doped sample; what remains to be computed needs the bands and the statistics of Chapter 3.*

Comprehension questions

No new physics here: answer by reading the figures and the equations of the chapter. The answers are at the end.

Q1. Why does the resistivity of a metal rise with temperature while that of a pure semiconductor falls? Answer with the Drude expression (2.1) and say which ingredient is a measured input.

Q2. Why is pure silicon an insulator at 0 K? Why is diamond, with the same structure, still an insulator at 300 K, while silicon is not?

Q3. The thermal energy kT is 26 meV and E_g is 1.12 eV. Why are there nevertheless free carriers at 300 K? Why does the probability $e^{-E_g/kT}$ not give their number?

Q4. A bond of a silicon crystal breaks. How many carriers appear, and of which kind? Under a field $\vec{\mathcal{E}}$ pointing to the right, in which direction do the bonding electrons that fill the vacancy move, and in which direction does the hole move?

Q5. Why is the hole less mobile than the electron in silicon? Is the relay picture enough to explain that μ_n/μ_p is 21 in GaAs?

Q6. Recombination needs two carriers and generation needs none. Use this to explain why G is taken independent of the carrier concentrations while R is not.

Q7. In an n-type sample the minority holes are created by heat at the same rate as in the pure material. Why are they nevertheless far fewer?

Q8. Why is the fifth electron of phosphorus about 25 times less bound than a bonding electron of silicon? Why are we allowed to use the permittivity ε_r of silicon in this estimate?

Q9. Does a phosphorus-doped silicon sample ($10^{16}\,\text{cm}^{-3}$) conduct at 0 K? At 77 K? At 300 K? Justify with figure 2.8.

Q10. $10^{16}\,\text{cm}^{-3}$ donors are added to pure silicon. Why has the hole concentration not stayed equal to n_i? Which of the three laws (2.13) does the naive answer violate, and which does it respect?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 2, with their answers; the worked solutions are in the **Solutions** document. The 8 exercises that used to close this chapter are Problems 2.9 to 2.16 there; the other problems of the set (2.1 to 2.8) go further.

Answers

Q1. In a metal n is fixed by the bonding; heating increases the scattering, so τ_c and μ fall and $\rho = 1/(nq\mu)$ rises. The fall of τ_c with T is a measured input: the Drude model does not predict it. In a semiconductor the scattering also increases, but n increases very fast because bonds break; this wins over the fall of μ, so ρ falls.

Q2. At 0 K all the electrons are in bonds, nobody is free, and nothing can move. At 300 K bonds break in silicon because $E_g/kT = 43$; in diamond $E_g/kT = 212$, the exponential factor is smaller by about 10^{73}, and there is practically no carrier.

Q3. Because the probability of a fluctuation of energy E_g is small, but not zero: $\propto \mathrm{e}^{-E_g/kT}$, and the crystal has 10^{23} bonds per cm^3. That probability governs the creation of pairs, not their number: pairs also recombine, and the number present is set by the balance $G = R$. Its value is computed in Chapter 3.

Q4. Two: a free electron and a hole, always together. The bonding electrons jump to the left, against the field; the vacancy moves to the right, along the field, as a positive charge.

Q5. The hole moves only by jumps of bound electrons from one bond to the next, while the free electron moves through the whole crystal: for the same force, the relay responds less readily. The picture gives the direction of the effect, not its size: the ratio 21 of GaAs comes from an exceptionally light electron, that is, from the effective mass (Chapter 3).

Q6. Thermal generation only needs energy from the lattice, whatever the number of carriers already present; recombination needs an electron and a hole to meet, so it grows with np. Doping changes the carriers, not (in the ideal model) the thermal energy supplied by the lattice.

Q7. Each hole finds a partner much sooner, because electrons are abundant: its recombination probability per unit time, αn_0, is enormous. The balance $\alpha n_0 p_0 = G_i = \alpha n_i^2$ then gives $p_0 = n_i^2/n_0$.

Q8. The fifth electron is not in a bond: it orbits a positive ion screened by the silicon ($\varepsilon_r^2 = 137$) and responds with a small effective mass ($0.26\, m_0$), which gives tens of meV (45 measured) against 1.12 eV for a bond. Using ε_r is justified because the orbit (2.4 nm) encloses thousands of atoms: the electron sees a continuous medium.

Q9. At 0 K, no: the fifth electron stays bound to its donor. At 77 K about a third of the donors are ionised: it conducts, but poorly. At 300 K 99.6% are ionised: $n_0 \simeq 10^{16}\,\text{cm}^{-3}$.

Q10. The holes, surrounded by abundant electrons, recombine a million times faster than in the pure material, until $\alpha n_0 p_0$ falls back to G_i. The naive answer respects neutrality (3) and ionisation (2), but violates mass action (1): $np = 10^{26}$ instead of 10^{20}.

References

- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — ch. 4: electrons and holes, dopants, hydrogen-like model, ionisation, charge neutrality (Table 4.2: intrinsic concentrations, Table 2.3); ch. 5, Table 5.1: mobilities (Table 2.4); ch. 6: generation and recombination.
- S. M. Sze and M. K. Lee, *Semiconductor Devices: Physics and Technology*, 3rd ed., Wiley, 2012 — ch. 1, Fig. 20: measured ionisation energies of impurities in Si and GaAs (Table 2.5).

- S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley, 2007 — ch. 1: physics and properties of semiconductors.
- C. Kittel, *Introduction to Solid State Physics*, 8th ed., Wiley, 2005 — ch. 8: semiconductor crystals, holes, hydrogen-like impurity states.
- N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Holt, Rinehart and Winston, 1976 — ch. 28: homogeneous semiconductors, impurity levels, law of mass action.
- AT&T Bell Laboratories, educational film with Walter Brattain on the properties of semiconductors (observations of Section 2.1).

WHERE THIS LEADS

We now have a complete qualitative picture of the semiconductor at equilibrium: bonds, pairs created by heat and destroyed by recombination, dopants that give or accept an electron, and three laws that fix the concentrations. Chapter 3 makes this picture quantitative: it introduces the band structure and Fermi–Dirac statistics, computes n_i and its temperature dependence, recovers the mass-action law, derives the ionisation law of dopants and locates the Fermi level. Chapter 4 will then take the sample out of equilibrium, by light or injection.

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 3

Pure and Doped Semiconductors at Equilibrium: Energy Bands and the Fermi Level

Bonds recalled · free electrons · bands and gaps · metal, insulator, semiconductor · Fermi–Dirac · effective mass and density of states · n, p and n_i · the intrinsic level · donor levels and ionisation · neutrality · the Fermi level against doping and temperature

---

WHERE WE ARE

Chapter 2 built the qualitative picture of a semiconductor at thermal equilibrium, without bands and without a Fermi level. From what is observed it identified the carriers, electrons and holes, created in pairs by heat and destroyed by recombination; from the balance of the two it obtained the mass-action law $n_0 p_0 = n_i^2$; with total ionisation and charge neutrality it gave n_0 and p_0 of a doped sample. To do so it took several numbers as inputs: the value $n_i = 10^{10}\ \mathrm{cm}^{-3}$ for silicon at 300 K, the binding energy of 45 meV of the fifth electron of phosphorus (estimated at 26 meV by the hydrogen model), the fact that holes are heavier and less mobile than electrons, and the curve of the ionised fraction of the donors against temperature, shown but not derived.

This chapter derives them. It needs two ingredients that chapter 2 named but did not use: the **energy structure of the crystal** (bands and gaps) and the **statistics of the electrons** (the Fermi–Dirac function). Part A builds the bands, Part B adds the statistics and treats the pure semiconductor, Part C treats the doped semiconductor at any temperature. Everything stays at thermal equilibrium; chapter 4 will take the sample out of it.

CHAPTER OUTLINE

3.1 Two kinds of bonding, recalled

PART A — Energy bands

3.2 The metal: free electrons in a wire

3.3 A periodic potential: bands and gaps

3.4 Metal, insulator, semiconductor

PART B — Statistics and the pure semiconductor

3.5 Fermi–Dirac statistics and the Fermi level

3.6 Effective mass and density of states

3.7 From bands to densities: n and p

3.8 The pure semiconductor: the mass-action law, n_i and E_i

3.9 Intrinsic density and temperature

3.10 Where is the Fermi level of a pure semiconductor, and how does it move with temperature?

PART C — The doped semiconductor

3.11 Donor and acceptor levels: the ionised donors N_D^+

3.12 Doping leaves the bands unchanged: only E_F moves
3.13 Neutrality, N_D^+, n and p: the doped sample at any temperature
3.14 The position of the Fermi level in doped material
3.15 Clarifications
Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Contrast metallic and covalent bonding, and say what chapter 2 left open and which two ingredients answer it.
- Introduce, in a few paragraphs and without solid-state formalism, the quantisation of k, the energy bands and the forbidden gap, and count the states of a band ($2N$ per band).
- Use the filling of the bands to classify metal, insulator and semiconductor, and define the Fermi level at 0 K.
- Introduce the Fermi–Dirac function, the effective mass and the density of states, and derive n and p in terms of the Fermi level.
- Derive the mass-action law from the bands, compute n_i and its temperature dependence, and locate the intrinsic level E_i at any temperature.
- Derive the number of ionised donors, combine it with neutrality to obtain n_0, p_0 and E_F of a doped sample at any temperature, and place E_F against doping.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- state the quantisation of k for a free electron in a wire, explain how a periodic potential creates bands and gaps, show that a band holds $2N$ states, and explain why the $8N$ valence electrons of silicon fill exactly four bands;
- classify metal, insulator and semiconductor from the band filling and the size of the gap ($E_g < 100\,kT$ or $> 200\,kT$);
- write the Fermi–Dirac function, say what the Fermi level is and is not, and derive $n = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$ and $p = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$ with their validity condition;
- define the effective mass from the curvature of $E(k)$ and explain with it why holes are less mobile than electrons in silicon;
- show that E_F cancels in np, compute n_i, explain the factor $\frac{1}{2}$ in $n_i \propto \mathrm{e}^{-E_g/2kT}$, and derive $E_i(T)$;
- derive $N_D^+ = N_D/[1 + 2\,\mathrm{e}^{(E_F-E_D)/kT}]$, write the neutrality equation in E_F, explain freeze-out, the extrinsic plateau and the intrinsic regime, and compute E_F of n- and p-type samples with a cross-check.

HOW THIS CHAPTER WORKS

- **Quantitative after qualitative.** Chapter 2 gave the picture and the laws; this chapter computes the numbers they contain. Each result of chapter 2 is recovered and its domain of validity is made explicit.
- **What belongs to the model is said.** Every statement is either derived, or a model input stated as such (a measured value, an imported result of solid-state or statistical physics), or a fact outside the model, given afterwards and marked.
- **A deliberately short solid-state course.** Part A gives a few ideas, the results we need stated as inputs where they are imported, and the references where the proofs are (D. Neamen, *Semiconductor Physics and Devices*, chapters 2 to 4, is the natural companion).
- **What is deliberately left out.** What happens when the equilibrium is disturbed (light, injection), how fast it comes back and how the carriers move in space: chapter 4.

Symbol	Meaning	Remark
$n, p; n_0, p_0$	electron and hole concentrations; at thermal equilibrium	cm^{-3}; subscript 0 = equilibrium
n_i	intrinsic concentration	$n_0 = p_0 = n_i$ in the pure material
k, L, a, N	wavenumber; wire length; lattice period; number of cells	$p = \hbar k$
E_C, E_V, E_g	lowest energy of the conduction band, highest energy of the valence band, gap	$E_g = E_C - E_V$
E_F, E_i	Fermi level; intrinsic level (Fermi level of the pure material)	E_i is near midgap
E_D, E_A	donor and acceptor energy levels	just below E_C, just above E_V
N_D, N_A, N_D^+	donor and acceptor concentrations; ionised donors	cm^{-3}
$f(E)$	Fermi–Dirac occupation probability	$1 - f$: probability that the state is empty
$g_c(E), g_v(E)$	densities of states per unit volume and energy	$\text{cm}^{-3}\,\text{eV}^{-1}$
N_C, N_V	effective densities of states of the two bands	cm^{-3}; $\propto T^{3/2}$
m_n^*, m_p^*	effective masses of electrons and holes	density-of-states or conductivity mass (Section 3.6)
μ_n, μ_p	mobilities	$\text{cm}^2/(\text{V·s})$

Table 3.1 – Notation of the chapter.

Notation. Silicon at 300 K throughout: $kT = 25.9\,\text{meV}$ (26 meV in words), $kT \ln 10 = 59.6\,\text{meV}$, $E_g = 1.12\,\text{eV}$, $N_C = 2.8 \times 10^{19}$ and $N_V = 1.04 \times 10^{19}\,\text{cm}^{-3}$, $\mu_n = 1350$ and

$\mu_p = 460\ \text{cm}^2/(\text{V·s})$, $\varepsilon_r = 11.7$. Reference value $n_i = 10^{10}\,\text{cm}^{-3}$, used in all examples of the course (the formula of Section 3.8 gives 6.9×10^9; published values range from 7×10^9 to 1.5×10^{10}). The values for germanium and gallium arsenide are in Table 3.3.

3.1 Two kinds of bonding, recalled

> ***THE QUESTION***
>
> *Chapter 2 compared a metal and a semiconductor at zero kelvin through their bonds. What, in each bond, decides how the electrons may move, and what does a quantitative theory need to add?*

Metallic bonding. In a metal such as copper or sodium, each atom gives up its outer electrons to the whole crystal. The positive ion cores sit on a lattice, bathed in a sea of electrons that belong to no atom in particular: the bonding itself provides free electrons, about one per atom, even at 0 K. This is the picture behind the Drude model of chapter 1. Sommerfeld made it quantum: in his **free-electron model** the electrons do not interact with the ions or with each other, only with the boundaries of the sample, and they obey quantum mechanics and the Pauli principle. Section 3.2 solves this model in one dimension.

Covalent bonding. In silicon, germanium, diamond or gallium arsenide, each atom shares its four valence electrons with four neighbours. The four bonds point to the corners of a tetrahedron (the sp^3 hybridisation), and the crystal is a perfectly periodic stacking of these tetrahedra:

- the **diamond structure** for the elements of column IV (C, Si, Ge);
- the **zinc-blende structure** for the III–V compounds (GaAs, InP) and some II–VI compounds (ZnS, CdS), where the two neighbours of a bond are of different kinds.

In both cases the smallest repeated unit, the primitive cell, contains **two atoms**; the crystal of length L along a direction is a chain of $N = L/a$ such cells. At 0 K all the valence electrons are in bonds: no electron is free, and the crystal is an insulator. Semiconductors and insulators share this bonding; chapter 2 showed that heat breaks some bonds and creates electron–hole pairs.

What chapter 2 left open. The qualitative picture gave the right laws but took their numbers as inputs:

- the value of n_i ($10^{10}\,\text{cm}^{-3}$ in silicon, 2.4×10^{13} in germanium): why these numbers, and why such different ones?
- how n_i, and therefore the resistivity of a pure sample, depends on temperature;
- the measured 45 meV of the phosphorus donor, against the 26 meV of the hydrogen estimate, and the curve of the ionised fraction against temperature;
- why holes are less mobile than electrons;
- how n_0 of a doped sample behaves when the temperature is far from 300 K.

Two ingredients answer all of these. The first is the **band structure**: which energies an electron may have in the periodic crystal, and how many states there are at each energy

(Part A, and Section 3.6). The second is the **Fermi–Dirac statistics**: how electrons are distributed among those states at a temperature T (Section 3.5).

PART A

Energy bands

Which energies an electron may have in a crystal, and how many states each band holds.

3.2 The metal: free electrons in a wire

THE QUESTION

What energies can an electron have if nothing but the walls of a wire confines it?

We start from Sommerfeld's model: electrons that do not interact with the ions or with each other, only with the boundaries of the sample. Take a one-dimensional wire of length L. In quantum mechanics an electron is described by a wave function $\psi(x)$, and with zero potential inside the wire its stationary states obey the Schrödinger equation

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = E\,\psi. \tag{3.1}$$

The solutions are plane waves, with a wavenumber k related to the momentum by the de Broglie relation $p = \hbar k$. Their energy is the kinetic energy $p^2/2m$, written as a function of k:

$$\psi_k(x) = A\,\mathrm{e}^{ikx}, \qquad E(k) = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m}, \qquad p = \hbar k. \tag{3.2}$$

The relation $E(k)$ is a **parabola**. It is the first example of what is called a **dispersion relation**, or an energy–momentum relation $E = f(p)$, the central object of the theory: knowing it tells us which energies an electron can have and how it responds to forces.

Not every k is allowed. The wave must fit the wire: if the wire is closed on itself (periodic boundary conditions), the wave must come back to the same value after a length L, $\mathrm{e}^{ikL} = 1$. This quantises the wavenumber:

$$\mathrm{e}^{ikL} = 1 \implies k_j = \frac{2\pi}{L}\,j, \qquad j = 0, \pm1, \pm2, \ldots \tag{3.3}$$

The allowed states are **equally spaced in** k, with a spacing $2\pi/L$. For a wire of one centimetre the spacing is tiny against the range of k that matters, so the states look like a continuum, but they are countable, and counting them is the whole point. The Pauli exclusion principle allows at most one electron per quantum state; a state is specified by k **and** the spin, so each value of k holds **two** electrons (figure 3.1).

At absolute zero the N_e electrons of the wire occupy the lowest states: all states below a limiting wavenumber k_F, none above. Counting two electrons per k,

$$2 \cdot \frac{2k_F}{2\pi/L} = N_e \implies k_F = \frac{\pi}{2}\frac{N_e}{L}, \qquad E_F = \frac{\hbar^2 k_F^2}{2m}. \tag{3.4}$$

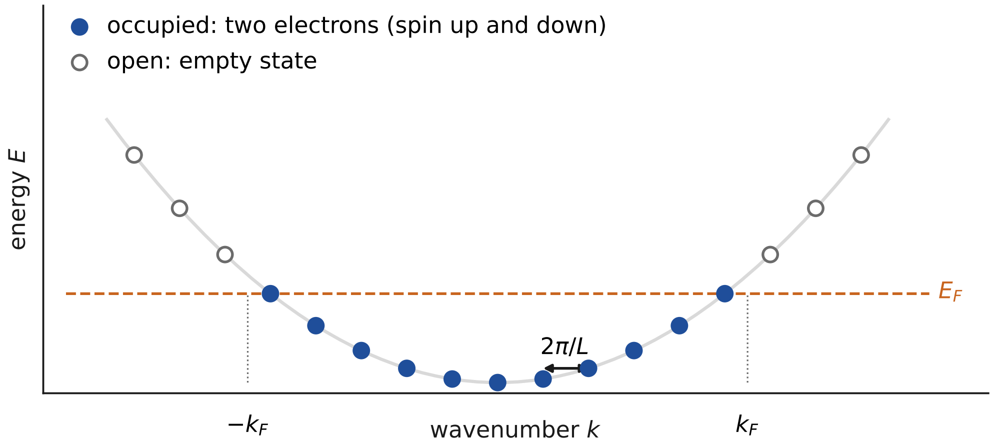


Figure 3.1 *– Free electrons in a wire at 0 K. The allowed wavenumbers are spaced by* $2\pi/L$*; each holds two electrons. The states are filled up to* k_F*;* E_F *is the energy of the last occupied state.*

EXAMPLE 3.1 — A ONE-DIMENSIONAL METAL

A chain of atoms spaced by $a = 0.30$ nm, one electron per atom: $N_e/L = 1/a$. Then $k_F = \pi/2a = 5.2 \times 10^9\ \text{m}^{-1}$ and $E_F = \hbar^2 k_F^2/2m = 1.04\,\text{eV}$. The wavenumber k_F is exactly **half** of π/a: remember it, it will mean “half-filled” in Section 3.4.

The Fermi level, first definition. The energy E_F is, at 0 K, the boundary between occupied and empty states: **the Fermi level is the energy of the highest occupied state at 0 K.** This is the first meaning of the Fermi level; Section 3.5 extends it to $T > 0$.

3.3 A periodic potential: bands and gaps

THE QUESTION

What changes when the electron moves not in empty space but among ions regularly spaced by a distance a*?*

In a crystal the ions repeat with period a, and the electron sees a **periodic potential** $V(x + a) = V(x)$. Solving the Schrödinger equation in such a potential is the subject of solid-state physics. We do not solve it here: we take two of its results as **inputs of our model** (Kittel, ch. 7; Neamen, ch. 3), and give for each its physical reason, not its proof.

First input: k is defined modulo $2\pi/a$. Two wavenumbers that differ by $2\pi/a$ describe the same state in the lattice, because the lattice cannot distinguish them (it only samples the wave at the sites). So it is enough to describe the electron by a k in one interval of width $2\pi/a$, taken symmetric around zero: the **first Brillouin zone**,

$$-\frac{\pi}{a} < k \le \frac{\pi}{a} \qquad \text{gap at } k = \pm\frac{\pi}{a}. \tag{3.5}$$

Second input: forbidden energies. A wave with $k = \pm\pi/a$ has a wavelength $2a$, exactly the condition for Bragg reflection by planes spaced by a: it is reflected by the lattice instead of travelling through it. Instead of a travelling wave there are two **standing waves**. One piles the electron density on the ions, where the potential energy is low; the other piles it between the ions, where it is high. At $k = \pm\pi/a$ these two waves have

different energies, and no state exists between the two values. The parabola of the free electron is cut: **allowed bands** separated by **forbidden gaps** (figure 3.2).

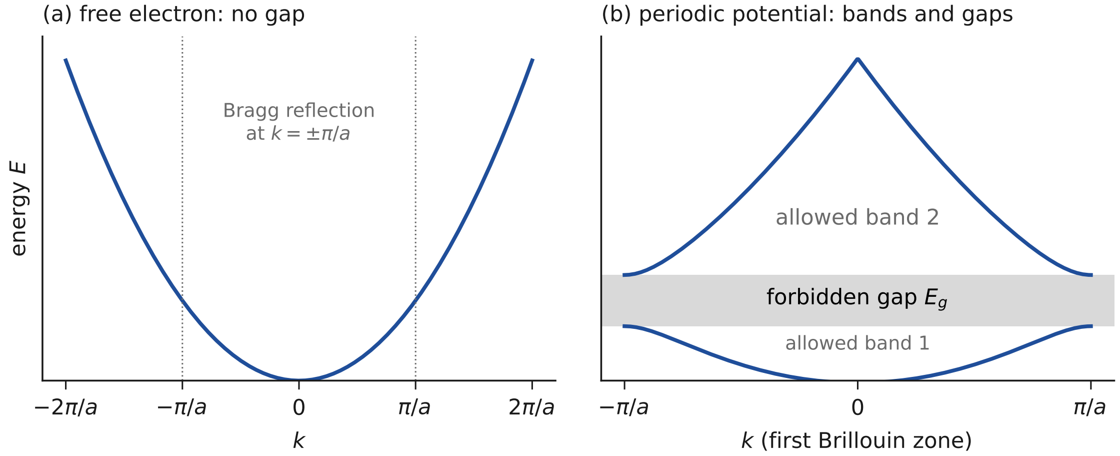


Figure 3.2 *– (a) The free electron: a single parabola. (b) In the periodic potential, the dispersion relation is reduced to the first Brillouin zone and a gap* E_g *opens at* $k = \pm\pi/a$*: allowed energies form bands, forbidden energies form gaps.*

How many states does a band hold? This one we can count ourselves. Take a chain of N cells of length a, so $L = Na$. The wavenumbers of the first zone are spaced by $2\pi/L$ and the zone has width $2\pi/a$. With two electrons (spin up and down) per value of k,

$$\frac{2\pi/a}{2\pi/L} = \frac{L}{a} = N \text{ values of } k \implies 2N \text{ states per band.} \qquad (3.6)$$

A band holds $2N$ states: two electrons per unit cell, the factor 2 coming from the spin. This simple counting is what classifies solids. It does not depend on the shape of the bands, only on the number of cells; it carries over to three dimensions, where each band again holds two states per primitive cell.

3.4 Metal, insulator, semiconductor

> ***THE QUESTION***
>
> *Why is a full band an insulator, and a half-filled band a metal? And why is silicon, at 0 K, exactly on the insulating side?*

Consider the chain of N cells again, one atom per cell, and suppose each atom brings z valence electrons. Each band holds $2N$ electrons. Fill the bands from the bottom, as in Section 3.2, and look at the result on the bands of the one-dimensional model itself (figure 3.3).

- $z = 1$: N electrons for $2N$ states, the band is **half full** (this is the $k_F = \pi/2a$ of Example 3.1). Right next to the last occupied state there are empty states. An electric field pushes every electron the same way and shifts the whole set of occupied k slightly (figure 3.3a): there are now more electrons going one way than the other, a net current. The material conducts: a **metal**.
- $z = 2$: $2N$ electrons, the band is **exactly full** (figure 3.3b). There is no empty state to move to inside the band, and to reach the next band an electron needs at least the

gap energy E_g. A full band also carries no current as a whole: for each occupied state k the state $-k$ is occupied too, and their contributions cancel. The material does not conduct: an **insulator** (at 0 K).

- $z = 3$: two electrons fill the first band, the third half-fills the second: a metal again.

Silicon. Silicon has 4 valence electrons per atom and 2 atoms per cell. A crystal of N cells therefore has

$$4 \text{ electrons/atom} \times 2 \text{ atoms/cell} \times N \text{ cells} = 8N \text{ electrons},$$

which fill exactly $8N/2N =$ **4 bands** at 0 K (figure 3.3c). The highest of these full bands is the **valence band**; the next allowed band, empty at 0 K, is the **conduction band**; between them lies the gap. The same count holds for germanium, diamond and, in the zinc-blende structure, for GaAs ($3+5$ electrons per pair of atoms).

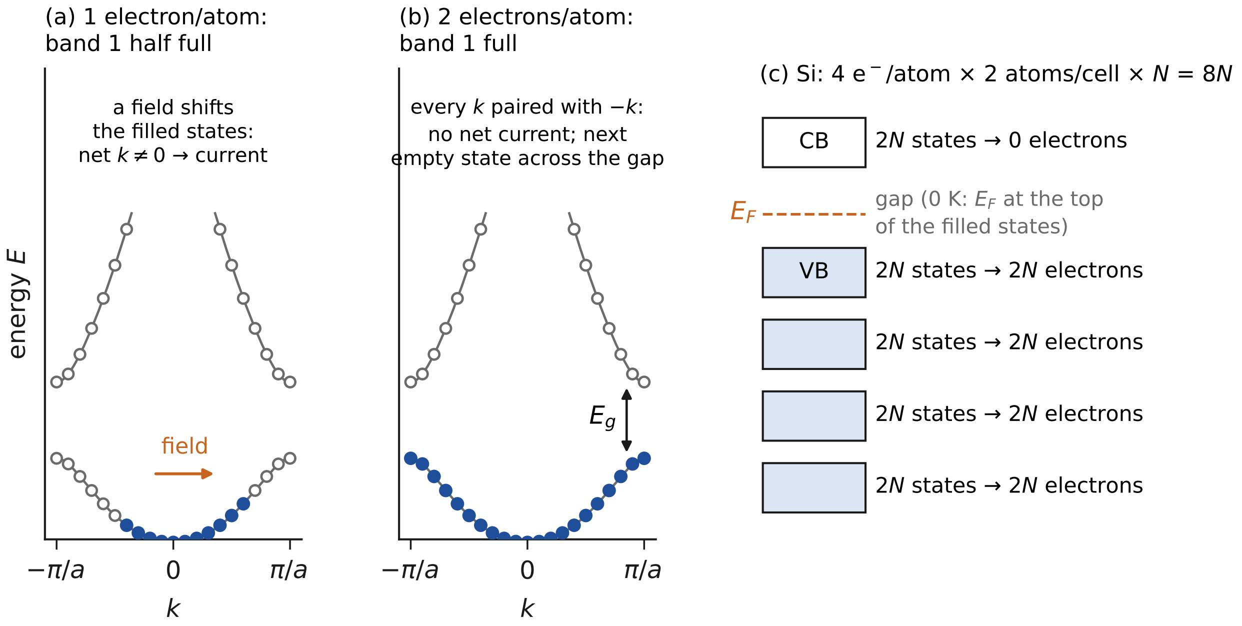

(a) 1 electron/atom:
band 1 half full
a field shifts
the filled states:
net k ≠ 0 → current
field
energy E
−π/a
0
π/a
k
(b) 2 electrons/atom:
band 1 full
every k paired with −k:
no net current; next
empty state across the gap
E_g
−π/a
0
π/a
k
(c) Si: 4 e⁻/atom × 2 atoms/cell × N = 8N
CB
2N states → 0 electrons
E_F
gap (0 K: E_F at the top
of the filled states)
VB
2N states → 2N electrons
2N states → 2N electrons
2N states → 2N electrons
2N states → 2N electrons

Figure 3.3 *– Filling the bands (filled circles: occupied k states, two electrons each; open circles: empty states). (a) One electron per atom: band 1 is half full; a field shifts the occupied states and a net current flows. (b) Two electrons per atom: band 1 is full, every k is paired with $-k$, and the first empty state lies across the gap. (c) Silicon: $8N$ electrons in bands of $2N$ states fill exactly four bands; at 0 K the Fermi level is at the top of the filled states, in the gap.*

The difference between an insulator and a semiconductor is only the size of the gap. At 0 K, both have a full valence band and an empty conduction band (figure 3.4). At a finite temperature, the thermal energy can lift an electron across the gap. This is the broken bond of chapter 2, and the gap E_g is the bond-breaking energy: the values 1.12, 0.66 and 5.48 eV are band gaps (Table 3.2). A useful working criterion compares E_g with kT:

- if $E_g < 100\,kT$ (below 2.6 eV at 300 K), enough pairs appear to matter: a **semiconductor** (silicon: $E_g/kT = 43$);
- if $E_g > 200\,kT$ (above 5.2 eV at 300 K), practically none do: an **insulator** (diamond: $E_g/kT = 212$).

The criterion is a rule of thumb, not a sharp boundary: chapter 2's Boltzmann factor shows how fast the number of pairs collapses between the two.

Material	C (diamond)	Si	Ge	GaAs	Sn (grey tin)
E_g at 300 K (eV)	5.48	1.12	0.66	1.42	0
E_g/kT	212	43	25	55	0
Type	insulator	semiconductor	semiconductor	semiconductor	conductor

Table 3.2 *– Gap energies of a few materials of the diamond and zinc-blende structures at 300 K (Neamen, Table B.4, for Si, Ge and GaAs; diamond after Sze & Ng).*

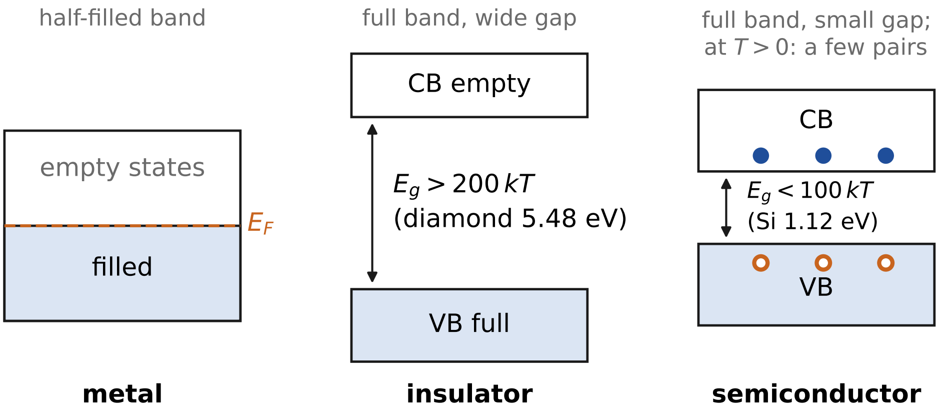
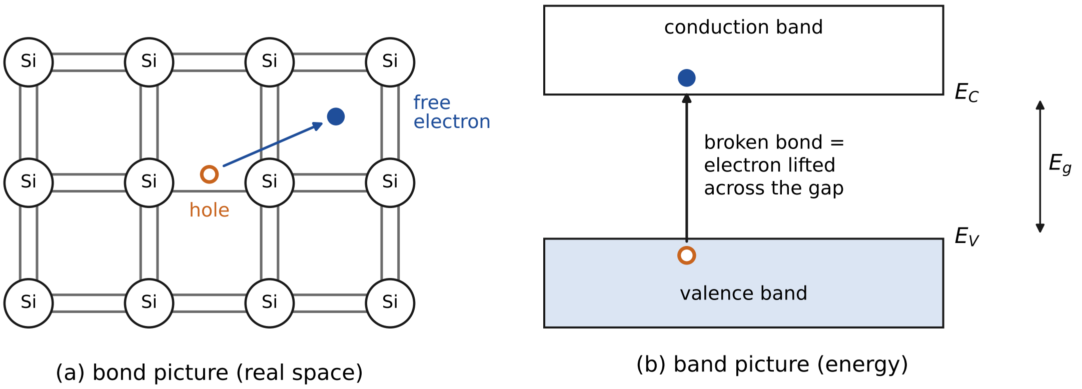


Figure 3.4 *– Three solids in the usual simplified drawing. (a) One electron per atom: the band is half full and empty states are adjacent, a metal; the Fermi level is inside the band. (b) A full valence band, an empty conduction band and a large gap: an insulator. (c) Same filling, small gap: at $T > 0$ a few electrons are promoted to the conduction band, leaving as many holes.*

The Fermi level at 0 K. With the definition of Section 3.2, the highest occupied state at 0 K, the Fermi level of a metal is inside a band. For silicon it is the top of the valence band, and nothing is occupied above it up to E_C: the boundary between occupied and empty states lies **in the gap** (figure 3.3c). Section 3.10 shows that at 0 K it sits exactly at midgap.

> **OUTSIDE THE MODEL: "EVEN NUMBER OF ELECTRONS" DOES NOT MEAN "INSULATOR"**
>
> The one-dimensional count suggests that an even number of valence electrons per cell always gives full bands, hence an insulator. In three dimensions this is not true: two bands can **overlap** in energy, and electrons then occupy the bottom of the upper band before the top of the lower one is full. Magnesium and calcium have two valence electrons per atom and are metals for this reason. The count of $2N$ states per band holds; what the one-dimensional picture misses is that the bands of a real crystal need not be separated by a gap.

The bond picture and the band picture of the same event. Breaking a bond (figure 3.5a) is, on the energy diagram (figure 3.5b), lifting an electron from the valence band to the conduction band, which leaves a hole in the valence band. The electron lifted to the conduction band finds there a large number of empty states: it moves freely, as a metal electron does. The hole it leaves in the valence band is an empty state in an otherwise full band: neighbouring electrons can move into it, which is the motion of the vacancy described in chapter 2.



(a) bond picture (real space)

(b) band picture (energy)

Figure 3.5 *– The bond picture and the band picture of the same event. Breaking a bond (a) is, on the energy diagram (b), lifting an electron from the valence band to the conduction band, which leaves a hole in the valence band.*

Which bands carry a current? Conduction is the motion of electrons from one state to another state of the same band, which requires empty states nearby. A **completely full band carries no current** (every k paired with $-k$, no empty state to move to), and a **completely empty band carries none either** (no electron). This is why semiconductors are insulators at 0 K. At $T > 0$ the few electrons of the conduction band and the few empty states of the valence band, the holes, are the only carriers: the two carriers of chapter 2 are now located on the energy diagram.

THE BAND PICTURE AND ITS NOTATION (USED FOR THE REST OF THE COURSE)

- $\mathbf{E_C}$: the lowest energy of the conduction band. $\mathbf{E_V}$: the highest energy of the valence band. $\mathbf{E_g = E_C - E_V}$ is the gap.
- Energy is plotted **upward**. An electron higher on the diagram has more energy; a hole has more energy when it is **lower** on the diagram.
- E_C and E_V are band **edges**, not the only energies: an electron in the conduction band can sit above E_C, the excess being kinetic energy (Section 3.6).
- The horizontal axis of the usual band diagram is **position**: the crystal is uniform, so the bands are horizontal lines; the diagram says nothing about where the electrons are, only about their energies.

WHAT PART A HAS ESTABLISHED

- A free electron in a wire has $E = \hbar^2 k^2/2m$ with $k = 2\pi j/L$; each k holds two electrons; at 0 K the states are filled up to E_F, the Fermi level, highest occupied energy.
- A periodic potential reduces k to the first Brillouin zone and opens gaps (inputs). A band holds $2N$ states, two per cell.

- Half-filled band: metal. Full bands and an empty band separated by a gap: insulator at 0 K. Silicon's $8N$ electrons fill exactly four bands: the valence band is full, the conduction band empty, E_F at 0 K is in the gap.
- Semiconductor and insulator differ only by the gap ($E_g < 100\,kT$ or $> 200\,kT$). A broken bond is an electron in the conduction band plus a hole in the valence band; full and empty bands carry no current.
- Still missing: how many electrons occupy the conduction band at $T > 0$. That needs the statistics (Part B).

PART B

Statistics and the pure semiconductor

How electrons fill the bands at a temperature T, and what this gives for the pure material.

3.5 Fermi–Dirac statistics and the Fermi level

THE QUESTION

The bands tell us which energies are allowed. How many electrons occupy them at a temperature T?

Electrons obey the Pauli principle, and electrons of the same kind are indistinguishable. The statistics that follows, which we take from statistical physics (an **input**), is the **Fermi–Dirac distribution**: the probability that a state of energy E is occupied is

$$f(E) = \frac{1}{1 + \exp[(E - E_F)/kT]}. \tag{3.7}$$

The parameter E_F is the **Fermi level**. Its meaning is read on the function itself (figure 3.6a). When $E = E_F$, $f = 1/2$. At absolute zero, f is a step: every state below E_F is occupied and every state above is empty, which is the situation of the wire of Section 3.2: the two definitions of E_F agree at 0 K. At a finite temperature the step is **smeared over an energy range of a few kT around E_F**: states just below E_F start to empty and states just above start to fill.

The holes: $1 - f$. A hole is an empty state of the valence band. The probability that a state of energy E is **empty** is

$$1 - f(E) = \frac{1}{1 + \exp[-(E - E_F)/kT]}. \tag{3.8}$$

The function is symmetric about E_F: a state at $E_F + \Delta$ is occupied with the same probability as a state at $E_F - \Delta$ is empty.

The Boltzmann limit. Far above E_F the exponential in the denominator dominates the 1, and f reduces to the Boltzmann factor of chapter 2; far below, $1 - f$ does the same:

$$f(E) \simeq \mathrm{e}^{-(E-E_F)/kT} \quad (E - E_F \gg kT), \qquad 1 - f(E) \simeq \mathrm{e}^{-(E_F-E)/kT} \quad (E_F - E \gg kT). \tag{3.9}$$

The error is below about 5 % as soon as $|E - E_F| > 3kT$ (figure 3.6b). This is the **non-degenerate** (Maxwell–Boltzmann) limit; it will make all the integrals of Section 3.7 elementary.

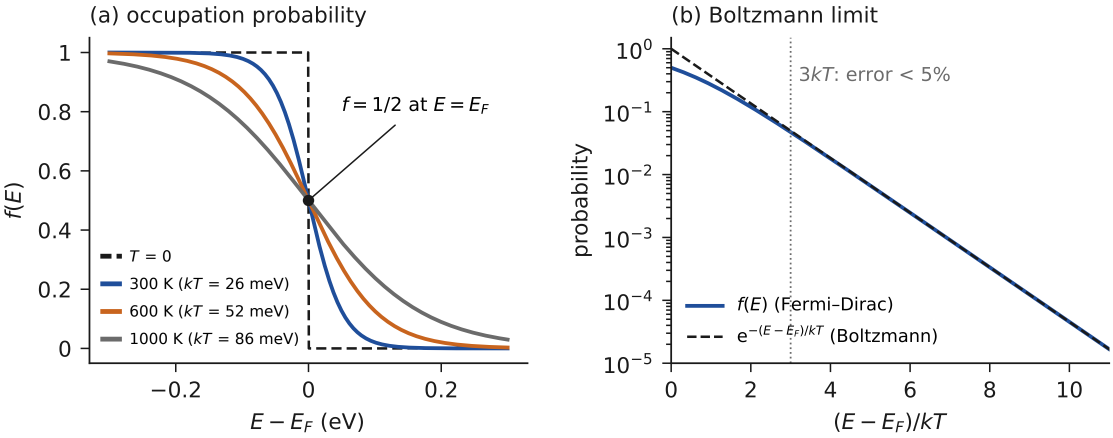


Figure 3.6 – *(a) The Fermi–Dirac function at several temperatures. At $T = 0$ it is a step at E_F. The width of the transition grows with kT. The occupation probability is $1/2$ at $E = E_F$ at every temperature. (b) Above E_F, on a logarithmic scale, f merges with the Boltzmann factor $e^{-(E-E_F)/kT}$ beyond about $3kT$.*

BE AWARE OF…

The Fermi level is a statistical parameter, not a band. It is fixed by the requirement that the total number of electrons is the right one (in a doped sample, through neutrality: Section 3.13). It may lie **in the gap**, where there are no states at all: nobody sits at E_F; it only says how the states on either side are populated. In a semiconductor we shall find E_F inside the gap, and the statistics works there: it is the probability *if there were a state*.

3.6 Effective mass and density of states

THE QUESTION

To count the electrons of a band we need the number of states at each energy. What does $E(k)$ look like near the band edges, and what does it say about how electrons and holes respond to a force?

3.6.1 The effective mass

Near the band edges, the bottom of the conduction band and the top of the valence band, the dispersion relation is again a parabola (an input of solid-state physics, like those of Section 3.3); any smooth curve looks like a parabola near its extremum:

$$E(k) \simeq E_C + \frac{\hbar^2 k^2}{2m_n^*} \qquad E(k) \simeq E_V - \frac{\hbar^2 k^2}{2m_p^*}. \tag{3.10}$$

Comparing with the free electron (3.2), the curvature of the band plays the role of $1/m$. This defines the **effective mass**:

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{\mathrm{d}^2 E}{\mathrm{d}k^2}. \tag{3.11}$$

An electron in the conduction band reacts to an applied force as a free particle of mass m_n^*, whatever the periodic potential does to it. This is how the whole effect of the lattice is hidden in a single number. A strongly curved band gives a light particle, a flat band a heavy one (figure 3.7a). The hole, the absence of an electron near the top of the valence band, has a positive effective mass m_p^*, defined by the curvature of the top of the valence band.

E_C is the bottom, not the place. Equation (3.10) says more than it seems. An electron of the conduction band with wavenumber k has the energy $E_C + \hbar^2 k^2/2m_n^*$: it sits **above** E_C, and the difference $E - E_C$ is its **kinetic energy**, exactly as $p^2/2m$ was for the free electron of Section 3.2. E_C is the energy of an electron of the band at rest, the potential energy of the band. Likewise a hole deeper in the valence band has kinetic energy $E_V - E$, counted downward (figure 3.7b). At thermal equilibrium most electrons stay within a few kT of E_C (Section 3.7). (Outside the model: experiment shows that an electron given more energy, by light for example, relaxes back towards E_C by handing its excess to the lattice; chapter 4 returns to this.)

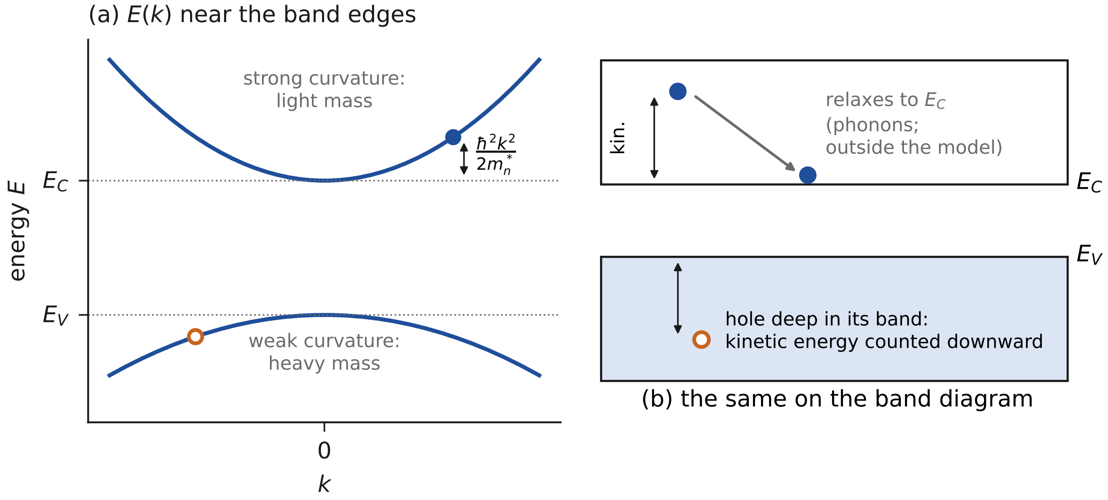

(a) E(k) near the band edges
strong curvature:
light mass
$\frac{\hbar^2 k^2}{2m_n^*}$
energy E
E_C
E_V
weak curvature:
heavy mass
0
k
kin.
relaxes to E_C
(phonons;
outside the model)
E_C
E_V
hole deep in its band:
kinetic energy counted downward
(b) the same on the band diagram

Figure 3.7 – Effective mass and kinetic energy in the bands. (a) Near the band edges $E(k)$ is a parabola whose curvature defines $1/m^*$: an electron with wavenumber k lies $\hbar^2 k^2/2m_n^*$ above E_C; a flatter band means a heavier particle. (b) The same on the band diagram: an electron high in the conduction band carries kinetic energy $E - E_C$; a hole deep in the valence band carries kinetic energy $E_V - E$.

OUTSIDE THE ONE-DIMENSIONAL MODEL: TWO EFFECTIVE MASSES

Our model has one parabola and therefore one mass. Real silicon is three-dimensional: its conduction band has six equivalent minima, and around each one the mass depends on the direction, $m_l = 0.98\,m_0$ along the axis of the minimum and $m_t = 0.19\,m_0$ across it (m_0 is the free-electron mass). The two quantities we need combine these numbers differently, and we take both as **inputs** (Table 3.3):

- **counting states** adds the six minima: the **density-of-states mass** $6^{2/3}(m_l m_t^2)^{1/3} = 1.08\,m_0$, used in $g_c(E)$ and N_C below (for holes, $0.56\,m_0$);
- **responding to a force** averages the directions: the **conductivity mass** $3/(1/m_l + 2/m_t) = 0.26\,m_0$, the m of the Drude mobility of chapter 1 and of the hydrogen-like estimate of chapter 2 (for holes, $0.37\,m_0$).

The two masses are not interchangeable; using one for the other changes estimates by a factor 2 to 4.

Why holes are less mobile: the reason promised in chapter 2. The Drude mobility of chapter 1 is $\mu = q\tau_c/m$, with the conductivity mass. In silicon the top of the valence band is **flatter** than the bottom of the conduction band: by (3.11) holes are heavier, $m_p^* = 0.37\,m_0$ against $m_n^* = 0.26\,m_0$. The same is true in germanium (0.21 against 0.12) and, much more strongly, in GaAs (0.34 against 0.067). A heavier carrier is accelerated less by the same field between two collisions, hence a smaller mobility. The mass ratio, $0.37/0.26 = 1.4$, accounts for part of the measured ratio $\mu_n/\mu_p = 1350/460 = 2.9$; the rest comes from a shorter collision time τ_c for holes, which is a measured input, outside our model. The intuitive picture of chapter 2, a hole moving by successive jumps of bound electrons, is now a statement about the curvature of the valence band.

BE AWARE OF… “HEAVIER HOLES” REFERS TO THE CONDUCTIVITY MASS

In silicon the density-of-states masses go the other way: $1.08\,m_0$ for electrons and $0.56\,m_0$ for holes, because the conduction band has six minima to fill. Mobility uses the conductivity mass, the counting of states the density-of-states mass. Always say which one you use.

3.6.2 The density of states

To add up the electrons we need the **density of states** $g(E)$: the number of states per unit volume and per unit energy. Counting the states in three dimensions follows Section 3.2: each direction is quantised as in (3.3), so each allowed $\vec{k}$ occupies a volume $(2\pi)^3/V$ in k-space. The number of states per unit volume with a wavenumber smaller than k is therefore $2\times\frac{4}{3}\pi k^3/(2\pi)^3 = k^3/3\pi^2$, with the factor 2 for spin. Using $E - E_C = \hbar^2 k^2/2m_n^*$ and differentiating with respect to E,

$$g_c(E) = \frac{1}{2\pi^2}\left(\frac{2m_n^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_C} \qquad (E > E_C), \tag{3.12}$$

$$g_v(E) = \frac{1}{2\pi^2}\left(\frac{2m_p^*}{\hbar^2}\right)^{3/2}\sqrt{E_V - E} \qquad (E < E_V). \tag{3.13}$$

The density of states is zero in the gap and grows as the square root of the distance to the band edge, faster for a heavier mass: the density-of-states mass is the one that enters here.

3.7 From bands to densities: n and p

THE QUESTION

Knowing $f(E)$ and $g(E)$, how do we get the number of electrons in the conduction band and of holes in the valence band?

We add up, over all energies of the band, the number of states per unit volume at each energy times the probability that they are occupied. The electron concentration is the integral over the conduction band of $g_c(E)f(E)$, and the hole concentration is the integral over the valence band of $g_v(E)$ times the probability $1 - f(E)$ that the state is **empty**:

$$n = \int_{E_C}^{\infty} g_c(E)\, f(E)\, \mathrm{d}E \qquad p = \int_{-\infty}^{E_V} g_v(E)\, [1 - f(E)]\, \mathrm{d}E. \tag{3.14}$$

Figure 3.8 shows the three factors: the density of states vanishes in the gap, f falls through E_F, and their product is non-zero only in two thin slices, just above E_C (electrons) and just below E_V (holes).

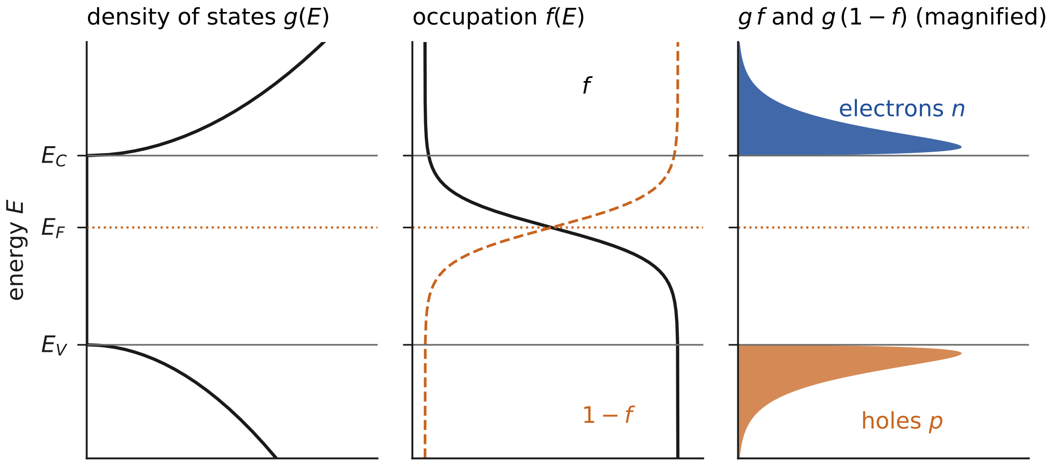


Figure 3.8 – *From density of states and occupation probability to carrier densities. The electrons are the part of $g\,f$ in the conduction band, the holes the part of $g\,(1-f)$ in the valence band. Schematic: kT is drawn much larger than its real value, otherwise the shaded areas would be invisible.*

These integrals can be done exactly (Fermi integrals), but a simplification is almost always valid. If the Fermi level is far from the band edges, $E_C - E_F \gg kT$ and $E_F - E_V \gg kT$, the Boltzmann forms (3.9) apply over the whole band. With $x = (E - E_C)/kT$ the electron integral becomes

$$n = \frac{1}{2\pi^2}\left(\frac{2m_n^* kT}{\hbar^2}\right)^{3/2} \mathrm{e}^{-(E_C - E_F)/kT} \int_0^{\infty} \sqrt{x}\, \mathrm{e}^{-x}\, \mathrm{d}x,$$

and the integral of $\sqrt{x}\,\mathrm{e}^{-x}$ is $\sqrt{\pi}/2$. The hole integral is the mirror image. The result is

$$n = N_C \exp\left[-\frac{E_C - E_F}{kT}\right] \qquad p = N_V \exp\left[-\frac{E_F - E_V}{kT}\right] \tag{3.15}$$

with the **effective densities of states**

$$N_C = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \qquad N_V = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}. \tag{3.16}$$

N_C and N_V depend on the temperature as $T^{3/2}$. For silicon at 300 K, with the density-of-states masses 1.08 and 0.56 m_0, they are 2.8×10^{19} and 1.04×10^{19} cm^{-3} (Table 3.3). The condition of validity is the nondegenerate condition: E_F more than about $3kT$ from both band edges.

Reading the result. The formula $n = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$ says that the conduction band, with all its states spread over an energy range, behaves as if it were **a single level of N_C states placed at E_C**, each occupied with the probability $f(E_C) = \mathrm{e}^{-(E_C-E_F)/kT}$. Likewise the valence band behaves as N_V states at E_V, each empty with probability $\mathrm{e}^{-(E_F-E_V)/kT}$. N_C and N_V are **numbers of states**, the exponentials are **probabilities**. Two consequences:

- the exponentials make the carrier concentrations **extremely sensitive** to the position of E_F in the gap: moving E_F by $kT \ln 10 = 59.6$ meV changes n by a factor 10 and p by a factor 1/10;
- N_C and N_V **carry the material**: through the effective masses they are different for silicon, germanium and gallium arsenide (Table 3.3).

Quantity	Si	Ge	GaAs
Gap E_g at 300 K (eV)	1.12	0.66	1.42
Type of gap (chapter 4)	indirect	indirect	direct
m_n^*/m_0, m_p^*/m_0 (density of states)	1.08, 0.56	0.55, 0.37	0.067, 0.48
m_n^*/m_0, m_p^*/m_0 (conductivity)	0.26, 0.37	0.12, 0.21	0.067, 0.34
N_C (cm^{-3})	2.8×10^{19}	1.04×10^{19}	4.7×10^{17}
N_V (cm^{-3})	1.04×10^{19}	6.0×10^{18}	7.0×10^{18}
n_i (cm^{-3})	10^{10} (reference)	2.4×10^{13}	1.8×10^{6}
μ_n, μ_p (cm^2/(V·s))	1350, 460	3900, 1900	8500, 400

Table 3.3 – Typical values at 300 K (low doping), after Neamen, Table B.4. The n_i of silicon is the reference value of the course (Section 3.8), and so is $\mu_p = 460$ (Neamen gives 480).

EXAMPLE 3.2 — WHERE DOES N_C COME FROM?

With $m_n^* = 1.08\,m_0$, $kT = 1.381 \times 10^{-23} \times 300$ J and $h = 6.626 \times 10^{-34}$ J·s, (3.16) gives $N_C = 2.8 \times 10^{25}$ m^{-3} $= 2.8 \times 10^{19}$ cm^{-3}; with $m_p^* = 0.56\,m_0$, $N_V = 1.05 \times 10^{19}$ cm^{-3}. Compare with the 5×10^{22} atoms per cm^3: the conduction band behaves like a level holding about one state per 2000 atoms, the states within kT of the edge. At $n = 10^{16}$ cm^{-3} only one of these effective states in 2800 is occupied.

3.8 The pure semiconductor: the mass-action law, n_i and E_i

THE QUESTION

What do the expressions of n and p say about the constant n_i^2 of chapter 2? For the pure material, where is E_F, how many carriers are there, and why 10^{10} in silicon?

The product. Multiply the two expressions (3.15). With $E_C - E_V = E_g$, the Fermi level disappears:

$$n\,p = N_C N_V \exp\left(-\frac{E_g}{kT}\right). \tag{3.17}$$

The product np does not depend on E_F. Doping moves E_F (Part C) and changes n and p by orders of magnitude, but not their product: for a given material at a given temperature it is the same in a pure and in a doped sample. Applied to the pure material, where $n = p = n_i$, it gives the value of the constant of chapter 2:

$$n_0\,p_0 = n_i^2 = N_C N_V \exp\left(-\frac{E_g}{kT}\right). \tag{3.18}$$

This is the **mass-action law, obtained a second time, and independently**. The two derivations do not do the same job:

- the **kinetic argument** of chapter 2 (the balance $G_0 = R_0$ with $R = \alpha np$) explains **why** a product law exists (recombination needs both carriers), but rests on the assumptions $G_0 = G_i$ and α unchanged by doping, and cannot give the value;
- the **statistical derivation** needs neither assumption; it needs only thermal equilibrium (one E_F for both bands) and nondegenerate statistics. It gives the **value** of n_i^2 and the **domain** of the law: E_F cancels only while it lies more than about $3kT$ from both band edges. For very heavy doping the law fails.

Both are needed: the first says what equilibrium is (a balance of two processes, which chapter 4 will disturb), the second says how many carriers that balance leaves.

BE AWARE OF...

Equilibrium only. The law $n_0 p_0 = n_i^2$ applies to thermal equilibrium, where a single Fermi level describes both carriers. Under illumination or electrical injection, in general, $np \neq n_i^2$: this is the subject of chapter 4.

The intrinsic level. In the pure material each broken bond gives one electron and one hole: **n = p**. Equating the two expressions (3.15) fixes the Fermi level of the pure material, called the **intrinsic level** E_i. Section 3.10 solves this equation; the result is that E_i lies very close to midgap, within kT in silicon.

The intrinsic concentration. Taking the square root of (3.18) gives

$$n = p = n_i = \sqrt{N_C N_V}\,\exp\left(-\frac{E_g}{2kT}\right). \tag{3.19}$$

Why $E_g/2$? Creating a pair costs E_g, and the product np carries the full factor $e^{-E_g/kT}$. But a pair is **two** particles: in the pure material the product is split into two equal factors, $n = p = \sqrt{np}$, and each one carries half of the exponent. This is the quantitative content that chapter 2 could not supply: the probability $e^{-E/kT}$ of a fluctuation was not the number of pairs; the number of pairs is the square root of a product fixed by the balance.

The numbers. With the values of Table 3.3 and $kT = 25.9\,\text{meV}$, (3.19) gives $n_i = 6.9 \times 10^9\,\text{cm}^{-3}$ for silicon at 300 K, within the range of published values (7×10^9 to 1.5×10^{10}). We keep the reference value $\mathbf{n_i = 10^{10}}\,\text{cm}^{-3}$ in all examples of the course. Two values of $E_C - E_i$ will therefore be met: computed from the formula of Section 3.10, $E_C - E_i = 0.573\,\text{eV}$; computed from the reference value with (3.15), $E_C - E_i = kT\ln(N_C/n_i) = 0.563\,\text{eV}$. The measured order of magnitude quoted in chapter 2 is now explained: 10^{10} pairs per cm^3 among 10^{23} bonds per cm^3, one bond in 10^{13}.

Densities referred to E_i. Writing (3.15) for $E_F = E_i$ gives $n_i = N_C\,e^{-(E_C-E_i)/kT} = N_V\,e^{-(E_i-E_V)/kT}$; dividing (3.15) by these makes N_C and N_V disappear:

$$n = n_i \exp\left[\frac{E_F - E_i}{kT}\right] \qquad p = n_i \exp\left[-\frac{E_F - E_i}{kT}\right]. \tag{3.20}$$

These two expressions hold for any position of E_F in the nondegenerate range, and their product is n_i^2 by construction. They will make Part C transparent: E_F above E_i means more electrons than holes, below means the reverse.

3.9 Intrinsic density and temperature

THE QUESTION

How does n_i vary with temperature, and what does this give for the resistivity of a pure sample, the observation of chapter 2?

In (3.19), N_C and N_V vary as $T^{3/2}$, so $\sqrt{N_C N_V}$ also varies as $T^{3/2}$, and the exponential dominates:

$$n_i(T) \propto T^{3/2} \exp\left(-\frac{E_g}{2kT}\right). \tag{3.21}$$

On a plot of $\ln n_i$ against $1/T$ the exponential gives a nearly straight line of slope $-E_g/2k$ (figure 3.9): the gap is measured by the slope. In what follows E_g is held at its 300 K value; its slow decrease with T is outside our model.

The conductivity of the pure material follows from the Drude expression of chapter 1 applied to both carriers:

$$\sigma_i = q\,n_i\,(\mu_n + \mu_p). \tag{3.22}$$

EXAMPLE 3.3 — HOW MANY PAIRS, AND WHAT RESISTIVITY? (300 K)

Silicon: $n_i = 10^{10}\,\text{cm}^{-3}$, i.e. one pair for 5×10^{12} atoms, or one broken bond in 10^{13}. The intrinsic conductivity (3.22) is $\sigma_i = 1.6{\times}10^{-19}{\times}10^{10}{\times}1810 = 2.9{\times}10^{-6}$ S/cm, so $\rho \approx 3.4 \times 10^5\ \Omega\cdot\text{cm}$. Germanium: $n_i = 2.4 \times 10^{13}$, $\mu_n + \mu_p = 5800$, $\rho \approx 45\ \Omega\cdot\text{cm}$. The gap of germanium is smaller by 0.46 eV: its n_i is 2400 times larger and its resistivity 7700 times smaller.

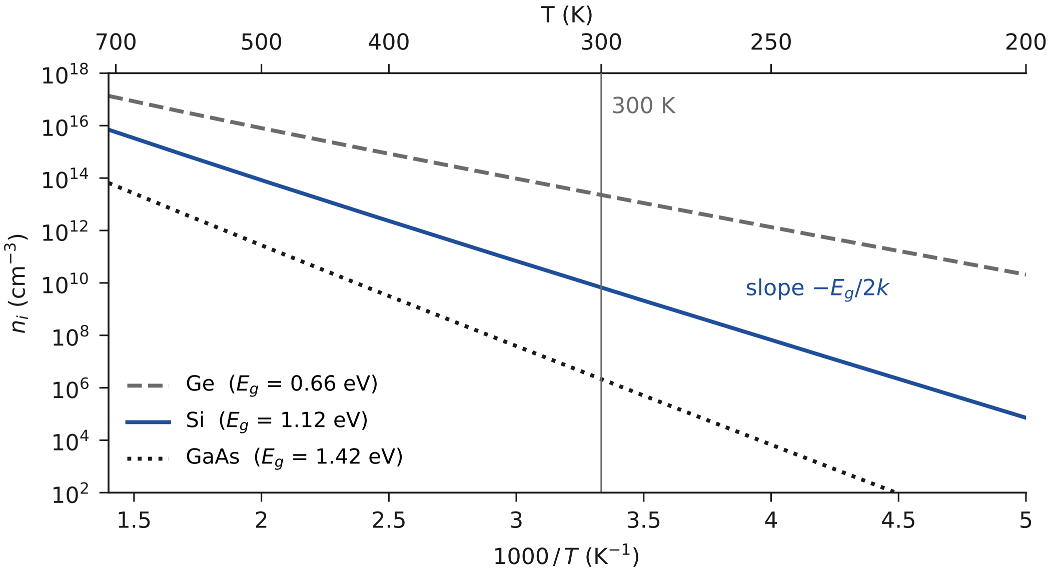


Figure 3.9 *– The intrinsic concentration against* $1000/T$ *for Ge, Si and GaAs, computed from (3.19) with the values of Table 3.3 and* $N_C, N_V \propto T^{3/2}$*. Nearly straight lines of slope* $-E_g/2k$*: the smaller the gap, the more carriers; the larger the gap, the steeper the line. For silicon at 300 K the formula gives about* $7 \times 10^9\,\mathrm{cm}^{-3}$ *(*6.9×10^9 *with* $kT = 25.9\,\mathrm{meV}$*).*

EXAMPLE 3.4 — THE FALL OF THE RESISTIVITY WITH TEMPERATURE

Two inputs: E_g held at 1.12 eV, and the measured trend that, in pure silicon near room temperature, the mobility varies roughly as $T^{-3/2}$ (lattice scattering; a regime-dependent approximation). Between 300 and 350 K, (3.21) multiplies n_i by $(350/300)^{3/2}\exp[(E_g/2k)(1/300 - 1/350)] = 1.26 \times 22.1 = 28$, while the mobility falls by only 20 %. The resistivity of the pure material falls by a factor of about $28/1.26 = 22$.

The resistivity of a pure semiconductor is dominated by $\exp(E_g/2kT)$: it falls with T because the number of carriers rises. This was observation (a) of chapter 2, explained there by the creation of pairs; it now comes **with its law**, and with the reason why the effect is so much stronger in silicon than in germanium.

3.10 Where is the Fermi level of a pure semiconductor, and how does it move with temperature?

THE QUESTION

In the pure material $n = p$*. Where does this put the Fermi level, exactly, and does it move when the temperature changes?*

Derivation. Write $n = p$ with the expressions (3.15) and $E_F = E_i$:

$$N_C\,\mathrm{e}^{-(E_C-E_i)/kT} = N_V\,\mathrm{e}^{-(E_i-E_V)/kT} \implies \mathrm{e}^{(2E_i-E_C-E_V)/kT} = \frac{N_V}{N_C}.$$

Taking the logarithm, and using (3.16), in which everything cancels in the ratio except the masses, $N_V/N_C = (m_p^*/m_n^*)^{3/2}$:

$$E_i = \frac{E_C + E_V}{2} + \frac{kT}{2}\ln\frac{N_V}{N_C} = \frac{E_C + E_V}{2} + \frac{3}{4}kT\ln\frac{m_p^*}{m_n^*}. \tag{3.23}$$

Reading. The intrinsic level is at the **middle of the gap**, shifted by a term that vanishes only if $N_C = N_V$, i.e. if the two density-of-states masses are equal. It moves **towards the band with the smaller density of states**: fewer states must be occupied with a higher probability to give the same number of carriers.

- **At 0 K** the shift vanishes: E_i is **exactly at midgap**. This is the Fermi level at 0 K that Section 3.4 placed "in the gap": halfway between the last full state and the first empty one.
- **At** $T > 0$ the ratio m_p^*/m_n^* does not depend on T (as long as the masses are constant), so the shift is **proportional to** T: E_i moves linearly with temperature away from midgap (figure 3.10).

Numbers at 300 K ($kT/2 = 12.95\,\text{meV}$):

- **Si**: $N_V/N_C = 1.04/2.8$, shift $= 12.95\ln(0.371) = -12.8\,\text{meV}$ (the mass form, $\frac{3}{4} \times 25.9\ln(0.56/1.08)$, gives the same). E_i lies 12.8 meV **below** midgap, less than kT;
- **Ge**: $12.95\ln(6.0/10.4) = -7.1\,\text{meV}$, below midgap;
- **GaAs**: $12.95\ln(7.0 \times 10^{18}/4.7 \times 10^{17}) = +35\,\text{meV}$, **above** midgap, because $m_n^* = 0.067\,m_0$ is so small that $N_C \ll N_V$ (the mass form gives 38 meV; the difference only reflects the rounding of the tabulated values).

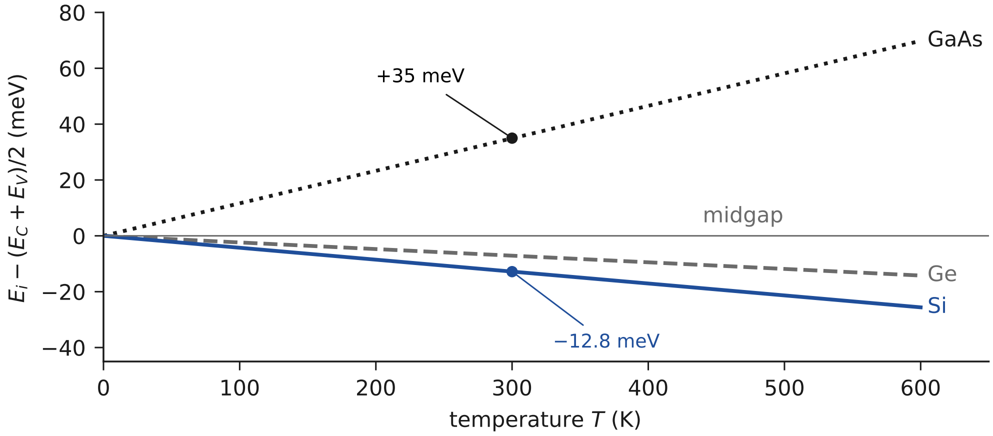


Figure 3.10 – *Position of the intrinsic level with respect to midgap against temperature, from (3.23) with the ratios N_V/N_C of Table 3.3. The shift is zero at 0 K and grows linearly with T. Approximation: the gaps and the effective masses are held at their 300 K values; in reality E_g decreases slowly with T, which moves the band edges but changes the shift with respect to midgap very little.*

"The Fermi level of a pure semiconductor is at midgap" is therefore an excellent approximation, not an identity: exact at 0 K, off by less than kT at 300 K in Si and Ge, about $1.4\,kT$ in GaAs. Even at 600 K the shift in silicon is only $-25.6\,\text{meV}$, against a half-gap of 560 meV.

WHAT PART B HAS ESTABLISHED

- Statistics: $f(E) = 1/(1 + \mathrm{e}^{(E-E_F)/kT})$; E_F is the energy where $f = 1/2$, a statistical parameter, not a band; holes occupy with probability $1 - f$; Boltzmann limit beyond $3kT$.
- Effective mass $1/m^* = (1/\hbar^2)\,\mathrm{d}^2E/\mathrm{d}k^2$; holes are heavier (conductivity mass) and less mobile. Above E_C the energy is kinetic. $g(E) \propto \sqrt{|E - E_{C,V}|}$.
- Concentrations: $n = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$, $p = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$, $N_{C,V} = 2(2\pi m^* kT/h^2)^{3/2}$, valid if E_F is more than $3kT$ from both edges.
- E_F cancels in the product: $n_0 p_0 = n_i^2 = N_C N_V\,\mathrm{e}^{-E_g/kT}$, the value and the domain of mass action.
- Pure material: $n_i = \sqrt{N_C N_V}\,\mathrm{e}^{-E_g/2kT}$ (factor $\frac{1}{2}$: a pair is two particles), $n_i \propto T^{3/2}\mathrm{e}^{-E_g/2kT}$ explains the fall of ρ; $E_i = (E_C + E_V)/2 + \frac{3}{4}kT\ln(m_p^*/m_n^*)$: midgap at 0 K, shifted linearly with T.
- Doped material: these expressions give n and p in terms of E_F; they do not say where E_F is. That requires counting the electrons the dopants deliver, and neutrality (Part C).

PART C

The doped semiconductor

How many electrons a donor really gives, what doping changes in the band picture, and where the Fermi level goes, at any temperature.

3.11 Donor and acceptor levels: the ionised donors N_D^+

THE QUESTION

Chapter 2 said that the fifth electron of phosphorus is held by only 45 meV. How does this appear on the energy diagram, and what fraction of the donors gives up its electron?

A donor atom adds, for its fifth electron, an **allowed energy inside the gap**, because this electron is bound, but only weakly. The level is called $\mathbf{E_D}$ and lies just below the conduction band. An electron on this level is attached to its donor; freeing it means lifting it to the conduction band, which costs $E_C - E_D$ (figure 3.11a). After this, the donor has lost its electron: it is **ionised** and carries a fixed charge $+q$. The donors are few and far apart (Section 3.12), so their level is a set of isolated, localised states, not a band.

The value of $E_C - E_D$, recalled. Chapter 2 modelled the bound electron as a hydrogen atom in silicon: the dielectric constant $\varepsilon_r = 11.7$ divides the binding energy by $\varepsilon_r^2 = 137$, and the electron responds to the force of the ion with the conductivity mass $0.26\,m_0$ of Section 3.6.1. The estimate $13.6\,\mathrm{eV} \times (m^*/m_0)/\varepsilon_r^2 \approx 26\,\mathrm{meV}$ gives the order of magnitude of the measured 45 meV (phosphorus), 54 meV (arsenic) and 39 meV (antimony); the differences come from the core of each impurity atom, which the model ignores. Here $E_C - E_D$ is a measured **input**: 45 meV for phosphorus.

What fraction of the donors is ionised? The donor level obeys the same statistics as the bands, with one modification that we take as an **input** from statistical physics: the level can hold one electron of either spin, but not two (the second would cost a large repulsion). This puts a **degeneracy factor 2** in its occupation: the probability that a donor keeps its electron is

$$f_D(E_D) = \frac{1}{1 + \frac{1}{2}\,\mathrm{e}^{(E_D - E_F)/kT}}.$$

The concentration of ionised donors is the concentration of donors that did **not** keep their electron, $N_D[1 - f_D(E_D)]$:

$$N_D^+ = N_D\left[1 - \frac{1}{1 + \frac{1}{2}\,\mathrm{e}^{(E_D - E_F)/kT}}\right] = \frac{N_D}{1 + 2\exp[(E_F - E_D)/kT]}. \tag{3.24}$$

Without the factor 2 this would be $N_D[1 - f(E_D)]$ with the ordinary Fermi–Dirac function. Its reading is on figure 3.11b:

- E_F well **below** E_D (a few kT): the exponential is small, $N_D^+ \simeq N_D$, the donors are **all ionised**;
- $E_F = E_D - kT\ln 2$: exactly half of the donors are ionised;
- E_F **above** E_D: the electrons stay on their donors, $N_D^+ \ll N_D$.

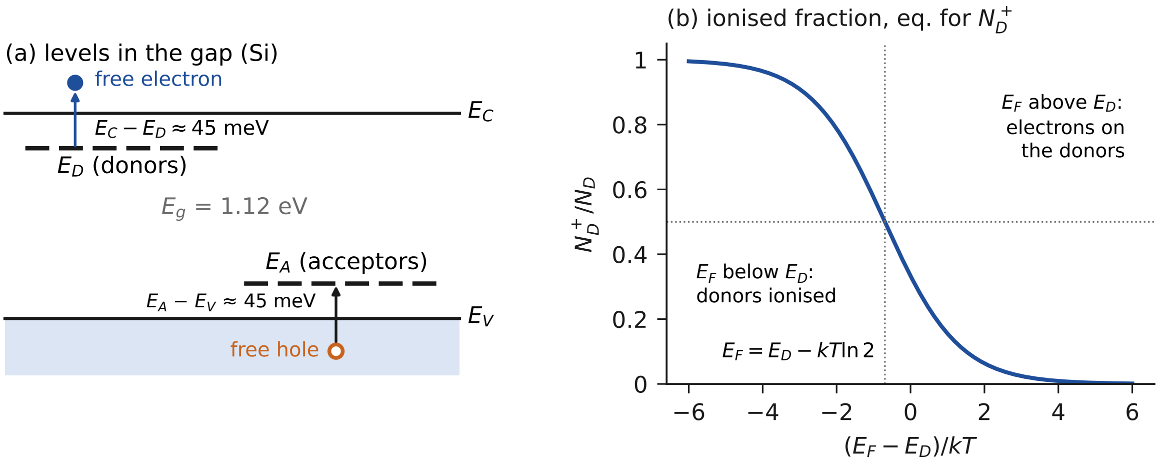


Figure 3.11 – *(a) Donor and acceptor levels in the gap. Ionising a donor lifts its electron by $E_C - E_D \approx 45$ meV into the conduction band; ionising an acceptor lifts an electron of the valence band by $E_A - E_V \approx 45$ meV onto the acceptor, which leaves a free hole. (b) The ionised fraction (3.24) against the position of the Fermi level with respect to the donor level.*

Acceptors are the mirror image. An acceptor adds an empty level E_A just above the valence band ($E_A - E_V \approx 45\,\text{meV}$ for boron in silicon). An electron of the valence band can jump onto it, which leaves a free hole in the valence band and makes the acceptor a fixed negative ion. The statistics is the same with the roles of electrons and holes exchanged, the acceptors being ionised when E_F lies a few kT **above** E_A; we do not need its formula, because the idea is entirely contained in the donor case.

What (3.24) does not yet give. The ionised fraction depends on E_F, which we do not know. Chapter 2 showed the curve of N_D^+/N_D against temperature; to compute it we need E_F at each temperature, and this requires one more relation, charge neutrality. We therefore draw that curve in Section 3.13, where it becomes a derived result, after the next section has checked that the band expressions still hold in a doped sample.

3.12 Doping leaves the bands unchanged: only E_F moves

> **THE QUESTION**
>
> *Chapter 2 described doping with bonds. In the band picture, what does a donor change, and what does it leave intact?*

The answer starts from an order of magnitude. Even a strongly doped sample with 10^{17} donors per cm^3 contains one phosphorus atom for 5×10^5 silicon atoms. The crystal structure does not change: same lattice, same period, same periodic potential, seen by the electrons almost everywhere. Therefore **the band structure is the same**: same E_C and E_V, same gap, same effective masses, same N_C and N_V.

What doping changes is the **number of electrons** that must be accommodated in these unchanged bands. A donor brings one extra electron; an acceptor takes one. The statistics of Section 3.5 then has to place more (or fewer) electrons in the same states, and the only parameter it can adjust is the Fermi level: **doping moves** E_F, up towards E_C for donors, down towards E_V for acceptors.

This has an important consequence. The derivation of Section 3.7 never used the fact that the crystal was pure: it used only the bands, the Fermi–Dirac function and the nondegenerate approximation. So, within the same constraints, **the expressions (3.15) and (3.20) remain valid in a doped semiconductor**, with E_F wherever the doping puts it, and so does their product, $n_0 p_0 = n_i^2$ (3.18). The constraints are:

- **thermal equilibrium**: one E_F for the whole sample and for both bands;
- **nondegenerate statistics**: E_F at least $3kT$ from the band edges, which in silicon means a doping below about $10^{18}\,\text{cm}^{-3}$ (Section 3.14);
- **dilute dopants**: few enough not to modify the bands.

What these expressions cannot tell us is **where** E_F **is**. To find it we must count the electrons that the dopants actually deliver to the bands: not one per donor, but N_D^+, given by (3.24) in terms of E_F itself.

3.13 Neutrality, N_D^+, n and p: the doped sample at any temperature

> **THE QUESTION**
>
> *We now have n, p and N_D^+, all three as functions of the unknown E_F. What relation fixes E_F, and what does it give at 300 K and at any other temperature?*

3.13.1 One equation for one unknown

The relation is **charge neutrality**, the third law of chapter 2. A homogeneous sample at equilibrium carries no net charge, so the negative charges balance the positive ones. Take an n-type sample doped with donors only. The negative charges are the electrons; the positive ones are the holes and the ionised donors N_D^+:

$$n_0 = p_0 + N_D^+. \tag{3.25}$$

With the band expressions (3.15) for n_0 and p_0 and the ionisation (3.24) for N_D^+, this is

$$N_C\,\mathrm{e}^{-(E_C-E_F)/kT} = N_V\,\mathrm{e}^{-(E_F-E_V)/kT} + \frac{N_D}{1+2\,\mathrm{e}^{(E_F-E_D)/kT}}, \tag{3.26}$$

one equation for one unknown, E_F. Its left side increases with E_F, its right side decreases: there is exactly one solution, which can always be found numerically (by bisection, a few lines of code). Figure 3.12 shows the solution graphically at 300 K: E_F is where the curve of n_0 crosses the curve of $p_0 + N_D^+$. Once E_F is known, (3.15) gives n_0 and p_0, and (3.24) gives the ionised fraction.

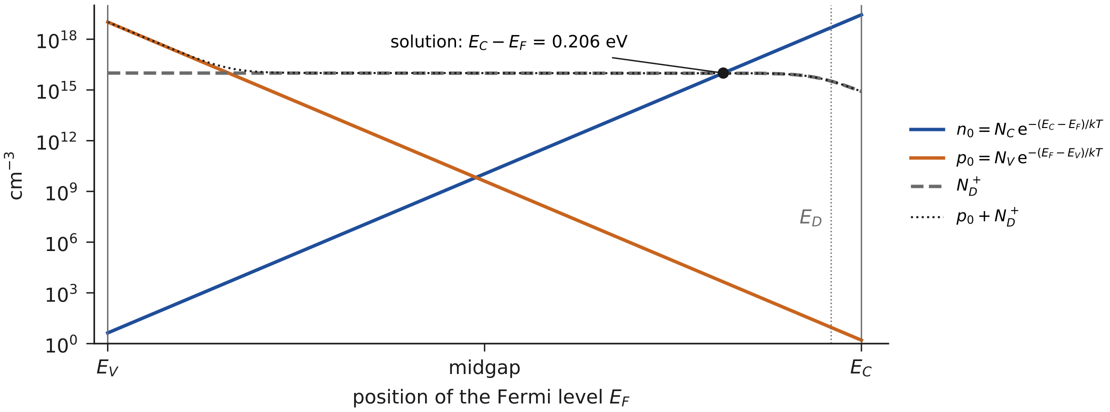


Figure 3.12 *– Graphical solution of the neutrality equation (3.26) for silicon with $N_D = 10^{16}\,\mathrm{cm}^{-3}$ at 300 K. Each term is drawn against the position of E_F in the gap (logarithmic scale). The electrons rise with E_F, the holes and the ionised donors fall; the crossing fixes E_F, 0.206 eV below E_C. At the crossing N_D^+ is still on its plateau N_D (complete ionisation) and p_0 is negligible.*

3.13.2 At 300 K: chapter 2's results recovered

At room temperature figure 3.12 shows that the crossing lies where $N_D^+ \simeq N_D$: **complete ionisation**. With $N_D^+ = N_D$, and with $p_0 = n_i^2/n_0$, which follows from the product of the band expressions (3.18), neutrality gives the quadratic $n_0^2 - N_D n_0 - n_i^2 = 0$ whose positive root is

$$n_0 = \frac{N_D}{2} + \sqrt{\frac{N_D^2}{4} + n_i^2}, \qquad p_0 = \frac{n_i^2}{n_0}, \tag{3.27}$$

and for $N_D \gg n_i$, $n_0 \simeq N_D$, $p_0 \simeq n_i^2/N_D$. These are exactly the results of chapter 2, obtained there from the three laws mass action + total ionisation + neutrality. The band theory adds two things: the **value** of n_i, and a way to **check** the assumption of total ionisation. For $N_D = 10^{16}\,\mathrm{cm}^{-3}$, $E_C - E_F = 0.206\,\mathrm{eV}$ (Section 3.14), so $E_F - E_D = -(0.206 - 0.045) = -0.161\,\mathrm{eV}$ and (3.24) gives $2\,\mathrm{e}^{-0.161/0.0259} = 4\times10^{-3}$: **99.6 % of the donors are ionised**. The assumption of chapter 2 is checked.

3.13.3 Three regimes of temperature

Everything so far was done at 300 K. Two assumptions were made on the way: complete ionisation ($N_D^+ = N_D$) and $N_D \gg n_i$. Temperature breaks the first one when it is too low and the second one when it is too high. In what follows E_g is held at 1.12 eV and $n_i(T)$ follows (3.21) normalised to $10^{10}\,\text{cm}^{-3}$ at 300 K (an approximation, as in Section 3.9). The numerical solution of (3.26) at every temperature gives figures 3.13 and 3.14; each regime can also be understood by hand.

The ionised fraction (chapter 2's curve, now derived). Figure 3.13 shows N_D^+/N_D against T for three dopings. Three observations follow from it.

- **At low temperature the donors are not ionised (freeze-out):** almost all the electrons stay on their donors and the sample conducts poorly. At 77 K (liquid nitrogen) only about a third (36 %) of $10^{16}\,\text{cm}^{-3}$ donors are ionised: $kT = 6.6$ meV is seven times smaller than 45 meV, and the solution of neutrality puts E_F close to E_D, where the factor $2\,e^{(E_F - E_D)/kT}$ of (3.24) is not small.
- **Above about 200 K, for $N_D \lesssim 10^{16}\,\text{cm}^{-3}$, ionisation is practically complete:** for $10^{16}\,\text{cm}^{-3}$ the fraction is 94 % at 150 K, 98 % at 200 K and 99.6 % at 300 K.
- **Heavier doping ionises later.** For $N_D = 10^{18}\,\text{cm}^{-3}$ the Fermi level approaches the donor level and the ionised fraction at 300 K is only about 76 %; in that range the nondegenerate formulas themselves start to fail, and the curve should be read as a trend.

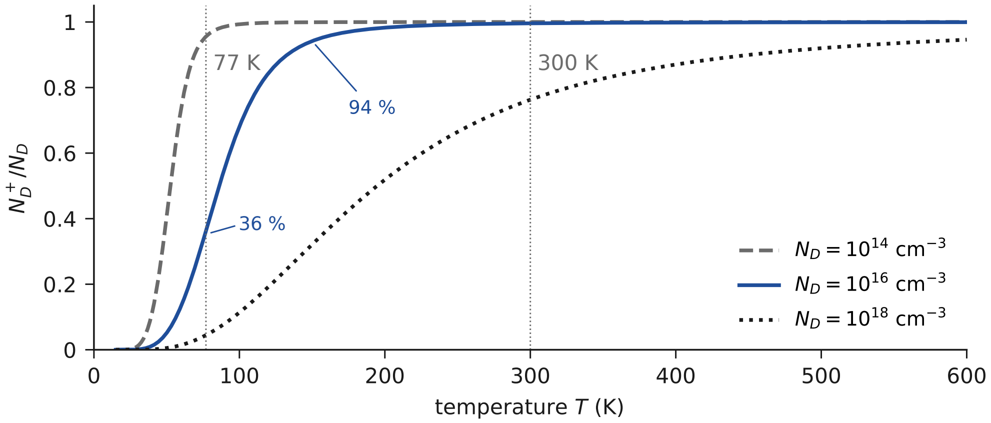


Figure 3.13 *– Ionised fraction of donors (45 meV) in silicon against temperature, for three doping levels, from the numerical solution of the neutrality equation (3.26) with (3.24). This is the curve chapter 2 showed without proof.*

Intermediate temperatures: the extrinsic plateau. Both assumptions hold, so $n_0 \simeq N_D$ and does not depend on temperature. For $N_D = 10^{16}\,\text{cm}^{-3}$ this plateau extends from about 200 K to about 580 K, a window over which n_i itself changes by about ten orders of magnitude (from 10^5 to $10^{15}\,\text{cm}^{-3}$). This is why the conductivity of a doped semiconductor is so stable around room temperature.

Low temperature: freeze-out. When kT becomes small compared with $E_C - E_D = 45$ meV the donors no longer all give up their electron: $N_D^+ < N_D$. Holes are negligible (n_i is astronomically small), so neutrality reads $n_0 = N_D^+$. With (3.15) the exponential of

(3.24) can be written

$$\exp\left[\frac{E_F - E_D}{kT}\right] = \frac{n_0}{N_C}\exp\left[\frac{E_C - E_D}{kT}\right],$$

and in the strongly frozen limit, where the denominator of (3.24) is large, $n_0 = N_D^+$ becomes $n_0^2 \simeq (N_C N_D/2)\,\mathrm{e}^{-(E_C-E_D)/kT}$:

$$n_0 \simeq \sqrt{\frac{N_C N_D}{2}}\exp\left[-\frac{E_C - E_D}{2kT}\right] \qquad (n_0 \ll N_D). \tag{3.28}$$

The electrons "freeze" back onto their donors. At 50 K this gives $5.3 \times 10^{14}\,\mathrm{cm}^{-3}$ (the numerical solution gives 5.1×10^{14}): only 5 % of the donors are ionised. The factor $\frac{1}{2}$ in the exponent has the same origin as in $n_i \propto \mathrm{e}^{-E_g/2kT}$: each free electron leaves behind one ionised donor, so the balance involves the product $n_0 N_D^+ = n_0^2$.

High temperature: the intrinsic regime returns. Complete ionisation now holds, but $n_i(T)$ keeps growing and eventually catches up with N_D: the condition $N_D \gg n_i$ fails. Nothing new is needed; the exact root (3.27), with $n_i(T)$ in it, does the job. It gives $n_0 \simeq N_D$ while $n_i \ll N_D$, and $n_0 \simeq p_0 \simeq n_i$ once $n_i \gg N_D$: thermally generated pairs swamp the dopants, and the sample behaves as pure silicon again. For $N_D = 10^{16}\,\mathrm{cm}^{-3}$, n_i reaches $N_D/10$ at about 580 K (≈ 310 °C); for $N_D = 10^{14}\,\mathrm{cm}^{-3}$ already at about 425 K. The wider the gap, the later this happens. (Outside the model: this is one reason why wide-gap materials such as silicon carbide and gallium nitride are used for high-temperature electronics.)

And the Fermi level? It is the unknown of (3.26), so the same solution gives it at every temperature (figure 3.14b); equivalently, $E_C - E_F = kT\ln[N_C(T)/n_0(T)]$. At very low temperature it sits between E_D and E_C, where the electrons are being exchanged between donors and band; on the plateau it descends towards midgap as N_C grows with T while n_0 stays at N_D; in the intrinsic regime it merges with E_i.

BE AWARE OF…

Four conditions stand behind the 300 K results of chapter 2: **thermal equilibrium** (one single E_F for the whole sample), **complete ionisation** (not too cold), $\mathbf{N_D \gg n_i}$ (not too hot), and **nondegenerate statistics** (not too heavily doped). Every formula of the form $n_0 \simeq N_D$, $p_0 = n_i^2/N_D$ must be read with this window in mind; outside it, go back to (3.26).

3.14 The position of the Fermi level in doped material

THE QUESTION

The sample is in equilibrium and we know n_0 and p_0. At which energy must the Fermi level lie to produce them, and how does it move with the doping?

The expressions (3.20) hold for any position of E_F, pure or doped (Section 3.12). Each one gives a one-to-one relation between a concentration and E_F, so knowing n_0 is the same as knowing E_F. The order of the argument matters: in an isolated sample the

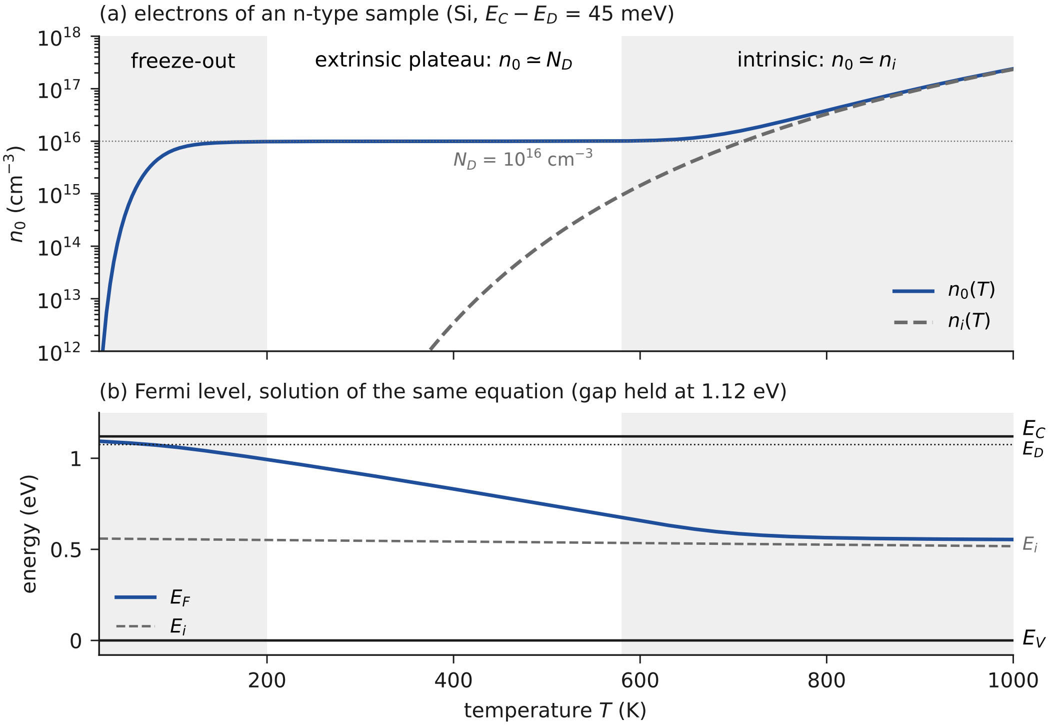


Figure 3.14 *– (a) Electron concentration of n-type silicon ($N_D = 10^{16}\,\mathrm{cm}^{-3}$) against temperature, from neutrality (3.26) with the ionised fraction (3.24) and $n_i(T)$; the freeze-out and intrinsic regimes are shaded. (b) The Fermi level, solution of the same equation, with $E_i(T)$ from (3.23). The band gap is held at 1.12 eV for simplicity.*

Fermi level is not a knob we turn, it is **imposed** by neutrality. We therefore obtained n_0 and p_0 first (Section 3.13) and now **invert** the relations:

$$E_F = E_i + kT \ln \frac{n_0}{n_i} = E_i - kT \ln \frac{p_0}{n_i}. \tag{3.29}$$

The two forms must agree, because $n_0 p_0 = n_i^2$ makes the two logarithms equal and opposite: this is a built-in cross-check. With complete ionisation and $N_D \gg n_i$ ($N_A \gg n_i$) the result is explicit, and the band edges give an equivalent form through (3.15):

$$\begin{aligned} &\text{n-type:} \quad E_F - E_i \simeq kT \ln \frac{N_D}{n_i}, \qquad E_C - E_F = kT \ln \frac{N_C}{N_D}; \\ &\text{p-type:} \quad E_i - E_F \simeq kT \ln \frac{N_A}{n_i}, \qquad E_F - E_V = kT \ln \frac{N_V}{N_A}. \end{aligned} \tag{3.30}$$

Each tenfold increase of the doping moves E_F by $kT \ln 10 = 59.6\,\mathrm{meV}$, towards the band of the majority carrier.

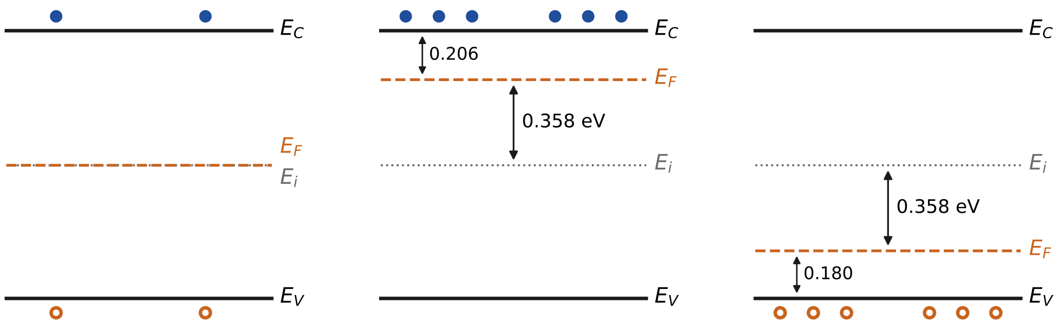
One formula for all dopings. The exact expressions (3.27) can also be inverted once and for all: subtracting the two relations (3.20) gives $n_0 - p_0 = 2n_i \sinh[(E_F - E_i)/kT]$, and neutrality with complete ionisation says $n_0 - p_0 = N_D$, hence

$$E_F - E_i = kT \operatorname{arcsinh} \frac{N_D}{2n_i}, \tag{3.31}$$

and the same with $-N_A$ for a p-type sample. This single formula contains both limits: for $N_D \gg n_i$ it reduces to the logarithm, for $N_D \ll n_i$ it gives $E_F \to E_i$ (figure 3.16). There is no switch between “pure” and “doped”, only a comparison between the doping and n_i.

EXAMPLE 3.5 — FERMI LEVEL OF THE $10^{16}\,\mathrm{cm}^{-3}$ N-TYPE SAMPLE

From n_0: $E_F - E_i = kT\ln(n_0/n_i) = 0.0259\ln(10^6) = 0.358\,\mathrm{eV}$.
Cross-check from p_0: $E_F - E_i = -kT\ln(p_0/n_i) = -0.0259\ln(10^4/10^{10}) = +0.358\,\mathrm{eV}$.
Cross-check from the band edge: $E_C - E_F = kT\ln(N_C/N_D) = 0.0259\ln 2800 = 0.206\,\mathrm{eV}$ and $E_C - E_i = kT\ln(N_C/n_i) = 0.563\,\mathrm{eV}$, so $E_F - E_i = 0.563 - 0.206 = 0.358\,\mathrm{eV}$. Three routes, one answer.
Interpretation: E_F lies above E_i (n-type) and $0.206/0.0259 = 7.9\,kT$ below E_C, which confirms the nondegenerate assumption. The p-type sample with $N_A = 10^{16}\,\mathrm{cm}^{-3}$ is the mirror image: $E_i - E_F = 0.358\,\mathrm{eV}$ and $E_F - E_V = kT\ln(N_V/N_A) = 0.180\,\mathrm{eV}$ (figure 3.15).



pure **n-type, $N_D = 10^{16}$ cm^{-3}** **p-type, $N_A = 10^{16}$ cm^{-3}**

Figure 3.15 *– Band diagrams to scale for silicon at 300 K: pure, n-type and p-type with $10^{16}\,\mathrm{cm}^{-3}$ of dopants. E_F moves from E_i toward the band of the majority carrier by $kT\ln(N/n_i) = 0.358$ eV; the filled and open circles indicate the majority carriers. Energies in eV.*

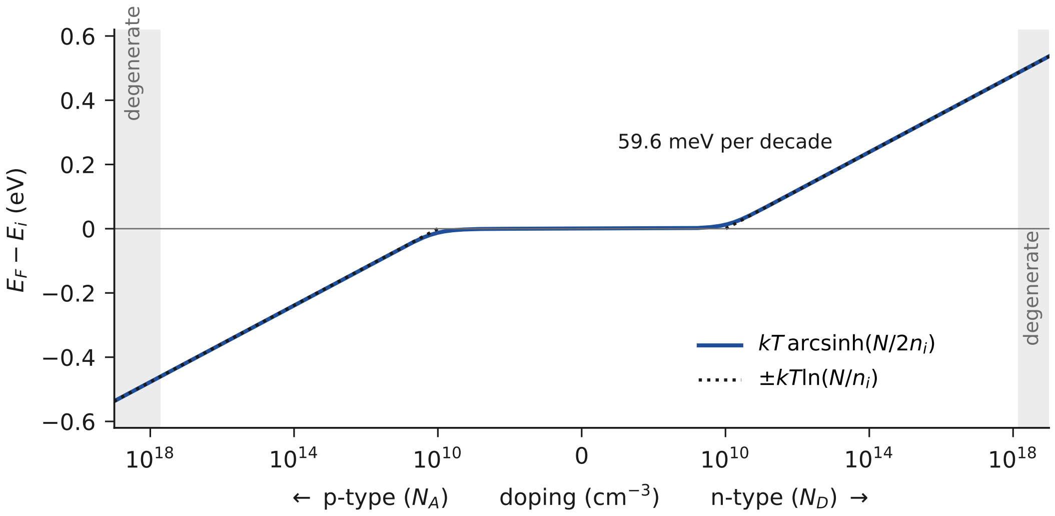


Figure 3.16 *– $E_F - E_i$ against doping for silicon at 300 K: exact expression (3.31) and logarithmic approximation. They merge once the doping exceeds about $3n_i$. In the shaded zones, beyond about $1.4\times10^{18}\,\mathrm{cm}^{-3}$ (n-type) and $5\times10^{17}\,\mathrm{cm}^{-3}$ (p-type), E_F is within $3kT$ of a band edge and the nondegenerate formulas no longer hold.*

The degenerate limit. The formulas assume E_F at least $3kT$ below E_C. With $E_C -$

$E_F = kT\ln(N_C/N_D)$ this requires $N_D \le N_C\,\mathrm{e}^{-3} \approx 1.4\times10^{18}\,\mathrm{cm}^{-3}$ in silicon (and $N_A \le N_V\,\mathrm{e}^{-3} \approx 5\times10^{17}\,\mathrm{cm}^{-3}$), shaded in figure 3.16. Above, the sample is **degenerate**: the full Fermi integrals are needed, mass action in the form $n_0p_0 = n_i^2$ no longer holds, and the dopants start to modify the bands. This is outside the model of this chapter.

3.15 Clarifications

THE QUESTION

Several words of this chapter look alike and are easily confused: equilibrium and steady state, the various energy levels, pure and doped, the temperature regimes. What exactly distinguishes them?

The four tables below gather distinctions that the chapter has made one at a time. They contain no new result.

(i) Three situations. This chapter treats only the first column. Chapter 4 will disturb the sample; the vocabulary must be fixed now, because a constant concentration does not prove equilibrium.

	Thermal equilibrium	Steady state	Transient
External excitation	none (no light, no bias, no net current)	yes, constant (light, applied voltage)	switched on or off
What is constant	everything; $G_0 = R_0$	concentrations, but $G \neq R_0$ balanced by extra recombination or transport	nothing: $n(t)$, $p(t)$ change
A single E_F?	yes, one for both bands and the whole sample	no	no
$n_0p_0 = n_i^2$?	yes (this chapter)	no, $np \neq n_i^2$	no
Where treated	chapters 2 and 3	chapter 4	chapter 4

Table 3.4 *– Equilibrium, steady state and transient. Thermal equilibrium is a particular steady state; a constant concentration alone does not prove equilibrium.*

(ii) The energy levels. Two of these levels belong to the crystal, two to the impurities, one is a reference, and one is not a level of states at all.

(iii) Pure, n-type and p-type at 300 K (silicon, complete ionisation, $N \gg n_i$, nondegenerate).

(iv) The three temperature regimes of an n-type sample ($N_D = 10^{16}\,\mathrm{cm}^{-3}$, silicon).

WHAT PART C HAS ESTABLISHED

- Donors add localised levels E_D just below E_C; with the degeneracy factor 2 (input), $N_D^+ = N_D/[1 + 2\,\mathrm{e}^{(E_F - E_D)/kT}]$. Acceptors are the mirror image.
- Doping changes neither the bands nor N_C, N_V, n_i: it moves E_F. The band expressions of n, p and $n_0p_0 = n_i^2$ remain valid (equilibrium, nondegenerate, dilute).

Level	What it is	Set by	Moves with
E_C	bottom of the conduction band (electron at rest)	the crystal (bands)	not with doping; slowly with T (outside the model)
E_V	top of the valence band (hole at rest)	the crystal	same as E_C
E_i	Fermi level of the pure material, where $n = p$	E_C, E_V and N_V/N_C, eq. (3.23)	linearly with T from midgap (figure 3.10)
E_F	statistical parameter: $f = 1/2$; not a state	neutrality, eq. (3.26)	with doping (59.6 meV per decade) and with T (figure 3.14)
E_D	localised level of the donor electron, $\approx$ 45 meV below E_C	the donor atom	with E_C only
E_A	localised empty level of the acceptor, $\approx$ 45 meV above E_V	the acceptor atom	with E_V only

Table 3.5 *– The energy levels of the band diagram.*

	Pure	n-type (N_D)	p-type (N_A)
Majority / minority	none: $n_0 = p_0$	electrons / holes	holes / electrons
Concentrations	$n_0 = p_0 = n_i$	$n_0 \simeq N_D$, $p_0 = n_i^2/N_D$	$p_0 \simeq N_A$, $n_0 = n_i^2/N_A$
Position of E_F	$E_F = E_i$, near midgap	above E_i: $E_F - E_i = kT\ln(N_D/n_i)$	below E_i: $E_i - E_F = kT\ln(N_A/n_i)$
Distance to a band	$E_C - E_i = 0.563$ eV	$E_C - E_F = kT\ln(N_C/N_D)$	$E_F - E_V = kT\ln(N_V/N_A)$
Example 10^{16} cm^{-3}	$n_0 = p_0 = 10^{10}$	$n_0 = 10^{16}$, $p_0 = 10^4$, $E_C - E_F = 0.206$ eV	$p_0 = 10^{16}$, $n_0 = 10^4$, $E_F - E_V = 0.180$ eV

Table 3.6 *– Pure, n-type and p-type silicon at 300 K.*

	Freeze-out	Extrinsic plateau	Intrinsic
Range (here)	below about 150–200 K	about 200–580 K	above about 580 K
What dominates	electrons held by donors ($kT \ll E_C - E_D$)	all donors ionised, $N_D \gg n_i$	thermal pairs, $n_i \gg N_D$
n_0	$\propto \mathrm{e}^{-(E_C-E_D)/2kT}$, eq. (3.28)	$\simeq N_D$, independent of T	$\simeq n_i(T)$
E_F	between E_D and E_C	$E_C - kT\ln(N_C/N_D)$, descending	$\to E_i$

Table 3.7 *– The three temperature regimes of a doped sample.*

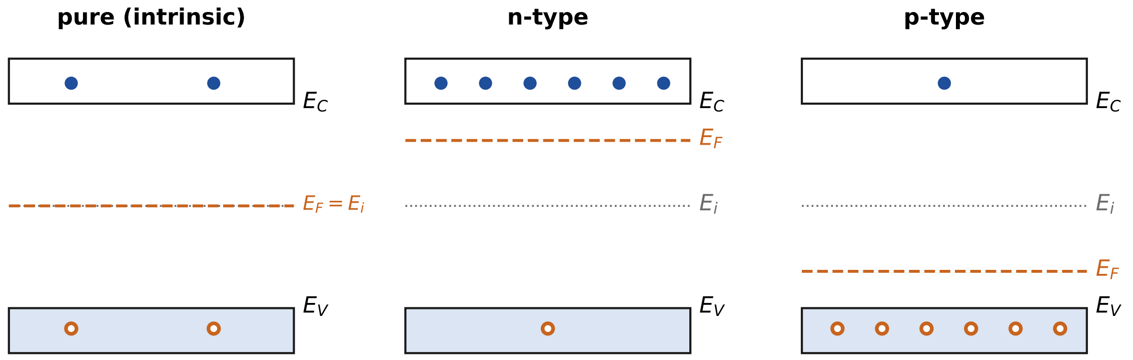
- Neutrality $n_0 = p_0 + N_D^+$ with the band expressions is one equation in E_F; at 300 K it reduces to the three laws of chapter 2 and checks complete ionisation (99.6 %).
- Temperature: freeze-out ($n_0 \propto \mathrm{e}^{-(E_C-E_D)/2kT}$), extrinsic plateau ($n_0 = N_D$, above about 200 K for 10^{16}), intrinsic regime when $n_i(T)$ reaches N_D; E_F follows, from near E_C down to E_i.

- Fermi level, deduced after the concentrations: $E_F = E_i + kT\ln(n_0/n_i) = E_i - kT\ln(p_0/n_i)$; $kT\,\mathrm{arcsinh}(N_D/2n_i)$ for all dopings; 59.6 meV per decade; degenerate above about $10^{18}\,\mathrm{cm}^{-3}$.

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Bonds and bands.** Metallic bonding provides free electrons; covalent bonding (diamond and zinc-blende structures, 2 atoms per cell, sp^3) does not. A free electron in a wire has $E = \hbar^2k^2/2m$, $k = 2\pi j/L$, two electrons per k. A periodic potential opens gaps at $k = \pm\pi/a$ (input). A band holds $2N$ states; silicon's $8N$ electrons fill exactly 4 bands.
- **Metal, insulator, semiconductor.** Half-filled band: metal. Full valence band and empty conduction band: insulator at 0 K; semiconductor if $E_g < 100\,kT$, insulator if $E_g > 200\,kT$. Full and empty bands carry no current. E_F at 0 K is the highest occupied energy, in the gap for a semiconductor.
- **Statistics.** $f(E) = 1/(1 + \mathrm{e}^{(E-E_F)/kT})$, holes $1 - f$; E_F is a statistical parameter, not a band; Boltzmann beyond $3kT$.
- **Effective mass and states.** $1/m^* = (1/\hbar^2)\,\mathrm{d}^2E/\mathrm{d}k^2$; above E_C the energy is kinetic; holes heavier (conductivity mass), hence less mobile; $g \propto \sqrt{|E - E_{C,V}|}$.
- **Densities.** $n = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$, $p = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$, $N_{C,V} = 2(2\pi m^*kT/h^2)^{3/2}$ (density-of-states masses), valid if E_F is more than $3kT$ from both edges.
- **Mass action and the pure material.** $n_0p_0 = n_i^2 = N_CN_V\,\mathrm{e}^{-E_g/kT}$, independent of E_F; $n_i = \sqrt{N_CN_V}\,\mathrm{e}^{-E_g/2kT}$ (6.9×10^9 by the formula; reference $10^{10}\,\mathrm{cm}^{-3}$ for Si); $n_i \propto T^{3/2}\mathrm{e}^{-E_g/2kT}$ explains the fall of ρ. $E_i = (E_C + E_V)/2 + \frac{3}{4}kT\ln(m_p^*/m_n^*)$: midgap at 0 K, −12.8 meV in Si and +35 meV in GaAs at 300 K.
- **Doped material.** Same bands, E_F moves. E_D just below E_C; $N_D^+ = N_D/[1 + 2\,\mathrm{e}^{(E_F-E_D)/kT}]$. Neutrality $n_0 = p_0 + N_D^+$ is one equation in E_F; at 300 K, $n_0 \simeq N_D$, $p_0 = n_i^2/N_D$. Freeze-out, extrinsic plateau (from about 200 K), intrinsic regime.
- **Fermi level.** $E_F - E_i = kT\ln(n_0/n_i)$, $E_C - E_F = kT\ln(N_C/N_D)$, $kT\,\mathrm{arcsinh}(N_D/2n_i)$; 59.6 meV per decade of doping; degenerate above about $10^{18}\,\mathrm{cm}^{-3}$.



$n_0 = p_0 = n_i$

$n_i = \sqrt{N_C N_V}\, \mathrm{e}^{-E_g/2kT}$

$E_F = E_i \approx$ midgap

$n_0 \simeq N_D, \quad p_0 = n_i^2/N_D$

$E_C - E_F = kT\ln(N_C/N_D)$

$E_F - E_i = kT\ln(N_D/n_i)$

$p_0 \simeq N_A, \quad n_0 = n_i^2/N_A$

$E_F - E_V = kT\ln(N_V/N_A)$

$E_i - E_F = kT\ln(N_A/n_i)$

Figure 3.17 *– The band representation at equilibrium, summarised. The bands are the same in the three samples; only the Fermi level moves, and with it the populations of the two bands (filled circles: electrons; open circles: holes). The formulas under each diagram hold at equilibrium, with complete ionisation and nondegenerate statistics.*

Comprehension questions

No new physics here: answer by reading the figures and the equations of the chapter. The answers are at the end.

Q1. Why does a metal have free electrons at 0 K while silicon has none? Answer first with bonds, then with bands.

Q2. Why does a band of a chain of N cells hold $2N$ states? Why do the valence electrons of silicon fill exactly four bands?

Q3. Why is pure silicon an insulator at 0 K? Why is diamond, with the same structure, still an insulator at 300 K, while silicon is not?

Q4. Can the Fermi level lie in the gap, where there are no states? What does E_F mean in that case?

Q5. An electron of the conduction band has an energy 0.1 eV above E_C. What is this 0.1 eV? What part of its fate does our model describe, and what part not?

Q6. Why are holes less mobile than electrons in silicon? Which effective mass do you use, and why is the density-of-states mass of holes nevertheless the smaller one?

Q7. Why does the Fermi level drop out of the product np? Why do the expressions of n and p remain valid in a doped sample, and what do they not tell us?

Q8. The mass-action law is derived in chapter 2 and again in this chapter. What does each derivation give that the other does not?

Q9. Why is E_i not exactly at midgap, in which direction does it shift in silicon and in gallium arsenide, and how does the shift depend on temperature?

Q10. Why does the ionised fraction of $10^{16}\,\mathrm{cm}^{-3}$ donors in silicon fall to about a third at 77 K? Between 200 K and 580 K n_i changes by ten orders of magnitude, yet n_0 hardly moves: why, and what happens outside this window?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 3, with their answers; the worked solutions are in the **Solutions** document. The 8 exercises that used to close this chapter are Problems 3.10 to 3.17 there; the other problems of the set (3.1 to 3.9) go further.

Answers

Q1. Bonds: in a metal the outer electrons are given to the whole crystal; in silicon they are all engaged in covalent bonds. Bands: the metal has a partly filled band, with empty states right above the last occupied one; silicon's $8N$ electrons fill four bands exactly, and the next empty state is across the gap.

Q2. The first Brillouin zone has width $2\pi/a$ and the allowed k are spaced by $2\pi/L$: $L/a = N$ values, two spin states each. Silicon: 4 electrons/atom × 2 atoms/cell × N cells $= 8N$ electrons $= 4 \times 2N$.

Q3. At 0 K the valence band is full and the conduction band empty; full and empty bands carry no current. At 300 K, $E_g/kT = 43$ in silicon (< 100): enough pairs appear. In diamond $E_g/kT = 212$ (> 200): the exponential factor is smaller by an enormous factor, and there is practically no carrier.

Q4. Yes. E_F is the energy at which the occupation probability would be $1/2$ if a state existed; it is a statistical parameter fixed by the number of electrons (neutrality) and may lie in the gap.

Q5. It is kinetic energy: $E - E_C = \hbar^2 k^2 / 2m_n^*$; the model describes this. That the electron then gives it to the lattice by emitting phonons and relaxes towards E_C (within about a picosecond) is an experimental fact outside the model of this chapter.

Q6. The mobility is $q\tau_c/m$ with the conductivity mass; the valence band of silicon is flatter, so $m_p^* = 0.37\,m_0 > m_n^* = 0.26\,m_0$; the shorter collision time of holes (an input) does the rest of the factor 2.9. The density-of-states mass of electrons (1.08) exceeds that of holes (0.56) because the conduction band has six minima to count.

Q7. In np the two exponents add up to $-(E_C - E_V)/kT = -E_g/kT$. The expressions remain valid because doping does not change the bands, only the number of electrons, hence the position of E_F. They do not tell where E_F is: that requires counting the ionised dopants and neutrality (Sections 3.11 and 3.13).

Q8. The kinetic argument of chapter 2 gives the reason: recombination needs both carriers, so equilibrium fixes a product; but it rests on the assumptions $G_0 = G_i$ and constant α and cannot give the value. The statistical one (Section 3.8) needs only equilibrium and nondegenerate statistics; it gives the value $N_C N_V\,\mathrm{e}^{-E_g/kT}$ and the domain (not too heavily doped).

Q9. E_i is where $n = p$; this requires $N_C\,\mathrm{e}^{-(E_C - E_i)/kT} = N_V\,\mathrm{e}^{-(E_i - E_V)/kT}$, so it moves toward the band with the smaller density of states. In silicon $N_V < N_C$ and E_i is 12.8 meV below midgap; in GaAs $N_C \ll N_V$ and it is about 35 meV above. The shift $(3/4)kT\ln(m_p^*/m_n^*)$ is zero at 0 K and proportional to T.

Q10. At 77 K, $kT = 6.6\,\text{meV}$, seven times smaller than 45 meV: the solution of neutrality puts E_F near E_D, the factor $2\,e^{(E_F - E_D)/kT}$ is not small, and only 36 % of the donors are ionised. On the plateau both assumptions hold, complete ionisation and $N_D \gg n_i$, so $n_0 = N_D$, independent of T. Below, the donors are not all ionised (freeze-out, $n_0 < N_D$); above, n_i catches up with N_D and the sample becomes intrinsic ($n_0 \approx n_i$).

References

- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — the reference for this chapter: ch. 2–3 (quantum mechanics, energy bands, effective mass, density of states), ch. 4 (Fermi–Dirac statistics, n and p, n_i, dopant levels and ionisation, charge neutrality, position of the Fermi level); Appendix B, Table B.4: the material constants of Table 3.3.
- S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley, 2007 — ch. 1 and appendices: properties of semiconductors (the gap of diamond in Table 3.2).
- C. Kittel, *Introduction to Solid State Physics*, 8th ed., Wiley, 2005 — ch. 1 (crystal structures), ch. 6–8: free-electron gas, energy bands, semiconductor crystals.
- N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Saunders, 1976 — ch. 2 (the Sommerfeld theory of metals) and ch. 28 (homogeneous semiconductors).

WHERE THIS LEADS

We now know, with numbers, what fixes the carriers of a semiconductor left to itself: the bands and the Fermi–Dirac statistics give n and p in terms of a single Fermi level; their product n_i^2 is a property of the material; neutrality fixes the Fermi level of any sample at any temperature. Chapter 4 disturbs this equilibrium. It asks where the concentrations go when light or injection adds carriers, how fast they come back, which physical processes recombine them, and how electrons and holes move in space when their concentrations are no longer uniform; the single Fermi level and the law $n_0 p_0 = n_i^2$ will no longer hold.

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 4

Carriers Out of Equilibrium

Equilibrium, steady state, transient · return to equilibrium · light and photoconductivity · recombination mechanisms · electron and hole currents · continuity · diffusion length · an illuminated surface, from switch-on to steady state

WHERE WE ARE

Chapter 1 gave the machinery of transport for a **generic carrier** in a metal: drift, the Drude model, diffusion, the Einstein relation, and the continuity equation of a species whose number does not change. Chapter 2 identified the carriers of a semiconductor and gave their qualitative picture **at equilibrium**: pairs created at a rate G and destroyed at a rate $R = \alpha np$, a balance $G_0 = R_0$ that gives the mass-action law $n_0 p_0 = n_i^2$, and charge neutrality, which gives n_0 and p_0. Chapter 3 founded these results on band theory: Fermi–Dirac statistics and the effective densities of states give n, p and $n_i(T)$, and the Fermi level places the equilibrium on the band diagram. This chapter **disturbs** that equilibrium. What happens when light or injection adds carriers? Where do the concentrations go, and how fast do they come back? What does light do to the conductivity? Which physical processes actually recombine the carriers? And what does the generic transport of Chapter 1 become for **electrons and holes**, two species of opposite charge that are created and destroyed together? Concentrations pushed away from equilibrium, moved by drift and diffusion and left to relax are the raw material of every device that follows.

CHAPTER OUTLINE

PART A — The uniform sample: time only

4.1 Three situations: equilibrium, steady state, transient
4.2 The return to equilibrium
4.2.1 Where do the concentrations end up?
4.2.2 How fast? The balance $dn/dt = G - R$
4.3 Light in: absorption and photoconductivity
4.3.1 A reminder: the photoelectric effect
4.3.2 Absorption in a semiconductor: a threshold, not a resonance
4.3.3 Case A — photoconductivity: the uniformly illuminated sample
4.4 Mechanisms: beyond the generic model
4.4.1 The five processes
4.4.2 Emission: direct and indirect gaps, and the colours of LEDs
4.4.3 Rates and lifetimes: which mechanism wins?

PART B — Carriers in space

4.5 Carriers in space: electrons, holes and the continuity equations
4.5.1 Electrons and holes: specifying the currents of Chapter 1
4.5.2 Continuity: Chapter 1's equation plus a source term
4.5.3 Excess minority carriers and the ambipolar diffusion coefficient
4.6 Case studies in space: the illuminated surface

4.6.1 Case B — light on the surface: what happens at the surface
4.6.2 The dark bulk in steady state
4.6.3 Discussion: lifetime, diffusion length and the electric field
Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Distinguish **thermal equilibrium, steady state and transient**, and describe what happens when equilibrium is disturbed: where the system returns, how fast, and what the **lifetime** of the excess carriers is.
- Explain why light above the gap creates pairs, and derive the **photoconductive** response of a uniformly illuminated sample.
- State the **real generation and recombination mechanisms** as physics beyond the generic model, and say which one sets the lifetime in a given material.
- Specify for **electrons and holes** the generic transport of Chapter 1, generalise its continuity equation with the source term $G - R$, and use the result for excess minority carriers in space: an illuminated surface, from switch-on (transient) to the steady profile in the bulk.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- tell an equilibrium from a steady state, and explain why $np \neq n_i^2$ under illumination;
- show that the state reached after a disturbance is unique, derive $\Delta n(t)$ from $dn/dt = G - R$, and distinguish the lifetime of an individual electron from that of the excess carriers;
- explain absorption as a threshold $h\nu \geq E_g$, compute a cut-off wavelength, and derive $\Delta n = \tau g_0$ and $\Delta\sigma = q(\mu_n + \mu_p)\tau g_0$;
- name the five generation and recombination processes, sketch them on a band diagram, explain why LEDs are made of direct-gap compounds and not of silicon, and say which process limits the lifetime;
- write J_n and J_p with the correct signs, and the continuity equations as Chapter 1's equation plus $G - R$;
- describe what happens under an illuminated surface: at the edge of the lit layer the excess rises as $(\tau g_0/2)(1 - \mathrm{e}^{-t/\tau})$, in the bulk a front advances as $\sqrt{D_p t}$ and then stops near L_p; derive the steady profile $\Delta p = (\tau g_0/2)\,\mathrm{e}^{-x/L_p}$, and explain why a shorter lifetime means a shorter diffusion length.

HOW THIS CHAPTER WORKS

- **Generic first, specific after.** Sections 4.1 to 4.3 use the generic recombination model of Chapter 2, so that the reasoning is self-consistent before any real mechanism appears; Section 4.4 then states the real mechanisms, marked as physics beyond the model, and Section 4.5 specifies the transport for electrons and holes.

- **Question before formula.** Each section opens with the question that the previous result leaves open.
- **Assumptions are stated where they start to matter**: uniformity, low-level injection, quasi-neutrality, constant coefficients, boundary conditions.

Symbol	Meaning	Remark
G, R, U	generation, recombination and net recombination rates	pairs per cm^3 per second; $U = R - G$
α	generic recombination coefficient (Chapter 2)	cm^3/s; an effective parameter
τ	lifetime of the excess carriers	decay time constant
n_0, p_0; n_i	equilibrium concentrations; intrinsic concentration	$n_0 p_0 = n_i^2$; $10^{10}\,\text{cm}^{-3}$ for Si (reference)
Δn, Δp	excess concentrations	$n = n_0 + \Delta n$, $p = p_0 + \Delta p$
g_0	optical generation rate	pairs per cm^3 per second, set by the light
$h\nu$, λ_g	photon energy; cut-off wavelength	λ_g (nm) $\approx 1240/E_g$ (eV)
B, C_n, τ_{SRH}	radiative and Auger coefficients; SRH lifetime	imported, mechanism-dependent
q, $\mathcal{E}$	elementary charge (positive); electric field	electron charge $-q$, hole charge $+q$
J_n, J_p	electron and hole current densities	conventional current, A/cm^2
μ_n, μ_p, D_n, D_p	mobilities and diffusion coefficients	Einstein relation: $D = \mu kT/q$
D_a, L	ambipolar diffusion coefficient, diffusion length	$L = \sqrt{D\tau}$
Δp_s	excess held at the edge of the dark bulk (Case B)	cm^{-3}
erfc	complementary error function	$\text{erfc}(0) = 1$, $\text{erfc}(+\infty) = 0$

Table 4.1 *– Notation of the chapter. Silicon at 300 K throughout unless stated:* $kT = 25.9\,\text{meV}$, $E_g = 1.12\,\text{eV}$, *reference value* $n_i = 10^{10}\,\text{cm}^{-3}$ *(Chapter 2),* $N_C = 2.8 \times 10^{19}$ *and* $N_V = 1.04 \times 10^{19}\,\text{cm}^{-3}$ *(Chapter 3),* $\mu_n = 1350$ *and* $\mu_p = 460$ *cm²/(V·s), hence* $D_n = 35$ *and* $D_p = 12$ *cm²/s. References to Chapters 2 (qualitative picture) and 3 (bands and Fermi level) are made by topic.*

PART A

The uniform sample: time only

Return to equilibrium, light, recombination mechanisms

4.1 Three situations: equilibrium, steady state, transient

THE QUESTION

Chapters 2 and 3 described a sample left to itself. What changes when something acts on it: light, an applied voltage, injected carriers?

Chapter 2 counted the carriers through two rates: thermal generation G, which creates pairs, and recombination R, which destroys them. For a uniform sample, with transport ignored for the moment, pairs are created and destroyed together, so for electrons and holes alike (the balance of Chapter 2)

$$\frac{dn}{dt} = \frac{dp}{dt} = G - R. \tag{4.1}$$

At thermal equilibrium the two rates balance, $G_0 = R_0$, and the concentrations take their equilibrium values n_0, p_0, with $n_0 p_0 = n_i^2$. When something acts on the sample, one of the terms changes: light adds a generation rate g, so that $G = G_0 + g$; injected carriers raise n and p and therefore R. The balance (4.1) then no longer vanishes, or vanishes at other values. It will be used twice in this chapter for a uniform sample: during the return to equilibrium (Section 4.2) and under light (Section 4.3); Section 4.5 adds transport. But the word "steady" covers more than one situation, and the difference matters for everything that follows.

THREE SITUATIONS NOT TO CONFUSE

- **Thermal equilibrium**: no external excitation (no light, no applied voltage, no net current). The concentrations have their equilibrium values n_0, p_0 and generation balances recombination.
- **Steady state**: no macroscopic quantity changes with time, but the sample may be driven. Under continuous illumination or applied bias the concentrations are constant while the sample is **not** in thermal equilibrium ($np \neq n_i^2$). The balance may involve not only generation and recombination but also transport.
- **Transient**: the concentrations are changing with time, for example just after the light is switched on or off.

Thermal equilibrium is therefore a particular steady state, but a constant concentration does not prove equilibrium. Figure 4.1 follows one sample through the three situations.

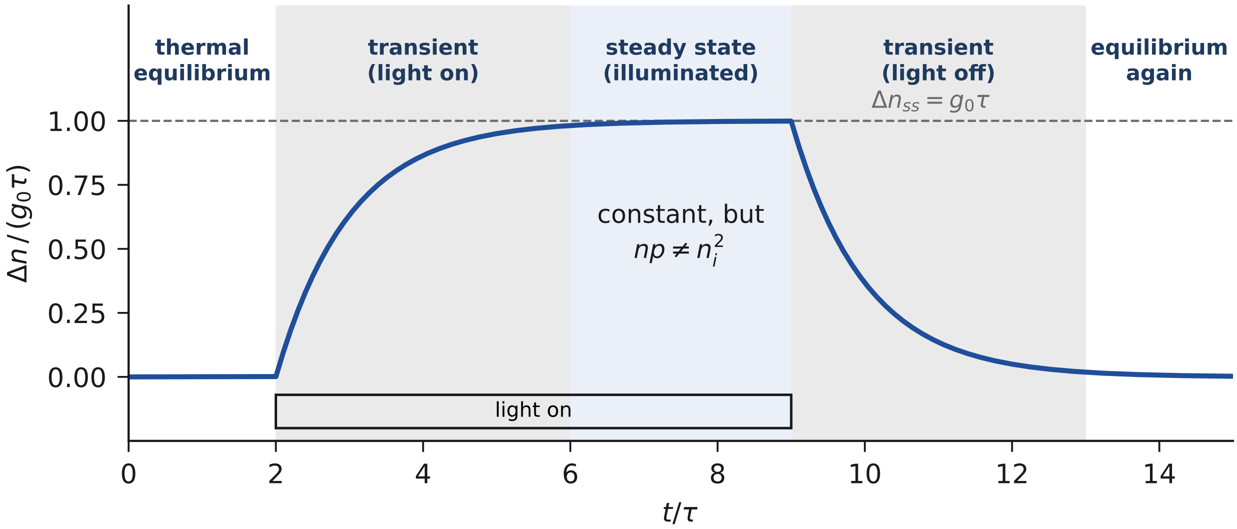


Figure 4.1 *– One sample through the three situations. At equilibrium the concentration is constant. When light is switched on (at 2τ) the sample goes through a transient and settles in a steady state where the concentration is constant again, yet generation, now larger, is balanced by recombination and $np \neq n_i^2$. When the light is switched off (at 9τ) a second transient brings the sample back to equilibrium. The two exponentials are derived in Sections 4.2 and 4.3.*

4.2 The return to equilibrium

THE QUESTION

A light pulse creates extra electron–hole pairs in an n-type sample, then the light is switched off. Where will the concentrations end up, and how fast do they get there?

This section is deliberately generic: one uniform sample, no transport, the generic recombination law $R = \alpha np$ of Chapter 2. The sample starts at equilibrium, $n = n_0$ and $p = p_0$. Light creates pairs, so extra electrons and extra holes appear **together**. In a uniform, quasi-neutral sample the two excesses are equal and the concentrations become

$$n = n_0 + \Delta n, \qquad p = p_0 + \Delta p, \qquad \Delta p = \Delta n. \tag{4.2}$$

BE AWARE OF...

$\Delta p = \Delta n$ **is a statement about pair generation in a quasi-neutral region.** It is not a universal property: when transport, contacts or spatially varying fields matter, the two excesses can differ locally. Section 4.5 shows how they are tied together in the situations of interest.

We say the sample is under **low-level injection** when $\Delta n \ll n_0$ (for n-type material: the excess is small compared with the **majority** concentration). Be careful about what this means: the minority carriers can change enormously. With $n_0 = 10^{16}$, $p_0 = 10^4$ and $\Delta n = 10^{14}\ \text{cm}^{-3}$, the electron concentration changes by 1% while the hole concentration is multiplied by 10^{10}.

4.2.1 Where do the concentrations end up?

We do not assume the answer. Call n_1 and p_1 the unknown concentrations after the light has been off for a long time, and use only what we know to be true. The doping has not changed, so neutrality still gives $n_1 - p_1 = n_0 - p_0$: both concentrations moved by the same amount s, $n_1 = n_0 + s$ and $p_1 = p_0 + s$. And a sample that has settled with nothing acting on it is at thermal equilibrium, so mass action holds, $n_1 p_1 = n_i^2 = n_0 p_0$. Expanding $(n_0 + s)(p_0 + s) = n_0 p_0$ gives

$$s\,(n_0 + p_0 + s) = 0. \tag{4.3}$$

Either $s = 0$, or $s = -(n_0 + p_0)$. The second root would give $n_1 = -p_0$ and $p_1 = -n_0$, negative concentrations, which cannot exist. So $s = 0$: **the only equilibrium the sample can reach is the original one**. Notice what this settles and what it does not: it says where the evolution must stop, not whether the sample gets there nor how fast. For that we need the rates.

4.2.2 How fast? The balance $dn/dt = G - R$

We go back to the balance (4.1) and simply write down its two terms, both already known from Chapter 2.

- **Generation.** The light is off: only thermal generation remains, and in the model it does not depend on the carriers present (the assumption made in Chapter 2 for thermal generation). It keeps its equilibrium value, $G = G_0 = \alpha\, n_0 p_0$.
- **Recombination.** It follows the actual concentrations: $R = \alpha\, n\, p = \alpha\,(n_0 + \Delta n)(p_0 + \Delta p)$.

Since n_0 is a constant, $dn/dt = d\Delta n/dt$, and with $\Delta p = \Delta n$ the balance reads

$$\frac{d\Delta n}{dt} = \alpha\, n_0 p_0 - \alpha\,(n_0 + \Delta n)(p_0 + \Delta n). \tag{4.4}$$

Expand the product: $(n_0 + \Delta n)(p_0 + \Delta n) = n_0 p_0 + (n_0 + p_0)\,\Delta n + \Delta n^2$. Its first term is exactly the generation, which is what equilibrium means, and it cancels. What is left is the recombination *caused by the excess*:

$$\frac{d\Delta n}{dt} = -\,\alpha\,(n_0 + p_0)\,\Delta n \;-\; \alpha\,\Delta n^2. \tag{4.5}$$

Neglect the second-order term. The ratio of the last term to the first is $\Delta n/(n_0 + p_0)$, which is small under low-level injection. Dropping it leaves a linear equation:

$$\frac{d\Delta n}{dt} = -\frac{\Delta n}{\tau}, \qquad \frac{1}{\tau} = \alpha\,(n_0 + p_0) \qquad \Longrightarrow \qquad \Delta n(t) = \Delta n(0)\,\mathrm{e}^{-t/\tau}. \tag{4.6}$$

The excess carriers decay exponentially and τ is the characteristic time over which the sample forgets the pulse. This gives a precise meaning to “the semiconductor returns to equilibrium”; the decay ends at $\Delta n = 0$, the unique equilibrium found above.

Which lifetime is this? In Chapter 2 an individual electron had the lifetime $\tau_n = 1/(\alpha p)$. In n-type material that number is enormous, because holes are rare ($p_0 = 10^4\ \mathrm{cm}^{-3}$; with the effective value $\alpha = 10^{-11}$ cm³/s of Problem 4.9 of the exercise collection it is 10^7 s). Yet the excess electrons disappear with $\tau \approx 1/(\alpha n_0)$, which is **much shorter** (10 µs with the same α). There is no contradiction. Each excess hole finds a partner almost immediately because electrons are abundant, and every such event removes one excess electron as well. The decay of the excess is governed by the **minority carrier**: τ is the minority-carrier lifetime, and it decreases when the majority doping increases.

BE AWARE OF…

The exponential law is a low-level result. Keeping the quadratic term of (4.5), the exact solution is, with $a = n_0 + p_0$ and $\tau = 1/(\alpha a)$, $\Delta n(t) = a\,\Delta n(0)\,\mathrm{e}^{-t/\tau}/[a + \Delta n(0)(1 - \mathrm{e}^{-t/\tau})]$ (Figure 4.2). For $\Delta n(0) \ll a$ it reduces to the exponential; for $\Delta n(0) \gg a$ the decay starts much faster (the two excess populations recombine with each other, $\propto \Delta n^2$). Under high injection the lifetime is no longer a constant.

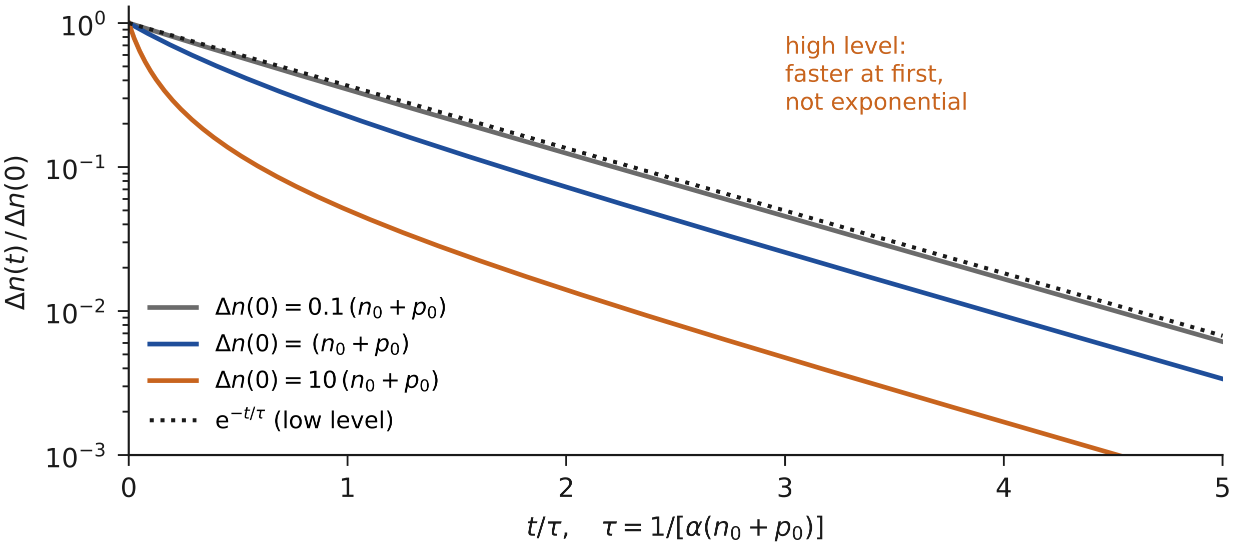


Figure 4.2 *– Decay of the excess carriers after switch-off. Low level: a straight line on a semi-log plot, slope* $-1/\tau$*. High level: curved, because the effective lifetime depends on* Δn *itself.*

And if the light stays on? Then a generation g_0 is added to the balance and the excess no longer decays: it builds up to a constant value. Before writing it, we need to know how light creates pairs.

4.3 Light in: absorption and photoconductivity

THE QUESTION

Which light creates pairs in a semiconductor, and how much does steady light change its conductivity?

4.3.1 A reminder: the photoelectric effect

The photoelectric effect is the well-understood starting point. Light falling on a metal ejects electrons, but only if its frequency exceeds a threshold, however intense the light. Einstein's explanation (1905): light is absorbed in quanta, photons of energy $h\nu$; one photon gives all its energy to one electron. To leave the metal, the electron needs at least the work function W, the energy that separates the highest occupied states (E_F of the metal) from the vacuum. If $h\nu \geq W$ it escapes, and **the excess becomes its kinetic energy**:

$$E_{\text{kin,max}} = h\nu - W. \tag{4.7}$$

Two features matter for what follows (Figure 4.3a). It is a **threshold**: every photon with $h\nu \geq W$ works, not only one particular energy. And the energy above the threshold is not lost: it goes into the motion of the electron.

Compare with an **isolated atom**, hydrogen for example (Figure 4.3b). Its electron can only occupy discrete levels E_1, E_2, ... A photon is absorbed only if its energy matches *exactly* a difference between two levels, $h\nu = E_2 - E_1$: this is a **resonance**, and the absorption spectrum is a set of narrow lines. A photon slightly above $E_2 - E_1$ is not absorbed by that transition, because there is no state to put the extra energy in. (Above the ionisation energy, 13.6 eV, the atom behaves again as a threshold: the electron is freed and the excess is kinetic energy, just as in the metal.)

4.3.2 Absorption in a semiconductor: a threshold, not a resonance

A semiconductor behaves like the metal, not like the atom, and the reason is the band structure. Above E_C the states form a **continuum** (Chapter 3). A photon of energy $h\nu$ can lift an electron from the valence band to the conduction band as soon as

$$h\nu \geq E_g, \tag{4.8}$$

and it does not need to match any particular value: if $h\nu > E_g$, the electron lands **above** E_C and the hole **below** E_V, and the excess $h\nu - E_g$ is shared between them as **kinetic energy** (the kinetic energy in the bands, Chapter 3, and Figure 4.3c). Both carriers then give this excess to the lattice by emitting phonons and settle near the band edges (**thermalisation**; experiment shows that it takes about a picosecond, a time scale outside our model); the excess ends as heat. What remains is one more free electron and one more free hole: an electron–hole pair, exactly like the broken bond of Chapter 2, except that the energy came from a photon instead of a thermal fluctuation.

This is the important fact: **in a semiconductor every photon above the gap can create a pair**. The threshold sets a cut-off wavelength,

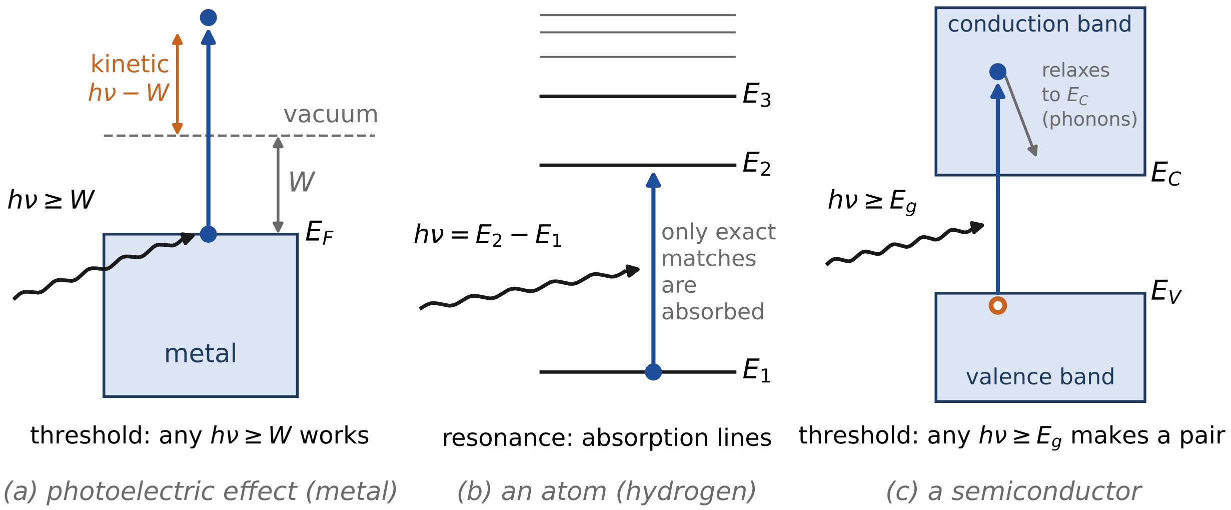
$$\lambda_g = \frac{hc}{E_g} \qquad \Longleftrightarrow \qquad \lambda_g\,(\text{nm}) \approx \frac{1240}{E_g\,(\text{eV})}, \tag{4.9}$$

1.1 µm for silicon, 1.9 µm for germanium, 0.87 µm for GaAs. Light of longer wavelength is not absorbed by band-to-band transitions: silicon is transparent in the infrared beyond 1.1 µm. Light of shorter wavelength, the whole visible spectrum for silicon, is absorbed.

BE AWARE OF...

Outside the model: two refinements. Photons are also absorbed below the gap by dopants, defects and free carriers, weakly. And because a photon carries almost no momentum, an electron must keep the same k when it is lifted; in an indirect-gap material such as silicon (Section 4.4.2) absorption near the threshold needs the help of a lattice vibration and is therefore weaker, which is why silicon solar cells must be thicker than GaAs ones. Neither refinement changes the threshold $h\nu \geq E_g$.

Optical generation. Each absorbed photon creates one pair. Under steady light, pairs are therefore created at a rate g_0 per cm³ and per second, set by the photon flux and


kinetic
hν − W
vacuum
W
hν ≥ W
E_F
metal
threshold: any hν ≥ W works
E_3
E_2
hν = E_2 − E_1
only exact matches are absorbed
E_1
resonance: absorption lines
conduction band
relaxes to E_C (phonons)
E_C
hν ≥ E_g
E_V
valence band
threshold: any hν ≥ E_g makes a pair

(a) photoelectric effect (metal) (b) an atom (hydrogen) (c) a semiconductor

Figure 4.3 *– Three ways of absorbing light. (a) Photoelectric effect in a metal: any photon with* $h\nu \geq W$ *frees an electron; the excess is kinetic energy. (b) An atom: only photons that match a level difference exactly are absorbed (lines). (c) A semiconductor: any photon with* $h\nu \geq E_g$ *creates an electron–hole pair; the excess is kinetic energy, then given to the lattice.*

by how strongly the material absorbs it; we take g_0 as given. Pairs created by light are indistinguishable from pairs created by heat: the balance (4.1) simply acquires the term g_0.

4.3.3 Case A — photoconductivity: the uniformly illuminated sample

> ***THE QUESTION***
>
> *Light falls uniformly on an n-type sample and stays on. How large does the excess become, how long does it take, and by how much does the conductivity change?*

The sample: silicon with $N_D = 10^{16}\,\text{cm}^{-3}$ ($n_0 = 10^{16}$, $p_0 = 10^4\,\text{cm}^{-3}$), minority-hole lifetime $\tau = 10\,\mu\text{s}$. The light generates pairs at the rate $g_0 = 10^{19}\,\text{cm}^{-3}\text{s}^{-1}$, **uniformly**: there is no gradient, hence no current, and each point behaves as the uniform sample of Section 4.2. The light is switched on at $t = 0$ and left on. We repeat the three steps of Section 4.2.2, with the light now on.

- **Generation.** The light and the lattice both create pairs, and their rates add: $G = g_0 + \alpha\, n_0 p_0$.
- **Recombination.** As before it follows the actual concentrations: $R = \alpha\, n(t)\, p(t)$, with $n(t) = n_0 + \Delta n(t)$, $p(t) = p_0 + \Delta p(t)$ and $\Delta p = \Delta n$.

Expanding R as in Section 4.2.2, its term $\alpha n_0 p_0$ cancels the thermal generation and

$$\frac{d\Delta n}{dt} = g_0 - \alpha\,(n_0 + p_0)\,\Delta n - \alpha\,\Delta n^2. \tag{4.10}$$

Neglecting the second-order term (low level) and using $1/\tau = \alpha(n_0 + p_0)$,

$$\frac{d\Delta n}{dt} = g_0 - \frac{\Delta n}{\tau} \qquad \Longrightarrow \qquad \Delta n(t) = \tau g_0\,(1 - \mathrm{e}^{-t/\tau}), \qquad \Delta n(+\infty) = \tau\, g_0. \tag{4.11}$$

The sample behaves as a reservoir filled at the constant rate g_0 and leaking at the rate $\Delta n/\tau$; it stops filling when the leak equals the supply. With our numbers $\Delta n(+\infty) = 10^{-5} \times 10^{19} = 10^{14}\,\text{cm}^{-3}$, reached within 1% after $\ln 100\,\tau \approx 4.6\,\tau = 46$ µs.

The conductivity. Each extra pair adds one electron and one hole. The Drude expression $\sigma = nq\mu$ (Chapters 1 and 2), applied to both carriers, gives the extra conductivity

$$\Delta\sigma = q\,(\mu_n + \mu_p)\,\Delta n \qquad \Longrightarrow \qquad \Delta\sigma(+\infty) = q\,(\mu_n + \mu_p)\,\tau g_0. \tag{4.12}$$

When the light is switched off, Δn decays as $e^{-t/\tau}$ (Section 4.2.2) and so does $\Delta\sigma$. **This is observation (b) of Chapter 2, photoconductivity, explained**: the resistance drops under light and recovers in the dark, both with the time constant τ.

EXAMPLE 4.1 — THE PHOTOCONDUCTIVE RESPONSE

With the numbers above, $\Delta\sigma = 1.6\times10^{-19}\times1810\times10^{14} = 2.9\times10^{-2}$ S/cm. The dark conductivity of the sample is $\sigma_0 \simeq qN_D\mu_n = 1.6\times10^{-19}\times10^{16}\times1350 = 2.2$ S/cm: the light raises it by 1.3%. In lightly doped material, where σ_0 is far smaller, the same Δn changes σ by orders of magnitude (Problem 4.12 of the exercise collection). The lifetime acts twice: a longer τ gives a **larger** response, $\propto \tau$, but a **slower** one, also $\propto \tau$.

Checks. (i) Low level: $\Delta n/n_0 = 1\%$, so dropping $\alpha\Delta n^2$ was legitimate; keeping it changes the plateau by 1% (Problem 4.13 of the exercise collection). (ii) This is a steady state and not equilibrium: $np \approx 10^{16} \times 10^{14} = 10^{30}$ cm^{-6}, ten orders of magnitude above $n_i^2 = 10^{20}$. (iii) A shorter τ gives a **lower** plateau, reached **sooner** (Figure 4.4).

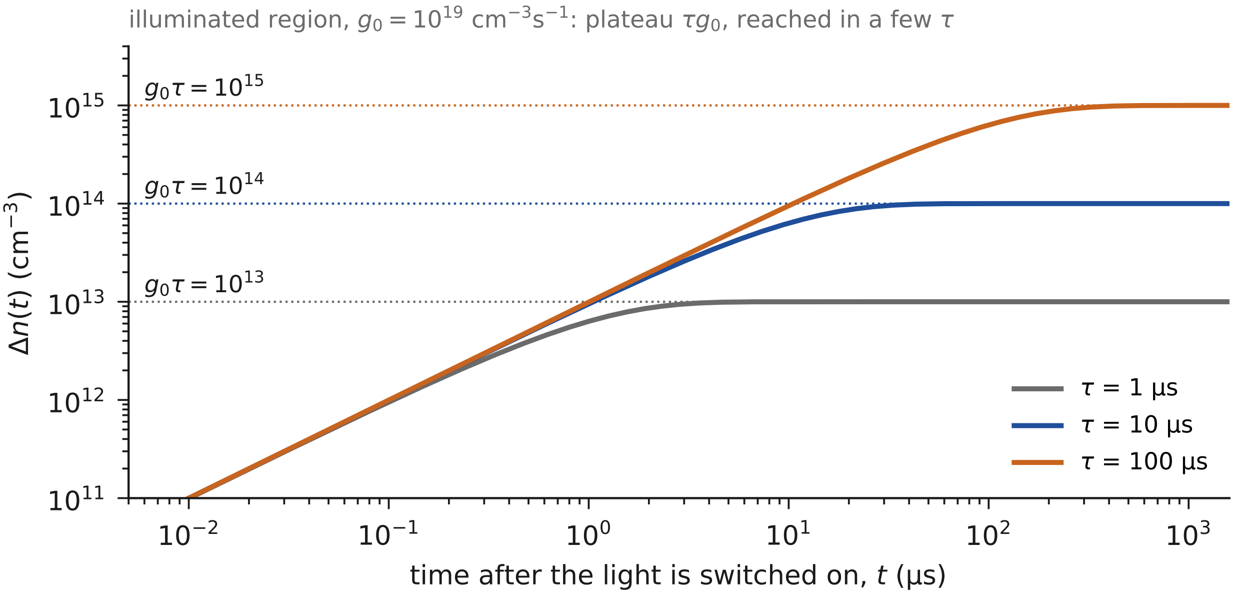


Figure 4.4 *– Case A: build-up of the excess in the uniformly illuminated sample for three lifetimes ($g_0 = 10^{19}$ cm^{-3}s^{-1}, n-type, $n_0 = 10^{16}$ cm^{-3}). Each curve rises as $1 - e^{-t/\tau}$ to the plateau τg_0: the longer the lifetime, the more carriers accumulate, and the longer it takes.*

The same pairs, separated by the field of a junction instead of recombining, give the photovoltaic effect of observation (e) of Chapter 2: that is for the junction chapters.

4.4 Mechanisms: beyond the generic model

THE QUESTION

The generic model needed no mechanism. What physically makes an electron and a hole recombine, where does the energy go, and which process sets the lifetime?

BEYOND THE GENERIC MODEL

This section leaves the model used so far. The processes are described physically; their rate formulas and coefficients are **imported** from the physics of each process (Neamen, ch. 6; Sze & Ng, ch. 1; Schubert for light emission). They are stated, not derived. Which mechanism dominates depends on the material, the doping, the defects, the temperature and the injection level: every statement of the kind “X dominates” below is material- and regime-dependent.

4.4.1 The five processes

Energy-band diagrams distinguish the principal processes without a full quantum treatment (Figure 4.5). Each one is an **energy transfer**, and the arrows are not literal paths through space.

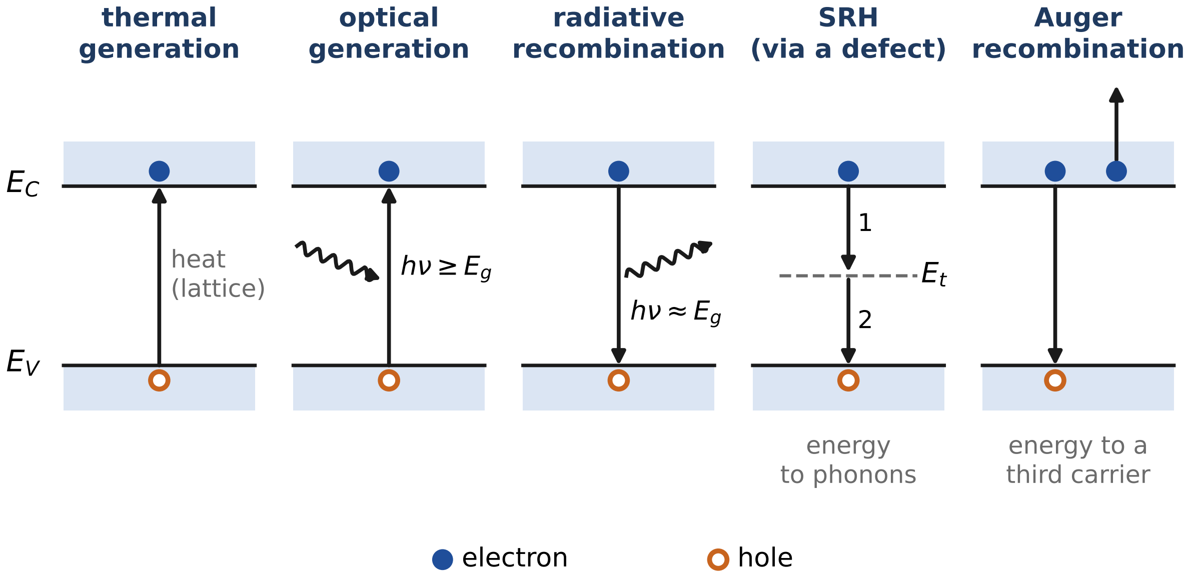

thermal generation
optical generation
radiative recombination
SRH (via a defect)
Auger recombination
E_C
E_V
heat (lattice)
$h\nu \geq E_g$
$h\nu \approx E_g$
1
E_t
2
energy to phonons
energy to a third carrier
electron
hole

Figure 4.5 – *The five processes on an energy-band diagram. Generation lifts an electron from E_V to E_C (thermal energy or a photon). Recombination removes an electron–hole pair, and the released energy leaves as a photon (radiative), as lattice vibrations through a defect level E_t in two steps (SRH), or as kinetic energy of a third carrier (Auger).*

Thermal generation. At a nonzero temperature the lattice has thermal energy, which can lift an electron across the gap and create a pair. It is the G_0 of Chapter 2, and defect levels can take part in it too. At equilibrium it is balanced by recombination. (For the junction chapters: in a region emptied of carriers, such as a reverse-biased junction, it is not balanced and contributes to the current.)

Optical generation. An absorbed photon with $h\nu \geq E_g$ lifts an electron from the valence band to the conduction band and leaves a hole: the g_0 of Section 4.3.

Radiative recombination. A conduction-band electron falls into a valence-band hole and a **photon** is emitted, $h\nu \approx E_g$ (the approximation neglects the kinetic energies of the carriers). It is the reverse of optical generation; Section 4.4.2 explains why it is efficient in some materials and weak in silicon.

Shockley–Read–Hall (SRH) recombination. Real crystals contain defects and impurities, which introduce allowed states inside the gap. Such a level acts as an intermediate step: a defect captures an electron, then a hole, completing the recombination (the order may be reversed). The energy goes to the lattice as vibrations (**phonons**), not to a photon. Its rate depends on the density of defects, the trap energy, the capture properties and the carrier concentrations.

Auger recombination. The electron and the hole recombine and the energy goes to a **third carrier**, an electron or a hole, which is lifted high into its band and then relaxes by emitting phonons. Because three carriers are involved, the rate contains terms in n^2p and np^2, and it matters only at high carrier concentrations (heavy doping, strong injection).

4.4.2 Emission: direct and indirect gaps, and the colours of LEDs

In radiative recombination both carriers sit near their band edges after thermalisation, so the emitted photon has $h\nu \approx E_g$, and **the colour of the light is set by the gap**. This is the principle of the LED (observation (d) of Chapter 2). But it works only in some semiconductors, and the reason is the one detail of the band structure that the one-dimensional model of Chapter 3 hid.

In the one-dimensional chain of Chapter 3, the top of band 1 and the bottom of band 2 sit at the same k. In real three-dimensional crystals this is not guaranteed (a fact of solid-state physics, outside our model):

- In a **direct-gap** material (GaAs, GaN, InP and their alloys), the minimum of the conduction band and the maximum of the valence band are at the same wavenumber (Figure 4.6a). An electron at the bottom of the conduction band can drop vertically into a hole at the top of the valence band. The photon carries away the energy but almost no momentum, and none is needed: the transition is allowed and fast, and the material emits light efficiently.
- In an **indirect-gap** material (Si, Ge), the minimum of the conduction band is at a different k from the maximum of the valence band (Figure 4.6b). The electron and the hole have different momenta, and a photon cannot make up the difference. The transition needs a third partner, a **phonon** that supplies the momentum; such three-body events are rare. The pair usually recombines in another way, through a defect (SRH), and its energy ends as heat.

Consequence: LEDs are not made of silicon. Silicon dominates electronics, detectors and solar cells (absorption works, Section 4.3), but it is a very poor light emitter. Light-emitting diodes and laser diodes are made of direct-gap compound semiconductors, chosen and alloyed so that their gap gives the desired colour through (4.9) (Table 4.2).

Red and infrared LEDs have existed since the 1960s, but efficient **blue** emission needs a wide direct gap and resisted for three decades, because good-quality GaN and InGaN layers could not be grown and p-doped. Its achievement in the early 1990s (Akasaki,

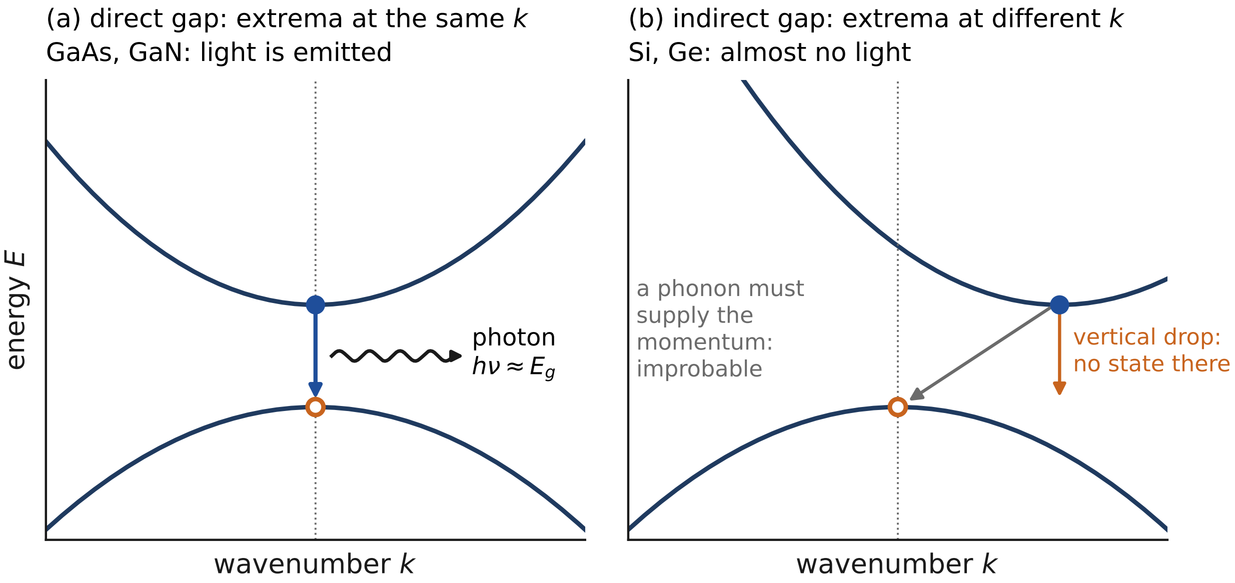

(a) direct gap: extrema at the same k
GaAs, GaN: light is emitted
(b) indirect gap: extrema at different k
Si, Ge: almost no light
energy E
photon
$h\nu \approx E_g$
a phonon must supply the momentum: improbable
vertical drop: no state there
wavenumber k
wavenumber k

Figure 4.6 *– Direct and indirect gaps. (a) Direct gap: the conduction-band minimum lies directly above the valence-band maximum, at the same k; recombination emits a photon. (b) Indirect gap: the extrema are at different k; a photon cannot supply the momentum difference, a phonon must take part, and light emission is very weak.*

Material	Gap (eV)	Type	Emission
Si	1.12	indirect	practically none
GaAs	1.42	direct	870 nm, infrared (remote controls)
AlGaInP alloys	1.9–2.2	direct	red to yellow, 650–570 nm
InGaN alloys	2.3–2.8	direct	green to blue, 540–450 nm
GaN	3.4	direct	ultraviolet, 365 nm

Table 4.2 *– Gap and emission of a few semiconductors,* $\lambda \approx 1240/E_g$ *(orders of magnitude; alloy compositions set the exact colour).*

Amano and Nakamura, Nobel Prize in Physics 2014) completed the three primary colours: **red, green and blue are all possible**, hence full-colour LED displays and white LEDs (a blue LED exciting a yellow phosphor), now the dominant light sources.

4.4.3 Rates and lifetimes: which mechanism wins?

It is convenient to work with the **net recombination rate** $U = R - G$. In the generic model, with thermal generation only, $G = G_0 = \alpha n_i^2$ and

$$U \equiv R - G = \alpha\,(np - n_i^2). \tag{4.13}$$

The imported formulas of the real mechanisms have the same structure: a factor $(np - n_i^2)$ times a coefficient that depends on the mechanism. Radiative, with the radiative coefficient B:

$$U_{\text{rad}} = B\,(np - n_i^2). \tag{4.14}$$

SRH, with capture lifetimes τ_{n0}, τ_{p0} and $n_1 = n_i\,\mathrm{e}^{(E_t - E_i)/kT}$, $p_1 = n_i\,\mathrm{e}^{-(E_t - E_i)/kT}$ the equilibrium concentrations that would exist if E_F were at the trap level E_t:

$$U_{\text{SRH}} = \frac{np - n_i^2}{\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1)}. \tag{4.15}$$

Auger, with coefficients C_n and C_p (one per type of third carrier):

$$U_{\text{Auger}} = (C_n n + C_p p)\,(np - n_i^2). \tag{4.16}$$

What the common factor does and does not mean. Each of these formulas was built so that its net rate vanishes at thermal equilibrium, where $np = n_i^2$ (the mass-action law, Chapters 2 and 3). The common factor is therefore a **consistency check** of the imported formulas with the mass-action law; it is not a new derivation of that law. Mass action was established kinetically in Chapter 2, under stated assumptions, and statistically, from the bands, in Chapter 3.

Rates add. The mechanisms act in parallel, so their net rates add. At low-level injection in n-type material each net rate reduces to $\Delta n/\tau_k$ (with $n \simeq n_0$, $p \simeq \Delta n$ and n_1, p_1 negligible), and therefore the inverse lifetimes add:

$$\frac{1}{\tau} = \frac{1}{\tau_{\text{rad}}} + \frac{1}{\tau_{\text{SRH}}} + \frac{1}{\tau_{\text{Auger}}}. \tag{4.17}$$

Process	Essential idea	Energy	Low-level lifetime (n-type)	When it matters (typical)
Thermal generation	the lattice lifts an electron across the gap	taken from the lattice	— (rate $G_0 = \alpha n_i^2$ in the generic model)	all semiconductors at $T > 0$
Optical generation	an absorbed photon creates a pair	taken from the photon, $h\nu \geq E_g$	— (rate g_0 set by the light)	photodetectors, solar cells, illumination experiments
Radiative	band-to-band transition, photon emitted	to a photon	$\tau_{\text{rad}} = 1/(Bn_0)$	direct-gap materials; $B \sim 10^{-10}$ (GaAs), 10^{-14} (Si) cm³/s
SRH	two-step capture on a defect level	to phonons (heat)	$\tau_{\text{SRH}} \approx \tau_{p0}$, independent of doping	silicon at moderate doping; set by crystal quality
Auger	energy given to a third carrier	to a hot carrier, then phonons	$\tau_{\text{Auger}} = 1/(C_n n_0^2)$	heavy doping, high injection; $C_n \sim 3 \times 10^{-31}$ cm^6/s (Si)

Table 4.3 *– The five processes. The low-level lifetimes follow from the imported net rates with $n_0 \gg p_0$ and n_1, p_1 negligible; the coefficients are orders of magnitude. These are five physical pathways that contribute to the terms G and R, not five unrelated rules.*

EXAMPLE 4.2 — WHICH MECHANISM SETS THE LIFETIME? ($n_0 = 10^{16}$ CM^{-3})

Silicon parameters (Table 4.3): radiative $1/(Bn_0) = 1/(10^{-14} \times 10^{16}) = 10$ ms; Auger $1/(C_n n_0^2) = 1/(3 \times 10^{-31} \times 10^{32}) = 33$ ms; SRH with $\tau_{\text{SRH}} = 50$ µs. The rates add, so $\tau \approx 50$ µs: **SRH wins by more than two orders of magnitude**.

In a direct-gap material with $B = 10^{-10}$ cm³/s the radiative lifetime at the same doping is $1/(10^{-10} \times 10^{16}) = 1$ µs, and it wins.

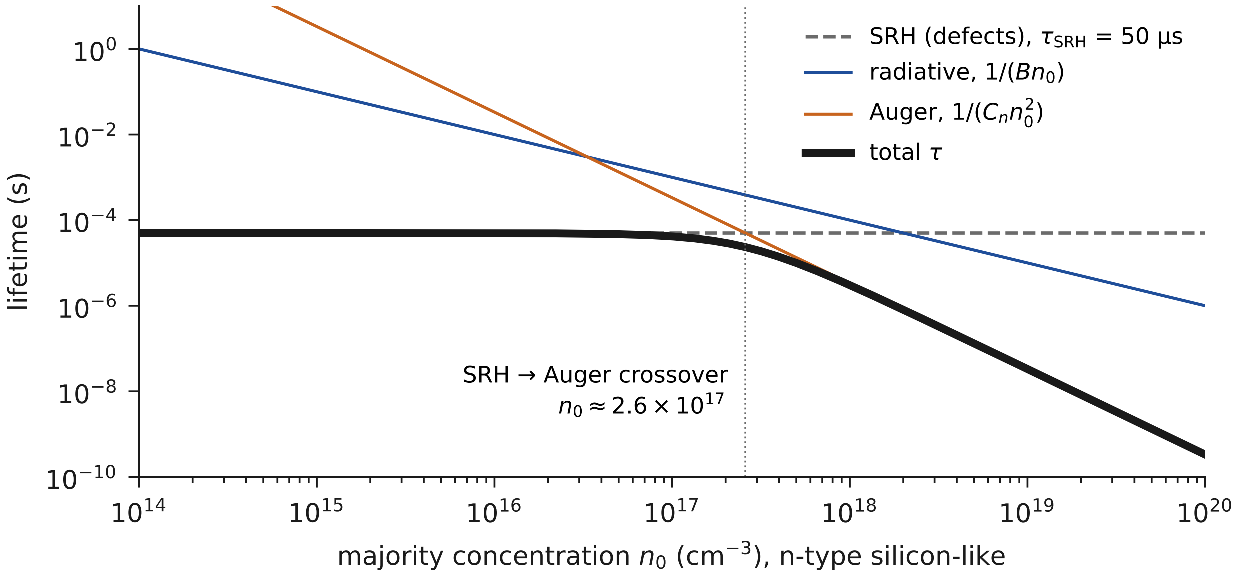


Figure 4.7 *– Lifetime of the excess carriers against doping for an n-type silicon-like material ($B = 10^{-14}$ cm³/s, $C_n = 3 \times 10^{-31}$ cm^6/s, $\tau_{\mathrm{SRH}} = 50$ μs). Rates add, so the total (thick line) follows the shortest of the three. SRH is flat, radiative falls as $1/n_0$, Auger as $1/n_0^2$; beyond about $2.6 \times 10^{17}\ \mathrm{cm}^{-3}$ Auger takes over.*

BE AWARE OF...

The lifetime is not a material constant. It depends on the dominant mechanism, hence on the material, the doping, the density of defects, the temperature and the injection level. The coefficient α of the generic model must be read as an effective parameter: $\alpha \approx B$ for radiative recombination, or $\alpha \approx 1/(\tau n_0)$ if the lifetime is measured. The reasoning of Sections 4.2 and 4.3 does not depend on which.

What survives of the generic model. The model is not discarded when the real mechanisms arrive. It carries the logic: generation, recombination, lifetime, equilibrium as a balance, mass action, relaxation. The mechanisms say what contributes to G and R. After the light is switched off, the excess carriers disappear through whichever mechanisms dominate; if a single effective lifetime and low-level injection are adequate approximations, the decay is a single exponential. If several mechanisms, strong injection, spatial variation or significant transport matter, the observed decay departs from it. Transport is the subject of the next part.

WHAT PART A HAS ESTABLISHED

- Equilibrium, steady state and transient are three distinct situations; a constant concentration does not prove equilibrium.
- After a disturbance the sample returns to the original equilibrium, the only possible one, as $\mathrm{e}^{-t/\tau}$ with $1/\tau = \alpha(n_0 + p_0)$: the minority-carrier lifetime.
- Light creates pairs above a threshold $h\nu \geq E_g$; under uniform illumination the excess rises to τg_0 and $\Delta\sigma = q(\mu_n + \mu_p)\tau g_0$ (Case A).
- The real mechanisms (radiative, SRH, Auger) are imported; their rates add, and the fastest sets τ.

PART B

Carriers in space

Electron and hole currents, continuity, the illuminated surface

4.5 Carriers in space: electrons, holes and the continuity equations

THE QUESTION

Until now the sample was uniform and carriers only appeared and disappeared. What happens when they also move, and what does the generic transport of Chapter 1 become for electrons and holes?

4.5.1 Electrons and holes: specifying the currents of Chapter 1

Chapter 1 treated a **generic carrier** without saying whether it was an electron or a hole: a species of concentration c, charge q_c, mobility μ and diffusion coefficient D moves with a drift velocity $\pm\mu\mathcal{E}$ in a field $\mathcal{E}$ (the sign is that of its charge) and diffuses down its concentration gradient. Its particle flux is $\Phi = c\,v_d - D\,\partial c/\partial x$ and its current density $J = q_c\Phi$. Electrons and holes are two such species with opposite charges. **Specifying** means choosing the signs once and for all (Table 4.4).

	Electrons	Holes
Charge q_c	$-q$	$+q$
Drift velocity	$-\mu_n\mathcal{E}$: against $\mathcal{E}$	$+\mu_p\mathcal{E}$: along $\mathcal{E}$
Particle flux Φ	$-\mu_n n\mathcal{E} - D_n\,\partial n/\partial x$	$\mu_p p\mathcal{E} - D_p\,\partial p/\partial x$

Table 4.4 *– The generic carrier of Chapter 1 specified for electrons and holes.*

Multiplying by the charge, $J_n = -q\Phi_n$ and $J_p = +q\Phi_p$, gives the two current densities:

$$J_n = q\mu_n n\mathcal{E} + qD_n\frac{\partial n}{\partial x} \tag{4.18}$$

$$J_p = q\mu_p p\mathcal{E} - qD_p\frac{\partial p}{\partial x} \tag{4.19}$$

Look at the signs, they are the whole point (Figure 4.8). In the **drift** term the electrons move against $\mathcal{E}$ but carry a negative charge, so their current flows along $\mathcal{E}$, like the hole current: **the two drift currents add**. In the **diffusion** term both species flow down the gradient of their own concentration, but the current follows the flux for holes and opposes it for electrons. The total current density is $J = J_n + J_p$.

The two species are tied to the field and to the temperature by the **Einstein relation** of Chapter 1, valid for each:

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q}. \tag{4.20}$$

In silicon at 300 K, $\mu_n = 1350$ and $\mu_p = 460$ cm²/(V·s) give $D_n = 35$ and $D_p = 12$ cm²/s: electrons are faster, in drift and in diffusion. At thermal equilibrium J_n and J_p vanish

separately, drift balancing diffusion for each species; non-uniform doping will exploit this in the chapters that follow.

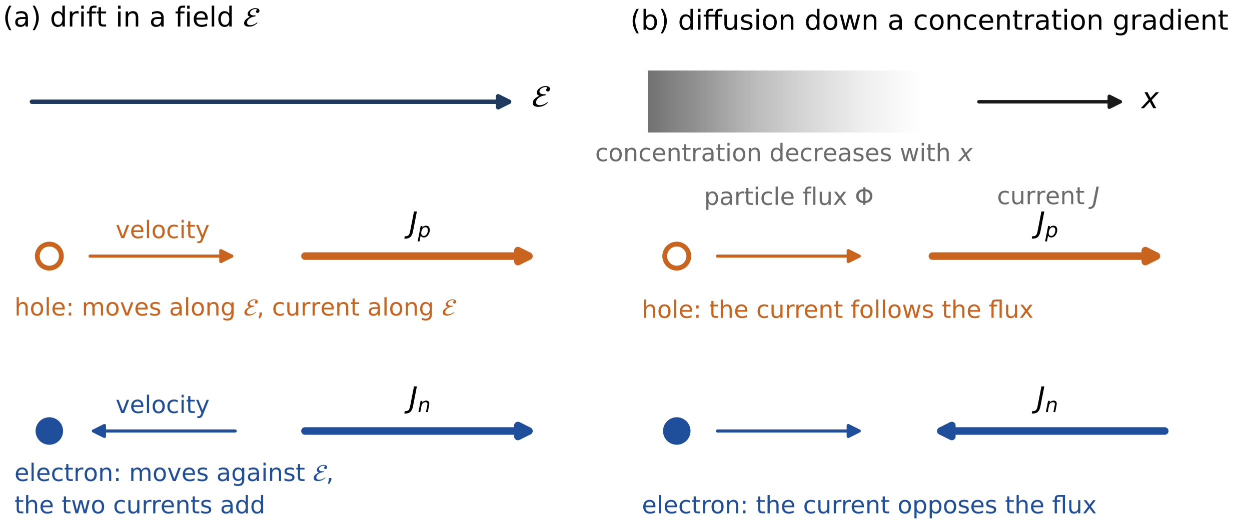

(a) drift in a field $\mathcal{E}$
$\mathcal{E}$
velocity
J_p
hole: moves along $\mathcal{E}$, current along $\mathcal{E}$
velocity
J_n
electron: moves against $\mathcal{E}$,
the two currents add
(b) diffusion down a concentration gradient
x
concentration decreases with x
particle flux Φ
current J
J_p
hole: the current follows the flux
J_n
electron: the current opposes the flux

Figure 4.8 – *The signs of the currents. (a) In a field $\mathcal{E}$ holes move along it and electrons against it, and both currents point along $\mathcal{E}$. (b) Down a concentration gradient both species flow towards low concentration; the hole current follows the flux, the electron current opposes it.*

4.5.2 Continuity: Chapter 1's equation plus a source term

Chapter 1 wrote the continuity equation of a **conserved** species: in one dimension, $\partial c/\partial t = -\partial\Phi/\partial x$, the concentration in a slice changes only by what enters and leaves it. Electrons and holes are not conserved: pairs are created and destroyed inside the slice (Figure 4.9). The balance on the slice between x and $x + dx$ therefore acquires a **source term**, the net number of pairs created per unit volume and time:

$$\frac{\partial c}{\partial t} = -\frac{\partial \Phi}{\partial x} + G - R. \tag{4.21}$$

This is the equation of Chapter 1, generalised; nothing else changes. Applied to holes, with $\Phi_p = J_p/q$,

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G - R. \tag{4.22}$$

For electrons the flux is $\Phi_n = -J_n/q$ because of their negative charge, so the divergence term changes sign; this is the only difference between the two equations:

$$\frac{\partial n}{\partial t} = +\frac{1}{q}\frac{\partial J_n}{\partial x} + G - R. \tag{4.23}$$

The generation and recombination terms are **the same** in both equations: pairs are created and destroyed together. Each equation says that a concentration changes because of transport (the divergence of the current) and because of generation minus recombination. For a uniform sample the currents drop out and we recover the balance (4.1).

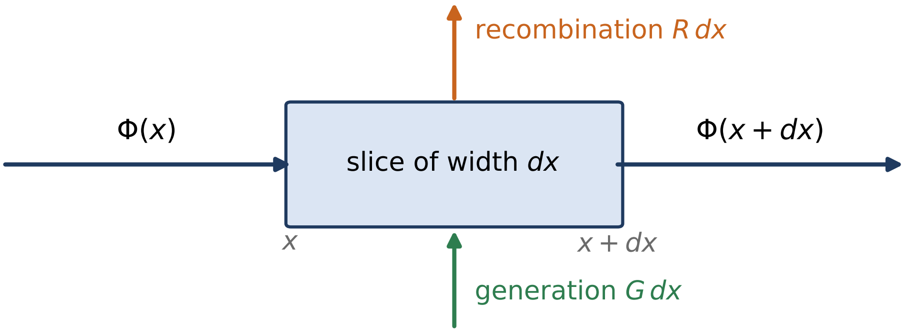


Figure 4.9 *– Balance of particles in a thin slice: what enters at x minus what leaves at $x+dx$ plus what is generated minus what recombines inside. The particle flux is $\Phi_p = J_p/q$ for holes and $\Phi_n = -J_n/q$ for electrons.*

BE AWARE OF…

Steady state does not mean $G = R$. In steady state the time derivatives vanish, but the divergence of the currents can balance a local difference between generation and recombination. In the dark bulk of Case B (Section 4.6.2) no light is absorbed, yet recombination goes on: it is fed by holes that diffusion brings in. The same caution applies to a contact that injects carriers: a steady state then requires the contact conditions, the geometry and the applied voltages to be specified as well.

4.5.3 Excess minority carriers and the ambipolar diffusion coefficient

Take an n-type sample with excess carriers created in pairs, $\Delta n = \Delta p$, in a quasi-neutral region. The hole equation contains the hole current, which contains the field $\mathcal{E}$, and this field is not imposed from outside: the carriers set it up themselves. Electrons diffuse faster than holes ($D_n > D_p$), run ahead, and create a small field that holds them back and pushes the holes forward, so that the two species move together. The two continuity equations are therefore coupled; the coupling can be solved in four steps.

- **Quasi-neutrality.** $\Delta n = \Delta p$, with uniform equilibrium concentrations, gives $\partial n/\partial x = \partial p/\partial x = \partial \Delta p/\partial x$.
- **No net current in the bulk.** $J_n + J_p = 0$: whatever flows in as hole current is compensated by the electron current.
- **Solve for the field.** Adding (4.18) and (4.19) and imposing $J_n + J_p = 0$ gives the field set up by the carriers:

$$\mathcal{E} = \frac{D_p - D_n}{\mu_n n + \mu_p p}\,\frac{\partial \Delta p}{\partial x}. \tag{4.24}$$

- **Eliminate it.** Substituting into J_p and using the Einstein relation (4.20) to replace the mobilities by the diffusion coefficients gives, after simplification,

$$J_p = -qD_a\frac{\partial p}{\partial x}, \qquad D_a = \frac{n+p}{n/D_p + p/D_n}. \tag{4.25}$$

The holes behave as if they diffused alone, with the **ambipolar diffusion coefficient** D_a, a concentration-weighted combination of D_n and D_p. **The field is not forgotten:**

it is inside D_a. Two limits matter. For $n \gg p$ (n-type, low-level injection) $D_a \approx D_p$: the minority carrier diffuses at its own rate and the abundant majority, hardly perturbed, adjusts the field. For $n = p$ (intrinsic material or strong injection) $D_a = 2D_nD_p/(D_n + D_p)$ = 18 cm²/s in silicon, between the two (Figure 4.10).

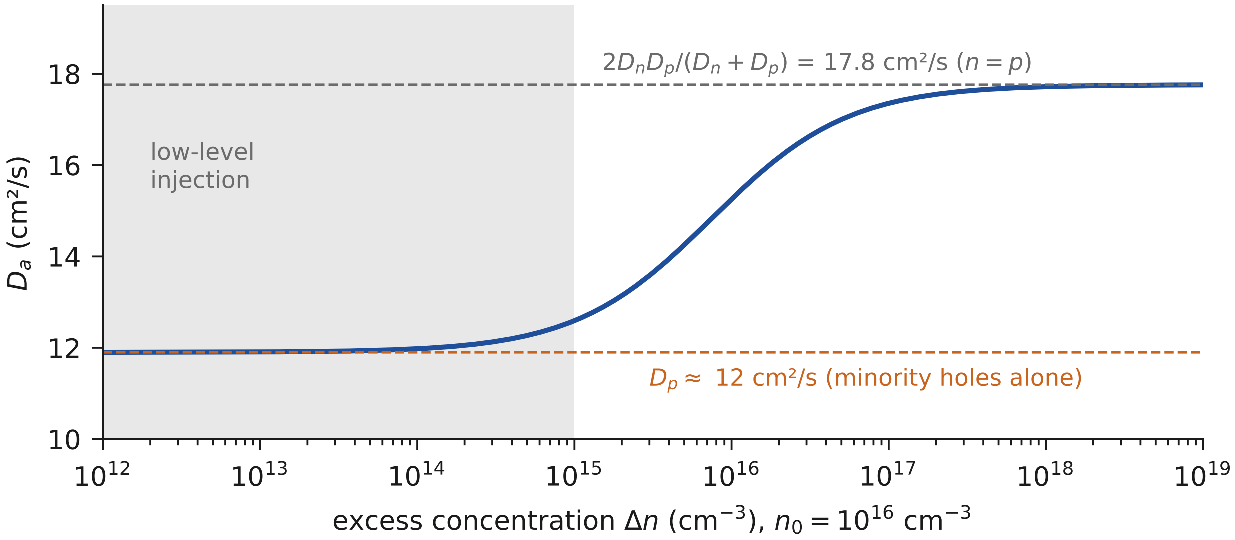


Figure 4.10 *– The ambipolar coefficient for n-type silicon with $n_0 = 10^{16}\,\mathrm{cm}^{-3}$ against the excess concentration. At low-level injection D_a equals the minority-hole value D_p; it rises towards $2D_nD_p/(D_n + D_p)$ at strong injection.*

Inserting $J_p = -qD_a\,\partial p/\partial x$ into the hole equation, with $R - G = \Delta p/\tau$ at low level (4.6), and adding an excess generation rate $g(x, t)$ such as the optical generation of Section 4.3, gives the **equation of the excess carriers**:

$$\frac{\partial \Delta p}{\partial t} = D_a \frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{\tau} + g(x, t). \tag{4.26}$$

In n-type material at low-level injection $D_a = D_p$ and $\tau = \tau_p$ is the minority-hole lifetime; in p-type material the roles of electrons and holes are exchanged. Without spatial dependence the diffusion term vanishes and we recover Case A and the decay of Section 4.2. In a steady state, diffusion and recombination compete, which defines a length, the **diffusion length** $L = \sqrt{D_a\tau}$. In between, a transient keeps both time and space. Section 4.6 uses this equation in space.

BE AWARE OF…

Assumptions behind the excess-carrier equation: quasi-neutral region, no net current in the bulk, low-level injection (so that D_a and τ are constants) and the Einstein relation of Chapter 1. An applied uniform field adds a drift term, $-\mu_p\mathcal{E}\,\partial\Delta p/\partial x$ (Section 4.6.3). Outside these assumptions, for example in the strong field of a depletion region, the equation changes.

4.6 Case studies in space: the illuminated surface

4.6.1 Case B — light on the surface: what happens at the surface

THE QUESTION

Light is switched on at $t = 0$ on the surface of a thick n-type silicon sample. What happens at the surface, and how does the excess reach the dark bulk?

The sample is that of Case A: $N_D = 10^{16}\,\mathrm{cm}^{-3}$ ($n_0 = 10^{16}$, $p_0 = 10^4\,\mathrm{cm}^{-3}$), $\tau = 10$ µs, $D_p = 12$ cm²/s. The light is now absorbed only in a layer at the surface, where it generates pairs at the rate $g_0 = 10^{19}\,\mathrm{cm}^{-3}\mathrm{s}^{-1}$; behind this layer the bulk is in the dark. Before switch-on the sample is at equilibrium: $\Delta p = 0$ everywhere.

The model. We take the illuminated layer to be uniform and thick compared with the distance carriers diffuse during their lifetime (this distance, L_p, is found below; which light gives such a layer depends on the absorption of the material, outside our model). Its boundary with the dark bulk is taken as $x = 0$, with $x < 0$ the lit side and $x > 0$ the dark side (Figure 4.12a). We suppose **quasi-neutrality**: the electrons follow the holes, $\Delta n = \Delta p$, and at low level in n-type material $D_a = D_p$. Everything can then be read from the excess-carrier equation (4.26), with $g = g_0$ for $x < 0$ and $g = 0$ for $x > 0$.

Deep in the layer: Case A. Far from the boundary, several L_p inside the layer, there is no gradient, hence no current, and each point behaves as the uniformly illuminated sample of Case A: $\Delta p = \tau g_0(1 - \mathrm{e}^{-t/\tau})$, which reaches the plateau $\tau g_0 = 10^{14}\,\mathrm{cm}^{-3}$ in a few τ (Section 4.3.3).

At the edge of the layer: half of Case A. Near the boundary the layer loses carriers to the dark side. A symmetry argument gives the edge value without calculation. Equation (4.26) is linear: if the side $x > 0$ were lit too, we would recover Case A exactly everywhere, and this uniform solution is the sum of two problems, light on $x < 0$ only (ours) and light on $x > 0$ only (its image in a mirror placed at $x = 0$). At $x = 0$ the two problems give the same value, since they are images of each other, and their sum is $\tau g_0(1 - \mathrm{e}^{-t/\tau})$. Each is therefore half of it, at all times:

$$\Delta p(0,t) = \frac{\tau g_0}{2}\,(1 - \mathrm{e}^{-t/\tau}). \tag{4.27}$$

The edge fills with the lifetime, exactly like Case A, but towards half of its plateau: $\tau g_0/2 = 5\times10^{13}\,\mathrm{cm}^{-3}$. (Problem 4.19 of the exercise collection recovers this steady value by matching the profiles on both sides.)

In the dark bulk: a front that advances, then stops. For $x > 0$, $g = 0$ and (4.26) becomes

$$\frac{\partial \Delta p}{\partial t} = D_p \frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{\tau}. \tag{4.28}$$

Let us first isolate what the bulk does by supposing the edge value held fixed from switch-on, $\Delta p(0,t) = \Delta p_s$, with $\Delta p(x,0) = 0$ for $x > 0$. The solution is

$$\Delta p(x,t) = \frac{\Delta p_s}{2}\left[\mathrm{e}^{-x/L_p}\,\mathrm{erfc}\!\left(\frac{x}{2\sqrt{D_p t}} - \sqrt{\frac{t}{\tau}}\right) + \mathrm{e}^{x/L_p}\,\mathrm{erfc}\!\left(\frac{x}{2\sqrt{D_p t}} + \sqrt{\frac{t}{\tau}}\right)\right], \tag{4.29}$$

with $L_p = \sqrt{D_p \tau}$, and where $\mathrm{erfc}(z) = (2/\sqrt{\pi})\int_z^\infty \mathrm{e}^{-u^2}du$ is the complementary error function: it goes from 2 for $z \to -\infty$ to 0 for $z \to +\infty$ and equals 1 at $z = 0$. We do not derive it; it is a solution that one **checks by substitution** (Problem 4.24 of the exercise collection): each of the two terms obeys (4.28), the sum equals Δp_s at $x = 0$ (because $\operatorname{erfc}(-z) + \operatorname{erfc}(z) = 2$), and it vanishes at $t = 0$ for every $x > 0$. Since the problem has a single solution, this is the one.

Reading the solution (Figure 4.11).

- **At short times**, $t \ll \tau$: $\sqrt{t/\tau} \to 0$, the two terms merge and

  $$\Delta p \approx \Delta p_s \operatorname{erfc}(x/2\sqrt{D_p t}).$$

  This is **pure diffusion**, without recombination: the excess penetrates over a distance of the order of $\sqrt{D_p t}$, and the point where it is half the edge value advances as $0.95\sqrt{D_p t}$ — 23 µm after 0.5 µs.

- **At long times**, $t \gg \tau$: the first erfc tends to 2, the second to 0, and $\Delta p \to \Delta p_s \, e^{-x/L_p}$. The profile no longer changes: **recombination has stopped the front** at the distance that pure diffusion reaches in one lifetime, $\sqrt{D_p \tau} = L_p$ (the half-value point stops at $L_p \ln 2 = 0.69\, L_p$).
- **The crossover** happens around $t \approx \tau$. At the depth $x = L_p$, the excess reaches 90% of its final value at $t \approx 1.1\,\tau$, that is 11 µs here (Figure 4.11b).

The real switch-on. At the edge the value is not fixed from $t = 0$: it builds up according to (4.27). The exact solution of our problem for $x \geq 0$ is obtained by taking (4.29) with $\Delta p_s = \tau g_0/2$ and subtracting a term that dies out with the lifetime:

$$\begin{aligned}\Delta p(x,t) = \frac{\tau g_0}{4}\left[e^{-x/L_p}\operatorname{erfc}\left(\frac{x}{2\sqrt{D_p t}} - \sqrt{\frac{t}{\tau}}\right) + e^{x/L_p}\operatorname{erfc}\left(\frac{x}{2\sqrt{D_p t}} + \sqrt{\frac{t}{\tau}}\right)\right] \\ - \frac{\tau g_0}{2}\, e^{-t/\tau}\operatorname{erfc}\left(\frac{x}{2\sqrt{D_p t}}\right).\end{aligned} \tag{4.30}$$

The subtracted term also obeys (4.28) (a solution of pure diffusion multiplied by $e^{-t/\tau}$), it vanishes at $t = 0$ for $x > 0$, and at $x = 0$ the whole gives back (4.27). The approach is slower — 90% of the final value at $x = L_p$ after $3.0\,\tau = 30$ µs, instead of $1.1\,\tau$ — because the point $x = L_p$ waits both for the edge to fill and for diffusion to bring the carriers. But the picture is the same: a front that advances as $\sqrt{D_p t}$ and that recombination stops near L_p. A numerical solution of (4.26) on both sides of the boundary reproduces (4.29) and (4.30) to within the numerical error.

OUTSIDE THE MODEL: STRONGLY ABSORBED LIGHT AND THE REAL SURFACE

If the light is absorbed in a layer thin compared with L_p (blue light in silicon, for example), the layer no longer reaches the plateau of Case A: the pairs diffuse out of it as soon as they are created. The boundary then imposes a flux, that of the pairs created per cm² and per second, instead of a value; the shape e^{-x/L_p} and the reach L_p of the steady profile do not change, only the amplitude at the edge does. Likewise, the real surface of the sample recombines carriers through its own defects; we have assumed it far away (thick layer) and inactive. Both refinements are treated in Neamen, ch. 6.

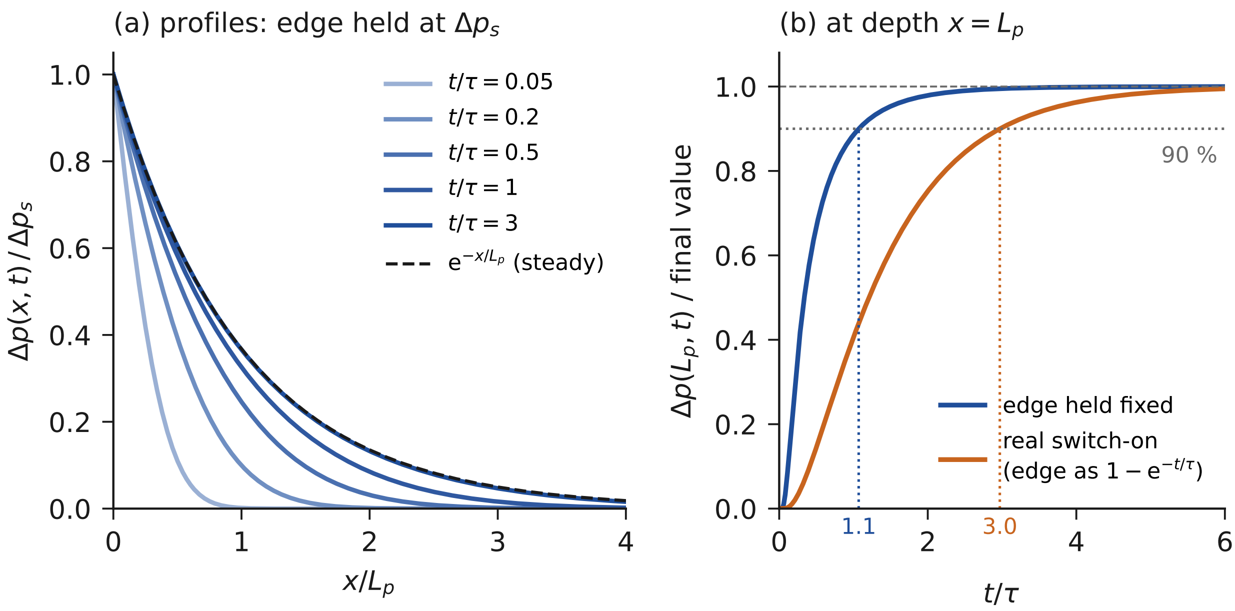


Figure 4.11 *– Case B: the transient under the illuminated surface, in reduced units ($\tau = 10$ μs and $L_p = 110$ μm for our sample). (a) Profiles (4.29) in the dark bulk, the edge value being held at Δp_s: at $t/\tau = 0.05$, 0.2, 0.5, 1 and 3, the front first advances as $\sqrt{D_p t}$ and then stops on the steady profile e^{-x/L_p} (dashed). (b) The excess at the depth $x = L_p$, relative to its final value: edge held fixed (90% at $1.1\,\tau$) and real switch-on (4.30), the edge rising as $1 - e^{-t/\tau}$ (90% at $3.0\,\tau$). After a few τ nothing changes any more: this is the steady state.*

4.6.2 The dark bulk in steady state

THE QUESTION

The light stays on. Once the steady state is reached, how are the excess carriers distributed inside the semiconductor?

In the steady state, $\partial/\partial t = 0$. In the dark bulk ($x > 0$) no light is absorbed, $g = 0$, yet there are excess holes, brought by diffusion from the layer. Transport is now essential; the excess-carrier equation (4.26), with $D_a = D_p$, becomes

$$D_p \frac{d^2 \Delta p}{dx^2} - \frac{\Delta p}{\tau} = 0. \tag{4.31}$$

Its solutions are $e^{\pm x/L_p}$ with $L_p = \sqrt{D_p \tau}$. The growing exponential is rejected (the excess cannot diverge deep inside), and the amplitude is fixed at $x = 0$ by the edge value found in (4.27) for $t \to \infty$, $\Delta p(0) = \tau g_0/2$:

$$\Delta p(x) = \frac{\tau g_0}{2} e^{-x/L_p}, \qquad L_p = \sqrt{D_p \tau}. \tag{4.32}$$

This is also the $t \to \infty$ limit of (4.30). L_p is the **diffusion length**. Here $L_p = \sqrt{12 \times 10^{-5}}$ cm $= 1.1 \times 10^{-2}$ cm $= 110$ μm. The excess is 5×10^{13} cm^{-3} at the edge, 1.8×10^{13} at $x = L_p$, 3.4×10^{11} at $5L_p$, and falls to p_0 at $x = L_p \ln(5 \times 10^9) = 22\,L_p \approx 2.4$ mm. The electrons follow, $n = n_0 + \Delta p$, and change by at most 0.5% (Figure 4.12).

BE AWARE OF…

The edge value is $\tau g_0/2$, not τg_0. It would be tempting to take the plateau of Case A at the edge: that would treat the illuminated layer as a reservoir that diffusion does not deplete. Near its edge the layer does lose carriers to the dark side, and the symmetry argument (4.27) gives exactly half; Problem 4.19 of the exercise collection recovers it by solving (4.26) on both sides and matching Δp and its slope at $x = 0$. The plateau τg_0 is reached only a few L_p inside the layer. The exponential shape and the length L_p do not depend on this point; only the amplitude does.

What flows. At $x = 0$ the hole current density is $J_p = qD_p\Delta p(0)/L_p = 1.6{\times}10^{-19}{\times}12{\times}5{\times}10^{13}/1.1 \times 10^{-2} \approx 8.8$ mA/cm². Since $D_p\tau g_0/(2L_p) = g_0L_p/2$, this is $J_p(0) = q\,g_0L_p/2$: **the holes entering the dark bulk are those that the light generates in a slab of thickness** $L_p/2$. The electron current is $J_n = -J_p$, so the total current is zero. In steady state the flux of holes entering at $x = 0$ is exactly what is lost by recombination over the whole depth (Problem 4.17 of the exercise collection): a steady state in which $G \neq R$ locally, balanced by transport.

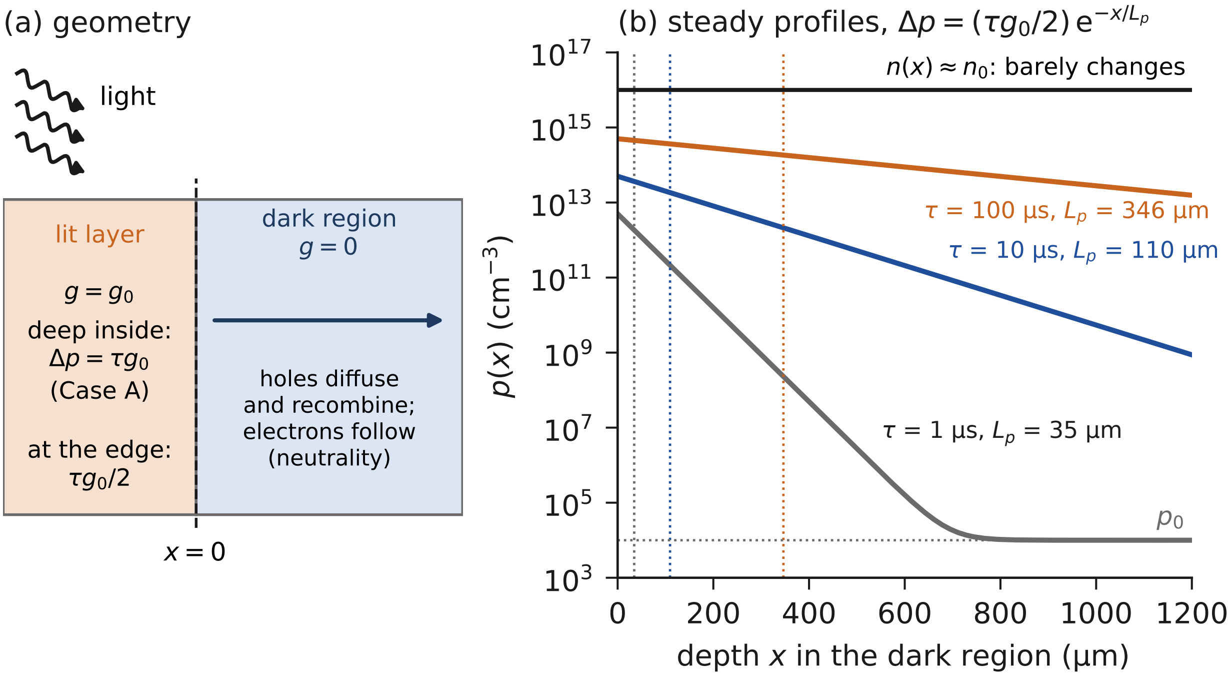


***Figure 4.12** – Case B in steady state. (a) The model: a uniform illuminated layer, at the Case A plateau τg_0 deep inside and at $\tau g_0/2$ at its edge, feeds the dark bulk. (b) Steady-state hole profiles (4.32) for three lifetimes (vertical dotted lines: $x = L_p$): the shorter the lifetime, the lower the edge value $\tau g_0/2$ and the shorter the diffusion length L_p. The majority electrons stay at n_0.*

4.6.3 Discussion: lifetime, diffusion length and the electric field

Why does a shorter lifetime give a shorter diffusion length? Three readings of the same fact.

- **The equation.** In (4.31) diffusion spreads the excess and recombination removes it at the rate $1/\tau$ per carrier. The profile falls over the length where the two compete on equal terms: $D_p/L^2 = 1/\tau$. The faster the carriers are removed, the steeper the profile must be for diffusion to keep feeding them, so the shorter L.

- **The carrier.** A hole survives on average a time τ and, during that time, random-walks over a typical distance $\sqrt{D_p\tau}$; the mean depth reached by the holes in (4.32) is exactly L_p. The transient of Figure 4.11 shows the same thing: the front advances as $\sqrt{D_p t}$ while $t < \tau$, then stops. A carrier cannot carry the excess farther than it lives.
- **The numbers.** With the same light, $g_0 = 10^{19}$ cm^{-3}s^{-1} (Table 4.5).

τ	$\Delta p(0) = \tau g_0/2$	$L_p = \sqrt{D_p\tau}$	$J_p(0) = qg_0L_p/2$
1 µs	5×10^{12} cm^{-3}	35 µm	2.8 mA/cm²
10 µs	5×10^{13} cm^{-3}	110 µm	8.8 mA/cm²
100 µs	5×10^{14} cm^{-3}; inside the layer $\tau g_0 = 10^{15}$, 10% of n_0: the edge of low-level injection, where the formula starts to fail (Problems 4.13 and 4.21 of the exercise collection)	346 µm	28 mA/cm²

Table 4.5 *– Case B: effect of the lifetime on the steady profile, for the same illumination.*

The lifetime acts twice: on the amplitude (proportional to τ) and on the reach (proportional to $\sqrt{\tau}$). It also sets the settling time: the steady state is reached in a few τ. A long lifetime is what a solar cell or a photodetector needs: carriers generated far from the collecting junction still reach it.

Should the electric field be added? It is already there. Section 4.5.3 showed that the faster electrons set up a small internal field (4.24), and that its effect is entirely contained in D_a. Its size can be checked directly. At $x = 0$, with $\partial\Delta p/\partial x = -\Delta p(0)/L_p$,

$$\mathcal{E} \approx \frac{(12-35)\times(-5\times10^{13}/1.1\times10^{-2})}{1350\times10^{16}} \approx 7.7 \text{ mV/cm},$$

pointing into the sample, so that it helps the holes forward and holds the electrons back. The ratio of the hole drift current to the hole diffusion current follows from (4.24) and the Einstein relation: $p\,(D_n - D_p)/(D_n n + D_p p) \approx (\Delta p/n_0)(1 - D_p/D_n) = 0.33\%$. At low level the field is a correction of the order of $\Delta p/n_0$; even at strong injection it only raises D_a from 12 to 18 cm²/s, which changes L by a factor $\sqrt{1.5} = 1.22$, whereas the lifetime varies by orders of magnitude from one sample to another. The picture is therefore governed by the lifetime, not by the internal field.

When does a field matter? When it is **imposed from outside**: an applied voltage, or the built-in field of a junction in the next chapters. It then adds a drift term $-\mu_p\mathcal{E}\,\partial\Delta p/\partial x$ to (4.26), and the holes are carried a distance of the order of $\mu_p\mathcal{E}\tau$ before recombining: 460 µm for $\mathcal{E} = 10$ V/cm, about four times L_p (Problem 4.18 of the exercise collection). Whether carriers diffuse or drift, the distance they cover is limited by their lifetime: $\sqrt{D\tau}$ for diffusion, $\mu\mathcal{E}\tau$ for drift.

BE AWARE OF…

One equation, three faces. Case A is (4.26) without space (uniform sample, no current): it gives the *time* τ and the *level* τg_0. The steady profile of Case B is (4.26) without time: it gives the *length* $L = \sqrt{D\tau}$. The transient of Section 4.6.1 keeps both, and shows how one turns into the other: a front as $\sqrt{D_p t}$ that stops at L_p after a few τ. The same lifetime appears everywhere, which is why measuring one (a decay time) predicts the other (a diffusion length).

WHAT PART B HAS ESTABLISHED

- The currents of Chapter 1 specified: $J_n = q\mu_n n\mathcal{E} + qD_n\,\partial n/\partial x$, $J_p = q\mu_p p\mathcal{E} - qD_p\,\partial p/\partial x$; continuity acquires the source term $G - R$.
- In a quasi-neutral region the internal field disappears into D_a; at low level in n-type material $D_a = D_p$ and the excess obeys (4.26).
- Under an illuminated surface the edge fills as $(\tau g_0/2)(1 - \mathrm{e}^{-t/\tau})$, the front advances as $\sqrt{D_p t}$ and stops near L_p; the steady state $\Delta p = (\tau g_0/2)\mathrm{e}^{-x/L_p}$ is reached in a few τ.

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Three situations**: thermal equilibrium (no excitation, $G_0 = R_0$), steady state (constant, possibly driven, $np \neq n_i^2$ under light), transient (changing). Steady state does not mean $G = R$ locally once transport is present.
- **Return to equilibrium**: the state reached after a disturbance is unique. From $dn/dt = G - R$ with $G = \alpha n_0 p_0$ and $R = \alpha(n_0 + \Delta n)(p_0 + \Delta n)$, dropping Δn^2: $\Delta n(t) = \Delta n(0)\mathrm{e}^{-t/\tau}$, $1/\tau = \alpha(n_0 + p_0)$, the lifetime of the **minority** carriers.
- **Light**: absorption is a threshold, $h\nu \geq E_g$ ($\lambda_g \approx 1240/E_g$ nm); every absorbed photon makes a pair, the excess energy becomes kinetic, then heat. Optical generation g_0.
- **Photoconductivity (Case A)**: $\Delta n(t) = \tau g_0(1 - \mathrm{e}^{-t/\tau})$, $\Delta\sigma = q(\mu_n + \mu_p)\tau g_0$; rise and decay with τ; larger but slower for a longer lifetime.
- **Mechanisms** (imported, beyond the model): thermal and optical generation; radiative, SRH and Auger recombination. Their net rates carry the factor $(np - n_i^2)$, a consistency check with mass action, not a proof of it; the rates $1/\tau$ add. Emission needs a direct gap: LEDs use GaAs, AlGaInP, InGaN, GaN, not Si.
- **Electrons and holes**: $J_n = q\mu_n n\mathcal{E} + qD_n\,\partial n/\partial x$, $J_p = q\mu_p p\mathcal{E} - qD_p\,\partial p/\partial x$; continuity = Chapter 1's equation plus $G - R$: $\partial p/\partial t = -\frac{1}{q}\partial J_p/\partial x + G - R$, $\partial n/\partial t = +\frac{1}{q}\partial J_n/\partial x + G - R$.
- **Excess carriers**: $\partial\Delta p/\partial t = D_a\,\partial^2\Delta p/\partial x^2 - \Delta p/\tau + g$; $D_a \approx D_p$ at low level in n-type material; the internal field is inside D_a.
- **Illuminated surface, transient (Case B)**: at the edge of the lit layer $\Delta p(0,t) = (\tau g_0/2)(1 - \mathrm{e}^{-t/\tau})$; in the bulk a front that advances as $\sqrt{D_p t}$ (diffusion), stopped near $L_p = \sqrt{D_p\tau}$ by recombination; nothing changes after a few τ.

- **Bulk in steady state (Case B)**: $\Delta p = (\tau g_0/2)\,\mathrm{e}^{-x/L_p}$, $J_p(0) = q g_0 L_p/2$; shorter lifetime: lower level, shorter reach.

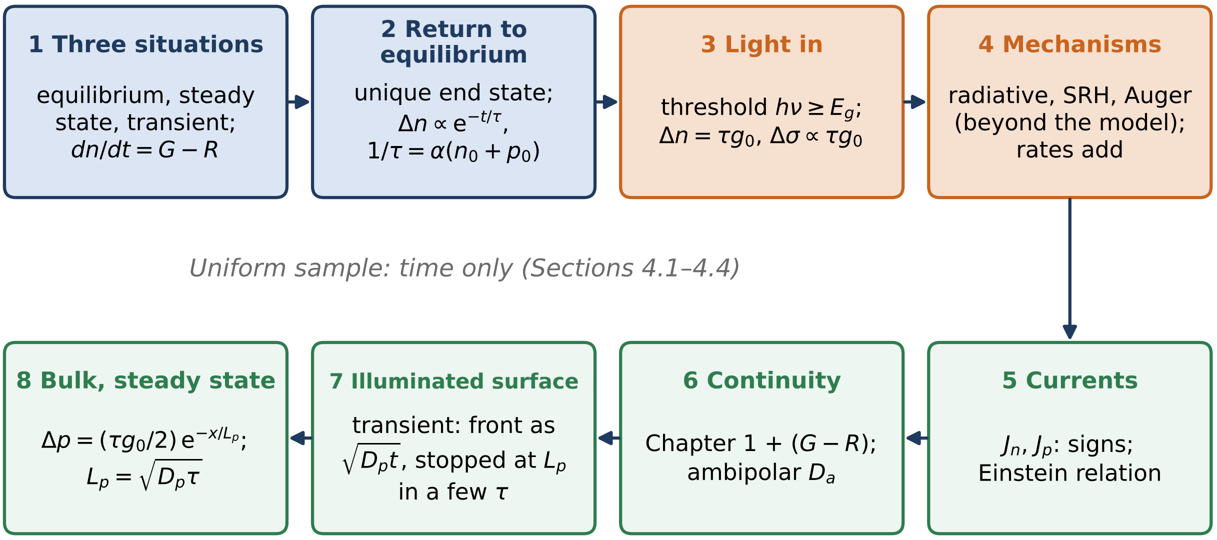

1 Three situations
equilibrium, steady state, transient; $dn/dt = G - R$
2 Return to equilibrium
unique end state; $\Delta n \propto \mathrm{e}^{-t/\tau}$, $1/\tau = \alpha(n_0 + p_0)$
3 Light in
threshold $h\nu \geq E_g$; $\Delta n = \tau g_0$, $\Delta\sigma \propto \tau g_0$
4 Mechanisms
radiative, SRH, Auger (beyond the model); rates add
Uniform sample: time only (Sections 4.1–4.4)
8 Bulk, steady state
$\Delta p = (\tau g_0/2)\,\mathrm{e}^{-x/L_p}$; $L_p = \sqrt{D_p \tau}$
7 Illuminated surface
transient: front as $\sqrt{D_p t}$, stopped at L_p in a few τ
6 Continuity
Chapter 1 + $(G - R)$; ambipolar D_a
5 Currents
J_n, J_p: signs; Einstein relation

Figure 4.13 – *The logic of the chapter. The first row (Part A) keeps the sample uniform and follows the concentrations in time: how they return to equilibrium, what light does, which physical processes recombine the carriers. The second row (Part B) lets the carriers move: the currents of Chapter 1 specified for electrons and holes, the continuity equation with its source term, then the illuminated surface, from its transient to its steady state in the bulk.*

Comprehension questions

No new physics here: answer by reading the figures and the equations of the chapter. The answers are at the end.

Q1. Is $np = n_i^2$ true in a sample under steady illumination? Justify with Figure 4.1 and the two-box picture of Chapter 2 (constant generation, recombination growing with np).

Q2. In n-type material an individual electron has a lifetime $1/(\alpha p_0)$, extremely long, yet the excess electrons decay with $\tau \approx 1/(\alpha n_0)$. Reconcile the two statements.

Q3. A pulse creates $\Delta n = 10^{14}\,\mathrm{cm}^{-3}$ in n-type silicon with $N_D = 10^{16}$. Is this low-level injection? By which factor do the electron and hole concentrations change?

Q4. In the argument of Section 4.2.1, why is the root $s = -(n_0 + p_0)$ rejected, and what does the argument not tell us about the return to equilibrium?

Q5. In Section 4.2.2, the term $\alpha n_0 p_0$ disappears from the balance. What does it represent physically, and why must it disappear?

Q6. A hydrogen atom absorbs only certain wavelengths; a silicon crystal absorbs all visible light. Explain the difference.

Q7. Why is a photoconductor with a longer lifetime more sensitive but slower?

Q8. Why are there no silicon LEDs, although silicon absorbs visible light very well?

Q9. Which mechanism limits the lifetime in silicon with $n_0 = 10^{16}\,\mathrm{cm}^{-3}$? In GaAs? What happens in silicon at $n_0 = 10^{19}\,\mathrm{cm}^{-3}$?

Q10. All the mechanism formulas contain the factor $(np - n_i^2)$. Why is this not a proof of the mass-action law?

Q11. Electrons move against the field $\mathcal{E}$. In which direction does their drift current flow? Do the drift currents of electrons and holes add or subtract? If both concentrations decrease with x, in which direction does each diffusion current flow?

Q12. In Case A, why can the currents be ignored, while in Case B they are essential?

Q13. Explain in one sentence, without equations, why a shorter lifetime gives a shorter diffusion length. Does the internal electric field change this conclusion?

Q14. In Case B, how far has the excess penetrated into the dark bulk after a time $t \ll \tau$? Why does the front stop advancing, and at what distance? Read the answer from Figure 4.11a.

Q15. Why is the excess at the edge of the lit layer, at every instant, half of that of Case A? Why, then, does the steady profile not start at τg_0?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 4, with their answers; the worked solutions are in the **Solutions** document. The 16 exercises that used to close this chapter are Problems 4.9 to 4.24 there; the other problems of the set (4.1 to 4.8) go further.

Answers

Q1. No. Illumination adds generation, the sample is out of equilibrium, and $np = (n_0 + \Delta n)(p_0 + \Delta n) > n_i^2$: in the two-box picture the working point sits above the crossing, recombination exceeds thermal generation and the excess is sustained by the light. The steady state is constant but is not equilibrium. Mass action holds only at thermal equilibrium.

Q2. The lifetime of one electron is long because holes are rare. But recombination removes an electron and a hole together, and the excess holes find electrons immediately (αn_0 per hole). The excess electrons disappear at the pace of the excess holes: the decay is set by the minority carrier.

Q3. $\Delta n / n_0 = 1\% \ll 1$: low level. n changes by 1%, p goes from 10^4 to $\approx 10^{14}$, a factor 10^{10}.

Q4. It gives $n_1 = n_0 + s = -p_0 < 0$ and $p_1 = -n_0 < 0$: concentrations cannot be negative. The argument fixes only the end state; it says nothing about whether it is reached or how fast, which needs the rates.

Q5. It is the recombination that already existed at equilibrium, exactly balanced by thermal generation $\alpha n_0 p_0$. Only the recombination caused by the excess, $\alpha(n_0 + p_0)\Delta n + \alpha \Delta n^2$, drives the evolution.

Q6. The atom has discrete levels: a photon is absorbed only if $h\nu$ equals a level difference (resonance). The crystal has a continuum of states above E_C: every photon with $h\nu \geq E_g$ (1.12 eV, i.e. $\lambda < 1.1$ µm, all the visible) creates a pair, the excess becoming kinetic energy (threshold).

Q7. The plateau $\Delta n = \tau g_0$, hence $\Delta\sigma$, is proportional to τ: more sensitive. But the rise and the decay both go as $e^{-t/\tau}$: the response takes a few τ, so it is slower.

Q8. Silicon has an indirect gap: an electron at the bottom of the conduction band and a hole at the top of the valence band have different k, and a photon cannot supply the momentum difference. Emission needs a phonon and is very improbable; the pair recombines through defects and the energy becomes heat. Absorption at higher photon energies does not suffer from this.

Q9. Silicon: SRH (about 50 µs against 10 ms radiative and 33 ms Auger). GaAs: radiative (about 1 µs). Silicon at 10^{19}: Auger, $\tau \approx 33$ ns (Problem 4.15 of the exercise collection).

Q10. Because the formulas are imported and were built to vanish at equilibrium, where $np = n_i^2$: the factor expresses their consistency with mass action, which they assume, rather than deriving it. Mass action itself comes from Chapters 2 (balance argument) and 3 (band statistics).

Q11. The electron drift current flows along $\mathcal{E}$ (negative charge moving against $\mathcal{E}$): drift currents of electrons and holes add. With both concentrations decreasing with x, both species diffuse towards $+x$; the hole current follows the flux (towards $+x$), the electron current opposes it (towards $-x$).

Q12. In the uniformly illuminated sample there is no gradient, so no diffusion current and no divergence; each point is a uniform sample. In the dark bulk there is no generation: the excess exists only because diffusion brings it, so the current divergence is the only source.

Q13. A carrier cannot carry the excess farther than it lives: during its lifetime τ it random-walks over about $\sqrt{D\tau}$. The internal field only changes D_p into D_a (at most ×1.5 in Si) and is negligible at low level; it does not change the conclusion.

Q14. Over a distance of the order of $\sqrt{D_p t}$: at short times (4.29) reduces to pure diffusion, $\Delta p \approx \Delta p_s \operatorname{erfc}(x/2\sqrt{D_p t})$, and the half-value point advances as $0.95\sqrt{D_p t}$ (curves $t/\tau = 0.05$ and 0.2). The front stops because the advancing holes recombine: it hardly goes beyond the distance covered in one lifetime, $\sqrt{D_p \tau} = L_p$. After a few τ the profile is e^{-x/L_p} and no longer changes (curves $t/\tau = 1$ and 3, almost on top of the dashed line).

Q15. The equation is linear: lighting $x < 0$ and lighting $x > 0$ are two problems that are mirror images of each other, and their sum is the uniform Case A. At $x = 0$ they give the same value, so each gives half of $\tau g_0(1 - e^{-t/\tau})$. The edge of the layer loses carriers to the dark side; only the depth of the layer, several L_p inside, reaches τg_0.

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- A. Einstein, "Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt", *Annalen der Physik* 17, 132–148 (1905) — the photoelectric effect.
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WHERE THIS LEADS

We now know how the concentrations of a semiconductor relax when disturbed, what light does to them, which physical processes recombine the carriers, and how electrons and holes move and are created or destroyed in space. Three observations of Chapter 2 are still open: rectification, light emission from a diode, and the photovoltaic effect. All three need what happens when the doping is not uniform, so that a Fermi level shared by two regions (Chapter 3) imposes fields and currents: the junction. The tools of this chapter, the lifetime, the continuity equations and the diffusion length, are the vocabulary in which that question will be asked.

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 5

The PN Junction at Equilibrium

The generic picture: two pieces, contact, diffusion and drift · the charged layer · the band picture: the Fermi level is constant, the barrier · carrier densities across the junction · the built-in voltage, two routes · electrostatics, width and peak field

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WHERE WE ARE

Chapter 2 gave the qualitative picture of the carriers of a semiconductor at equilibrium: pairs created and destroyed, the mass-action law $n_0 p_0 = n_i^2$, n- and p-doped materials with their majority and minority carriers. Chapter 3 founded these results on band theory and placed the Fermi level E_F on the band diagram. Chapter 4 disturbed the equilibrium: lifetime, electron and hole currents $J_n = q\mu_n n\mathcal{E} + qD_n\, dn/dx$ and $J_p = q\mu_p p\mathcal{E} - qD_p\, dp/dx$, the Einstein relation, the diffusion length $L = \sqrt{D\tau}$. Three observations of Chapter 2 are still open: **rectification**, **light emission** from a diode, and the **photovoltaic effect**. All three happen in one object, a crystal doped P on one side and N on the other: the **PN junction**. This chapter builds the junction **at equilibrium**. It follows the path used at every step of Chapters 5 to 8: first the **generic picture** of Chapters 2 and 4 (carriers, diffusion, drift, charges and fields), then the **band picture** of Chapter 3 (energies, the Fermi level), then the **carrier densities**, and only then the **equations**. Three linked figures carry this path through all the junction chapters: the junction cartoon, the band diagram and the density profile, drawn on one horizontal axis. Chapter 6 takes the junction out of equilibrium (bias, current, light); Chapter 7 treats its capacitances.

CHAPTER OUTLINE

5.1 What we want to explain
PART A — The generic picture
5.2 Two pieces apart
5.3 Contact: the depletion layer forms
PART B — The band picture
5.4 The band diagrams of the two pieces
5.5 At equilibrium, the Fermi level is constant
5.6 The junction in bands
PART C — Carrier densities
5.7 Carrier densities across the junction
PART D — The equations
5.8 The built-in voltage: two routes, one result
5.9 The electrostatics of the layer: width and peak field
5.10 Clarifications
Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Follow the **formation of the junction** in a thought experiment, first in the generic picture: two pieces apart, then in contact; diffusion, recombination near the interface, uncovered ions, a field that opposes diffusion, and equilibrium between drift and diffusion for each carrier.
- Show that **the Fermi level is constant** across any system at equilibrium, and use it to draw the **band diagram** of the junction.
- Give a **rough but useful picture** — a conductor, a charged insulator, a conductor — and read the same equilibrium as a **barrier** that only the Boltzmann tail of the majority carriers can climb.
- Read the **carrier densities** across the junction: majority and minority levels, and the Boltzmann factor $\mathrm{e}^{-qV_d/kT}$ that links the two sides.
- Derive the **built-in voltage** V_d by two routes, the cornerstone of the chapter: integrating the current equation $J_n = 0$ across the SCR, and aligning the Fermi levels. Then derive the **width** of the space-charge region and its **peak field** from Poisson's equation; Chapters 6 and 7 need both.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- draw the band diagrams of separate P and N pieces, and compute $E_F(N) - E_F(P)$;
- derive $J_n = \mu_n n\, dE_F/dx$ and $J_p = \mu_p p\, dE_F/dx$, and conclude that E_F is flat at equilibrium; say where a current flows;
- describe how the space-charge region forms and what stops it growing, in the generic picture and in the band picture;
- sketch ρ, $\mathcal{E}$, V and the bands across the junction, derive $N_A W_p = N_D W_n$ and $W = \sqrt{(2\varepsilon V_d/q)(1/N_A + 1/N_D)}$, and compute V_d, W, W_n, W_p and $\mathcal{E}_{max}$;
- derive $V_d = (kT/q)\ln(N_A N_D/n_i^2)$ from $J_n = 0$ and from $qV_d = E_F(N) - E_F(P)$, and write $n_{p0} = N_D\,\mathrm{e}^{-qV_d/kT}$ and $p_{n0} = N_A\,\mathrm{e}^{-qV_d/kT}$;
- explain why a voltmeter does not read V_d and why $qV_d < E_g$.

HOW THIS CHAPTER WORKS

- **Generic picture, bands, densities, equations, in this order.** Part A tells the story with carriers, charges and fields only, at a novice level. Part B tells it again with bands and the Fermi level. Part C reads the carrier densities. Part D writes the equations: it contains only what the pictures have prepared, and proves it.
- **Three linked figures.** The junction cartoon (electrons and fixed donor ions on the N side, holes and fixed acceptor ions on the P side, uncovered ions in the depletion region), the band diagram and the density profile are drawn on one horizontal axis (Figure 5.5). Chapters 6, 7 and 8 reuse them with the same colours: electrons blue and filled, holes orange and open, ions grey.

- **The rough picture first.** Whenever a picture or a calculation gives up precision for clarity, a box says so and names where the point is done properly.
- **Equilibrium only.** No voltage is applied and no current flows. Bias, the current–voltage law and light are the subject of Chapter 6; capacitances, of Chapter 7.
- **One running example**: silicon at 300 K, $N_A = 10^{18}\,\text{cm}^{-3}$ on the P side, $N_D = 10^{16}\,\text{cm}^{-3}$ on the N side. Every number in the chapter refers to it, and Chapters 6 and 7 use the same numbers.

Symbol	Meaning	Remark
N_A, N_D	acceptor (P side) and donor (N side) doping	10^{18} and $10^{16}\,\text{cm}^{-3}$ in the example
p_{p0}, n_{p0}	holes and electrons at equilibrium, neutral P side	$p_{p0} = N_A$, $n_{p0} = n_i^2/N_A$
n_{n0}, p_{n0}	electrons and holes at equilibrium, neutral N side	$n_{n0} = N_D$, $p_{n0} = n_i^2/N_D$
SCR	space-charge region (depletion layer)	from $x_p = -W_p$ to $x_n = W_n$; junction at $x = 0$
W, W_p, W_n	total width of the SCR and its parts on the P and N sides	$W = W_p + W_n$
ρ, $\mathcal{E}$, V	charge density, electric field, potential	N side grounded: $V = 0$ in neutral N
E_C, E_V, E_i, E_F	band edges, intrinsic level, Fermi level (Chapter 3)	$E_C(x)$ follows $-qV(x)$
ε	permittivity of silicon	$\varepsilon = \varepsilon_r\varepsilon_0$, $\varepsilon_r = 11.7$
V_d	built-in (diffusion) voltage	$V_N - V_P$ at equilibrium; 0.835 V in the example

Table 5.1 – *Notation of the chapter. Silicon at 300 K:* $kT = 25.9\,\text{meV}$, $kT/q = 25.9$ *mV,* $E_g = 1.12\,\text{eV}$, $n_i = 10^{10}\,\text{cm}^{-3}$ *(reference value),* $\mu_n = 1350$ *and* $\mu_p = 460$ *cm²/(V·s),* $D_n = 35$ *and* $D_p = 12$ *cm²/s,* $\varepsilon_0 = 8.854 \times 10^{-14}$ *F/cm.*

5.1 What we want to explain

Three observations from Chapter 2 are still open: **rectification**, **light emission** and the **photovoltaic effect**. The diode, a PN junction, lets the current pass in one direction only; semiconductor diodes also have photoelectric properties, used in light-emitting diodes, solar cells, photodetectors and laser diodes. The junction is also the building block of nearly every other semiconductor component: bipolar transistors, junction field-effect transistors, thyristors. These effects concern a device, not a material, and they need **non-uniform** doping. The simplest structure is the **abrupt PN junction**: a single crystal, doped P (N_A) for $x < 0$ and N (N_D) for $x > 0$, with two metal contacts. The plane $x = 0$ is the **metallurgical junction**. Before we can understand what a voltage or light does to it (Chapter 6), we must understand the junction left alone, at equilibrium: that is this chapter.

How a junction is made, and its symbol (Figure 5.1). A PN junction joins a P region and an N region of the **same** semiconductor: it is a **homojunction**. It cannot be made by pressing a P piece against an N piece: the surface of each piece is covered by a thin

native oxide, an insulator, which stops the current, and two surfaces never match atom for atom. The junction is made **inside the volume** of one crystal, away from the surfaces: donor atoms are brought into a P crystal (by diffusion or ion implantation, or an N layer is grown on it by epitaxy), so that part of it becomes N; metal contacts are then deposited on the two regions. In a circuit, the diode is drawn as a triangle against a bar: the P side is the **anode**, on the triangle side, and the N side the **cathode**, on the bar side. On the component, a **marking ring** is printed at the cathode end.

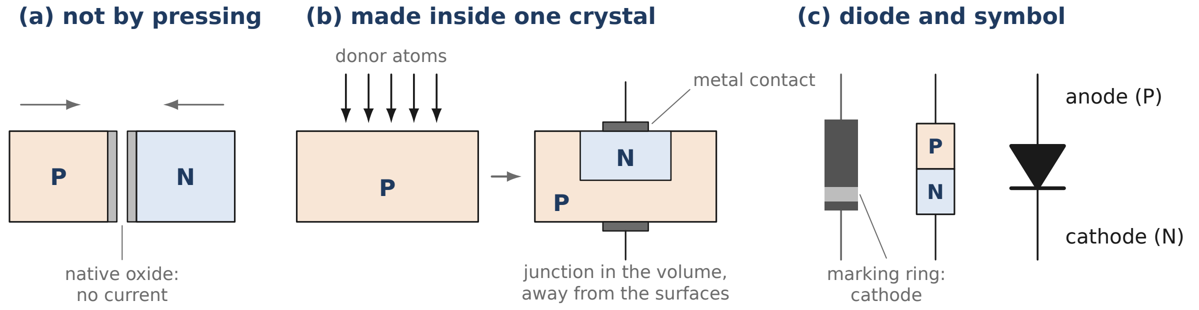

(a) not by pressing
(b) made inside one crystal
(c) diode and symbol
P
N
native oxide:
no current
donor atoms
P
metal contact
N
P
junction in the volume,
away from the surfaces
P
N
marking ring:
cathode
anode (P)
cathode (N)

Figure 5.1 – How a junction is made, and its symbol (schematic, not to scale). (a) Two pieces pressed together do not form a working junction: the native oxide on their surfaces blocks the current. (b) The junction is made inside one crystal: donor atoms are brought into a P crystal, which becomes N near its surface; metal contacts are added on the N region and on the P region; the junction lies in the volume, away from the surfaces. (c) A packaged diode, with its marking ring at the cathode end; its structure, P on the anode side and N on the cathode side; and its circuit symbol.

EXAMPLE 5.1 — THE REFERENCE JUNCTION OF THE COURSE

Silicon at 300 K, $N_A = 10^{18}\,\text{cm}^{-3}$ on the P side, $N_D = 10^{16}\,\text{cm}^{-3}$ on the N side, $n_i = 10^{10}\,\text{cm}^{-3}$, $\varepsilon_r = 11.7$. The P side is a hundred times more heavily doped than the N side: this is a so-called P^+N junction, the most common kind in diodes. The results that this chapter will establish:

$$V_d = 0.835\ \text{V}, \quad W_n = 0.33\ \mu\text{m}, \quad W_p = 3.3\ \text{nm}, \quad \mathcal{E}_{max} = 51\ \text{kV/cm}.$$

INPUT: A DOPING AT THE LIMIT

$N_A = 10^{18}\,\text{cm}^{-3}$ is just beyond the non-degeneracy limit (Chapter 3): E_F is only about $2kT$ above E_V. We keep the non-degenerate formulas: the error on V_d is a few hundredths of a volt.

PART A

The generic picture

Two pieces apart, contact, diffusion, drift and the charged layer

A THOUGHT EXPERIMENT

A junction is not made by sticking two pieces together (Section 5.1, Figure 5.1): it is made inside a single crystal. “Two pieces apart, then contact at $t = 0$” is a thought experiment. It does not describe fabrication, but it leads to the right final state and shows why that state is what it is. We follow it in four steps, drawn with the same bar in every figure: apart ($t < 0$), contact ($t = 0$), the transient ($t > 0$) and equilibrium ($t \to \infty$). Part A tells it with carriers, ions and fields only; Part B tells it again in bands.

5.2 Two pieces apart

THE QUESTION

What do the two pieces, P and N, contain before they touch?

The generic picture. Take the two pieces apart, each at equilibrium. Chapter 2 gave their content: complete ionisation, neutrality and mass action.

$$\text{P side:}\quad p_{p0} = N_A, \quad n_{p0} = \frac{n_i^2}{N_A}; \qquad \text{N side:}\quad n_{n0} = N_D, \quad p_{n0} = \frac{n_i^2}{N_D}. \tag{5.1}$$

For Example 5.1: $p_{p0} = 10^{18}$ and $n_{p0} = 10^2\,\text{cm}^{-3}$ on the P side; $n_{n0} = 10^{16}$ and $p_{n0} = 10^4\,\text{cm}^{-3}$ on the N side (Figure 5.2, top; the bottom of the figure is for Part B). The majority carriers are the holes on the P side and the electrons on the N side; the others are the minority carriers. Only the dopant ions are drawn in the cartoon, not the silicon atoms. The subscripts read as follows: the letter gives the side (p or n), the 0 means equilibrium. Each piece is **neutral at every point**: on the P side, each mobile hole balances the charge of a fixed acceptor ion N_A^-; on the N side, each mobile electron balances a fixed donor ion N_D^+. Nothing flows, and there is no field.

One thing stands out in these numbers: there are 10^{16} electrons per cm³ on the N side and 10^2 on the P side, a ratio of 10^{14}; there are 10^{18} holes on the P side and 10^4 on the N side, again a ratio of 10^{14}. This is no coincidence: both ratios equal $N_A N_D / n_i^2$. That number will set the voltage of the junction (Section 5.8). Something else stands out: **the N piece is full of electrons and the P piece almost empty of them; the P piece is full of holes and the N piece almost empty of them.** Put the two pieces in contact, and the carriers will not stay where they are.

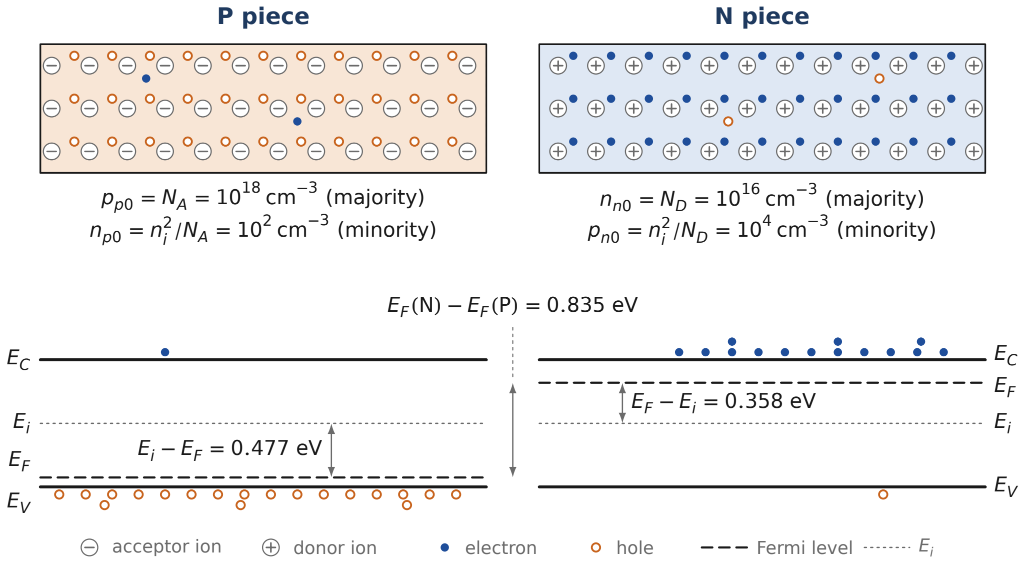

P piece
N piece
$p_{p0} = N_A = 10^{18}\ \text{cm}^{-3}$ (majority)
$n_{p0} = n_i^2/N_A = 10^2\ \text{cm}^{-3}$ (minority)
$n_{n0} = N_D = 10^{16}\ \text{cm}^{-3}$ (majority)
$p_{n0} = n_i^2/N_D = 10^4\ \text{cm}^{-3}$ (minority)
$E_F(\text{N}) - E_F(\text{P}) = 0.835$ eV
E_C
E_i
E_F
E_V
$E_i - E_F = 0.477$ eV
$E_F - E_i = 0.358$ eV
E_C
E_F
E_i
E_V
⊖ acceptor ion
⊕ donor ion
● electron
○ hole
Fermi level
E_i

Figure 5.2 *– The two pieces apart ($t < 0$): the cartoon above its band picture. Top: the P piece holds fixed acceptor ions (⊖) and as many mobile holes (open circles); the N piece holds fixed donor ions (⊕) and as many mobile electrons (filled dots); two minority carriers on each side are a reminder that they exist (the numbers of carriers drawn are not to scale; the densities of Example 5.1 are written below each piece). Bottom: the band diagrams of the same two pieces, energies to scale. Same material, so the same E_C, E_V and E_i (dotted, at mid-gap). The Fermi level (dashed) is near E_V in the P piece and near E_C in the N piece; the two differ by $E_F(\text{N}) - E_F(\text{P}) = kT\ln(N_A N_D/n_i^2) = 0.835$ eV.*

5.3 Contact: the depletion layer forms

THE QUESTION

Stick the two pieces together. What do the carriers do, and why does the motion stop?

Diffusion starts (Figure 5.3b). Figure 5.3 draws the whole thought experiment with one bar: (a) is the pair of pieces of Figure 5.2, brought together in (b). At contact, the interface separates a region rich in holes from a region with 10^{14} times fewer, and a region rich in electrons from a region almost empty of them. A concentration gradient produces a diffusion flux, from the dense region to the sparse one (Chapters 1 and 4). **Holes diffuse from P to N, electrons from N to P.** At the first instant there is no field yet: **diffusion is the only motion**.

The carriers that cross recombine (Figure 5.3c). A hole that arrives on the N side becomes a minority carrier among 10^{16} electrons per cm³; it soon finds a partner and recombines (Chapter 4: the minority-carrier lifetime). The same holds for an electron that reaches the P side. The carriers that leave are therefore not replaced, and recombination takes place near the interface, on the far side of it for each carrier: holes recombine in the N region, electrons in the P region.

Ions are uncovered. Each hole that leaves P leaves behind an acceptor ion N_A^- that is no longer balanced; each electron that leaves N leaves a donor ion N_D^+. Near the interface a **fixed negative charge on the P side** and a **fixed positive charge on the N side** appear. These charges create a field $\mathcal{E}$, pointing from positive to negative, that is, from N to P (towards negative x).

The field opposes diffusion: drift is born. This field pushes holes ($+q$) along it, towards P, and electrons ($-q$) the other way, towards N: exactly against their diffusion. It therefore produces a **drift** current that opposes the diffusion current, for each carrier. The more ions diffusion uncovers, the stronger the field, and the more carriers drift brings back.

Equilibrium (Figure 5.3d). The process stops when, for each carrier, drift exactly balances diffusion:

$$J_n = \underbrace{q\mu_n n\mathcal{E}}_{\text{drift}} + \underbrace{qD_n\frac{dn}{dx}}_{\text{diffusion}} = 0, \qquad J_p = \underbrace{q\mu_p p\mathcal{E}}_{\text{drift}} - \underbrace{qD_p\frac{dp}{dx}}_{\text{diffusion}} = 0. \tag{5.2}$$

Both equalities are needed, not just $J_n + J_p = 0$. If we had $J_n = -J_p \neq 0$, electrons would circulate for ever one way and holes the other: this would need a source that creates them on one side and a sink that destroys them on the other, which is not an equilibrium. Part B will show that these two equalities are exactly the statement that the Fermi level is flat (Section 5.5).

The region emptied of free carriers is called the **space-charge region** (SCR), or **depletion layer**. It runs from $x_p = -W_p$ on the P side to $x_n = W_n$ on the N side. The transient is extremely short: its order of magnitude is the dielectric relaxation time ε/σ, about 0.5 ps on the N side (an estimate, outside the model).

A COMMON MISTAKE

“The field opposes recombination”: no. The field opposes **diffusion**. Recombination is only the fate of the carriers that have crossed; it explains why the ions stay uncovered, but it is not what stops at equilibrium. What balances is the diffusion flux and the drift flux, for each carrier.

The rough picture: conductor, charged insulator, conductor. Far from the interface, each side has kept its carriers: it conducts, it is a conductor. The SCR has almost no free carriers left: it does not conduct, it is an insulator, but a **charged** one, $-Q$ on the P side and $+Q$ on the N side (Figure 5.3d, labels under the bar). Where is the field? In a conductor at equilibrium the field is zero, otherwise it would drive a current; the field is therefore **confined to the SCR**. The picture is that of a capacitor whose insulator itself carries the charges; Chapter 7 develops the capacitances, and this chapter only computes the width W of the layer and its peak field.

The field and the potential, qualitatively. The field points from the positive ions (N side) to the negative ions (P side): from N to P. A field points from high to low potential, so **the N side ends at a higher potential than the P side**. The rise of the potential from P to N is the **built-in voltage** V_d; Part D computes it ($V_d = 0.835$ V in the example), together with the width of the layer and its largest field.

WHAT THE PICTURE SIMPLIFIES

The picture assumes sharp edges and a completely empty SCR (the **depletion approximation**). In reality the carrier densities fall continuously across the layer (Figure 5.8, Section 5.7) and the edges are spread over a short distance. The approximation is excellent for the charge and the field, because the remaining carriers are very few compared with the ions in almost all of the layer.

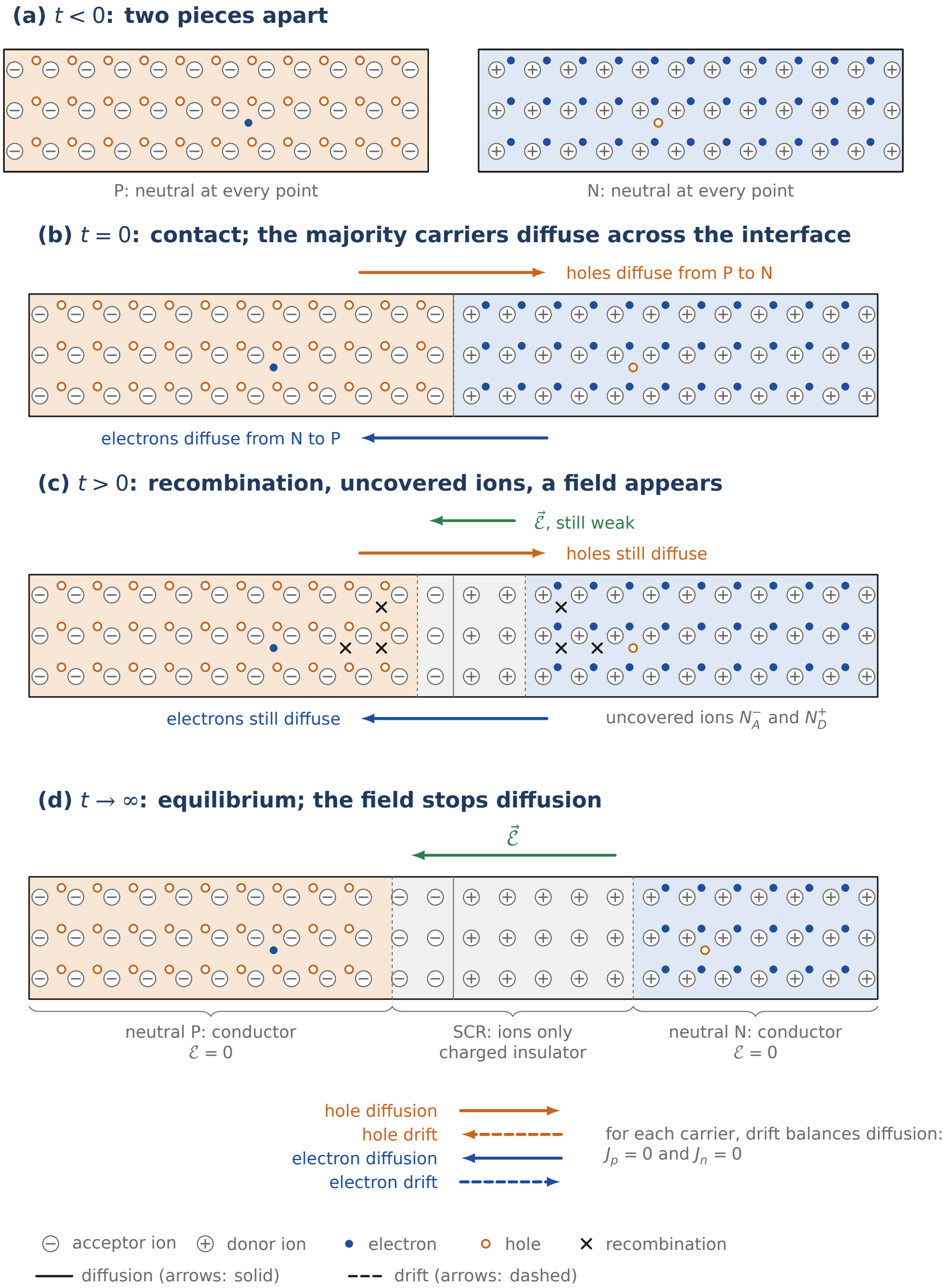

(a) $t < 0$: two pieces apart
P: neutral at every point
N: neutral at every point
(b) $t = 0$: contact; the majority carriers diffuse across the interface
holes diffuse from P to N
electrons diffuse from N to P
(c) $t > 0$: recombination, uncovered ions, a field appears
$\vec{\mathcal{E}}$, still weak
holes still diffuse
electrons still diffuse
uncovered ions N_A^- and N_D^+
(d) $t \to \infty$: equilibrium; the field stops diffusion
$\vec{\mathcal{E}}$
neutral P: conductor
$\mathcal{E} = 0$
SCR: ions only
charged insulator
neutral N: conductor
$\mathcal{E} = 0$
hole diffusion
hole drift
electron diffusion
electron drift
for each carrier, drift balances diffusion:
$J_p = 0$ and $J_n = 0$
acceptor ion
donor ion
electron
hole
recombination
diffusion (arrows: solid)
drift (arrows: dashed)

Figure 5.3 *– The thought experiment in the generic picture, one bar drawn four times with the same geometry (sketch: numbers of ions and carriers not to scale; the P part of the SCR is exaggerated, its real width in Example 5.1 is $W_p = 3.3$ nm against $W_n = 0.33$ μm). (a) $t < 0$: the two pieces apart, each neutral at every point (Figure 5.2). (b) $t = 0$: at contact, the majority carriers diffuse across the interface, holes from P to N and electrons from N to P. (c) $t > 0$: the carriers that have crossed recombine near the interface (×); the ions they leave behind stay uncovered (grey zone) and create a field $\vec{\mathcal{E}}$, still weak, pointing from N to P. (d) $t \to \infty$: equilibrium. The SCR contains only ions; its field makes the drift of each carrier (dashed arrows) balance its diffusion (solid arrows). Rough picture: the neutral regions are conductors with $\mathcal{E} = 0$, the SCR is a charged insulator, $-Q$ on the P side and $+Q$ on the N side.*

WHAT PART A HAS ESTABLISHED

- Apart, each piece is neutral at every point; the N piece is rich in electrons, the P piece rich in holes, and the density ratios $n_{n0}/n_{p0} = p_{p0}/p_{n0} = N_A N_D/n_i^2$ are huge.
- At contact, diffusion only: holes go from P to N, electrons from N to P, and recombine on the other side. The ions they leave uncovered form a charged layer and a field, from N to P, which drives a drift against diffusion. The field grows until drift balances diffusion, for each carrier: $J_n = J_p = 0$.
- Result: a charged depletion region (SCR) between two neutral sides; the field is confined to the SCR; the potential rises from P to N by V_d. Rough picture: conductor, charged insulator, conductor.

PART B

The band picture

The bands of the two pieces, the constant Fermi level, the junction in bands and the barrier

5.4 The band diagrams of the two pieces

THE QUESTION

What do the band diagrams of the two pieces look like before they touch?

What we need from Chapter 3.

- The densities follow from the position of the Fermi level with respect to the intrinsic level:
$$n = n_i\,\mathrm{e}^{(E_F - E_i)/kT}, \qquad p = n_i\,\mathrm{e}^{(E_i - E_F)/kT}, \qquad np = n_i^2, \tag{5.3}$$
or, equivalently, $n = N_C\,\mathrm{e}^{-(E_C - E_F)/kT}$ and $p = N_V\,\mathrm{e}^{-(E_F - E_V)/kT}$.
- The energies on a band diagram are those of an electron. An electron in a region at electrostatic potential V has a potential energy $-qV$: **all the levels E_C, E_V, E_i move together by $-qV(x)$.** A region at a higher potential has its bands lower.

The two diagrams (Figure 5.2, bottom). Both pieces are the same material, silicon: far from any field, they have the same E_C, the same E_V and the same E_i, which we draw at the same heights. Only the Fermi level differs. By (5.3):

$$\text{N piece:} \quad E_F - E_i = kT \ln \frac{N_D}{n_i} = 0.358\,\text{eV}; \qquad \text{P piece:} \quad E_i - E_F = kT \ln \frac{N_A}{n_i} = 0.477\,\text{eV}. \tag{5.4}$$

The Fermi level of the N piece is close to E_C; that of the P piece close to E_V. Their difference is

$$E_F(N) - E_F(P) = kT \ln \frac{N_D}{n_i} + kT \ln \frac{N_A}{n_i} = kT \ln \frac{N_A N_D}{n_i^2} = 0.835\,\text{eV}. \tag{5.5}$$

The same number $N_A N_D/n_i^2$ appears: the Fermi levels differ by kT times its logarithm.

What the difference means. The Fermi level measures how "full" the states are. With E_F higher on the N side, an electron at a given energy is more likely to be found on the N side than on the P side. If the pieces are allowed to exchange electrons, electrons will pass from N to P, and holes from P to N: this is the diffusion of Part A, read in bands. The next section makes this precise: it shows that the exchange stops only when the two Fermi levels are equal.

5.5 At equilibrium, the Fermi level is constant

> ***THE QUESTION***
>
> *What does the Fermi level have to do with the current? What must happen to the two different Fermi levels when the pieces touch?*

Recall: the two currents of Chapters 1 and 4. A carrier current has two components. The **drift** (or conduction) current is due to the field: $J_n = q\mu_n n\mathcal{E}$ and $J_p = q\mu_p p\mathcal{E}$, where the mobility μ measures how easily a carrier moves in a field. The **diffusion** current is due to a density gradient (Fick's law: the particle flux is $-D\,dn/dx$, from the dense region towards the sparse one): $J_n = +qD_n\,dn/dx$ for electrons, whose charge is $-q$, and $J_p = -qD_p\,dp/dx$ for holes. The two coefficients are linked by the **Einstein relation** $D/\mu = kT/q$, the same for electrons and holes.

The electron current written with E_F. Take a region in which the Fermi level may vary with x. This is possible as long as each small slice is close to equilibrium; the densities are then given by (5.3) with $E_F(x)$ and $E_i(x)$. Start from the total electron current,

$$J_n = q\mu_n n\mathcal{E} + qD_n\frac{dn}{dx}.$$

- **The field.** The bands follow $-qV(x)$, so $dE_i/dx = -q\,dV/dx = q\mathcal{E}$.
- **The gradient.** Differentiating $n = n_i\,\mathrm{e}^{(E_F-E_i)/kT}$ gives $\dfrac{dn}{dx} = \dfrac{n}{kT}\left(\dfrac{dE_F}{dx} - \dfrac{dE_i}{dx}\right) = \dfrac{n}{kT}\dfrac{dE_F}{dx} - \dfrac{qn\mathcal{E}}{kT}$.
- **Einstein.** With $D_n = \mu_n kT/q$, the diffusion term is $qD_n\,dn/dx = \mu_n n\,dE_F/dx - q\mu_n n\mathcal{E}$.

The drift term cancels against part of the diffusion term, and only one term is left:

$$\boxed{J_n = \mu_n\,n\,\frac{dE_F}{dx}} \qquad \text{and likewise} \qquad \boxed{J_p = \mu_p\,p\,\frac{dE_F}{dx}}. \tag{5.6}$$

The hole relation follows in the same way from $J_p = q\mu_p p\mathcal{E} - qD_p\,dp/dx$ and $p = n_i\,\mathrm{e}^{(E_i-E_F)/kT}$ (Problem 5.10 of the exercise collection). One may equally write $p = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$ and $q\mathcal{E} = dE_V/dx$: all the levels move together, so the result does not depend on which band edge is used.

Reading. Drift and diffusion are two halves of a single tendency: carriers move when the Fermi level has a slope, and the current is proportional to that slope, to the density and to the mobility. The bands may be tilted or bent; what matters is E_F.

TO REMEMBER

A current flows wherever the Fermi level has a gradient. Conversely, if no current flows, the Fermi level is flat.

An example with a slope: a bar and a battery (Figure 5.4a). Connect a uniform N bar to a battery. In steady state the bar stays neutral, $n = N_D$ everywhere, and there is no diffusion. The field is uniform and the bands tilt with the potential, lowest at the + terminal. Since $E_C - E_F$ is fixed by n, which is the same everywhere, E_F tilts parallel to the bands, and its slope gives the current: $J_n = \mu_n N_D \, dE_F/dx = \sigma\mathcal{E}$, which is Ohm's law. Strictly, the bar is not at equilibrium; E_F here is the local Fermi level of each slice, close to equilibrium. Chapter 6 will call it a quasi-Fermi level.

At equilibrium, E_F is flat. At thermal equilibrium no species flows: $J_n = 0$ and $J_p = 0$, separately (Section 5.3 said why both are needed). By (5.6), since n, p and the mobilities are positive,

$$\frac{dE_F}{dx} = 0 \qquad \text{everywhere.} \tag{5.7}$$

In any system at thermal equilibrium, however non-uniform, the Fermi level is the same at every point (Figure 5.4b). The bands may bend; E_F does not.

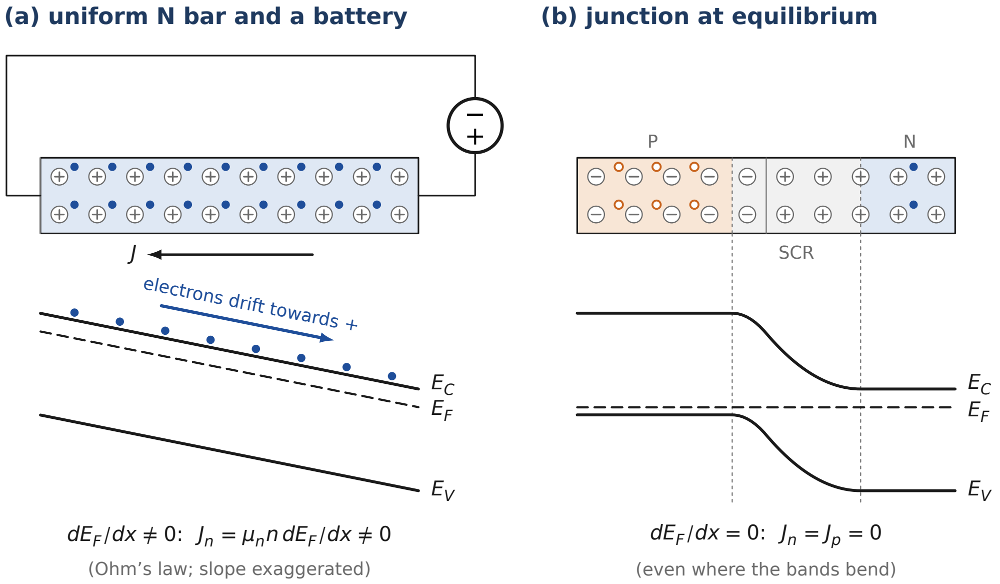

(a) uniform N bar and a battery
(b) junction at equilibrium
J
electrons drift towards +
E_C
E_F
E_V
$dE_F/dx \neq 0$: $J_n = \mu_n n \, dE_F/dx \neq 0$
(Ohm's law; slope exaggerated)
P
N
SCR
E_C
E_F
E_V
$dE_F/dx = 0$: $J_n = J_p = 0$
(even where the bands bend)

Figure 5.4 – *The Fermi level and the current. (a) A uniform N bar connected to a battery (+ terminal on the right): the bands are lowest at the + terminal and E_F tilts parallel to them; electrons drift towards +, the current J flows the other way, and the slope of E_F gives it, $J_n = \mu_n n \, dE_F/dx$ (Ohm's law; the slope is greatly exaggerated: about 0.5 meV over 1 mm in Problem 5.11 of the exercise collection). Strictly, E_F in (a) is the local Fermi level of each slice. (b) A junction at equilibrium, with nothing connected: the bands bend in the SCR, but E_F is flat, and no current flows. Energies in (b) to scale for Example 5.1; SCR width schematic.*

Consequence for the thought experiment. Before contact, the two Fermi levels differ by 0.835 eV (Figure 5.2). Once the pieces touch, they form one system, and at its new equilibrium a single, flat Fermi level must run from the P contact to the N contact. Something has to move: carriers must flow until the Fermi levels are aligned. The higher E_F of the N side tells us the direction: electrons go from N to P, holes from P to N. Part

A followed this flow in the generic picture (Section 5.3); Section 5.6 draws its result in bands.

WHAT THE PROOF ASSUMES

Relation (5.6) needs the densities to be given by (5.3) at each point, that is, each slice close to equilibrium, and the Einstein relation. Both hold at equilibrium, which is all we need here. There is also a general thermodynamic argument: two systems that can exchange electrons are in equilibrium only when their Fermi levels (their electrochemical potentials) are equal. The proof above is the same statement, written with the drift and diffusion currents of Chapter 4.

5.6 The junction in bands

THE QUESTION

What does the band diagram of the junction look like at equilibrium, and how does it show the balance between drift and diffusion?

At the moment of contact. At $t = 0$ nothing has moved yet: the bands of the two sides are still at the same height, and the two Fermi levels still differ by 0.835 eV across the interface. A Fermi level with a step is a Fermi level with a gradient: by Section 5.5, a current flows, electrons from N to P and holes from P to N, the diffusion of Figure 5.3b. It stops only when the Fermi level is flat.

Two rules fix the final diagram.

1. E_F **is flat** across the whole device (Section 5.5).
2. **Far from the junction, each side is as it was before contact**: neutral, with the same densities, so the same distance between E_F and the bands as in Figure 5.2.

To satisfy both, the bands of the P side must sit higher than those of the N side, by exactly the difference of the Fermi levels before contact. The bands **bend** in between, in the SCR, and only there (Figure 5.5b, under the cartoon of the same junction). The height of the step is written qV_d:

$$qV_d = E_{Cp} - E_{Cn} = E_{Vp} - E_{Vn} = E_F(N) - E_F(P)\big|_{\text{before contact}} = kT \ln \frac{N_A N_D}{n_i^2}. \quad (5.8)$$

Since the bands follow $-qV$, a step qV_d down from P to N is a potential higher on the N side by V_d: $V_N - V_P = V_d$. This is the **built-in voltage** that Part A announced; Section 5.8 finds it again from the current equation.

The field is where the bands bend. With $E_C = \text{const} - qV$, the field is $\mathcal{E} = -dV/dx = (1/q)\, dE_C/dx$. Where the bands are flat, in the neutral regions, $\mathcal{E} = 0$: these are the conductors of Figure 5.5a. Where they bend, in the SCR, there is a field: the charged insulator. The two pictures say the same thing. The slope of the bands is negative (E_C falls from P to N), so $\mathcal{E} < 0$: the field points towards negative x, from N to P, as found in Section 5.3.

Electrons and holes on the diagram. An electron tends to roll **down** the conduction band, towards lower energy; a hole tends to float **up** the valence band. In the SCR, the

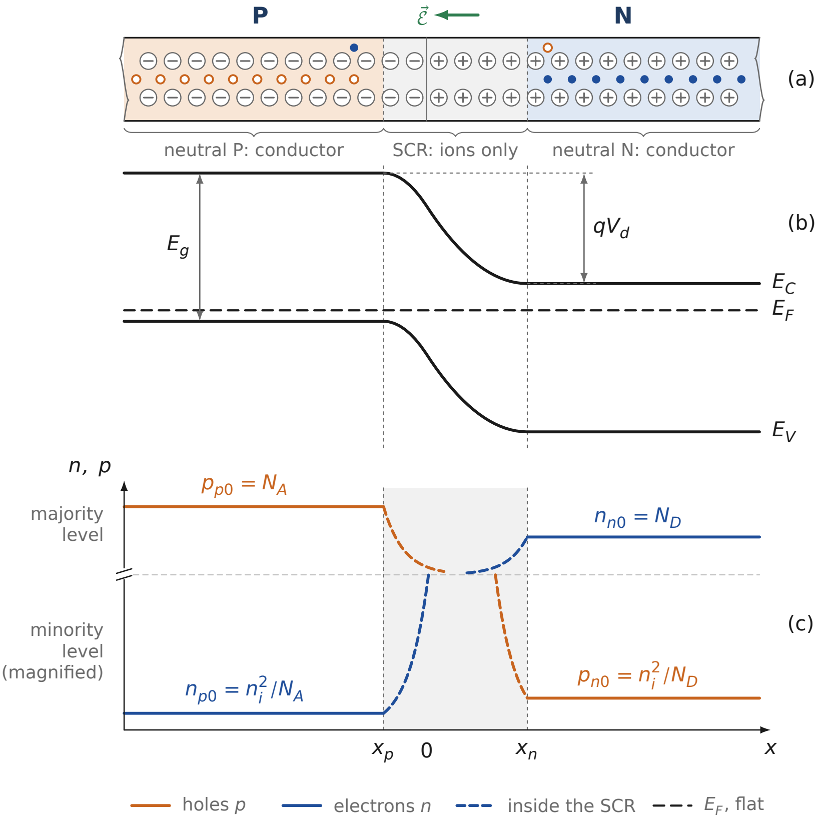


***Figure 5.5** – The three linked building blocks of the junction at equilibrium, on one horizontal axis. Chapters 6, 7 and 8 reuse them with the same colours and the same geometry. (a) The junction cartoon (Figure 5.3d): holes (open circles) and fixed acceptor ions (⊖) on the P side, electrons (filled dots) and fixed donor ions (⊕) on the N side; the SCR holds only uncovered ions; one minority carrier is drawn on each side; the field $\vec{\mathcal{E}}$ points from N to P. Only the region around the junction is framed (wavy ends): nothing happens at the device ends. (b) The band diagram, computed for the reference junction with energies to scale ($E_g = 1.12$ eV, $qV_d = 0.835$ eV): E_F (dashed) is flat; the bands bend only in the SCR, between x_p and x_n. (c) The density profile (linear, not to scale): a scale break separates the majority levels (top) from the minority levels (bottom, magnified), so that both read clearly. Solid: neutral regions; dashed: inside the SCR, where each density goes from one level to the other (Section 5.7). Widths schematic: the share of the SCR on the P side is exaggerated (in the example $W_p = 3.3$ nm, a hundredth of W_n); Figure 5.8 draws the junction to scale.*

step pushes electrons towards N and holes towards P: this is the drift of Section 5.3, read on the diagram. At equilibrium two currents coexist for each carrier and cancel: the **diffusion** current, carried by the majority carriers of the other side, and the **drift** (conduction) current, carried by the minority carriers that the field sweeps across. The barrier reading makes this precise.

The barrier reading of the balance (Figure 5.6). Look at the electrons.

- **Diffusion, read as climbing.** On the N side, electrons are plentiful (N_D), but to reach the P side they must climb the step qV_d. Their energies above E_C follow the Boltzmann tail of Chapter 3: the fraction with enough energy to climb a step qV_d is $e^{-qV_d/kT}$, which is 10^{-14} here. Only those cross: this is the diffusion flux, from N to P.
- **Drift, read as rolling down.** On the P side, the electrons are minority carriers, n_{p0}

per cm³. Every one of them that reaches the edge of the SCR rolls down the step to the N side, whatever its energy: this is the drift flux, from P to N.

- **The balance.** The two fluxes are equal when the densities that feed them are equal: $N_D \,\mathrm{e}^{-qV_d/kT} = n_{p0}$. With $n_{p0} = n_i^2/N_A$, this gives (5.8) once more.

For holes, the same reading holds with the diagram upside down: the many holes of the P side must “climb down” the valence band step, and only the fraction $\mathrm{e}^{-qV_d/kT}$ of them can; the minority holes of the N side float up freely. Balance: $N_A \,\mathrm{e}^{-qV_d/kT} = p_{n0}$.

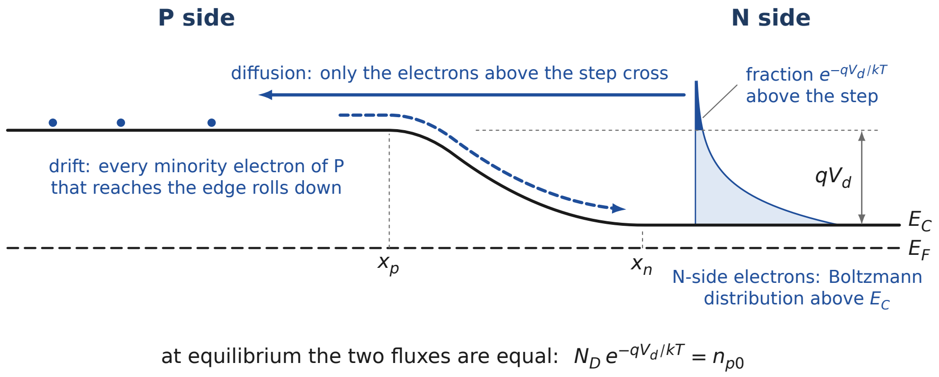

P side
N side
diffusion: only the electrons above the step cross
fraction $e^{-qV_d/kT}$ above the step
drift: every minority electron of P that reaches the edge rolls down
qV_d
E_C
E_F
x_p
x_n
N-side electrons: Boltzmann distribution above E_C
at equilibrium the two fluxes are equal: $N_D\, e^{-qV_d/kT} = n_{p0}$

Figure 5.6 – *The barrier reading of the equilibrium, for electrons: the conduction band of Figure 5.5b with the same x geometry and the step* qV_d *to scale. Right: the electrons of the N side are distributed in energy above* E_C *(Boltzmann, Chapter 3; the horizontal width of the shaded shape is the number of electrons at each energy, drawn with an enlarged* kT*); only the fraction* $\mathrm{e}^{-qV_d/kT}$ *above the top of the step (dark;* 10^{-14} *in the example, greatly magnified here) can climb it: this is the diffusion flux (solid arrow). Left: the minority electrons of the P side that reach the edge of the SCR roll down the step: this is the drift flux (dashed arrow). At equilibrium the two are equal. For holes the picture is the same, upside down, on the valence band.*

WHAT THE BARRIER READING SIMPLIFIES

The picture of a carrier “jumping” the whole step at once is a reading, not a calculation: in the SCR the carriers collide many times and climb by many small steps. The result does not depend on this, because at equilibrium the Boltzmann factor holds at every point (it is (5.3) with a flat E_F). The reading is useful because it shows at once what a voltage will do (Chapter 6): change the height of the step, and therefore the climbing flux, exponentially, while the rolling-down flux stays the same.

WHAT PART B HAS ESTABLISHED

- Apart, the two pieces have the same bands and different Fermi levels: $E_F(N) - E_F(P) = kT \ln(N_A N_D/n_i^2) = 0.835$ eV.
- $J_n = \mu_n n \, dE_F/dx$ and $J_p = \mu_p p \, dE_F/dx$: a current flows wherever E_F has a gradient; at equilibrium E_F is flat.
- A flat E_F forces the bands to bend in the SCR by $qV_d = E_F(N) - E_F(P)$; the field is where the bands bend. Barrier reading: only the Boltzmann fraction $\mathrm{e}^{-qV_d/kT}$ of the majority carriers climbs the step, and this flux equals the minority carriers

that roll down.

PART C

Carrier densities

The minority and majority levels across the junction

5.7 Carrier densities across the junction

THE QUESTION

How do n and p vary from one side of the junction to the other?

At the edges, from the bands. Use $n = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$ with the single, flat E_F. In the neutral P region, $E_C = E_{Cp}$; in the neutral N region, $E_C = E_{Cn}$ and $n = n_{n0} = N_D$. Then

$$\begin{aligned} n_{p0} &= N_C\,\mathrm{e}^{-(E_{Cp}-E_F)/kT} = \underbrace{N_C\,\mathrm{e}^{-(E_{Cn}-E_F)/kT}}_{n_{n0}=N_D}\,\mathrm{e}^{-(E_{Cp}-E_{Cn})/kT} \\ &\Longrightarrow \quad \boxed{n_{p0} = N_D\,\mathrm{e}^{-qV_d/kT}}. \end{aligned} \tag{5.9}$$

In the same way, with $p = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$ and $E_{Vp} - E_{Vn} = qV_d$,

$$\begin{aligned} p_{n0} &= N_V\,\mathrm{e}^{-(E_F-E_{Vn})/kT} = \underbrace{N_V\,\mathrm{e}^{-(E_F-E_{Vp})/kT}}_{p_{p0}=N_A}\,\mathrm{e}^{-(E_{Vp}-E_{Vn})/kT} \\ &\Longrightarrow \quad \boxed{p_{n0} = N_A\,\mathrm{e}^{-qV_d/kT}}. \end{aligned} \tag{5.10}$$

Part D finds them again from the current equation (5.12). Read on the band diagram, they say: **the minority density on one side is the majority density on the other side, reduced by the Boltzmann factor of the step**. In the example, $\mathrm{e}^{-qV_d/kT} = \mathrm{e}^{-32.2} = 10^{-14}$. Chapter 6 will use exactly these relations at the edges of the SCR, with a step changed by the applied voltage.

Across the layer: the density profile (Figure 5.5c). Between the two edges, the step of the bands rises progressively, and the distance between E_F and the band edges with it: each density changes by the Boltzmann factor of the local height of the step. Read from left to right, the hole density stays at N_A in neutral P, then falls inside the SCR, by fourteen orders of magnitude, to p_{n0} at x_n; the electron density rises the other way, from n_{p0} at x_p to N_D at x_n. On a linear axis no single scale can show 10^{18} and 10^2 together: the profile is therefore drawn with a **scale break**, majority levels above, minority levels below and magnified. Inside the SCR (dashed) each density leaves one level and reaches the other. Two features of the profile will be used in every later chapter.

- **Flat profiles in the neutral regions**: no gradient, so no diffusion; and no field, so no drift: no current, as equilibrium requires.

- **The minority level on each side is set by the majority level on the other side**, through the factor $\mathrm{e}^{-qV_d/kT}$ of the step. Chapter 6 will lower or raise the step: the minority levels at the edges will follow, and the profile will no longer be flat.

WHAT PART C HAS ESTABLISHED

- $n_{p0} = N_D\,\mathrm{e}^{-qV_d/kT}$ and $p_{n0} = N_A\,\mathrm{e}^{-qV_d/kT}$: the minority density on one side is the majority density on the other side, reduced by the Boltzmann factor of the step (10^{-14} here).
- The density profile is flat in the neutral regions and changes only across the SCR; drawn on a linear axis with a scale break, it shows the majority and the minority levels together.

PART D

The equations

The built-in voltage by two routes, the width and the peak field

5.8 The built-in voltage: two routes, one result

THE QUESTION

What internal voltage is needed for drift to balance diffusion exactly across the SCR? Does it agree with the step of the band diagram?

The convention. From now on the N side is the reference, **grounded**: $V = 0$ in the neutral N region. The neutral P region is then at $V_P = -V_d$ (Part A: the potential rises from P to N by V_d).

Route 1: integrate the current equation $J_n = 0$ (the generic picture). At equilibrium the electron current is zero at every point, in particular all across the SCR: drift exactly balances diffusion (Section 5.3). Start from this electron balance (5.2) and replace D_n by the Einstein relation, $D_n = \mu_n kT/q$, and the field by $\mathcal{E} = -dV/dx$:

$$q\mu_n n\left(-\frac{dV}{dx}\right) + q\,\frac{\mu_n kT}{q}\,\frac{dn}{dx} = 0 \qquad \Longrightarrow \qquad \frac{1}{n}\frac{dn}{dx} = \frac{q}{kT}\,\frac{dV}{dx}. \tag{5.11}$$

The mobility has dropped out: the result does not depend on how easily electrons move, only on the balance between the two tendencies. Integrate from x_p (P edge: $n = n_{p0}$, $V = -V_d$) to x_n (N edge: $n = n_{n0}$, $V = 0$):

$$\ln\frac{n_{n0}}{n_{p0}} = \frac{qV_d}{kT} \qquad \Longleftrightarrow \qquad n_{p0} = n_{n0}\,\mathrm{e}^{-qV_d/kT}. \tag{5.12}$$

With $n_{n0} = N_D$ and $n_{p0} = n_i^2/N_A$:

$$\boxed{V_d = \frac{kT}{q}\ln\frac{N_A N_D}{n_i^2}} \tag{5.13}$$

THE CURRENT EQUATION, NOT THE CONTINUITY EQUATION

Route 1 integrates the **current** equation $J_n = q\mu_n n\mathcal{E} + qD_n\, dn/dx$, set to zero. The continuity equation of Chapter 4 would not help here: at equilibrium it reduces to $0 = 0$ (nothing changes in time, generation equals recombination, and the current, being zero, has no divergence). The information is in the current itself: drift and diffusion are each enormous across the SCR, and they cancel exactly at every point.

Holes give the same result. The same calculation with $J_p = 0$ gives $(1/p)\,dp/dx = -(q/kT)\,dV/dx$, hence $p_{n0} = p_{p0}\,\mathrm{e}^{-qV_d/kT}$, and exactly (5.13). The two carriers balance separately in the same field, and they can because their two density ratios are equal (Section 5.2).

Route 2: aligning the Fermi levels (the band picture). Part B (Section 5.6) found the step of the bands equal to the difference of the Fermi levels before contact, (5.8):

$$qV_d = E_F(N) - E_F(P) = kT\ln\frac{N_D}{n_i} + kT\ln\frac{N_A}{n_i} = kT\ln\frac{N_A N_D}{n_i^2}.$$

The same route can be written with the densities of the neutral regions. Since E_C follows $-qV$, $E_{Cn} = \text{const} - qV_N$ and $E_{Cp} = \text{const} - qV_P$, so $E_{Cp} - E_{Cn} = q(V_N - V_P) = qV_d$. With the single, flat E_F, $n_{n0} = N_C\,\mathrm{e}^{-(E_{Cn}-E_F)/kT} = N_D$ on the N side and $n_{p0} = N_C\,\mathrm{e}^{-(E_{Cp}-E_F)/kT} = n_i^2/N_A$ on the P side; dividing the two, $\mathrm{e}^{qV_d/kT} = N_A N_D/n_i^2$, the same formula.

The same result. This is no coincidence: route 2 is route 1 written with E_F. Route 1 integrates $J_n = 0$; by (5.6), $J_n = 0$ is $dE_F/dx = 0$, and integrating "E_F flat" across the layer is exactly what aligning the Fermi levels does.

EXAMPLE 5.2 — THE BUILT-IN VOLTAGE OF THE REFERENCE JUNCTION

$V_d = 0.0259 \times \ln(10^{18} \times 10^{16}/10^{20}) = 0.0259 \times \ln(10^{14}) = 0.0259 \times 32.24 = 0.835$ V. By route 2: $0.358 + 0.477 = 0.835$ eV, the two shares of (5.4). Each decade of doping, on either side, adds $kT\ln 10/q = 59.6$ mV: with $N_A = 10^{17}$, $V_d = 0.775$ V.

CORNERSTONE — THE BUILT-IN VOLTAGE, TWO ROUTES

Route 1, current equation: $J_n = 0$ across the SCR $\Rightarrow \dfrac{1}{n}\dfrac{dn}{dx} = \dfrac{q}{kT}\dfrac{dV}{dx} \Rightarrow \dfrac{n_{n0}}{n_{p0}} = \mathrm{e}^{qV_d/kT}$.

Route 2, bands: E_F flat $\Rightarrow qV_d = E_F(N) - E_F(P)|_{\text{apart}} = kT\ln\dfrac{N_D}{n_i} + kT\ln\dfrac{N_A}{n_i}$.

Result: $V_d = \dfrac{kT}{q}\ln\dfrac{N_A N_D}{n_i^2} = 0.835$ V for the reference junction.

Inside the layer. Integrating (5.11) to any point x of the SCR instead of the edge gives

$$n(x) = n_{n0}\,\mathrm{e}^{qV(x)/kT}, \qquad p(x) = p_{p0}\,\mathrm{e}^{-q[V(x)+V_d]/kT}. \tag{5.14}$$

In band language: $E_C(x) - E_F$ grows by $-qV(x)$, and n falls by the Boltzmann factor. Their product is $n_{n0}p_{p0}\,\mathrm{e}^{-qV_d/kT} = n_i^2$ at every point: mass action holds everywhere, as it must at equilibrium. Once the potential $V(x)$ is known (Section 5.9), Figure 5.8 plots these profiles for the example, to scale and on a linear axis, under the band diagram drawn on the same x axis. Each majority density keeps its full value up to the edge of its neutral region and falls to practically nothing inside the SCR: the holes within a few tens of nanometres of x_p, the electrons over about 0.1 µm before x_n. This is what the depletion approximation assumes. The minority densities, 10^2 and $10^4\ \mathrm{cm}^{-3}$, are invisible at this scale: across 0.33 µm each density changes by fourteen orders of magnitude, and at every point the distance between E_F and a band edge sets the density by the Boltzmann factor. The two densities are equal, $n = p = n_i$, where $E_F = E_i$ on the band diagram, about 0.11 µm into the N side (Problem 5.13 of the exercise collection).

> **WHY** $qV_d < E_g$
>
> With $n = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$ and $p = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$, route 2 can also be written $qV_d = E_g-(E_C-E_F)_N-(E_F-E_V)_P$: the gap minus the two distances between E_F and the band edges far from the junction. As long as both sides are non-degenerate these distances are positive, and $qV_d < E_g$. With the course values of N_C and N_V this gives $1.12 - 0.206 - 0.061 = 0.853$ eV, 18 meV more than 0.835 eV: the difference only reflects the rounded reference value $n_i = 10^{10}\ \mathrm{cm}^{-3}$ (Chapter 3). We keep $V_d = 0.835$ V.

5.9 The electrostatics of the layer: width and peak field

> ***THE QUESTION***
>
> *How do the charge, the field and the potential vary across the junction, and how wide is the SCR?*

We keep the N side grounded: $V_N = 0$ in the neutral N region and $V_P = -V_d$ in the neutral P region, with $V_d = 0.835$ V from Section 5.8. Figure 5.7 stacks the profiles.

The charge: four regions (Figure 5.7). In the depletion approximation the junction falls into four regions, numbered from left to right:

- region 1, $x < x_p$: neutral P, $\rho = 0$;
- region 2, $x_p < x < 0$: the P side of the SCR, $\rho = -qN_A$ (uncovered acceptor ions);
- region 3, $0 < x < x_n$: the N side of the SCR, $\rho = +qN_D$ (uncovered donor ions);
- region 4, $x > x_n$: neutral N, $\rho = 0$.

The tools. The field is $\mathcal{E} = -dV/dx$, and Poisson's equation $d^2V/dx^2 = -\rho/\varepsilon$ reads $d\mathcal{E}/dx = \rho/\varepsilon$. In each region ρ is constant, so we integrate once for $\mathcal{E}$ and once more for V. The constants come from the neighbouring regions: the field and the potential are **continuous** at every boundary, since the charge densities are finite (there is no sheet of charge).

Regions 1 and 4: the neutral regions. There $\rho = 0$, so $\mathcal{E}$ is constant; it is zero, because these regions are conductors at equilibrium (a field would drive a current), and it stays zero up to the contacts. Then V is constant too:

$$\mathcal{E} = 0, \quad V = V_P \quad \text{(region 1)}; \qquad \mathcal{E} = 0, \quad V = V_N \quad \text{(region 4)}.$$

Region 2. Integrate $d\mathcal{E}/dx = -qN_A/\varepsilon$ from x_p, where the field is continuous with region 1, $\mathcal{E}(x_p) = 0$:

$$\mathcal{E}(x) = -\frac{qN_A}{\varepsilon}(x - x_p).$$

Integrate $dV/dx = -\mathcal{E}$ from x_p, where $V(x_p) = V_P$ (continuity with region 1); this gives the first line of (5.15).

Region 3. In the same way, integrate $d\mathcal{E}/dx = +qN_D/\varepsilon$ back from x_n, where $\mathcal{E}(x_n) = 0$ and $V(x_n) = V_N$ (continuity with region 4):

$$\mathcal{E}(x) = \frac{qN_D}{\varepsilon}(x - x_n),$$

and the second line of (5.15):

$$\begin{aligned} V(x) &= \frac{qN_A}{2\varepsilon}(x - x_p)^2 + V_P \qquad \text{region 2: } x_p < x < 0, \\ V(x) &= -\frac{qN_D}{2\varepsilon}(x - x_n)^2 + V_N \quad \text{region 3: } 0 < x < x_n. \end{aligned} \tag{5.15}$$

In both regions the field is negative: it points towards negative x, from N to P. The potential is made of two parabolic arcs. So far x_p and x_n are unknown: the two conditions that fix them come from the last boundary, $x = 0$, between regions 2 and 3.

- **Continuity of the field** at $x = 0$ (the electric displacement $\varepsilon\mathcal{E}$ is continuous, and ε is the same on both sides): $-\frac{qN_A}{\varepsilon}(0 - x_p) = \frac{qN_D}{\varepsilon}(0 - x_n)$, that is, $N_A x_p = -N_D x_n$, or, with $W_p = -x_p$ and $W_n = x_n$,

  $$N_A W_p = N_D W_n. \tag{5.16}$$

  This is also the **neutrality** of the junction as a whole: the negative charge equals the positive charge.

- **Continuity of the potential** at $x = 0$: $\frac{qN_A}{2\varepsilon}W_p^2 + V_P = -\frac{qN_D}{2\varepsilon}W_n^2 + V_N$, and with $V_N - V_P = V_d$,

  $$N_A W_p^2 + N_D W_n^2 = \frac{2\varepsilon}{q} V_d. \tag{5.17}$$

The widths. Solving (5.16) and (5.17) together:

$$\begin{gathered} W_n = \sqrt{\frac{2\varepsilon V_d}{qN_D}\frac{1}{1 + N_D/N_A}}, \qquad W_p = \sqrt{\frac{2\varepsilon V_d}{qN_A}\frac{1}{1 + N_A/N_D}}, \\ W = W_p + W_n = \sqrt{\frac{2\varepsilon V_d}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}. \end{gathered} \tag{5.18}$$

The layer extends mostly into the more lightly doped side. With $N_A = 100\,N_D$, $W_p = W_n/100$: the SCR of a P$^+$N junction lies almost entirely on the N side, and $W \simeq W_n \simeq \sqrt{2\varepsilon V_d/qN_D}$.

The field. It is a triangle, pointing towards negative x, with its largest magnitude at $x = 0$:

$$\mathcal{E}_{max} = \frac{qN_D W_n}{\varepsilon} = \frac{qN_A W_p}{\varepsilon}, \qquad V_d = \tfrac{1}{2}\,\mathcal{E}_{max}\,W. \tag{5.19}$$

The potential rises from $-V_d$ to 0 along the two arcs (5.15); the total rise is the area of the field triangle.

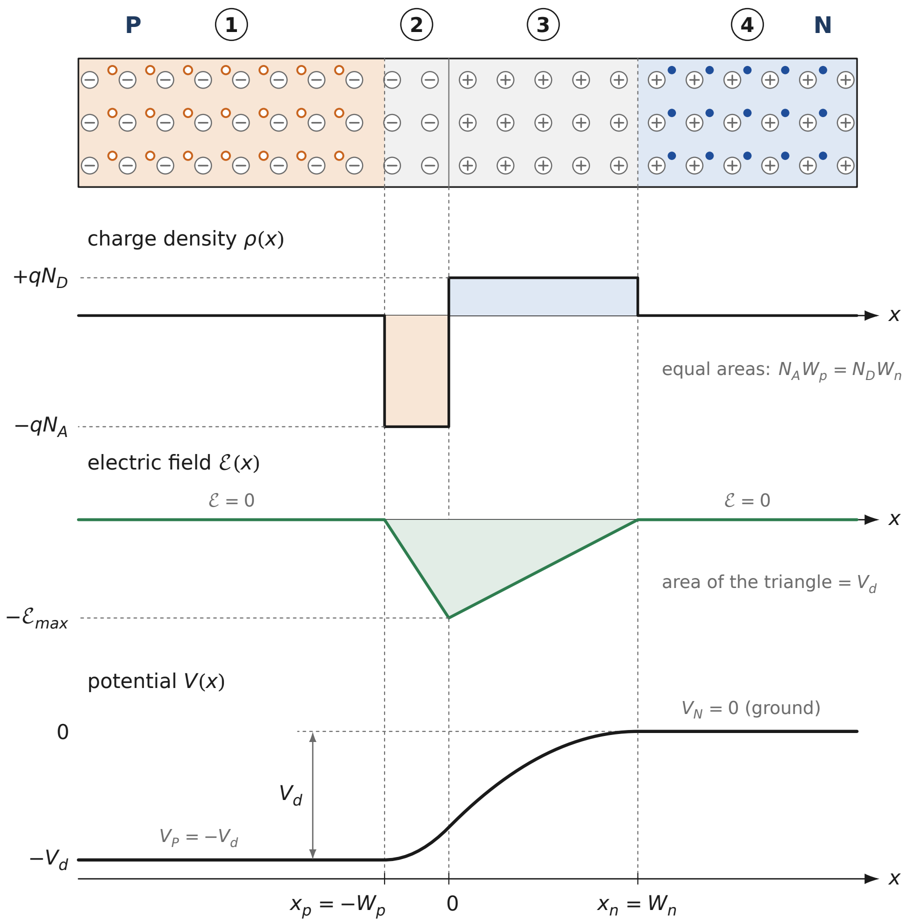


Figure 5.7 *– The electrostatics of the junction at equilibrium, N side grounded, depletion approximation. From top to bottom, on the same x scale: the junction with its four regions (1: neutral P; 2: P side of the SCR; 3: N side of the SCR; 4: neutral N), the charge density $\rho(x)$ (blocks $-qN_A$ and $+qN_D$ of equal areas), the field $\mathcal{E}(x)$ (zero in regions 1 and 4, linear in regions 2 and 3, largest in magnitude at $x = 0$; the shaded area is V_d) and the potential $V(x)$ (constant in regions 1 and 4, two parabolic arcs in regions 2 and 3, from $-V_d$ to 0). Drawn with $N_A \approx 3N_D$ so that region 2 is visible: in Example 5.1, $N_A = 100\,N_D$ and $W_p = 3.3$ nm is a hundred times smaller than W_n. The band edge $E_C(x)$ of Figure 5.5b is $-qV(x)$ shifted: the same curve upside down.*

EXAMPLE 5.3 — THE WIDTHS AND THE FIELD OF THE REFERENCE JUNCTION

$\varepsilon = 11.7 \times 8.854 \times 10^{-14} = 1.036 \times 10^{-12}$ F/cm and $V_d = 0.835$ V (Example 5.2). Then $W = \sqrt{2 \times 1.036 \times 10^{-12} \times 0.835 \times 1.01 \times 10^{-16}/1.602 \times 10^{-19}} = 3.30 \times 10^{-5}$ cm = 0.33 µm, $W_n = W \times 100/101 = 0.327$ µm and $W_p = W/101 = 3.3$ nm. Largest field: $\mathcal{E}_{max} = qN_D W_n/\varepsilon = 1.602 \times 10^{-19} \times 10^{16} \times 3.27 \times 10^{-5}/1.036 \times 10^{-12} = 5.1 \times 10^4$ V/cm, that is, 51 kV/cm. Check: $\frac{1}{2}\mathcal{E}_{max} W = \frac{1}{2} \times 5.06 \times 10^4 \times 3.30 \times 10^{-5} = 0.835$ V $= V_d$. Charge per unit area in each layer: $qN_D W_n = 5.2 \times 10^{-8}$ C/cm². Almost all the voltage drops on the N side: by (5.15), the share carried by each side is $qN_D W_n^2/2\varepsilon$ and $qN_A W_p^2/2\varepsilon$, in the ratio W_n/W_p: 0.827 V on the N side and 8 mV on the P side.

A field of 51 kV/cm is huge on the scale of devices: 5 V per micrometre. It is there with no battery connected, and yet no current flows: it is precisely the field needed to cancel diffusion (Section 5.8).

> **WHERE THESE RESULTS GO NEXT**
>
> Chapter 6 applies a voltage V: the voltage across the SCR becomes $V_d - V$, and every width in (5.18) and the peak field (5.19) follow with V_d replaced by $V_d - V$. Chapter 7 reads W as the thickness of a capacitor and develops the junction capacitances.

The junction to scale. With $V(x)$ known, the profiles (5.14) and the bands can be computed and drawn to scale (Figure 5.8). This is the check of the building blocks of Figure 5.5: the same shapes, but at the real scale the P part of the SCR disappears and the minority densities are invisible.

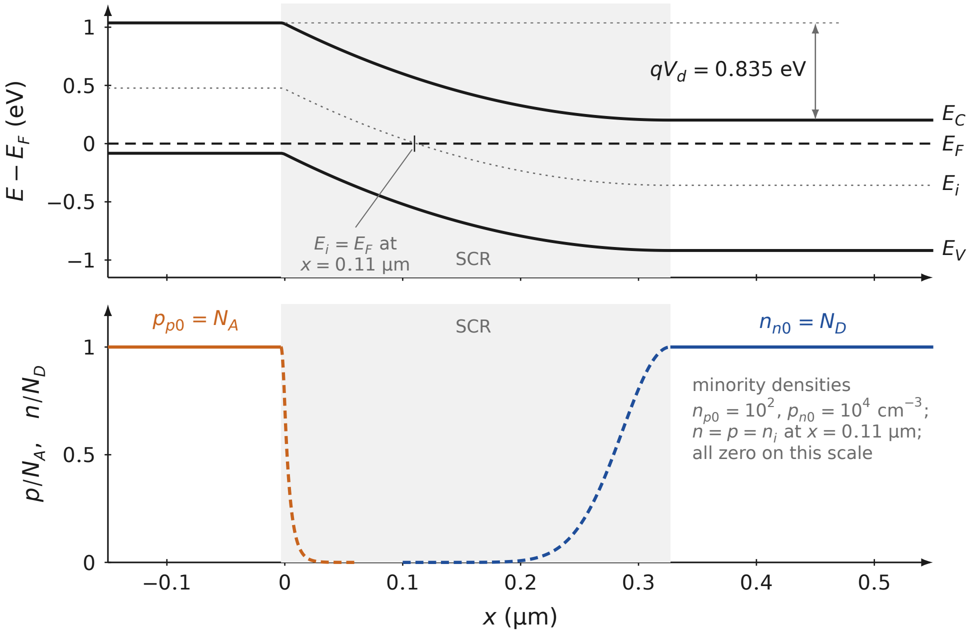


Figure 5.8 *– The reference junction at equilibrium, to scale, computed in the depletion approximation with (5.15) and (5.14) ($V_d = 0.835$ V, $x_p = -3.3$ nm, $x_n = 0.327$ µm, $kT = 25.9$ meV; SCR shaded). Top: the band diagram, energies measured from the flat E_F; E_i (dotted) is placed at mid-gap, so that $E_i - E_F = 0.477$ eV on the P side and $E_F - E_i = 0.358$ eV on the N side. At this scale the P part of the SCR is invisible: the bands bend almost entirely on the N side. Bottom: the majority densities on a linear scale, each in units of its value in its neutral region, p/N_A (orange) and n/N_D (blue); solid in the neutral regions, dashed inside the SCR. They fall to practically zero inside the layer, the holes within a few nanometres, the electrons over about 0.1 µm. The minority densities ($n_{p0} = 10^2$, $p_{n0} = 10^4$ cm^{-3}) and the point where $n = p = n_i$ ($x = 0.11$ µm, where $E_i = E_F$ above) are zero on this scale; everywhere, $np = n_i^2$.*

Where are the carriers really? Far inside the SCR, n and p are both tiny compared with the ions N_D and N_A: the layer is indeed “empty”, and the depletion approximation holds. Near the edges, over a few Debye lengths (about 40 nm on the N side), the majority density falls from its neutral value to almost zero: this is the softening of the edges mentioned in Section 5.3.

5.10 Clarifications

THE QUESTION

Three questions that students often ask about V_d: can it be measured with a voltmeter, can it exceed the gap, and is it the voltage at which a diode "turns on"?

A VOLTMETER DOES NOT READ V_d

Connect a voltmeter across an unbiased diode: it reads zero, not 0.835 V. A voltmeter measures the difference of the Fermi levels of its two leads, divided by q. At equilibrium the Fermi level is the same everywhere, in the semiconductor, in the metal contacts and in the leads (Section 5.5): the reading is zero. The voltage V_d is really there inside, as an electrostatic step, but the contacts between the metal and each side have their own steps, and the sum around the loop is zero. Otherwise a diode at rest would be a battery delivering current for ever, with no energy source.

BE AWARE OF…

- **V_d is not the "threshold voltage" of the diode.** A silicon diode "turns on" at about 0.5–0.7 V; V_d is 0.835 V here, and it depends on the doping. The practical threshold depends on what current one calls "significant"; Chapter 6 will compute it from the current–voltage law.
- **qV_d is smaller than E_g** as long as neither side is degenerate (Section 5.8).
- **The bands bend, E_F does not.** A bent band diagram does not mean a current; a sloping E_F does.

Step	Generic picture	Band picture
Apart ($t < 0$)	two neutral pieces; $n_{n0}/n_{p0} = p_{p0}/p_{n0} = N_A N_D/n_i^2$	same bands, $E_F(N) - E_F(P) = kT\ln(N_A N_D/n_i^2)$
What drives the exchange	density gradients: diffusion	different Fermi levels; $J = \mu n\, dE_F/dx$
Contact	diffusion, recombination, uncovered ions, field	electrons leave N, holes leave P, until the E_F are aligned
Equilibrium	drift = diffusion for each carrier: $J_n = J_p = 0$	E_F flat; bands bend by qV_d
Field	confined to the SCR (charged insulator between conductors)	where the bands bend, $\mathcal{E} = (1/q)\,dE_C/dx$
Balance	diffusion flux = drift flux	Boltzmann fraction climbing the step = minority carriers rolling down
Built-in voltage	$J_n = 0$ integrated: $V_d = (kT/q)\ln(N_A N_D/n_i^2)$	$qV_d = E_F(N) - E_F(P)$, same value
Densities	$n_{p0} = n_{n0}\mathrm{e}^{-qV_d/kT}$, $p_{n0} = p_{p0}\mathrm{e}^{-qV_d/kT}$	minority = majority of the other side times $\mathrm{e}^{-qV_d/kT}$

Table 5.2 *– Clarification: each step of the chapter in the two pictures.*

WHAT PART D HAS ESTABLISHED

- Cornerstone: $V_d = (kT/q)\ln(N_A N_D/n_i^2) = 0.835$ V by two routes, integrating the current equation $J_n = 0$ across the SCR, and $qV_d = E_F(N) - E_F(P)$. The two are one: $J_n = 0$ means E_F flat.
- Electrostatics, N side grounded: block charges, triangular field, potential from $-V_d$ to 0 along two parabolas. Continuity of the field gives $N_A W_p = N_D W_n$; continuity of the potential gives $N_A W_p^2 + N_D W_n^2 = 2\varepsilon V_d/q$; hence $W = \sqrt{(2\varepsilon V_d/q)(1/N_A + 1/N_D)}$ and $\mathcal{E}_{max} = qN_D W_n/\varepsilon$. Example: $W_n = 0.33$ µm, $W_p = 3.3$ nm, $\mathcal{E}_{max} = 51$ kV/cm.
- $np = n_i^2$ everywhere. A voltmeter reads zero; $qV_d < E_g$; V_d is not the threshold.

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Generic picture**: apart, two neutral pieces; at contact, majority carriers diffuse and recombine on the other side; the uncovered ions (N_A^- on the P side, N_D^+ on the N side) create a field, from N to P, that drives a drift against diffusion. Equilibrium: drift = diffusion for each carrier, $J_n = J_p = 0$. Rough picture: conductor, charged insulator (SCR), conductor; field confined to the SCR; potential higher on the N side by V_d.
- **Bands**: apart, same bands and $E_F(N) - E_F(P) = kT\ln(N_A N_D/n_i^2)$. $J_n = \mu_n n\, dE_F/dx$, $J_p = \mu_p p\, dE_F/dx$: at equilibrium E_F is flat everywhere, and the bands bend in the SCR only, by qV_d. Barrier reading: the Boltzmann fraction $\mathrm{e}^{-qV_d/kT}$ of the majority carriers climbs the step; it balances the minority carriers that roll down.
- **Densities**: $n_{p0} = N_D \mathrm{e}^{-qV_d/kT}$, $p_{n0} = N_A \mathrm{e}^{-qV_d/kT}$, $np = n_i^2$; flat in the neutral regions; majority and minority levels read together with a scale break.
- **Built-in voltage** (cornerstone): $V_d = (kT/q)\ln(N_A N_D/n_i^2) = 0.835$ V, by integrating the current equation $J_n = 0$ across the SCR or by aligning E_F. A voltmeter does not read it; $qV_d < E_g$.
- **Electrostatics** (N grounded): Poisson, $N_A W_p = N_D W_n$, $N_A W_p^2 + N_D W_n^2 = 2\varepsilon V_d/q$, $W = \sqrt{(2\varepsilon V_d/q)(1/N_A + 1/N_D)}$, $\mathcal{E}_{max} = qN_D W_n/\varepsilon$; the layer lies in the lightly doped side.

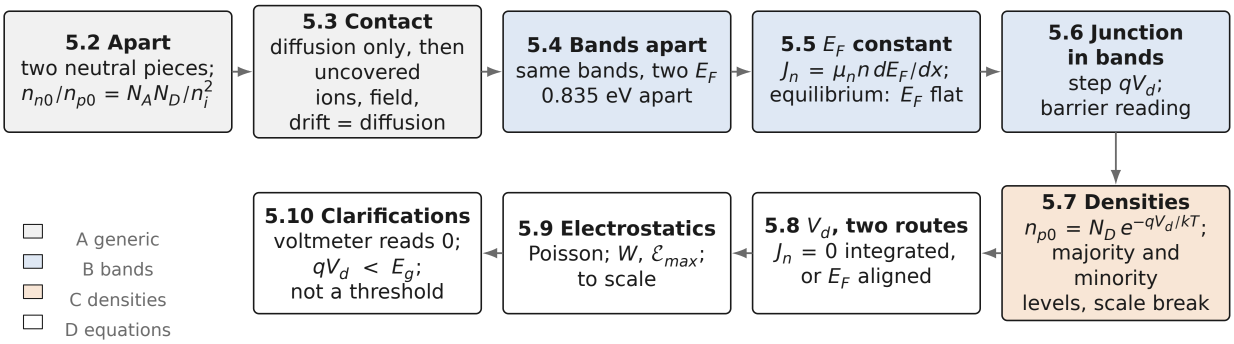

5.2 Apart
two neutral pieces; $n_{n0}/n_{p0} = N_A N_D / n_i^2$
5.3 Contact
diffusion only, then uncovered ions, field, drift = diffusion
5.4 Bands apart
same bands, two E_F 0.835 eV apart
5.5 E_F constant
$J_n = \mu_n n \, dE_F/dx$; equilibrium: E_F flat
5.6 Junction in bands
step qV_d; barrier reading
5.7 Densities
$n_{p0} = N_D \, e^{-qV_d/kT}$; majority and minority levels, scale break
5.8 V_d, two routes
$J_n = 0$ integrated, or E_F aligned
5.9 Electrostatics
Poisson; W, $\mathcal{E}_{max}$; to scale
5.10 Clarifications
voltmeter reads 0; $qV_d < E_g$; not a threshold
A generic
B bands
C densities
D equations

Figure 5.9 *– The logic of the chapter, section by section, in the order used in every junction chapter: the generic picture (A, grey), the band picture (B, blue), the carrier densities (C, orange) and the equations (D, white). The bottom row reads from right to left.*

Comprehension questions

No new physics here: answer from the figures and equations of the chapter. The answers are at the end.

Q1. Before contact, the two pieces have the same E_C and E_V but different E_F. Why the same bands? Which Fermi level is higher, and by how much?

Q2. State in words what $J_n = \mu_n n \, dE_F/dx$ means. Why is E_F flat at equilibrium even where the bands bend?

Q3. At contact, why do carriers diffuse across the interface, and what stops the motion? Why are $J_n = 0$ and $J_p = 0$ needed separately?

Q4. Correct: "at equilibrium, the field of the junction prevents recombination".

Q5. Why is the field confined to the SCR? Answer once with the conductor, charged insulator, conductor picture and once with the band diagram.

Q6. Why must the bands of the P side sit higher than those of the N side after contact? By how much?

Q7. In the barrier reading, which electrons cross from N to P, and which cross from P to N? Why are the two fluxes equal at equilibrium?

Q8. In the reference junction, why is the SCR almost entirely on the N side? Where does the voltage V_d drop?

Q9. Which two conditions at $x = 0$ give the two equations for W_p and W_n?

Q10. Explain why the two routes to V_d (zero current, alignment of E_F) must give the same result.

Q11. A voltmeter connected across a diode at rest reads zero, although $V_d = 0.835$ V. Explain with the Fermi level.

Q12. Why is $qV_d < E_g$ for non-degenerate doping? Is V_d the threshold voltage of the diode?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 5, with their answers; the worked solutions are in the **Solutions** document. The 8 exercises that used to close this chapter are Problems 5.9 to 5.16 there; the other problems of the set (5.1 to 5.8) go further.

Answers

Q1. Same material: far from any field the bands are those of silicon. $E_F(N)$ is higher, by $kT\ln(N_AN_D/n_i^2) = 0.835$ eV.

Q2. Carriers move when the Fermi level has a slope; the current is the slope times the density times the mobility. Drift and diffusion are two parts of this one term. At equilibrium no species flows, so the slope is zero; where the bands bend, the field and the density gradient are both present and cancel exactly.

Q3. The interface separates regions where each carrier has very different densities: diffusion goes from dense to sparse. The carriers that cross recombine and uncover ions; the field of these ions drives a drift opposite to diffusion, and the motion stops when they balance. $J_n = -J_p \neq 0$ would be a permanent flow of electrons one way and holes the other, which needs a source and a sink.

Q4. The field opposes **diffusion**: it makes each carrier drift opposite to its diffusion. Recombination is only the fate of the carriers that have crossed.

Q5. The neutral regions are full of free carriers: they are conductors, and a field there would drive a current, impossible at equilibrium. On the band diagram the bands are flat in the neutral regions and bend only in the SCR; the field is $(1/q)\,dE_C/dx$.

Q6. E_F must be flat, and far from the junction each side keeps its distance between E_F and the bands. The P side, whose E_F was lower by 0.835 eV, must have its bands raised by that amount: $qV_d = 0.835$ eV.

Q7. From N to P: the N-side electrons in the Boltzmann tail above the top of the step, a fraction $\mathrm{e}^{-qV_d/kT}$ of N_D (diffusion). From P to N: the minority electrons of the P side that reach the edge and roll down (drift). They are equal because $N_D\mathrm{e}^{-qV_d/kT} = n_{p0}$.

Q8. Neutrality $N_AW_p = N_DW_n$: on the N side the ions are a hundred times sparser, so the layer is a hundred times wider there. The voltage on each side is $qNW^2/2\varepsilon$: 0.827 V on the N side, 8 mV on the P side.

Q9. Continuity of the field ($N_AW_p = N_DW_n$) and continuity of the potential ($N_AW_p^2 + N_DW_n^2 = 2\varepsilon V_d/q$).

Q10. $J_n = 0$ is, by $J_n = \mu_n n\,dE_F/dx$, the statement that E_F is flat. Integrating it across the SCR is aligning the Fermi levels of the two sides.

Q11. A voltmeter measures the difference of the Fermi levels of its leads. At equilibrium E_F is the same everywhere, leads included: the reading is zero. The internal step V_d is compensated by the contact steps.

Q12. $qV_d = E_g - (E_C - E_F)_N - (E_F - E_V)_P$, and both distances are positive for non-degenerate doping. No: the threshold depends on the current chosen (Chapter 6).

References

- S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley, 2007 — Ch. 2: the PN junction at equilibrium (band diagram, depletion region).
- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — Ch. 7: the PN junction at equilibrium, built-in potential, electric field and space-charge width.
- R. F. Pierret, *Semiconductor Device Fundamentals*, Addison-Wesley, 1996 — Ch. 5: junction electrostatics, with a very careful qualitative presentation of the band diagram.

WHERE THIS LEADS

We now have the junction at equilibrium in two pictures: a charged insulator between two conductors, and a step qV_d in the bands under a flat Fermi level. **Chapter 6** applies a voltage: the step becomes $q(V_d - V)$, the climbing flux of the barrier reading changes by $e^{qV/kT}$ while the rolling-down flux does not, the Fermi level splits into two quasi-Fermi levels, and the current–voltage law of the diode follows from the minority carriers that diffuse in the neutral regions; light and the LED come at the end. **Chapter 7** takes the capacitor picture seriously: the charge stored in the junction changes with the voltage.

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 6

The PN Junction Out of Equilibrium

The generic picture: the barrier and the crossing fluxes · bands under bias and quasi-Fermi levels · carrier densities, forward and reverse · from the lamp to the junction · the Shockley law of the long diode · the short diode · the current inside the device · the characteristic · photovoltaic effect · LED · a short base, 1947

WHERE WE ARE

Chapter 5 built the PN junction at equilibrium, along one path: the generic picture, then the bands, then the densities, then the equations. In the generic picture, a charged insulator, the space-charge region (SCR), sits between two conductors, and its field makes drift balance diffusion for each carrier. In the band picture, the Fermi level is flat and the bands bend by a step qV_d, which only the Boltzmann tail $\mathrm{e}^{-qV_d/kT}$ of the majority carriers can climb. The densities are flat in the neutral regions, with $n_{p0} = N_D\,\mathrm{e}^{-qV_d/kT}$ and $p_{n0} = N_A\,\mathrm{e}^{-qV_d/kT}$ linking the two sides. The equations gave $V_d = 0.835$ V and $W = 0.33$ µm. Chapter 4 gave the tools for excess carriers: the lifetime τ, the diffusion length $L = \sqrt{D\tau}$, and the excess-carrier equation with its solutions.

This chapter **takes the junction out of equilibrium**, along the same path. A voltage is applied: what does it change in the generic picture, in the bands, in the densities, and what current results? The answer is the current–voltage law of the diode, derived from the minority carriers that diffuse in the neutral regions. Light comes at the end: the solar cell and the light-emitting diode.

CHAPTER OUTLINE

6.1 What changes when a voltage is applied
PART A — The generic picture
6.2 Where the voltage drops
6.3 The barrier and the four crossing fluxes
PART B — The band picture
6.4 The bands under bias
6.5 Low injection, quasi-equilibrium and quasi-Fermi levels
PART C — Carrier densities
6.6 The minority densities at the edges
6.7 Densities across the junction, forward and reverse
PART D — The current: the Shockley derivation
6.8 From the lamp to the junction
6.9 The neutral regions: minority carriers diffuse
6.10 The Shockley law of the long diode
6.11 Worked example: the short diode
6.12 The current inside the device
6.13 The characteristic and its limits

6.14 The bias scenario: everything together
PART E — Light, and a look back to 1947
6.15 The photovoltaic effect
6.16 The light-emitting diode
6.17 A short neutral region, and the transistor of 1947
Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Explain **rectification** in the generic picture: a voltage changes only the barrier of the SCR; the climbing diffusion flux of majority carriers is multiplied by $e^{qV/kT}$, the supply-limited drift flux of minority carriers is not.
- Draw the **bands under bias**: the step becomes $q(V_d - V)$, the gap is unchanged, the quasi-Fermi levels separate by qV.
- Obtain the **minority densities at the edges**, $n_{p0}e^{qV/kT}$ and $p_{n0}e^{qV/kT}$, and draw the density profiles for several forward and reverse voltages.
- Derive the **Shockley law** $J = J_0(e^{qV/kT} - 1)$ for the long diode, understand why two minority gradients make a current where the lamp of Chapter 4 does not, and treat the short diode as a variant.
- Explain the principle of the **solar cell** and of the **LED**.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- say how each of the four crossing fluxes changes in forward and reverse bias, and why; tell them apart from the huge local drift and diffusion currents inside the SCR;
- draw the band diagram for $V = 0$, $V > 0$ and $V < 0$, with the step $q(V_d - V)$ and the quasi-Fermi levels; compute $W(V)$;
- derive $n_p(x_p) = n_{p0}e^{qV/kT}$ and $p_n(x_n) = p_{n0}e^{qV/kT}$, and sketch the densities across the junction at several voltages;
- explain why the excess under a lamp carries no current while the excesses of a junction do;
- solve the minority-carrier profile in a long neutral region, derive $J = J_0(e^{qV/kT} - 1)$ and compute J_0; adapt the result to a short region;
- draw the electron and hole currents across the whole device, say who carries the current far from the junction, tell V_d apart from the practical threshold, and state the limits of the model;
- compute V_{oc} of a solar cell and say why $V_{oc} < V_d$; say where an LED emits and why not in silicon.

HOW THIS CHAPTER WORKS

- **The path of Chapter 5**: generic picture (Part A), bands (Part B), densities (Part C), equations (Part D). The three linked figures of Chapter 5, the cartoon, the band diagram and the density profile, are reused with the same colours and geometry, now for several voltages.
- **Equilibrium, forward, reverse, always together.** Every figure compares the equilibrium with forward and reverse biases, so that what moves is what the voltage changes.
- **The same running example** as Chapter 5, now as a **long diode**: both neutral regions are much longer than the diffusion lengths (Table 6.1). A short diode is treated as a worked example (Section 6.11); Chapter 8 builds on it.

Symbol	Meaning	Remark
V	applied voltage, source on P, N grounded	$V > 0$: forward; $V < 0$: reverse
x_p, x_n	edges of the SCR	$x_p = -W_p$, $x_n = W_n$
τ_n, τ_p	minority lifetimes (P side, N side)	0.1 µs and 10 µs
L_n, L_p	diffusion lengths, $L = \sqrt{D\tau}$	18.7 µm and 110 µm
d_p, d_n	widths of the neutral P and N regions	reference diode: $d \gg L$ (long); short example: $d_p = 2$, $d_n = 10$ µm
E_{Fn}, E_{Fp}	quasi-Fermi levels of electrons and holes	equal to E_F at equilibrium
J, J_0	current density; saturation current density	$J > 0$ from P to N inside the device; $J_0 = 2.1 \times 10^{-12}$ A/cm² (long)
J_{ph}, V_{oc}	photocurrent; open-circuit voltage	solar cell

Table 6.1 *– Notation added in this chapter; the others are those of Chapter 5. Reference junction: silicon at 300 K,* $N_A = 10^{18}$, $N_D = 10^{16}\,\mathrm{cm}^{-3}$, $n_i = 10^{10}\,\mathrm{cm}^{-3}$, $V_d = 0.835$ *V,* $kT/q = 25.9$ *mV,* $D_n = 35$ *and* $D_p = 12$ *cm²/s,* $\mu_n = 1350$ *and* $\mu_p = 460$ *cm²/(V·s).*

6.1 What changes when a voltage is applied

THE QUESTION

What do we keep from Chapter 5, and what is new?

At equilibrium, three statements held together: no species flows ($J_n = J_p = 0$), the Fermi level is flat, and the step of the bands is qV_d. A voltage breaks all three. A current now flows, so the Fermi level can no longer be flat everywhere (Chapter 5: a current flows wherever E_F has a gradient). The question of this chapter is therefore: **where** does the Fermi level change, **how much**, and what current results?

We keep from Chapter 5 the electrostatics (Poisson's equation and the widths), the relations between the densities on the two sides of a step, and the two pictures. Four facts will be used again and again; we recall them here.

- **Bands and potential.** The potential energy of an electron is $-qV(x)$, and the bands follow it. At equilibrium the N side is at the higher potential: $V_N - V_P = V_d$, so the bands of the P side lie higher: $E_{Cp} - E_{Cn} = qV_d$ and $E_{Vp} - E_{Vn} = qV_d$.

- **Densities from the bands** (Chapter 3): the electron density is $n = N_C\,\mathrm{e}^{-(E_C-E_F)/kT}$, the hole density is $p = N_V\,\mathrm{e}^{-(E_F-E_V)/kT}$, and at equilibrium $np = n_i^2$.
- **Across the step** (Chapter 5): the minority densities are $n_{p0} = N_D\,\mathrm{e}^{-qV_d/kT} = n_i^2/N_A$ on the P side and $p_{n0} = N_A\,\mathrm{e}^{-qV_d/kT} = n_i^2/N_D$ on the N side; for the reference junction $n_{p0} = 10^2$ and $p_{n0} = 10^4\,\mathrm{cm}^{-3}$.
- **Low injection** will mean that the majority densities are not affected by the bias: $n_n = N_D$ and $p_p = N_A$ in the neutral regions. Section 6.5 states it precisely, together with its companion, quasi-equilibrium.

We keep from Chapter 4 the behaviour of an excess of minority carriers in a neutral region: it diffuses and recombines, with the profile set by the excess-carrier equation.

PART A

The generic picture

Where the voltage drops; the barrier; the four fluxes that cross the SCR

6.2 Where the voltage drops

THE QUESTION

Connect a voltage source V to the P side, with the N side still grounded. Where does the applied voltage drop?

The rough picture. The device is a chain in series: contact, neutral P, SCR, neutral N, contact. The neutral regions are conductors and the SCR an insulator; we take the metal contacts as perfect (ohmic). A voltage applied to elements in series divides like their resistances: **it drops almost entirely across the SCR**, by far the most resistive element.

The voltage across the SCR, $V_N - V_P$, was V_d at equilibrium. With the source on P and N grounded, the neutral P region goes from $-V_d$ to $-V_d + V$, and

$$V_N - V_P = V_d - V. \tag{6.1}$$

The bias changes only the barrier. Everything in the neutral regions stays as it was; what changes is the height of the potential step that the SCR opposes to the majority carriers, $V_d - V$: lower in forward bias, higher in reverse bias. All the electrostatics of Chapter 5 still holds, with V_d replaced by $V_d - V$: the derivation of the widths used only Poisson's equation and the voltage across the layer, so

$$W(V) = \sqrt{\frac{2\varepsilon\,(V_d - V)}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}, \qquad W_n = \frac{N_A}{N_A + N_D}\,W, \qquad \mathcal{E}_{max} \propto \sqrt{V_d - V}. \tag{6.2}$$

In forward bias the SCR is thinner and its field weaker; in reverse bias it is wider and its field stronger (Figure 6.1, Table 6.2).

V (V)	−5	−4	−2	0	+0.3	+0.5
$V_d - V$ (V)	5.84	4.84	2.84	0.84	0.54	0.34
W (µm)	0.87	0.79	0.61	0.33	0.26	0.21
$\mathcal{E}_{max}$ (kV/cm)	134	122	93	51	40	32

Table 6.2 *– Width of the SCR and largest field of the reference junction under bias, from (6.2) ($W_n \simeq W$, $W_p = W/101$).*

WHAT THE PICTURE SIMPLIFIES

The SCR is not an ordinary resistor: its current depends exponentially on the voltage (Part B). But the argument holds as long as the current is modest: the voltage drop in the neutral regions and the contacts (the **series resistance**) is then negligible. At large forward currents it is not (Problem 6.14 of the exercise collection). Note also that a current now flows, so the neutral regions are not exactly at equilibrium; Section 6.5 says how close they stay.

6.3 The barrier and the four crossing fluxes

THE QUESTION

How does the voltage act on the diffusion and drift of electrons and holes across the SCR, and why is the effect so different in forward and in reverse bias?

Four particle fluxes cross the SCR: electron diffusion (N to P), electron drift (P to N), hole diffusion (P to N) and hole drift (N to P). At equilibrium they balance in pairs (Chapter 5). Figure 6.1 shows them under the cartoon of the same junction, at equilibrium, in forward and in reverse bias.

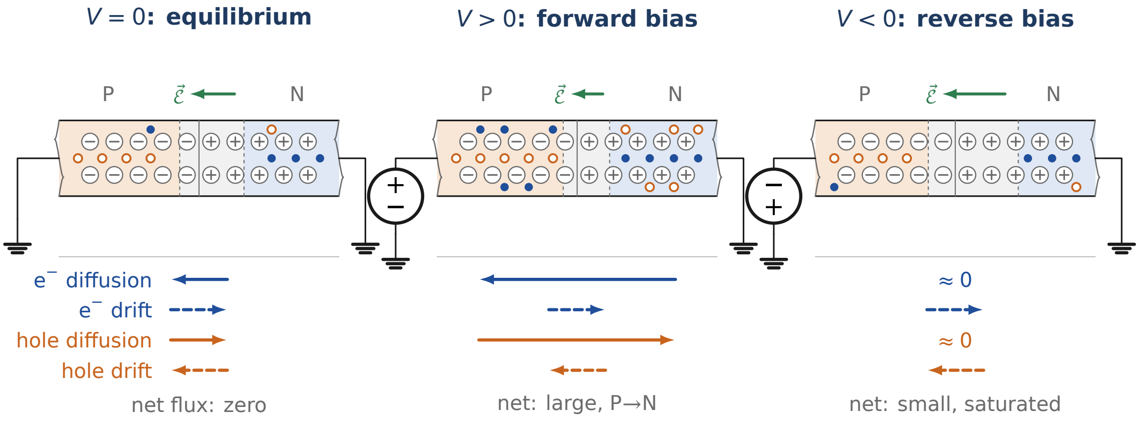


Figure 6.1 *– The generic picture under bias: the junction cartoon (as in Chapter 5) at equilibrium, in forward bias ($V = +0.4$ V) and in reverse bias ($V = -0.8$ V), source on P, N grounded. The SCR, which holds only ions, is thinner in forward bias and wider in reverse bias; the field arrow is longer when the field is stronger. Forward bias: injected minority carriers appear near the edges (electrons in P, holes in N). Reverse bias: the edges are emptied of minority carriers. Below, the four particle fluxes that cross the SCR (blue: electrons; orange: holes; solid: diffusion; dashed: drift; lengths not to scale): the two diffusion fluxes grow enormously in forward bias and vanish in reverse bias; the two drift fluxes do not change.*

The voltage changes the barrier, an activation energy. The majority carriers that diffuse across the SCR must climb the step of potential energy $q(V_d - V)$: it plays the part of an **activation energy**, and only the Boltzmann fraction $e^{-q(V_d - V)/kT} = e^{-qV_d/kT}\, e^{qV/kT}$ of them has the energy to climb it (Figure 6.2).

- **The climbing flux (majority diffusion)** is therefore **multiplied by** $e^{qV/kT}$ compared with equilibrium: by 10^5 at +0.3 V, by 10^{-5} at −0.3 V. In forward bias the majority carriers are countless on each side (10^{16} electrons per cm³ on the N side, 10^{18} holes on the P side): even a small extra share, taken from such a large reservoir, makes a **large net current**, from P to N. Electrons and holes move in opposite directions but have opposite charges, so both carry current the same way.
- **The rolling-down flux (minority drift) is unchanged.** Drift can only sweep across the carriers that present themselves at the edge of the layer: the minority carriers. A minority electron of the P side that reaches the edge is carried across in a time very short compared with its lifetime, whether the field is 40 or 130 kV/cm. The flux depends only on how many minority carriers **arrive at the edge**: those that thermal generation creates nearby and that diffusion brings there before they recombine (Chapter 4). It is a waterfall: the flow over it is set by the river that feeds it, not by the height of the fall. It is **supply-limited**: small, because minority carriers are rare, and independent of the voltage.

The net flux of electrons is therefore proportional to $e^{qV/kT} - 1$: zero at equilibrium, huge in forward bias, and tending to the constant rolling-down flux in reverse bias. This gives the **shape** of the diode law and its asymmetry, and rough orders of magnitude; it does not give the value of the constant J_0, which depends on how many minority carriers arrive at the edges. Part D computes it.

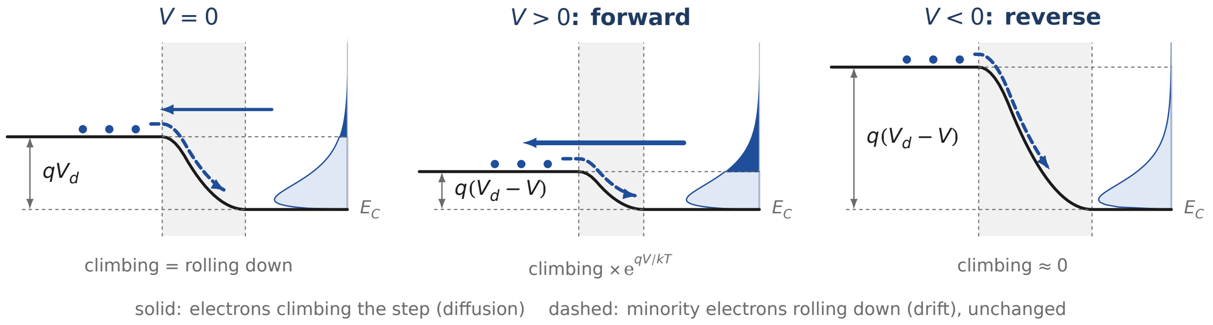


Figure 6.2 *– The barrier reading under bias, for electrons: conduction band only, with the steps of Figure 6.3 ($V = 0$, +0.4 V, −0.8 V; SCR tinted). At the right of each panel, the energy distribution of the electrons of the N side, $n(E) \propto \sqrt{E - E_C}\, e^{-(E-E_C)/kT}$; those above the top of the step (dark) can climb it. Not to scale: the spread of the distribution is drawn with kT about ten times too large, so that the dark fraction is visible at all; with the real kT it is $e^{-qV_d/kT} \approx 10^{-14}$ at equilibrium. In forward bias the step is lower and the climbing flux (solid arrow) grows as $e^{qV/kT}$; in reverse bias no part of the distribution reaches the top. The minority electrons of the P side (dots) roll down the step whatever its height (dashed arrow): that flux is unchanged.*

The asymmetry is rectification. A positive voltage opens the diffusion valve; a negative voltage closes it; drift always flows at the same small rate.

CROSSING FLUXES ARE NOT THE LOCAL CURRENTS OF THE SCR

The four arrows of Figure 6.1 are the fluxes that **cross** the whole SCR. Do not confuse them with the **local** drift and diffusion currents at a point inside the SCR. Those are enormous: at equilibrium, in the N part of the reference SCR, where $n \approx 4\times10^{14}\ \text{cm}^{-3}$ and $\mathcal{E} \approx 16$ kV/cm, the electron drift current density is $q\mu_n n\mathcal{E} \approx 1.4$ kA/cm², and the diffusion current is the same in the opposite direction. They nearly cancel at every point; under bias they cancel slightly less, and their small difference is the current of the diode (Section 6.5). The crossing fluxes count carriers that go all the way across; the local currents describe the ceaseless back-and-forth inside the layer.

WHAT PART A HAS ESTABLISHED

- The applied voltage drops almost entirely across the SCR: V_d becomes $V_d - V$; only the barrier changes. $W \propto \sqrt{V_d - V}$ (0.33 µm at equilibrium, 0.87 µm at −5 V).
- The barrier is an activation energy: the climbing diffusion flux of majority carriers is multiplied by $\mathrm{e}^{qV/kT}$; the supply-limited drift flux of minority carriers is unchanged. Net flux $\propto \mathrm{e}^{qV/kT} - 1$: rectification. This gives the shape of the law, not J_0.
- The crossing fluxes are small differences between local drift and diffusion currents of the order of kA/cm².

PART B

The band picture

The step $q(V_d - V)$, the unchanged gap, quasi-equilibrium and the quasi-Fermi levels

6.4 The bands under bias

THE QUESTION

What happens to the band diagram when a voltage is applied?

The guide (Figure 6.3, top). The bands follow $-qV(x)$. The neutral N region is grounded: its bands stay where they are, and it is the reference. The neutral P region follows the source: its bands move by $-qV$, down in forward bias, up in reverse bias. In between, the SCR takes the whole voltage.

- **The step** between the two sides becomes $q(V_d - V)$: lower in forward bias, higher in reverse bias. This is the barrier of Part A, now drawn.
- **The gap is unchanged.** E_C and E_V move together at every point: $E_C - E_V = E_g$ all along the device, across the SCR too. The voltage bends the bands more or less; it never squeezes them.
- **The Fermi levels of the neutral regions** are no longer equal. Each neutral region is still close to equilibrium, with its own Fermi level at the same distance from its bands as before; the P one has moved with its bands: $E_F(N) - E_F(P) = qV$. This is what a voltmeter connected to the two contacts reads (Chapter 5): the applied voltage is the difference of the Fermi levels of the contacts, divided by q.

Figure 6.3 draws the three band diagrams, computed for the reference junction and to scale in energy, at $V = 0$, $+0.4$ V and -0.8 V. What the dashed lines in the biased diagrams are, the quasi-Fermi levels, is the subject of the next section.

6.5 Low injection and quasi-equilibrium in the SCR

THE QUESTION

A current flows, so the Fermi level must have a gradient somewhere. Where? And can we still use the equilibrium relations inside the SCR?

Two hypotheses make the problem solvable. They are those of the classical theory of the diode.

1. **Low injection.** The injected minority carriers remain far fewer than the majority carriers: the majority densities keep their equilibrium values, $p_p = N_A$ on the P side and $n_n = N_D$ on the N side, right up to the edges of the SCR.
2. **Quasi-equilibrium in the SCR.** Inside the SCR, the drift and diffusion terms of each current remain much larger than their difference, the net current. So, for each carrier,

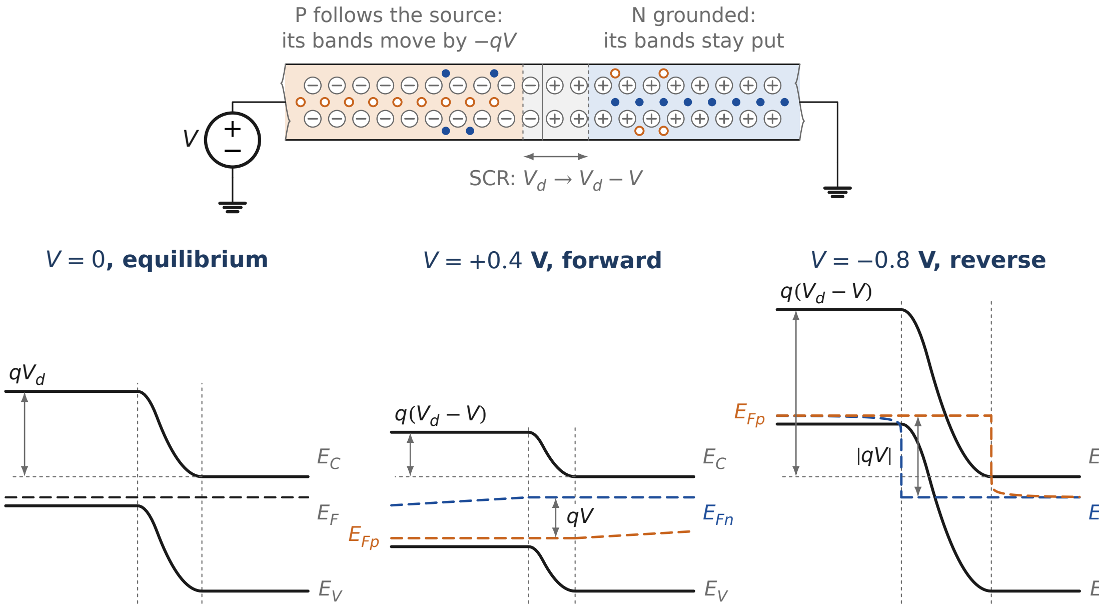


Figure 6.3 *– The bands under bias. Top, the guide: the junction cartoon with its circuit (source on P, N grounded). The neutral N region is the reference; the neutral P region follows the source and its bands move by $-qV$; the SCR takes the voltage, $V_d \to V_d - V$. Bottom: the band diagrams of the reference junction, computed, energies to scale (1 eV = 1 cm), N side as reference, at $V = 0$, +0.4 V and −0.8 V. The step is $q(V_d - V)$: 0.835, 0.435 and 1.635 eV; the gap $E_C - E_V = 1.12$ eV is the same everywhere. Dashed: at $V = 0$ the single flat E_F; under bias the quasi-Fermi levels E_{Fn} (blue) and E_{Fp} (orange), flat across the SCR and separated there by qV (Section 6.5). In the neutral regions of this long diode, the minority quasi-Fermi level returns slowly towards the majority one in forward bias (by kT per diffusion length, far beyond the frame), and drops abruptly at the edge in reverse bias, where the minority carriers are extracted. Widths schematic, P share of the SCR exaggerated.*

$J \approx 0$ still holds there *term by term*, as at equilibrium:

$$J_p = q\mu_p p\mathcal{E} - qD_p \frac{dp}{dx} \approx 0, \qquad J_n = q\mu_n n\mathcal{E} + qD_n \frac{dn}{dx} \approx 0 \qquad \text{inside the SCR.}$$

EXAMPLE 6.1 — WHY QUASI-EQUILIBRIUM HOLDS

Take $V = +0.6$ V. The net current will be (Section 6.10) $J = 2.1\times10^{-12}\times e^{23.2} = 0.024$ A/cm^2. In the middle of the N part of the SCR, the field is about $\mathcal{E}_{max}/2 = 13$ kV/cm, and the hole density is about ten times its value at the edge, $p \approx 10^{15}\,\text{cm}^{-3}$. The hole drift term alone is $q\mu_p p\mathcal{E} = 1.602\times10^{-19}\times460\times10^{15}\times1.3\times10^{4} \approx 1000$ A/cm^2, and the diffusion term is almost the same in the opposite direction. Their difference, 0.024 A/cm^2, is forty thousand times smaller than each of them. Setting $J_p \approx 0$ inside the SCR is therefore an excellent approximation. It is the same reason as in Chapter 5: drift and diffusion are each intense, and the bias only tips their balance slightly.

The quasi-Fermi levels. Out of equilibrium, electrons and holes need not share one Fermi level. In each slice that is close to equilibrium for each species, we define the **quasi-Fermi level of electrons**, E_{Fn}, and **of holes**, E_{Fp}, by

$$n = n_i\,e^{(E_{Fn}-E_i)/kT}, \qquad p = n_i\,e^{(E_i-E_{Fp})/kT}, \qquad np = n_i^2\,e^{(E_{Fn}-E_{Fp})/kT}. \tag{6.3}$$

At equilibrium $E_{Fn} = E_{Fp} = E_F$. The derivation of Chapter 5 goes through unchanged for each species: $J_n = \mu_n n\, dE_{Fn}/dx$ and $J_p = \mu_p p\, dE_{Fp}/dx$. **A current of electrons flows wherever E_{Fn} has a gradient; a current of holes wherever E_{Fp} has one.**

In the SCR, the quasi-Fermi levels are flat. Quasi-equilibrium says that, inside the SCR, each current is negligible compared with its drift and diffusion terms. With $J_n = \mu_n n\, dE_{Fn}/dx$, a finite current through a region where n is large needs only a tiny slope; the slope of E_{Fn} across the thin SCR is negligible. Hence:

- E_{Fn} keeps across the SCR the value it has in the neutral N region, where electrons are majority carriers: $E_{Fn} = E_F(N)$;
- E_{Fp} keeps across the SCR the value it has in the neutral P region: $E_{Fp} = E_F(P)$;
- they are separated by the applied voltage: $E_{Fn} - E_{Fp} = qV$ **everywhere in the SCR** (Figure 6.3).

Note what quasi-equilibrium does *not* say: it does not say that electrons and holes share one Fermi level in the SCR. Each quasi-Fermi level is constant across the layer, which is the content of $J_n \approx 0$ and $J_p \approx 0$ term by term; the two constants differ by qV. Only at $V = 0$ do they coincide. By (6.3), $np = n_i^2\, \mathrm{e}^{qV/kT}$ throughout the SCR: more than n_i^2 in forward bias, less in reverse bias.

Where does the Fermi level change, then? In the **neutral regions**, for the **minority** carriers. There the minority quasi-Fermi level goes from its value at the edge of the SCR back to the majority one at the contact. The majority quasi-Fermi levels stay practically flat, because the majority density is so large that the current needs almost no slope. Section 6.9 computes what happens there.

THE CORRECT VERSION OF A TEMPTING SHORTCUT

It is tempting to take the equilibrium relation $n_{p0} = N_D\, \mathrm{e}^{-qV_d/kT}$ and simply replace V_d by $V_d - V$, “because the voltage across the SCR is now $V_d - V$”. The relation was derived for zero current, so this is not allowed in general. It is allowed here, at the edges of the SCR, **because of quasi-equilibrium**: inside the SCR the currents are negligible compared with their drift and diffusion terms. The next section does the substitution with that justification, in band language.

WHAT PART B HAS ESTABLISHED

- Under bias the step of the bands is $q(V_d - V)$; the gap is unchanged; the N side is the reference and the bands of the P side move by $-qV$.
- Low injection and quasi-equilibrium: inside the SCR each current is negligible compared with its drift and diffusion terms, so E_{Fn} and E_{Fp} are flat there, separated by qV; $np = n_i^2 \mathrm{e}^{qV/kT}$ in the SCR.
- The Fermi level changes only for the minority carriers in the neutral regions: that is where the current will be computed.

PART C

Carrier densities

The edge densities; the profiles at equilibrium, in forward and in reverse bias

6.6 The minority densities at the edges

THE QUESTION

What are the minority-carrier densities at the edges of the SCR under bias?

Electrons at the P edge, x_p. At x_p the conduction band is at E_{Cp} and the electrons have the quasi-Fermi level of the N side, flat across the SCR. As in Chapter 5,

$$\begin{aligned} n_p(x_p) &= N_C\,\mathrm{e}^{-(E_{Cp}-E_{Fn})/kT} = \underbrace{N_C\,\mathrm{e}^{-(E_{Cn}-E_{Fn})/kT}}_{n_{n0}=N_D}\,\mathrm{e}^{-(E_{Cp}-E_{Cn})/kT} \\ &= N_D\,\mathrm{e}^{-q(V_d-V)/kT}, \end{aligned} \tag{6.4}$$

since the step is now $E_{Cp} - E_{Cn} = q(V_d - V)$ and, by low injection, $n_n = N_D$ in the neutral N region. With $n_{p0} = N_D\,\mathrm{e}^{-qV_d/kT}$ (Chapter 5):

$$\boxed{n_p(x_p) = n_{p0}\,\mathrm{e}^{qV/kT}} \qquad \text{and, in the same way,} \qquad \boxed{p_n(x_n) = p_{n0}\,\mathrm{e}^{qV/kT}}. \tag{6.5}$$

The hole relation follows from $p = N_V\,\mathrm{e}^{-(E_{Fp}-E_V)/kT}$ at x_n, with E_{Fp} flat at the level of the P side and $E_{Vp} - E_{Vn} = q(V_d - V)$ (Problem 6.11 of the exercise collection). Both also follow from $np = n_i^2\,\mathrm{e}^{qV/kT}$ at the edges, with the majority density unchanged.

The minority densities at the edges of the SCR are multiplied by $\mathrm{e}^{qV/kT}$ relative to equilibrium (Figure 6.4). In forward bias, excess minority carriers are **injected** on both sides; in reverse bias, the edges are **emptied**: minority carriers are extracted.

EXAMPLE 6.2 — THE DENSITIES AT THE EDGES

At $V = +0.6$ V, $\mathrm{e}^{qV/kT} = \mathrm{e}^{23.2} = 1.15\times10^{10}$: at the N edge, $p_n = 10^4\times1.15\times10^{10} = 1.2\times10^{14}\,\mathrm{cm}^{-3}$, still small compared with $N_D = 10^{16}$ (low injection); at the P edge, $n_p = 1.2\times10^{12}\,\mathrm{cm}^{-3}$. At $V = -0.3$ V, $\mathrm{e}^{qV/kT} = 9\times10^{-6}$: the edges are practically empty. Low injection on the N side ends when $p_{n0}\mathrm{e}^{qV/kT}$ approaches N_D, that is, near $V = (kT/q)\ln(N_D/p_{n0}) = 0.72$ V.

6.7 Densities across the junction, forward and reverse

THE QUESTION

What do the density profiles look like across the whole junction, and how do they change with the voltage?

The density block of Chapter 5 (Figure 5.5c) is now drawn for seven voltages (Figure 6.4): equilibrium, three forward and three reverse biases, for the long reference diode. The scale break separates the majority levels (top) from the minority levels (bottom, magnified).

- **The majority densities do not move** (low injection): N_A on the P side, N_D on the N side, at every voltage.
- **Forward bias: the edges rise.** At each step of +0.1 V the edge densities are multiplied by $\mathrm{e}^{0.1/0.0259} \approx 48$ (the drawing is not to scale: it only shows them rise). Away from the edge, the excess decays back to equilibrium over a few diffusion lengths, as an exponential (Section 6.9). The higher the edge, the **steeper the gradient**: the gradient will carry the current.
- **Reverse bias: the edges are emptied**, and they are emptied at once: at −0.1 V the edge density is already $\mathrm{e}^{-3.9} = 2$ % of its equilibrium value. The profile becomes a deficit that recovers over a few diffusion lengths. Between −0.5, −2 and −5 V **nothing changes** in the neutral regions: the gradient, hence the current, is the same; only the SCR widens. This is the saturation of the reverse current, seen in the densities.
- **Inside the SCR** (dashed) each density goes from one level to the other, following the Boltzmann factor of the local step (quasi-equilibrium).

WHAT PART C HAS ESTABLISHED

- Edges: $n_p(x_p) = n_{p0}\mathrm{e}^{qV/kT}$, $p_n(x_n) = p_{n0}\mathrm{e}^{qV/kT}$, from the bands with the step $q(V_d - V)$ and quasi-equilibrium.
- Forward bias injects: the edge densities rise exponentially and the gradients steepen. Reverse bias extracts: the edges are emptied at once and the profiles stop changing; only the SCR widens.
- The majority densities do not move (low injection).

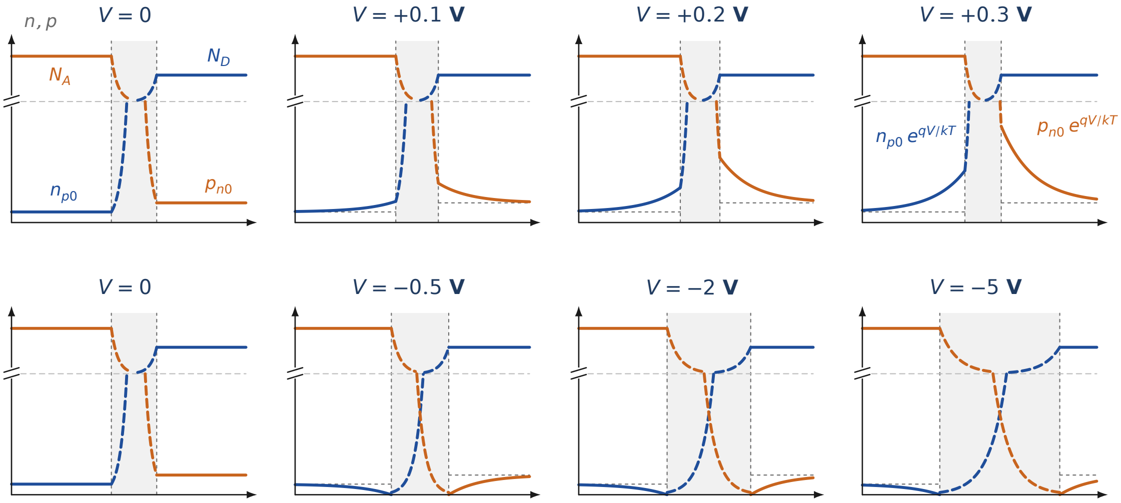


Figure 6.4 *– Carrier densities across the long reference diode, linear cartoon with a scale break (majority levels above the break, minority levels below, magnified; levels not to scale), same geometry in every panel; SCR tinted, its width* $\propto \sqrt{V_d - V}$*. Holes orange, electrons blue; dashed inside the SCR; dotted: the equilibrium minority levels* n_{p0} *and* p_{n0}*. Top row: equilibrium, +0.1, +0.2 and +0.3 V. The edge densities* $n_{p0}e^{qV/kT}$ *and* $p_{n0}e^{qV/kT}$ *rise (each +0.1 V multiplies them by 48), and the excess decays exponentially over a diffusion length: the gradients at the edges steepen. Bottom row: equilibrium, −0.5, −2 and −5 V. The edges are emptied, the minority densities recover over a diffusion length; the profiles in the neutral regions are the same at all three voltages, only the SCR widens. Low injection: the majority levels never move.*

PART D

The current: the Shockley derivation

The lamp and the junction, the assumptions, the long diode, the short diode, the current inside the device, the characteristic

6.8 From the lamp to the junction

> ***THE QUESTION***
>
> *Chapter 4 already had excess carriers with a gradient, under a lamp, and no current flowed. Why should the gradients of a junction carry a current?*

The lamp of Chapter 4 (Figure 6.5a). Light on one surface of an N bar creates pairs: the excess electrons and the excess holes are equal, $\Delta n = \Delta p$ (neutrality), and they have the same profile, decaying into the dark bulk over L_p. Both diffuse **in the same direction**, away from the surface. Their charges are opposite, so their currents are opposite: $J_n = +qD\,d\Delta n/dx$ and $J_p = -qD\,d\Delta p/dx$ cancel. With no closed circuit, there is **no net current**: a gradient, and yet no current.

The junction (Figure 6.5b). In forward bias there are **two** minority gradients, at the **two** edges of the SCR: electrons injected into the P region diffuse *towards the left*, away from the junction; holes injected into the N region diffuse *towards the right*. The two fluxes go in **opposite directions**, and the charges are opposite: the two currents **add**, both from

P to N. A net current flows.

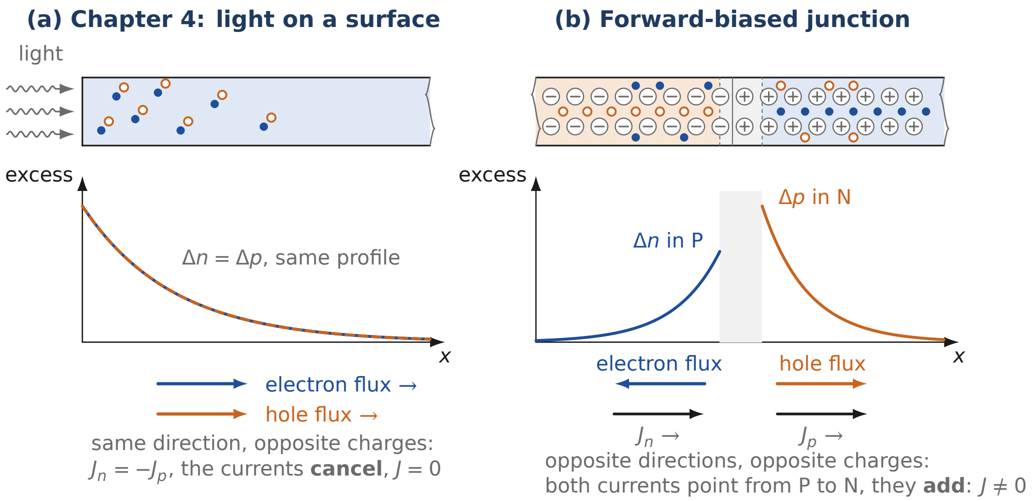

(a) Chapter 4: light on a surface
light
excess
$\Delta n = \Delta p$, same profile
x
electron flux →
hole flux →
same direction, opposite charges:
$J_n = -J_p$, the currents **cancel**, $J = 0$
(b) Forward-biased junction
excess
Δn in P
Δp in N
x
electron flux
hole flux
J_n →
J_p →
opposite directions, opposite charges:
both currents point from P to N, they **add**: $J \neq 0$

Figure 6.5 *– A gradient with no current, and gradients with a current. (a) The lamp of Chapter 4: light on the surface of an N bar creates equal excesses of electrons and holes with the same profile; both diffuse in the same direction; their charges are opposite, so their currents cancel: $J = 0$. (b) The forward-biased junction (cartoon of Figure 6.1): electrons injected into P diffuse to the left, holes injected into N diffuse to the right; opposite directions and opposite charges: both currents point from P to N and add.*

WHY THIS MATTERS

A minority gradient is not, by itself, a current: under the lamp, the majority carriers diffuse with the minority carriers and cancel their current. In the neutral region of a junction the majority carriers also follow the minority carriers to keep neutrality, but they do not cancel the current: they are supplied by the contact and carry the current further by drift (Section 6.12). The current of the diode is fixed by the two minority gradients at the edges of the SCR.

The Shockley assumptions. The calculation of the current rests on four assumptions, those of the classical theory of the diode (W. Shockley, 1949).

1. **All the voltage is across the SCR**: the field in the neutral regions is negligible (Section 6.2).
2. **Low injection and quasi-equilibrium in the SCR** (Section 6.5), which give the edge densities $n_{p0}e^{qV/kT}$ and $p_{n0}e^{qV/kT}$.
3. **The minority currents are pure diffusion**, and are taken at the two edges of the SCR: the electron current at x_p, the hole current at x_n.
4. **No generation or recombination in the SCR**: each current crosses the thin SCR unchanged, so the total current is the sum of the two edge currents.

6.9 The neutral regions: minority carriers diffuse

THE QUESTION

How do the injected minority carriers move in the neutral regions, and what profile do they take?

The two transport mechanisms, recalled from Chapters 1 and 4.

- **Diffusion (Fick's law).** A carrier gradient drives a flux from the denser to the sparser region, flux $= -D\,dn/dx$. The currents are $J_n = qD_n\,dn/dx$ for electrons and $J_p = -qD_p\,dp/dx$ for holes (opposite signs because the charges are opposite).
- **Conduction (drift).** A field drives $J_n = q\mu_n n\mathcal{E}$ and $J_p = q\mu_p p\mathcal{E}$. The total currents are the sums, with $D/\mu = kT/q$ (Einstein).

In a neutral region, minority carriers move by diffusion only. The region stays neutral: an excess of injected minority holes Δp is matched within picoseconds by an equal excess of majority electrons from the contact (Chapter 4). The field there is the small ohmic field that carries the majority current. Times the tiny minority density, it gives a negligible minority drift (Example 6.3). So the minority current is pure diffusion, and the excess obeys the excess-carrier equation of Chapter 4, in steady state:

$$D_p \frac{d^2\Delta p}{dx^2} - \frac{\Delta p}{\tau_p} = 0 \qquad (x_n < x < x_d), \tag{6.6}$$

with two boundary conditions:

- at the edge of the SCR, (6.5): $\Delta p(x_n) = p_n(x_n) - p_{n0} = p_{n0}\left(e^{qV/kT} - 1\right)$;
- at the ohmic contact, where recombination is so fast that the carriers are at equilibrium: $\Delta p(x_d) = 0$.

The same holds for the electrons injected into the P region, with D_n, τ_n, $\Delta n(x_p) = n_{p0}(e^{qV/kT} - 1)$ and $\Delta n(x_g) = 0$.

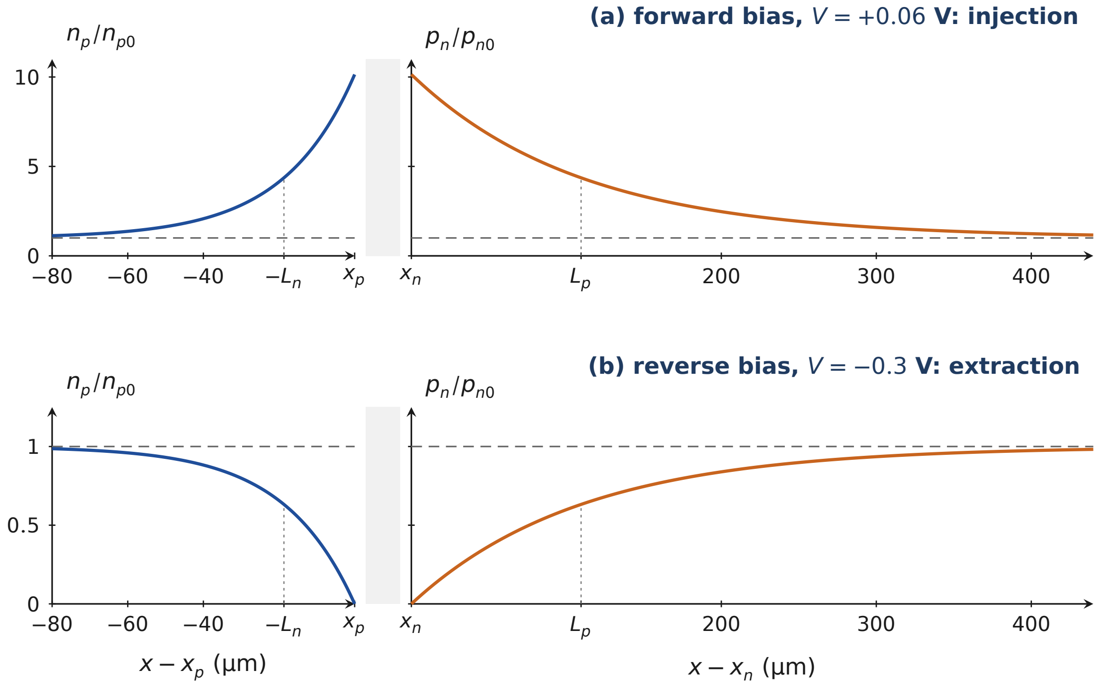
The long region: an exponential. In the reference diode both neutral regions are much longer than the diffusion lengths ($d_n \gg L_p = 110$ µm, $d_p \gg L_n = 18.7$ µm). The solution of Chapter 4 (case B) that vanishes far from the edge is

$$\Delta p(x) = p_{n0}\left(e^{qV/kT} - 1\right) e^{-(x-x_n)/L_p} \qquad (x > x_n), \tag{6.7}$$

and in the same way $\Delta n(x) = n_{p0}(e^{qV/kT} - 1)\, e^{-(x_p - x)/L_n}$ for $x < x_p$. The injected holes all recombine within a few L_p of the edge (Figure 6.6); in reverse bias the same formula, with $e^{qV/kT} - 1 \approx -1$, is a deficit that recovers over L_p.

EXAMPLE 6.3 — THE MINORITY DRIFT REALLY IS NEGLIGIBLE

At $J = 10$ mA/cm², the reference diode is biased at $V = (kT/q)\ln(J/J_0) = 0.58$ V and the hole density at the N edge is $p_{n0}e^{qV/kT} = 4.9 \times 10^{13}\ \text{cm}^{-3}$. The ohmic field of the N region is $\mathcal{E} = J/\sigma = 0.46\ \Omega\cdot\text{cm} \times 10^{-2}$ A/cm² $= 4.6$ mV/cm. The hole drift current is then $q\mu_p p\mathcal{E} = 1.602 \times 10^{-19} \times 460 \times 4.9 \times 10^{13} \times 4.6 \times 10^{-3} = 1.6 \times 10^{-5}$ A/cm², two thousandths of the hole current (0.85×10^{-2} A/cm²), and this is at the



(a) forward bias, $V = +0.06$ V: injection

(b) reverse bias, $V = -0.3$ V: extraction

Figure 6.6 *– Minority densities around the SCR of a long diode (the reference doping and lifetimes, $L_n = 19$ µm, $L_p = 110$ µm, neutral regions much longer than L), computed, linear scale, in units of their equilibrium values n_{p0} and p_{n0}; dashed: the equilibrium level. The two sides have different x scales; the grey gap stands for the SCR, far too thin to be seen at these scales. (a) A small forward bias, $V = +0.06$ V ($e^{qV/kT} = 10$, chosen so that the equilibrium level stays visible): the injected excess decays as $e^{-(x_p - x)/L_n}$ and $e^{-(x - x_n)/L_p}$; dotted: one diffusion length. (b) Reverse bias, $V = -0.3$ V: the edges are emptied and the minority carriers nearby diffuse towards the SCR, which sweeps them across.*

> edge, where the holes are densest; one diffusion length further it is e times smaller again. The electrons ($10^{16}\,\text{cm}^{-3}$) in the same field carry the majority current.

6.10 The Shockley law of the long diode

> **THE QUESTION**
>
> *What current flows through the diode at a voltage V?*

The current at each edge is pure minority diffusion. At x_n the hole current is the diffusion current of the injected holes; at x_p the electron current is the diffusion current of the injected electrons. The slope of the exponential (6.7) at the edge is $-\Delta p(x_n)/L_p$:

$$J_p(x_n) = -qD_p \frac{dp}{dx}\Big|_{x_n} = qD_p \frac{p_{n0}}{L_p}\left(e^{qV/kT} - 1\right),$$
$$J_n(x_p) = qD_n \frac{dn}{dx}\Big|_{x_p} = qD_n \frac{n_{p0}}{L_n}\left(e^{qV/kT} - 1\right). \tag{6.8}$$

The total current. In the thin SCR nothing recombines (assumption 4), so J_n has the same value at x_n as at x_p. The total current, the same at every point of the device, is therefore the sum of the two minority currents taken at their own edges:

$$J = J_p(x_n) + J_n(x_p) = \left(qD_p \frac{p_{n0}}{L_p} + qD_n \frac{n_{p0}}{L_n}\right)\left(\mathrm{e}^{qV/kT} - 1\right),$$

$$\boxed{J = J_0\left(\mathrm{e}^{qV/kT} - 1\right)} \tag{6.9}$$

This is the **diode law**, or **Shockley law**. With $p_{n0} = n_i^2/N_D$ and $n_{p0} = n_i^2/N_A$, the **saturation current density** is

$$\boxed{J_0 = q\,n_i^2\left(\frac{D_p}{N_D L_p} + \frac{D_n}{N_A L_n}\right)} \quad \text{(long diode).} \tag{6.10}$$

EXAMPLE 6.4 — THE SATURATION CURRENT OF THE REFERENCE DIODE

With all lengths in cm, $qn_i^2 = 1.602 \times 10^{-19} \times 10^{20} = 16.0$ (C·cm^{-6}), $L_p = \sqrt{12 \times 10^{-5}} = 1.10 \times 10^{-2}$ cm and $L_n = \sqrt{35 \times 10^{-7}} = 1.87 \times 10^{-3}$ cm. Hole term $16.0 \times 12/(10^{16} \times 1.10 \times 10^{-2}) = 1.75 \times 10^{-12}$ A/cm²; electron term $16.0 \times 35/(10^{18} \times 1.87 \times 10^{-3}) = 3.0 \times 10^{-13}$ A/cm². Total $J_0 = 2.1 \times 10^{-12}$ A/cm², 85 % of it carried by holes. For a diode of area 10^{-4} cm² (100 µm × 100 µm), $I_0 = 2.1 \times 10^{-16}$ A.

Reading the law.

- **Zero at equilibrium**, as it must be.
- **Reverse bias:** $J \to -J_0$. This is the waterfall current of Section 6.3: the minority carriers generated within about a diffusion length of the edges diffuse to them and are swept across. It does not depend on V (Figure 6.4, bottom row).
- **Forward bias:** $J \approx J_0\,\mathrm{e}^{qV/kT}$, multiplied by 10 for every $(kT/q)\ln 10 = 59.6$ mV added to V.
- **The lightly doped side dominates.** J_0 is proportional to n_i^2/N: the side with the fewer majority carriers has the more minority carriers. In the P$^+$N reference diode the holes injected into the N side carry 85 % of the current. The current of a junction is controlled by its more lightly doped side.
- **The value of** J_0 depends on $n_i^2 \propto \mathrm{e}^{-E_g/kT}$: it is tiny in silicon, much larger in germanium, and it grows very fast with temperature.

6.11 Worked example: the short diode

THE QUESTION

What changes when a neutral region is much shorter than the diffusion length, so that its ohmic contact is close to the junction?

EXAMPLE 6.5 — THE SHORT DIODE

Keep the doping and the lifetimes of the reference diode, but make both neutral regions short: $d_p = 2$ µm $\ll L_n = 18.7$ µm and $d_n = 10$ µm $\ll L_p = 110$ µm, each ending on an ohmic contact.

Profile. Over a distance $d \ll L$ almost no carrier recombines: the term $\Delta p/\tau_p$ of (6.6) is negligible, $d^2\Delta p/dx^2 = 0$, and the profile is a **straight line** from the edge value to zero at the contact x_d, where the minority carriers are held at equilibrium (Figure 6.7):

$$\Delta p(x) = p_{n0}\left(e^{qV/kT} - 1\right)\frac{x_d - x}{d_n}, \qquad \Delta n(x) = n_{p0}\left(e^{qV/kT} - 1\right)\frac{x - x_g}{d_p}. \qquad (6.11)$$

Current. The slope is $\Delta p(x_n)/d_n$ instead of $\Delta p(x_n)/L_p$: everything is as in the long diode with L **replaced by** d:

$$J = J_0\left(e^{qV/kT} - 1\right), \qquad J_0 = q\,n_i^2\left(\frac{D_p}{N_D d_n} + \frac{D_n}{N_A d_p}\right) \qquad \text{(short diode)}. \qquad (6.12)$$

Numbers: hole term $16.0 \times 12/(10^{16} \times 10^{-3}) = 1.92 \times 10^{-11}$, electron term $16.0 \times 35/(10^{18} \times 2 \times 10^{-4}) = 2.8 \times 10^{-12}$; $J_0 = 2.2 \times 10^{-11}$ A/cm², 87 % holes: **about ten times the long diode**. The gradients are steeper because the carriers are removed at a contact only d away instead of recombining over L. In between, the general solution of (6.6) gives $L_p \tanh(d_n/L_p)$ in place of L_p, which tends to d_n for a short region and to L_p for a long one (Problem 6.13 of the exercise collection).

Voltage. For the same current, the long diode needs a higher voltage, but only slightly: $V_{long} - V_{short} = (kT/q)\ln(J_{0,short}/J_{0,long}) \approx (kT/q)\ln(L_p/d_n) = (kT/q)\ln 11 = 62$ mV, the hole term dominating both. A factor ten in current costs only 60 mV.

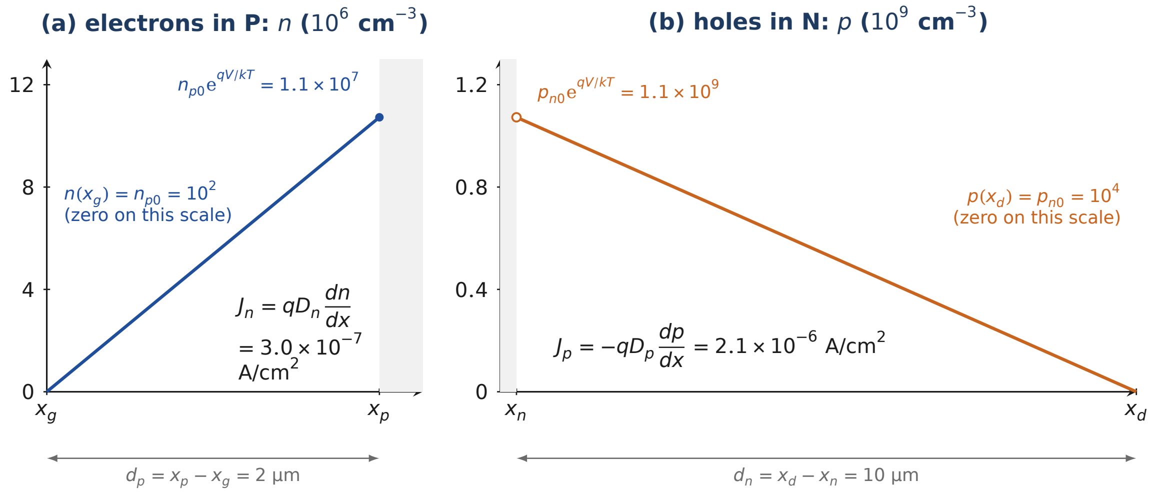


Figure 6.7 *– The minority profiles of the short diode at $V = +0.3$ V, computed, linear scale (the two panels have their own x and density scales; grey strips: the SCR, to scale). (a) Electrons in the P region, from $n_{p0}e^{qV/kT} = 1.1 \times 10^7$ cm^{-3} at the edge x_p to $n_{p0} = 10^2$ cm^{-3} at the contact x_g, over $d_p = x_p - x_g$. (b) Holes in the N region, from $p_{n0}e^{qV/kT} = 1.1 \times 10^9$ cm^{-3} at x_n to p_{n0} at the contact x_d, over $d_n = x_d - x_n$. The slopes give the diffusion currents, $J_n = 3.0 \times 10^{-7}$ and $J_p = 2.1 \times 10^{-6}$ A/cm²; their sum is the current of the diode at 0.3 V, 2.4×10^{-6} A/cm².*

SHORT AND LONG: A KEY IDEA FOR CHAPTER 8

In a long region, the injected minority carriers all recombine on the way; in a short region, almost all of them reach the far side. Chapter 8 makes this the cornerstone of the bipolar transistor: its base is a neutral region much shorter than L, so the minority carriers injected by one junction cross it without recombining and reach the neighbouring junction. The straight-line profile (6.11) and the current (6.12) are the results that Chapter 8 uses.

6.12 The current inside the device

THE QUESTION

The current is the same at every point of the device. Who carries it, and where?

Figure 6.8 draws the electron and hole current densities across the whole forward-biased diode.

- **The total current is constant** everywhere, from contact to contact: charge does not accumulate in steady state.
- **Across the SCR, the electron and hole currents are both flat**: nothing recombines there (assumption 4). This is why the two edge currents can simply be added.
- **In each neutral region, the majority carriers take over.** Near the edge the current is carried by the injected **minority** carriers, by diffusion; their current decays like the diffusion profile, $e^{-(x-x_n)/L_p}$ for the holes, because they recombine. Each recombination consumes a majority carrier, supplied by the contact: the **majority** current rises by exactly as much, and far from the junction the majority carriers carry the whole current, by drift.

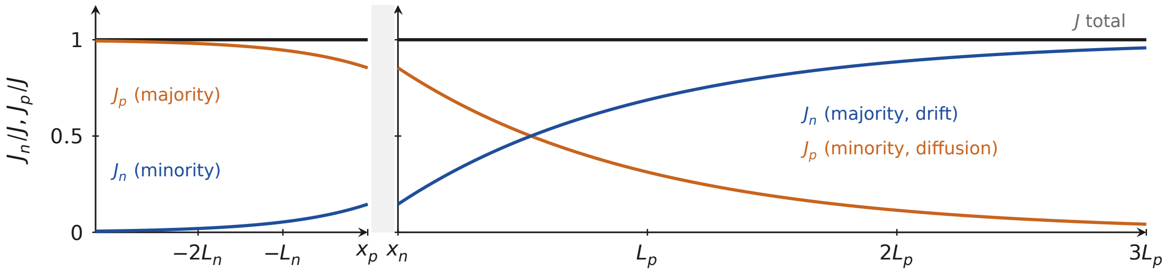
The majority carriers drift in a tiny field. Far from the junction the majority current is drift, $J = q\mu n E$, in the small ohmic field of the neutral region. With 10^{16} electrons per cm^3 on the N side of the reference diode, 1 A/cm^2 needs only $\mathcal{E} = J/(q\mu_n N_D) = 0.46$ V/cm; in a P region doped $N_A = 10^{17}\,\text{cm}^{-3}$, $J/(q\mu_p N_A) = 0.14$ V/cm with the course mobility (≈ 0.2 V/cm with the real hole mobility at this doping, about 320 cm^2/(V·s)). Such a field is **negligible for the minority carriers**: multiplied by their tiny density it gives a negligible drift current (Example 6.3). But it is not zero, and the small voltage drop it produces in the neutral regions is exactly what the Shockley assumption 1 neglects. At high current it becomes the **series resistance** of the diode (Section 6.13).

A COMMON MISTAKE

“There is no drift in the neutral regions.” This is true **for minority carriers only**. The field there is weak but not zero: it makes the majority carriers drift, and they carry the current far from the junction. For minority carriers, the same field times a tiny density gives a negligible drift current (Example 6.3): they move by diffusion only.

The quasi-Fermi levels tell the same story (Figure 6.3). The minority quasi-Fermi level has a slope in the neutral region: that slope, times the small minority density, gives

(a) long diode: gradual hand-over



(b) short diode: hand-over at the contacts

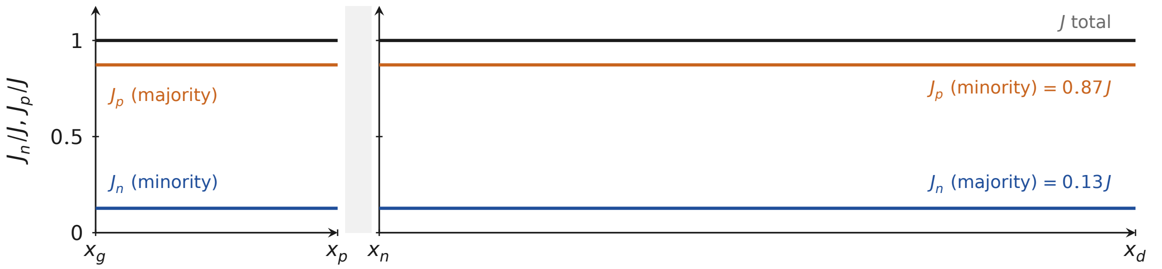


Figure 6.8 *– Electron (blue) and hole (orange) current densities across a forward-biased diode, in units of the total current J (black), which is the same everywhere; computed. The grey gaps stand for the SCR (not to scale), across which both currents are constant. (a) Long diode (reference doping and lifetimes, $d \gg L$): the minority currents decay over L_n and L_p and the majority currents take over; at the edges the holes carry 85 % of J. (b) The short diode of Section 6.11 ($d_p = 2$ µm, $d_n = 10$ µm): each current is constant across its neutral region (bulk recombination removes only $d_n^2/2L_p^2 \approx 0.4$ %, neglected), holes 87 %; the hand-over happens at the contacts.*

the diffusion current. The majority quasi-Fermi level is almost flat: a tiny slope, times the large majority density, gives the same current by drift.

6.13 The characteristic and its limits

> ***THE QUESTION***
>
> *What does the characteristic look like, what is the "threshold" of a diode, and where does the model stop?*

No threshold in the law. Law (6.9) has no threshold: the current is multiplied by 10 each time V increases by 59.6 mV (Figure 6.9). On a linear scale it seems to take off suddenly, because J_0 is tiny. The practical "threshold" is the voltage at which the current becomes significant: with $J_0 = 2.1 \times 10^{-12}$ A/cm², $V = (kT/q)\ln(J/J_0)$ gives **0.52 V for 1 mA/cm²** and **0.70 V for 1 A/cm²**. It depends on the chosen current, on the area and on J_0; $V_d = 0.835$ V is an internal equilibrium quantity, of a different nature.

The limits of the model.

- **High injection** (forward, beyond about 0.7 V here): the injected minority carriers approach the majority density (Example 6.2); the slope becomes $e^{qV/2kT}$.
- **Series resistance** (large forward currents): part of the voltage drops in the neutral

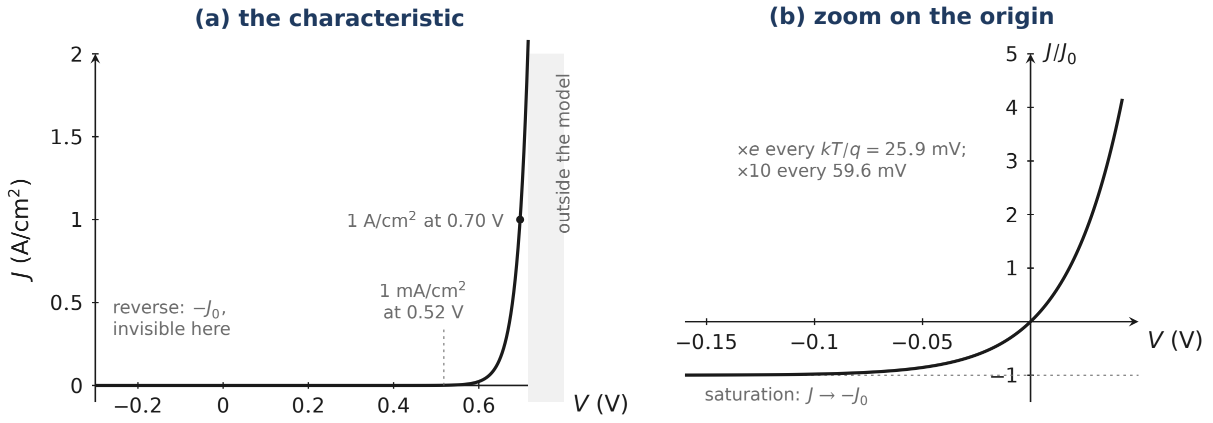


Figure 6.9 *– The characteristic* $J = J_0(\mathrm{e}^{qV/kT} - 1)$ *of the long reference diode* ($J_0 = 2.1 \times 10^{-12}$ *A/cm², kT/q = 25.9 mV), computed, linear scales. (a) In A/cm²: the current seems to start near 0.5 V; it reaches 1 mA/cm² at 0.52 V and 1 A/cm² at 0.70 V; the reverse current* $-J_0$ *is invisible. Shaded: beyond about 0.72 V, where low injection fails (Example 6.2) and the model no longer applies. (b) The same law around* $V = 0$*, in units of* J_0*: in reverse bias* J *saturates at* $-J_0$ *within a few* kT/q*; in forward bias it is multiplied by e every 25.9 mV, by 10 every 59.6 mV.* V_d *is the threshold of neither plot.*

regions and the contacts (Problem 6.14 of the exercise collection); the characteristic bends over and becomes linear.

- **Generation and recombination inside the SCR**: neglected here. In real silicon, generation in the SCR often dominates the reverse current, which then grows slowly with $|V|$ (with the width of the SCR); recombination in the SCR adds a forward component in $\mathrm{e}^{qV/2kT}$ at low current.
- **Breakdown** under strong reverse bias: when the field reaches a few hundred kV/cm, carriers gain enough energy to create pairs (avalanche), and the reverse current rises abruptly (about −29 V for the reference junction, Problem 6.10 of the exercise collection).

6.14 The bias scenario: everything together

THE QUESTION

What does the junction look like, piece by piece, as the voltage goes from reverse to forward bias?

Figure 6.10 puts the three building blocks together at three voltages, −2 V, 0 and +0.4 V, with the same geometry in every column, so that what moves from one column to the next is what the voltage changes. Each column is read from top to bottom, on one horizontal axis: the cartoon with its field, the computed band diagram with the quasi-Fermi levels, and the densities.

What changes, and what does not. From one column to the next only three things move: the width of the SCR ($\propto \sqrt{V_d - V}$), the height of the step ($q(V_d - V)$) and the separation of the quasi-Fermi levels (qV). Everything else follows. The edge densities follow the step through the Boltzmann factor; the diffusion fluxes follow the edge densities; the

	Equilibrium ($V = 0$)	Forward ($V > 0$)	Reverse ($V < 0$)
Voltage across SCR	V_d	$V_d - V$, smaller	$V_d + \lvert V \rvert$, larger
Step of the bands	qV_d	$q(V_d - V)$, lower	$q(V_d + \lvert V \rvert)$, higher
Width W	0.33 µm	narrower (0.21 µm at 0.5 V)	wider (0.87 µm at −5 V)
Fermi levels	one flat E_F	$E_{Fn} - E_{Fp} = qV > 0$ in the SCR	$E_{Fn} - E_{Fp} = qV < 0$ in the SCR
Diffusion current	balances drift	$\times\, e^{qV/kT}$: very large	$\times\, e^{qV/kT} \to 0$
Drift current	limited by supply	unchanged	unchanged
Minority carriers at the edges	n_{p0}, p_{n0}	$\times\, e^{qV/kT}$: injection	≈ 0: extraction
Total current	0	$J_0(e^{qV/kT} - 1)$, large, from P to N	$\approx -J_0$, small, from N to P
Rough picture	conductor, charged insulator, conductor	thinner insulator	thicker insulator

Table 6.3 – *Clarification: the junction in the three situations.*

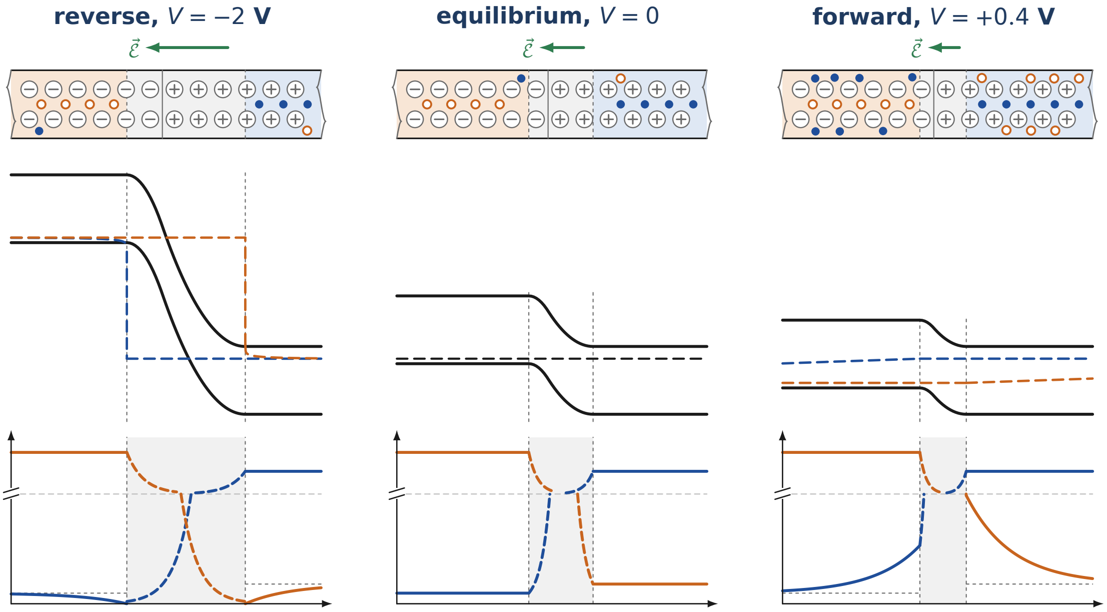


Figure 6.10 – *The bias scenario of the long reference diode, built from the three blocks: one column per voltage (reverse −2 V, equilibrium, forward +0.4 V), same geometry and scales in all three. Top: the cartoon; the SCR widens in reverse bias, minority carriers are injected near the edges in forward bias and extracted in reverse bias, and the field arrow follows* $\mathcal{E}_{max} \propto \sqrt{V_d - V}$. *Middle: the bands, computed (1 eV = 0.8 cm, N side as reference), with* E_F *or the quasi-Fermi levels* E_{Fn} *(blue) and* E_{Fp} *(orange). Bottom: the densities (linear cartoon with a scale break, as in Figure 6.4).* $W = 0.61$, 0.33 *and* 0.24 µm; $J = -J_0$, 0 *and* 1.1×10^{-5} *A/cm².*

drift fluxes do not move at all. Reverse bias empties the edges, forward bias fills them: that is rectification, seen at once in the cartoon, in the bands and in the densities.

WHY THE SEQUENCE STOPS BELOW +0.5 V

One could continue the sequence beyond +0.4 V, to +1 V and +2 V and draw the step and the SCR shrinking to nothing. Those panels would be wrong: both voltages exceed V_d = 0.835 V, and the picture of this chapter does not survive there. If the voltage across the SCR really approached V_d, the step and the SCR would vanish and the diode would be just a resistor: the two conductors of the rough picture would touch. This never happens. Long before, at about 0.72 V in our example, the injected minority carriers become as numerous as the majority carriers (low injection fails, Example 6.2), and the current is so large that a growing part of the applied voltage drops in the series resistance of the neutral regions and contacts instead of across the SCR (Section 6.13, Problem 6.14 of the exercise collection). A source of +1 or +2 V therefore does not put 1 or 2 V across the junction; in a real circuit a series resistor limits the current, and the voltage across the junction stays near 0.7 V. The model of this chapter is safe up to about 0.65–0.7 V.

WHAT PART D HAS ESTABLISHED

- Under a lamp, electrons and holes diffuse the same way and their currents cancel; in a junction, the two minority gradients point in opposite directions and their currents add.
- Shockley assumptions: voltage across the SCR only, low injection and quasi-equilibrium, minority diffusion currents taken at the SCR edges, no recombination in the SCR.
- Long diode: exponential profiles over L; $J = J_0(\mathrm{e}^{qV/kT} - 1)$, $J_0 = qn_i^2(D_p/N_D L_p + D_n/N_A L_n) = 2.1 \times 10^{-12}$ A/cm², 85 % holes. Short diode (worked example): straight lines, $L \to d$, J_0 ten times larger; a long diode needs only $(kT/q)\ln(L/d)$ more voltage.
- Total current constant; electron and hole currents flat across the SCR; in the neutral regions the majority carriers take over, drifting in a field of a fraction of a V/cm. The threshold (0.52 V at 1 mA/cm², 0.70 V at 1 A/cm²) is not V_d.

PART E

Light, and a look back to 1947

The solar cell, the light-emitting diode, and why a short region matters

6.15 The photovoltaic effect

THE QUESTION

A junction under light, with no battery at all, delivers a voltage and power. Where do they come from?

Light creates pairs wherever it is absorbed (Chapter 4: any photon with $h\nu \geq E_g$ creates a pair). Three cases, depending on where (Figure 6.11):

- **In the SCR**, the field separates the pair at once: the electron goes to N, the hole to P.
- **In a neutral region, within a diffusion length of the layer**, the minority carrier of the pair has a good chance of reaching the edge by diffusion before it recombines; the field then sweeps it to the other side. This is the supply-limited drift current of Section 6.3, with a supply provided by light.
- **Further away**, the pair recombines before arriving: it is lost.

The field sorts the charges: electrons to N, holes to P. This flux is a current from N to P inside the device, the **photocurrent** J_{ph}; it is proportional to the number of pairs created per second in the layer and within L of it. A solar cell is therefore made with long, thick neutral regions that absorb the light: it is a long diode.

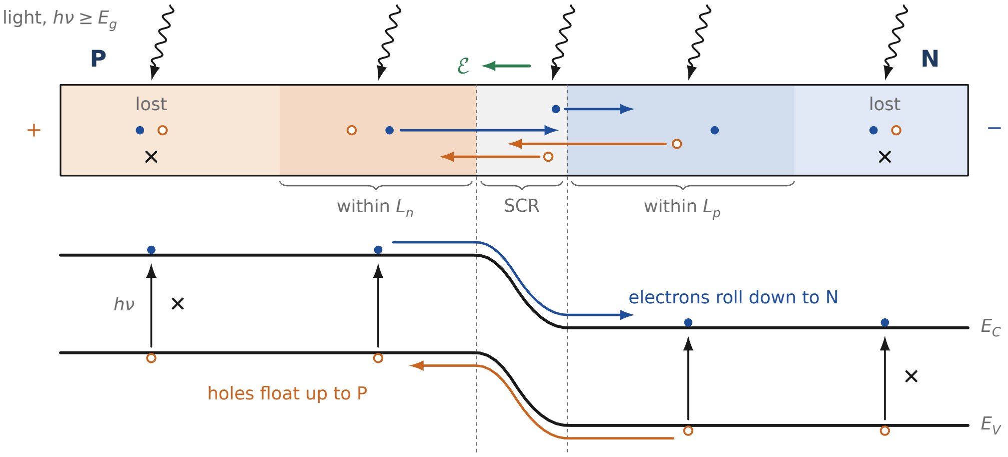


Figure 6.11 *– The field sorts the pairs (sketch, not to scale). Top: the device under light, in open circuit. Pairs created in the SCR are separated on the spot; those created within a diffusion length of it (tinted zones) reach it by diffusion of their minority carrier, which the field $\mathcal{E}$ then sweeps across; the others recombine (×, lost). Holes pile up on the P side and electrons on the N side: the P side charges positively, which forward-biases the junction. Bottom: the same device in bands, on the same horizontal axis (the step is drawn at its equilibrium height for clarity; in open circuit it is lowered by qV_{oc}). Each photon lifts an electron from E_V to E_C (vertical arrows); electrons roll down the step towards N, holes float up it towards P; far from the SCR the pair recombines.*

The sorted charges forward-bias the junction. If the circuit is open, holes pile up on the P side and electrons on the N side: the P side becomes positive with respect to N. This is exactly a forward bias, and it wakes up the diffusion current of Part B, which flows from P to N, **opposite to the photocurrent**. The pile-up continues until the two balance. Assuming that light simply adds its photocurrent to the diode characteristic (superposition, good in low injection),

$$J = J_0 \left(e^{qV/kT} - 1 \right) - J_{ph}. \tag{6.13}$$

In open circuit, $J = 0$, and the voltage that appears is the **open-circuit voltage**:

$$V_{oc} = \frac{kT}{q} \ln \left(1 + \frac{J_{ph}}{J_0} \right). \tag{6.14}$$

In short circuit, $V = 0$ and $J_{sc} = -J_{ph}$. In between, $V > 0$ and $J < 0$: the product VJ is negative, and the junction **delivers** power to the circuit (Figure 6.12).

In bands. Light raises both n and p; the electron and hole quasi-Fermi levels split. In open circuit their separation at the contacts is qV_{oc}: the voltage of the cell is the splitting of the quasi-Fermi levels that light produces.

EXAMPLE 6.6 — A SILICON SOLAR CELL

Under the midday sun, a silicon cell typically collects $J_{ph} = 35$ mA/cm², that is, 2.2×10^{17} pairs per cm² per second. A thick cell is a long diode; take $J_0 = 10^{-12}$ A/cm² for it (another device than the reference diode): $V_{oc} = 0.0259 \times \ln(1 + 3.5 \times 10^{10}) = 0.0259 \times 24.3 = 0.63$ V. Under light ten times weaker, $V_{oc} = 0.57$ V: −59.6 mV per decade of light. The largest power of the ideal characteristic (6.13) is reached near $V = 0.55$ V, $J = 33$ mA/cm²: about 18 mW/cm², or 0.83 of the product $V_{oc} J_{sc}$.

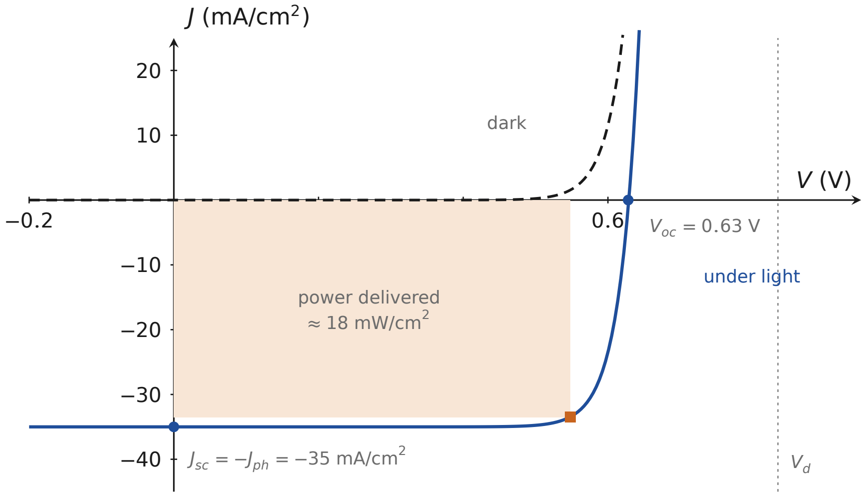


Figure 6.12 – *Characteristics of a silicon solar cell in the dark (dashed) and under light (solid), computed with* $J_{ph} = 35$ *mA/cm² and* $J_0 = 10^{-12}$ *A/cm². This is another device than the reference diode of the chapter: a thick cell, a long diode, hence its smaller* J_0*. Light shifts the curve by* $-J_{ph}$*. Dots:* J_{sc} *and* V_{oc}*; square: the maximum-power point (0.55 V, −33.5 mA/cm²); the tinted rectangle is the power delivered, 18 mW/cm².* $V_{oc} = 0.63$ *V stays well below* V_d *(dotted).*

BE AWARE OF…

- $V_{oc} < V_d$**, always.** V_{oc} is a forward bias; as it approaches V_d, the step and its field disappear, and nothing is left to sort the pairs. Reaching V_d would need $J_{ph} = J_0 \, \mathrm{e}^{qV_d/kT}$, about 100 A/cm² or 3000 suns, and low injection would be lost long before.
- V_{oc} **is not "0.7 V".** It grows as the logarithm of the light and falls when J_0 rises (poorer material, higher temperature): 0.57 V at 0.1 sun, and 0.57 V too with $J_0 = 10^{-11}$ A/cm² at one sun (Problem 6.16 of the exercise collection).
- **The field does not "become" the electromotive force.** It sorts the charges; the sorted charges bias the junction, and the measured voltage is that forward bias.

THE PHOTODIODE

Under reverse bias, the same junction produces no power but measures light: (6.13) gives $J \approx -J_0 - J_{ph} \approx -J_{ph}$, a current proportional to the illumination and almost independent of the voltage. The wider SCR collects more pairs and responds faster.

6.16 The light-emitting diode

THE QUESTION

A suitable diode, forward-biased, emits light. Where, and why not a silicon diode?

In forward bias, minority carriers are **injected** on both sides: electrons into the P region, holes into the N region (Figure 6.4, top row). In a long region these excess minority carriers recombine with the majority carriers around them as they diffuse away from the edge, over a distance of the order of their diffusion length. **Recombination therefore takes place near the edges of the SCR, within L_n on the P side and within L_p on the N side** (Figure 6.13). Each electron that passes through the diode feeds at most one recombination.

In a **direct-gap** material (GaAs and its alloys, InGaN), a large share of these recombinations is **radiative**: each emits a photon of energy $h\nu \approx E_g$, which sets the colour. In silicon, an **indirect-gap** material, radiative recombination is very unlikely; the pairs recombine through defects and their energy becomes heat (Chapter 4). This is why a silicon diode warms up but does not glow. In band language: in the regions of injection $E_{Fn} - E_{Fp} > 0$, so $np > n_i^2$, and the excess pairs recombine.

A COMMON MISTAKE

"The LED emits in the depletion layer." No, not in a simple PN junction. Carriers cross the SCR in a few picoseconds, whereas they live for nanoseconds or more; they have no time to recombine there. They recombine afterwards, in the neutral regions, within a diffusion length of the edges. The SCR is also very thin under forward bias (0.2 µm in our example), compared with diffusion lengths of a micrometre or more.

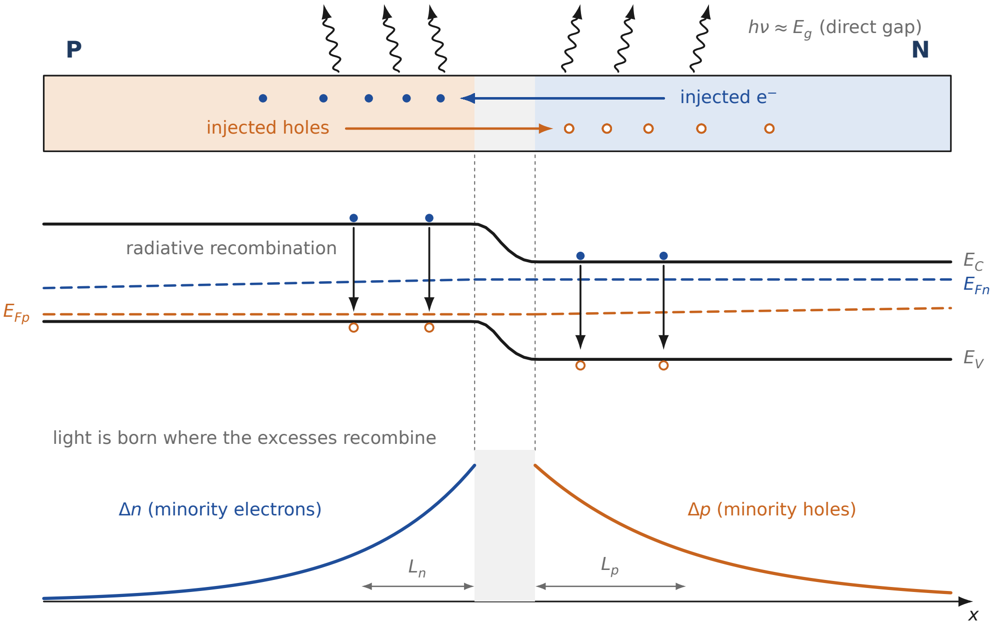

P
N
hν ≈ E_g (direct gap)
injected e⁻
injected holes
radiative recombination
E_C
E_{Fn}
E_{Fp}
E_V
light is born where the excesses recombine
Δn (minority electrons)
Δp (minority holes)
L_n
L_p
x

Figure 6.13 *– The homojunction LED under forward bias (sketch; one horizontal axis for the three rows; a long diode with drawn diffusion lengths* $L_n < L_p$*). Top: electrons are injected into the P region and holes into the N region; light is born where they recombine. Middle: the same device in bands (*$qV = 0.4$ *eV, to the energy scale of Figure 6.3). The injected electrons sit in the conduction band of the P side and the injected holes in the valence band of the N side; each radiative recombination drops an electron from* E_C *to* E_V *and emits* $h\nu \approx E_g$*. The quasi-Fermi levels, flat across the SCR, are computed for these long profiles: the minority one falls by only about* kT *per diffusion length, so it rejoins the majority one far beyond the drawn region. Bottom: the minority excesses decay over* L_n *and* L_p *on either side of the SCR; this is where, and not in the SCR, radiative recombination takes place in a homojunction.*

OUTSIDE THE MODEL: REAL LEDS

Modern LEDs are not homojunctions. They place at the junction a very thin active layer of a smaller-gap material (heterostructure, quantum wells), where the injected electrons and holes are trapped together: there, the emission does come from a thin layer at the junction, but by design, not by the mechanism of this chapter. The principle is the same: injection under forward bias, then radiative recombination in a direct-gap material.

6.17 A short neutral region, and the transistor of 1947

THE QUESTION

What happens if a short neutral region is placed between two junctions? And what has it to do with a gold foil and a razor blade?

Take a short N region between two P regions. Forward-bias the first junction: it injects holes into the N region, with the edge density $p_{n0}e^{qV/kT}$ of Section 6.6. The region is short, so the holes cross it by diffusion without recombining (Section 6.11). Reverse-bias the second junction: its edge is emptied, $p \approx 0$, exactly like an ohmic contact for the minority carriers, and its field sweeps every hole that arrives into the second P region. The current of the second junction is then set by the voltage of the **first**: one junction controls another. This is the **bipolar transistor**, a PNP here; the short N region is its **base**. Chapter 8 studies the mirror device, the NPN transistor, in which the roles of electrons and holes are exchanged.

A story to end with. Chapter 0 told how, in December 1947, Bardeen and Brattain obtained the first transistor at Bell Laboratories. The way it was built is worth a second look now. Brattain wrapped a strip of gold foil around the edge of a small plastic (polystyrene) triangle, then slit the foil at the tip with a razor blade, and pressed the triangle onto a germanium crystal with a spring: two gold contacts, a few tens of micrometres apart (about 50 µm). Why go to such trouble to bring two contacts so close? Bardeen and Brattain knew it from experiment: the effect appeared only when the contacts were very close together. They did not yet have a theory that said why. That theory came in 1949, when Shockley published his analysis of the p–n junction and of the junction transistor, the analysis of this chapter: and in it, the width of the region between the two junctions, compared with the diffusion length, became a **design rule**.

At this point you cannot yet see why shortness should matter so much: the short diode of Section 6.11 only carried ten times more current than the long one. Be patient. In Chapter 8 the short region becomes the cornerstone of the transistor, and the razor blade will make sense.

WHAT PART E HAS ESTABLISHED

- Solar cell: light creates pairs; those in the SCR and within L of it are sorted by the field (electrons to N, holes to P); the sorted charges forward-bias the junction. $J = J_0(e^{qV/kT} - 1) - J_{ph}$; $V_{oc} = (kT/q)\ln(1 + J_{ph}/J_0) \approx 0.63\text{ V} < V_d$; qV_{oc} is the splitting of the quasi-Fermi levels.
- LED: forward bias injects minority carriers, which recombine within L of the edges of the SCR; in a direct-gap material the recombination emits photons $h\nu \approx E_g$; not in silicon.
- Bipolar transistor: a short region injected by one junction and emptied by another. The 1947 point-contact transistor already needed contacts a few tens of micrometres apart; Shockley's 1949 theory made this a design rule (Chapter 8).

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Generic picture**: the voltage drops across the SCR and changes only the barrier, $V_d \to V_d - V$, $W \propto \sqrt{V_d - V}$. The climbing (diffusion) flux of majority carriers is multiplied by $e^{qV/kT}$; the rolling-down (drift) flux of minority carriers is unchanged, limited by supply. Rectification. Local drift and diffusion currents in the SCR are of the order of kA/cm² and nearly cancel.
- **Bands**: the step is $q(V_d - V)$, the gap unchanged. Quasi-equilibrium: E_{Fn} and E_{Fp} are flat across the SCR, $E_{Fn} - E_{Fp} = qV$, $np = n_i^2 e^{qV/kT}$.
- **Densities**: $n_p(x_p) = n_{p0}e^{qV/kT}$, $p_n(x_n) = p_{n0}e^{qV/kT}$: injection in forward bias (edges rise, gradients steepen), extraction in reverse bias (edges emptied, profiles saturate). Majority densities unchanged.
- **Lamp and junction**: same-direction gradients cancel; opposite gradients at the two edges add.
- **Shockley law**: $J = J_0(e^{qV/kT} - 1)$; long diode $J_0 = qn_i^2(D_p/N_D L_p + D_n/N_A L_n) = 2.1 \times 10^{-12}$ A/cm², 85 % holes; short diode $L \to d$ (2.2×10^{-11} A/cm²). ×10 per 59.6 mV; threshold 0.52–0.70 V, not V_d.
- **Inside the device**: total current constant; flat electron and hole currents across the SCR; majority takeover in the neutral regions, by drift in a field of a fraction of a V/cm (series resistance at high current).
- **Light**: solar cell, $V_{oc} = (kT/q)\ln(1 + J_{ph}/J_0) < V_d$; LED, radiative recombination within L of the edges, direct gap. A short region between two junctions is the base of a bipolar transistor.

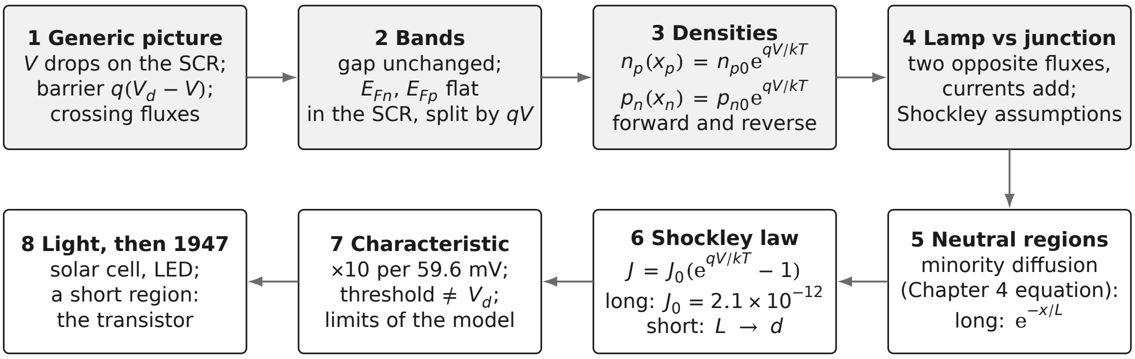

1 Generic picture
V drops on the SCR;
barrier $q(V_d - V)$;
crossing fluxes
2 Bands
gap unchanged;
E_{Fn}, E_{Fp} flat
in the SCR, split by qV
3 Densities
$n_p(x_p) = n_{p0}e^{qV/kT}$
$p_n(x_n) = p_{n0}e^{qV/kT}$
forward and reverse
4 Lamp vs junction
two opposite fluxes,
currents add;
Shockley assumptions
5 Neutral regions
minority diffusion
(Chapter 4 equation):
long: $e^{-x/L}$
6 Shockley law
$J = J_0(e^{qV/kT} - 1)$
long: $J_0 = 2.1 \times 10^{-12}$
short: $L \to d$
7 Characteristic
×10 per 59.6 mV;
threshold $\neq V_d$;
limits of the model
8 Light, then 1947
solar cell, LED;
a short region:
the transistor

Figure 6.14 – *The logic of the chapter. Top row (grey): the generic picture; the bands with the quasi-Fermi levels; the densities, forward and reverse; the lamp against the junction and the assumptions. Bottom row, from right to left: diffusion in the neutral regions, the Shockley law of the long diode (the short diode as a variant), the characteristic, then light and the first transistor.*

Comprehension questions

No new physics here: answer from the figures and equations of the chapter. The answers are at the end.

Q1. Why does the applied voltage drop almost entirely across the SCR? When does this stop being true?

Q2. Draw the band diagram in forward bias. What is the height of the step, and how far apart are the Fermi levels of the two neutral regions?

Q3. In forward bias the diffusion current becomes very large, but the drift current hardly changes. Explain both facts in the generic picture and in the barrier picture.

Q4. Why does the reverse current hardly depend on the voltage? Why is it so small?

Q5. State the quasi-equilibrium hypothesis. Why does it hold? What does it imply for the quasi-Fermi levels in the SCR?

Q6. Why can we replace V_d by $V_d - V$ in $n_{p0} = N_D \mathrm{e}^{-qV_d/kT}$ at the edge of the SCR, although the relation was derived for zero current?

Q7. Why do the minority carriers in a neutral region move by diffusion only? What boundary condition holds at an ohmic contact?

Q8. Why is the minority profile a straight line in a short region? What is it in a long region?

Q9. Why does the lightly doped side dominate J_0? Which carriers carry most of the current in the reference diode?

Q10. "In the neutral regions there is no drift." Correct this, and say who carries the current far from the junction, in a long and in a short diode.

Q11. Is the "threshold voltage" of a silicon diode V_d? What is it, and what does it depend on?

Q12. Why is $V_{oc} < V_d$? Where does a homojunction LED emit its light?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 6, with their answers; the worked solutions are in the **Solutions** document. The 8 exercises that used to close this chapter are Problems 6.10 to 6.17 there; the other problems of the set (6.1 to 6.9) go further.

Answers

Q1. The elements are in series and the SCR, with no carriers, is by far the most resistive. It stops being true at large forward currents, when the ohmic drop in the neutral regions and the contacts becomes comparable (Problem 6.14 of the exercise collection).

Q2. The P side moves down by qV: the step is $q(V_d - V)$, and $E_F(N) - E_F(P) = qV$.

Q3. Generic: the weaker field no longer holds back the countless majority carriers; drift is a waterfall, set by the supply of minority carriers, not by the field. Barrier: the fraction of majority carriers above a step $q(V_d - V)$ is multiplied by $\mathrm{e}^{qV/kT}$; every minority carrier reaching the top rolls down whatever the height.

Q4. Only the drift of minority carriers remains, set by how many reach the edge, not by the field: it saturates. It is small because minority carriers are rare ($p_{n0} = 10^4$, $n_{p0} = 10^2\,\mathrm{cm}^{-3}$).

Q5. Inside the SCR, the drift and diffusion terms of each current remain much larger than the net current (about 1000 A/cm^2 against 0.024 A/cm^2 at 0.6 V), so $J \approx 0$ term by term. Hence E_{Fn} and E_{Fp} are flat across the SCR, separated by qV.

Q6. Because of quasi-equilibrium: inside the SCR the densities still follow the Boltzmann factor of the step, with the flat quasi-Fermi levels. The step is $q(V_d - V)$, so $n_p(x_p) = N_D \mathrm{e}^{-q(V_d - V)/kT}$.

Q7. The region is neutral; the small ohmic field times the tiny minority density gives a negligible drift (Example 6.3). At an ohmic contact recombination is very fast: the excess is zero.

Q8. Almost no carrier recombines over $d \ll L$, so $d^2\Delta p/dx^2 = 0$. In a long region, $\Delta p \propto \mathrm{e}^{-x/L}$.

Q9. $J_0 \propto n_i^2/N$: fewer majority carriers means more minority carriers to inject and collect. The holes injected into the N side carry 85 %.

Q10. True for minority carriers only. Far from the edges, the majority carriers carry the current by drift in the small ohmic field. In a long diode they take over gradually, over L; in a short diode, at the contact.

Q11. No. $V_d = 0.835$ V is an internal equilibrium voltage. The threshold is the voltage at which the current becomes significant: 0.52 V for 1 mA/cm^2, 0.70 V for 1 A/cm^2. It depends on the chosen current, the area and J_0 (doping, geometry, material, temperature).

Q12. V_{oc} is a forward bias created by the sorted charges; as it approached V_d, the step and its field would disappear and sorting would stop. The LED emits within a diffusion length of the edges of the SCR, in the neutral regions, not in the SCR.

References

- S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley, 2007 — Ch. 2: current–voltage characteristics of the PN junction, generation–recombination, breakdown; Ch. 13: solar cells and photodetectors.
- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — Ch. 8: the PN junction diode (minority-carrier distribution, ideal current, short diode); Ch. 14: optical devices.
- R. F. Pierret, *Semiconductor Device Fundamentals*, Addison-Wesley, 1996 — Ch. 6: the ideal diode, with quasi-Fermi levels and the short-diode case.

WHERE THIS LEADS

We now know why a PN junction rectifies, what current it carries, how it turns light into power and how it can emit light. One thing is still missing for circuits: speed. A junction stores charge, in its SCR and in its injected minority carriers, and this charge changes with the voltage. **Chapter 7** computes these two stores: the transition capacitance (reverse bias) and the diffusion capacitance (forward bias).

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 7

The Capacitance Effects of the Junction

Two stores of charge · the SCR under reverse bias · the transition capacitance · the $1/C^2$ measurement · the varicap · the stored minority charge · the diffusion capacitance · the small-signal diode · switching a diode off · transistors and logic

WHERE WE ARE

Chapter 5 built the junction at equilibrium: a space-charge region (SCR) of width $W = 0.33$ µm, made of fixed ions, between two neutral regions, with W obtained from Poisson's equation. Chapter 6 applied a voltage: V_d became $V_d - V$, so $W \propto \sqrt{V_d - V}$, and in forward bias minority carriers were injected at the edges of the SCR. In the long neutral regions of the reference diode they diffuse away and recombine, with an exponential profile that decays over a diffusion length. From that profile came the Shockley law, $J = J_0(\mathrm{e}^{qV/kT} - 1)$ with $J_0 = 2.1 \times 10^{-12}$ A/cm² for the reference diode.

Both results hide a **charge that depends on the voltage**. The ions of the SCR are uncovered or covered again when W changes; the injected minority carriers pile up or disappear when the edge densities change. A charge that changes with the voltage is a **capacitance**. This chapter computes the two of them, says when each one dominates, and shows why they set the speed of every junction device.

CHAPTER OUTLINE

7.1 Why a junction stores charge
PART A — The transition capacitance
7.2 The SCR under reverse bias
7.3 The transition capacitance C_T
7.4 Measuring a junction with $1/C_T^2$
7.5 Engineering $C(V)$: the varicap
PART B — The diffusion capacitance
7.6 The stored minority charge
7.7 The diffusion capacitance C_D
7.8 C_T and C_D across the bias range
PART C — Why it matters
7.9 The small-signal diode
7.10 Switching a diode off
7.11 Transistors and logic circuits
Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

- Identify the **two stores of charge** of a junction: the ions of the SCR and the injected minority carriers.
- Derive the **transition capacitance** $C_T = \varepsilon S/W$ from the widths of Chapter 5, and its dependence $C_T = C_{T0}/\sqrt{1 - V/V_d}$ for a one-sided junction.
- Use the straight line $1/C_T^2$ against V to **measure** V_d and the doping, and explain the **varicap**.
- Derive the **diffusion capacitance** $C_D = (q/kT)\,\bar{\tau}\,I$ from the minority profiles of Chapter 6, and interpret $\bar{\tau}$ as the time a carrier stays: the lifetime in a long region, the transit time in a short one.
- Build the **small-signal model** of the diode and explain why stored charge slows down **switching**.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- compute C_T at any reverse bias, and say why it is the capacitance of a parallel-plate capacitor of thickness W;
- extract V_d and N_D from two or more $C(V)$ measurements;
- choose the doping profile of a varicap for a given tuning law;
- compute the stored charge $Q = \bar{\tau} I$ and C_D at a given current, and compare C_D (nanofarads) with C_T (picofarads);
- draw the small-signal model $r_d \parallel (C_T + C_D)$ and compute its elements and its time constant;
- estimate the storage time of a diode and say how fast diodes are made.

HOW THIS CHAPTER WORKS

- **No new physics.** Everything comes from two results already derived: the width $W(V)$ (Chapters 5 and 6) and the minority profiles of the neutral regions (Chapter 6). This chapter only asks how the charge they describe changes with V.
- **Two pictures again.** The transition capacitance is told with the charge density $\rho(x)$; the diffusion capacitance with the minority-carrier profiles.
- **The same reference diode**, the long diode of Chapter 6, now with an area: $S = 10^{-4}$ cm² (Table 7.1). The currents stay at or below 0.1 mA, inside low injection.

Symbol	Meaning	Value for the reference diode
S	area of the junction	10^{-4} cm² (100 µm × 100 µm)
C_T, C_{T0}	transition capacitance; its value at $V = 0$	$C_{T0} = 3.15$ pF
Q_p, Q_n	excess minority charge stored in the N and P regions	$\propto (e^{qV/kT} - 1)$
τ_p, τ_n	lifetimes of holes in N, electrons in P	10 µs; 0.1 µs
$\bar{\tau}$	mean lifetime, weighted by the currents	8.56 µs
C_D	diffusion capacitance	$(q/kT)\,\bar{\tau} I$
r_d	small-signal (dynamic) resistance	kT/qI
I	current, $I = SJ$	$I_0 = SJ_0 = 2.1 \times 10^{-16}$ A

Table 7.1 *– Notation added in this chapter; the others are those of Chapters 5 and 6. Reference diode: silicon at 300 K,* $\mathrm{P^+N}$, $N_A = 10^{18}$, $N_D = 10^{16}\,\mathrm{cm^{-3}}$, $V_d = 0.835$ *V,* $\varepsilon = 11.7\,\varepsilon_0 = 1.036 \times 10^{-12}$ *F/cm, long neutral regions (d ≫ L),* $D_p = 12$ *and* $D_n = 35$ *cm²/s,* $L_p = 110$ *µm,* $L_n = 18.7$ *µm,* $J_0 = 2.1 \times 10^{-12}$ *A/cm², 85 % of it carried by holes.*

7.1 Why a junction stores charge

THE QUESTION

Where is there charge in a junction, and which of it changes when the voltage changes?

A capacitor is any device whose stored charge depends on the voltage across it. Its **capacitance** is the derivative

$$C = \left|\frac{dQ}{dV}\right|, \tag{7.1}$$

which for a junction depends on the operating point: we use the derivative, not the ratio Q/V, because only the change of charge matters for a small change of voltage. A PN junction has two stores of charge.

- **The ions of the SCR.** The SCR holds $-qN_AW_pS$ on the P side and $+qN_DW_nS$ on the N side, equal by neutrality (Chapter 5). When V changes, W changes and ions are uncovered or covered again at the two edges. This is the **transition capacitance** C_T, also called depletion or junction capacitance. It exists at every voltage, and it is the only one in reverse bias.
- **The injected minority carriers.** In forward bias, holes are injected into the N region and electrons into the P region (Chapter 6). Each excess is neutralised by an equal excess of majority carriers, so the region stays neutral; but the amount of injected charge grows like $e^{qV/kT}$. This is the **diffusion capacitance** C_D. It matters only in forward bias.

The two stores are in different places and obey different laws: Part A treats the first, Part B the second, and Part C uses both.

PART A

The transition capacitance

The SCR under reverse bias, $C_T = \varepsilon S/W$, the $1/C^2$ measurement, the varicap

7.2 The SCR under reverse bias

THE QUESTION

What happens to the charge of the SCR when the reverse voltage is increased?

The widths under bias. The electrostatics of Chapter 5 did not depend on how the step V_d was created. Under a voltage V (source on P, N grounded), only the total drop changes, $V_N - V_P = V_d - V$ (Chapter 6). Poisson's equation still gives a parabola on each side of the junction,

$$V(x) = \frac{qN_A}{2\varepsilon}(x - x_p)^2 + V_P \quad (x_p < x < 0),$$
$$V(x) = -\frac{qN_D}{2\varepsilon}(x - x_n)^2 + V_N \quad (0 < x < x_n),$$

with $V_N = 0$ and now $V_P = V - V_d$: in reverse bias the P side is lowered by $|V|$ below its equilibrium value $-V_d$ (the potential profiles at two biases are those of Chapter 6). Writing that the field and the potential are continuous at $x = 0$, with $W_p = -x_p$ and $W_n = x_n$, gives the two conditions

$$\begin{aligned} N_A W_p &= N_D W_n \qquad \text{(continuity of the field)}, \\ N_A W_p^2 + N_D W_n^2 &= \frac{2\varepsilon}{q}(V_d - V) \qquad \text{(continuity of the potential)}, \end{aligned} \tag{7.2}$$

and the widths of Chapter 5 hold with V_d replaced by $V_d - V$:

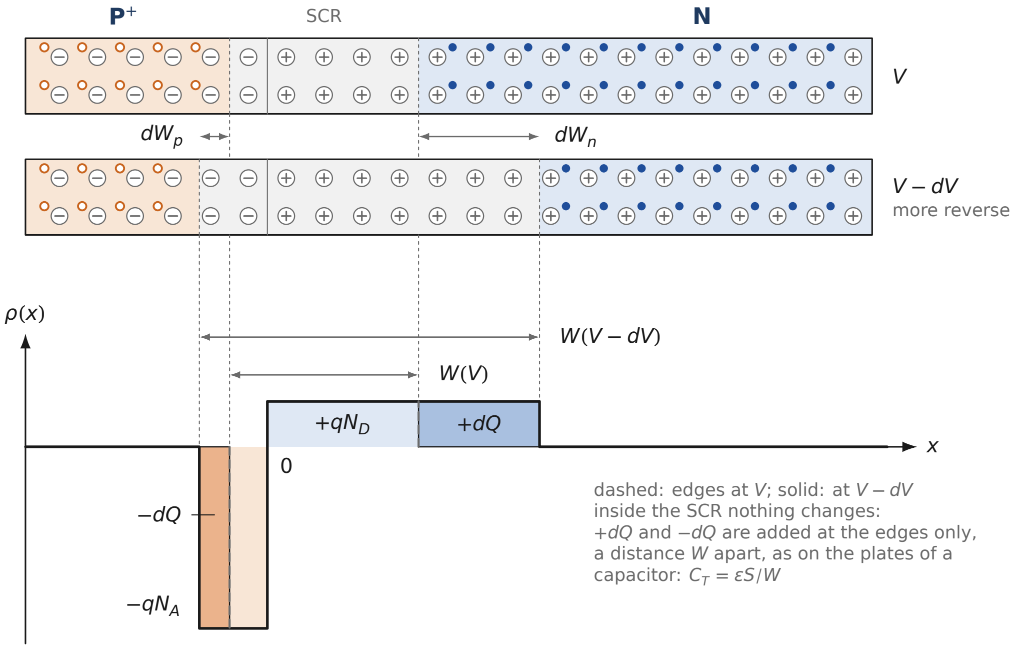
$$W_n = \sqrt{\frac{2\varepsilon(V_d - V)}{qN_D}\,\frac{1}{1 + N_D/N_A}}, \qquad W_p = \sqrt{\frac{2\varepsilon(V_d - V)}{qN_A}\,\frac{1}{1 + N_A/N_D}}, \qquad W = W_p + W_n. \tag{7.3}$$

The one-sided junction. When $N_A \gg N_D$ (a P$^+$N junction, as in the reference diode), $W_p \approx 0$ and

$$W \simeq W_n \simeq \sqrt{\frac{2\varepsilon}{qN_D}(V_d - V)}. \tag{7.4}$$

The whole SCR lies in the lightly doped side, and its width is set by that side alone.

The edges move. In reverse bias ($V < 0$), $V_d - V$ grows, so W grows. Going from V to $V - dV$ (with $dV > 0$, a little more reverse) moves the N edge out by dW_n and the P edge out by dW_p (Figure 7.1, top). The newly uncovered ions are a slab of charge $+dQ = qN_D S\,dW_n$ at the N edge and a slab $-dQ = -qN_A S\,dW_p$ at the P edge, equal by the first condition (7.2) since $N_A\,dW_p = N_D\,dW_n$ (Figure 7.1, bottom). Nothing changes inside the SCR: the charge is added **only at its edges**, a distance W apart. In a P$^+$N junction the P slab is very thin and very dense: the same charge in a layer $N_A/N_D = 100$ times thinner.



dashed: edges at V; solid: at $V - dV$
inside the SCR nothing changes:
$+dQ$ and $-dQ$ are added at the edges only,
a distance W apart, as on the plates of a
capacitor: $C_T = \varepsilon S/W$

Figure 7.1 *– The SCR of a* P^+N *junction at two reverse biases,* V *and* $V - dV$ *(schematic). Top: the junction at* V *and at* $V - dV$*; fixed ions (⊖ acceptors, ⊕ donors), holes (open circles) and electrons (dots) in the neutral regions, the SCR in grey; its edges move out by* dW_p *and* dW_n*. Bottom, aligned with the bars: the charge density* $\rho(x)$*,* $-qN_A$ *on the P side and* $+qN_D$ *on the N side, with the edges at* V *dashed. The step* dV *adds the slabs* $+dQ$ *and* $-dQ$ *(darker), at the two edges only, a distance* W *apart: the plates of a capacitor,* $C_T = \varepsilon S/W$*. Not to scale: the step* dV *is drawn large, the ion symbols do not show the doping, and the ratio* N_A/N_D *is drawn as 4 instead of 100 so that the P part is visible (the areas of* ρ *are equal on the two sides, as neutrality requires).*

EXAMPLE 7.1 — THE REFERENCE JUNCTION UNDER REVERSE BIAS

From (7.3) with $V_d = 0.835$ V: $W = 0.33$ µm at $V = 0$, 0.49 µm at −1 V, 0.79 µm at −4 V, 0.87 µm at −5 V and 1.19 µm at −10 V. Going from −4 to −5 V moves the N edge by 78 nm and uncovers $qN_D S\, dW_n = 1.602 \times 10^{-19} \times 10^{16} \times 10^{-4} \times 7.8 \times 10^{-6} = 1.2$ pC, while the P edge moves by 0.78 nm, a hundred times less.

7.3 The transition capacitance C_T

THE QUESTION

How large is the capacitance of the SCR, and how does it depend on the voltage?

A parallel-plate capacitor. The bottom of Figure 7.1 is exactly the picture of a capacitor: a change dV adds $+dQ$ on one face and $-dQ$ on the other, and the two faces are a distance W apart in a material of permittivity ε. The inside, between the faces, does not change. So we expect the parallel-plate formula. Let us derive it. The charge on the N side is

$$Q = qN_D S W_n = qS\,\frac{N_A N_D}{N_A + N_D}\,W,$$

and by (7.3), $W^2 = \dfrac{2\varepsilon}{q}\,\dfrac{N_A + N_D}{N_A N_D}\,(V_d - V)$. Since $Q \propto W \propto \sqrt{V_d - V}$, $dQ/dV = -Q/[2(V_d -$

$V)]$, and replacing $V_d - V$ by its expression in terms of W:

$$\boxed{C_T = \left|\frac{dQ}{dV}\right| = \frac{\varepsilon S}{W}}. \tag{7.5}$$

It holds for any doping, not only for the one-sided junction: the SCR behaves as a capacitor whose thickness is its own width, a thickness **that changes with the voltage**.

The one-sided junction. With (7.4),

$$\boxed{C_T = \frac{C_{T0}}{\sqrt{1 - V/V_d}}} \qquad \text{with} \qquad C_{T0} = S\sqrt{\frac{\varepsilon q N_D}{2V_d}}. \tag{7.6}$$

C_T decreases in reverse bias, as the plates move apart, and increases in forward bias, as they come closer (Figure 7.2a).

EXAMPLE 7.2 — THE CAPACITANCE OF THE REFERENCE DIODE

With $S = 10^{-4}$ cm², $\varepsilon = 1.036 \times 10^{-12}$ F/cm, $N_D = 10^{16}\,\text{cm}^{-3}$ and $V_d = 0.835$ V:

$$C_{T0} = 10^{-4}\sqrt{\frac{1.036 \times 10^{-12} \times 1.602 \times 10^{-19} \times 10^{16}}{2 \times 0.835}} = 3.15\ \text{pF}.$$

The parallel-plate formula $\varepsilon S/W$ with $W = 0.330$ µm gives 3.14 pF; the small difference is the thin P part, neglected in (7.6). Across the bias range, from (7.4) and (7.6):

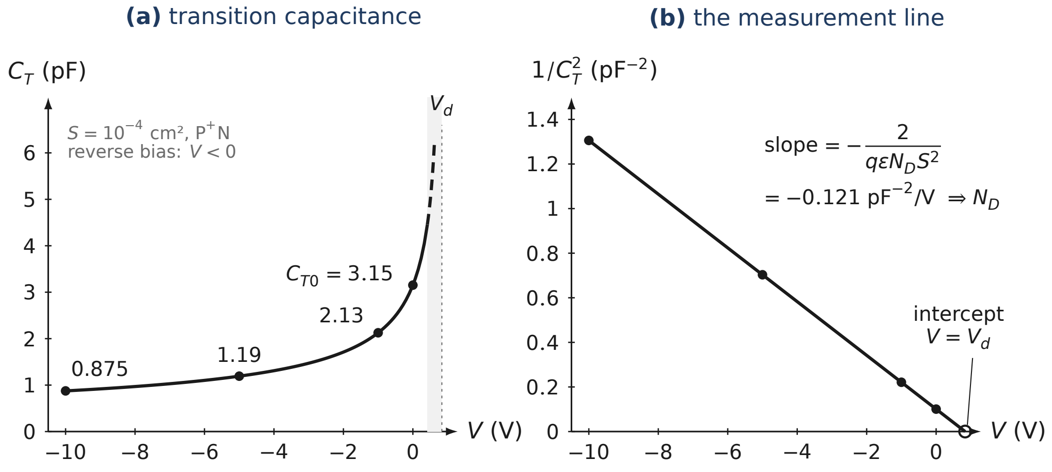
V (V)	−10	−5	−1	0	+0.2	+0.4
W (µm)	1.19	0.87	0.49	0.33	0.29	0.24
C_T (pF)	0.87	1.19	2.12	3.15	3.6	4.4

A capacitance of a few picofarads for a 100 µm square: small, but not negligible at radio frequencies. Keep the order of magnitude in mind: Part B will find nanofarads.

A note on the permittivity. Some tables give $\varepsilon_r = 11.9$ for silicon. With that value (and V_d rounded to 0.83 V) the same calculation gives $C_{T0} = 3.18$ pF, 1 % more. This course keeps $\varepsilon_r = 11.7$, the value used since Chapter 1; the difference is smaller than the spread of real devices.

WHAT THE FORMULA ASSUMES

The depletion approximation of Chapter 5: an SCR completely empty of carriers, with sharp edges. It is very good in reverse bias. In forward bias, close to V_d, the SCR is thin, the edges are no longer sharp on the scale of W, and many carriers cross it: (7.6) would give an infinite C_T at $V = V_d$, which is not physical. In practice (7.6) is used up to about $V_d/2$; above that, C_D dominates anyway (Section 7.8).


(a) transition capacitance

(b) the measurement line

Figure 7.2 *– The transition capacitance of the reference diode ($S = 10^{-4}$ cm², $N_D = 10^{16}\,\mathrm{cm}^{-3}$, $V_d = 0.835$ V), computed from (7.6); the dots are the values of Example 7.2, in pF. (a) $C_T(V)$ rises steeply as V approaches V_d. In the grey band, from $V_d/2$ to V_d, the depletion approximation is doubtful and the curve is dashed (see the box below). (b) $1/C_T^2$ is a straight line in V (7.7): its slope gives N_D, and it reaches zero at $V = V_d$ (open circle).*

7.4 Measuring a junction with $1/C_T^2$

THE QUESTION

Can a capacitance measurement tell us the doping and the built-in voltage of a junction?

Square and invert (7.6):

$$\boxed{\frac{1}{C_T^2} = \frac{2}{q\varepsilon N_D S^2}\,(V_d - V)}. \tag{7.7}$$

Plotted against V, $1/C_T^2$ is a **straight line** (Figure 7.2b). Its slope, $-2/(q\varepsilon N_D S^2)$, gives the doping of the lightly doped side; it crosses the axis $1/C_T^2 = 0$ at $V = V_d$. This $C(V)$ measurement is the standard way of measuring doping and built-in voltage in a laboratory: it needs only a capacitance meter, which applies a small alternating voltage on top of the reverse bias.

The line is straight only for uniform doping. For a non-uniform $N_D(x)$, the local slope of $1/C^2$ gives the doping at the edge of the SCR, at the depth $W = \varepsilon S/C$. Sweeping the reverse voltage moves this edge through the material and draws the **doping profile**: the same measurement turns into a depth probe.

EXAMPLE 7.3 — THE MEASUREMENT LINE OF THE REFERENCE DIODE

The slope of (7.7) is $2/(q\varepsilon N_D S^2) = 1.21 \times 10^{23}$ F^{-2}/V, that is, 0.121 pF^{-2} per volt. At $V = 0$, $1/C_{T0}^2 = 0.101$ pF^{-2}, and $0.101/0.121 = 0.835$ V: the line reaches zero at $V = V_d$, as it must. Problem 7.10 of the exercise collection goes the other way, from measured capacitances to N_D and V_d.

7.5 Engineering $C(V)$: the varicap

THE QUESTION

A capacitance that changes with a voltage: is that a defect or a resource?

For a circuit that must stay linear, $C_T(V)$ is a nuisance. For a circuit that must be **tuned**, it is a resource: a reverse-biased diode is a capacitor whose value is set by a DC voltage, with no moving part. A diode designed for this is a **varicap** (or varactor). It is always used in reverse bias, where it draws only the tiny current $-I_0$.

The exponent depends on the doping profile. The general law is

$$C \propto (V_d - V)^{-m}, \tag{7.8}$$

with $m = 1/2$ for the abrupt junction of this course, $m = 1/3$ for a linearly graded junction (the doping changes sign linearly across the junction), and m larger than 1, up to about 2, for a **hyperabrupt** junction, whose doping decreases away from the junction (Figure 7.4a). The larger m, the wider the range of capacitance for a given range of voltage.

Tuning a circuit. With an inductance L, the varicap forms a resonant circuit of frequency

$$f_0 = \frac{1}{2\pi\sqrt{LC_T(V)}} \ \propto\ (V_d - V)^{m/2}. \tag{7.9}$$

A hyperabrupt diode with $m = 2$ gives $f_0 \propto (V_d - V)$: the frequency is **linear in the tuning voltage**. (Figure 7.4b). This is how radio and television receivers, and the voltage-controlled oscillators of phase-locked loops, are tuned.

A varicap in its circuit. The diode must be held in reverse bias by a DC voltage, while the tuned circuit sees only its capacitance. Figure 7.3 shows the usual arrangement. The tuning voltage V_R reaches node A through a large resistor R_B. The varicap goes from ground to A, **cathode at A**: A is positive, so the diode is reverse biased by V_R and draws only $-I_0$; no current flows in R_B, and the full V_R appears across the diode. A coupling capacitor C_c links A to node B, where the inductance L goes to ground and the radio-frequency signal v_{ac} is taken. C_c blocks the DC, so the inductance, a short circuit for DC, does not short-circuit V_R. For the signal, R_B is large enough to be an open circuit, and $C_c \gg C_T$ is nearly a short circuit: the tank is L with C_T in parallel, the series combination $C_T C_c/(C_T + C_c) \simeq C_T$. Turning V_R tunes f_0. The signal must stay small compared with V_R: a swing that brings the diode near forward bias would change C_T during the cycle and, beyond, make it conduct.

EXAMPLE 7.4 — TUNING WITH THE REFERENCE DIODE

With $L = 10\ \mu\text{H}$ and $C_c = 1$ nF ($\gg C_T$), (7.9) gives $f_0 = 34.5$ MHz at $V_R = 1$ V ($C_T = 2.13$ pF) and 53.8 MHz at $V_R = 10$ V (0.875 pF). Between these voltages, C changes by a factor 2.4 and f_0 by $\sqrt{2.4} = 1.56$. A hyperabrupt diode would cover a far wider range for the same voltages.

Why only in reverse bias? A forward-biased diode also has a capacitance, and a much larger one. But it comes with a resistor. Its small-signal model (Section 7.9) is the dynamic resistance $r_d = V_T/I$ in parallel with C_D (plus C_T), and both are set by the same current: $r_d C_D = \bar{\tau}$. Two consequences rule it out for tuning.

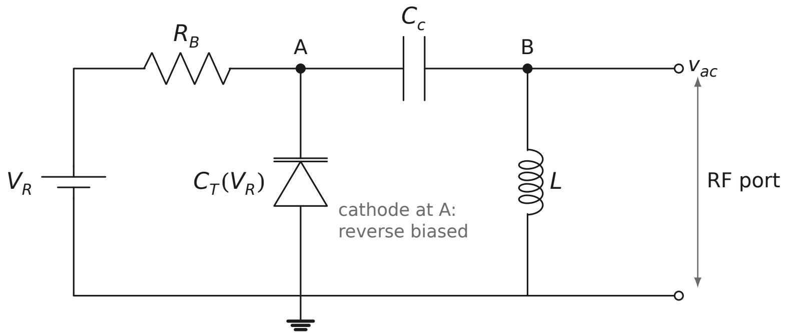
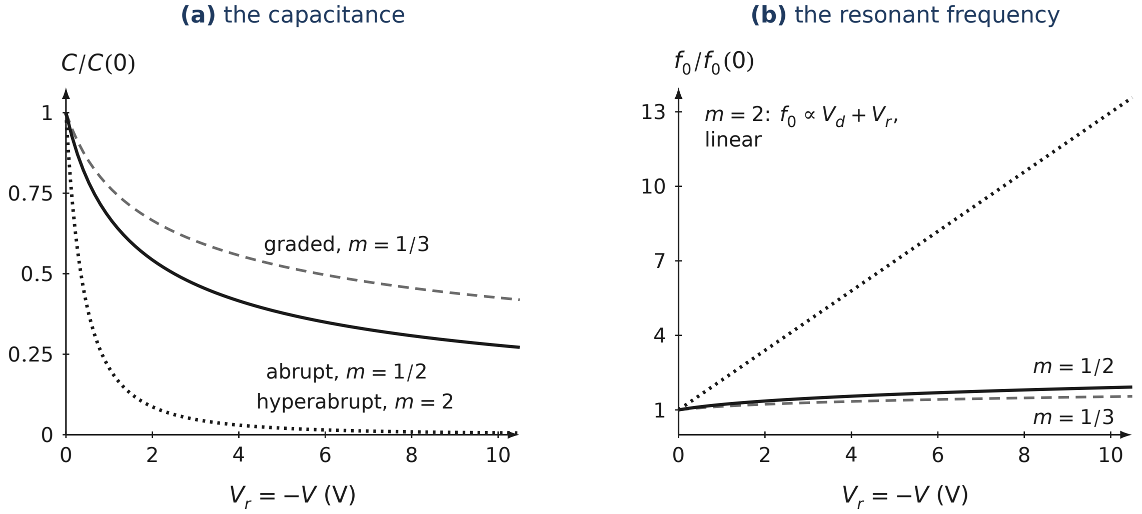


Figure 7.3 *– A varicap tuning circuit. The DC tuning voltage V_R reaches node A through R_B and reverse biases the varicap (ground to A, cathode at A). The coupling capacitor C_c (from A to B) blocks the DC; the inductance L (from B to ground) forms the tank with $C_T(V_R)$; the radio-frequency port v_{ac} is at B. For the signal: R_B open, C_c short, tank = $L \parallel C_T$.*


(a) the capacitance

(b) the resonant frequency

Figure 7.4 *– The varicap laws, computed, linear scales, against the reverse voltage $V_r = -V$, for three doping profiles: graded ($m = 1/3$, grey dashed), abrupt ($m = 1/2$, solid) and hyperabrupt ($m = 2$, dotted); $V_d = 0.835$ V for all three, to compare the exponents only. (a) $C/C(0) = (1 + V_r/V_d)^{-m}$: the larger m, the faster the capacitance falls. (b) The resonant frequency $f_0/f_0(0) = (1 + V_r/V_d)^{m/2}$ with a fixed L: a straight line for $m = 2$, a slow rise for the abrupt and graded junctions.*

- **The resistor kills the resonance.** At 1 mA, $r_d = 26\ \Omega$. Across a tank of $L = 0.25$ µH at 100 MHz, whose reactance is $\omega L = 157\ \Omega$, a parallel 26 Ω nearly short-circuits the circuit: the quality factor $r_d/\omega L$ falls to about 0.2, and there is no resonance left.
- **The capacitance is not set by a voltage alone.** $C_D \propto I$, and I depends exponentially on V and on the temperature: the tuning would drift with every change of current.

In reverse bias almost no current flows: r_d is practically infinite and C_T is a clean, nearly lossless capacitor, set by a DC voltage alone. Its values, a few picofarads, are exactly those that radio-frequency tanks need: $f_0 = 1/(2\pi\sqrt{LC})$ gives about 100 MHz for 10 pF with 0.25 µH, and a 10:1 swing of C gives a $\sqrt{10} \approx 3:1$ swing of f_0.

WHAT PART A HAS ESTABLISHED

- A change of voltage uncovers or covers again ions at the two edges of the SCR, a distance W apart: the SCR is a parallel-plate capacitor, $C_T = \varepsilon S/W$.
- For a one-sided junction, $C_T = C_{T0}/\sqrt{1 - V/V_d}$ with $C_{T0} = S\sqrt{\varepsilon q N_D/2V_d} = 3.15$ pF for the reference diode.
- $1/C_T^2$ is linear in V: its slope gives N_D, its intercept V_d; for a non-uniform doping, it gives the profile.
- The varicap, always in reverse bias, uses $C \propto (V_d - V)^{-m}$ to tune circuits with a voltage; a resistor brings the DC bias, a coupling capacitor isolates it from the tank.

PART B

The diffusion capacitance

The stored minority charge, $Q = \bar{\tau} I$, $C_D = (q/kT)\,\bar{\tau} I$, comparison with C_T

7.6 The stored minority charge

THE QUESTION

In forward bias, how much minority charge is stored in the neutral regions?

The holes of the N region. Take the junction in forward bias, $0 < V < V_d$, at low injection (Chapter 6). In the long N region of the reference diode, the hole profile is the exponential of Chapter 6, from $p_n(x_n) = p_{n0}\mathrm{e}^{qV/kT}$ at the edge of the SCR back to p_{n0} over a few diffusion lengths (Figure 7.5):

$$\Delta p(x) = p_{n0}\left(\mathrm{e}^{qV/kT} - 1\right)\mathrm{e}^{-(x-x_n)/L_p}.$$

Do not confuse the two values at the edge: the *density* is $p_n(x_n) = p_{n0}\mathrm{e}^{qV/kT}$, the *excess* is $\Delta p_n(x_n) = p_{n0}(\mathrm{e}^{qV/kT} - 1)$; only the excess counts as stored charge, since p_{n0} is there at equilibrium too. The excess holes carry a charge

$$Q_p = qS\int_{x_n}^{\infty} \Delta p(x)\,dx = qS\,L_p\,p_{n0}\left(\mathrm{e}^{qV/kT} - 1\right), \tag{7.10}$$

the area under the exponential times qS: the area of an exponential is its height times its decay length, as if the excess filled a rectangle of width L_p. The -1 matters only near $V = 0$: in forward bias at useful currents $\mathrm{e}^{qV/kT} \gg 1$ and $Q_p \simeq qSL_p p_{n0}\mathrm{e}^{qV/kT}$, the form often quoted; but only (7.10) gives $Q_p = 0$ at equilibrium and a small deficit, $Q_p \to -qSL_p p_{n0}$, in reverse bias, where the edge is emptied. The region stays neutral: an equal excess of electrons, supplied by the contact, sits with the holes. The stored charge is therefore not a charge separated by a distance, as in the SCR; it is a quantity of injected carriers, which must be supplied to raise V and removed to lower it.

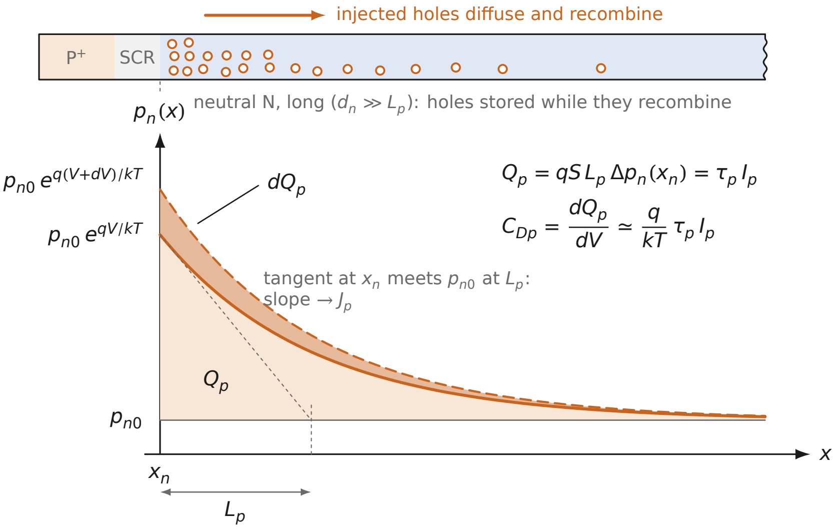

injected holes diffuse and recombine
P+
SCR
neutral N, long ($d_n \gg L_p$): holes stored while they recombine
$p_n(x)$
$p_{n0}\, e^{q(V+dV)/kT}$
$p_{n0}\, e^{qV/kT}$
dQ_p
$Q_p = qS\,L_p\,\Delta p_n(x_n) = \tau_p I_p$
$C_{Dp} = \frac{dQ_p}{dV} \simeq \frac{q}{kT}\tau_p I_p$
tangent at x_n meets p_{n0} at L_p:
slope → J_p
Q_p
p_{n0}
x
x_n
L_p

Figure 7.5 *– The stored minority charge in forward bias (P^+N long diode, schematic). Top: the holes injected into the N region (open circles) are stored there while they diffuse away from the SCR and recombine; their density falls over a few L_p (the wavy end: the region goes on, much longer than L_p). Bottom, aligned: the hole profile $p_n(x)$, linear scale, from $p_{n0}e^{qV/kT}$ at x_n back to p_{n0}. The light area above p_{n0}, times qS, is the stored charge Q_p; it equals the rectangle of height $\Delta p_n(x_n)$ and width L_p, where the tangent at x_n meets p_{n0}; the slope at x_n is the hole current. Raising V to $V + dV$ raises the whole profile (dashed) and adds the darker area dQ_p, which gives the hole part C_{Dp} of the diffusion capacitance. Not to scale: p_{n0} is exaggerated so that it is visible.*

The charge is proportional to the current. The hole current at x_n is the slope of the same exponential (Chapter 6):

$$I_p = qSD_p\,\frac{p_{n0}}{L_p}\left(e^{qV/kT} - 1\right).$$

The two expressions have the same voltage dependence. Their ratio is a time, L_p^2/D_p, which is the lifetime:

$$\boxed{Q_p = \tau_p I_p} \qquad \text{since} \qquad \frac{L_p^2}{D_p} = \tau_p. \tag{7.11}$$

The stored charge is the current multiplied by the time each hole spends in the region, like the water in a pipe, equal to the flow times the time of passage. Here a hole stays until it recombines, a lifetime on average. The same result follows from a balance, without any profile: in steady state, the holes stored must be renewed at the rate they recombine, Q_p/τ_p, and this rate is the hole current supplied at the edge, I_p.

The electrons of the P region give in the same way $Q_n = \tau_n I_n$. The total stored charge is

$$Q = \tau_p I_p + \tau_n I_n = \bar{\tau}\, I, \qquad \bar{\tau} = \frac{\tau_p I_p + \tau_n I_n}{I_p + I_n}, \tag{7.12}$$

where $\bar{\tau}$ is the mean of the two lifetimes weighted by the shares of the current. The shares are those of J_0 (Chapter 6), independent of V, so $\bar{\tau}$ is a constant of the diode.

EXAMPLE 7.5 — THE STORED CHARGE OF THE REFERENCE DIODE

Holes carry 85.4 % of the current and live $\tau_p = 10$ µs; electrons carry 14.6 % and live $\tau_n = 0.1$ µs. So $\bar{\tau} = 0.854 \times 10 + 0.146 \times 0.1 = 8.56$ µs: the lightly doped side, which carries the current, also stores the charge. At $I = 10$ µA, $Q = 8.56$ µs × 10 µA = 86 pC, of which the electrons hold only 0.15 pC.

THE SHORT DIODE STORES LITTLE

In the short diode of Chapter 6 (Section 6.11), the injected holes reach the contact before they recombine. The profile is a triangle of base d_n, $Q_p = \frac{1}{2} q S d_n \Delta p_n(x_n)$, and the ratio Q_p/I_p is the **transit time** $d_n^2/2D_p = 41.7$ ns instead of the lifetime: for the same current the 10 µm region stores 240 times less charge (Problem 7.13 of the exercise collection). A short region stores little because its carriers leave by the contact before they can accumulate. This is the key to fast diodes, and it will be the key to the base of the transistor (Chapter 8).

7.7 The diffusion capacitance C_D

THE QUESTION

What capacitance does the stored minority charge give?

From (7.12) and $I = I_0(e^{qV/kT} - 1)$, $dI/dV = (q/kT)\, I_0 e^{qV/kT} = (q/kT)(I + I_0)$, exactly. In forward bias $I \gg I_0$ (already at $V = 0.2$ V, $I = 2300\, I_0$), so:

$$\boxed{C_D = \frac{dQ}{dV} = \frac{q}{kT}\,\bar{\tau}\, I}. \tag{7.13}$$

Region by region, per unit area, the derivative of (7.10) gives, with no approximation,

$$\frac{C_{Dp}}{S} = \frac{1}{S}\frac{dQ_p}{dV} = \frac{q^2 L_p p_{n0}}{kT} e^{qV/kT}, \qquad \frac{C_{Dn}}{S} = \frac{q^2 L_n n_{p0}}{kT} e^{qV/kT}.$$

With $p_{n0} = n_i^2/N_D$ and $n_{p0} = n_i^2/N_A$ their sum reads

$$\frac{C_D}{S} = \frac{q^2 n_i^2}{kT}\left(\frac{L_n}{N_A} + \frac{L_p}{N_D}\right) e^{qV/kT}, \tag{7.14}$$

in which the two terms are the stores of the P and N regions. Here no −1 appears, and none is dropped: the −1 of Q is a constant, and its derivative is zero. C_D **is proportional to the current**: it is negligible in reverse bias and at small forward bias, and it grows by a factor 10 every 60 mV, like the current.

EXAMPLE 7.6 — THE DIFFUSION CAPACITANCE OF THE REFERENCE DIODE

With $\bar{\tau} = 8.56$ µs and $kT/q = 25.9$ mV, $C_D = I \times 0.330$ nF/µA. The currents stay at or below 0.1 mA, where low injection holds for $S = 10^{-4}$ cm² (Chapter 6: it ends near 0.72 V, about 0.2 mA):

I	1 µA	10 µA	100 µA
V (V)	0.578	0.637	0.697
$Q = \bar{\tau} I$	8.6 pC	86 pC	0.86 nC
C_D	0.33 nF	3.3 nF	33 nF
C_T	5.7 pF	6.5 pF	7.7 pF (doubtful)
C_D/C_T	58	510	4300

C_T is counted in picofarads, C_D in nanofarads: three orders of magnitude apart at a modest 0.1 mA. (Using the hole lifetime 10 µs for the whole current would overestimate C_D by 17 %: the electrons, which carry 15 % of the current, live only 0.1 µs.)

WHAT THE CHARGE-CONTROL PICTURE SIMPLIFIES

(7.13) assumes that the stored charge follows the voltage, keeping at every instant the steady-state profile: it is a quasi-static picture. A full small-signal solution of the diffusion equation gives, at low frequency, a smaller value: about 1/2 of (7.13) for a long region (and 2/3 for a short one), because part of the change of stored charge recombines instead of flowing back through the junction. The dependence $C_D \propto I$ and the order of magnitude are unchanged; we keep (7.13), which is the usual engineering formula.

7.8 C_T and C_D across the bias range

THE QUESTION

Which of the two capacitances matters at a given voltage?

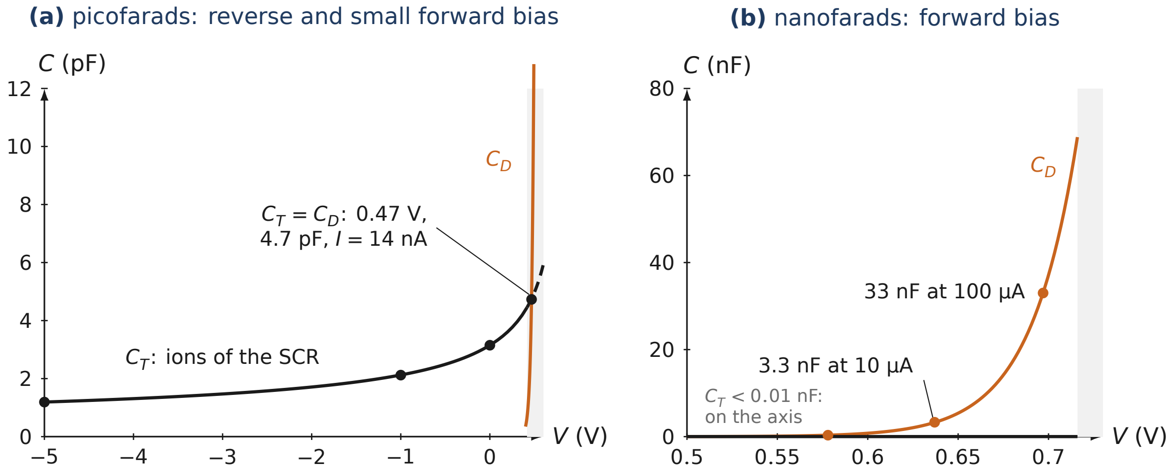
The two laws are very different (Figure 7.6). C_T varies slowly, as $(V_d - V)^{-1/2}$: by a factor 3.7 between −5 and +0.4 V. C_D varies like the current, by a factor 10 every 60 mV. The two cannot even be drawn on one linear scale: Figure 7.6a is in picofarads, Figure 7.6b in nanofarads. The result is a sharp division of roles:

- **in reverse bias and at small forward bias**, C_T is alone;
- **in forward bias at useful currents**, C_D dominates, by far.

For the reference diode the two are equal near $V = 0.47$ V, at about 4.7 pF, where the current is only 14 nA. This crossing lies a little above $V_d/2$, where (7.6) is only an estimate (Section 7.3); the uncertainty on C_T moves the crossing by a few millivolts only, because C_D changes by a factor 10 every 60 mV.

WHAT PART B HAS ESTABLISHED

- In forward bias the neutral regions store an injected minority charge, the area of the exponential profiles of Chapter 6: $Q_p = qSL_p p_{n0}(e^{qV/kT} - 1)$.
- This charge is the current multiplied by the time a carrier stays: $Q = \bar{\tau} I$, with the lifetimes in a long diode ($\bar{\tau} = 8.56$ µs here) and the much shorter transit time


(a) picofarads: reverse and small forward bias

(b) nanofarads: forward bias

Figure 7.6 *– C_T (black) and C_D (orange) of the long reference diode ($S = 10^{-4}$ cm²), computed from (7.6) and from (7.13) with $\bar{\tau} = 8.56$ µs; linear scales. (a) In picofarads, from −5 to +0.6 V: C_T alone in reverse bias (dots: −5, −1 and 0 V); C_D rises from nothing and crosses C_T at 0.47 V (4.7 pF, I = 14 nA), then leaves the scale within a few millivolts. In the grey band, above $V_d/2$, the depletion approximation behind C_T is doubtful (dashed). (b) In nanofarads, from 0.5 to 0.72 V, where low injection ends (grey): C_D at 1, 10 and 100 µA (0.33, 3.3 and 33 nF); C_T, below 0.01 nF, lies on the axis.*

$d^2/2D$ in a short one.

- $C_D = (q/kT)\,\bar{\tau}\,I$ is proportional to the current: nanofarads, against picofarads for C_T. C_T rules in reverse bias, C_D in forward bias; they cross near 0.47 V for the reference diode.

PART C

Why it matters

The small-signal diode, switching, transistors and logic circuits

7.9 The small-signal diode

THE QUESTION

How does a forward-biased diode respond to a small alternating voltage superposed on its bias?

The dynamic resistance. Around an operating point Q of current I, a small voltage δv produces a current δi given by the slope of the characteristic (Figure 7.7a):

$$\frac{1}{r_d} = \frac{dI}{dV} = \frac{q}{kT}\,I \qquad \Longrightarrow \qquad r_d = \frac{kT}{qI}. \tag{7.15}$$

At 0.1 mA, $r_d = 259\ \Omega$, whatever the diode: the slope of an exponential depends only on the current.

The capacitances. At the same time, the voltage changes both stores of charge, which draws the currents $C_T\, d\delta v/dt$ and $C_D\, d\delta v/dt$. The three effects are in parallel, since they share the same voltage (Figure 7.7b):

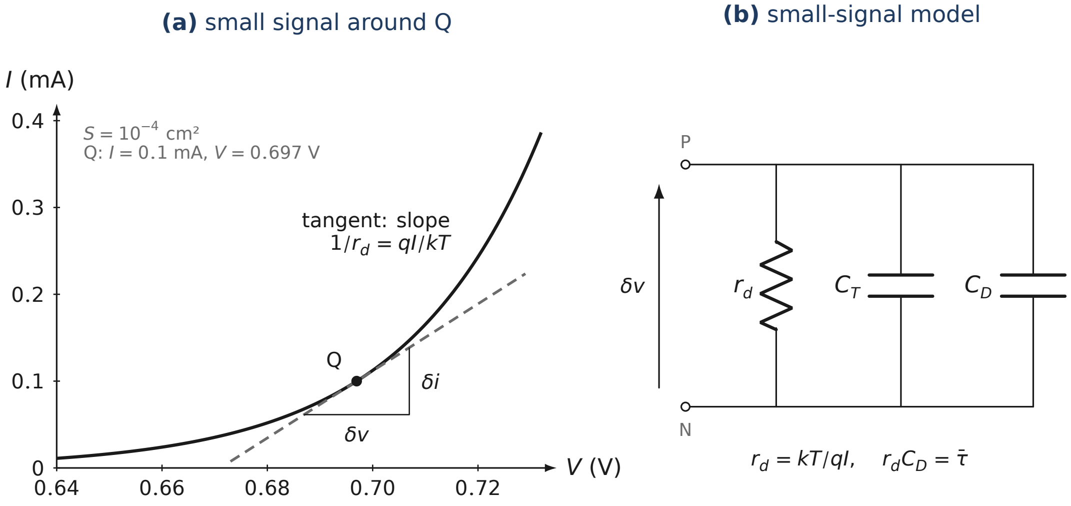
$$\text{small-signal diode} \;=\; r_d \;\|\; (C_T + C_D). \tag{7.16}$$

A series resistance r_s (neutral regions and contacts, Chapter 6) completes the model at large currents.

A time constant independent of the current. In forward bias, where C_D dominates,

$$r_d C_D = \frac{kT}{qI} \times \frac{q}{kT}\,\bar{\tau} I = \bar{\tau}. \tag{7.17}$$

The diode stops following the signal near $f = 1/(2\pi\bar{\tau})$, whatever its current: a larger current lowers r_d but raises C_D in the same proportion. Only a shorter storage time makes the diode faster: a shorter lifetime, or a short neutral region where the transit time replaces the lifetime.


(a) small signal around Q

(b) small-signal model

Figure 7.7 *– The small-signal diode. (a) The characteristic of the reference diode ($S = 10^{-4}$ cm²), computed, linear scale, around the operating point Q ($I = 0.1$ mA, $V = 0.697$ V). A small step δv (here 20 mV, drawn large) gives a current step δi along the tangent (dashed), of slope $1/r_d = qI/kT$. (b) The small-signal model between the P and N terminals: r_d in parallel with C_T and C_D; the series resistance r_s of the neutral regions and contacts is not drawn.*

EXAMPLE 7.7 — THE SMALL-SIGNAL MODEL OF THE REFERENCE DIODE

I	1 µA	10 µA	0.1 mA
V (V)	0.578	0.637	0.697
r_d	25.9 kΩ	2.59 kΩ	259 Ω
C_D	0.33 nF	3.3 nF	33 nF
C_T	5.7 pF	6.5 pF	($\ll C_D$)

In each column $r_d C_D = 8.56$ µs, and the corner frequency of the diffusion part is $1/(2\pi \times 8.56\ \mu\text{s}) = 18.6$ kHz: a long silicon diode stops following a signal in the audio range already. The short diode of Chapter 6 (transit time 36 ns) would reach 4.4 MHz.

7.10 Switching a diode off

THE QUESTION

A diode carries a forward current I_F. The circuit suddenly reverses the voltage. Does the current stop at once?

No: the stored charge must leave first. Take the circuit of Figure 7.8a: a source v_s in series with a resistor R, on the P side, the N side grounded; v_s jumps from $+V_F$ to $-V_R$ at $t = 0$. At $t = 0$ the neutral regions contain $Q = \bar{\tau} I_F$ of injected carriers. The edge densities, and therefore the junction voltage, stay set by this charge (Chapter 6: $p_n(x_n) = p_{n0}\mathrm{e}^{qV/kT}$): while carriers remain in excess at the edges, the junction is still forward biased, whatever the source does. The external circuit therefore sees a **reverse current** $-I_R$, set by the source and the resistor, which pulls the stored carriers back across the SCR (Figure 7.8). In the hole profile (Figure 7.8b) the slope at x_n changes sign: the holes near the edge now flow back towards the SCR, while those further away keep diffusing outwards and recombining. This lasts a **storage time** t_s, until the edge densities fall to their equilibrium values. Only then does the junction voltage turn negative; the SCR widens, the transition capacitance charges, and the current decays to $-I_0$.

An estimate: the charge-control equation. The stored charge changes for two reasons: the terminal current brings carriers in (or takes them out), and recombination removes $Q/\bar{\tau}$ per second:

$$\frac{dQ}{dt} = i(t) - \frac{Q}{\bar{\tau}}. \tag{7.18}$$

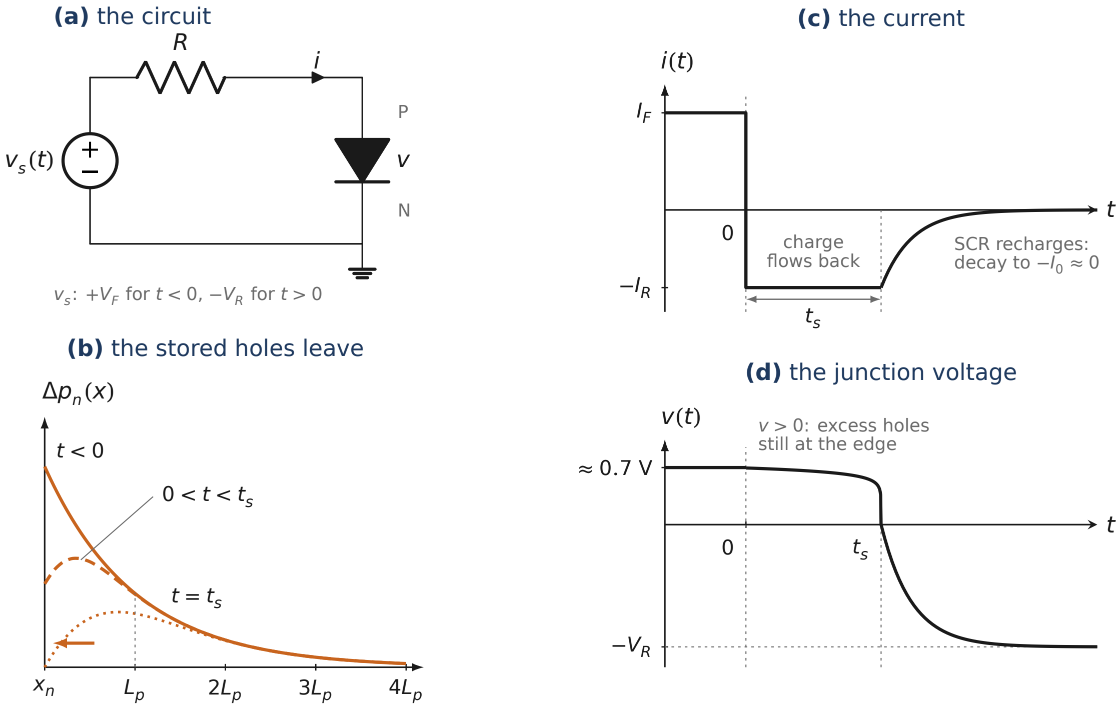
In steady state $dQ/dt = 0$ and $Q = \bar{\tau} I_F$, as in Part B. After $t = 0$, $i = -I_R$, and the solution starting from $\bar{\tau} I_F$ is $Q(t) = -\bar{\tau} I_R + \bar{\tau}(I_F + I_R)\mathrm{e}^{-t/\bar{\tau}}$: the charge would vanish after

$$t_Q = \bar{\tau}\ln\left(1 + \frac{I_F}{I_R}\right). \tag{7.19}$$

Recombination helps: the charge disappears faster than Q/I_R. The junction voltage, however, is set by the edge density, not by the total charge, and the edge empties first (Figure 7.8b): the full solution of the diffusion equation gives a storage time t_s about three times shorter than t_Q ($0.23\,\bar{\tau}$ against $0.69\,\bar{\tau}$ for $I_F = I_R$). Either way **the scale is the lifetime**.

EXAMPLE 7.8 — STORAGE TIME

With $I_F = I_R = 0.1$ mA, the reference diode stores $Q = 0.86$ nC. By (7.19) it would all be gone after $t_Q = 8.56\times\ln 2 = 5.9$ µs; the edge empties first, after $t_s \approx 0.23\times 8.56 = 2$ µs. The short diode of Chapter 6, whose carriers leave by the contact after 36 ns, would switch in tens of nanoseconds. This is why **fast-recovery diodes** have thin neutral regions or a lifetime reduced on purpose (by adding recombination centres such as gold or platinum, Chapter 4), and why the fastest rectifiers are Schottky diodes, metal–semiconductor junctions that inject almost no minority carriers.


(a) the circuit

(c) the current

(b) the stored holes leave

(d) the junction voltage

Figure 7.8 – *Switching a diode off ((a), (c), (d): sketches, arbitrary units). (a) The circuit: the source jumps from* $+V_F$ *to* $-V_R$ *at* $t = 0$. *(b) The excess holes in the long N region, computed from the diffusion equation with recombination (x in units of* L_p*) and the current at* x_n *held at* $-I_R = -0.8\,I_F$*: before* $t = 0$ *(the exponential), during the storage time (dashed, at* $0.3\,t_s$*: the edge excess falls and the slope at* x_n *is reversed, the same on every curve, so holes flow back towards the SCR, arrow), and at* $t = t_s$ *(dotted: no excess left at the edge). Every profile stays below the initial one; far from the edge the holes keep recombining. Here* $t_s = 0.29\,\tau_p$*, less than the time* $t_Q = \tau_p \ln(1 + I_F/I_R) = 0.81\,\tau_p$ *at which the whole stored charge would be gone. (c) The current:* I_F*, then* $-I_R$ *during* t_s *while the stored charge is removed, then a decay to* $-I_0 \approx 0$ *as the SCR recharges. (d) The junction voltage stays positive during* t_s*, because excess holes remain at the edge; then it turns negative and tends to* $-V_R$.

7.11 Transistors and logic circuits

THE QUESTION

Where do these two capacitances appear in real circuits?

The bipolar transistor. Chapter 6 ended with a short N region between two junctions. In a transistor, the forward-biased emitter–base junction injects carriers into a thin base, a short region, where they are stored while they cross it: a diffusion capacitance, set by the base transit time $W_B^2/2D$ (D is the diffusion coefficient of the minority carriers of the base), of exactly the form of the short diode. The reverse-biased collector–base junction gives a transition capacitance. A thin base is the key to a fast transistor, for the same reason as a thin N region for a fast diode. Chapter 8 ends with these two capacitance effects.

MOS logic circuits. In an integrated circuit, the source and drain of every MOS transistor are PN junctions with the substrate, held in reverse bias. Their transition capacitances, with those of the gates and the wires, form the load C that each logic gate must charge or discharge. With a current I, a voltage swing ΔV takes a time

$$t \approx \frac{C\,\Delta V}{I}. \tag{7.20}$$

A 20 fF node charged by 50 µA over 1 V takes 0.4 ns. Smaller junctions mean smaller capacitances and faster circuits: one of the reasons transistors keep shrinking.

WHAT PART C HAS ESTABLISHED

- The small-signal diode is $r_d \parallel (C_T+C_D)$, with $r_d = kT/qI$; in forward bias $r_d C_D = \bar{\tau}$, independent of the current (18.6 kHz corner for the reference diode).
- Switching a diode off takes a storage time of the order of the lifetime (charge control: $dQ/dt = i - Q/\bar{\tau}$); fast diodes have thin regions or short lifetimes.
- In a bipolar transistor, the base stores charge (a diffusion capacitance at the emitter junction) and the collector junction adds a transition capacitance; in MOS logic, reverse-biased junctions are part of the load C of each gate, $t \approx C\Delta V/I$.

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Two stores**: the ions of the SCR (all voltages) and the injected minority carriers (forward bias). Capacitance $C = |dQ/dV|$ at the operating point.
- **Transition capacitance**: $C_T = \varepsilon S/W$, a parallel-plate capacitor of thickness $W(V)$; one-sided junction, $C_T = C_{T0}/\sqrt{1 - V/V_d}$, $C_{T0} = S\sqrt{\varepsilon q N_D/2V_d}$ (3.15 pF for the reference diode).
- **Measurement**: $1/C_T^2 = 2(V_d - V)/(q\varepsilon N_D S^2)$, a straight line: slope → N_D, intercept → V_d. **Varicap**: $C \propto (V_d - V)^{-m}$, $m = 1/3$, $1/2$, up to 2.
- **Stored charge**: $Q = \bar{\tau} I$; long region, the lifetime ($\bar{\tau} = 8.56$ µs for the reference diode); short region, the transit time $d^2/2D$, far shorter.
- **Diffusion capacitance**: $C_D = (q/kT)\,\bar{\tau} I$, proportional to the current: nanofarads against picofarads; $C_D \gg C_T$ in forward bias at useful currents.
- **Small signal**: $r_d = kT/qI$ in parallel with $C_T + C_D$; $r_d C_D = \bar{\tau}$. **Switching**: storage time of the order of $\bar{\tau}$, from $dQ/dt = i - Q/\bar{\tau}$.

Comprehension questions

No new physics here: answer from the figures and equations of the chapter. The answers are at the end.

Q1. Why is the capacitance defined as dQ/dV and not as Q/V for a junction?

Q2. When the reverse voltage increases, where are the new charges placed, and why does this make the SCR a parallel-plate capacitor?

Q3. Why does C_T decrease in reverse bias?

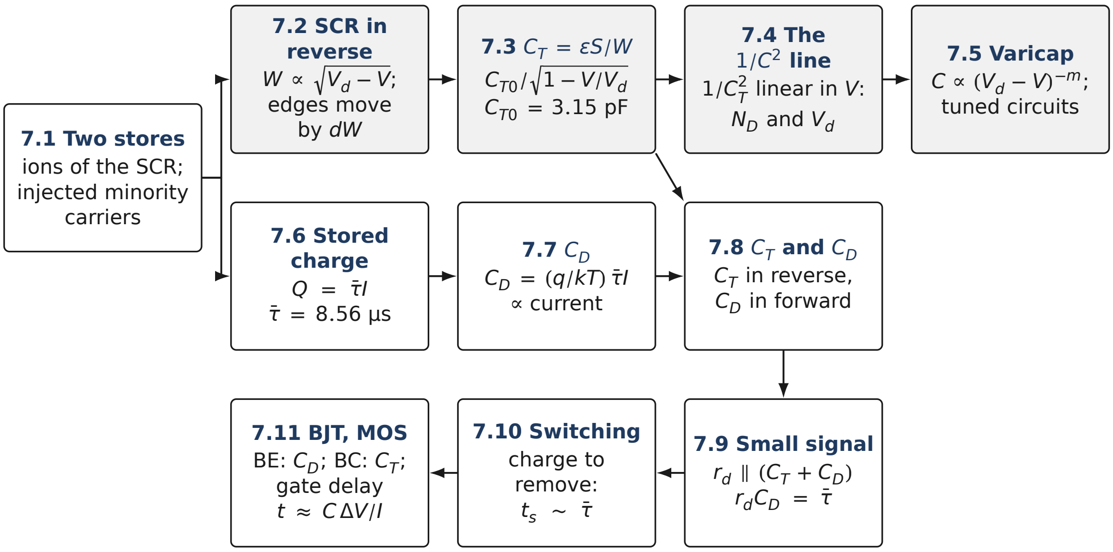

7.1 Two stores
ions of the SCR; injected minority carriers
7.2 SCR in reverse
$W \propto \sqrt{V_d - V}$; edges move by dW
7.3 $C_T = \varepsilon S/W$
$C_{T0}/\sqrt{1 - V/V_d}$
$C_{T0} = 3.15$ pF
7.4 The $1/C^2$ line
$1/C_T^2$ linear in V: N_D and V_d
7.5 Varicap
$C \propto (V_d - V)^{-m}$; tuned circuits
7.6 Stored charge
$Q = \bar{\tau} I$
$\bar{\tau} = 8.56$ µs
7.7 C_D
$C_D = (q/kT)\,\bar{\tau} I$
∝ current
7.8 C_T and C_D
C_T in reverse, C_D in forward
7.9 Small signal
$r_d \parallel (C_T + C_D)$
$r_d C_D = \bar{\tau}$
7.10 Switching
charge to remove: $t_s \sim \bar{\tau}$
7.11 BJT, MOS
BE: C_D; BC: C_T; gate delay $t \approx C\Delta V/I$

Figure 7.9 – *The logic of the chapter, section by section. The two stores (7.1) lead to Part A (top row, grey: the ions of the SCR, 7.2 to 7.5) and to Part B (middle row: the stored minority carriers, 7.6 to 7.8), which meet in the comparison of C_T and C_D. Part C (bottom row, read from right to left) uses both: the small-signal model, switching, and transistors and logic circuits.*

Q4. In a P$^+$N junction, which doping does C_T depend on? Why?

Q5. Why is $1/C_T^2$, and not C_T, plotted against V to analyse a measurement?

Q6. Why must a varicap be used in reverse bias?

Q7. In forward bias, the N region stays neutral. Then in what sense does it store a charge?

Q8. Why is the stored charge proportional to the current? What is the meaning of the proportionality factor?

Q9. Why does a short region store less charge than a long one, for the same current?

Q10. Why is C_D negligible in reverse bias?

Q11. Why is $r_d C_D$ independent of the current? What does it mean for the speed of a diode?

Q12. After a forward current is switched off, why does the diode voltage stay positive for a while?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 7, with their answers; the worked solutions are in the **Solutions** document. The 8 exercises that used to close this chapter are Problems 7.9 to 7.16 there; the other problems of the set (7.1 to 7.8) go further.

Answers

Q1. The charge of a junction is not proportional to V: what matters for a small signal is the change of charge for a small change of voltage, which depends on the operating point.

Q2. At the two edges of the SCR, $+dQ$ on the N side and $-dQ$ on the P side, a distance W apart, with nothing changed in between: exactly the charges added to the plates of a capacitor of thickness W.

Q3. The SCR widens as $\sqrt{V_d - V}$: the plates move apart.

Q4. On N_D, the lightly doped side: the SCR lies almost entirely in it, and its width is set by it.

Q5. Because $1/C_T^2$ is linear in V; a straight line gives the slope (N_D) and the intercept (V_d) directly and shows at once if the doping is not uniform.

Q6. In reverse bias it draws only the tiny current $-I_0$ and its capacitance is a well-defined C_T; in forward bias it would conduct, and C_D, proportional to a current that depends exponentially on V, would dominate.

Q7. It stores a quantity of injected carriers (holes, neutralised by extra electrons) that must be supplied to raise V and removed to lower it; it is a stored charge, not a separated one.

Q8. The charge is the area of the profile and the current its slope at the edge; both are proportional to $\mathrm{e}^{qV/kT} - 1$. The factor is the time a carrier spends in the region: the lifetime (long region, $L^2/D = \tau$) or the transit time (short region).

Q9. Its carriers reach the contact after a transit time $d^2/2D$, much shorter than the lifetime: they leave before they accumulate (41.7 ns against 10 µs for the holes of the reference material).

Q10. The edges are emptied: there is no injected charge, and the small changes of the minority densities, proportional to I_0, are negligible.

Q11. $r_d \propto 1/I$ and $C_D \propto I$. The product is $\bar{\tau}$: the speed of a diode in forward bias is set by the time its carriers stay (lifetime or transit time), not by its current.

Q12. Carriers in excess remain at the edges of the SCR; by the relation of Chapter 6, they fix a positive junction voltage until they have been removed.

References

- S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley, 2007 — Ch. 2: depletion capacitance, C–V profiling, diffusion capacitance, transient behaviour.
- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — Ch. 7: junction capacitance and varactor; Ch. 8: small-signal model and charge storage, diode switching transient.
- R. F. Pierret, *Semiconductor Device Fundamentals*, Addison-Wesley, 1996 — Ch. 7 and 8: C–V characteristics and small-signal admittance of the PN junction.

WHERE THIS LEADS

The three chapters on the PN junction are complete: equilibrium (Chapter 5), current under bias (Chapter 6), and stored charge (this chapter). The same ingredients, a step in the bands, injection at the edges of an SCR, diffusion across a short region and stored charge, are the building blocks of the bipolar transistor of Chapter 8, which ends with its two capacitance effects, and, beyond this course, of the MOS transistor and of optoelectronic devices.

SEMICONDUCTOR DEVICES PHYSICS

CHAPTER 8

The Bipolar Junction Transistor

Technology and the three design constraints · two junctions sharing a base · the physical picture · bands under bias · long base and short base · the two edge densities · the Ebers–Moll equations · I_C against V_{CB} and V_{CE} · the Early effect · capacitances

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WHERE WE ARE

Chapters 5, 6 and 7 built the PN junction step by step. At equilibrium (Chapter 5), a space-charge region (SCR) separates two neutral regions and the bands bend by qV_d. Under a voltage V (Chapter 6), the step becomes $q(V_d - V)$, the minority densities at the edges of the SCR are multiplied by $\mathrm{e}^{qV/kT}$, and the injected minority carriers diffuse across the neutral regions; in a **short** region their profile is a straight line, and its slope gives the current. Chapter 7 added the two stores of charge, the ions of the SCR (C_T) and the injected minority carriers ($C_D = (q/kT)\,\bar{\tau}_{tr} I$, with a transit time $d^2/2D$ for a short region).

Chapter 6 ended with a sketch: a short N region between two P regions, injected by one junction and emptied by the other. One junction controls the current of the other. This chapter builds that device completely. We study its mirror image, the **NPN transistor**, in which the roles of electrons and holes are exchanged: the carriers that cross the base are electrons. Nothing in the physics is new. What is new is that two junctions share one short region, and that the current of one is set by the voltage of the other.

CHAPTER OUTLINE

8.1 Technology and design constraints
Part A — The structure, the physical picture and the energy landscape
8.2 The transistor at equilibrium: two junctions sharing a base
8.3 How the two junctions work together: the physical picture
8.4 The band diagram at equilibrium
8.5 Forward-active bias: barriers, SCRs and fields
Part B — Carrier densities: long base and short base
8.6 Long base and short base, side by side
8.7 Several V_{BE}, several V_{CB}
8.8 From carrier transport to transistor action
Part C — Currents and the Ebers–Moll equations
8.9 Assumptions and conventions
8.10 The currents in forward-active operation
8.11 Both junctions at once: the Ebers–Moll equations
Part D — The characteristics
8.12 I_C against V_{CB}: the plateau
8.13 The BC junction forward biased: the knee
8.14 I_C against V_{CE}: a change of axis
8.15 The operating regions

Part E — The Early effect
8.16 Base-width modulation
8.17 The Early voltage and the output resistance
Part F — Capacitances
8.18 The two capacitances of the transistor
Summary · Comprehension questions · Exercises · Answers · References

LEARNING OBJECTIVES — WHAT THIS CHAPTER SETS OUT TO DO

1. Explain the **physical operation** of the bipolar transistor: injection by the emitter–base junction, transport across a short base, collection by the base–collector junction.
2. Derive its **electrical characteristics** from the carrier profiles.
3. Deduce the **Ebers–Moll model** from the same analysis.
4. Read the **characteristics** $I_C(V_{CB})$ and $I_C(V_{CE})$, the Early effect and the two capacitances from the two edge densities of the base.

LEARNING OUTCOMES — BY THE END OF THE CHAPTER YOU CAN

- draw the NPN transistor at equilibrium and in forward-active bias, as a cartoon and as a band diagram, and say what each junction does;
- draw the electron profile in a long and in a short base for any pair of voltages V_{BE}, V_{CB}, and read the currents from its slopes;
- explain why the emitter must be heavily doped, the base lightly doped, and the base short, with numbers;
- derive $I_C = I_S\,\mathrm{e}^{V_{BE}/V_T}$, the emitter efficiency, the base transport factor and β_F, then the Ebers–Moll equations, and say what each term means;
- explain the plateau of $I_C(V_{CB})$, the flow at $V_{CB} = 0$, the knee and the zero when the BC junction is forward biased, and obtain $I_C(V_{CE})$ by a change of axis;
- explain the Early effect with the ruler picture, and why all plateaus meet at $-V_A$;
- compute the diffusion capacitance of the EB junction and the transition capacitance of the BC junction, and explain why a short base is fast.

HOW THIS CHAPTER WORKS

- **The order of Chapters 5 and 6 again**: first the structure and the physical picture, then the bands, then the carrier densities, and only then the equations. The Ebers–Moll equations come at the end of a reasoning that you can follow with the figures, not at the beginning.
- **Two voltages, one per junction.** The transistor is described with V_{BE} (positive: EB forward) and V_{CB} (positive: BC reverse), each imposed by its own source, because the physics is told junction by junction. The BC junction is forward biased

when $V_{CB} < 0$. The common-emitter description used in electronics, with $V_{CE} = V_{CB} + V_{BE}$, is then a change of axis.

- **One reference transistor** (Table 8.1). Its emitter–base junction is the reference junction of Chapters 5 to 7, mirrored: the same dopings, the same $V_d = 0.835$ V and the same area. Every number in the chapter refers to it.

Symbol	Meaning	Reference transistor
N_E, N_B, N_C	dopings of emitter (N^+), base (P), collector (N)	10^{18}, 10^{16}, 10^{15} cm^{-3}
d_E, W_B, d_C	neutral widths of emitter, base, collector	0.5 µm, 1.0 µm (at $V_{CB} = 5$ V), 10 µm
S	area of the junctions	10^{-4} cm² (100 µm × 100 µm)
V_{dE}, V_{dC}	built-in voltages of the EB and BC junctions	0.835 V and 0.656 V
n_{B0}, p_{E0}, p_{C0}	equilibrium minority densities in B, E, C	10^4, 10^2, 10^5 cm^{-3}
V_{BE}, V_{CB}, V_{CE}	junction and terminal voltages (Figure 8.10)	$V_{CE} = V_{CB} + V_{BE}$
I_E, I_B, I_C	terminal currents: I_E out of E, I_B into B, I_C into C	$I_E = I_B + I_C$
V_T	thermal voltage kT/q	25.9 mV, so that $e^{qV/kT} = e^{V/V_T}$
I_S	saturation current of the transistor	5.6×10^{-14} A
τ_n, L_n	lifetime and diffusion length of electrons in the base	0.1 µs, 18.7 µm
τ_p, L_p	lifetime and diffusion length of holes in the emitter and in the collector	emitter 0.1 µs, 11 µm; collector 10 µs, 110 µm
τ_B	base transit time $W_B^2/2D_n$	0.143 ns
α_F, β_F; α_R, β_R	forward and reverse current gains	0.992 and 121; 0.744 and 2.9

Table 8.1 – *Notation added in this chapter; the others are those of Chapters 5 to 7. Silicon at 300 K,* $n_i = 10^{10}$ cm^{-3}, $\varepsilon = 11.7\,\varepsilon_0 = 1.036 \times 10^{-12}$ *F/cm,* $D_n = 35$ *cm²/s for the electrons of the base and* $D_p = 12$ *cm²/s for the holes of the emitter and of the collector, as in the earlier chapters.*

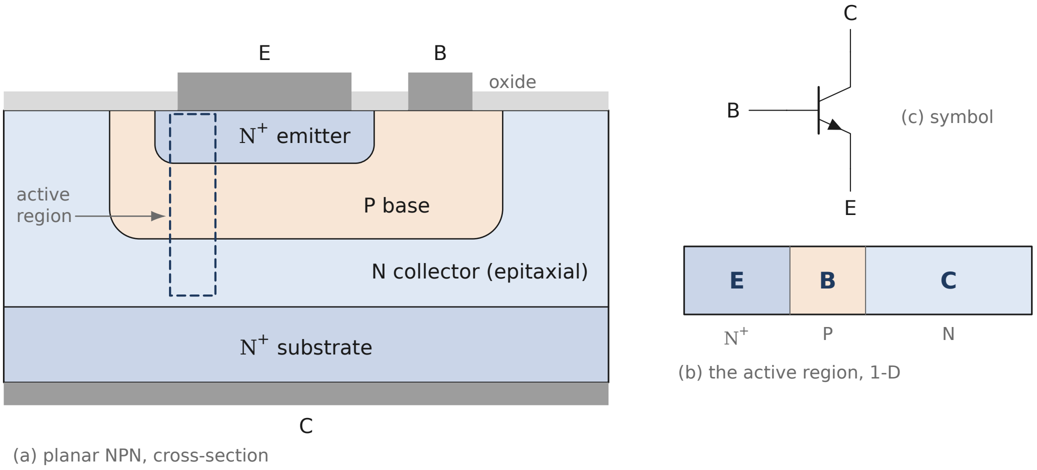
8.1 Technology and design constraints

THE QUESTION

What is a bipolar transistor made of, and which constraints does its technology impose?

The electronics course uses the bipolar transistor as a component, described by its measured characteristics. This chapter explains where those characteristics come from, starting from the structure of the device. The **bipolar junction transistor** (BJT) is made of three regions of alternating doping, N–P–N or P–N–P, and therefore of **two PN junctions** that share the middle region. The word “bipolar” says that both kinds of carriers take part: in an NPN transistor the main current is carried by electrons, but holes supply the control current. We develop the physics with the NPN; the PNP of Chapter 6 works in the same way with the carriers and all polarities exchanged.

How a transistor is made (Figure 8.1). As for the diode (Chapter 5), the junctions are made inside one crystal. In the usual planar process, an N layer is grown by epitaxy on a heavily doped N^+ substrate; acceptors are diffused or implanted into it to form a P well, the **base**; donors are then brought into the base, at a much higher concentration, to form an N^+ island, the **emitter**. The N layer under the base is the **collector**, contacted through the substrate. Windows in the oxide carry the metal contacts of the emitter and the base. The transistor action happens in the **active region**, the vertical slice under the emitter. Along that slice the device is a one-dimensional sequence N^+–P–N, and this is the device we study: emitter on the left, base in the middle, collector on the right.



(a) planar NPN, cross-section

(b) the active region, 1-D

(c) symbol

Figure 8.1 – *How a bipolar transistor is made (schematic, not to scale). (a) Cross-section of a planar NPN transistor: N epitaxial collector on an* N^+ *substrate, P base diffused into it,* N^+ *emitter diffused into the base; oxide (light grey) with windows for the metal contacts (dark grey) of the emitter and the base; the collector is contacted at the back. The dashed box is the active region under the emitter. (b) The active region as a one-dimensional device, drawn horizontally: this is the transistor of the whole chapter. (c) The circuit symbol: the arrow on the emitter points in the direction of the conventional current in normal operation.*

Three design constraints. A good transistor is not any N–P–N stack. Its technology imposes three constraints on the three regions. We state them here as constraints; their physical reasons appear one by one as the chapter follows the carriers.

1. **A heavily doped emitter**, $N_E \gg N_B$. Its reason appears in Section 8.3 (Step 2) and is made quantitative in Sections 8.6 and 8.10 (the emitter efficiency).
2. **A lightly doped base**, the same inequality seen from the base. Its reasons appear in Section 8.3 (Steps 2 and 3) and Section 8.10; its price, the Early effect, in Part E.
3. **A short base**, much thinner than the distance L_n that an electron travels in it before recombining. Its reason appears in Section 8.3 (Step 3) and is made quantitative in Sections 8.6 and 8.10 (the base transport factor); Part F shows that it also makes the transistor fast.

EXAMPLE 8.1 — THE REFERENCE TRANSISTOR OF THIS CHAPTER

Silicon at 300 K. Emitter N^+, $N_E = 10^{18}\,\text{cm}^{-3}$, neutral width $d_E = 0.5$ µm; base P, $N_B = 10^{16}\,\text{cm}^{-3}$, neutral width $W_B = 1.0$ µm at the reference bias; collector N, $N_C = 10^{15}\,\text{cm}^{-3}$; area $S = 10^{-4}$ cm². The EB junction is the reference junction of Chapters 5–7 with N and P exchanged: $V_{dE} = 0.835$ V and $W = 0.33$ µm at equilibrium, almost entirely in the base. The BC junction is one-sided the other way, with its SCR mostly in the collector. The results that the chapter will establish:

$$I_S = 5.6 \times 10^{-14}\ \mathrm{A}, \quad \beta_F = 121, \quad V_A \approx 44\ \mathrm{V}, \quad C_D = 5.5\ \mathrm{pF\ at}\ I_C = 1\ \mathrm{mA}.$$

A MODEST EMITTER DOPING, A LARGE AREA

Real emitters are doped 10^{19} to $10^{20}\,\mathrm{cm}^{-3}$, where silicon is degenerate and the gap narrows; the base is often thinner than 0.1 µm, and integrated transistors are a few micrometres wide. The reference transistor keeps $N_E = 10^{18}\,\mathrm{cm}^{-3}$, the area of the reference diode and a 1 µm base so that every formula of Chapters 5–7 applies unchanged and every number can be checked by hand. The trends it shows are those of real devices; its values of β_F, V_A and C_D are those of a modest discrete transistor.

PART A

The structure, the physical picture and the energy landscape

Two junctions sharing a base, how they work together, the band diagram at equilibrium and under forward-active bias

8.2 The transistor at equilibrium: two junctions sharing a base

THE QUESTION

Before any voltage is applied, what does the device contain?

Leave the three terminals unconnected and wait. Each junction forms exactly as in the thought experiment of Chapter 5: near each metallurgical junction, the majority carriers diffuse across, recombine, and uncover fixed ions, until the field of the uncovered charge stops the diffusion. The result is drawn in Figure 8.2. Read it from left to right.

- **Three conducting regions**: the neutral emitter, crowded with electrons; the neutral base, with its holes; the neutral collector, with fewer electrons than the emitter. Each is a conductor, as in Chapter 5.
- **Two junctions**: the emitter–base junction (EB) and the base–collector junction (BC).
- **Two SCRs**, which contain only fixed ions: donors on the N side, acceptors on the P side, in equal numbers on the two sides of each junction. Each SCR extends mostly into its more lightly doped side: the EB SCR lies almost entirely in the base, the BC SCR mostly in the collector.
- **Two fields**, each pointing from the N side to the P side, that is, towards the base on both sides: $\vec{\mathcal{E}}_{EB}$ in the EB SCR and $\vec{\mathcal{E}}_{BC}$ in the BC SCR.
- **Two built-in voltages**, V_{dE} and V_{dC}, one for each junction, given by the formula of

Chapter 5 with the dopings of the two sides.

$$V_{dE} = V_T \ln \frac{N_E N_B}{n_i^2} = 0.835\ \text{V}, \qquad V_{dC} = V_T \ln \frac{N_B N_C}{n_i^2} = 0.656\ \text{V}. \tag{8.1}$$

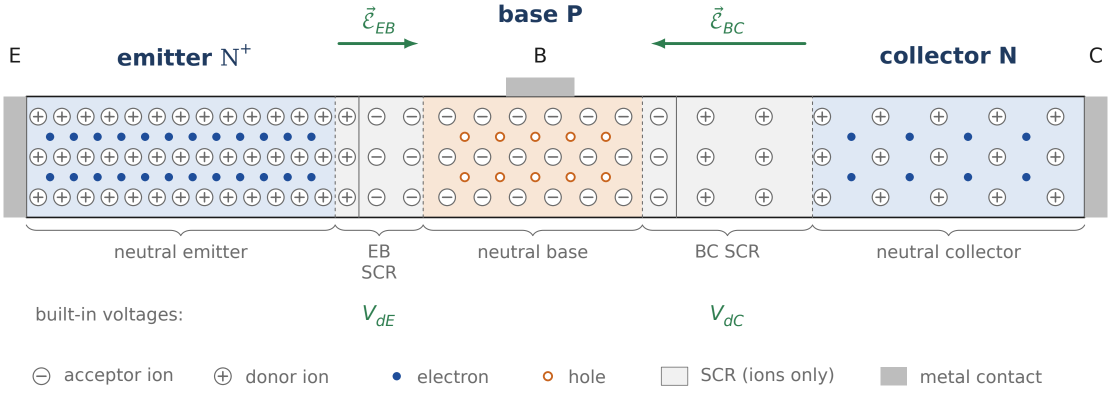


Figure 8.2 *– The NPN transistor at equilibrium (schematic: widths not to scale). Donor ions (⊕) in the emitter and the collector, acceptor ions (⊖) in the base; electrons (dots) and holes (open circles) in the three neutral regions; the two SCRs (grey) hold only ions. The spacing of the ions suggests the doping: dense in the N$^+$ emitter, sparse in the collector. Each SCR lies mostly in its lightly doped side (the emitter part of the EB SCR, in reality 1 % of it, is drawn as one column of ions). The two fields point towards the base. No terminal current flows.*

This is not yet a transistor. At equilibrium the device is only two junctions placed back to back. In each SCR, drift and diffusion balance for each carrier, as in Chapter 5, and no current flows through any terminal. Nothing yet connects the two junctions: each one is the junction of Chapter 5. What turns this structure into a transistor is the bias, and the fact that the base between the two SCRs is **short**. The next section shows how the two junctions then work together, before any band diagram or equation.

EXAMPLE 8.2 — THE TWO SCRS AT EQUILIBRIUM

With the widths of Chapter 5, the EB junction has $W = 0.330$ µm, of which 0.327 µm in the base and 3.3 nm in the emitter, and $\mathcal{E}_{max} = 51$ kV/cm. The BC junction, one-sided towards the collector, has $W = 0.97$ µm, of which only 88 nm in the base, and $\mathcal{E}_{max} = 14$ kV/cm. The base, 1.43 µm between the two metallurgical junctions, keeps a neutral width of about 1.0 µm between the two SCRs.

8.3 How the two junctions work together: the physical picture

THE QUESTION

How can a reverse-biased junction carry a large current, and how can the other junction control it?

We build the operation of the transistor from the junctions of Chapter 6, in three steps, without bands and without equations (Figure 8.3). We start from the junction that will carry the large current: the BC junction.

Step 1: the BC junction alone, reverse biased. Leave the EB junction at equilibrium ($V_{BE} = 0$) and reverse-bias the BC junction: the collector is made positive with respect to the base. This is the reverse-biased diode of Chapter 6. Its SCR widens and its field grows. The field points from the collector towards the base, so it drives into the collector every electron that enters the SCR from the base. Such electrons are minority carriers of the base, and there are very few of them: only the thermal minority electrons that happen to reach the edge of the SCR. The current is the tiny reverse current of Chapter 6. The collector is ready to collect, but it has almost nothing to collect.

Step 2: supply electrons through the EB junction. Now forward-bias the EB junction: the base is made positive with respect to the emitter. Chapter 6 showed what a forward-biased junction does: each side injects its majority carriers into the other. The emitter injects electrons into the base, and the base injects holes into the emitter. The two injections are not equal. Each side receives in proportion to its equilibrium minority density n_i^2/N, so the more lightly doped side receives the more. Because the emitter is much more heavily doped than the base, the current through the EB junction is carried almost entirely by electrons entering the base. This is the reason for the first design constraint.

Step 3: transport across the base, collection. The injected electrons are minority carriers in the neutral base. No field acts on them there: they spread by diffusion, from the EB edge, where they are numerous, towards the BC edge (Chapter 4). Each electron then has two possible fates. It may meet a hole of the base and recombine; or it may reach the edge of the BC SCR, where the field of Step 1 sweeps it into the collector. Which fate dominates depends on the base.

- If the base is **thin** compared with the diffusion length L_n, the distance an electron travels by diffusion before it recombines (Chapter 4), almost every electron reaches the BC SCR. This is the third design constraint.
- A **lightly doped** base helps in two ways: it receives most of the injection (Step 2), and it holds few holes for the injected electrons to recombine with. This is the second design constraint.

The collector current is therefore large: it is made of almost all the electrons injected by the emitter.

What the base current does. Each electron that recombines in the base takes one of its holes, and each hole injected into the emitter is a hole lost by the base. The base terminal supplies these holes: this is the base current. It is small compared with the collector current, because both losses are small. It accompanies the collector current; it does not cause it. The collector current is caused by the injection at the EB junction.

What controls the collector current. The number of electrons that reach the BC SCR is set by the injection at the EB junction, that is, by V_{BE}. The BC junction only has to collect them. Once its reverse bias holds its field, a larger reverse voltage hardly changes the number of electrons it receives. The collector current is therefore controlled by the EB junction and is relatively insensitive to the BC reverse voltage over the normal operating range. It is not perfectly independent of it: the widening BC SCR slightly shortens the base, which is the Early effect of Part E.

Forward-active operation. The bias of normal operation now follows from the roles of the two junctions:

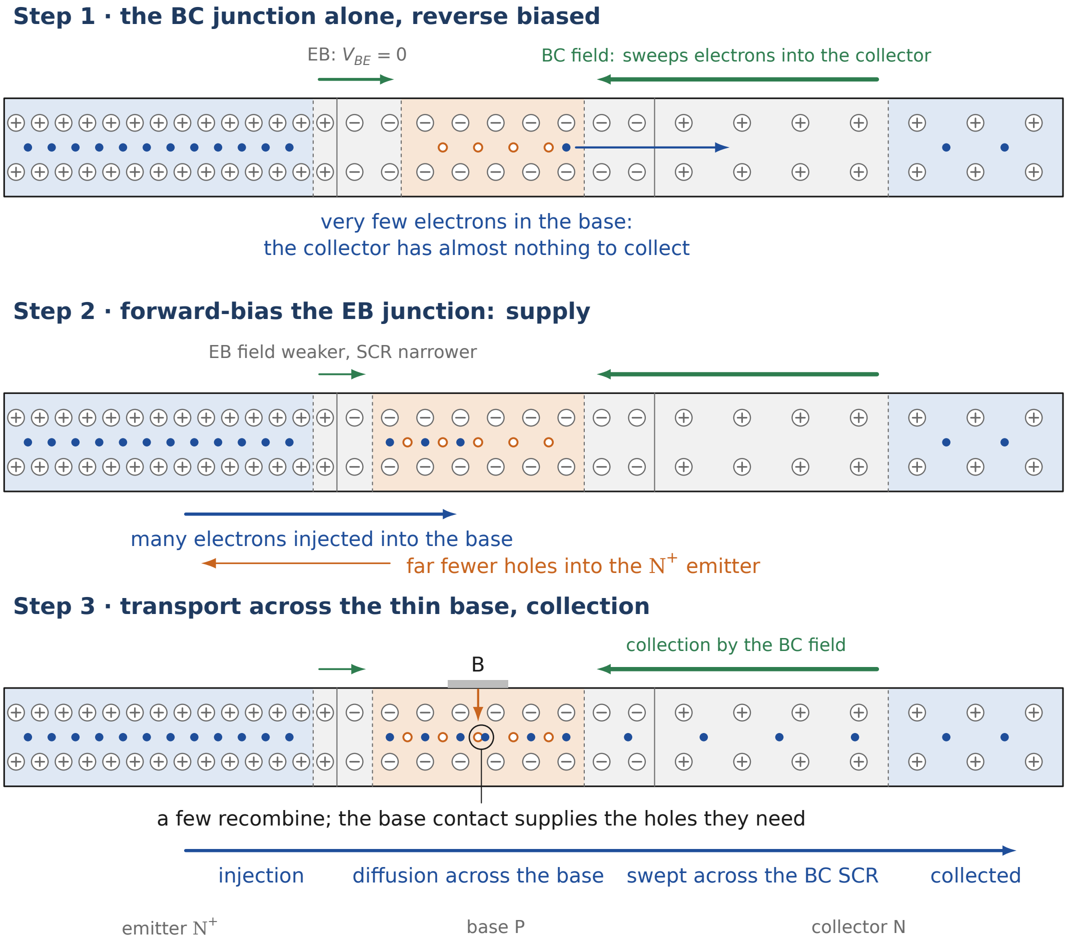

Step 1 · the BC junction alone, reverse biased
EB: $V_{BE} = 0$
BC field: sweeps electrons into the collector
very few electrons in the base:
the collector has almost nothing to collect
Step 2 · forward-bias the EB junction: supply
EB field weaker, SCR narrower
many electrons injected into the base
far fewer holes into the N^+ emitter
Step 3 · transport across the thin base, collection
collection by the BC field
B
a few recombine; the base contact supplies the holes they need
injection
diffusion across the base
swept across the BC SCR
collected
emitter N^+
base P
collector N

Figure 8.3 *– The physical picture of the transistor in three steps (qualitative: widths as in Figure 8.5, numbers of carriers not to scale). Ions, electrons (dots) and holes (open circles) as in Figure 8.2; green arrows: the fields of the two SCRs; blue and orange arrows: particle fluxes. Step 1: the BC junction is reverse biased alone ($V_{BE} = 0$); its SCR is wide and its field strong, ready to sweep into the collector any electron arriving from the base, but the base holds very few electrons. Step 2: the EB junction is forward biased; its SCR narrows, the emitter injects many electrons into the base, and the base injects far fewer holes into the heavily doped emitter. Step 3: the injected electrons diffuse across the thin base; a few recombine (circle) with holes, which the base contact replaces; the others reach the BC SCR and are swept into the collector.*

- the **EB junction forward biased**: it supplies the electrons;
- the **BC junction reverse biased**: it collects them;
- between them, a **thin base** that the electrons cross before they recombine.

The rest of the chapter formalises this picture. The bands (Sections 8.4 and 8.5) show the barrier lowered at the EB junction and the fall that collects at the BC junction. The carrier densities (Part B) show the injected electrons and their profile across the base. The equations (Part C) count them.

8.4 The band diagram at equilibrium

THE QUESTION

How do the bands look across the whole transistor at equilibrium?

The Fermi level is flat. At equilibrium the Fermi level is constant through any system (Chapter 5). The band diagram of the transistor is therefore built as for one junction: draw a flat E_F, place the bands of each neutral region at the right distance from it, and join them across each SCR (Figure 8.4).

The position of E_F in each region follows from its doping (Chapter 3): $E_F - E_i = V_T \ln(N_D/n_i)$ in an N region and $E_i - E_F = V_T \ln(N_A/n_i)$ in a P region.

- **Emitter**, N^+: E_F is 0.48 eV above E_i, only 0.08 eV (about $3kT$) below E_C, because the emitter is heavily doped.
- **Base**, P: E_F is 0.36 eV below E_i, 0.20 eV above E_V.
- **Collector**, N: E_F is 0.30 eV above E_i, 0.26 eV below E_C, further from E_C than in the emitter because the collector is lightly doped.

Two steps. Since E_F is flat and the bands of the base lie higher than those of the N regions, the conduction band rises by $qV_{dE} = 0.835$ eV from the emitter to the base, and falls by $qV_{dC} = 0.656$ eV from the base to the collector. For the electrons of the conduction band, the base is a **barrier** between two wells. The two steps are not equal: the emitter well is deeper than the collector well by $q(V_{dE} - V_{dC}) = 0.18$ eV, which is the difference of the positions of E_F below E_C in the two N regions. Across each SCR the bands follow the parabolic potential of Chapter 5.

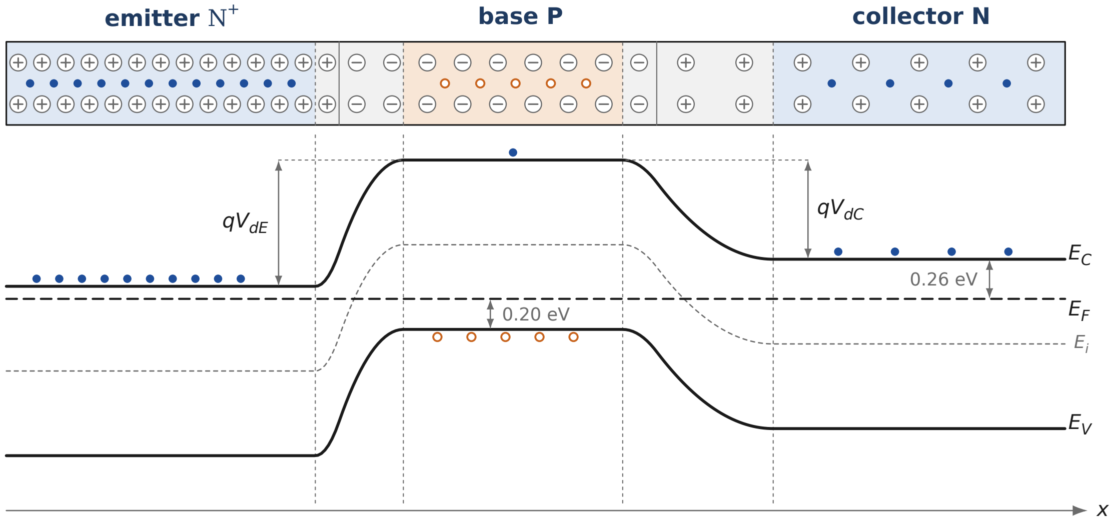


Figure 8.4 – *The NPN transistor at equilibrium: the cartoon of Figure 8.2 aligned with its band diagram. Energies to scale for the reference transistor (1 eV = 2 cm); widths schematic, the same as in the cartoon. E_F (dashed) is flat; E_i (grey dashed) follows the bands. The conduction band rises by qV_{dE} into the base and falls by qV_{dC} into the collector. The distances of E_F to the band edges are given in the base (0.20 eV to E_V) and in the collector (0.26 eV to E_C); in the emitter E_F is only 0.08 eV below E_C. Electrons (dots) sit on E_C, holes (open circles) under E_V; the single dot on the base E_C stands for the rare minority electrons of the base.*

Reading the barrier. The barrier reading of Chapters 5 and 6 applies at each junction: the electrons of the emitter can enter the base only if they climb the step qV_{dE}, which only the Boltzmann tail $e^{-qV_{dE}/kT}$ of their distribution can do; the minority electrons of the base roll down either step. At equilibrium the two fluxes balance at each junction separately. The bias will lower the first step and raise the second.

8.5 Forward-active bias: barriers, SCRs and fields

> **THE QUESTION**
>
> *How do the bands, the SCRs and the fields show the picture of Section 8.3?*

The forward-active condition. Section 8.3 found the bias of normal operation from the roles of the two junctions: the EB junction forward biased to supply electrons, the BC junction reverse biased to collect them. In a circuit, ground the emitter, apply $V_{BE} > 0$ to the base and a larger positive voltage V_{CE} to the collector (Figure 8.5b). Then

- the EB junction is **forward biased**: the P side (base) is at $+V_{BE}$ with respect to the N side (emitter);
- the BC junction is **reverse biased**: its N side (collector) is at $V_{CB} = V_{CE} - V_{BE} > 0$ above its P side (base).

This is the rule of Chapter 6 (source on P, N grounded) applied to the EB junction. For the BC junction the source is on the N side, so a positive V_{CB} is a reverse bias; we keep V_{CB} as the variable of this junction throughout the chapter.

Where the voltages drop. As in Chapter 6, the neutral regions are conductors and each applied voltage drops across an SCR: V_{BE} across the EB SCR, V_{CB} across the BC SCR. Each junction obeys Chapter 6 with its own voltage:

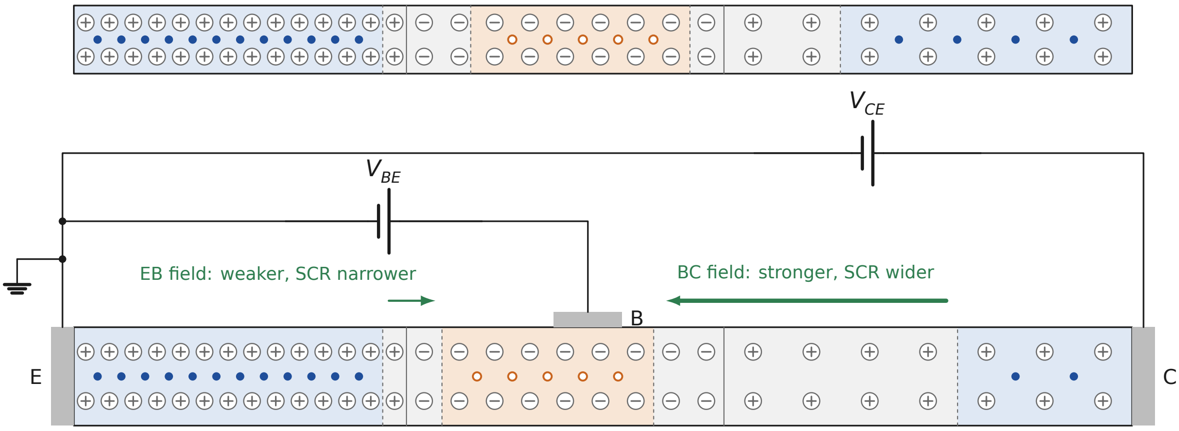
$$\text{EB step: } q(V_{dE} - V_{BE}), \qquad \text{BC step: } q(V_{dC} + V_{CB}). \tag{8.2}$$

In the band diagram (Figure 8.5c), the emitter stays where it is; the bands and the Fermi level of the base move down by qV_{BE}; those of the collector move down by $q(V_{BE} + V_{CB}) = qV_{CE}$. The Fermi levels of the three neutral regions are no longer aligned: they are separated by qV_{BE} (emitter to base) and qV_{CB} (base to collector), which is what voltmeters connected to the terminals read (Chapter 6).

The EB step: the supply. The step that the emitter electrons must climb falls from qV_{dE} to $q(V_{dE} - V_{BE})$. As in Chapter 6, only the Boltzmann tail of the emitter electrons can climb it, and the lowered step lets e^{V_{BE}/V_T} times more of them over. This is the injection of Step 2, read in the bands.

The BC step: the collection. The conduction band now falls from the base into the collector by $q(V_{dC} + V_{CB})$, much more than at equilibrium. For an electron of the base this fall is not an obstacle but a slope: an electron that arrives at the base edge of the BC SCR rolls down into the collector, whatever its energy (Figure 8.5c). This is the collection of Step 1, read in the bands; the field of the BC SCR is the slope of the band. Between the two steps the bands of the neutral base are flat: no field acts on the injected electrons there, and they cross the base by diffusion (Step 3).

(a) equilibrium


V_{CE}
V_{BE}
EB field: weaker, SCR narrower
BC field: stronger, SCR wider
B
E
C

(b) forward-active: V_{BE} = +0.6 V, V_{CB} = +1.0 V

(c) bands under bias (solid) and at equilibrium (light dashed)

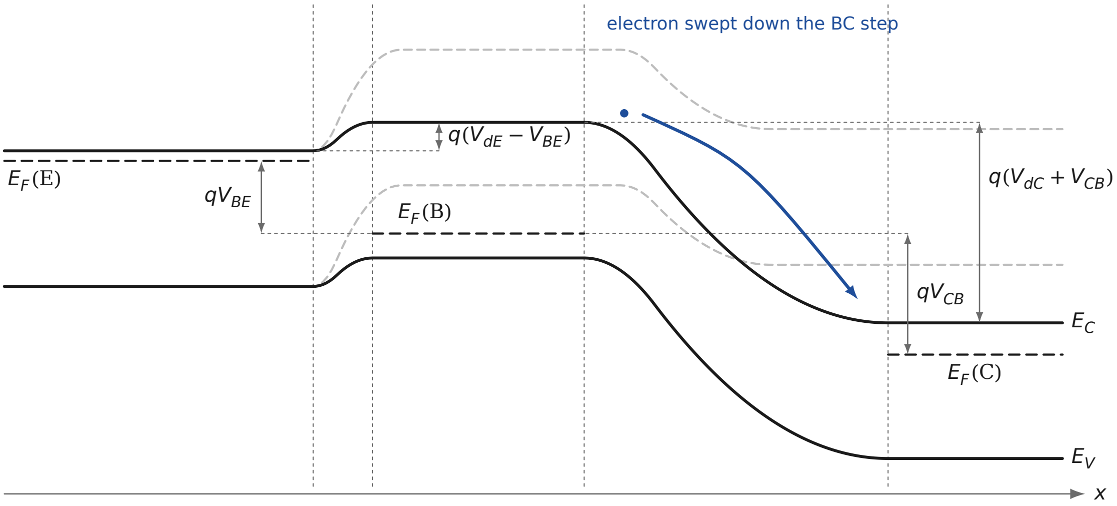

electron swept down the BC step
$q(V_{dE} - V_{BE})$
E_F(E)
qV_{BE}
E_F(B)
$q(V_{dC} + V_{CB})$
qV_{CB}
E_C
E_F(C)
E_V
x

Figure 8.5 – *Forward-active bias, V_{BE} = +0.6 V and V_{CB} = +1.0 V. (a) The transistor at equilibrium, for comparison. (b) Under bias, with its circuit (emitter grounded): the EB SCR is narrower and its field weaker; the BC SCR is wider and its field stronger. (c) The bands under bias (solid) over the bands at equilibrium (light dashed), energies to scale (1 eV = 1.6 cm), widths schematic, scaled as $\sqrt{V_d - V}$ from (a). The Fermi levels E_F(E), E_F(B), E_F(C) of the neutral regions are separated by qV_{BE} and qV_{CB}. The EB step falls to $q(V_{dE} - V_{BE})$ = 0.235 eV; the BC step grows to $q(V_{dC} + V_{CB})$ = 1.66 eV. An electron that reaches the BC SCR from the base rolls down into the collector.*

The two SCRs do not react in the same way. The widths follow Chapter 6, $W \propto \sqrt{V_d - V}$, with $V = V_{BE}$ for the EB junction and $V = -V_{CB}$ for the BC junction (Figure 8.5b):

- the **EB SCR contracts**: fewer ions are needed to hold the smaller step, and its field weakens;
- the **BC SCR expands**: more ions must be uncovered to hold the larger step, mostly on the collector side, but also on the base side, and its field strengthens.

EXAMPLE 8.3 — WIDTHS AND FIELDS UNDER BIAS

At V_{BE} = 0.6 V the EB SCR shrinks from 0.33 to 0.175 µm. At V_{CB} = 1 V the BC SCR grows from 0.97 to 1.53 µm, and at V_{CB} = 5 V to 2.84 µm, with a largest field of 40 kV/cm; its base side then takes 0.26 µm of the base. These two numbers, the shrinking of the EB SCR and the growth of the BC SCR into the base, both change the neutral width W_B; the second one will be the Early effect of Part E.

The same mechanism as the reverse current of a diode. The BC junction collects exactly as the reverse-biased diode of Chapter 6 extracts its minority carriers: every minority carrier that reaches the edge of a reverse-biased SCR is swept across it. In the transistor, the electrons that arrive at that edge are not only the rare thermal minority carriers of the base but those supplied by the EB junction. Part B shows this with the carrier densities.

WHAT PART A HAS ESTABLISHED

- At equilibrium the transistor is two junctions sharing a base: three neutral regions, two SCRs (mostly in the lightly doped sides), two fields pointing towards the base, two built-in voltages V_{dE} = 0.835 V and V_{dC} = 0.656 V. No current flows.
- The physical picture: the reverse-biased BC junction collects every electron that reaches it, but alone it has almost nothing to collect; the forward-biased EB junction supplies electrons to the base, mostly electrons because $N_E \gg N_B$; a thin base lets them reach the collector. The base current supplies the holes lost by recombination and by injection into the emitter.
- The flat Fermi level sits close to E_C in the N$^+$ emitter, 0.20 eV above E_V in the base, 0.26 eV below E_C in the collector. For conduction electrons the base is a barrier between two wells.
- In forward-active bias, the EB step falls to $q(V_{dE} - V_{BE})$ and its SCR contracts: this is the supply. The BC step grows to $q(V_{dC} + V_{CB})$ and its SCR expands: an electron that reaches its edge rolls down into the collector. This is the collection.

PART B

Carrier densities: long base and short base

The two edge densities of the base, the long base that does nothing, the short base that makes a transistor, several V_{BE} and several V_{CB}

8.6 Long base and short base, side by side

THE QUESTION

What does the forward bias of the EB junction change in the electron density of the base, and why does the length of the base decide whether anything reaches the collector?

At equilibrium. The densities follow from the bands exactly as in Chapter 5. In each neutral region the majority density equals the doping, and the minority density follows from the mass-action law:

$$n_{B0} = \frac{n_i^2}{N_B} = 10^4\,\mathrm{cm}^{-3}, \qquad p_{E0} = \frac{n_i^2}{N_E} = 10^2\,\mathrm{cm}^{-3}, \qquad p_{C0} = \frac{n_i^2}{N_C} = 10^5\,\mathrm{cm}^{-3}. \tag{8.3}$$

Follow the electrons, which will carry the main current: $10^{18}\,\mathrm{cm}^{-3}$ in the emitter, only $10^4\,\mathrm{cm}^{-3}$ in the base, $10^{15}\,\mathrm{cm}^{-3}$ in the collector. The base is nearly empty of electrons not because they are scarce in the device, but because the EB step holds the emitter electrons back: across each SCR the electron density falls by the Boltzmann factor of the step,

$$n_{B0} = N_E\,\mathrm{e}^{-V_{dE}/V_T} = N_C\,\mathrm{e}^{-V_{dC}/V_T}, \tag{8.4}$$

which says the same thing as the flat Fermi level. The holes are the majority carriers of the base, $N_B = 10^{16}\,\mathrm{cm}^{-3}$, and the minority carriers of the emitter and of the collector.

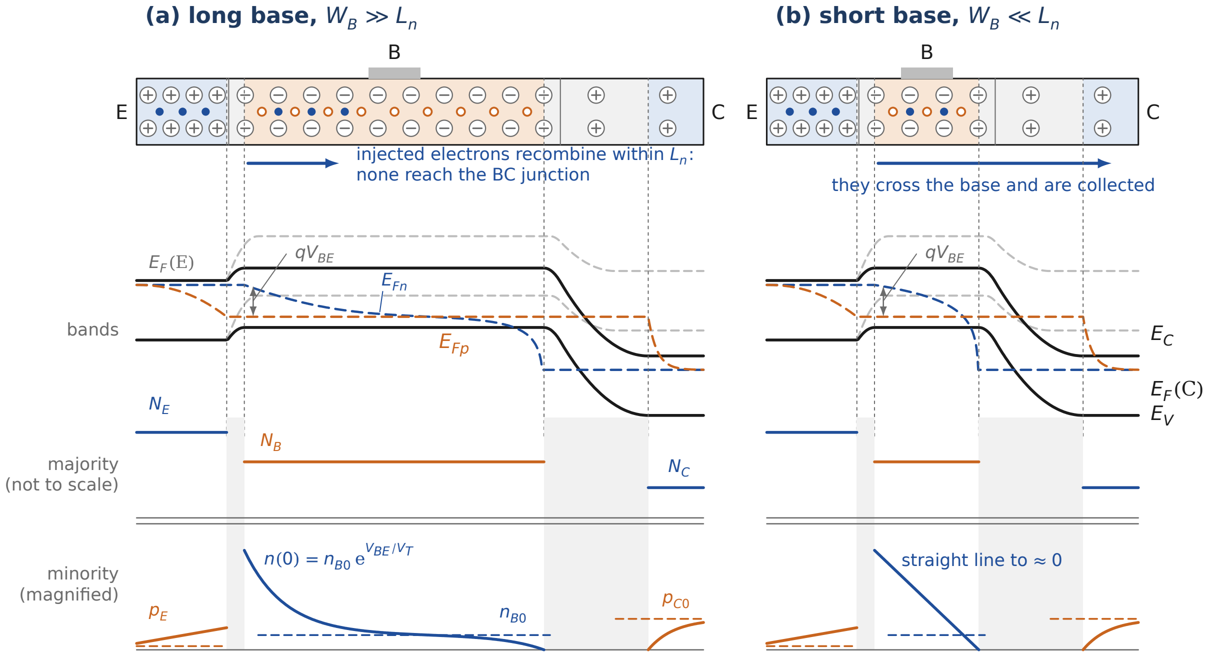
The two edges of the base. Each junction is the junction of Chapter 6 with its own voltage, and each one sets the minority density at its own edge of the neutral base by the Boltzmann factor of its step (quasi-equilibrium):

$$\boxed{n(0) = n_{B0}\,\mathrm{e}^{V_{BE}/V_T}, \qquad n(W_B) = n_{B0}\,\mathrm{e}^{-V_{CB}/V_T}}. \tag{8.5}$$

The minus sign is the only thing to remember: $V_{CB} > 0$ reverse biases the BC junction, raises its step by qV_{CB} and empties its edge. These two densities are the tools of the whole chapter. Between them the electrons diffuse across the base (Chapter 4), and what happens on the way depends on one length: the base width W_B compared with the diffusion length L_n.

Figure 8.6: the same transistor with a long base and with a short base, in forward-active bias, built from the three blocks of Chapters 5 and 6: cartoon, bands, densities. The emitter, the junctions and the voltages are the same in both columns; only the base width differs.

A long base: two diodes back to back (Figure 8.6a). Let the base be many diffusion lengths long. Near the emitter, the EB junction is the forward-biased diode of Chapter 6: it raises the density to $n(0)$, and the injected electrons diffuse and recombine, the excess



(a) long base, $W_B \gg L_n$

(b) short base, $W_B \ll L_n$

Figure 8.6 *– Long base (a) and short base (b), forward-active (V_{BE} = 0.6 V, V_{CB} = 1 V). Top: the cartoon (ions, majority carriers, the injected electrons of the base). Middle: the bands, energies to scale (1 eV = 0.72 cm), under bias (solid) over equilibrium (light dashed); quasi-Fermi levels E_{Fn} (blue) and E_{Fp} (orange), separated by qV_{BE} at the EB edge. Bottom: the densities, linear, with a scale break (majority densities above, minority densities below, magnified and not to scale); dashed: equilibrium. (a) The injected electrons recombine within a few L_n; the BC edge only sees the equilibrium n_{B0}, which it pulls down to zero. (b) No room to recombine: one straight line from $n(0)$ to ≈ 0, and E_{Fn} stays above E_{Fp} across the whole base.*

dying out as e^{-x/L_n}. Near the collector, the BC junction is the reverse-biased diode of Chapter 6: it empties its edge, and the base electrons nearby, at the equilibrium density n_{B0}, diffuse towards it and are swept across. Between the two, the base is back at n_{B0}, and the quasi-Fermi levels have merged. The two pulls never meet. The collector carries only the tiny reverse current of its own junction, whatever V_{BE}; the EB current flows in through the emitter and out through the base contact. **There is no transistor effect**: just two diodes sharing a P region, each obeying Chapter 6 on its own. In numbers, the diffusion equation with recombination gives the fraction of the injected electrons that reaches the BC edge:

$$n(x) \simeq n(0)\,\frac{\sinh[(W_B - x)/L_n]}{\sinh(W_B/L_n)}, \qquad \frac{I_n(W_B)}{I_n(0)} = \frac{1}{\cosh(W_B/L_n)}, \tag{8.6}$$

10 % for $W_B = 3L_n$, 0.009 % for $W_B = 10L_n$.

A short base: the two pulls join (Figure 8.6b). Now let the base be much shorter than L_n (the reference base: W_B = 1 µm, L_n = 18.7 µm). There is no room for the injected excess to die out: the profile is the straight line of the short diode (Chapter 6, Section 6.11), from $n(0)$ at the EB edge to $n(W_B) \approx 0$ at the BC edge, where the field of the BC SCR sweeps every arriving electron into the collector. By (8.6), $1/\cosh(W_B/L_n) \simeq 1 - W_B^2/2L_n^2 = 99.86$ % of the injected electrons arrive. In the bands, E_{Fn} stays above E_{Fp} all the way across the base and drops to the Fermi level of the collector only at the BC edge. This is the third design constraint, and it is the condition that couples the two junctions: only because the base is short does the density raised by one junction reach the other. Here the razor blade of Chapter 6 makes sense: Brattain's two gold contacts, about 50 µm apart on germanium, had to be closer than the distance a hole injected by one contact travels before recombining, so that the second contact could collect it. Too far apart, they were two diodes back to back.

The gradient sets the current. For a straight line the gradient is the difference of the two edge densities divided by the width, and Fick's law (Chapter 4) gives the electron current that crosses the base:

$$I_n = qSD_n \frac{n(0) - n(W_B)}{W_B} \simeq qSD_n \frac{n(0)}{W_B}. \tag{8.7}$$

At $V_{BE} = 0.611$ V, $n(0) = 1.8 \times 10^{14}\,\text{cm}^{-3}$ and $I_n = 1.0$ mA. Note that $n(0)$ remains far below $N_B = 10^{16}\,\text{cm}^{-3}$: low injection, the condition of all the Chapter 6 results, holds up to a few milliamperes for the reference transistor.

8.7 Several V_{BE}, several V_{CB}

THE QUESTION

In the short base, what does each voltage move, and what does it leave alone?

Equation (8.5) gives one voltage to each end of the line. Figure 8.7 moves them one at a time, in the spirit of the bias sweeps of Chapter 6.

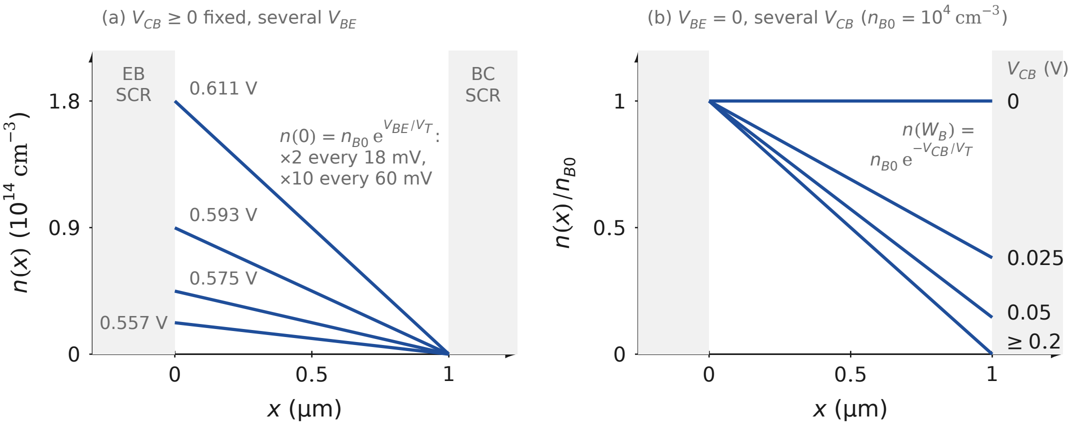


Figure 8.7 *– The electron line across the short base of the reference transistor ($W_B = 1$ µm fixed here), computed from (8.5). (a) The BC junction reverse biased, four values of V_{BE}: the EB end rises as e^{V_{BE}/V_T}, the BC end stays at ≈ 0; the slope, hence the current, doubles every 18 mV. (b) The EB junction unbiased, several values of V_{CB}, in units of n_{B0}: the BC end falls from n_{B0} at $V_{CB} = 0$ to ≈ 0 after a few V_T, and then stays there.*

V_{BE} moves the emitter end (Figure 8.7a). Every 18 mV doubles $n(0)$, every 60 mV multiplies it by 10, and the slope and the current follow. The collector end does not move.

V_{CB} moves the collector end (Figure 8.7b). With the EB junction unbiased, the line starts at n_{B0}. At $V_{CB} = 0$ the BC junction is at equilibrium and the line is flat: no current. A reverse V_{CB} pulls the collector end down as e^{-V_{CB}/V_T}: 37 % of n_{B0} at 25 mV, 14 % at

50 mV, practically zero beyond 0.2 V. The current that flows, $qSD_n n_{B0}/W_B = 5.6\times10^{-14}$ A, is the reverse saturation current of the BC junction (its electron part). Past a few V_T, a larger reverse voltage cannot pull the end below zero: the current saturates, exactly as the reverse current of the diode in Chapter 6. This is the waterfall picture: once the edge is held empty, a deeper fall does not change the flow of the river that feeds it.

Why the zero sits at the BC depletion edge. It is the reverse-biased junction of Chapter 6 again. Inside the BC SCR the electron density follows the Boltzmann factor of the local step, so at the base edge of the SCR it is $n_{B0}\,e^{-V_{CB}/V_T}$: the step of the junction has been raised by qV_{CB}, and the carriers on the high side see the whole extra fall. Wherever the edge of the SCR sits (Part E will move it), the density there is ≈ 0.

The collector current flows at $V_{CB} = 0$. Now forward bias the EB junction and wire the collector straight to the base (Figure 8.8). The BC junction is at equilibrium, its edge density is $n_{B0} = 10^4\,\text{cm}^{-3}$; but the EB end is at $n(0) = 1.8\times10^{14}\,\text{cm}^{-3}$, 1.8×10^{10} times higher. On the scale of the line, n_{B0} is zero. The slope, and the collector current, are the same as with a reverse bias of several volts: $I_C = I_S(e^{V_{BE}/V_T} - 1)$ instead of $I_S e^{V_{BE}/V_T}$, a difference of 5.6×10^{-14} A on 1 mA. No collector voltage is needed: the built-in field of the BC SCR collects. What the collector needs is not a voltage that pulls, but an edge that stays empty.

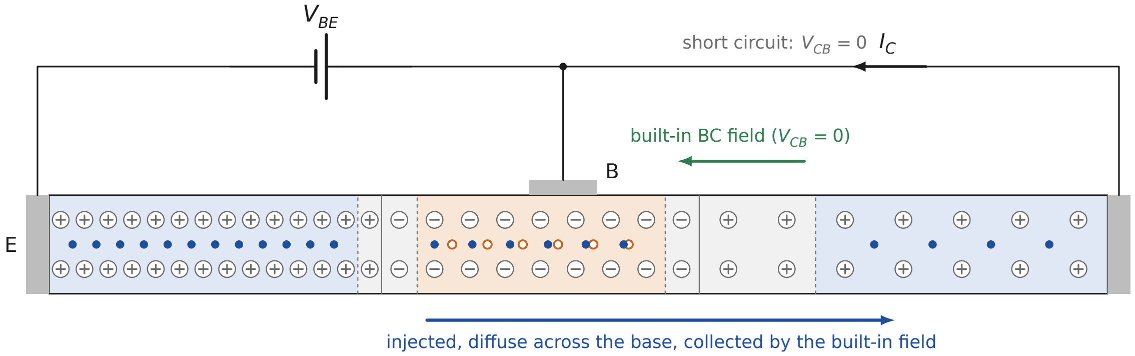

V_{BE}
short circuit: $V_{CB} = 0$ I_C
built-in BC field ($V_{CB} = 0$)
B
E
C
injected, diffuse across the base, collected by the built-in field

Figure 8.8 *– The collector current flows with the collector wired to the base ($V_{CB} = 0$). The EB junction is forward biased; the BC SCR keeps its equilibrium width and its built-in field, which sweeps into the collector every electron that diffuses across the base. The collector edge of the base stays at n_{B0}, negligible beside $n(0)$: the current is the one of forward-active operation.*

HOW EMPTY IS THE COLLECTING EDGE?

The condition $n(W_B) \approx 0$ is the boundary condition of the diffusion problem, not a literally empty edge. The electrons cross the BC SCR by drift, with $J_n = qnv_n$. In its field, tens of kilovolts per centimetre, they move at about their saturation velocity, $v_{sat} \approx 10^7$ cm/s, so their density is $n \approx J_C/qv_{sat} \approx 6\times10^{12}\,\text{cm}^{-3}$ at 1 mA, 3 % of $n(0)$. It lowers the gradient, and the current, by about 3 %; the ideal model and the figures neglect it.

A weak drift current made large. At the BC edge the electrons enter the SCR and its field carries them across: in every case the current through the BC junction is a **drift** current of minority carriers, the same mechanism as the reverse current of a diode. Forward biasing the EB junction changes neither the field, nor the SCR, nor the mechanism:

it multiplies the supply of minority electrons by e^{V_{BE}/V_T}, and the short base delivers them to the BC edge. The transistor turns the weak reverse current of a junction into a large one by turning up its supply.

8.8 From carrier transport to transistor action

> ***THE QUESTION***
>
> *Putting together injection, transport and collection: what does the transistor do, and what does each region contribute?*

We return to the three steps of Section 8.3, now read on Figure 8.6b; Figure 8.9 sums them up as fluxes.

1. **Injection** (Step 2). The forward-biased EB junction raises the electron density at the EB edge of the base by e^{V_{BE}/V_T}, to $1.8 \times 10^{14}\,\mathrm{cm}^{-3}$ at 0.611 V. The emitter, crowded with electrons, is the reservoir. It also raises the hole density at the edge of the emitter, to $p_{E0}e^{V_{BE}/V_T} = 1.8 \times 10^{12}\,\mathrm{cm}^{-3}$, a hundred times less than $n(0)$ because $N_E = 100\,N_B$.
2. **Transport** (Step 3). The electrons diffuse across the short base along a straight line whose slope is the collector current; only 0.14 % recombine on the way.
3. **Collection** (Step 1). The BC junction holds the other end of the base low and sweeps every arriving electron into the collector. It supplies none.

The **base current** supplies holes, for two purposes: the holes injected into the emitter, and the holes that recombine with injected electrons in the base. Both are proportional to e^{V_{BE}/V_T}, like the collector current, and both are small: the first because the emitter is much more heavily doped than the base, the second because the base is short.

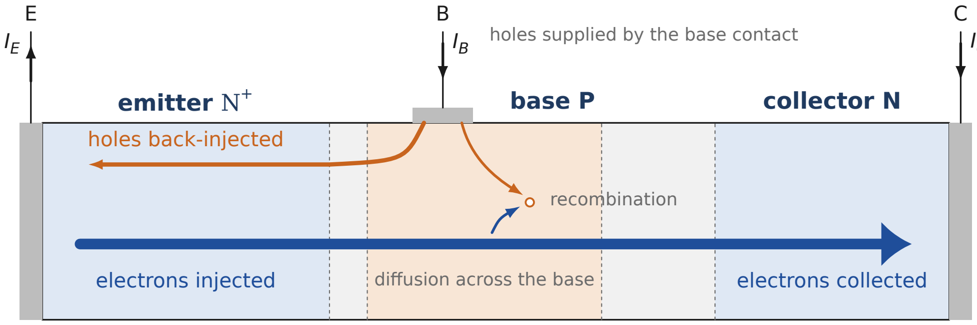


Figure 8.9 *– Transistor action in forward-active operation (schematic). Coloured arrows: particle fluxes; electrons (blue) are injected into the base, diffuse across it and are collected; a few recombine in the base with holes (orange) supplied by the base contact; other holes from the base contact are injected back into the emitter. Black arrows: the terminal currents, in the conventional direction: I_C enters the collector, I_B enters the base, $I_E = I_C + I_B$ leaves the emitter. Arrow widths only suggest the magnitudes: in the reference transistor the hole back-injection is 0.7 % of I_C and the recombination 0.14 %.*

Why this is amplification. The collector current is set by V_{BE}, which changes it by a factor 10 every 60 mV. The base current that accompanies it is about a hundred times smaller, so a small input current controls a large output current. And the collector current flows into a junction whose edge is held empty, so it hardly depends on the collector voltage (Part D): the collector behaves almost as a current source whose value is set at the input. A small power at the input controls a much larger power at the output, taken from the supply of the collector circuit.

WHAT PART B HAS ESTABLISHED

- Each junction sets the electron density at its own edge of the base: $n(0) = n_{B0}\mathrm{e}^{V_{BE}/V_T}$ and $n(W_B) = n_{B0}\mathrm{e}^{-V_{CB}/V_T}$.
- In a long base ($W_B \gg L_n$) the injected electrons recombine before reaching the BC junction: two diodes back to back, no transistor effect.
- In a short base the line runs straight from $n(0)$ to $n(W_B)$; its slope is the current, $qSD_n[n(0) - n(W_B)]/W_B$.
- V_{BE} moves the emitter end (×10 every 60 mV); a reverse V_{CB} moves the collector end from n_{B0} to ≈ 0 and then nothing more. At $V_{CB} = 0$ the collector end is n_{B0}, negligible: the collector current already flows.
- The BC junction supplies no electrons; it keeps its edge empty. The base current supplies the holes lost by injection into the emitter and by recombination.

PART C

Currents and the Ebers–Moll equations

Assumptions and conventions, the currents in forward-active operation, both junctions at once, the meaning of each term

8.9 Assumptions and conventions

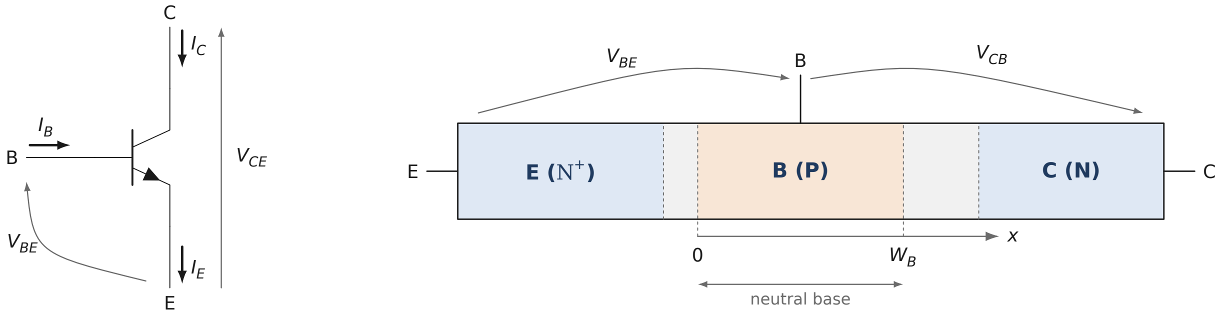
THE QUESTION

What exactly do we assume, and how are the currents and voltages counted?

Conventions (Figure 8.10).

- **Two voltages only**: $V_{BE} = V_B - V_E$ across the EB junction, positive in forward bias, and $V_{CB} = V_C - V_B$ across the BC junction, positive in reverse bias. The BC junction is forward biased when $V_{CB} < 0$. Each voltage is imposed by its own source; nothing is split. The terminal voltage of the common-emitter circuit is $V_{CE} = V_{CB} + V_{BE}$.
- **Terminal currents**, positive in normal operation: I_C **into** the collector, I_B **into** the base, I_E **out of** the emitter. Kirchhoff's law gives $I_E = I_B + I_C$.
- **The base axis**: $x = 0$ at the EB edge of the neutral base, $x = W_B$ at its BC edge. Electrons diffuse towards $+x$; their conventional current flows towards $-x$, from the collector to the emitter.

Assumptions. They are those of the short diode of Chapter 6 (Section 6.11), applied to each junction and to each neutral region:



(a) $I_E = I_B + I_C$, $V_{CE} = V_{CB} + V_{BE}$

(b) $V_{BE} > 0$: EB forward; $V_{CB} > 0$: BC reverse ($V_{CB} < 0$: BC forward)

Figure 8.10 *– The conventions of the chapter. (a) The NPN symbol with its terminal currents (I_B and I_C in, I_E out) and the voltages V_{BE} and V_{CE}; a voltage arrow points to the terminal taken as positive. (b) The one-dimensional transistor: V_{BE} across the EB junction (positive: forward), V_{CB} across the BC junction (positive: reverse); the axis x runs across the neutral base, from 0 at its EB edge to W_B at its BC edge.*

1. **One dimension**: the active region only (Figure 8.1); the edges of the device and the lateral flow of the base current are ignored.
2. **Uniform doping** in each region: no built-in field in the neutral regions, so the minority carriers move by diffusion only.
3. **Low injection**: the injected minority densities remain much smaller than the majority densities ($n(0) \ll N_B$).
4. **Quasi-equilibrium in both SCRs**, and no generation or recombination inside them: the edge densities are given by (8.5) and its analogues for the holes, and the currents are the same on the two sides of each SCR.
5. **Short neutral regions**: each neutral region is much shorter than the diffusion length of its minority carriers, and the ohmic contacts hold the minority carriers at equilibrium: $W_B = 1$ µm $\ll L_n = 18.7$ µm in the base; $d_C = 10$ µm $\ll L_p = 110$ µm in the collector; $d_E = 0.5$ µm $\ll L_p = 11$ µm in the emitter. Recombination in the base is first neglected, then added as a small correction.
6. **Fixed neutral widths**: W_B is taken at the operating point. Part E lifts this assumption.

8.10 The currents in forward-active operation

> ***THE QUESTION***
>
> *How large are the collector, emitter and base currents, and what sets their ratios?*

The collector current. In forward-active operation, $n(0) = n_{B0}\mathrm{e}^{V_{BE}/V_T}$ and $n(W_B) \approx 0$. The slope of the straight line gives the electron current that reaches the collector:

$$\boxed{I_C = qSD_n\frac{n(0)}{W_B} = I_S\,\mathrm{e}^{V_{BE}/V_T}}, \qquad I_S = \frac{qSD_n n_i^2}{N_B W_B}. \tag{8.8}$$

This is the law that the electronics course uses for the transistor, and we now know where each factor comes from: e^{V_{BE}/V_T} from the injection at the EB edge, D_n/W_B from the diffusion across the short base, n_i^2/N_B from the equilibrium minority density of the base.

EXAMPLE 8.4 — THE SATURATION CURRENT AND THE OPERATING POINT

$I_S = 1.602 \times 10^{-19} \times 10^{-4} \times 35 \times 10^{20}/(10^{16} \times 10^{-4}) = 5.6 \times 10^{-14}$ A. For $I_C = 1$ mA, $V_{BE} = V_T \ln(I_C/I_S) = 25.9\text{ mV} \times \ln(1.78 \times 10^{10}) = 0.611$ V, and $n(0) = 1.8 \times 10^{14}\,\text{cm}^{-3}$, 1.8 % of N_B. This operating point, $I_C = 1$ mA and $V_{CB} = 5$ V, is used throughout the chapter.

The emitter current: two components. The EB junction carries the electrons injected into the base, $I_{nE} \simeq I_C$, and the holes injected into the emitter. The holes form a straight-line profile across the short neutral emitter, from $p_{E0}e^{V_{BE}/V_T}$ at the edge to p_{E0} at the contact (Chapter 6):

$$I_{pE} = qSD_p \frac{p_{E0}}{d_E}\left(e^{V_{BE}/V_T} - 1\right) = \frac{qSD_p n_i^2}{N_E d_E}\left(e^{V_{BE}/V_T} - 1\right). \tag{8.9}$$

Only the electron component is useful. The **emitter efficiency** is the useful fraction:

$$\gamma = \frac{I_{nE}}{I_{nE} + I_{pE}} = \frac{1}{1 + \dfrac{D_p}{D_n}\dfrac{N_B}{N_E}\dfrac{W_B}{d_E}}. \tag{8.10}$$

The first design constraint in numbers: γ is close to 1 because $N_B/N_E \ll 1$. For the reference transistor, $(12/35) \times (1/100) \times (1/0.5) = 6.9 \times 10^{-3}$ and $\gamma = 0.9932$.

Recombination in the base: a charge-control argument. The injected electrons form a stored charge in the base, the area of the triangle of Figure 8.6b times qS. Chapter 7 (Section 7.6) showed that such a charge equals the current multiplied by the time a carrier spends in the region, here the transit time:

$$Q_B = \tfrac{1}{2}\, qS\, n(0)\, W_B = \tau_B I_C, \qquad \boxed{\tau_B = \frac{W_B^2}{2D_n}}, \tag{8.11}$$

the transit time $d_n^2/2D_p$ of the short diode of Chapter 7 with the base in place of the N region. The stored electrons recombine at the rate Q_B/τ_n (Chapter 4), which the base must replace with holes:

$$I_{rB} = \frac{Q_B}{\tau_n} = \frac{\tau_B}{\tau_n} I_C. \tag{8.12}$$

The **base transport factor** is the fraction of the injected electrons that reach the collector:

$$\alpha_T = \frac{I_C}{I_{nE}} = 1 - \frac{\tau_B}{\tau_n} = 1 - \frac{W_B^2}{2L_n^2}, \tag{8.13}$$

the expansion of $1/\cosh(W_B/L_n)$ of (8.6). The third design constraint in numbers: $\tau_B = (10^{-4})^2/70 = 0.143$ ns, seven hundred times shorter than $\tau_n = 0.1$ µs, and $\alpha_T = 0.99857$. An electron crosses the base long before it has a chance to recombine.

The current gains. The base current is the sum of the two hole currents, $I_B = I_{pE} + I_{rB}$. The two classical gains follow:

$$\alpha_F = \frac{I_C}{I_E} = \gamma\,\alpha_T, \tag{8.14}$$

$$\boxed{\frac{1}{\beta_F} = \frac{I_B}{I_C} = \frac{D_p}{D_n}\frac{N_B}{N_E}\frac{W_B}{d_E} + \frac{W_B^2}{2L_n^2}}, \qquad \beta_F = \frac{\alpha_F}{1 - \alpha_F}. \tag{8.15}$$

Each term of $1/\beta_F$ is one loss, and each is the ratio of a hole current to the electron current. Both are proportional to e^{V_{BE}/V_T}, so β_F does not depend on the current, within our assumptions.

EXAMPLE 8.5 — THE GAINS OF THE REFERENCE TRANSISTOR

$1/\beta_F = 6.86 \times 10^{-3} + 1.43 \times 10^{-3} = 8.29 \times 10^{-3}$, so $\beta_F = 121$ and $\alpha_F = 0.9918$. At $I_C = 1$ mA, $I_B = 8.3$ µA: 6.9 µA of holes injected into the emitter and 1.4 µA of recombination in the base. In this transistor the emitter efficiency limits the gain; a tenfold emitter doping would raise β_F to about 470. This is why real emitters are doped so heavily.

α_F AND β_F ARE NOT FREE PARAMETERS

The electronics course measures β and notes that it varies from one device to the next. Formula (8.15) says why: β_F depends on W_B squared and on the ratio N_B/N_E, which the fabrication controls only within some tolerance. It also explains why β falls at large currents, where low injection fails, and at very small currents, where recombination in the EB SCR, neglected here, adds to the base current.

8.11 Both junctions at once: the Ebers–Moll equations

THE QUESTION

What are the currents for any pair of voltages V_{BE} and V_{CB}, and what does each term mean?

Electrons: one line, two ends. With both edge densities (8.5), the straight line across a short base without recombination carries the electron current

$$I_n = \frac{qSD_n}{W_B}\left[n(0) - n(W_B)\right] = I_S\left(e^{V_{BE}/V_T} - e^{-V_{CB}/V_T}\right). \tag{8.16}$$

It is **the difference of two terms, each set by one junction**. The diffusion equation is linear, and so are the boundary conditions in n: the profile is the sum of a line pushed up from the emitter end and a line pushed up from the collector end. This superposition is the heart of the Ebers–Moll model.

Holes: one short region on each side. Each junction also injects holes into its N side when it is forward biased, and extracts them when it is reverse biased, with the short-region formula of Chapter 6:

$$\begin{aligned} I_{pE} &= I_{pE0}\left(e^{V_{BE}/V_T} - 1\right), & I_{pE0} &= \frac{qSD_p n_i^2}{N_E d_E}; \\ I_{pC} &= I_{pC0}\left(e^{-V_{CB}/V_T} - 1\right), & I_{pC0} &= \frac{qSD_p n_i^2}{N_C d_C}. \end{aligned} \tag{8.17}$$

The hole current I_{pE} flows from the base into the emitter and leaves through the emitter terminal; I_{pC} flows from the base into the collector and leaves through the collector terminal, so it counts negatively in I_C. Understanding the characteristics needs both gradients: the electron line in the base and the hole lines in the emitter and the collector.

The terminal currents. Adding the components at each terminal, with the conventions of Figure 8.10:

$$I_C = I_S\left(e^{V_{BE}/V_T} - e^{-V_{CB}/V_T}\right) - \frac{I_S}{\beta_R}\left(e^{-V_{CB}/V_T} - 1\right), \tag{8.18}$$

$$I_B = \frac{I_S}{\beta_F}\left(e^{V_{BE}/V_T} - 1\right) + \frac{I_S}{\beta_R}\left(e^{-V_{CB}/V_T} - 1\right), \tag{8.19}$$

$$I_E = I_C + I_B = I_S\left(e^{V_{BE}/V_T} - e^{-V_{CB}/V_T}\right) + \frac{I_S}{\beta_F}\left(e^{V_{BE}/V_T} - 1\right), \tag{8.20}$$

where, without recombination, $I_S/\beta_F = I_{pE0}$ and $I_S/\beta_R = I_{pC0}$. These are the **Ebers–Moll equations**, the form used by the electronics course, here written with V_{CB}. With recombination in the base, the full solution keeps exactly the same form: the forward recombination adds to I_S/β_F, the reverse recombination to I_S/β_R, and I_S changes by a relative amount $\tau_B/3\tau_n$, here 0.05 %, which we neglect:

$$\frac{I_S}{\beta_F} = I_{pE0} + \frac{\tau_B}{\tau_n} I_S, \qquad \frac{I_S}{\beta_R} = I_{pC0} + \frac{\tau_B}{\tau_n} I_S. \tag{8.21}$$

EXAMPLE 8.6 — THE REVERSE GAIN

$I_{pC0} = 1.602\times10^{-19}\times10^{-4}\times12\times10^{20}/(10^{15}\times10^{-3}) = 1.9\times10^{-14}$ A, a third of I_S: the lightly doped collector receives many holes when its junction is forward biased. With the recombination term, $\beta_R = 2.9$ and $\alpha_R = \beta_R/(\beta_R + 1) = 0.744$, against $\beta_F = 121$. The transistor is not symmetric: the collector, lightly doped, is a poor emitter.

The meaning of each term. Regroup (8.18) by junction, using $1 + 1/\beta_R = 1/\alpha_R$:

$$\boxed{I_C = \underbrace{I_S\left(e^{V_{BE}/V_T} - 1\right)}_{\text{electrons sent by the emitter end}} - \underbrace{\frac{I_S}{\alpha_R}\left(e^{-V_{CB}/V_T} - 1\right)}_{\text{what the collector end sends back}}} \tag{8.22}$$

- $I_S(e^{V_{BE}/V_T} - 1)$: the electrons that the raised emitter end of the line sends across the base, minus the equilibrium share. It depends on V_{BE} only.
- $I_S(e^{-V_{CB}/V_T} - 1)$: the electrons that the collector end sends back towards the emitter. Reverse bias ($V_{CB} > 0$) makes it $-I_S$, a constant: the collector end is empty and nothing is sent back. Forward bias ($V_{CB} < 0$) makes it grow exponentially.
- The factor $1/\alpha_R = 1 + 1/\beta_R$ adds to each electron sent back the holes that the base injects into the collector at the same time, $I_{pC} = (I_S/\beta_R)(e^{-V_{CB}/V_T} - 1)$. These holes also leave through the collector terminal, so they also subtract from I_C. $1/\alpha_R$ **counts the holes the base injects into the collector**: 1.34 for the reference transistor, because its collector is lightly doped.
- In I_B (8.19), the two terms are the holes that the base supplies to each junction: to the emitter and to recombination (I_S/β_F), and to the collector when the BC junction is forward biased (I_S/β_R).

The injection form. Defining the two junction currents $I_F = I_{ES}(e^{V_{BE}/V_T} - 1)$ and $I_R = I_{CS}(e^{-V_{CB}/V_T} - 1)$, the currents that each junction would carry if the other were short-circuited, (8.18) and (8.20) become

$$I_E = I_F - \alpha_R I_R, \qquad I_C = \alpha_F I_F - I_R, \tag{8.23}$$

with $I_{ES} = I_S/\alpha_F$, $I_{CS} = I_S/\alpha_R$, and the **reciprocity relation** $\alpha_F I_{ES} = \alpha_R I_{CS} = I_S$, which comes straight from (8.16): the electron current that crosses the base depends only on $n(0) - n(W_B)$, with the same coefficient in both directions. Each junction is a diode, and each one, when it injects, sends the fraction α of its current to the other junction: a controlled current source (Figure 8.11).

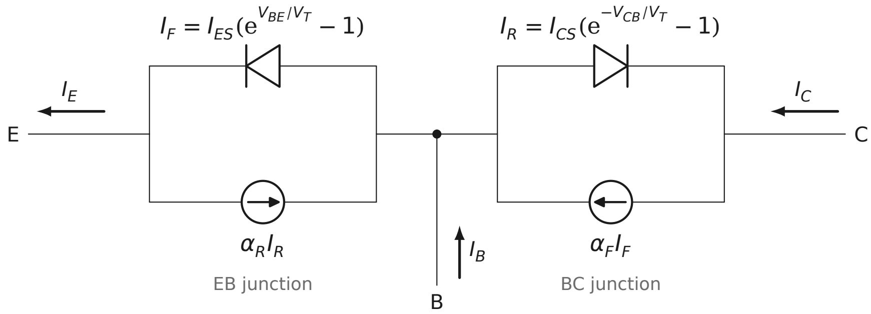

$I_F = I_{ES}(e^{V_{BE}/V_T} - 1)$
$I_R = I_{CS}(e^{-V_{CB}/V_T} - 1)$
I_E
I_C
E
C
$\alpha_R I_R$
I_B
$\alpha_F I_F$
EB junction
BC junction
B

Figure 8.11 *– The Ebers–Moll model of the NPN transistor, injection form (8.23). Each junction is a diode carrying its junction current (I_F for EB, I_R for BC); each controlled source sends to the other side the fraction of the injected current that crosses the base ($\alpha_F I_F$ to the collector, $\alpha_R I_R$ to the emitter).*

The two edge densities are the tools. Everything in this part comes from two numbers, $n(0)$ and $n(W_B)$, and from the hole densities that the same voltages set in the emitter and the collector. To read a characteristic, ask what the voltage does to the two ends of the line, and to the hole lines beside them. Part D does exactly this.

WHAT PART C HAS ESTABLISHED

- Forward-active: $I_C = I_S e^{V_{BE}/V_T}$, $I_S = qSD_n n_i^2/(N_B W_B) = 5.6 \times 10^{-14}$ A, the slope of the straight line.
- The base current supplies the holes injected into the emitter ($\gamma = 0.9932$) and those lost by recombination ($\alpha_T = 1 - \tau_B/\tau_n = 0.99857$, $\tau_B = W_B^2/2D_n = 0.143$ ns): $1/\beta_F = (D_p N_B W_B)/(D_n N_E d_E) + W_B^2/2L_n^2$, $\beta_F = 121$.
- Ebers–Moll, in V_{BE} and V_{CB}: $I_C = I_S(e^{V_{BE}/V_T} - 1) - (I_S/\alpha_R)(e^{-V_{CB}/V_T} - 1)$. The first term is what the emitter end sends, the second what the collector end sends back; $1/\alpha_R = 1 + 1/\beta_R$ counts the holes injected into the collector. Reference transistor: $\beta_R = 2.9$, $\alpha_R = 0.744$.

PART D

The characteristics

I_C against V_{CB}, the cornerstone; the knee when the BC junction is forward biased; I_C against V_{CE} by a change of axis; the operating regions

8.12 I_C against V_{CB}: the plateau

THE QUESTION

With V_{BE} and V_{CB} each imposed by its own source, how does the collector current depend on V_{CB}?

Connect one source between base and emitter, another between collector and base, and sweep V_{CB} for several fixed values of V_{BE}. Figure 8.12 shows the result, computed from (8.18) with the base width of Part E.

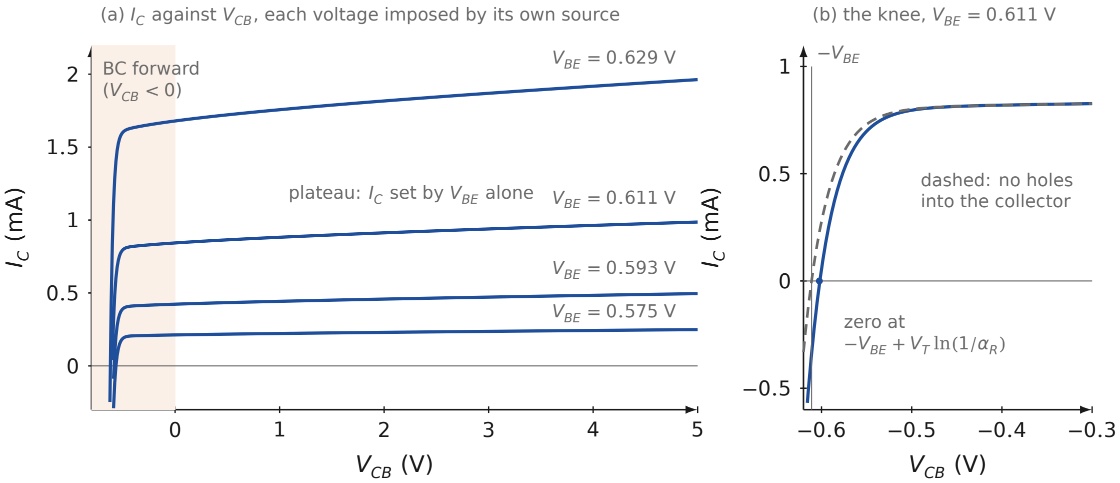


Figure 8.12 *– I_C against V_{CB} for the reference transistor, computed from the Ebers–Moll equations with base-width modulation. (a) Four values of V_{BE}, 18 mV apart (each curve twice the one below; 60 mV would give a factor 10). For $V_{CB} > 0$ the current is a plateau, set by V_{BE} alone; its slight tilt is the Early effect (Part E). In the tinted band the BC junction is forward biased; the current holds almost unchanged, then falls steeply in the last ≈100 mV before $V_{CB} = -V_{BE}$. (b) Zoom on the knee at $V_{BE} = 0.611$ V: the current crosses zero at $V_{CB} = -V_{BE} + V_T \ln(1/\alpha_R) = -0.602$ V; dashed, the same transistor without holes injected into the collector, whose zero is exactly at $-V_{BE}$.*

The plateau: the transfer-resistance effect. For V_{CB} above a few V_T, the collector current does not depend on V_{CB}. In the short base, the electron line runs from $n_{B0}e^{V_{BE}/V_T}$ down to about zero; a reverse V_{CB} only moves the collector end between n_{B0} and zero (Figure 8.7b), and n_{B0} is 10^{10} times smaller than $n(0)$. The slope, and the current, stay the same. The current is **supply-limited**: the same mechanism as the reverse saturation current of a diode, except that the emitter turns the supply up exponentially. In one sentence:

The collector does not push the current; it only collects what the emitter sends.

A current set at the input and independent of the voltage at the output: the input voltage is transferred to an output current, through a path of very high output resistance. This **trans**fer-re**sistor** effect, which gave the transistor its name, is the heart of amplification. The curves are labelled by V_{BE}, and every 60 mV multiplies the plateau by 10. The plateau is slightly tilted: the BC SCR eats into the base as V_{CB} grows (Part E).

8.13 The BC junction forward biased: the knee

THE QUESTION

What happens when V_{CB} becomes negative, and why does the current fall so late and so steeply?

The plateau holds, then a knee. Read (8.22) with $V_{CB} < 0$. The second term, $(I_S/\alpha_R)(e^{-V_{CB}/V_T} - 1)$, is negligible as long as $e^{-V_{CB}/V_T} \ll e^{V_{BE}/V_T}$: at $V_{CB} = -0.3$ V and $V_{BE} = 0.611$ V it is about 10^{-5} of the first. It reaches 10 % of the first term only when $-V_{CB} = V_{BE} - V_T \ln(10/\alpha_R)$, that is, 67 mV before $-V_{BE}$. The plateau therefore holds almost unchanged over most of the forward range, then bends in a sharp knee in the last ≈100 mV, then falls steeply (Figure 8.12).

Two pumps. The cause is not a weaker field: in forward bias the BC junction still has a field, and a weaker field never slowed a supply-limited current (Chapter 6). The cause is that the collector end of the line rises: $n(W_B) = n_{B0}e^{-V_{CB}/V_T}$. The BC junction, now forward biased, injects electrons back into the base. There are **two pumps**, one at each end of the line, and the current flows from the higher end to the lower: the line flattens and the current falls (Figure 8.13).

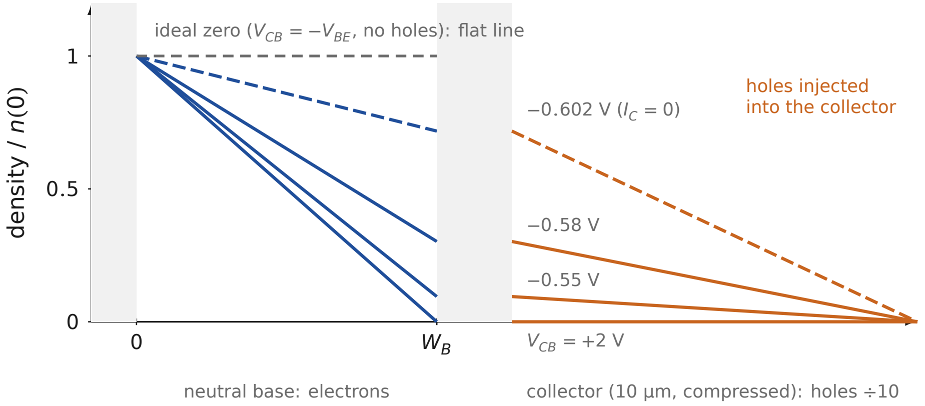


Figure 8.13 *– The two pumps, V_{BE} = 0.611 V fixed, computed. Left: the electron line across the base for V_{CB} = +2 V (to ≈ 0), −0.55 V, −0.58 V and −0.602 V; the collector end rises as $n_{B0}e^{-V_{CB}/V_T}$ and the line flattens. Right: the holes injected into the lightly doped collector at the same time (drawn ten times smaller), with their own straight line to the collector contact. At V_{CB} = −0.602 V (dashed) the electron current left by the residual slope is exactly cancelled by the hole current into the collector: I_C = 0. Grey dashed: the ideal case without holes, where the zero needs a perfectly flat line, at $V_{CB} = -V_{BE}$.*

Ideal case, no holes: zero at $V_{CB} = -V_{BE}$. If the collector injected no holes ($\beta_R \to \infty$, $\alpha_R = 1$), I_C would be the electron current alone, $I_S(e^{V_{BE}/V_T} - e^{-V_{CB}/V_T})$. It vanishes when the two ends are equal, $n(W_B) = n(0)$, that is, exactly at $V_{CB} = -V_{BE}$: a flat electron line, two pumps of equal strength, no flow.

Real case: zero a few tens of millivolts early. The forward-biased BC junction also injects holes into the collector, and their current leaves through the collector terminal. The collector current is zero when the residual electron slope, still pointing towards the collector, carries exactly as much as the holes into the collector. Setting (8.22) to zero, with the −1 terms negligible:

$$\boxed{V_{CB}\big|_{I_C=0} = -V_{BE} + V_T \ln\frac{1}{\alpha_R}}. \tag{8.24}$$

The shift is 7.7 mV for the reference transistor ($\alpha_R = 0.744$); about 18 mV for $\alpha_R = 0.5$, 60 mV for $\alpha_R = 0.1$, 120 mV for $\alpha_R = 0.01$. The worse the collector is as an emitter, the earlier the zero. Since the shift does not depend on V_{BE}, the zero points of the curves of Figure 8.12 are spaced like the values of V_{BE}. Beyond the zero, I_C changes sign: the collector end has become the higher one, and the transistor works upside down.

EXAMPLE 8.7 — THE KNEE IN NUMBERS

At $V_{BE} = 0.611$ V the plateau is 0.91 mA at $V_{CB} = 2$ V and still 0.84 mA at $V_{CB} = 0$ (the difference is the Early effect, not the knee). The current crosses zero at $V_{CB} = -0.602$ V. At $V_{CB} = -0.55$ V the collector end is at $e^{-0.061/0.0259} = 9.5\ \%$ of $n(0)$, and the current has fallen by about an eighth (electrons and holes sent back together, $1.34 \times 9.5\ \%$); at −0.58 V, 30 % of $n(0)$ and I_C has lost more than a third.

8.14 I_C against V_{CE}: a change of axis

THE QUESTION

The electronics course draws I_C against V_{CE} at constant I_B. Is this new physics?

No. With the emitter common to input and output, the output voltage is $V_{CE} = V_{CB} + V_{BE}$. At fixed V_{BE} this is an **algebraic change of axis**: each curve of Figure 8.12 slides to the right by its own V_{BE}, about 0.6 V. The knees, which sat near $V_{CB} \approx -V_{BE}$, all land near $V_{CE} \approx 0$: they bunch near the origin (Figure 8.14). The zero of I_C moves to $V_{CE} = V_T \ln(1/\alpha_R)$, a few millivolts.

Labelled by I_B. In the plateau $I_B = I_C/\beta_F$, so each value of V_{BE} corresponds to one value of I_B, and the curves can be labelled by the input current, as in the electronics course: 2, 4, 8 and 17 µA here.[1]

Saturation. Below the knee the BC junction is forward biased and the collector current falls: this is **saturation**. Measured transistors show $V_{CE,sat} \approx 0.1$–0.2 V, roughly independent of I_B. In the zoom, the rise is steepest at the start and then bends over (Figure 8.14b): it is the knee of Section 8.13 seen from the other side, not a diode curve. The ideal model places the knees at a few tens of millivolts; in a real transistor, the resistance of the neutral collector and of the contacts adds a voltage proportional to I_C, which pushes the knees of the higher curves to the right.

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[1] The curves are computed at fixed V_{BE}. At fixed I_B, V_{BE} is not exactly constant along a curve: in saturation part of the base current goes to the forward-biased BC junction, and the Ebers–Moll equations give a slightly lower V_{BE} for the same I_B. The larger $V_{BE,sat}$ of data sheets, about 0.75 V, refers to an overdriven base, $I_B \gg I_C/\beta_F$.

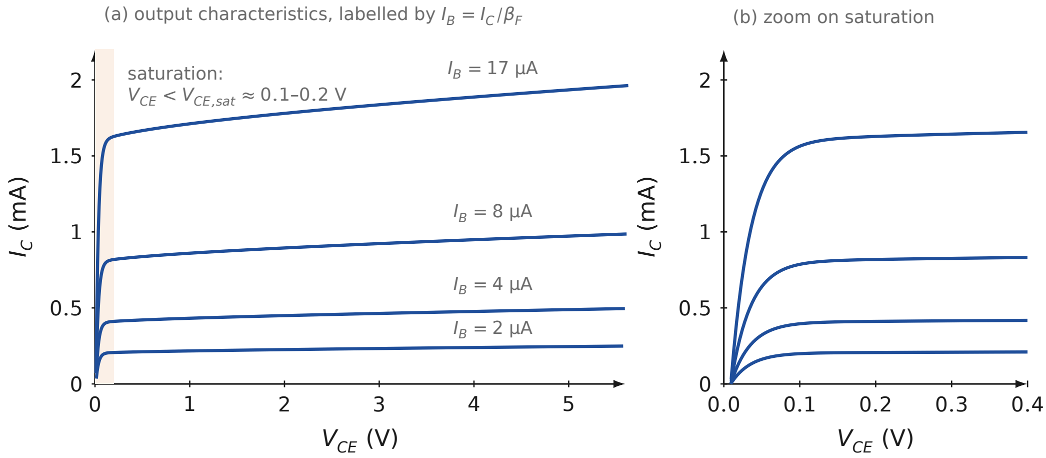


Figure 8.14 *– I_C against V_{CE}: the curves of Figure 8.12 shifted by their own V_{BE} ($V_{CE} = V_{CB} + V_{BE}$). (a) Labelled by the base current of the plateau, $I_B = I_C/\beta_F$: one I_B for each V_{BE}. Tinted: saturation. (b) Zoom on saturation: the rise is steepest at the start and then bends over; it is not the shape of a diode characteristic.*

8.15 The operating regions

THE QUESTION

What happens in each combination of junction biases?

The signs of V_{BE} and V_{CB} define four regions (Figure 8.15); the two ends of the electron line tell what happens in each.

- **Forward-active** ($V_{BE} > 0$, $V_{CB} > 0$). Injection at the emitter end, $n \approx 0$ at the collector end, $I_C = \beta_F I_B$: amplification.
- **Saturation** ($V_{BE} > 0$, $V_{CB} < 0$). Both ends raised: the two pumps of Section 8.13. $I_C < \beta_F I_B$, small V_{CE}; the base also supplies the holes of the second injection and stores much more charge.
- **Cut-off** ($V_{BE} < 0$, $V_{CB} > 0$). Both ends emptied; only the tiny reverse currents of the two junctions remain. An open switch.
- **Reverse-active** ($V_{BE} < 0$, $V_{CB} < 0$). The collector injects and the emitter collects, with the poor gain β_R. Rarely used.

EXAMPLE 8.8 — A SATURATED TRANSISTOR

The reference transistor is driven with $I_B = 0.1$ mA while its collector circuit allows only $I_C = 1$ mA, ten times less than $\beta_F I_B$. Solving (8.18)–(8.19) for the two voltages gives $V_{BE} = 0.619$ V and $V_{CB} = -0.576$ V: both junctions are forward biased, and $V_{CE} = 43$ mV. The voltage of a real saturated transistor is larger, typically 0.1 to 0.2 V, because of the resistance of the neutral collector, which these equations ignore.

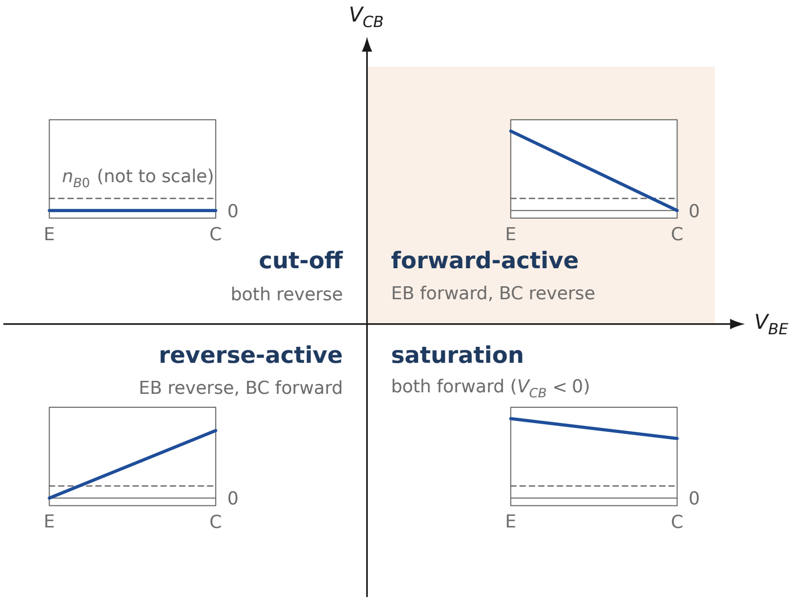

V_{CB}
V_{BE}
n_{B0} (not to scale)
0
E
C
cut-off
both reverse
forward-active
EB forward, BC reverse
reverse-active
EB reverse, BC forward
saturation
both forward ($V_{CB} < 0$)

Figure 8.15 – *The four operating regions in the plane (V_{BE}, V_{CB}), with the electron line across the neutral base in each (schematic; EB end at E, BC end at C). Grey line: $n = 0$. Dashed: the equilibrium level n_{B0},* ***not to scale*** *(it is 10^{10} times smaller than the injected densities). Forward-active (tinted, $V_{CB} > 0$): injection at E, $n \approx 0$ at C. Saturation ($V_{CB} < 0$): both ends raised, smaller slope. Reverse-active: injection at C only. Cut-off: both ends emptied.*

WHAT PART D HAS ESTABLISHED

- I_C against V_{CB} at fixed V_{BE} is a plateau: the collector does not push the current; it only collects what the emitter sends. Labelled by V_{BE}, ×10 every 60 mV.
- The BC edge density is $n_{B0}e^{-V_{CB}/V_T} \approx 0$ in reverse bias; at $V_{CB} = 0$ it is n_{B0}, still negligible, and I_C flows.
- BC forward: the plateau holds, then a knee in the last ≈100 mV and a steep fall: two pumps, the line flattens. Zero at $V_{CB} = -V_{BE}$ without holes, at $-V_{BE} + V_T \ln(1/\alpha_R)$ in reality.
- I_C against V_{CE}: the same curves slid by V_{BE}; labelled by $I_B = I_C/\beta_F$; saturation below $V_{CE,sat} \approx 0.1$–0.2 V.

PART E

The Early effect

Base-width modulation, the ruler picture, the common intercept $-V_A$, the output resistance

8.16 Base-width modulation

THE QUESTION

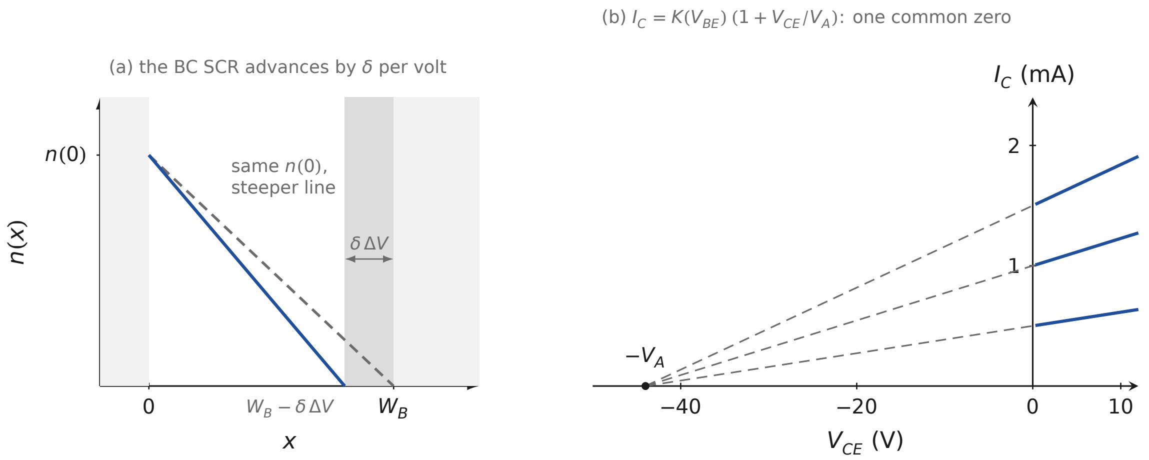
Why is the plateau slightly tilted, even in the forward-active region?

An inherent effect, never zero. The BC SCR widens when V_{CB} grows, mostly into the lightly doped collector but partly into the base. Its edge, where the electron line reaches zero, moves towards the emitter; the neutral base shortens. The emitter end, $n(0)$, is set by V_{BE} alone and does not move. Same height, shorter base: the line is steeper, and since $I_C = qSD_n\,n(0)/W_B$, the collector current rises. For the reference transistor, from $V_{CB} = 1$ V to 10 V, the neutral base shrinks from 1.12 to 0.90 µm and I_C rises from 0.89 to 1.11 mA at $V_{BE} = 0.611$ V. The effect is reduced by a heavily doped base, which the SCR hardly penetrates, and by a lightly doped collector, which takes most of the widening.

The ruler picture (Figure 8.16a). Draw the triangle of the base electrons by hand. When V_{CB} rises by ΔV, the zero point creeps towards the emitter by $\delta\,\Delta V$, where $\delta = |dW_B/dV_{CB}|$ is the advance of the SCR edge per volt. The current is inversely proportional to the length of the triangle:

$$I_C \propto \frac{1}{W_B - \delta\,\Delta V} \simeq \frac{1}{W_B}\left(1 + \frac{\delta\,\Delta V}{W_B}\right), \qquad \text{so} \qquad V_A \simeq \frac{W_B}{\delta}. \tag{8.25}$$

A 1 µm base whose edge creeps by 10 nm per volt gives $V_A \approx 100$ V. The height $n(0)$ never enters: V_A is a ruler measurement on the base, not a property of the current.


(a) the BC SCR advances by δ per volt

(b) $I_C = K(V_{BE})\,(1 + V_{CE}/V_A)$: one common zero

Figure 8.16 – *The Early effect. (a) The ruler picture: same $n(0)$, the BC SCR advances by δ per volt, the line becomes steeper. (b) The linear model $I_C = K(V_{BE})(1+V_{CE}/V_A)$ for three values of V_{BE} ($V_A = 44$ V, the reference transistor at $V_{CB} = 5$ V): extended backwards, all the plateaus meet the axis at the same point, $-V_A$.*

8.17 The Early voltage and the output resistance

THE QUESTION

Why do all the extended plateaus meet at the same point, and what does the slope give for the circuit?

Factorization. At fixed V_{BE}, I_C depends on V_{CB} only through W_B. Write the base width as $W_B(V_{CB}) \simeq W_{B0} - \delta V_{CB}$ near the operating range; then

$$I_C = \underbrace{\frac{qSD_n n_{B0}}{W_{B0}} \mathrm{e}^{V_{BE}/V_T}}_{K(V_{BE})\text{: injection}} \times \underbrace{\left(1 + \frac{V_{CB}}{V_A}\right)}_{\text{geometry of the base}}, \qquad V_A = \frac{W_{B0}}{\delta}. \tag{8.26}$$

The current is a product: an injection factor that depends only on V_{BE}, times a base-width factor that depends only on V_{CB} and not on the current. This is why the extended plateaus all cross zero at the same voltage, $V_{CB} = -V_A$, and all have the same relative slope $1/V_A$. In the variables of the electronics course, $V_{CE} = V_{CB} + V_{BE}$ with $V_{BE} \ll V_A$:

$$\boxed{I_C \simeq I_{C0}\left(1 + \frac{V_{CE}}{V_A}\right), \qquad r_o = \left(\frac{\partial I_C}{\partial V_{CE}}\right)^{-1} \simeq \frac{V_A}{I_C}}. \tag{8.27}$$

Typical values of V_A are 50 to 150 V for common discrete transistors (Neamen quotes 100 to 300 V): r_o is tens to hundreds of kilohms at a milliampere, and the transistor is still a good current source. The factorization breaks down at high injection, where $n(0)$ approaches N_B and the base charge itself changes with the current.

EXAMPLE 8.9 — THE EARLY VOLTAGE OF THE REFERENCE TRANSISTOR

At $V_{CB} = 5$ V the BC SCR is 2.84 µm wide, of which $N_C/(N_B + N_C) = 1/11$ in the base; it grows by 0.251 µm per volt, so its base edge advances by $\delta = 22.8$ nm/V. With $W_B = 1.0$ µm, $V_A = W_B/\delta = 44$ V and $r_o = 44$ kΩ at 1 mA. This modest value comes from the lightly doped base of the reference transistor; real transistors reach 50 to 300 V. Since the SCR widens as $\sqrt{V_{dC} + V_{CB}}$, δ is not constant: the same calculation gives 32 V at $V_{CB} = 2$ V and 54 V at 10 V. The single V_A read from a graph is an average over the range used.

THE SAME MOVING EDGE, TWO READINGS

When V_{CB} grows by dV_{CB}, the base edge of the BC SCR advances by $\delta\, dV_{CB}$ and uncovers acceptors carrying the charge $qSN_B\delta\, dV_{CB}$. Chapter 7 called this charge $C_T\, dV$. Hence $\delta = C_{jc}/(qSN_B)$ and $V_A = qSN_B W_B/C_{jc}$: the charge of the majority holes of the base divided by the BC transition capacitance. A base with much majority charge resists modulation; a small BC capacitance helps too. The same edge appears again in Part F, as a capacitance.

WHAT PART E HAS ESTABLISHED

- The BC SCR advances into the base when V_{CB} grows: same $n(0)$, shorter base, steeper line, larger I_C. Inherent, never zero; reduced by a heavily doped base and a lightly doped collector.

- Ruler: $V_A \simeq W_B/\delta$, with δ the advance per volt (1 µm and 10 nm/V give 100 V); $n(0)$ never enters.
- Factorization $I_C = K(V_{BE})(1+V/V_A)$: a common zero at $-V_A$, the same relative slope for every curve. $I_C \simeq I_{C0}(1+V_{CE}/V_A)$, $r_o \simeq V_A/I_C$. Reference: 44 V, 44 kΩ at 1 mA.

PART F

Capacitances

The diffusion capacitance of the forward-biased EB junction, the transition capacitance of the reverse-biased BC junction

8.18 The two capacitances of the transistor

THE QUESTION

Which charges inside the transistor change when its voltages change, and why does the short base make it fast?

Chapter 7 found two stores of charge in a junction: the ions of the SCR (transition capacitance C_T) and the injected minority carriers (diffusion capacitance C_D). The transistor has one junction of each kind.

The EB junction, forward biased: a diffusion capacitance. The injected electrons of the base store the charge (8.11), $Q_B = \tau_B I_C$, kept neutral by an equal excess of holes that the base terminal supplies. To change I_C, this charge must change first. Exactly as for the diode of Chapter 7,

$$\boxed{C_D = \frac{dQ}{dV_{BE}} = \tau_F \frac{dI_C}{dV_{BE}} = \frac{\tau_F I_C}{V_T}}, \tag{8.28}$$

where τ_F, the **forward transit time**, is essentially the base transit time $\tau_B = W_B^2/2D_n$ (the holes stored in the emitter and the crossing of the BC SCR add a little). For the reference transistor, $\tau_F \approx \tau_B = 0.143$ ns and $C_D = 5.5$ pF at 1 mA; for a typical transistor with $\tau_F \approx 0.5$ ns, $C_D \approx 20$ pF at 1 mA. The forward-biased EB junction also has its transition capacitance, a few picofarads, in parallel.

The twist: a transit time, not a lifetime. In the diode of Chapter 7, the stored charge was $Q = \bar{\tau} I$, with $\bar{\tau}$ close to the lifetime for a long region. In the transistor the time is τ_B, the time to cross the base, seven hundred times shorter than τ_n. The reason is the **two exits**. An injected electron can leave the base in two ways: by recombining, which takes about τ_n, or by reaching the collector, which takes about τ_B. The faster exit wins. In a long base the only exit is recombination, and the charge per unit of current is τ_n; in a short base the collector drains the electrons long before they recombine, and the charge is τ_B per unit of current. For the same current the short base stores $\tau_n/\tau_B = 700$ times less charge in the reference transistor, thousands of times less in real thin bases: it charges and discharges that much faster. The same short base that gives the transistor its gain also makes it fast.

The BC junction, reverse biased: a transition capacitance. The reverse-biased BC junction injects no minority carriers and stores no diffusion charge. It has only the transition capacitance of Chapter 7, in the reverse bias where the depletion approximation is excellent:

$$\boxed{C_{jc} = \frac{\varepsilon S}{W_{BC}} = \frac{C_{jc0}}{\sqrt{1 + V_{CB}/V_{dC}}}}, \tag{8.29}$$

with $C_{jc0} = 1.07$ pF for the reference transistor and $C_{jc} = 0.365$ pF at $V_{CB} = 5$ V; discrete transistors have 1 to 5 pF. It is the moving edge of the Early effect (Part E), seen as a capacitance.

EXAMPLE 8.10 — THE CAPACITANCES OF THE REFERENCE TRANSISTOR AT 1 MA

$C_D = \tau_B I_C/V_T = 0.143 \text{ ns} \times 1 \text{ mA}/25.9 \text{ mV} = 5.5$ pF, growing in proportion to the current; $C_{jc} = 0.365$ pF at $V_{CB} = 5$ V, falling slowly as V_{CB} grows. With a 0.1 µm base, τ_B would be a hundred times shorter and C_D a hundred times smaller.

Where this leads. These two capacitances, with the slopes of the characteristics ($g_m = I_C/V_T$ from the injection, the input resistance from the base current, r_o from the Early effect), are the physical content of the high-frequency small-signal model of the transistor, the **Giacoletto** model. Its assembly and its use in amplifier stages belong to the analog electronics course.

WHAT PART F HAS ESTABLISHED

- EB forward: diffusion capacitance $C_D = \tau_F I_C/V_T$, with $\tau_F \approx \tau_B = W_B^2/2D_n$, the base transit time, not the lifetime: 5.5 pF at 1 mA for the reference transistor, about 20 pF for $\tau_F = 0.5$ ns.
- Two exits: recombination (τ_n) or the collector (τ_B); the faster wins. The short base stores hundreds to thousands of times less charge than a long one: it is fast.
- BC reverse: transition capacitance $C_{jc} = \varepsilon S/W_{BC}$, 0.37 pF for the reference transistor, 1 to 5 pF for discrete transistors.

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Structure**: two junctions sharing a base. Design constraints: emitter heavily doped (injection into the base), base lightly doped and short (little recombination). At equilibrium: two SCRs, two built-in voltages, flat E_F, the base a barrier for conduction electrons.
- **Physical picture**: the reverse-biased BC junction collects every electron that reaches it but alone has almost nothing to collect; the forward-biased EB junction supplies electrons; a short base lets them reach the collector. The base current replaces the holes lost; it does not cause I_C.
- **Two edge densities**: $n(0) = n_{B0}\mathrm{e}^{V_{BE}/V_T}$, $n(W_B) = n_{B0}\mathrm{e}^{-V_{CB}/V_T}$. Long base: two diodes back to back. Short base: a straight line, slope = current.

- **Currents**: $I_C = I_S e^{V_{BE}/V_T}$, $I_S = qSD_n n_i^2/(N_B W_B)$; $1/\beta_F = (D_p N_B W_B)/(D_n N_E d_E) + W_B^2/2L_n^2$; $\tau_B = W_B^2/2D_n$.
- **Ebers–Moll**: $I_C = I_S(e^{V_{BE}/V_T} - 1) - (I_S/\alpha_R)(e^{-V_{CB}/V_T} - 1)$, $I_B = (I_S/\beta_F)(e^{V_{BE}/V_T} - 1) + (I_S/\beta_R)(e^{-V_{CB}/V_T} - 1)$; $1/\alpha_R = 1 + 1/\beta_R$ counts the holes injected into the collector.
- **I_C against V_{CB}**: a plateau set by V_{BE} (the collector only collects what the emitter sends); I_C flows at $V_{CB} = 0$; BC forward: knee in the last ≈100 mV, two pumps, zero at $-V_{BE} + V_T \ln(1/\alpha_R)$.
- **I_C against V_{CE}**: the same curves slid by V_{BE}; labelled by I_B; $V_{CE,sat} \approx 0.1$–0.2 V.
- **Early effect**: $V_A \simeq W_B/\delta$; $I_C \simeq I_{C0}(1 + V_{CE}/V_A)$, common intercept $-V_A$, $r_o \simeq V_A/I_C$.
- **Capacitances**: $C_D = \tau_F I_C/V_T$ (EB, $\tau_F \approx W_B^2/2D_n$), $C_{jc} = \varepsilon S/W_{BC}$ (BC).

Comprehension questions

No new physics here: answer from the figures and equations of the chapter. The answers are at the end.

Q1. At equilibrium, why does no current flow through the transistor, although it contains two fields?

Q2. Why does the EB SCR lie almost entirely in the base, and the BC SCR mostly in the collector?

Q3. Forward bias multiplies both edge densities of the EB junction by the same factor. Why, then, are many more electrons injected into the base than holes into the emitter?

Q4. A reverse-biased diode carries almost no current. Why does the reverse-biased BC junction of a transistor carry a large one?

Q5. Why is a transistor with a base ten diffusion lengths long just two diodes back to back?

Q6. Why does the collector current flow with $V_{CB} = 0$? What would stop it?

Q7. Why does the plateau of $I_C(V_{CB})$ hold almost unchanged when the BC junction becomes slightly forward biased, and then fall so steeply?

Q8. Without holes injected into the collector, where is the zero of I_C? Why does the real zero come earlier?

Q9. Why do all the extended plateaus meet at the same voltage $-V_A$?

Q10. Why does the diffusion capacitance of the EB junction use the base transit time and not the lifetime?

Q11. Why is C_{jc} only a transition capacitance?

Exercises

The practice exercises of this chapter are in the companion **Exercise collection**, Chapter 8, with their answers; the worked solutions are in the **Solutions** document. The 8 exercises that used to close this chapter are Problems 8.14 to 8.21 there; the other problems of the set (8.1 to 8.13) go further.

Answers

Q1. In each SCR drift and diffusion balance for each carrier, as in Chapter 5; the Fermi level is flat, and a current flows only where it has a gradient.

Q2. Each SCR holds equal charges on its two sides; the lightly doped side must be wider to hold the same charge. $N_B \ll N_E$, and $N_C \ll N_B$.

Q3. The factor multiplies the equilibrium minority densities, $n_{B0} = n_i^2/N_B$ and $p_{E0} = n_i^2/N_E$, and $N_E = 100N_B$.

Q4. The field of a reverse-biased junction sweeps every minority carrier that reaches its edge; in the diode these are rare, in the transistor the EB junction delivers 10^{10} times more to the edge, through a short base.

Q5. The injected excess dies out as e^{-x/L_n}; after ten diffusion lengths only e^{-10} of it is left. The BC junction sees the equilibrium base and carries only its own reverse current, whatever V_{BE}.

Q6. The BC edge is at n_{B0}, 10^{10} times below $n(0)$: on the scale of the line it is empty, and the built-in field collects. Only a forward bias of the BC junction large enough to raise its edge density towards $n(0)$ (the knee) stops it.

Q7. The collector end rises as e^{-V_{CB}/V_T} but stays negligible until $-V_{CB}$ is within a few V_T of V_{BE}; then it rises exponentially, and the line flattens within about 100 mV.

Q8. At $V_{CB} = -V_{BE}$: equal ends, flat line. The forward-biased BC junction also injects holes into the collector, which subtract from I_C; the zero is reached when the residual electron slope balances them, $V_T \ln(1/\alpha_R)$ earlier.

Q9. $I_C = K(V_{BE})(1+V/V_A)$: an injection factor times a base-width factor that does not depend on the current; the second factor vanishes at the same voltage for every curve.

Q10. The injected electrons have two exits, recombination (about τ_n) and the collector (about τ_B); in a short base the collector drains them first, so the charge per unit current is τ_B.

Q11. The BC junction is reverse biased: it injects no minority carriers, so only the charge of its SCR changes.

References

- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — Ch. 12: transistor action (§12.1), minority-carrier distributions (§12.2), current gain factors (§12.3), base-width modulation and other non-ideal effects (§12.4), the Ebers–Moll model (§12.5.1), frequency limitations and delay times (§12.6).
- S. M. Sze and M. K. Lee, *Semiconductor Devices: Physics and Technology*, 3rd ed., Wiley, 2012 — Ch. 4: transistor action and static characteristics, including the Early effect (§§4.1–4.2), frequency response (§4.3).
- B. Chaabane, *Électronique analogique 1*, course notes, ENISO, 2026/2027 — Ch. 4: the Ebers–Moll equations, the Early effect and the hybrid-π model as used in circuits; Ch. 5: the Giacoletto model and f_T.

SEMICONDUCTOR DEVICES PHYSICS

ANNEX A

Energy: One Thread Through the Course

State and transfer · the zero of energy · hydrogen and negative energies · quantum states · from atomic levels to bands · the band edge · allowed and occupied states · where the energy goes

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WHERE WE ARE

The word **energy** appears on almost every page of this course: the energy needed to break a bond (Chapter 2), energy bands and the Fermi energy (Chapter 3), the photon energy and the energy released by recombination (Chapter 4), the barrier qV_d of the junction (Chapter 5), and its lowering by a forward bias (Chapters 6 and 8). Each chapter used the idea as far as it needed it. This annex follows the idea from beginning to end, in a single argument.

The aim is a habit of thought rather than new results. Whenever an energy appears, you should be able to say **what it belongs to** (which state of which system), **relative to what** it is measured, and, when something changes, **where the energy goes**. The annex can be read after Chapter 3; Section A.7 uses the recombination mechanisms of Chapter 4.

ANNEX OUTLINE

A.1 Energy describes a state; work describes a transfer
A.2 The zero is chosen: what a negative energy means
A.3 From trajectories to quantum states
A.4 From atomic levels to crystal bands
A.5 The band edge: what $E - E_C$ means
A.6 Allowed states are not occupied states
A.7 Transitions: where does the energy go?
A.8 What to retain
Summary · Comprehension questions · Exercises · Answers · References

BY THE END OF THIS ANNEX YOU CAN

- distinguish an energy that describes a state (K, U, the energy of a quantum state) from a transfer of energy (work, heat, a photon, a phonon);
- explain a negative energy as a value below a chosen zero, and read ionisation energies, gaps and barriers as energy **differences**;
- estimate the size and the energy of the hydrogen atom from the cost of confining an electron, and transpose the estimate to a donor in silicon;
- say what a wave function gives (probabilities) and what it does not (a trajectory), and use $E = \hbar\omega$ and $p = \hbar k$ in their proper conditions;

- explain how the levels of isolated atoms become bands, and why an $E(k)$ diagram is not a map of positions;
- interpret $E - E_C$ as a kinetic-energy-like excess, and follow the energy of a drifting electron from the field to the lattice;
- keep apart the states that **exist** (bands) and the states that are **occupied** (statistics, E_F);
- for any transition, name the initial state, the final state, the interaction and the recipient of the energy.

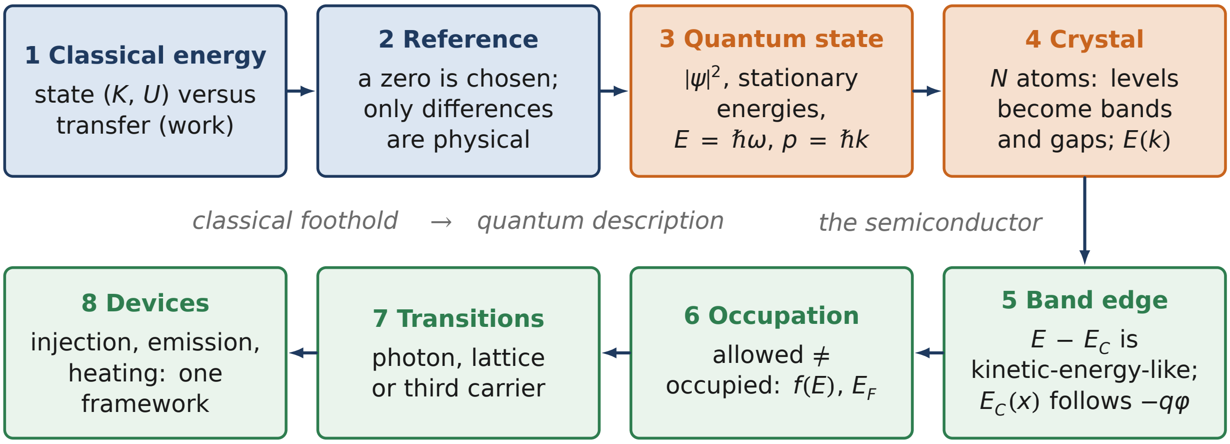

1 Classical energy
state (K, U) versus transfer (work)
2 Reference
a zero is chosen; only differences are physical
3 Quantum state
|ψ|², stationary energies, E = ħω, p = ħk
4 Crystal
N atoms: levels become bands and gaps; E(k)
classical foothold → quantum description
the semiconductor
8 Devices
injection, emission, heating: one framework
7 Transitions
photon, lattice or third carrier
6 Occupation
allowed ≠ occupied: f(E), E_F
5 Band edge
E − E_C is kinetic-energy-like; E_C(x) follows −qφ

Figure A.1 *– The argument of this annex in one view. Classical energy gives a physical foothold (1–2); quantum mechanics changes the description of states (3); crystal periodicity organises those states into bands (4–5); statistics decides which are occupied (6); interactions account for the energy when a carrier changes state (7). The device chapters then use one framework (8).*

A.1 Energy describes a state; work describes a transfer

THE QUESTION

A stone falls from a table to the floor. What changes, what is transferred, and what stays the same?

Let us start from situations everyone has seen: a body is accelerated, a weight is raised, a spring is compressed. In each of them something changes (a speed, a height, a length), and in each of them a force acts along a displacement.

Work changes the kinetic energy. Take a body of mass m moving along x under a net force F. Newton's law, $F = m\,\mathrm{d}v/\mathrm{d}t$, multiplied by the small displacement $\mathrm{d}x = v\,\mathrm{d}t$, gives

$$F\,\mathrm{d}x = m\,\frac{\mathrm{d}v}{\mathrm{d}t}\,v\,\mathrm{d}t = m\,v\,\mathrm{d}v = \mathrm{d}\!\left(\tfrac{1}{2}mv^2\right).$$

Adding these contributions along the path,

$$W_{net} = \int F\,\mathrm{d}x = \Delta K, \qquad K = \tfrac{1}{2}mv^2. \tag{A.1}$$

The quantity K can be computed from a single snapshot of the body (its mass and its speed at one instant): it describes the **state** of the body. The work W exists only while the body moves under the force: it describes a **process**, a transfer.

Potential energy. The weight mg does the work mgh when the body falls by h, whatever the path. Because this work depends only on the initial and final positions, it can be written as the decrease of a function of position, the **potential energy**: $W_{grav} = -\Delta U$, with $U_{grav} = mgh$. A spring of stiffness k compressed by x does the same with $U_{spring} = \frac{1}{2}kx^2$. Combining with (A.1), when only such forces act,

$$\Delta K + \Delta U = 0 \qquad \Longleftrightarrow \qquad E = K + U = \text{constant}. \tag{A.2}$$

Conservation is not immobility. During the fall, K grows and U decreases, continuously; only their sum stays fixed (Figure A.2a). A stone of 1 kg released from 2 m has $U = 19.6$ J and $K = 0$ at the top, 9.8 J of each halfway, and $K = 19.6$ J, a speed of 6.3 m/s, just before it touches the floor. When it hits the floor, the floor exerts a force over a very short distance and the 19.6 J are transferred again, now to the vibrations of the stone and of the floor, which is to say heat. The energy has not disappeared; it has changed hands.

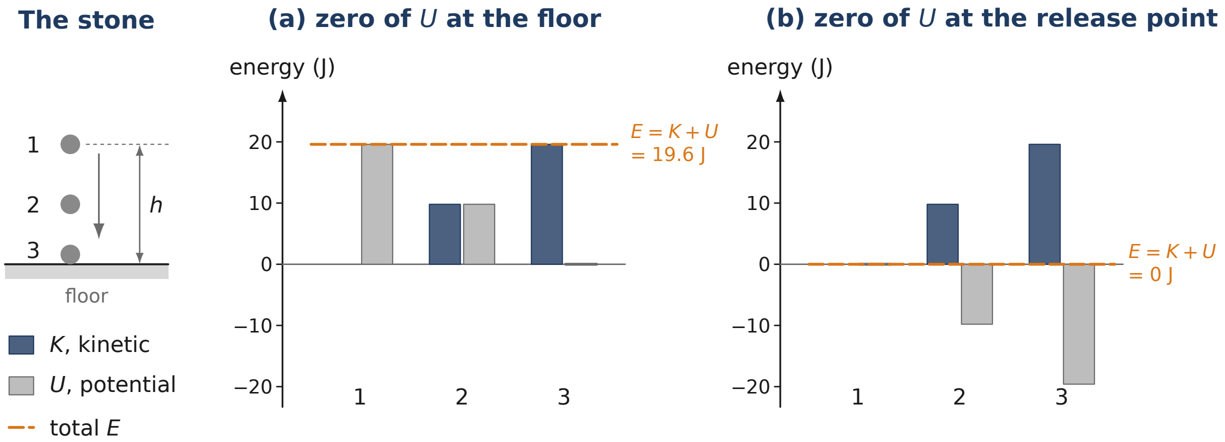


Figure A.2 – A 1 kg stone released at rest from $h = 2$ m (instant 1), halfway (2) and just before it reaches the floor (3). (a) With the zero of potential energy at the floor, U goes from 19.6 J to 0 while K goes from 0 to 19.6 J; the total stays at 19.6 J. (b) With the zero at the release point, every value of U is lowered by 19.6 J and the total is 0. The kinetic energies, the speeds and every change of U are the same in both panels.

BE AWARE OF… ENERGY IS NOT A SUBSTANCE

It is tempting to imagine energy as a fluid stored inside the stone and poured into the floor. Prefer this: energy is a **number attached to the state** of a system, computed from its positions and velocities; work (and later heat, a photon, a phonon) is a **mechanism of transfer** between systems. In the semiconductor, the same distinction separates the energy of an electron state from the processes, absorption, emission, collisions, that move an electron from one state to another.

The classical examples fix what energy means and how it is transferred. They do not prove that an electron in an atom behaves like a small planet. The next steps keep what is useful, the meaning of energy and of a reference, and change the model where it fails.

A.2 The zero is chosen: what a negative energy means

THE QUESTION

If the zero of the stone's potential energy is moved to the release point, has the physics changed? And what does the energy -13.6 *eV of the hydrogen atom mean?*

A.2.1 A reference is a convention

Only the work, the *change* of U, was defined in Section A.1. The value of U itself depends on where we put $U = 0$, and this choice is ours. In Figure A.2b the zero is at the release point: the potential energy is 0, then -9.8 J, then -19.6 J, and the total energy is 0 throughout. Nothing observable has changed: the speeds are the same, the force $-\mathrm{d}U/\mathrm{d}x$ is the same, and every difference of energy is the same. **Energy values depend on the chosen zero; energy differences do not.** A negative value only says that the state lies below the chosen zero.

A.2.2 Hydrogen: why the electron does not fall onto the proton

For the hydrogen atom the natural zero is the electron and the proton **infinitely far apart, with the electron at rest**. The Coulomb potential energy,

$$U(r) = -\frac{q^2}{4\pi\varepsilon_0 r},$$

is then negative at every finite distance, and it decreases without limit as $r \to 0$. Classically, nothing prevents the electron from falling onto the proton. Something must, since atoms have a size. That something is quantum, and an elementary argument shows its effect.

Confinement costs kinetic energy. An electron described by a wave confined to a region of size r has a wavelength of order r, hence, through the de Broglie relation $p = \hbar k$ used in Chapter 3, a momentum of order $\hbar/r$ and a kinetic energy of order

$$K(r) \approx \frac{\hbar^2}{2mr^2}.$$

The tighter the confinement, the higher the kinetic energy. Squeezing the electron closer to the proton lowers U but raises K; the atom settles where the total

$$E(r) \approx \frac{\hbar^2}{2mr^2} - \frac{q^2}{4\pi\varepsilon_0 r} \tag{A.3}$$

is lowest (Figure A.3a). Setting $\mathrm{d}E/\mathrm{d}r = 0$,

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{mq^2} = 0.053\ \text{nm}, \qquad E = -\frac{\hbar^2}{2ma_0^2} = -13.6\,\text{eV}, \tag{A.4}$$

and at this size $K = +13.6$ eV and $U = -27.2$ eV. The kinetic energy is positive, the potential energy negative, and the total is negative: the bound state lies 13.6 eV below the chosen zero. The exact solution of the Schrödinger equation (Neamen, §2.4.1) gives exactly these values, as averages over the state: $\langle K\rangle = +13.6$ eV, $\langle U\rangle = -27.2$ eV, $\langle E\rangle = -13.6$ eV. Our estimate is lucky in its numerical factor, not in its physics: the size of the atom is set by the competition between the Coulomb attraction and the kinetic cost of confinement.

Ionisation energy is a difference. To take the electron from the ground state to the zero, the free electron at rest far away, one must supply $0-(-13.6) = 13.6$ eV: the **ionisation energy**. The exact solution also gives the excited levels, $E_n = -13.6\,\text{eV}/n^2$ (-3.4, -1.51, -0.85 eV, ...). When the atom goes from $n = 2$ to $n = 1$ it releases $-3.4-(-13.6) = 10.2$ eV, as a photon of wavelength $1240/10.2 = 122$ nm. Spectroscopy measures such differences, never the absolute values. Shift the zero to the ground state and the levels read 0, 10.2, 12.1, 12.8 and 13.6 eV (Figure A.3b): the same atom, the same lines.

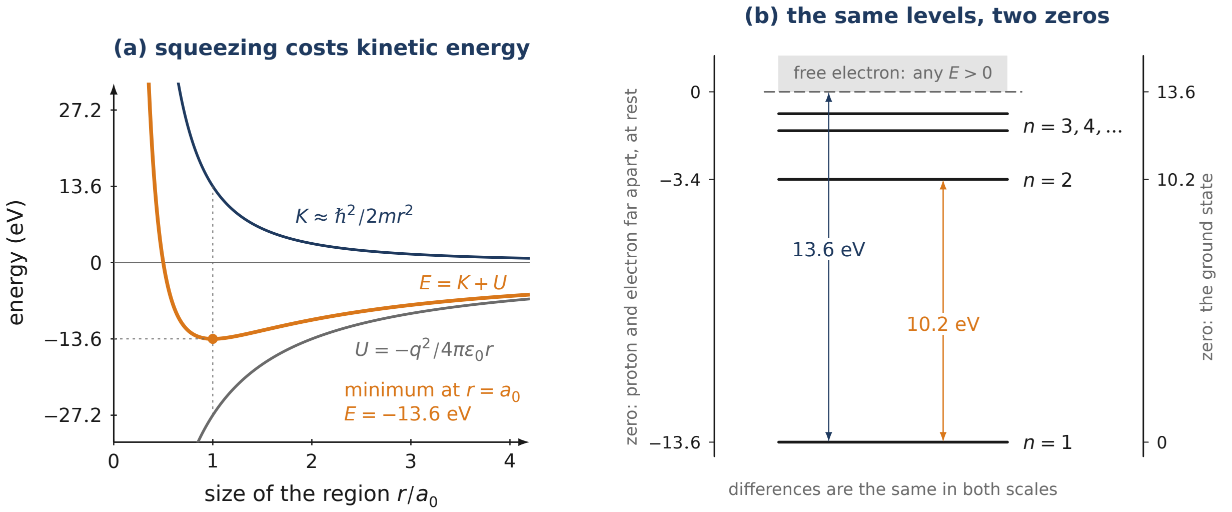


Figure A.3 – *(a) The hydrogen estimate (A.3): the kinetic cost of confinement $K \approx \hbar^2/2mr^2$ (blue) rises steeply at small size, the Coulomb energy U (grey) falls; their sum (orange) has a minimum at $r = a_0 = 0.053$ nm, $E = -13.6$ eV, where $K = 13.6$ eV and $U = -27.2$ eV. (b) The levels of hydrogen read with two zeros: on the left, the electron at rest far from the proton; on the right, the ground state. Every difference (13.6 eV to ionise, 10.2 eV for the first line) is the same on both scales.*

EXAMPLE A.1 — THE SAME ARGUMENT INSIDE SILICON: THE DONOR OF CHAPTER 2

In Chapter 2 the fifth electron of a phosphorus atom was modelled as a hydrogen atom inside silicon. Estimate (A.3) shows at once what changes. The Coulomb term is divided by $\varepsilon_r = 11.7$, and the kinetic term contains the effective mass $m^* = 0.26\,m_0$ instead of m_0 (Chapter 3: in the crystal the electron responds to forces with m^*). Redoing the minimisation, a_0 is multiplied by $\varepsilon_r m_0/m^*$ and the energy by $(m^*/m_0)/\varepsilon_r^2$:

$$a_d = 0.053 \times \frac{11.7}{0.26} = 2.4\ \text{nm}, \qquad E_d = 13.6 \times \frac{0.26}{11.7^2} = 26\ \text{meV},$$

the values of Chapter 2. Note the zero. In vacuum the zero was the electron at rest far away; in the crystal the free electron ‘at rest’ is the electron at the bottom of the conduction band. The donor level therefore lies 26 meV (measured: 45 meV for P) **below** E_C: E_C plays the role of the zero of the hydrogen problem.

CONCEPTUAL CHECKPOINT

A negative energy is not 'less than nothing'. It is a value below a chosen zero. Explain the ionisation energy of hydrogen, the ionisation energy of a donor, the gap E_g and the barrier qV_d as **differences** between two states; the band diagrams of this course never need an absolute zero of energy.

A.3 From trajectories to quantum states

THE QUESTION

The estimate of Section A.2.2 never mentioned an orbit. What replaces the trajectory of classical mechanics?

The conceptual change. Classical mechanics describes a particle by its position and momentum at each instant, and follows its trajectory. Quantum mechanics describes the system by a **state**. In the position representation the state is a wave function $\psi(x,t)$, and its meaning is statistical: the probability of finding the particle between x and $x + \mathrm{d}x$ is

$$|\psi(x,t)|^2\,\mathrm{d}x. \tag{A.5}$$

The wave function is not a material substance of which the electron is made, nor a ripple in space. Its complex amplitude carries the information needed to compute the probabilities of the results of measurements (Neamen, §2.2.2).

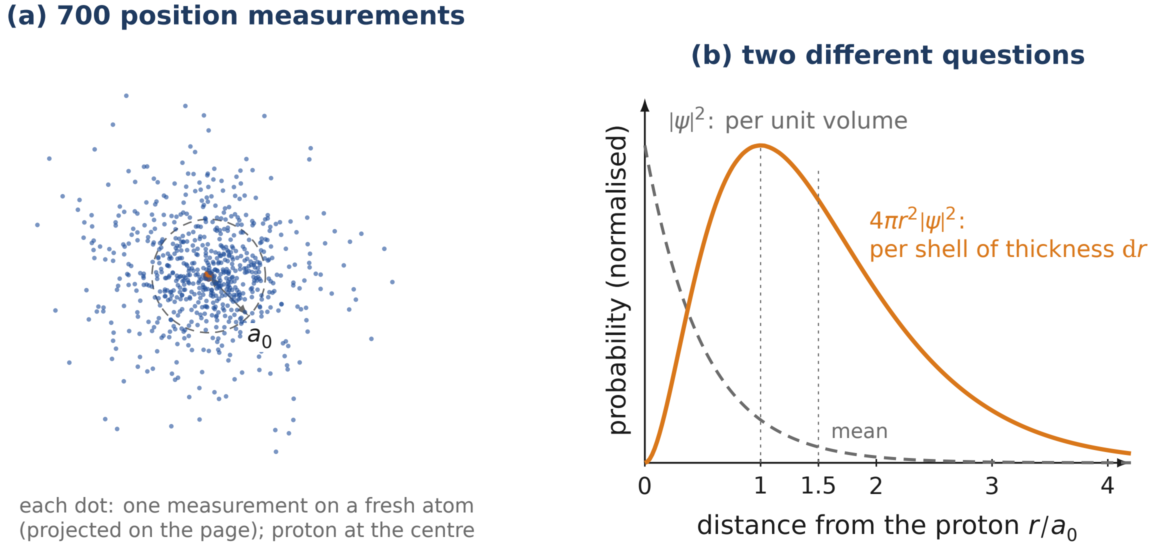
What a probability density looks like. For the ground state of hydrogen, $|\psi|^2 \propto \mathrm{e}^{-2r/a_0}$. Imagine preparing many identical atoms and measuring the position of the electron once in each (Figure A.4a). The results scatter around the proton; there is no curve along which the electron travels. Two different questions then have two different answers (Figure A.4b). The probability *per unit volume*, $|\psi|^2$, is largest at the proton. The probability of finding the electron at a distance between r and $r + \mathrm{d}r$, in a thin shell, is $4\pi r^2|\psi|^2\,\mathrm{d}r$; it vanishes at the centre (a shell of zero radius has zero volume) and peaks at $r = a_0$. The mean distance is $1.5\,a_0$. The 'size' a_0 of Section A.2.2 is the most probable distance, not the radius of an orbit.

Stationary states and $E = \hbar\omega$. A state of definite energy E evolves in time only through a phase,

$$\psi(x,t) = \varphi(x)\,\mathrm{e}^{-iEt/\hbar}, \qquad \text{so that} \qquad E = \hbar\omega. \tag{A.6}$$

Its probability density $|\psi|^2 = |\varphi|^2$ does not change with time: hence the name **stationary state**. This is why an atom in its ground state is stable, whereas a classical electron on a circular orbit, an accelerated charge, would radiate and spiral onto the proton. The relation $E = \hbar\omega$ links energy to the rate of phase evolution. For light it is the photon energy $E = h\nu = \hbar\omega$ that Chapter 4 used with $\lambda = hc/E$, that is $\lambda(\mathrm{nm}) \approx 1240/E(\mathrm{eV})$.

Momentum and $p = \hbar k$. A plane wave $\mathrm{e}^{i(kx-\omega t)}$ has a single wavenumber k, and for it $p = \hbar k$ (the de Broglie relation of Chapter 3). A bound state is not a single plane wave. The ground state of hydrogen has a **definite energy** but **no definite momentum**: its average momentum is zero (it goes nowhere), while the average of p^2 is $\hbar^2/a_0^2$, which is precisely the kinetic energy $\langle p^2\rangle/2m = 13.6$ eV found in Section A.2.2. A state of definite energy may combine many momenta.



(a) 700 position measurements

(b) two different questions

each dot: one measurement on a fresh atom (projected on the page); proton at the centre

Figure A.4 *– The ground state of hydrogen. (a) Results of 700 position measurements, each on a freshly prepared atom (computed from* $|\psi|^2$ *and projected onto the page); the dashed circle has radius* a_0*. (b) Two probabilities: per unit volume,* $|\psi|^2$ *(dashed), largest at the proton; per shell of radius* r *and thickness* $\mathrm{d}r$, $4\pi r^2|\psi|^2$ *(orange), largest at* $r = a_0$*, with a mean of* $1.5\,a_0$*. Both are normalised to 1 at their maximum.*

Keep	**Do not imply**
A quantum state can have a definite energy.	That the electron follows a classical orbit.
$|\psi|^2$ gives a probability density for the position.	That ψ itself is a probability, or a material wave.
$E = \hbar\omega$ links energy to the phase evolution of a stationary state.	That the electron is a classical oscillator storing energy in a physical wave.
$p = \hbar k$ holds for a plane-wave component.	That every bound state has one definite momentum.

Once energy is understood as a property of a quantum state, the question for a solid becomes: which states are allowed when the electron is not in one atom but in a periodic crystal?

A.4 From atomic levels to crystal bands

> ***THE QUESTION***
>
> *An isolated atom has discrete levels. What happens to them when* 5×10^{22} *atoms per* cm^3 *are assembled into a crystal?*

Chapter 3 answered this question from one side: it started from free electrons in a wire and added the periodic potential of the ions, which opens gaps where the electron waves are Bragg-reflected. The same bands can be reached from the other side, starting from the atoms. The two routes meet, and seeing both prevents a common confusion.

Two atoms. Bring two identical atoms close enough for the states of their outer electrons to overlap. The electron is no longer bound to one atom or the other; the two atomic states combine into two states of the pair, one lower in energy and one higher (the lower one, with the electron density piled up between the nuclei, is the covalent bond of Chapter 2).

One level of the isolated atom has become two levels of the pair.

N **atoms.** With N atoms each atomic level becomes N levels of the crystal (Figure A.5). The spread of these levels is set by how strongly neighbouring atoms overlap, a few electronvolts, not by N. The N levels therefore crowd into a fixed energy range: in a cubic centimetre of silicon, about 10^{22} levels in a few eV, a spacing of order 10^{-22} eV, immeasurably smaller than $kT = 0.026$ eV. For every practical purpose the levels form a continuous **band**. Outer atomic levels overlap more and give wide bands; deep inner levels hardly overlap and stay narrow. Between two bands there may be an energy range with no state at all: a **gap**.

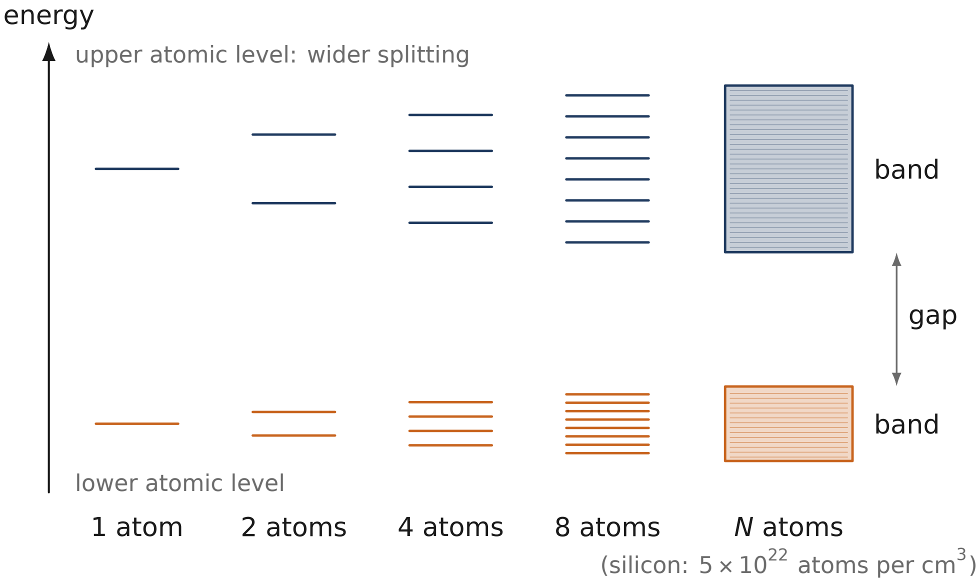

energy
upper atomic level: wider splitting
band
gap
band
lower atomic level
1 atom
2 atoms
4 atoms
8 atoms
N atoms
(silicon: 5 × 10^22 atoms per cm^3)

Figure A.5 – *From levels to bands (generic, not to scale). Each level of an isolated atom becomes 2, 4, 8, …, N levels when 2, 4, 8, …, N atoms interact. The spread is fixed by the overlap between neighbours, so the N levels crowd into a band of a few eV; the upper (outer) level overlaps more and gives a wider band. A gap separates the two bands. In silicon the outer s and p levels mix before forming the bands; Chapter 3 counts the result: four full bands.*

Counting. Each atomic level, with its two spin states, gives a band of $2N$ states, exactly the count obtained in Chapter 3 from the N allowed values of k in the Brillouin zone. The two routes, free electrons with a periodic potential and atoms brought together, agree on the essential: bands hold a fixed number of states, two per cell and per band, and they are separated by gaps.

Labelling the states: $E(k)$**.** The states of a band are labelled by the band index and the crystal wavenumber k; the dispersion relation $E(k)$ gives their energy. An E–k diagram is therefore a **catalogue of states**: it plots the energy of each allowed state against its wavenumber. It is not a map of where the electrons are in the crystal, and a band is not one atomic level blurred into a smear: it is a very large number of distinct states.

> **THE KEY TRANSITION**
>
> Do not jump from the discrete levels of hydrogen to the band diagram of a device. The physical change is this: many atoms form a periodic system, their states interact, and the allowed states belong to the whole crystal. Only then does a horizontal line

labelled E_C make sense, as the lowest energy of a band of crystal states.

A.5 The band edge: what $E - E_C$ means

THE QUESTION

An electron of the conduction band has the energy $E = E_C + 0.10$ eV. What is this 0.10 eV, and does it say where the electron is?

A kinetic-energy-like excess. Near the minimum k_0 of the conduction band, the dispersion relation is a parabola (Chapter 3, an input of solid-state physics):

$$E(k) \simeq E_C + \frac{\hbar^2 |k - k_0|^2}{2m^*}. \qquad \text{(A.7)}$$

This has exactly the form of the kinetic energy $p^2/2m$ of a free particle, with $\hbar(k - k_0)$ in the place of p and the effective mass m^* in the place of m. The energy above the band minimum is therefore **kinetic-energy-like**. The answer to the question is: the electron occupies a **higher allowed state** of the band, farther from k_0 (Figure A.6a), not a higher place in the crystal. With $m^* = 0.26\, m_0$, its speed is $v = \sqrt{2 \times 0.10\,\text{eV}/m^*} = 3.7 \times 10^5$ m/s.

How unusual is such an electron? At equilibrium the conduction electrons have, on average, the kinetic energy $\frac{3}{2}kT = 39$ meV (Section A.6). An excess of 0.10 eV is 2.6 times the average: a 'hot' electron. It will not stay hot: collisions with the lattice take its excess energy within about a picosecond (Chapter 4, thermalisation), and it settles back among the electrons within a few kT of E_C.

WHERE THE ANALOGY STOPS

The electron is not a small ball rolling on a shelf. A larger E means a different quantum state, not a different position. The kinetic-energy analogy is reliable near a parabolic extremum; far from it the band bends away from the parabola. The **crystal momentum** $\hbar k$ is not the ordinary mechanical momentum of the electron: it labels the state and changes under an applied force as a momentum would, but the periodic potential of the ions, which is already contained in $E(k)$ and in m^*, is not one of those forces.

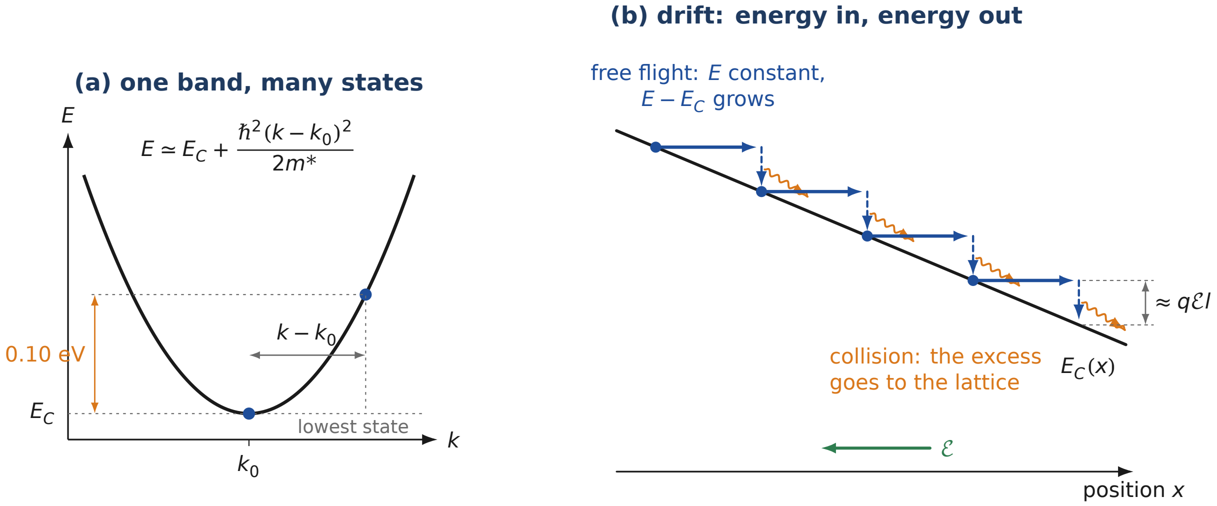
What E_C is, and what it is not. E_C is the energy of the lowest state of the conduction band. It is **not** the microscopic electrostatic energy of an electron at a point of the crystal: that energy varies strongly over each atomic cell, and its whole effect is already inside the band structure, which it created. What E_C does follow is a **slowly varying** potential $\varphi(x)$ added to the crystal, one that changes little over many cells, such as the potential of a battery or of a space-charge region. Such a potential shifts the whole band, rigidly:

$$E_C(x) = E_{C0} - q\varphi(x), \qquad E_V(x) = E_{V0} - q\varphi(x). \qquad \text{(A.8)}$$

In this sense, and only in it, $E_C(x)$ plays the role of the potential energy of a conduction electron and $E - E_C(x)$ the role of its kinetic energy. This is the meaning of 'the potential energy of the band' in Chapter 3, and the reason the bands of Chapter 5 bend with the potential.

Energy bookkeeping of drift. Put the two ideas together for an electron in a uniform field (Figure A.6b). The bands are tilted by (A.8). Between two collisions the electron moves freely: its total energy E is constant while $E_C(x)$ falls beneath it, so its kinetic-energy-like part $E - E_C$ grows, by about $q\mathcal{E}l$ over a mean free path l. At the collision it hands this excess to the lattice and starts again near the band edge. The field supplies energy, the lattice receives it: this is the microscopic picture of **Joule heating**, $J\mathcal{E}$ per unit volume.

How much per flight? In silicon, the Drude relation of Chapter 1 with $\mu_n = 1350\ \mathrm{cm}^2/(\mathrm{V{\cdot}s})$ and $m^* = 0.26\, m_0$ gives a collision time $\tau_c = \mu_n m^*/q = 0.20$ ps; with the thermal speed $\sqrt{3kT/m^*} = 2.3 \times 10^5$ m/s, the mean free path is $l \approx 46$ nm. In a field of 1 kV/cm the electron gains $q\mathcal{E}l \approx 4.6$ meV per flight, far below kT: it stays essentially thermal, and the linear laws of Chapter 1 hold. At 100 kV/cm the gain per flight would be 0.46 eV, far above kT: the electrons become hot, the collisions change character, and the drift velocity stops growing with the field (the saturation velocity, about 10^7 cm/s, met in Chapter 8).


(a) one band, many states

(b) drift: energy in, energy out

Figure A.6 – *(a) Near the minimum k_0 of the conduction band, the state with $E = E_C + 0.10$ eV is a state farther from k_0; the excess $\hbar^2(k - k_0)^2/2m^*$ has the form of a kinetic energy. (b) An electron in a uniform field (generic, not to scale; the field points to the left, E_C falls to the right). During each free flight its energy is constant and its height above E_C grows by about $q\mathcal{E}l$; at each collision it gives this excess to the lattice (wavy arrows, phonons). For holes the same picture holds upside down, in the valence band.*

A.6 Allowed states are not occupied states

THE QUESTION

The band structure says which states exist. Which of them hold an electron?

Two separate questions. Figure A.7 shows a pure silicon sample and a sample doped with donors. The allowed states, drawn as thin lines, are exactly the same in both: doping, which replaces at most a few atoms in a million, does not change the bands (Chapter 3). What differs is **which states are occupied**. That is decided by statistics: at thermal equilibrium the probability that an allowed state of energy E is occupied is the Fermi–Dirac function

$$f(E) = \frac{1}{1+\exp[(E-E_F)/kT]}. \tag{A.9}$$

The Fermi level E_F is the energy at which this probability would be one half. Its position is fixed by the number of electrons to accommodate (in a doped sample, through neutrality).

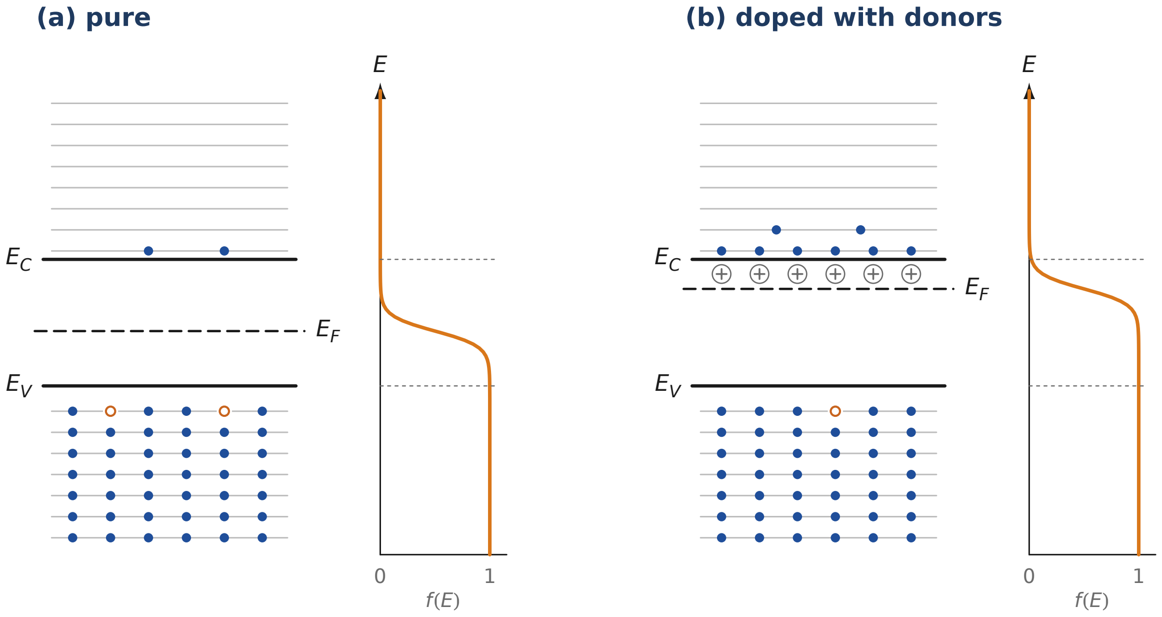

(a) pure **(b) doped with donors**

Figure A.7 – Allowed states (thin lines) and occupied states (filled dots: electrons; open circles: empty states of the valence band, holes) in (a) pure silicon and (b) silicon doped with donors (ionised donors ⊕ just below E_C). The bands, hence the allowed states, are the same; only the Fermi level moves, and with it the occupation probability $f(E)$ drawn on the right of each panel. Generic cartoon: the width of $f(E)$ around E_F is exaggerated, and the numbers of carriers are not to scale.

How different the occupations are. The probability that a state at E_C is occupied is $e^{-(E_C-E_F)/kT}$, which Chapter 3 showed to be n/N_C. In pure silicon it is $n_i/N_C = 10^{10}/2.8\times10^{19} = 3.6\times10^{-10}$: about four states in ten billion. With $N_D = 10^{16}\ \mathrm{cm^{-3}}$, E_F is 0.206 eV below E_C and the probability is $N_D/N_C = 3.6\times10^{-4}$. The same state is a million times more likely to be occupied; nothing about the state has changed.

Where are the electrons? Not at E_F. In a semiconductor E_F usually lies in the gap, where there are no states at all. The conduction electrons are spread over the first few kT above E_C, where the density of states is not yet small and the probability f not yet negligible. In the nondegenerate case their average kinetic-energy-like excess is $\frac{3}{2}kT =$

39 meV at 300 K, whatever the position of E_F: the Fermi level sets **how many** electrons there are; the temperature sets **how they are spread** in energy.

What E_F is. Statistical physics identifies the Fermi level with the **electrochemical potential** of the electrons: the increase of the free energy of the crystal when one electron is added, at constant temperature (an input). Three statements of the course follow from this meaning.

- At equilibrium E_F is the same everywhere (Chapter 5): if it were higher in one place, moving an electron from there to elsewhere would lower the free energy, and electrons would flow until the difference disappeared.
- A voltmeter reads the difference of the Fermi levels of its two leads divided by q (Chapter 5), not a difference of electrostatic potential.
- Out of equilibrium the electrons and the holes each keep a local level of their own, the quasi-Fermi levels E_{Fn} and E_{Fp} (Chapter 6). Their separation is the energy available per electron–hole pair, qV in a forward-biased junction.

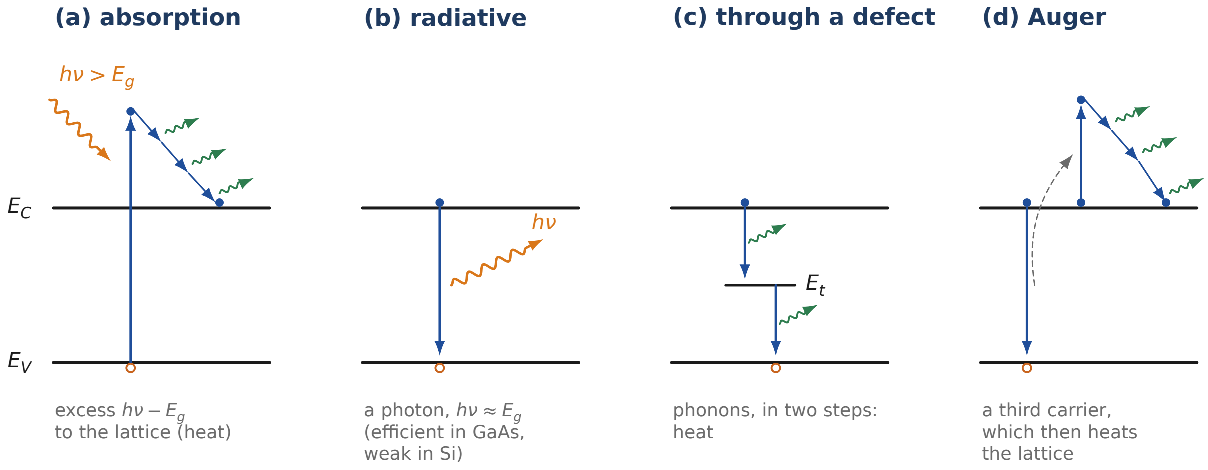
> **A USEFUL ORDER OF EXPLANATION**
>
> First the **available states** (the bands). Then the question **which are occupied** (the statistics). Only then the Fermi level, to interpret the populations. The Fermi level is not a shelf on which electrons sit, and it is not the energy of every electron; it is the parameter of the occupation probability.

A.7 Transitions: where does the energy go?

> ***THE QUESTION***
>
> *When a conduction electron drops into a hole, about 1.12 eV of electronic energy disappears from the electron system. Where does it go?*

When an electron changes state, the energy difference between the initial and the final state must be accounted for by the interaction that drives the change. Before any rate is computed (Chapter 4 does that), it pays to ask four questions every time: **the initial state, the final state, the interaction, and the recipient of the energy.** Figure A.8 and the table below apply them to the processes of Chapter 4.

Mechanism	Where the energy goes	Intuition to keep
Radiative	A photon is emitted when an electron recombines with a hole.	Part of the electronic energy appears as light; $h\nu \approx E_g$ in a direct-gap material.
Through a defect (SRH)	To lattice vibrations (phonons), in two smaller steps via a level in the gap.	No photon is needed; the energy has not disappeared, it has become heat.
Auger	To a third carrier, lifted high into its band.	Energy passes between carriers; the third one then relaxes and heats the lattice.
Thermalisation	The excess $h\nu - E_g$ of an absorbed photon goes to the lattice.	The light was absorbed, but not all its energy is available to the carriers.



excess $h\nu - E_g$ to the lattice (heat)

a photon, $h\nu \approx E_g$ (efficient in GaAs, weak in Si)

phonons, in two steps: heat

a third carrier, which then heats the lattice

Figure A.8 *– Energy accounting for four transitions (generic band diagrams; filled dots: electrons, open circles: holes; orange wavy arrows: photons; green wavy arrows: phonons, the lattice vibrations). (a) A photon with $h\nu > E_g$ creates a pair; the electron lands high in the band and loses the excess to the lattice in steps (thermalisation; the hole does the same in the valence band, not drawn). (b) Radiative recombination: a photon $h\nu \approx E_g$ carries the energy. (c) Recombination through a defect level E_t: two steps, the energy goes to the lattice. (d) Auger recombination: the energy lifts a third carrier high into its band, which then heats the lattice.*

Why steps help. A lattice vibration of silicon carries at most about 60 meV (an input of solid-state physics). Handing 1.12 eV directly to the lattice would require about 19 phonons at once, a very improbable event. A level near the middle of the gap splits the descent into two smaller steps; this is one way to see why defects are such efficient recombination centres in silicon, where the radiative path is slow (indirect gap, Chapter 4).

EXAMPLE A.2 — TWO ENERGY BUDGETS

A solar cell. A blue photon of 2.8 eV (443 nm) absorbed in silicon creates one pair; the excess $2.8 - 1.12 = 1.68$ eV, 60 % of the photon energy, becomes heat within about a picosecond. A red photon of 1.8 eV loses 0.68 eV, 38 %. An infrared photon of 1.0 eV, below the gap, is not absorbed at all. Thermalisation and transmission together are the main losses of a single-junction solar cell (Chapter 6).
A silicon diode. A forward-biased silicon diode at 0.7 V carrying 10 mA receives $0.7 \times 0.010 = 7$ mW from the source. In silicon nearly all the injected pairs recombine through defects; practically all of the 7 mW ends as lattice heat, and the diode warms without glowing. In a GaAs light-emitting diode a large part of the recombination is radiative, and a large part of the same kind of budget leaves as photons of about E_g.

DIAGNOSTIC QUESTION

A silicon diode warms up while emitting almost no visible light. Has energy been lost? No: it has been transferred, predominantly through non-radiative paths, and it ends as lattice heat. Whenever an energy seems to vanish, find its recipient.

Once you can track the states, their occupations and the transfers of energy, the topics of the device chapters, injection, emission, thermalisation, recombination rates and heating, become applications of one framework rather than separate facts.

A.8 What to retain

THE QUESTION

Which wrong pictures does this annex replace, and which distinctions should come back in every chapter?

Misconception	**Replacement idea**
Energy is a substance inside an object.	Energy is a number attached to the state of a system; work, heat, photons and phonons transfer it.
A negative energy means negative ‘stuff’.	The sign is relative to a chosen zero; only differences are observed.
A quantum electron is a tiny planet on an orbit.	A quantum state is described by a wave function and gives probabilities of measurement results.
Every state of definite energy has a definite momentum.	A state can have a definite energy and a spread of momenta.
A band is a blurred atomic level.	A band is a dense set of distinct crystal states, $2N$ per band.
An E–k plot shows where the electrons are.	It plots the energy of each state against its wavenumber.
E_C is a shelf, or the microscopic electrostatic potential.	E_C is the bottom of the conduction band; it follows only slowly varying potentials, $E_C(x) = E_{C0} - q\varphi(x)$.
The Fermi level is where the electrons sit.	It sets the occupation probabilities; often no state lies at E_F.
Recombination energy disappears if no photon is emitted.	It is transferred to the lattice or to a third carrier.

Distinction	**Why it matters in semiconductors**
State versus transfer	Separates energy from work, absorption, emission and heating.
Value versus difference	Makes band edges, gaps, binding energies and barriers interpretable.
Trajectory versus quantum state	Prevents orbit pictures of electrons in atoms and donors.
Allowed versus occupied	Separates the band structure from the Fermi–Dirac statistics.
Energy versus position	Makes $E(k)$ diagrams readable without treating them as maps.
Radiative versus non-radiative	Explains why electronic energy becomes light in GaAs and heat in Si.

QUESTIONS TO ASK YOURSELF IN ANY CHAPTER

- Relative to which zero is this energy given?
- Am I dealing with an energy value, an energy difference, or a transfer of energy?
- Which states are allowed in this model, and which of them are occupied?
- Does this diagram describe position, the energy of states, or occupation?
- If energy leaves a carrier or a state, which system receives it?
- Which part of the mechanical analogy is useful here, and where does it break down?

FINAL SYNTHESIS

The central thread is not that ‘energy appears everywhere’. It is a disciplined way of asking: what state is the system in, relative to what reference, which states are allowed and which are occupied, and how is energy transferred when the system changes?

Summary — essential knowledge

WHAT YOU MUST KNOW

- **State and transfer.** $W_{net} = \Delta K$; U defined by $W = -\Delta U$; $K + U$ constant when only such forces act. K and U describe a state; work describes a transfer.
- **Reference.** The zero of U is chosen; values change with it, differences do not. A negative energy lies below the zero.
- **Hydrogen.** Confinement costs $K \approx \hbar^2/2mr^2$; minimising $K + U$ gives $a_0 = 0.053$ nm, $E = -13.6$ eV, $K = 13.6$ eV, $U = -27.2$ eV. With m^* and ε_r: the donor, 26 meV below E_C, radius 2.4 nm.
- **Quantum state.** $|\psi|^2 \mathrm{d}x$ is a probability; stationary states, $E = \hbar\omega$; $p = \hbar k$ for a plane wave; definite energy does not mean definite momentum.
- **Bands.** N atoms turn each level into N crystal levels within a few eV: a band of $2N$ states (with spin), spacing $\sim 10^{-22}$ eV; gaps between bands. $E(k)$ is a catalogue of states, not a map.
- **Band edge.** $E - E_C \simeq \hbar^2|k-k_0|^2/2m^*$ is kinetic-energy-like; $E_C(x) = E_{C0} - q\varphi(x)$ for slowly varying φ. In drift the field gives $\approx q\mathcal{E}l$ per flight, the lattice takes it back (Joule heating).
- **Occupation.** $f(E) = 1/(1 + \mathrm{e}^{(E-E_F)/kT})$; same bands, different E_F; electrons within a few kT of E_C, mean excess $\frac{3}{2}kT$; E_F is the electrochemical potential.
- **Transitions.** Initial state, final state, interaction, recipient. Photon (radiative), phonons (SRH, thermalisation), third carrier (Auger).

Comprehension questions

No new physics here: answer by reading the figures and the equations of the annex. The answers are at the end.

Q1. A stone is held at rest 1 m above a table that is 1 m above the floor. Give its potential energy with three different zeros. What is the same in the three cases?

Q2. Why can the electron of a hydrogen atom not sit on the proton, although the Coulomb energy decreases without limit as $r \to 0$?

Q3. The ground state of hydrogen has $K = +13.6$ eV and $E = -13.6$ eV. How can the kinetic energy be positive and the total energy negative?

Q4. The probability per unit volume of the hydrogen ground state is largest at the proton, yet the most probable distance is a_0. Explain.

Q5. Why is the ground state of hydrogen stable, while a classical electron on an orbit would radiate?

Q6. A cubic centimetre of silicon contains 5×10^{22} atoms. Why do we speak of continuous bands rather than of separate levels?

Q7. An E–k diagram shows the conduction-band minimum and the valence-band maximum at different k. Does this mean that the electrons and the holes are in different places of the crystal?

Q8. Two silicon samples, one pure and one doped with 10^{16} donors per cm^3: are the allowed states the same? Are the occupations the same? By what factor does the occupation of a state at E_C differ?

Q9. Why does the energy given by the field to a drifting electron not accumulate in the electron?

Q10. A silicon diode carrying a forward current gets warm but emits almost no light. Where does the energy supplied by the source go?

Exercises

The practice exercises of this annex are in the companion **Exercise collection**, Annex A (Problems A.1 to A.6), with their answers; the worked solutions are in the **Solutions** document.

Answers

Q1. With m the mass: $mg \times 2$ m (zero at the floor), $mg \times 1$ m (zero at the table), 0 (zero at the stone). The differences between two positions, the force, the speed reached after a given fall: none depends on the zero.

Q2. Confining the electron to a region of size r costs a kinetic energy of order $\hbar^2/2mr^2$, which grows faster than the Coulomb energy decreases as $r \to 0$. The total is lowest at $r = a_0$.

Q3. The total is $K + U = 13.6 - 27.2 = -13.6$ eV; the potential energy is negative and twice as large as K. A negative total means a bound state, below the zero chosen at infinite separation.

Q4. A thin shell at distance r has a volume $4\pi r^2 \mathrm{d}r$, which vanishes at the centre. The probability of the shell, $4\pi r^2 |\psi|^2 \mathrm{d}r$, combines the density, largest at the centre, with the volume, largest far away; their product peaks at a_0.

Q5. In a stationary state $|\psi|^2$ does not change with time: there is no oscillating charge to radiate. The classical orbit is an accelerated charge, which would radiate and lose energy.

Q6. The N levels coming from one atomic level crowd into a range of a few eV; the spacing, of order 10^{-22} eV, is far below kT and below any measurable resolution.

Q7. No. The diagram plots the energy of states against their wavenumber; it says nothing about positions. The difference of k matters for momentum conservation in transitions (Chapter 4), not for location.

Q8. The allowed states are the same; the occupations differ. The probability for a state at E_C is $n_i/N_C = 3.6 \times 10^{-10}$ in the pure sample and $N_D/N_C = 3.6 \times 10^{-4}$ in the doped one: a factor 10^6.

Q9. Because at each collision the electron gives its excess back to the lattice. In a moderate field the gain per flight is a few meV, far below kT, so the electron stays close to thermal equilibrium; the energy ends as heat in the lattice.

Q10. Into the lattice, as heat: in silicon almost every injected pair recombines through defects, non-radiatively.

References

- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — ch. 2: energy quanta, wave–particle duality and the uncertainty principle (§2.1, p. 26), the physical meaning of the wave function (§2.2.2, p. 32), the one-electron atom (§2.4.1, p. 46); ch. 3: formation of energy bands (§3.1.1, p. 59), the k-space diagram (§3.1.3, p. 67), effective mass (§3.2.3, p. 75), the Fermi–Dirac function (§3.5.2, p. 91); ch. 4: the hydrogen-like ionisation energy of donors (§4.2.2, p. 120); ch. 6: generation and recombination (§6.1, p. 193).
- S. M. Sze and M. K. Lee, *Semiconductor Devices: Physics and Technology*, 3rd ed., Wiley, 2012 — ch. 1: valence bonds and energy bands (§§1.3–1.4, p. 22–29); ch. 2: generation and recombination processes (§2.3, p. 56).

WHERE THIS IS USED

Chapter 2: the energy to break a bond and the hydrogen-like donor (Section A.2). Chapter 3: bands, $E - E_C$ as kinetic energy, the Fermi level (Sections A.4–A.6). Chapter 4: absorption, thermalisation and the recombination mechanisms (Section A.7). Chapters 5 to 8: bands that follow the potential, $E_C(x) = E_{C0} - q\varphi(x)$, a flat E_F at equilibrium and quasi-Fermi levels out of it, and the energy balance of diodes, LEDs and solar cells. Annex B applies the same ideas to the simplest device of all, a uniform rod carrying a current.

SEMICONDUCTOR DEVICES PHYSICS

ANNEX B

A Uniformly Biased Intrinsic Silicon Rod

Equilibrium · the bands tilt · boundary conditions at the contacts · continuity · drift and diffusion together · why the profiles are uniform · the drift current · quasi-Fermi levels

WHERE WE ARE

Chapter 1 derived Ohm's law for carriers of uniform density, and Chapter 5 used the same picture for an N bar connected to a battery: "the bar stays neutral, $n = N_D$ everywhere, and there is no diffusion". Why, exactly, is there no diffusion? It is tempting to answer "because the bar is uniform", but a voltage is applied, the bands are tilted, and nothing yet tells us that the carrier densities stay uniform.

This annex settles the question for the simplest semiconductor, a pure (intrinsic) silicon rod, which has electrons and holes in equal numbers. It identifies which assumption does the work. The answer is: not neutrality alone, and not continuity alone, but continuity and the drift–diffusion equations together with a **boundary condition at the contacts**, which must be stated. The tools are those of Chapter 1 (drift, diffusion, Einstein relation, continuity), Chapter 3 (n and p in terms of the band edges), Chapter 4 (electron and hole currents, continuity with generation and recombination) and Chapter 5 (the bands follow $-q\varphi$). Section B.9 uses the quasi-Fermi levels of Chapter 6.

ANNEX OUTLINE

B.1 The question and a physical picture
B.2 Start from thermal equilibrium
B.3 Apply a voltage: the band edges tilt
B.4 State the boundary condition
B.5 Continuity: a steady current cannot accumulate charge
B.6 Combine continuity with drift–diffusion
B.7 The current is a drift current
B.8 What each assumption does
B.9 What happens to the Fermi level?
Summary · Comprehension questions · Exercises · Answers · References

BY THE END OF THIS ANNEX YOU CAN

- explain why tilted bands alone do not tell whether the carrier densities vary along the rod;
- state the boundary condition at the contacts and say why it is an assumption, not a consequence of neutrality;
- derive from steady-state continuity that J_n and J_p are constant, and say what

this does not prove;

- solve the drift–diffusion equations in a uniform field, interpret the two terms of the solution, and show that the end conditions select the uniform profile;
- compute the drift current $J = qn_i(\mu_n + \mu_p)\mathcal{E}$ and the resistance of an intrinsic rod;
- show that the quasi-Fermi levels are parallel to the band edges, and relate their total drop to the applied voltage.

B.1 The question and a physical picture

THE QUESTION

A voltage V is applied across a rod of pure silicon. In steady state, is the current carried by drift, by diffusion, or by both? Are the electron and hole densities uniform along the rod?

The rod extends from $x = 0$ to $x = L$ between two metal contacts; the + terminal of the source is connected at $x = 0$ (Figure B.1a). The field then points from $x = 0$ towards $x = L$, the electrons drift towards $x = 0$ and the holes towards $x = L$, and both carry current in the $+x$ direction.

A picture before the equations. Think of the electrons as a crowd on a conveyor belt moving at the drift velocity $v_n = \mu_n\mathcal{E}$. A **uniform** crowd delivers the same flux at every point: nothing piles up anywhere, and the situation can last. Could a **non-uniform** crowd also last? Drift carries any bump along, and diffusion spreads it out; a non-uniform shape can stay unchanged only if, at every point, diffusion exactly cancels drift. There is one such shape. The field raises the electron energy $E_C(x)$ steadily along the rod, and the electrons can crowd towards the low-energy end the way air crowds at the bottom of the atmosphere, their density falling by a factor e over the distance

$$\ell = \frac{kT}{q\mathcal{E}}, \tag{B.1}$$

over which the field changes their energy by kT. This 'barometric' layer is an equilibrium-like arrangement in which drift and diffusion balance: it carries **no current of its own**. The general steady state is therefore a uniform drifting crowd, which carries the current, plus possibly such a layer, which carries nothing. Whether a layer exists is not decided inside the rod; it is decided at its ends. The rest of this annex turns this picture into equations, step by step.

For a rod of 1 cm under 10 V, $\mathcal{E}$ = 10 V/cm and ℓ = 0.0259/10 = 26 µm, about $L/390$: any such layer would be a thin skin at one end.

B.2 Start from thermal equilibrium

With no applied voltage and a uniform temperature, the rod is at thermal equilibrium. There is no macroscopic field inside it; the band edges E_C and E_V and the Fermi level E_F are flat (Figure B.1b), and the pure material has

$$n = p = n_i \tag{B.2}$$

everywhere. This is the reference state.

B.3 Apply a voltage: the band edges tilt

Apply the voltage and wait for the end of the initial transient (Chapter 1: in silicon it lasts a dielectric relaxation time, a tiny fraction of a second). In the idealised bulk of the rod, take the field to be uniform,

$$\mathcal{E} = -\frac{\mathrm{d}\varphi}{\mathrm{d}x} = \text{constant}. \tag{B.3}$$

With the + terminal at $x = 0$, φ decreases along x and $\mathcal{E} > 0$. The band edges follow the electrostatic potential (Chapter 5; Annex A, Section A.5):

$$E_C(x) = E_{C0} - q\varphi(x), \qquad E_V(x) = E_{V0} - q\varphi(x), \qquad \frac{\mathrm{d}E_C}{\mathrm{d}x} = \frac{\mathrm{d}E_V}{\mathrm{d}x} = q\mathcal{E}. \tag{B.4}$$

The two band edges tilt with the same slope, and the gap is unchanged (Figure B.1c). From $x = 0$ to $x = L$ they rise by $q\mathcal{E}L = qV$.

> **BE AWARE OF…**
>
> The tilt of the bands, by itself, does **not** tell us whether n and p vary with position. The bands say what energy a carrier would have at each point; how many carriers are there is a separate question, which needs the transport equations and the boundary conditions.

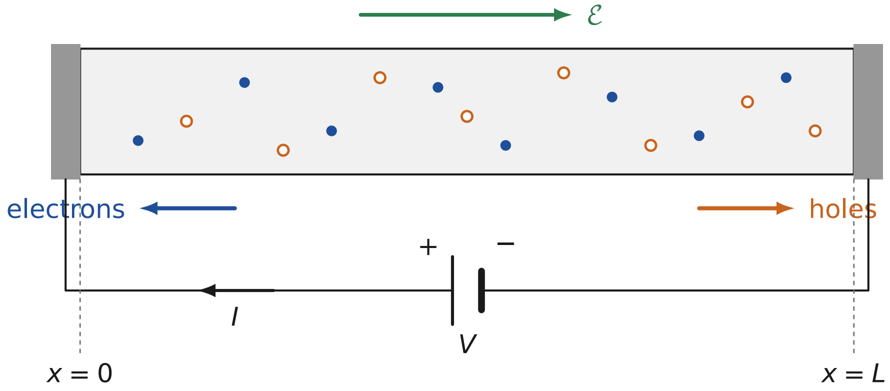
B.4 State the boundary condition

For the idealised problem we impose **equal carrier densities at the two ends**:

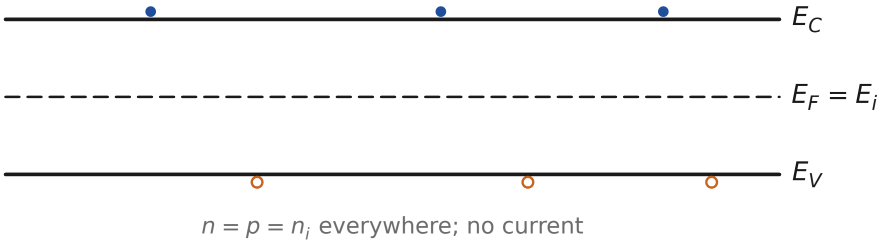
$$n(0) = n(L) \qquad \text{and} \qquad p(0) = p(L). \tag{B.5}$$

These are conditions on the rod **and its contacts**, and they must be stated explicitly. They describe ideal contacts that impose no difference of concentration between the two ends; ohmic contacts that hold the densities at their equilibrium values, $n = p = n_i$, as in the convention used throughout the course, are the natural example. They do not follow from charge neutrality alone: neutrality relates n and p at the same point, and says nothing about how the density at one end compares with the density at the other. The next two sections use steady-state continuity and the drift–diffusion equations to find the profiles inside the rod.

(a) the rod and its source


$\mathcal{E}$
electrons
holes
+
−
I
V
$x = 0$
$x = L$

(b) no voltage: equilibrium


E_C
$E_F = E_i$
E_V
$n = p = n_i$ everywhere; no current

(c) voltage V: the bands tilt

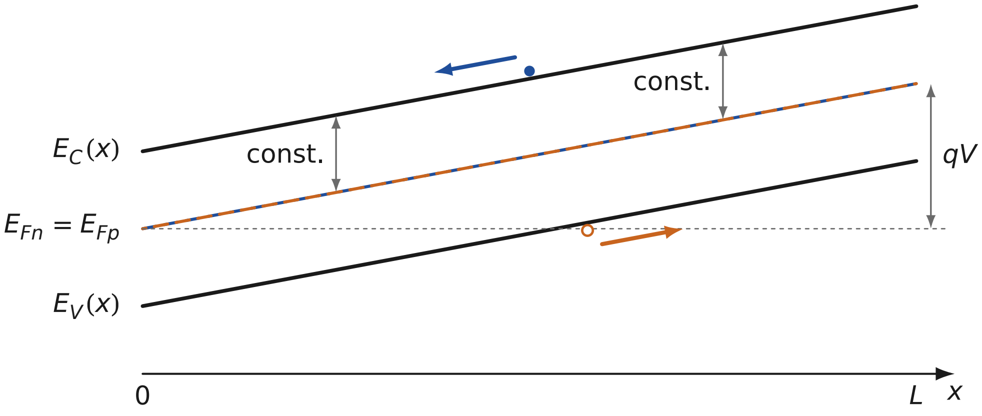

const.
$E_C(x)$
const.
qV
$E_{Fn} = E_{Fp}$
$E_V(x)$
0
L
x

Figure B.1 – *(a) An intrinsic silicon rod between two metal contacts, the + terminal at $x = 0$: the field points towards $x = L$, electrons drift towards $x = 0$, holes towards $x = L$, and the current I flows through the rod from $x = 0$ to $x = L$. (b) At equilibrium the bands and E_F are flat and $n = p = n_i$. (c) Under the voltage V, $E_C(x)$ and $E_V(x)$ tilt with the same slope $q\mathcal{E}$ and rise by qV from $x = 0$ to $x = L$; the quasi-Fermi levels coincide and tilt parallel to them, at a constant distance from each band edge (the result of Section B.9). Generic: the tilt is greatly exaggerated.*

B.5 Continuity: a steady current cannot accumulate charge

The one-dimensional continuity equations of Chapter 4, written with the conventional current densities, are

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G_n - R_n, \qquad \frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G_p - R_p. \tag{B.6}$$

In steady state the densities no longer change with time. If, in addition, there is no net generation or recombination in the bulk for either carrier, then

$$\frac{\mathrm{d}J_n}{\mathrm{d}x} = 0, \quad \frac{\mathrm{d}J_p}{\mathrm{d}x} = 0 \qquad \Longrightarrow \qquad J_n,\ J_p \text{ and } J = J_n + J_p \text{ constant along the rod.} \tag{B.7}$$

The physical meaning is direct: in steady operation, the current entering a slice must equal the current leaving it, otherwise charge would accumulate in the slice and the state would not be steady.

> **WHAT CONTINUITY DOES NOT PROVE**
>
> This argument establishes that each current is **constant**. It does not establish that the **concentrations** are uniform: a constant current could, in principle, be the sum of a drift part and a diffusion part that vary in opposite ways along the rod.

B.6 Combine continuity with drift–diffusion

For a uniform temperature and constant transport coefficients, the electron and hole currents are (Chapters 1 and 4)

$$J_n = q\mu_n n\mathcal{E} + qD_n\frac{\mathrm{d}n}{\mathrm{d}x}, \qquad J_p = q\mu_p p\mathcal{E} - qD_p\frac{\mathrm{d}p}{\mathrm{d}x}. \tag{B.8}$$

With a uniform field, the conditions $\mathrm{d}J_n/\mathrm{d}x = 0$ and $\mathrm{d}J_p/\mathrm{d}x = 0$ of (B.7) give one equation for each profile:

$$D_n\frac{\mathrm{d}^2 n}{\mathrm{d}x^2} + \mu_n\mathcal{E}\frac{\mathrm{d}n}{\mathrm{d}x} = 0, \qquad D_p\frac{\mathrm{d}^2 p}{\mathrm{d}x^2} - \mu_p\mathcal{E}\frac{\mathrm{d}p}{\mathrm{d}x} = 0. \tag{B.9}$$

Their general solutions are a constant plus an exponential. With the Einstein relation $D/\mu = kT/q$, the same for both carriers, the decay length of the exponential is the length $\ell = kT/q\mathcal{E}$ of (B.1) for both:

$$n(x) = C_1 + C_2\,\mathrm{e}^{-x/\ell}, \qquad p(x) = C_1' + C_2'\,\mathrm{e}^{+x/\ell}, \qquad \ell = \frac{D_n}{\mu_n\mathcal{E}} = \frac{D_p}{\mu_p\mathcal{E}} = \frac{kT}{q\mathcal{E}}. \tag{B.10}$$

The two terms are the two parts of the physical picture. Put the exponential term alone into the electron current (B.8): its derivative is $-C_2\mathrm{e}^{-x/\ell}/\ell$, and

$$q\mu_n\mathcal{E}\,C_2\mathrm{e}^{-x/\ell} - qD_n\frac{C_2\mathrm{e}^{-x/\ell}}{\ell} = q\mu_n\mathcal{E}\,C_2\mathrm{e}^{-x/\ell}\left(1 - \frac{D_n}{\mu_n\mathcal{E}\ell}\right) = 0.$$

The exponential term carries **no current**: drift and diffusion cancel in it exactly. It is the barometric layer of Section B.1: since $E_C(x)$ rises as $q\mathcal{E}x$, $\mathrm{e}^{-x/\ell} = \mathrm{e}^{-[E_C(x)-E_C(0)]/kT}$, the Boltzmann distribution of electrons in the tilted band. The holes have the mirror layer,

$e^{+x/\ell}$, crowded at the other end, where E_V is highest and the hole energy lowest. The current is carried entirely by the constant terms: $J_n = q\mu_n \mathcal{E}\, C_1$ and $J_p = q\mu_p \mathcal{E}\, C'_1$.

The ends decide. The boundary condition $n(0) = n(L)$ gives $C_2\left(1 - e^{-L/\ell}\right) = 0$. For a nonzero field, $e^{-L/\ell} \neq 1$, so $C_2 = 0$ and n is uniform. The hole equation with $p(0) = p(L)$ likewise gives $C'_2 = 0$ (Figure B.2):

$$n(x) = n(0) = n(L), \qquad p(x) = p(0) = p(L). \tag{B.11}$$

The uniform profile is not inserted to delete the diffusion term. It **follows** from the stated boundary conditions together with the steady-state transport equations, under the assumptions listed so far.

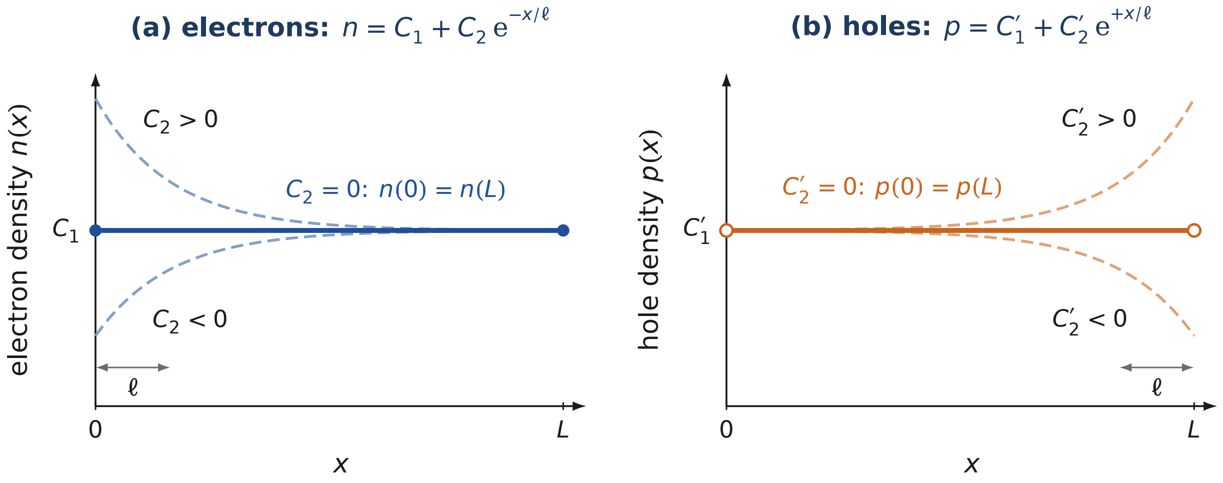


Figure B.2 – *The steady-state profiles allowed by drift and diffusion in a uniform field (generic, linear axes; ℓ drawn much larger than in a real rod, where it is a few hundredths of L or less). (a) Electrons: a uniform part C_1, which carries the current, plus a layer $C_2\, e^{-x/\ell}$ at the low-energy end $x = 0$, which carries none. (b) Holes: the mirror layer at $x = L$. Only the uniform profiles (thick lines) satisfy $n(0) = n(L)$ and $p(0) = p(L)$.*

B.7 The current is a drift current

Since the concentrations are uniform, their derivatives vanish, the diffusion terms of (B.8) are zero, and

$$J_n = q\mu_n n \mathcal{E}, \qquad J_p = q\mu_p p \mathcal{E}, \qquad J = q(\mu_n n + \mu_p p)\mathcal{E}. \tag{B.12}$$

With ideal ohmic contacts the end densities are the equilibrium ones, so $n = p = n_i$ in the whole rod, and

$$\boxed{J = q n_i (\mu_n + \mu_p)\, \mathcal{E}} \tag{B.13}$$

This is the drift current of the idealised intrinsic rod: Ohm's law of Chapter 1, $J = \sigma\mathcal{E}$, with $\sigma = q n_i(\mu_n + \mu_p)$, now justified for a material with two carriers and a tilted band diagram.

Two consistency checks close the argument. First, the assumption of no net generation or recombination is satisfied by the result: with $n = p = n_i$ everywhere, $np = n_i^2$, which is exactly the condition for $G = R$ (Chapter 4). Second, $n = p$ everywhere means zero net charge, so Gauss's law gives $\mathrm{d}\mathcal{E}/\mathrm{d}x = 0$: the field is uniform, as assumed. A barometric layer would not pass this test. The electron layer would sit at $x = 0$ and the hole layer at $x = L$; they could not neutralise each other, and their charge would bend the field, contradicting the starting assumption.

EXAMPLE B.1 — A ONE-CENTIMETRE ROD OF PURE SILICON

Take $L = 1$ cm, a cross-section $S = 1\ \mathrm{mm}^2 = 0.01\ \mathrm{cm}^2$, $V = 10$ V, and the values of the course: $n_i = 10^{10}\ \mathrm{cm}^{-3}$, $\mu_n = 1350$ and $\mu_p = 460\ \mathrm{cm}^2/(\mathrm{V{\cdot}s})$. The field is $\mathcal{E} = 10$ V/cm.

- Resistivity $\rho = 1/[qn_i(\mu_n + \mu_p)] = 3.45 \times 10^5\ \Omega{\cdot}\mathrm{cm}$; resistance $R = \rho L/S = 35\ \mathrm{M}\Omega$.
- $J = 1.602 \times 10^{-19} \times 10^{10} \times 1810 \times 10 = 2.9 \times 10^{-5}\ \mathrm{A/cm}^2$; $I = JS = 0.29\ \mu\mathrm{A}$.
- The electrons carry $\mu_n/(\mu_n + \mu_p) = 75$ % of the current, the holes 25 %. The electrons drift at $\mu_n\mathcal{E} = 1.35 \times 10^4$ cm/s and cross the rod in 74 µs.
- $\ell = kT/q\mathcal{E} = 26\ \mu\mathrm{m} \approx L/390$.
- The source delivers $VI = 2.9\ \mu\mathrm{W}$, all of it given to the lattice as heat (Annex A, Section A.5).

For comparison, the N bar of Chapter 5 with $N_D = 10^{16}\ \mathrm{cm}^{-3}$ carries, in the same field, $qN_D\mu_n\mathcal{E} = 22\ \mathrm{A/cm}^2$, 7.5×10^5 times more: pure silicon is a poor conductor because it has few carriers, not because they move slowly.

B.8 What each assumption does

The equal-end condition describes the idealised contact situation in which the two ends sustain the same carrier populations in the steady state. Continuity explains why a steady through-current is possible: if the current entering a slice differed from the current leaving it, carriers would accumulate and the state would change. Combined with the uniform field and the transport equations, the equal end concentrations rule out the exponential layer and select the uniform solution.

Keep the logic precise. **Non-accumulation by itself implies a constant current, not $n(0) = n(L)$ or $p(0) = p(L)$.** The equality of the end concentrations is a boundary condition of this idealised problem, physically appropriate when the contacts impose no difference of concentration. The continuity and transport equations then show that this condition leads to uniform concentrations in the rod (Figure B.3).

SCOPE OF THE MODEL

The derivation assumes a homogeneous rod, a uniform temperature and field, constant transport coefficients, steady state, no net generation or recombination in the bulk, and ideal contacts that impose equal carrier concentrations at the two ends. Outside the model: contacts that inject or extract carriers create a layer of thickness about ℓ at one end (Problem B.4 of the exercise collection); strong fields make the mobility field-dependent (velocity saturation, Annex A, Section A.5); a large current heats the rod and raises n_i; spatially varying doping or composition changes the bands themselves. Each of these can invalidate the uniform-profile conclusion.

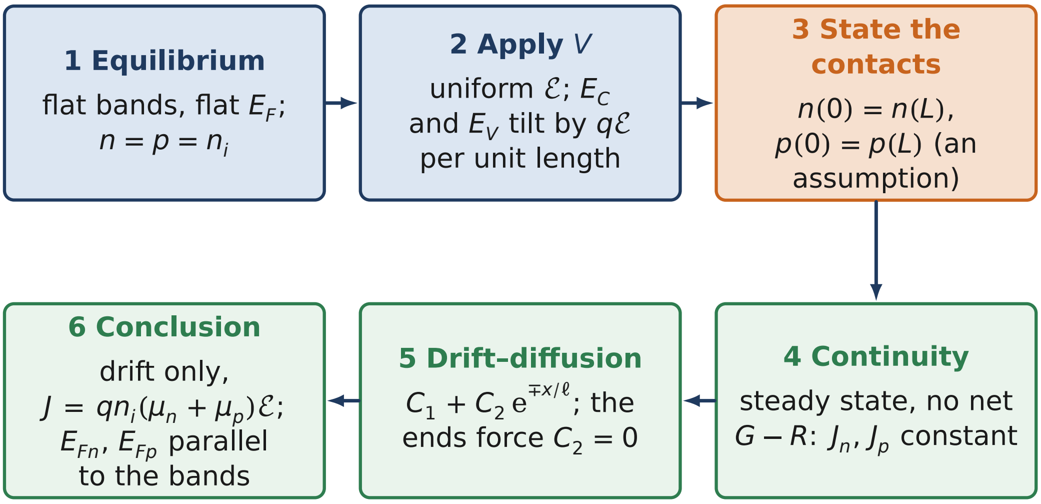

1 Equilibrium
flat bands, flat E_F; $n = p = n_i$
2 Apply V
uniform $\mathcal{E}$; E_C and E_V tilt by $q\mathcal{E}$ per unit length
3 State the contacts
$n(0) = n(L)$, $p(0) = p(L)$ (an assumption)
4 Continuity
steady state, no net $G - R$: J_n, J_p constant
5 Drift–diffusion
$C_1 + C_2\,\mathrm{e}^{\mp x/\ell}$; the ends force $C_2 = 0$
6 Conclusion
drift only, $J = qn_i(\mu_n + \mu_p)\mathcal{E}$; E_{Fn}, E_{Fp} parallel to the bands

Figure B.3 *– The logical sequence of this annex. The orange step is an assumption about the contacts; the green steps are consequences of continuity and of the drift–diffusion equations once that assumption is made.*

B.9 What happens to the Fermi level?

> ***THE QUESTION***
>
> *At equilibrium the Fermi level was flat. A current now flows: what becomes of it?*

Under current the rod is not at thermal equilibrium, and the occupation of each band is described by its own level, the quasi-Fermi levels E_{Fn} and E_{Fp} of Chapter 6. For nondegenerate carriers at uniform temperature and constant effective density of states,

$$n = N_C \exp\left[-\frac{E_C - E_{Fn}}{kT}\right], \qquad p = N_V \exp\left[-\frac{E_{Fp} - E_V}{kT}\right]. \tag{B.14}$$

Taking the logarithm of the first and differentiating with respect to x,

$$\frac{\mathrm{d}E_{Fn}}{\mathrm{d}x} = \frac{\mathrm{d}E_C}{\mathrm{d}x} + \frac{kT}{n}\frac{\mathrm{d}n}{\mathrm{d}x}. \tag{B.15}$$

Since n is uniform, $\mathrm{d}n/\mathrm{d}x = 0$ and, with (B.4),

$$\frac{\mathrm{d}E_{Fn}}{\mathrm{d}x} = \frac{\mathrm{d}E_C}{\mathrm{d}x} = q\mathcal{E}, \qquad \text{and similarly} \qquad \frac{\mathrm{d}E_{Fp}}{\mathrm{d}x} = \frac{\mathrm{d}E_V}{\mathrm{d}x} = q\mathcal{E}. \tag{B.16}$$

The quasi-Fermi levels tilt parallel to their band edges. Their distances to the band edges stay constant, which is just another way of saying that the concentrations are uniform (Figure B.1c).

In the intrinsic rod with ohmic contacts, $n = p = n_i$ gives $E_{Fn} = E_{Fp} = E_i(x)$: the two quasi-Fermi levels coincide, and the rod has a single local Fermi level that tilts with the bands. This is why Chapter 5 could speak of “the local Fermi level of each slice” for its N bar. Three readings follow.

- **The slope gives the current.** Chapter 5 wrote the current as $J_n = \mu_n n\,\mathrm{d}E_{Fn}/\mathrm{d}x$; here it gives $J_n = q\mu_n n_i\mathcal{E}$, the same as (B.12); likewise for holes.
- **The total drop is the applied voltage.** From $x = 0$ to $x = L$ the level rises by $q\mathcal{E}L = qV$; a voltmeter connected to the two contacts reads the difference of the Fermi levels divided by q, that is V.

- **The energy balance.** Each electron that crosses the rod descends qV in energy, and each hole does the same in its own band; the field gives this energy and the lattice receives it (Annex A).

Summary — essential knowledge

WHAT YOU MUST KNOW

- **Equilibrium:** flat bands, flat E_F, $n = p = n_i$.
- **Apply V:** uniform $\mathcal{E}$; E_C and E_V tilt with the same slope $\mathrm{d}E_C/\mathrm{d}x = q\mathcal{E}$; the gap is unchanged. The tilt alone says nothing about $n(x)$.
- **Boundary condition (assumption):** $n(0) = n(L)$, $p(0) = p(L)$, e.g. ohmic contacts holding $n = p = n_i$. Not a consequence of neutrality.
- **Continuity:** steady state, no net $G - R$: J_n, J_p constant. Not, by itself, uniform densities.
- **Drift–diffusion:** $n = C_1 + C_2 \mathrm{e}^{-x/\ell}$, $p = C'_1 + C'_2 \mathrm{e}^{+x/\ell}$, $\ell = kT/q\mathcal{E}$. The exponential terms are barometric layers that carry no current; the end conditions force $C_2 = C'_2 = 0$.
- **Result:** uniform densities, no diffusion, $J = qn_i(\mu_n + \mu_p)\mathcal{E}$; consistent with $np = n_i^2$ ($G = R$) and with Gauss (no charge, uniform field).
- **Fermi levels:** $\mathrm{d}E_{Fn}/\mathrm{d}x = \mathrm{d}E_C/\mathrm{d}x + (kT/n)\mathrm{d}n/\mathrm{d}x = q\mathcal{E}$; quasi-Fermi levels parallel to the bands; in the intrinsic rod $E_{Fn} = E_{Fp} = E_i(x)$, total drop qV.

Comprehension questions

Q1. Why does the tilt of the bands under bias not tell us, by itself, whether the electron density varies along the rod?

Q2. What does steady-state continuity establish, and what does it not establish?

Q3. What does the term $C_2\,\mathrm{e}^{-x/\ell}$ represent physically, and why does it carry no current? At which end of the rod would it sit, and why?

Q4. Why is $n(0) = n(L)$ an assumption rather than a consequence of neutrality? Which contacts realise it?

Q5. In this rod E_{Fn} and E_{Fp} coincide. Why? Why are they parallel to the band edges?

Q6. Where does the energy delivered by the source go?

Exercises

The practice exercises of this annex are in the companion **Exercise collection**, Annex B (Problems B.1 to B.6), with their answers; the worked solutions are in the **Solutions** document.

Answers

Q1. The bands give the energy a carrier would have at each point; the number of carriers depends on transport and on the boundary conditions. A tilted band is compatible with a uniform density (the result here) and with a barometric layer.

Q2. That J_n and J_p are constant along the rod. Not that n and p are uniform: a constant current could be the sum of drift and diffusion parts varying in opposite ways.

Q3. An equilibrium-like (Boltzmann) arrangement of electrons in the tilted band, in which drift and diffusion cancel exactly. It would sit at $x = 0$, where E_C is lowest, within about $\ell = kT/q\mathcal{E}$ of the contact.

Q4. Neutrality links n and p at one point, not the values at the two ends. Contacts that impose no concentration difference, for instance ohmic contacts holding the equilibrium densities, realise it.

Q5. Because $n = p = n_i$: both levels coincide with $E_i(x)$. They are parallel to the band edges because their distance to each edge is fixed by a density that does not vary.

Q6. To the lattice, as heat: the carriers gain energy from the field during their free flights and give it up at each collision; $P = VI$.

References

- D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed., McGraw-Hill, 2012 — ch. 5: drift current density and conductivity (§5.1, p. 157–169), diffusion and total current density (§5.2, p. 172–175), the Einstein relation (§5.3.2, p. 178); ch. 6: continuity equations (§6.2.1, p. 198), quasi-Fermi energy levels (§6.4, p. 219).
- S. M. Sze and M. K. Lee, *Semiconductor Devices: Physics and Technology*, 3rd ed., Wiley, 2012 — ch. 2: carrier drift (§2.1, p. 43), carrier diffusion (§2.2, p. 53), the continuity equation (§2.4, p. 62).

WHERE THIS IS USED

Chapter 1 (Ohm’s law for a uniform conductor), Chapter 5 (the N bar connected to a battery, Figure 5.4a, and the current written with the slope of E_F), Chapter 6 (the small ohmic field that carries the majority current in the neutral regions of a diode), and every neutral region of the diode and the transistor that is treated as a resistor. The method, state the boundary conditions, use continuity, then solve drift–diffusion, is the method of the device chapters.